

johnston-linear-matrix-algebra: equation evidence

Display equations

Identifier	Page	Conf.	LaTeX source	Rendered	Image
johnston-linear-matrix-algebra-EQ0528 [conf 0.001]	212	0.001	$\begin{array}{r} \leftarrow \begin{array}{c} S_2 \quad S_3 \\ R_3 + R_4 \\ R_4 + R_4 \end{array} \begin{array}{c} z \\ R_1 \end{array} \rightarrow \left[\begin{array}{c ccc ccc} 1 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 & 1 & 1 & 0 & 2 \\ 0 & 0 & 1 & 0 & -1 & -1 & -1 & -1 \\ 0 & 0 & 0 & 1 & -2 & 1 & -1 & -2 \end{array} \right] \begin{array}{c} 2 \\ \text{pivot} \end{array} \end{array}$		
johnston-linear-matrix-algebra-EQ0486 [conf 0.011]	196	0.011	$\begin{array}{r} \begin{array}{cccccccc} v & + & w & + & x & + & y & - \\ v & - & 2w & + & 2x & + & 2y & + \\ -v & + & w & + & x & & & + \\ & & & + & z & & & \\ 2w & & - & x & + & y & - & 3z \\ 2v & & & - & 2y & + & z & - \end{array} \end{array}$	$\begin{array}{r} \begin{array}{cccccccc} v & + & w & + & x & + & y & - & z & - & 1 \\ v & - & 2w & + & 2x & + & 2y & + & 2z & + & 2 \\ -v & + & w & + & x & & & + & z & & \\ & & 2w & - & x & + & y & - & 3z & + & 3 \\ 2v & + & w & & & - & 2y & + & z & - & 4 \end{array} \end{array}$	
johnston-linear-matrix-algebra-EQ1276 [conf 0.022]	427	0.022	$\left(3x^2 - 5x - 2 \right) \frac{x^2 + 3x + 1}{3x^4 + 4x^3 - 14x^2 - 11x - 2} \frac{3x^4 - 5x^3 - 2x^2}{9x^3 - 12x^2 - 11x - 2} \frac{9x^3 - 15x^2 - 6x}{3x^2 - 5x - 2} \frac{3x^2 - 5x - 2}{0} \right)$	$\left(3x^2 - 5x - 2 \right) \frac{x^2 + 3x + 1}{3x^4 + 4x^3 - 14x^2 - 11x - 2} \frac{3x^4 - 5x^3 - 2x^2}{9x^3 - 12x^2 - 11x - 2} \frac{9x^3 - 15x^2 - 6x}{3x^2 - 5x - 2} \frac{3x^2 - 5x - 2}{0} \right)$	

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johnston-linear-matrix-algebra-EQ0114 [conf 0.025]	63	0.025	<pre>\begin{figure} \[R^{\theta}\left(\mathbf{e}_{2}\right)=(-\sin (\theta), \cos (\theta)) \overbrace{\cos (\theta)}^{\cos (\theta)} \overbrace{\sin (\theta)}^{\sin (\theta)} \times \text { - } \underbrace{\cos (\theta)}_{R^{\theta}\left(\mathbf{e}_{1}\right)\left(\cos (\theta), \sin (\theta)\right)} \] \captionsetup{labelformat=empty} \caption{Figure 1.19: \((R^{\theta})\) rotates the standard basis vectors \((\mathbf{e}_1)\) and \((\mathbf{e}_2)\) to \((R^{\theta}(\mathbf{e}_1))\) and \((R^{\theta}(\mathbf{e}_2))\) respectively.} \end{figure}</pre>	(not rendered)	
johnston-linear-matrix-algebra-EQ0014 [conf 0.071]	27	0.071	<pre>\left.\begin{array}{rl} (\mathbf{v}+\mathbf{w}) \cdot(\mathbf{x}+\mathbf{y}) & =(\mathbf{v}+\mathbf{w}) \cdot \mathbf{x}+(\mathbf{v}+\mathbf{w}) \cdot \mathbf{y} \\ & =\mathbf{x} \cdot(\mathbf{v}+\mathbf{w})+\mathbf{y} \cdot(\mathbf{v}+\mathbf{w}) \\ & =\mathbf{x} \cdot \mathbf{v}+\mathbf{x} \cdot \mathbf{w}+\mathbf{y} \cdot \mathbf{v}+\mathbf{y} \cdot \mathbf{w} \\ & =\mathbf{v} \cdot \mathbf{x}+\mathbf{w} \cdot \mathbf{x}+\mathbf{v} \cdot \mathbf{y}+\mathbf{w} \cdot \mathbf{y} \end{array}\right) \quad \left(\begin{array}{l} \text { property (a) } \\ \text { property (b) } \end{array}\right)</pre>	$\left.\begin{array}{rl} (\mathbf{v}+\mathbf{w}) \cdot(\mathbf{x}+\mathbf{y}) & =(\mathbf{v}+\mathbf{w}) \cdot \mathbf{x}+(\mathbf{v}+\mathbf{w}) \cdot \mathbf{y} \\ & =\mathbf{x} \cdot(\mathbf{v}+\mathbf{w})+\mathbf{y} \cdot(\mathbf{v}+\mathbf{w}) \\ & =\mathbf{x} \cdot \mathbf{v}+\mathbf{x} \cdot \mathbf{w}+\mathbf{y} \cdot \mathbf{v}+\mathbf{y} \cdot \mathbf{w} \\ & =\mathbf{v} \cdot \mathbf{x}+\mathbf{w} \cdot \mathbf{x}+\mathbf{v} \cdot \mathbf{y}+\mathbf{w} \cdot \mathbf{y} \end{array}\right) \quad \left(\begin{array}{l} \text { property (a) } \\ \text { property (b) } \end{array}\right)$	$\begin{aligned} (\mathbf{v}+\mathbf{w}) \cdot(\mathbf{x}+\mathbf{y}) & =(\mathbf{v}+\mathbf{w}) \cdot \mathbf{x}+(\mathbf{v}+\mathbf{w}) \cdot \mathbf{y} && \text { (property (b))} \\ & =\mathbf{x} \cdot(\mathbf{v}+\mathbf{w})+\mathbf{y} \cdot(\mathbf{v}+\mathbf{w}) && \text { (property (a))} \\ & =\mathbf{x} \cdot \mathbf{v}+\mathbf{x} \cdot \mathbf{w}+\mathbf{y} \cdot \mathbf{v}+\mathbf{y} \cdot \mathbf{w} && \text { (property (b))} \\ & =\mathbf{v} \cdot \mathbf{x}+\mathbf{w} \cdot \mathbf{x}+\mathbf{v} \cdot \mathbf{y}+\mathbf{w} \cdot \mathbf{y} . && \text { (property (a))} \end{aligned}$
johnston-linear-matrix-algebra-EQ0005 [conf 0.092]	22	0.092	<pre>\begin{array}{lrl} & \& \& \mathbf{x}-(3,2,1) & = & (1,2,3)-3 \mathbf{x} \\ \mathbf{x} & & & & & \\ 4 \mathbf{x} & & & & & \\ \mathbf{x} & & & & & \\ \text { (divide both sides by 4) } & & & & & \end{array}</pre>	$\begin{aligned} \mathbf{x}-(3,2,1) & = (1,2,3)-3 \mathbf{x} \\ \Rightarrow \mathbf{x} & = (4,4,4)-3 \mathbf{x} \quad \text { (add } (3,2,1) \text { to both sides)} \\ \Rightarrow 4 \mathbf{x} & = (4,4,4) \quad \text { (add } 3 \mathbf{x} \text { to both sides)} \\ \Rightarrow \mathbf{x} & = (1,1,1) . \quad \text { (divide both sides by 4)} \end{aligned}$	$\begin{aligned} \mathbf{x}-(3,2,1) & = (1,2,3)-3 \mathbf{x} \\ \Rightarrow \mathbf{x} & = (4,4,4)-3 \mathbf{x} \quad \text { (add } (3,2,1) \text { to both sides)} \\ \Rightarrow 4 \mathbf{x} & = (4,4,4) \quad \text { (add } 3 \mathbf{x} \text { to both sides)} \\ \Rightarrow \mathbf{x} & = (1,1,1) . \quad \text { (divide both sides by 4)} \end{aligned}$

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johnston-linear-matrix-algebra- EQ0947 [conf 0.100]	336	0.100	$\begin{array}{l} \mathbf{p}_1 \mathbf{q}_1^* = \frac{1}{3} \begin{bmatrix} 3 \\ 1 \\ 0 \end{bmatrix} \begin{bmatrix} 1 & 0 & -1 \end{bmatrix} = \frac{1}{3} \begin{bmatrix} 3 & 0 & -3 \\ 1 & 0 & -1 \\ 0 & 0 & 0 \end{bmatrix} \\ \mathbf{p}_2 \mathbf{q}_2^* = \frac{1}{3} \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} \begin{bmatrix} -1 & 3 & 1 \end{bmatrix} = \frac{1}{3} \begin{bmatrix} 0 & 0 & 0 \\ -1 & 3 & 1 \\ 0 & 0 & 0 \end{bmatrix}, \text{ and} \\ \mathbf{p}_3 \mathbf{q}_3^* = \frac{1}{3} \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix} \begin{bmatrix} 0 & 0 & 3 \end{bmatrix} = \frac{1}{3} \begin{bmatrix} 0 & 0 & 3 \\ 0 & 0 & 0 \\ 0 & 0 & 3 \end{bmatrix}. \end{array}$	$\begin{array}{l} \mathbf{p}_1 \mathbf{q}_1^* = \frac{1}{3} \begin{bmatrix} 3 \\ 1 \\ 0 \end{bmatrix} \begin{bmatrix} 1 & 0 & -1 \end{bmatrix} = \frac{1}{3} \begin{bmatrix} 3 & 0 & -3 \\ 1 & 0 & -1 \\ 0 & 0 & 0 \end{bmatrix} \\ \mathbf{p}_2 \mathbf{q}_2^* = \frac{1}{3} \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} \begin{bmatrix} -1 & 3 & 1 \end{bmatrix} = \frac{1}{3} \begin{bmatrix} 0 & 0 & 0 \\ -1 & 3 & 1 \\ 0 & 0 & 0 \end{bmatrix}, \text{ and} \\ \mathbf{p}_3 \mathbf{q}_3^* = \frac{1}{3} \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix} \begin{bmatrix} 0 & 0 & 3 \end{bmatrix} = \frac{1}{3} \begin{bmatrix} 0 & 0 & 3 \\ 0 & 0 & 0 \\ 0 & 0 & 3 \end{bmatrix}. \end{array}$	$\begin{array}{l} \mathbf{p}_1 \mathbf{q}_1^* = \frac{1}{3} \begin{bmatrix} 3 \\ 1 \\ 0 \end{bmatrix} \begin{bmatrix} 1 & 0 & -1 \end{bmatrix} = \frac{1}{3} \begin{bmatrix} 3 & 0 & -3 \\ 1 & 0 & -1 \\ 0 & 0 & 0 \end{bmatrix} \\ \mathbf{p}_2 \mathbf{q}_2^* = \frac{1}{3} \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} \begin{bmatrix} -1 & 3 & 1 \end{bmatrix} = \frac{1}{3} \begin{bmatrix} 0 & 0 & 0 \\ -1 & 3 & 1 \\ 0 & 0 & 0 \end{bmatrix}, \text{ and} \\ \mathbf{p}_3 \mathbf{q}_3^* = \frac{1}{3} \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix} \begin{bmatrix} 0 & 0 & 3 \end{bmatrix} = \frac{1}{3} \begin{bmatrix} 0 & 0 & 3 \\ 0 & 0 & 0 \\ 0 & 0 & 3 \end{bmatrix}. \end{array}$
johnston-linear-matrix-algebra- EQ0543	218	0.103	$\begin{array}{l} \operatorname{minimize}: 3y_1 + 3y_2 + 3y_3 + 3y_4 + 3y_5 \\ \text{subject to: } y_1 + y_4 + y_5 \geq 1 \text{ \& } y_1 + y_2 + y_3 \geq 1 \text{ \& } y_1 + y_2 + y_5 \geq 1 \text{ \& } y_1 + y_2 + y_3 \geq 1 \\ y_2 + y_3 + y_4 \geq 1 \text{ \& } y_2 + y_3 + y_4 \geq 1 \text{ \& } y_2 + y_3 + y_4 \geq 1 \text{ \& } y_2 + y_3 + y_4 \geq 1 \\ y_3 + y_4 + y_5 \geq 1 \text{ \& } y_3 + y_4 + y_5 \geq 1 \text{ \& } y_3 + y_4 + y_5 \geq 1 \text{ \& } y_3 + y_4 + y_5 \geq 1 \\ y_2, y_3, y_4, y_5 \geq 0 \end{array}$	$\begin{array}{l} \operatorname{minimize}: 3y_1 + 3y_2 + 3y_3 + 3y_4 + 3y_5 \\ \text{subject to: } y_1 + y_4 + y_5 \geq 1 \\ y_1 + y_2 + y_3 \geq 1 \\ y_1 + y_2 + y_5 \geq 1 \\ y_2 + y_3 + y_4 \geq 1 \\ y_2 + y_3 + y_4 \geq 1 \\ y_2 + y_3 + y_4 \geq 1 \\ y_3 + y_4 + y_5 \geq 1 \\ y_3 + y_4 + y_5 \geq 1 \\ y_3 + y_4 + y_5 \geq 1 \\ y_2, y_3, y_4, y_5 \geq 0 \end{array}$	$\begin{array}{l} \operatorname{minimize}: 3y_1 + 3y_2 + 3y_3 + 3y_4 + 3y_5 \\ \text{subject to: } y_1 + y_4 + y_5 \geq 1 \\ y_1 + y_2 + y_3 \geq 1 \\ y_1 + y_2 + y_5 \geq 1 \\ y_2 + y_3 + y_4 \geq 1 \\ y_2 + y_3 + y_4 \geq 1 \\ y_2 + y_3 + y_4 \geq 1 \\ y_3 + y_4 + y_5 \geq 1 \\ y_3 + y_4 + y_5 \geq 1 \\ y_3 + y_4 + y_5 \geq 1 \\ y_2, y_3, y_4, y_5 \geq 0 \end{array}$

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johnston-linear-matrix-algebra-EQ0750 (0)	283	■ 0.105	<pre>\begin{gathered} \left[\begin{array}{cccc} a_{1,1} & \cdots & a_{1,n-1} & a_{1,n} \\ \vdots & \ddots & \vdots & \vdots \\ 0 & \cdots & a_{n-1,n-1} & a_{n-1,n} \\ 0 & \cdots & 0 & a_{n,n} \end{array}\right] \xrightarrow{\frac{1}{a_{n,n}}R_n} \left[\begin{array}{cccc} a_{1,1} & \cdots & a_{1,n-1} & a_{1,n} \\ \vdots & \ddots & \vdots & \vdots \\ 0 & \cdots & a_{n-1,n-1} & a_{n-1,n} \\ 0 & \cdots & 0 & 1 \end{array}\right] \\ \begin{array}{c} R_1 - a_{1,n}R_n \\ R_2 - a_{2,n}R_n \\ \vdots \\ R_{n-1} - a_{n-1,n}R_n \\ 0 \end{array} \left \begin{array}{cccc} a_{1,1} & \cdots & a_{1,n-1} & 0 \\ \vdots & \ddots & \vdots & \vdots \\ 0 & \cdots & a_{n-1,n-1} & 0 \\ 0 & 1 & & \end{array} \right]</pre>	$\begin{bmatrix} a_{1,1} & \cdots & a_{1,n-1} & a_{1,n} \\ \vdots & \ddots & \vdots & \vdots \\ 0 & \cdots & a_{n-1,n-1} & a_{n-1,n} \\ 0 & \cdots & 0 & a_{n,n} \end{bmatrix} \xrightarrow{\frac{1}{a_{n,n}}R_n} \begin{bmatrix} a_{1,1} & \cdots & a_{1,n-1} & a_{1,n} \\ \vdots & \ddots & \vdots & \vdots \\ 0 & \cdots & a_{n-1,n-1} & a_{n-1,n} \\ 0 & \cdots & 0 & 1 \end{bmatrix}$ $\begin{array}{c} R_1 - a_{1,n}R_n \\ R_2 - a_{2,n}R_n \\ \vdots \\ R_{n-1} - a_{n-1,n}R_n \end{array} \begin{bmatrix} a_{1,1} & \cdots & a_{1,n-1} & 0 \\ \vdots & \ddots & \vdots & \vdots \\ 0 & \cdots & a_{n-1,n-1} & 0 \\ 0 & 1 & & \end{bmatrix}$	$\begin{bmatrix} a_{1,1} & \cdots & a_{1,n-1} & a_{1,n} \\ \vdots & \ddots & \vdots & \vdots \\ 0 & \cdots & a_{n-1,n-1} & a_{n-1,n} \\ 0 & \cdots & 0 & 1 \end{bmatrix} \xrightarrow{\frac{1}{a_{n,n}}R_n} \begin{bmatrix} a_{1,1} & \cdots & a_{1,n-1} & a_{1,n} \\ \vdots & \ddots & \vdots & \vdots \\ 0 & \cdots & a_{n-1,n-1} & a_{n-1,n} \\ 0 & \cdots & 0 & 1 \end{bmatrix}$ $\begin{array}{c} R_1 - a_{1,n}R_n \\ R_2 - a_{2,n}R_n \\ \vdots \\ R_{n-1} - a_{n-1,n}R_n \end{array} \begin{bmatrix} a_{1,1} & \cdots & a_{1,n-1} & 0 \\ \vdots & \ddots & \vdots & \vdots \\ 0 & \cdots & a_{n-1,n-1} & 0 \\ 0 & 1 & & \end{bmatrix}$
johnston-linear-matrix-algebra-EQ0774	289	■ 0.111	<pre>\begin{aligned} m_{1,2} &= \operatorname{det} \left(\left[\begin{array}{ccc} 4 & 5 & 3 \\ 4 & 6 & 6 \\ 7 & 9 & 9 \end{array} \right] \right) = \operatorname{det} \left(\left[\begin{array}{cc} 4 & 6 \\ 7 & 9 \end{array} \right] \right) = 4 \times 9 - 6 \times 7 = 36 - 42 \\ &= -6 \end{aligned}</pre>	$m_{1,2} = \det \left(\left[\begin{array}{ccc} 4 & 5 & 3 \\ 4 & 6 & 6 \\ 7 & 9 & 9 \end{array} \right] \right) = \det \left(\left[\begin{array}{cc} 4 & 6 \\ 7 & 9 \end{array} \right] \right) = 4 \times 9 - 6 \times 7 = 36 - 42 = -6.$	$m_{1,2} = \det \left(\left[\begin{array}{ccc} 4 & 5 & 3 \\ 4 & 6 & 6 \\ 7 & 9 & 9 \end{array} \right] \right) = \det \left(\left[\begin{array}{cc} 4 & 6 \\ 7 & 9 \end{array} \right] \right) = 4 \times 9 - 6 \times 7 = 36 - 42 = -6.$
johnston-linear-matrix-algebra-EQ0748	283	■ 0.116	<pre>\begin{aligned} &\left[\begin{array}{ccc} 1 & 1 & 1 \\ 1 & 2 & 4 \\ 1 & 3 & 9 \end{array}\right] \xrightarrow{\begin{array}{c} R_2 - R_1 \\ R_3 - R_1 \end{array}} \left[\begin{array}{ccc} 1 & 1 & 1 \\ 0 & 1 & 3 \\ 0 & 2 & 8 \end{array}\right] \\ &\xrightarrow{\begin{array}{c} R_1 - R_2 \\ R_3 - 2R_2 \end{array}} \left[\begin{array}{ccc} 1 & 0 & -2 \\ 0 & 1 & 3 \\ 0 & 0 & 2 \end{array}\right] \\ &\xrightarrow{\frac{1}{2}R_3} \left[\begin{array}{ccc} 1 & 0 & -2 \\ 0 & 1 & 3 \\ 0 & 0 & 1 \end{array}\right] \xrightarrow{\begin{array}{c} R_1 + 2R_3 \\ R_2 - 3R_3 \end{array}} \left[\begin{array}{ccc} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{array}\right]. \end{aligned}</pre>	$\begin{bmatrix} 1 & 1 & 1 \\ 1 & 2 & 4 \\ 1 & 3 & 9 \end{bmatrix} \xrightarrow{\begin{array}{c} R_2 - R_1 \\ R_3 - R_1 \end{array}} \begin{bmatrix} 1 & 1 & 1 \\ 0 & 1 & 3 \\ 0 & 2 & 8 \end{bmatrix} \xrightarrow{\begin{array}{c} R_1 - R_2 \\ R_3 - 2R_2 \end{array}} \begin{bmatrix} 1 & 0 & -2 \\ 0 & 1 & 3 \\ 0 & 0 & 2 \end{bmatrix}$ $\xrightarrow{\frac{1}{2}R_3} \begin{bmatrix} 1 & 0 & -2 \\ 0 & 1 & 3 \\ 0 & 0 & 1 \end{bmatrix} \xrightarrow{\begin{array}{c} R_1 + 2R_3 \\ R_2 - 3R_3 \end{array}} \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}.$	$\begin{bmatrix} 1 & 1 & 1 \\ 1 & 2 & 4 \\ 1 & 3 & 9 \end{bmatrix} \xrightarrow{\begin{array}{c} R_2 - R_1 \\ R_3 - R_1 \end{array}} \begin{bmatrix} 1 & 1 & 1 \\ 0 & 1 & 3 \\ 0 & 2 & 8 \end{bmatrix} \xrightarrow{\begin{array}{c} R_1 - R_2 \\ R_3 - 2R_2 \end{array}} \begin{bmatrix} 1 & 0 & -2 \\ 0 & 1 & 3 \\ 0 & 0 & 2 \end{bmatrix}$ $\xrightarrow{\frac{1}{2}R_3} \begin{bmatrix} 1 & 0 & -2 \\ 0 & 1 & 3 \\ 0 & 0 & 1 \end{bmatrix} \xrightarrow{\begin{array}{c} R_1 + 2R_3 \\ R_2 - 3R_3 \end{array}} \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}.$

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johnston-linear-matrix-algebra-E09439	170	0.131	<pre>\begin{figure} \[\begin{aligned} {[A \mid I] } &= \left[\begin{array}{cccccc cccc} * & * & * & * & * & * & * & * & 1 & 0 & 0 & 0 \\ * & * & * & * & * & * & * & * & 0 & 1 & 0 & 0 \\ * & * & * & * & * & * & * & * & 0 & 0 & 1 & 0 \\ * & * & * & * & * & * & * & * & 0 & 0 & 0 & 1 \end{array} \right] \\ \xrightarrow{\text{row-reduce}} & \left[\begin{array}{cccccc cccc} \star & * & * & * & * & * & * & * & * & * & * & * \\ 0 & 0 & \star & * & * & * & * & * & * & * & * & * \\ 0 & 0 & 0 & 0 & 0 & \star & * & * & * & * & * & * \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & * & * & * & * \end{array} \right] = [R \mid E] \\ & \leftarrow \text{null}(A^T) \end{aligned} \]</pre> range(A^T)	(not rendered)	
johnston-linear-matrix-algebra-E09765	287	0.133	<pre>\begin{aligned} & \operatorname{det} \left(\left[\begin{array}{ccc} a & b & c \\ d & e & f \\ g & h & i \end{array} \right] \right) \\ &= \operatorname{det} \left(\left[\begin{array}{ccc} a & b & c \\ 0 & e & f \\ 0 & h & i \end{array} \right] \right) + \operatorname{det} \left(\left[\begin{array}{ccc} 0 & b & c \\ d & e & f \\ 0 & h & i \end{array} \right] \right) + \operatorname{det} \left(\left[\begin{array}{ccc} 0 & b & c \\ 0 & e & f \\ g & h & i \end{array} \right] \right), \end{aligned}</pre>	$\det \left(\begin{bmatrix} a & b & c \\ d & e & f \\ g & h & i \end{bmatrix} \right) = \det \left(\begin{bmatrix} a & b & c \\ 0 & e & f \\ 0 & h & i \end{bmatrix} \right) + \det \left(\begin{bmatrix} 0 & b & c \\ d & e & f \\ 0 & h & i \end{bmatrix} \right) + \det \left(\begin{bmatrix} 0 & b & c \\ 0 & e & f \\ g & h & i \end{bmatrix} \right),$	$\det \left(\begin{bmatrix} a & b & c \\ d & e & f \\ g & h & i \end{bmatrix} \right) = \det \left(\begin{bmatrix} a & b & c \\ 0 & e & f \\ 0 & h & i \end{bmatrix} \right) + \det \left(\begin{bmatrix} 0 & b & c \\ d & e & f \\ 0 & h & i \end{bmatrix} \right) + \det \left(\begin{bmatrix} 0 & b & c \\ 0 & e & f \\ g & h & i \end{bmatrix} \right),$
johnston-linear-matrix-algebra-E09963	45	0.136	<pre>\begin{aligned} A B &= \left[\begin{array}{cc} I_3 & I_3 \\ 0 & C \end{array} \right] \left[\begin{array}{cc} D & O \\ O & D \end{array} \right] \\ &= \left[\begin{array}{cc} I_3 & D + I_3 \\ 0 & O \end{array} \right] \left[\begin{array}{cc} O & O + I_3 \\ O & D + C \end{array} \right] = \left[\begin{array}{cc} D & D \\ O & CD \end{array} \right]. \end{aligned}</pre>	$AB = \left[\begin{array}{cc} I_3 & I_3 \\ O & C \end{array} \right] \left[\begin{array}{cc} D & O \\ O & D \end{array} \right] = \left[\begin{array}{cc} I_3 & D + I_3 \\ 0 & O \end{array} \right] \left[\begin{array}{cc} O & O + I_3 \\ O & D + C \end{array} \right] = \left[\begin{array}{cc} D & D \\ O & CD \end{array} \right].$	$AB = \left[\begin{array}{cc} I_3 & I_3 \\ O & C \end{array} \right] \left[\begin{array}{cc} D & O \\ O & D \end{array} \right] = \left[\begin{array}{cc} I_3 D + I_3 O & I_3 O + I_3 D \\ O D + C O & O O + C D \end{array} \right] = \left[\begin{array}{cc} D & D \\ O & CD \end{array} \right].$

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johnston-linear-matrix-algebra-EQ0204	98	0.137	$\begin{aligned} & \text{(equation 3)} \quad 2x + 4y + 5z = 12 \\ & -2(\text{equation 1}) - 2x - 6y + 4z = -10 \\ \hline & \text{(new equation 3)} \quad -2y + 9z = 2 \end{aligned}$	$\begin{array}{rcl} \text{(equation 3)} & 2x + 4y + 5z = 12 & \\ -2(\text{equation 1}) & -2x - 6y + 4z = -10 & \\ \hline \text{(new equation 3)} & -2y + 9z = 2 & \end{array}$	(equation 3) $2x + 4y + 5z = 12$ $-2(\text{equation 1})$ $-2x - 6y + 4z = -10$ (new equation 3) $-2y + 9z = 2$
johnston-linear-matrix-algebra-EQ0610	235	0.138	$\begin{aligned} & R_m + \ell_{m,m-1} R_{m-1}, \\ & R_m + \ell_{m,m-2} R_{m-2}, \\ & \vdots \\ & R_m + \ell_{m,1} R_1, \end{aligned} \quad R_{m-1} + \ell_{m-1,m-2} R_{m-2} = \dots$	$\begin{array}{rcl} R_m + \ell_{m,m-1} R_{m-1}, & & \\ R_m + \ell_{m,m-2} R_{m-2}, & R_{m-1} + \ell_{m-1,m-2} R_{m-2} & \\ \vdots & \vdots & \ddots \\ R_m + \ell_{m,1} R_1, & R_{m-1} + \ell_{m-1,1} R_1, & \cdots R_2 + \ell_{2,1} R_1. \end{array}$	$R_m + \ell_{m,m-1} R_{m-1},$ $R_m + \ell_{m,m-2} R_{m-2},$ $\vdots$ $R_m + \ell_{m,1} R_1,$ $R_{m-1} + \ell_{m-1,m-2} R_{m-2} = \dots$
johnston-linear-matrix-algebra-EQ0006	22	0.141	$\begin{aligned} & \mathbf{x} + 2(\mathbf{v} + \mathbf{w}) = -\mathbf{v} - 3(\mathbf{x} - \mathbf{w}) \\ \Rightarrow & \mathbf{x} + 2\mathbf{v} + 2\mathbf{w} = -\mathbf{v} - 3\mathbf{x} + 3\mathbf{w} \quad (\text{expand parentheses}) \\ \Rightarrow & 4\mathbf{x} = -3\mathbf{v} + \mathbf{w} \quad (\text{add } 3\mathbf{x}, \text{ subtract } 2\mathbf{v} + 2\mathbf{w}) \\ \Rightarrow & \mathbf{x} = \frac{1}{4}(\mathbf{w} - 3\mathbf{v}). \quad (\text{divide both sides by 4}) \end{aligned}$	$\begin{aligned} \Rightarrow & \mathbf{x} + 2\mathbf{v} + 2\mathbf{w} = -\mathbf{v} - 3\mathbf{x} + 3\mathbf{w} \\ \Rightarrow & 4\mathbf{x} = -3\mathbf{v} + \mathbf{w} \\ \Rightarrow & \mathbf{x} = \frac{1}{4}(\mathbf{w} - 3\mathbf{v}). \end{aligned}$	$\mathbf{x} + 2(\mathbf{v} + \mathbf{w}) = -\mathbf{v} - 3(\mathbf{x} - \mathbf{w})$ $\Rightarrow \mathbf{x} + 2\mathbf{v} + 2\mathbf{w} = -\mathbf{v} - 3\mathbf{x} + 3\mathbf{w}$ (expand parentheses) $\Rightarrow 4\mathbf{x} = -3\mathbf{v} + \mathbf{w}$ (add 3x, subtract 2v + 2w) $\Rightarrow \mathbf{x} = \frac{1}{4}(\mathbf{w} - 3\mathbf{v}).$ (divide both sides by 4)
johnston-linear-matrix-algebra-EQ1327	446	0.172	$\overbrace{(1,2)}^{\substack{y \\ C}} x$	$\overbrace{(1,2)}^{\substack{y \\ C}} x$	
johnston-linear-matrix-algebra-EQ0510 (2.B.9)	205	0.175	$\begin{aligned} & \text{maximize : } z \\ & \text{subject to: } z \end{aligned}$	$\begin{array}{rcl} \text{maximize :} & z & \\ \text{subject to:} & z & \end{array}$	maximize: z subject to: z $-2x_1 - x_2 + x_3 = 0$ $-3x_1 - 2x_2 + 2x_3 = 1$ $+x_1 - x_2 + x_3 = 2$ $s_3 - x_1 + 3x_2 - 3x_3 = 3$ $s_1, \quad s_2, \quad s_3, \quad x_1, \quad x_2, \quad x_3 \geq 0$

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johnston-linear-matrix-algebra-EQ0308	123	■ 0.177	$\begin{array}{lll} \{\left[\begin{array}{lll} 0 & 1 & 1 \\ 1 & 0 & 1 \\ 1 & 1 & 0 \end{array}\right] \left[\begin{array}{ccc} -1 & 1 & 1 \\ 1 & 1 & -1 \\ 1 & 1 & -1 \end{array}\right] \right\} & = & \left[\begin{array}{lll} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{array}\right] \\ \left[\begin{array}{lll} 1 & 0 & 0 \\ 1 & 0 & 0 \\ 1 & 0 & 0 \end{array}\right] & & \\ \left[\begin{array}{ccc} -1 & 1 & 1 \\ 1 & 1 & -1 \\ 1 & 1 & -1 \end{array}\right] & & \end{array}$	$\begin{array}{lll} \left[\begin{array}{lll} 0 & 1 & 1 \\ 1 & 0 & 1 \\ 1 & 1 & 0 \end{array}\right] \left(\frac{1}{2} \left[\begin{array}{lll} -1 & 1 & 1 \\ 1 & -1 & 1 \\ 1 & 1 & -1 \end{array}\right]\right) & = & \left[\begin{array}{lll} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{array}\right] \\ \left(\frac{1}{2} \left[\begin{array}{lll} -1 & 1 & 1 \\ 1 & -1 & 1 \\ 1 & 1 & -1 \end{array}\right]\right) \left[\begin{array}{lll} 0 & 1 & 1 \\ 1 & 0 & 1 \\ 1 & 1 & 0 \end{array}\right] & = & \left[\begin{array}{lll} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{array}\right]. \end{array}$	$\begin{array}{ll} \left[\begin{array}{lll} 0 & 1 & 1 \\ 1 & 0 & 1 \\ 1 & 1 & 0 \end{array}\right] \left(\frac{1}{2} \left[\begin{array}{lll} -1 & 1 & 1 \\ 1 & -1 & 1 \\ 1 & 1 & -1 \end{array}\right]\right) = \left[\begin{array}{lll} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{array}\right] & \text{and} \\ \left(\frac{1}{2} \left[\begin{array}{lll} -1 & 1 & 1 \\ 1 & -1 & 1 \\ 1 & 1 & -1 \end{array}\right]\right) \left[\begin{array}{lll} 0 & 1 & 1 \\ 1 & 0 & 1 \\ 1 & 1 & 0 \end{array}\right] = \left[\begin{array}{lll} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{array}\right]. \end{array}$
johnston-linear-matrix-algebra-EQ0936	334	■ 0.185	$A^k = \underbrace{(P D \overbrace{\left(P^{-1} \right)}^{P=I} \overbrace{\left(P^{-1} \right)}^{P=I} \cdots \left(P D P^{-1} \right))}_k = P D^k P^{-1} \quad \text{for all } k \geq 0,$	$A^k = \underbrace{(P D P^{-1}) (P D P^{-1}) (P D P^{-1}) \cdots (P D P^{-1})}_k = P D^k P^{-1} \quad \text{for all } k \geq 0,$	$A^k = \underbrace{(P D P^{-1}) (P D P^{-1}) (P D P^{-1}) \cdots (P D P^{-1})}_{k \text{ times}} = P D^k P^{-1} \quad \text{for all } k \geq 0,$
johnston-linear-matrix-algebra-EQ1376	452	■ 0.204	$\begin{aligned} A^2 &= (2\mathbf{u}\mathbf{u}^T - I)^2 \\ &= (2\mathbf{u}\mathbf{u}^T - I)(2\mathbf{u}\mathbf{u}^T - I) \\ &= 4\mathbf{u}\mathbf{v}^T\mathbf{u}\mathbf{u}^T - 4\mathbf{u}\mathbf{v}^T + I \\ &= 4\mathbf{u}\mathbf{v}^T - 4\mathbf{u}\mathbf{u}^T + I = I, \end{aligned}$	$\begin{aligned} A^2 &= (2\mathbf{u}\mathbf{u}^T - I)^2 \\ &= (2\mathbf{u}\mathbf{u}^T - I)(2\mathbf{u}\mathbf{u}^T - I) \\ &= 4\mathbf{u}\mathbf{u}^T\mathbf{u}\mathbf{u}^T - 4\mathbf{u}\mathbf{u}^T + I \\ &= 4\mathbf{u}\mathbf{u}^T - 4\mathbf{u}\mathbf{u}^T + I = I, \end{aligned}$	$\begin{aligned} A^2 &= (2\mathbf{u}\mathbf{u}^T - I)^2 \\ &= (2\mathbf{u}\mathbf{u}^T - I)(2\mathbf{u}\mathbf{u}^T - I) \\ &= 4\mathbf{u}\mathbf{u}^T\mathbf{u}\mathbf{u}^T - 4\mathbf{u}\mathbf{u}^T + I \\ &= 4\mathbf{u}\mathbf{u}^T - 4\mathbf{u}\mathbf{u}^T + I = I, \end{aligned}$
johnston-linear-matrix-algebra-EQ0092	54	■ 0.216	$\begin{aligned} [T] \mathbf{v} &= [T(\mathbf{e}_1) \mid T(\mathbf{e}_2) \mid \cdots \mid T(\mathbf{e}_n)] \begin{bmatrix} v_1 \\ v_2 \\ \vdots \\ v_n \end{bmatrix} \\ &= v_1 T(\mathbf{e}_1) + v_2 T(\mathbf{e}_2) + \cdots + v_n T(\mathbf{e}_n) \quad (\text{block matrix multiplication}) \\ &= T(v_1 \mathbf{e}_1 + v_2 \mathbf{e}_2 + \cdots + v_n \mathbf{e}_n) \quad (\text{since } T \text{ is linear}) \\ &= T(\mathbf{v}). \quad (\mathbf{v} = v_1 \mathbf{e}_1 + v_2 \mathbf{e}_2 + \cdots + v_n \mathbf{e}_n) \end{aligned}$	$\begin{aligned} [T] \mathbf{v} &= [T(\mathbf{e}_1) \mid T(\mathbf{e}_2) \mid \cdots \mid T(\mathbf{e}_n)] \begin{bmatrix} v_1 \\ v_2 \\ \vdots \\ v_n \end{bmatrix} \\ &= v_1 T(\mathbf{e}_1) + v_2 T(\mathbf{e}_2) + \cdots + v_n T(\mathbf{e}_n) \quad (\text{block matrix multiplication}) \\ &= T(v_1 \mathbf{e}_1 + v_2 \mathbf{e}_2 + \cdots + v_n \mathbf{e}_n) \quad (\text{since } T \text{ is linear}) \\ &= T(\mathbf{v}). \quad (\mathbf{v} = v_1 \mathbf{e}_1 + v_2 \mathbf{e}_2 + \cdots + v_n \mathbf{e}_n) \end{aligned}$	$\begin{aligned} [T] \mathbf{v} &= [T(\mathbf{e}_1) \mid T(\mathbf{e}_2) \mid \cdots \mid T(\mathbf{e}_n)] \begin{bmatrix} v_1 \\ v_2 \\ \vdots \\ v_n \end{bmatrix} \quad (\text{definition of } [T]) \\ &= v_1 T(\mathbf{e}_1) + v_2 T(\mathbf{e}_2) + \cdots + v_n T(\mathbf{e}_n) \quad (\text{block matrix multiplication}) \\ &= T(v_1 \mathbf{e}_1 + v_2 \mathbf{e}_2 + \cdots + v_n \mathbf{e}_n) \quad (\text{since } T \text{ is linear}) \\ &= T(\mathbf{v}). \quad (\mathbf{v} = v_1 \mathbf{e}_1 + v_2 \mathbf{e}_2 + \cdots + v_n \mathbf{e}_n) \end{aligned}$

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johnston-linear-matrix-algebra EQ0800	295	0.268	<pre> \begin{aligned} \operatorname{det}\left(\left[\begin{array}{cccc} a & b & c & d \\ e & f & g & h \\ i & j & k & \ell \\ m & n & o & p \end{array} \right]\right) = & \text{afkp} - \text{af}\ell o - \text{agjp} + \text{agln} + \text{ahjo} - \text{ahkn} \\ & - \text{bekp} + \text{belo} + \text{bgip} - \text{bg}\ell m - \text{bhio} + \text{bhkm} \\ & + \text{cejp} - \text{celn} - \text{cfip} + \text{cflm} + \text{chin} - \text{chjm} \\ & - \text{dejo} + \text{dekn} + \text{dfio} - \text{dfkm} - \text{dgin} + \text{dgjm}, \end{aligned} </pre>	$\det \left(\begin{bmatrix} a & b & c & d \\ e & f & g & h \\ i & j & k & \ell \\ m & n & o & p \end{bmatrix} \right) = \text{afkp} - \text{af}\ell o - \text{agjp} + \text{agln} + \text{ahjo} - \text{ahkn} \\ - \text{bekp} + \text{belo} + \text{bgip} - \text{bg}\ell m - \text{bhio} + \text{bhkm} \\ + \text{cejp} - \text{celn} - \text{cfip} + \text{cflm} + \text{chin} - \text{chjm} \\ - \text{dejo} + \text{dekn} + \text{dfio} - \text{dfkm} - \text{dgin} + \text{dgjm},$	$\det \left(\begin{bmatrix} a & b & c & d \\ e & f & g & h \\ i & j & k & \ell \\ m & n & o & p \end{bmatrix} \right) = \text{afkp} - \text{af}\ell o - \text{agjp} + \text{agln} + \text{ahjo} - \text{ahkn} \\ - \text{bekp} + \text{belo} + \text{bgip} - \text{bg}\ell m - \text{bhio} + \text{bhkm} \\ + \text{cejp} - \text{celn} - \text{cfip} + \text{cflm} + \text{chin} - \text{chjm} \\ - \text{dejo} + \text{dekn} + \text{dfio} - \text{dfkm} - \text{dgin} + \text{dgjm},$
johnston-linear-matrix-algebra EQ0469	191	0.269	<pre> \begin{aligned} & \& \left[\begin{array}{llll l} 1 & 0 & 1 & 2 & 2 \\ 0 & 1 & 1 & 1 & 1 \\ 2 & 1 & 1 & 1 & 0 \\ 1 & 0 & 1 & 0 & 1 \end{array} \right] \xrightarrow{\substack{R_3-2R_1 \\ R_4-R_1}} \left[\begin{array}{llll l} 1 & 0 & 1 & 2 & 2 \\ 0 & 1 & 1 & 1 & 1 \\ 0 & 1 & 2 & 0 & 2 \\ 0 & 0 & 0 & 1 & 2 \end{array} \right] \\ & \xrightarrow{\substack{R_3-R_2 \\ R_2-R_3}} \left[\begin{array}{llll l} 1 & 0 & 1 & 2 & 2 \\ 0 & 1 & 1 & 1 & 1 \\ 0 & 0 & 1 & 2 & 1 \\ 0 & 0 & 0 & 1 & 2 \end{array} \right] \xrightarrow{\substack{R_1-2R_4 \\ R_2-R_4 \\ R_3-2R_4}} \left[\begin{array}{llll l} 1 & 0 & 1 & 0 & 1 \\ 0 & 1 & 1 & 0 & 2 \\ 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 & 2 \end{array} \right] \\ & \xrightarrow{\substack{R_1-R_3 \\ R_2-R_3}} \left[\begin{array}{llll l} 1 & 0 & 0 & 0 & 1 \\ 0 & 1 & 0 & 0 & 2 \\ 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 & 2 \end{array} \right] . \end{aligned} </pre>	$\begin{aligned} & \left[\begin{array}{llll l} 1 & 0 & 1 & 2 & 2 \\ 0 & 1 & 1 & 1 & 1 \\ 2 & 1 & 1 & 1 & 0 \\ 1 & 0 & 1 & 0 & 1 \end{array} \right] \xrightarrow{\substack{R_3-2R_1 \\ R_4-R_1}} \left[\begin{array}{llll l} 1 & 0 & 1 & 2 & 2 \\ 0 & 1 & 1 & 1 & 1 \\ 0 & 1 & 2 & 0 & 2 \\ 0 & 0 & 0 & 1 & 2 \end{array} \right] \\ & \xrightarrow{\substack{R_3-R_2 \\ R_2-R_3}} \left[\begin{array}{llll l} 1 & 0 & 1 & 2 & 2 \\ 0 & 1 & 1 & 1 & 1 \\ 0 & 0 & 1 & 2 & 1 \\ 0 & 0 & 0 & 1 & 2 \end{array} \right] \xrightarrow{\substack{R_1-2R_4 \\ R_2-R_4 \\ R_3-2R_4}} \left[\begin{array}{llll l} 1 & 0 & 1 & 0 & 1 \\ 0 & 1 & 1 & 0 & 2 \\ 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 & 2 \end{array} \right] \\ & \xrightarrow{\substack{R_1-R_3 \\ R_2-R_3}} \left[\begin{array}{llll l} 1 & 0 & 0 & 0 & 1 \\ 0 & 1 & 0 & 0 & 2 \\ 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 & 2 \end{array} \right] . \end{aligned}$	$\begin{aligned} & \xrightarrow{R_3-R_2} \left[\begin{array}{llll l} 1 & 0 & 1 & 2 & 2 \\ 0 & 1 & 1 & 1 & 1 \\ 0 & 0 & 1 & 2 & 1 \\ 0 & 0 & 0 & 1 & 2 \end{array} \right] \xrightarrow{\substack{R_1-2R_4 \\ R_2-R_4 \\ R_3-2R_4}} \left[\begin{array}{llll l} 1 & 0 & 1 & 0 & 1 \\ 0 & 1 & 1 & 0 & 2 \\ 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 & 2 \end{array} \right] \\ & \xrightarrow{\substack{R_1-R_3 \\ R_2-R_3}} \left[\begin{array}{llll l} 1 & 0 & 0 & 0 & 1 \\ 0 & 1 & 0 & 0 & 2 \\ 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 & 2 \end{array} \right] . \end{aligned}$

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johnston-linear-matrix-algebra- EQ0887	319	0.283	$\begin{aligned} P^{-1} A P &= P^{-1} A \left[\mathbf{v}_1 \cdots \mathbf{v}_k V \right] \\ &= P^{-1} [A \mathbf{v}_1 \cdots A \mathbf{v}_k AV] \\ &= P^{-1} [\lambda_1 \mathbf{v}_1 \cdots \lambda_1 \mathbf{v}_k AV] \\ &= [\lambda_1 P^{-1} \mathbf{v}_1 \cdots \lambda_1 P^{-1} \mathbf{v}_k P^{-1} AV] \\ &= [\lambda_1 \mathbf{e}_1 \cdots \lambda_1 \mathbf{e}_k P^{-1} AV]. \end{aligned}$	$\begin{aligned} P^{-1} A P &= P^{-1} A [\mathbf{v}_1 \cdots \mathbf{v}_k V] \\ &= P^{-1} [A \mathbf{v}_1 \cdots A \mathbf{v}_k AV] \\ &= P^{-1} [\lambda_1 \mathbf{v}_1 \cdots \lambda_1 \mathbf{v}_k AV] \\ &= [\lambda_1 P^{-1} \mathbf{v}_1 \cdots \lambda_1 P^{-1} \mathbf{v}_k P^{-1} AV] \\ &= [\lambda_1 \mathbf{e}_1 \cdots \lambda_1 \mathbf{e}_k P^{-1} AV]. \end{aligned}$	$\begin{aligned} P^{-1} A P &= P^{-1} A [\mathbf{v}_1 \cdots \mathbf{v}_k V] && \text{(definition of } P) \\ &= P^{-1} [A \mathbf{v}_1 \cdots A \mathbf{v}_k AV] && \text{(block matrix mult.)} \\ &= P^{-1} [\lambda_1 \mathbf{v}_1 \cdots \lambda_1 \mathbf{v}_k AV] && (\mathbf{v}_1, \dots, \mathbf{v}_k \text{ are eigenvecs.}) \\ &= [\lambda_1 P^{-1} \mathbf{v}_1 \cdots \lambda_1 P^{-1} \mathbf{v}_k P^{-1} AV] && \text{(block matrix mult.)} \\ &= [\lambda_1 \mathbf{e}_1 \cdots \lambda_1 \mathbf{e}_k P^{-1} AV]. && (P \mathbf{e}_j = \mathbf{v}_j, \text{ so } P^{-1} \mathbf{v}_j = \mathbf{e}_j) \end{aligned}$
johnston-linear-matrix-algebra- EQ0779	290	0.290	$m_{1,1} = \det \left(\begin{bmatrix} 2 & 1 \\ 2 & 1 \\ 0 & 0 \end{bmatrix} \right) = \det \left(\begin{bmatrix} -1 & 1 \\ 1 & 0 \end{bmatrix} \right) = -1,$	$m_{1,1} = \det \left(\begin{bmatrix} 2 & 1 \\ -1 & 1 \\ 1 & 0 \end{bmatrix} \right) = \det \left(\begin{bmatrix} -1 & 1 \\ 1 & 0 \end{bmatrix} \right) = -1,$	$m_{1,1} = \det \left(\begin{bmatrix} 2 & 1 \\ -1 & 1 \\ 1 & 0 \end{bmatrix} \right) = \det \left(\begin{bmatrix} -1 & 1 \\ 1 & 0 \end{bmatrix} \right) = -1,$
johnston-linear-matrix-algebra- EQ0440	170	0.296	$\begin{aligned} &\left[\begin{array}{cccc ccc} 1 & 1 & 1 & -1 & 1 & 0 & 0 \\ 0 & 1 & 1 & 0 & 0 & 1 & 0 \\ -1 & 1 & 1 & 1 & 0 & 0 & 1 \end{array} \right] \xrightarrow{R_3+R_1} \left[\begin{array}{cccc ccc} 1 & 1 & 1 & -1 & 1 & 0 & 0 \\ 0 & 1 & 1 & 0 & 0 & 1 & 0 \\ 0 & 2 & 2 & 0 & 1 & 0 & 1 \end{array} \right] \\ &\xrightarrow{R_1-R_2} \left[\begin{array}{cccc ccc} 1 & 0 & 0 & -1 & 1 & -1 & 0 \\ 0 & 1 & 1 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 1 & -2 & 1 \end{array} \right]. \end{aligned}$	$\begin{aligned} &\left[\begin{array}{cccc ccc} 1 & 1 & 1 & -1 & 1 & 0 & 0 \\ 0 & 1 & 1 & 0 & 0 & 1 & 0 \\ -1 & 1 & 1 & 1 & 0 & 0 & 1 \end{array} \right] \xrightarrow{R_3+R_1} \left[\begin{array}{cccc ccc} 1 & 1 & 1 & -1 & 1 & 0 & 0 \\ 0 & 1 & 1 & 0 & 0 & 1 & 0 \\ 0 & 2 & 2 & 0 & 1 & 0 & 1 \end{array} \right] \\ &\xrightarrow{R_1-R_2, R_3-2R_2} \left[\begin{array}{cccc ccc} 1 & 0 & 0 & -1 & 1 & -1 & 0 \\ 0 & 1 & 1 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 1 & -2 & 1 \end{array} \right]. \end{aligned}$	$\begin{aligned} &\left[\begin{array}{cccc ccc} 1 & 1 & 1 & -1 & 1 & 0 & 0 \\ 0 & 1 & 1 & 0 & 0 & 1 & 0 \\ -1 & 1 & 1 & 1 & 0 & 0 & 1 \end{array} \right] \xrightarrow{R_3+R_1} \left[\begin{array}{cccc ccc} 1 & 1 & 1 & -1 & 1 & 0 & 0 \\ 0 & 1 & 1 & 0 & 0 & 1 & 0 \\ 0 & 2 & 2 & 0 & 1 & 0 & 1 \end{array} \right] \\ &\xrightarrow{R_1-R_2, R_3-2R_2} \left[\begin{array}{cccc ccc} 1 & 0 & 0 & -1 & 1 & -1 & 0 \\ 0 & 1 & 1 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 1 & -2 & 1 \end{array} \right]. \end{aligned}$
johnston-linear-matrix-algebra- EQ1487	464	0.306	$\begin{aligned} &\left[\begin{array}{ccc c} 1 & 2 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 1 & 3 & 0 & 0 \end{array} \right] \xrightarrow{R_3-R_1} \left[\begin{array}{ccc c} 1 & 2 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 1 & 0 & 0 \end{array} \right] \\ &\xrightarrow{R_3-R_2} \left[\begin{array}{ccc c} 1 & 2 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{array} \right]. \end{aligned}$	$\begin{aligned} &\left[\begin{array}{ccc c} 1 & 2 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 1 & 3 & 0 & 0 \end{array} \right] \xrightarrow{R_3-R_1} \left[\begin{array}{ccc c} 1 & 2 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 1 & 0 & 0 \end{array} \right] \\ &\xrightarrow{R_3-R_2} \left[\begin{array}{ccc c} 1 & 2 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{array} \right]. \end{aligned}$	$\begin{aligned} &\left[\begin{array}{ccc c} 1 & 2 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 1 & 3 & 0 & 0 \end{array} \right] \xrightarrow{R_3-R_1} \left[\begin{array}{ccc c} 1 & 2 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 1 & 0 & 0 \end{array} \right] \\ &\xrightarrow{R_3-R_2} \left[\begin{array}{ccc c} 1 & 2 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{array} \right]. \end{aligned}$
johnston-linear-matrix-algebra- EQ1069	365	0.310	$\left(\begin{array}{cccc} 3 & 4 & 2 & 1 \end{array} \right) \rightarrow (14223) \rightarrow (1243) \rightarrow (1234).$	$(3 \ 4 \ 2 \ 1) \rightarrow (14223) \rightarrow (1243) \rightarrow (1234).$	$(3 \ 4 \ 2 \ 1) \rightarrow (14223) \rightarrow (1243) \rightarrow (1234).$

Identifier	Page	Conf.	LaTeX source	Rendered	Image
johnston-linear-matrix-algebra-EQ1055	361	■ 0.311	$\begin{array}{llll} \left(\begin{array}{lll} 1 & 2 & 3 \\ 1 & 3 & 2 \end{array} \right) & \left(\begin{array}{lll} 2 & 1 & 3 \\ 2 & 3 & 1 \end{array} \right) & \left(\begin{array}{lll} 3 & 1 & 2 \\ 3 & 2 & 1 \end{array} \right) \\ \end{array}$	$\begin{pmatrix} 1 & 2 & 3 \\ 1 & 3 & 2 \end{pmatrix} \begin{pmatrix} 2 & 1 & 3 \\ 2 & 3 & 1 \end{pmatrix} \begin{pmatrix} 3 & 1 & 2 \\ 3 & 2 & 1 \end{pmatrix}$	
johnston-linear-matrix-algebra-EQ1211	409	■ 0.314	$\begin{aligned} x_k &= \sum_{n=0}^{k-1} a_n x_n && \text{(recurrence that defines } x_k) \\ &= \sum_{n=0}^{k-1} a_n \left(\sum_{j=0}^{k-1} q_j(n) \lambda_j^n \right) && \left(\text{since } x_n = \sum_{j=0}^{k-1} q_j(n) \lambda_j^n \text{ when } n < k \right) \\ &= \sum_{n=0}^{k-1} a_n \sum_{j=0}^{k-1} \left(\sum_{\ell=0}^{r_j-1} c_{j,\ell} n^\ell \right) \lambda_j^n && \left(\text{since } q_j(n) = \sum_{\ell=0}^{r_j-1} c_{j,\ell} n^\ell \right) \\ &= \sum_{j=0}^{k-1} \sum_{\ell=0}^{r_j-1} c_{j,\ell} \left(\sum_{n=0}^{k-1} a_n n^\ell \lambda_j^n \right) && \text{(swap the order of the sums)} \\ &= \sum_{j=0}^{k-1} \sum_{\ell=0}^{r_j-1} c_{j,\ell} (k^\ell \lambda_j^k) && \text{(Theorem B.6.2: } \lambda_j \text{ is root of char. poly.)} \\ &= \sum_{j=0}^{k-1} q_j(k) \lambda_j^k, && \left(\text{since } q_j(k) = \sum_{\ell=0}^{r_j-1} c_{j,\ell} k^\ell \right) \end{aligned}$	$\begin{aligned} x_k &= \sum_{n=0}^{k-1} a_n x_n && \text{(recurrence that defines } x_k) \\ &= \sum_{n=0}^{k-1} a_n \left(\sum_{j=0}^{k-1} q_j(n) \lambda_j^n \right) && \left(\text{since } x_n = \sum_{j=0}^{k-1} q_j(n) \lambda_j^n \text{ when } n < k \right) \\ &= \sum_{n=0}^{k-1} a_n \sum_{j=0}^{k-1} \left(\sum_{\ell=0}^{r_j-1} c_{j,\ell} n^\ell \right) \lambda_j^n && \left(\text{since } q_j(n) = \sum_{\ell=0}^{r_j-1} c_{j,\ell} n^\ell \right) \\ &= \sum_{j=0}^{k-1} \sum_{\ell=0}^{r_j-1} c_{j,\ell} \left(\sum_{n=0}^{k-1} a_n n^\ell \lambda_j^n \right) && \text{(swap the order of the sums)} \\ &= \sum_{j=0}^{k-1} \sum_{\ell=0}^{r_j-1} c_{j,\ell} (k^\ell \lambda_j^k) && \text{(Theorem B.6.2: } \lambda_j \text{ is root of char. poly.)} \\ &= \sum_{j=0}^{k-1} q_j(k) \lambda_j^k, && \left(\text{since } q_j(k) = \sum_{\ell=0}^{r_j-1} c_{j,\ell} k^\ell \right) \end{aligned}$	
johnston-linear-matrix-algebra-EQ1292	436	■ 0.320	$\begin{array}{l} 1+3+5+\cdots+(2n-1) = n^2 \\ \Rightarrow 1+3+5+\cdots+(2n-1)+(2n+1) = n^2+(2n+1) \\ \Rightarrow 1+3+5+\cdots+(2(n+1)-1) = n^2+2n+1 \\ \Rightarrow 1+3+5+\cdots+(2(n+1)-1) = (n+1)^2 \end{array}$	$\begin{aligned} 1+3+5+\cdots+(2n-1) &= n^2 && (P(n) \text{ is true}) \\ \Rightarrow 1+3+5+\cdots+(2n-1)+(2n+1) &= n^2+(2n+1) && \text{(add } 2n+1) \\ \Rightarrow 1+3+5+\cdots+(2(n+1)-1) &= n^2+2n+1 && \text{(rewrite slightly)} \\ \Rightarrow 1+3+5+\cdots+(2(n+1)-1) &= (n+1)^2 && \text{(factor RHS)} \end{aligned}$	$\begin{aligned} &1+3+5+\cdots+(2n-1) = n^2 && (P(n) \text{ is true}) \\ \Rightarrow &1+3+5+\cdots+(2n-1)+(2n+1) = n^2+(2n+1) && \text{(add } 2n+1) \\ \Rightarrow &1+3+5+\cdots+(2(n+1)-1) = n^2+2n+1 && \text{(rewrite slightly)} \\ \Rightarrow &1+3+5+\cdots+(2(n+1)-1) = (n+1)^2 && \text{(factor RHS)} \end{aligned}$

Identifier	Page	Conf.	LaTeX source	Rendered	Image
johnston-linear-matrix-algebra-EQ0434	168	0.322	<pre>\begin{aligned} & \left[\begin{array}{cccc} 1 & 1 & & \\ -1 & 2 & & \\ & & 0 & 1 \end{array} \right] & \begin{array}{l} R_3 - R_1 \\ R_5 + R_1 \end{array} & \left[\begin{array}{cccc} 1 & 1 & -1 & 2 \\ 0 & 1 & 0 & 1 \\ 1 & 0 & -1 & 1 \\ 0 & 0 & 1 & -1 \\ -1 & 1 & 4 & -3 \end{array} \right] \\ & \left[\begin{array}{cccc} 1 & 1 & -1 & 2 \\ 0 & 1 & 0 & 1 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & -1 \\ 0 & 0 & 3 & -3 \end{array} \right] & \begin{array}{l} R_1 - R_2 \\ R_3 + R_2 \\ R_5 - 2R_2 \end{array} & \left[\begin{array}{cccc} 1 & 0 & -1 & 1 \\ 0 & 1 & 0 & 1 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & -1 \\ 0 & 0 & 3 & -3 \end{array} \right] \\ & \left[\begin{array}{cccc} 1 & 0 & -1 & 1 \\ 0 & 1 & 0 & 1 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & -1 \\ 0 & 0 & 3 & -3 \end{array} \right] & \begin{array}{l} R_3 \leftrightarrow R_4 \\ R_1 + R_3 \\ R_5 - 3R_3 \end{array} & \left[\begin{array}{cccc} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 1 \\ 0 & 0 & 1 & -1 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{array} \right] . \end{aligned}</pre>		
johnston-linear-matrix-algebra-EQ0575	224	0.324	<pre>\begin{figure} \[\left. A = \left[\begin{array}{cccccc} * & * & * & * & * & * \\ * & * & * & * & * & * \\ * & * & * & * & * & * \\ * & * & * & * & * & * \\ * & * & * & * & * & * \\ * & * & * & * & * & * \\ * & * & * & * & * & * \end{array} \right] = \left[\begin{array}{ccc} * & * & * \\ * & * & * \\ * & * & * \\ * & * & * \\ * & * & * \\ * & * & * \\ * & * & * \end{array} \right] \left[\begin{array}{cccccc} * & * & * & * & * & * \\ * & * & * & * & * & * \\ * & * & * & * & * & * \\ * & * & * & * & * & * \end{array} \right] \right. \\ & \left. \right] \right. \] \captionsetup{labelformat=empty} \\ \caption{Figure 2.39: Theorem 2.C. 1 says that every matrix can be written as a product of a tall and skinny matrix and a short and fat matrix. The rank determines exactly how skinny and short the matrices in the product can be (as shown here, \((A)\) has rank 3, so it can be written as a product of a matrix \((C)\) with 3 columns and a matrix \((R)\) with 3 rows).} \end{figure}</pre>	(not rendered)	

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johnston-linear-matrix-algebra-EQ0564	222	0.340	$\begin{aligned} & \operatorname{maximize}: & & x_1 + x_2 + x_3 + x_4 \\ & \text{subject to: } & & 4x_1 + 3x_2 + 2x_3 \leq 2 \\ & & & x_1 + 4x_2 + x_3 + 2x_4 \leq 2 \\ & & & 2x_1 + x_2 + 3x_3 + 2x_4 \leq 2 \\ & & & x_1, x_2, x_3, x_4 \geq 0 \end{aligned}$	$\begin{aligned} & \operatorname{maximize}: & & x_1 + x_2 + x_3 + x_4 \\ & \text{subject to: } & & 4x_1 + 3x_2 + 2x_3 \leq 2 \\ & & & x_1 + 4x_2 + x_3 + 2x_4 \leq 2 \\ & & & 2x_1 + x_2 + 3x_3 + 2x_4 \leq 2 \\ & & & x_1, x_2, x_3, x_4 \geq 0 \end{aligned}$	$\begin{aligned} \text{maximize:} & & & x_1 + x_2 + x_3 + x_4 \\ \text{subject to:} & & & 4x_1 + 3x_2 + 2x_3 \leq 1 \\ & & & x_1 + 4x_2 + x_3 + 2x_4 \leq 2 \\ & & & 2x_1 + x_2 + 3x_3 + 2x_4 \leq 2 \\ & & & x_1, x_2, x_3, x_4 \geq 0 \end{aligned}$
johnston-linear-matrix-algebra-EQ1352	449	0.341	$\mathbf{I}_m \mathbf{A} = \left[\begin{array}{c} \frac{\mathbf{e}_1^T \mathbf{A}}{\mathbf{e}_2^T \mathbf{A}} \\ \vdots \\ \frac{\mathbf{e}_m^T \mathbf{A}}{\mathbf{e}_m^T \mathbf{A}} \end{array} \right] = \left[\begin{array}{c} \frac{\mathbf{a}_1}{\mathbf{a}_2} \\ \vdots \\ \mathbf{a}_m \end{array} \right] = \mathbf{A}.$	$\mathbf{I}_m \mathbf{A} = \left[\begin{array}{c} \frac{\mathbf{e}_1^T \mathbf{A}}{\mathbf{e}_2^T \mathbf{A}} \\ \vdots \\ \frac{\mathbf{e}_m^T \mathbf{A}}{\mathbf{e}_m^T \mathbf{A}} \end{array} \right] = \left[\begin{array}{c} \frac{\mathbf{a}_1}{\mathbf{a}_2} \\ \vdots \\ \mathbf{a}_m \end{array} \right] = \mathbf{A}.$	$\mathbf{I}_m \mathbf{A} = \left[\begin{array}{c} \frac{\mathbf{e}_1^T \mathbf{A}}{\mathbf{e}_2^T \mathbf{A}} \\ \vdots \\ \frac{\mathbf{e}_m^T \mathbf{A}}{\mathbf{e}_m^T \mathbf{A}} \end{array} \right] = \left[\begin{array}{c} \frac{\mathbf{a}_1}{\mathbf{a}_2} \\ \vdots \\ \mathbf{a}_m \end{array} \right] = \mathbf{A}.$
johnston-linear-matrix-algebra-EQ0183	89	0.345	$\frac{\sum_{j=1}^n w_j^2}{\sum_{j=1}^n w_j} \quad (\text{average number of friends of friends} = \frac{\text{total names on lists divided by number of lists}}{\text{total names on lists divided by number of lists}})$	$\frac{\sum_{j=1}^n w_j^2}{\sum_{j=1}^n w_j} \quad (\text{average number of friends of friends} = \frac{\text{total names on lists divided by number of lists}}{\text{total names on lists divided by number of lists}})$	$\frac{\sum_{j=1}^n w_j^2}{\sum_{j=1}^n w_j} \quad (\text{average number of friends of friends} = \frac{\text{total names on lists divided by number of lists}}{\text{total names on lists divided by number of lists}})$
johnston-linear-matrix-algebra-EQ0889	319	0.348	$\begin{aligned} & p_A(\lambda) = \det(P^{-1}AP - \lambda I) & & (\text{since } p_A(\lambda) = p_{P^{-1}AP}(\lambda)) \\ & = \det \left(\begin{bmatrix} \lambda_1 I_k - \lambda I_k & B \\ O & C - \lambda I_{n-k} \end{bmatrix} \right) & & (\text{block form of } P^{-1}AP) \\ & = \det((\lambda_1 - \lambda) I_k) \det(C - \lambda I_{n-k}) & & (\text{by Exercise 3.2.16(a)}) \\ & = (\lambda_1 - \lambda)^k p_C(\lambda) & & (\text{since } (\lambda_1 - \lambda) I_k \text{ is diagonal}) \end{aligned}$	$\begin{aligned} p_A(\lambda) &= \det(P^{-1}AP - \lambda I) & & (\text{ since } p_A(\lambda) = p_{P^{-1}AP}(\lambda)) \\ &= \det \left(\begin{bmatrix} \lambda_1 I_k - \lambda I_k & B \\ O & C - \lambda I_{n-k} \end{bmatrix} \right) & & (\text{block form of } P^{-1}AP) \\ &= \det((\lambda_1 - \lambda) I_k) \det(C - \lambda I_{n-k}) & & (\text{by Exercise 3.2.16(a)}) \\ &= (\lambda_1 - \lambda)^k p_C(\lambda) & & (\text{since } (\lambda_1 - \lambda) I_k \text{ is diagonal}) \end{aligned}$	$\begin{aligned} p_A(\lambda) &= \det(P^{-1}AP - \lambda I) & & (\text{since } p_A(\lambda) = p_{P^{-1}AP}(\lambda)) \\ &= \det \left(\begin{bmatrix} \lambda_1 I_k - \lambda I_k & B \\ O & C - \lambda I_{n-k} \end{bmatrix} \right) & & (\text{block form of } P^{-1}AP) \\ &= \det((\lambda_1 - \lambda) I_k) \det(C - \lambda I_{n-k}) & & (\text{by Exercise 3.2.16(a)}) \\ &= (\lambda_1 - \lambda)^k p_C(\lambda) & & (\text{since } (\lambda_1 - \lambda) I_k \text{ is diagonal}) \end{aligned}$

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johnston-linear-matrix-algebra-EQ0211	100	0.351	$\begin{aligned} & \left[\begin{array}{ccc c} 1 & 3 & -2 & 5 \\ 0 & 2 & -6 & 4 \\ 0 & 0 & 3 & 6 \end{array} \right] \xrightarrow{\frac{1}{3}R_3} \left[\begin{array}{ccc c} 1 & 3 & -2 & 5 \\ 0 & 2 & -6 & 4 \\ 0 & 0 & 1 & 2 \end{array} \right] \\ & \xrightarrow{\begin{smallmatrix} R_1+2R_3 \\ R_2+6R_3 \end{smallmatrix}} \left[\begin{array}{ccc c} 1 & 3 & 0 & 9 \\ 0 & 2 & 0 & 16 \\ 0 & 0 & 1 & 2 \end{array} \right] \xrightarrow{\frac{1}{2}R_2} \left[\begin{array}{ccc c} 1 & 3 & 0 & 9 \\ 0 & 1 & 0 & 8 \\ 0 & 0 & 1 & 2 \end{array} \right] \\ & \xrightarrow{R_1-3R_2} \left[\begin{array}{ccc c} 1 & 0 & 0 & -15 \\ 0 & 1 & 0 & 8 \\ 0 & 0 & 1 & 2 \end{array} \right]. \end{aligned}$		
johnston-linear-matrix-algebra-EQ1076	367	0.366	$\begin{aligned} & \sum_{1 \leq j_1 \leq \dots \leq j_m \leq n} \left(\sum_{\sigma \in S_m} b_{j_{\sigma(1)},1} \dots b_{j_{\sigma(m)},m} \det(A_{j_{\sigma(1)}, \dots, j_{\sigma(m)}}) \right) \\ &= \sum_{1 \leq j_1 \leq \dots \leq j_m \leq n} \left(\sum_{\sigma \in S_m} \operatorname{sgn}(\sigma) b_{j_{\sigma(1)},1} \dots b_{j_{\sigma(m)},m} \right) \det(A_{j_1, \dots, j_m}) \\ &= \sum_{1 \leq j_1 \leq \dots \leq j_m \leq n} \det(B_{j_1, \dots, j_m}) \det(A_{j_1, \dots, j_m}). \end{aligned}$		
johnston-linear-matrix-algebra-EQ0595	229	0.367	$\begin{aligned} & \left[\begin{array}{ccccc} 1 & 1 & 1 & 1 & 1 \\ 1 & 2 & 4 & 8 & 16 \\ 1 & 2 & 5 & 1 & 4 \\ 1 & 3 & 9 & 27 & 81 \end{array} \right] \longrightarrow \left[\begin{array}{ccc} 1 & 1 & 1 \\ 1 & 2 & 4 \\ 1 & 3 & 9 \end{array} \right]. \end{aligned}$		

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johnston-linear-matrix-algebra- EQ1593	476	0.385	$\begin{aligned} & \operatorname{det}\left(\left[\begin{array}{cc} I_n & \mathbf{0} \\ \mathbf{w}^T & 1 \end{array}\right]\left[\begin{array}{cc} I_n+\mathbf{v}\mathbf{w}^T & \mathbf{v} \\ \mathbf{0}^T & 1 \end{array}\right]\left[\begin{array}{cc} I_n & \mathbf{0} \\ -\mathbf{w}^T & 1 \end{array}\right]\right) \\ & =\operatorname{det}\left(\left[\begin{array}{cc} I_n+\mathbf{v}\mathbf{w}^T & \mathbf{v} \\ \mathbf{0}^T & 1 \end{array}\right]\right)=\operatorname{det}\left(I_n+\mathbf{v}\mathbf{w}^T\right) . \end{aligned}$	$\det \left(\left[\begin{array}{cc} I_n & \mathbf{0} \\ \mathbf{w}^T & 1 \end{array} \right] \left[\begin{array}{cc} I_n + \mathbf{v}\mathbf{w}^T & \mathbf{v} \\ \mathbf{0}^T & 1 \end{array} \right] \left[\begin{array}{cc} I_n & \mathbf{0} \\ -\mathbf{w}^T & 1 \end{array} \right] \right)$ $= \det \left(\left[\begin{array}{cc} I_n + \mathbf{v}\mathbf{w}^T & \mathbf{v} \\ \mathbf{0}^T & 1 \end{array} \right] \right) = \det (I_n + \mathbf{v}\mathbf{w}^T) .$	$\det \left(\left[\begin{array}{cc} I_n & \mathbf{0} \\ \mathbf{w}^T & 1 \end{array} \right] \left[\begin{array}{cc} I_n + \mathbf{v}\mathbf{w}^T & \mathbf{v} \\ \mathbf{0}^T & 1 \end{array} \right] \left[\begin{array}{cc} I_n & \mathbf{0} \\ -\mathbf{w}^T & 1 \end{array} \right] \right)$ $= \det \left(\left[\begin{array}{cc} I_n + \mathbf{v}\mathbf{w}^T & \mathbf{v} \\ \mathbf{0}^T & 1 \end{array} \right] \right) = \det(I_n + \mathbf{v}\mathbf{w}^T) .$
johnston-linear-matrix-algebra- EQ0495	200	0.390	$\begin{aligned} & \text { maximize: } x_1+2 x_2 \\ & \text { subject to: }-3 x_1+4 x_2 \leq-1 \\ & \quad 2 x_1+x_2 \leq 5 \\ & \quad-2 x_1-x_2 \leq-5 \\ & \quad x_1, \quad x_2 \geq 0 \end{aligned}$	$\begin{aligned} & \text { maximize: } x_1+2 x_2 \\ & \text { subject to: }-3 x_1+4 x_2 \leq-1 \\ & \quad 2 x_1+x_2 \leq 5 \\ & \quad-2 x_1-x_2 \leq-5 \\ & \quad x_1, \quad x_2 \geq 0 \end{aligned}$	$\begin{aligned} & \text { maximize: } x_1+2 x_2 \\ & \text { subject to: }-3 x_1+4 x_2 \leq-1 \\ & \quad 2 x_1+x_2 \leq 5 \\ & \quad-2 x_1-x_2 \leq-5 \\ & \quad x_1, \quad x_2 \geq 0 \end{aligned}$
johnston-linear-matrix-algebra- EQ0686 (2)	260	0.399	$\begin{aligned} & \left[\begin{array}{ccc c} 1 & 1 & 2 & 2 \\ -1 & 1 & 3 & 1 \\ 1 & 2 & 1 & 0 \\ 0 & 0 & 1 & 1 \\ 2 & -1 & -1 & 2 \end{array}\right] \xrightarrow{\substack{R_2+R_1 \\ R_3-R_1 \\ R_5-2R_1}} \left[\begin{array}{ccc c} 1 & 1 & 2 & 2 \\ 0 & 2 & 5 & 3 \\ 0 & 1 & -1 & -2 \\ 0 & 0 & 1 & 1 \\ 0 & -3 & -5 & -2 \end{array}\right] \\ & \xrightarrow{\substack{R_2-2R_3 \\ R_5+3R_3}} \left[\begin{array}{ccc c} 1 & 1 & 2 & 2 \\ 0 & 0 & 7 & 7 \\ 0 & 1 & -1 & -2 \\ 0 & 0 & 1 & 1 \\ 0 & 0 & -8 & -8 \end{array}\right] \xrightarrow{\substack{R_2-7R_4 \\ R_5+8R_4}} \left[\begin{array}{ccc c} 1 & 1 & 2 & 2 \\ 0 & 0 & 0 & 0 \\ 0 & 1 & -1 & -2 \\ 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 0 \end{array}\right] . \end{aligned}$	$\left[\begin{array}{ccc c} 1 & 1 & 2 & 2 \\ -1 & 1 & 3 & 1 \\ 1 & 2 & 1 & 0 \\ 0 & 0 & 1 & 1 \\ 2 & -1 & -1 & 2 \end{array}\right] \xrightarrow{\substack{R_2+R_1 \\ R_3-R_1 \\ R_5-2R_1}} \left[\begin{array}{ccc c} 1 & 1 & 2 & 2 \\ 0 & 2 & 5 & 3 \\ 0 & 1 & -1 & -2 \\ 0 & 0 & 1 & 1 \\ 0 & -3 & -5 & -2 \end{array}\right]$ $\xrightarrow{\substack{R_2-2R_3 \\ R_5+3R_3}} \left[\begin{array}{ccc c} 1 & 1 & 2 & 2 \\ 0 & 0 & 7 & 7 \\ 0 & 1 & -1 & -2 \\ 0 & 0 & 1 & 1 \\ 0 & 0 & -8 & -8 \end{array}\right] \xrightarrow{\substack{R_2-7R_4 \\ R_5+8R_4}} \left[\begin{array}{ccc c} 1 & 1 & 2 & 2 \\ 0 & 0 & 0 & 0 \\ 0 & 1 & -1 & -2 \\ 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 0 \end{array}\right] .$	$\left[\begin{array}{ccc c} 1 & 1 & 2 & 2 \\ -1 & 1 & 3 & 1 \\ 1 & 2 & 1 & 0 \\ 0 & 0 & 1 & 1 \\ 2 & -1 & -1 & 2 \end{array}\right] \xrightarrow{\substack{R_2+R_1 \\ R_3-R_1 \\ R_5-2R_1}} \left[\begin{array}{ccc c} 1 & 1 & 2 & 2 \\ 0 & 2 & 5 & 3 \\ 0 & 1 & -1 & -2 \\ 0 & 0 & 1 & 1 \\ 0 & -3 & -5 & -2 \end{array}\right]$ $\xrightarrow{\substack{R_2-2R_3 \\ R_5+3R_3}} \left[\begin{array}{ccc c} 1 & 1 & 2 & 2 \\ 0 & 0 & 7 & 7 \\ 0 & 1 & -1 & -2 \\ 0 & 0 & 1 & 1 \\ 0 & 0 & -8 & -8 \end{array}\right] \xrightarrow{\substack{R_2-7R_4 \\ R_5+8R_4}} \left[\begin{array}{ccc c} 1 & 1 & 2 & 2 \\ 0 & 0 & 0 & 0 \\ 0 & 1 & -1 & -2 \\ 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 0 \end{array}\right] .$
johnston-linear-matrix-algebra- EQ0218	102	0.407	$\begin{aligned} & \left[\begin{array}{ccccc} 0 & 2 & -2 & 6 & -3 \\ 0 & -2 & 1 & -5 & 2 \\ 0 & 0 & -1 & 1 & 0 \\ 0 & -4 & 2 & -10 & 5 \end{array}\right] \xrightarrow{\substack{R_2+R_1 \\ R_4+2R_1}} \left[\begin{array}{ccccc} 0 & 2 & -2 & 6 & -3 \\ 0 & 0 & -1 & -3 & 2 \\ 0 & 0 & -1 & 1 & 0 \\ -2 & 0 & 0 & -2 & 1 \end{array}\right] \\ & \xrightarrow{\substack{R_2 \leftrightarrow R_3 \\ R_4+R_2}} \left[\begin{array}{ccccc} 0 & 2 & -2 & 6 & -3 \\ 0 & 0 & -1 & -3 & 2 \\ 0 & 0 & -1 & 1 & 0 \\ -2 & 0 & 0 & -2 & 1 \end{array}\right] \end{aligned}$	$\left[\begin{array}{ccccc} 0 & 2 & -2 & 6 & -3 \\ 0 & -2 & 1 & -5 & 2 \\ 0 & 0 & -1 & 1 & 0 \\ 0 & -4 & 2 & -10 & 5 \end{array}\right] \xrightarrow{\substack{R_2+R_1 \\ R_4+2R_1}} \left[\begin{array}{ccccc} 0 & 2 & -2 & 6 & -3 \\ 0 & 0 & -1 & -3 & 2 \\ 0 & 0 & -1 & 1 & 0 \\ -2 & 0 & 0 & -2 & 1 \end{array}\right]$ $\xrightarrow{\substack{R_2 \leftrightarrow R_3 \\ R_4+R_2}} \left[\begin{array}{ccccc} 0 & 2 & -2 & 6 & -3 \\ 0 & 0 & -1 & -3 & 2 \\ 0 & 0 & -1 & 1 & 0 \\ -2 & 0 & 0 & -2 & 1 \end{array}\right]$	$\left[\begin{array}{ccccc} 0 & 2 & -2 & 6 & -3 \\ 0 & -2 & 1 & -5 & 2 \\ 0 & 0 & -1 & 1 & 0 \\ 0 & -4 & 2 & -10 & 5 \end{array}\right] \xrightarrow{\substack{R_2+R_1 \\ R_4+2R_1}} \left[\begin{array}{ccccc} 0 & 2 & -2 & 6 & -3 \\ 0 & 0 & -1 & -3 & 2 \\ 0 & 0 & -1 & 1 & 0 \\ 0 & 0 & 0 & -2 & 1 \end{array}\right]$ $\xrightarrow{\substack{R_2 \leftrightarrow R_3 \\ R_4+R_2}} \left[\begin{array}{ccccc} 0 & 2 & -2 & 6 & -3 \\ 0 & 0 & -1 & -3 & 2 \\ 0 & 0 & -1 & 1 & 0 \\ 0 & 0 & 0 & -2 & 1 \end{array}\right]$

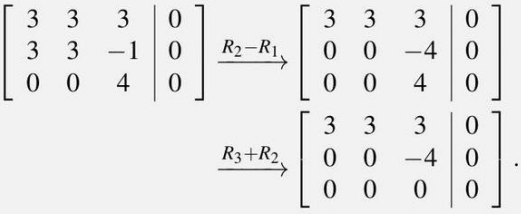
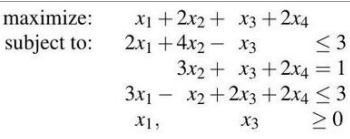
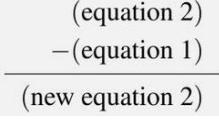
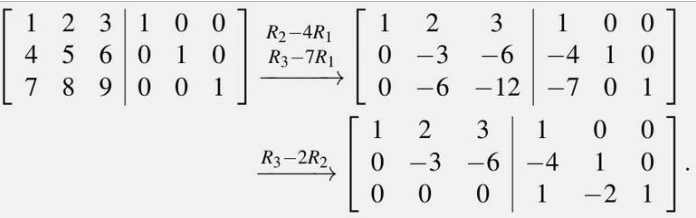
Identifier	Page	Conf.	LaTeX source	Rendered	Image
johnston-linear-matrix-algebra-EQ0481	196	■ 0.436	$\begin{aligned} 2w+x+y &=2 \\ w^2+y^2+z^2 &=1 \\ x+y+2z &=0 \end{aligned}$	$\begin{aligned} 2w+x+y &=2 \\ w^2+y^2+z^2 &=1 \\ x+y+2z &=0 \\ 2x+2y+z &=0 \end{aligned}$	$\begin{aligned} 2w+x+y &=2 \\ w^2+y^2+z^2 &=1 \\ x+y+2z &=0 \\ 2x+2y+z &=0 \end{aligned}$
johnston-linear-matrix-algebra-E0656	251	■ 0.447	$\begin{aligned} c_1-d_1 &=0, & c_2-d_2 &=0, & \dots, & c_k-d_k &=0, & \text{ so} \\ c_1 &=d_1, & c_2 &=d_2, & \dots, & c_k &=d_k. \end{aligned}$	$\begin{aligned} c_1-d_1 &=0, & c_2-d_2 &=0, & \dots, & c_k-d_k &=0, & \text{ so} \\ c_1 &=d_1, & c_2 &=d_2, & \dots, & c_k &=d_k. \end{aligned}$	$\begin{aligned} c_1-d_1 &=0, & c_2-d_2 &=0, & \dots, & c_k-d_k &=0, & \text{ so} \\ c_1 &=d_1, & c_2 &=d_2, & \dots, & c_k &=d_k. \end{aligned}$
johnston-linear-matrix-algebra-E0884	318	■ 0.451	$\begin{aligned} \operatorname{rank}(A) &= \\ \operatorname{rank}(B) &=3, & \operatorname{tr}(A) &= \\ \operatorname{tr}(B) &=0, \dots \end{aligned}$	$\operatorname{rank}(A) = \operatorname{rank}(B) = 3, \quad \operatorname{tr}(A) = \operatorname{tr}(B) = 0, \dots$	$\begin{aligned} \operatorname{rank}(A) &= \operatorname{rank}(B) = 3, & \operatorname{tr}(A) &= \operatorname{tr}(B) = 0, \\ \det(A) &= \det(B) = 2, & \text{ and } p_A(\lambda) &= p_B(\lambda) = -\lambda^3 + 3\lambda + 2. \end{aligned}$
johnston-linear-matrix-algebra-EQ1017	351	■ 0.458	$V = \begin{bmatrix} 1 & 1 & \cdots & 1 \\ \lambda_1 & \lambda_2 & \cdots & \lambda_n \\ \lambda_1^2 & \lambda_2^2 & \cdots & \lambda_n^2 \\ \vdots & \vdots & \ddots & \vdots \\ \lambda_1^{n-1} & \lambda_2^{n-1} & \cdots & \lambda_n^{n-1} \end{bmatrix}$	$V = \begin{bmatrix} 1 & 1 & \cdots & 1 \\ \lambda_1 & \lambda_2 & \cdots & \lambda_n \\ \lambda_1^2 & \lambda_2^2 & \cdots & \lambda_n^2 \\ \vdots & \vdots & \ddots & \vdots \\ \lambda_1^{n-1} & \lambda_2^{n-1} & \cdots & \lambda_n^{n-1} \end{bmatrix}$	$V = \begin{bmatrix} 1 & 1 & \cdots & 1 \\ \lambda_1 & \lambda_2 & \cdots & \lambda_n \\ \lambda_1^2 & \lambda_2^2 & \cdots & \lambda_n^2 \\ \vdots & \vdots & \ddots & \vdots \\ \lambda_1^{n-1} & \lambda_2^{n-1} & \cdots & \lambda_n^{n-1} \end{bmatrix}$
johnston-linear-matrix-algebra-E0793	293	■ 0.462	$\begin{aligned} \begin{bmatrix} a_{1,1} & a_{1,2} & \cdots & a_{1,n} \\ 0 & a_{2,2} & \cdots & a_{2,n} \\ \vdots & \vdots & \ddots & \vdots \\ 0 & a_{n,2} & \cdots & a_{n,n} \end{bmatrix} & \text{ has row echelon form } \begin{bmatrix} a_{1,1} & a_{1,2} & \cdots & a_{1,n} \\ 0 & & & \\ \vdots & & R & \\ 0 & & & \end{bmatrix}, \\ \begin{bmatrix} a_{1,1} & a_{1,2} & \cdots & a_{1,n} \\ 0 & a_{2,2} & \cdots & a_{2,n} \\ \vdots & \vdots & \ddots & \vdots \\ 0 & a_{n,2} & \cdots & a_{n,n} \end{bmatrix} & \text{ has row echelon form } \begin{bmatrix} a_{1,1} & a_{1,2} & \cdots & a_{1,n} \\ 0 & & & \\ \vdots & & R & \\ 0 & & & \end{bmatrix}, \end{aligned}$	$\begin{bmatrix} a_{1,1} & a_{1,2} & \cdots & a_{1,n} \\ 0 & a_{2,2} & \cdots & a_{2,n} \\ \vdots & \vdots & \ddots & \vdots \\ 0 & a_{n,2} & \cdots & a_{n,n} \end{bmatrix} \text{ has row echelon form } \begin{bmatrix} a_{1,1} & a_{1,2} & \cdots & a_{1,n} \\ 0 & & & \\ \vdots & & R & \\ 0 & & & \end{bmatrix},$	$\begin{bmatrix} a_{1,1} & a_{1,2} & \cdots & a_{1,n} \\ 0 & a_{2,2} & \cdots & a_{2,n} \\ \vdots & \vdots & \ddots & \vdots \\ 0 & a_{n,2} & \cdots & a_{n,n} \end{bmatrix} \text{ has row echelon form } \begin{bmatrix} a_{1,1} & a_{1,2} & \cdots & a_{1,n} \\ 0 & & & \\ \vdots & & R & \\ 0 & & & \end{bmatrix},$
johnston-linear-matrix-algebra-E0994	347	■ 0.463	$\begin{aligned} e^x &= \sum_{k=0}^{\infty} \frac{1}{k!} x^k & \text{ for all } x \in \mathbb{R}, \\ \sin(x) &= \sum_{k=0}^{\infty} \frac{(-1)^k}{(2k+1)!} x^{2k+1} & \text{ for all } x \in \mathbb{R}, \\ \frac{1}{1-x} &= \sum_{k=0}^{\infty} x^k & \text{ for all } x \in (-1, 1), \text{ and } \\ -\ln(1-x) &= \sum_{k=0}^{\infty} \frac{1}{k} x^k & \text{ for all } x \in (-1, 1). \end{aligned}$	$\begin{aligned} e^x &= \sum_{k=0}^{\infty} \frac{1}{k!} x^k & \text{ for all } x \in \mathbb{R}, \\ \sin(x) &= \sum_{k=0}^{\infty} \frac{(-1)^k}{(2k+1)!} x^{2k+1} & \text{ for all } x \in \mathbb{R}, \\ \frac{1}{1-x} &= \sum_{k=0}^{\infty} x^k & \text{ for all } x \in (-1, 1), \text{ and } \\ -\ln(1-x) &= \sum_{k=0}^{\infty} \frac{1}{k} x^k & \text{ for all } x \in (-1, 1). \end{aligned}$	$\begin{aligned} e^x &= \sum_{k=0}^{\infty} \frac{1}{k!} x^k & \text{ for all } x \in \mathbb{R}, \\ \sin(x) &= \sum_{k=0}^{\infty} \frac{(-1)^k}{(2k+1)!} x^{2k+1} & \text{ for all } x \in \mathbb{R}, \\ \frac{1}{1-x} &= \sum_{k=0}^{\infty} x^k & \text{ for all } x \in (-1, 1), \text{ and } \\ -\ln(1-x) &= \sum_{k=0}^{\infty} \frac{1}{k} x^k & \text{ for all } x \in (-1, 1). \end{aligned}$

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johnston-linear-matrix-algebra-EQ0501	201	■ 0.469	$\begin{array}{cc} \operatorname{maximize}: & -3x_1 + 2x_2 \\ \text{subject to:} & x_1 + 3x_2 \leq 4 \\ & x_1 - 3x_2 \leq -4 \\ & x_1 - 2x_2 \leq -3 \\ & x_2 \geq 0 \end{array}$	$\begin{array}{ll} \text{maximize :} & -3x_1 + 2x_2 \\ \text{subject to:} & x_1 + 3x_2 \leq 4 \\ & x_1 - 3x_2 \leq -4 \\ & x_1 - 2x_2 \leq -3 \\ & x_2 \geq 0 \end{array}$	$\begin{array}{ll} \text{maximize:} & -3x_1 + 2x_2 \\ \text{subject to:} & x_1 + 3x_2 \leq 4 \\ & x_1 - 3x_2 \leq -4 \\ & x_1 - 2x_2 \leq -3 \\ & x_2 \geq 0 \end{array}$
johnston-linear-matrix-algebra-EQ0558	222	■ 0.471	$\begin{aligned} & \operatorname{minimize}: \quad \quad \quad 3x_1 + x_2 \\ & \text{subject to:} \quad 2x_1 - x_2 \geq 1 \\ & \quad \quad \quad x_1, x_2 \geq 0 \end{aligned}$	$\begin{array}{ll} \text{minimize :} & 3x_1 + x_2 \\ & x_1 + x_2 \geq 2 \\ \text{subject to:} & 2x_1 - x_2 \geq 1 \\ & x_1, x_2 \geq 0 \end{array}$	$\begin{array}{ll} \text{minimize:} & 3x_1 + x_2 \\ \text{subject to:} & x_1 + x_2 \geq 2 \\ & 2x_1 - x_2 \geq 1 \\ & x_1, x_2 \geq 0 \end{array}$
johnston-linear-matrix-algebra-EQ0023	31	■ 0.475	$\begin{aligned} & \ \mathbf{v} + \mathbf{w}\ ^2 = (\mathbf{v} + \mathbf{w}) \cdot (\mathbf{v} + \mathbf{w}) \quad (\text{definition of length}) \\ & = (\mathbf{v} \cdot \mathbf{v}) + 2(\mathbf{v} \cdot \mathbf{w}) + (\mathbf{w} \cdot \mathbf{w}) \quad (\text{dot product properties (FOIL)}) \\ & = \ \mathbf{v}\ ^2 + 2(\mathbf{v} \cdot \mathbf{w}) + \ \mathbf{w}\ ^2 \quad (\text{definition of length}) \\ & \leq \ \mathbf{v}\ ^2 + 2\ \mathbf{v}\ \ \mathbf{w}\ + \ \mathbf{w}\ ^2 \quad (\text{Cauchy-Schwarz inequality}) \\ & = (\ \mathbf{v}\ + \ \mathbf{w}\)^2 \quad (\text{factor cleverly}) \end{aligned}$	$\begin{array}{ll} \ \mathbf{v} + \mathbf{w}\ ^2 = (\mathbf{v} + \mathbf{w}) \cdot (\mathbf{v} + \mathbf{w}) & (\text{definition of length}) \\ = (\mathbf{v} \cdot \mathbf{v}) + 2(\mathbf{v} \cdot \mathbf{w}) + (\mathbf{w} \cdot \mathbf{w}) & (\text{dot product properties (FOIL)}) \\ = \ \mathbf{v}\ ^2 + 2(\mathbf{v} \cdot \mathbf{w}) + \ \mathbf{w}\ ^2 & (\text{definition of length}) \\ \leq \ \mathbf{v}\ ^2 + 2\ \mathbf{v}\ \ \mathbf{w}\ + \ \mathbf{w}\ ^2 & (\text{Cauchy-Schwarz inequality}) \\ = (\ \mathbf{v}\ + \ \mathbf{w}\)^2 & (\text{factor cleverly}) \end{array}$	$\begin{array}{ll} \ \mathbf{v} + \mathbf{w}\ ^2 = (\mathbf{v} + \mathbf{w}) \cdot (\mathbf{v} + \mathbf{w}) & (\text{definition of length}) \\ = (\mathbf{v} \cdot \mathbf{v}) + 2(\mathbf{v} \cdot \mathbf{w}) + (\mathbf{w} \cdot \mathbf{w}) & (\text{dot product properties (FOIL)}) \\ = \ \mathbf{v}\ ^2 + 2(\mathbf{v} \cdot \mathbf{w}) + \ \mathbf{w}\ ^2 & (\text{definition of length}) \\ \leq \ \mathbf{v}\ ^2 + 2\ \mathbf{v}\ \ \mathbf{w}\ + \ \mathbf{w}\ ^2 & (\text{Cauchy-Schwarz inequality}) \\ = (\ \mathbf{v}\ + \ \mathbf{w}\)^2. & (\text{factor cleverly}) \end{array}$
johnston-linear-matrix-algebra-EQ0040	39	■ 0.479	$\begin{aligned} & AB = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix} \begin{bmatrix} 5 & 6 & 7 \\ 8 & 9 & 10 \end{bmatrix} \\ & = \begin{bmatrix} (1,2) \cdot (5,8) & (1,2) \cdot (6,9) & (1,2) \cdot (7,10) \\ (3,4) \cdot (5,8) & (3,4) \cdot (6,9) & (3,4) \cdot (7,10) \end{bmatrix} \\ & = \begin{bmatrix} 1 \times 5 + 2 \times 8 & 1 \times 6 + 2 \times 9 & 1 \times 7 + 2 \times 10 \\ 3 \times 5 + 4 \times 8 & 3 \times 6 + 4 \times 9 & 3 \times 7 + 4 \times 10 \end{bmatrix} \\ & = \begin{bmatrix} 21 & 24 & 27 \\ 47 & 54 & 61 \end{bmatrix}. \end{aligned}$	$\begin{aligned} AB &= \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix} \begin{bmatrix} 5 & 6 & 7 \\ 8 & 9 & 10 \end{bmatrix} \\ &= \begin{bmatrix} (1,2) \cdot (5,8) & (1,2) \cdot (6,9) & (1,2) \cdot (7,10) \\ (3,4) \cdot (5,8) & (3,4) \cdot (6,9) & (3,4) \cdot (7,10) \end{bmatrix} \\ &= \begin{bmatrix} 1 \times 5 + 2 \times 8 & 1 \times 6 + 2 \times 9 & 1 \times 7 + 2 \times 10 \\ 3 \times 5 + 4 \times 8 & 3 \times 6 + 4 \times 9 & 3 \times 7 + 4 \times 10 \end{bmatrix} \\ &= \begin{bmatrix} 21 & 24 & 27 \\ 47 & 54 & 61 \end{bmatrix}. \end{aligned}$	$\begin{aligned} AB &= \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix} \begin{bmatrix} 5 & 6 & 7 \\ 8 & 9 & 10 \end{bmatrix} \\ &= \begin{bmatrix} (1,2) \cdot (5,8) & (1,2) \cdot (6,9) & (1,2) \cdot (7,10) \\ (3,4) \cdot (5,8) & (3,4) \cdot (6,9) & (3,4) \cdot (7,10) \end{bmatrix} \\ &= \begin{bmatrix} 1 \times 5 + 2 \times 8 & 1 \times 6 + 2 \times 9 & 1 \times 7 + 2 \times 10 \\ 3 \times 5 + 4 \times 8 & 3 \times 6 + 4 \times 9 & 3 \times 7 + 4 \times 10 \end{bmatrix} \\ &= \begin{bmatrix} 21 & 24 & 27 \\ 47 & 54 & 61 \end{bmatrix}. \end{aligned}$
johnston-linear-matrix-algebra-EQ0197	95	■ 0.483	$\begin{bmatrix} 3x - 2y + z \\ 2x + 3y - 2z \end{bmatrix} = \begin{bmatrix} -3 \\ 5 \end{bmatrix}.$	$\begin{bmatrix} 3x - 2y + z \\ 2x + 3y - 2z \end{bmatrix} = \begin{bmatrix} -3 \\ 5 \end{bmatrix}.$	$\begin{bmatrix} 3x - 2y + z \\ 2x + 3y - 2z \end{bmatrix} = \begin{bmatrix} -3 \\ 5 \end{bmatrix}.$

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johnston-linear-matrix-algebra-EQ0416	161	0.483	$\begin{aligned} & \left[\begin{array}{cc c} 1 & 3 & x \\ -1 & 3 & y \\ -1 & -1 & z \end{array} \right] \xrightarrow[R_3+R_1]{R_2+R_1} \left[\begin{array}{cc c} 1 & 3 & x \\ 0 & 6 & x+y \\ 0 & 2 & x+z \end{array} \right] \\ & \xrightarrow{R_3-\frac{1}{3}R_2} \left[\begin{array}{cc c} 1 & 3 & x \\ 0 & 6 & x+y \\ 0 & 0 & \frac{2}{3}x-\frac{1}{3}y+z \end{array} \right]. \end{aligned}$	$\begin{aligned} & \left[\begin{array}{cc c} 1 & 3 & x \\ -1 & 3 & y \\ -1 & -1 & z \end{array} \right] \xrightarrow[R_3+R_1]{R_2+R_1} \left[\begin{array}{cc c} 1 & 3 & x \\ 0 & 6 & x+y \\ 0 & 2 & x+z \end{array} \right] \\ & \xrightarrow{R_3-\frac{1}{3}R_2} \left[\begin{array}{cc c} 1 & 3 & x \\ 0 & 6 & x+y \\ 0 & 0 & \frac{2}{3}x-\frac{1}{3}y+z \end{array} \right]. \end{aligned}$	
johnston-linear-matrix-algebra-EQ0559	222	0.486	$\begin{aligned} & \operatorname{operatorname{maximize}}: \quad 2x_1+x_2 \\ & \text{subject to: } \quad x_1+x_2 \geq 2 \\ & \quad \quad \quad x_1+2x_2 \geq 3 \\ & \quad \quad \quad 2x_1+2x_2 \leq 5 \\ & \quad \quad \quad x_1, \quad x_2 \geq 0 \end{aligned}$	$\begin{aligned} & \operatorname{maximize}: \quad 2x_1+x_2 \\ & \text{subject to: } \quad x_1+x_2 \geq 2 \\ & \quad \quad \quad x_1+2x_2 \geq 3 \\ & \quad \quad \quad 2x_1+2x_2 \leq 5 \\ & \quad \quad \quad x_1, \quad x_2 \geq 0 \end{aligned}$	
johnston-linear-matrix-algebra-EQ0554	221	0.491	$\begin{aligned} & \operatorname{operatorname{maximize}}: \quad x_2 \\ & \text{subject to: } \quad x_1+x_2 = -3 \\ & \quad \quad \quad 2x_1-x_2 \geq 1 \\ & \quad \quad \quad x_1 \geq 0 \end{aligned}$	$\begin{aligned} & \operatorname{maximize}: \quad x_2 \\ & \text{subject to: } \quad x_1+x_2 = -3 \\ & \quad \quad \quad 2x_1-x_2 \geq 1 \\ & \quad \quad \quad x_1 \geq 0 \end{aligned}$	
johnston-linear-matrix-algebra-EQ0547 (2.B.14)	220	0.493	$\begin{aligned} & \operatorname{operatorname{minimize}}: \quad 3y_1-3y_2+y_3 \\ & \text{subject to: } \quad y_1-y_2-y_3 \geq 1 \\ & \quad \quad \quad y_1-y_2+y_3 \geq 2 \\ & \quad \quad \quad y_1, \quad y_2, \quad y_3 \geq 0 \end{aligned}$	$\begin{aligned} & \operatorname{minimize}: \quad 3y_1-3y_2+y_3 \\ & \text{subject to: } \quad y_1-y_2-y_3 \geq 1 \\ & \quad \quad \quad y_1-y_2+y_3 \geq 2 \\ & \quad \quad \quad y_1, \quad y_2, \quad y_3 \geq 0 \end{aligned}$	
johnston-linear-matrix-algebra-EQ0673	48	0.493	$\begin{aligned} & A B = \left[\begin{array}{cc} 1 & 2 \\ 3 & 4 \end{array} \right] \left[\begin{array}{ccc} -1 & 1 & 1 \\ 0 & 1 & 0 \end{array} \right] \\ & = \left[\begin{array}{ccc} (1,2) \cdot (-1,0) & (1,2) \cdot (1,1) & (1,2) \cdot (1,0) \\ (3,4) \cdot (-1,0) & (3,4) \cdot (1,1) & (3,4) \cdot (1,0) \end{array} \right] \\ & = \left[\begin{array}{ccc} -1 & 3 & 1 \\ -3 & 7 & 3 \end{array} \right]. \end{aligned}$	$\begin{aligned} & AB = \left[\begin{array}{cc} 1 & 2 \\ 3 & 4 \end{array} \right] \left[\begin{array}{ccc} -1 & 1 & 1 \\ 0 & 1 & 0 \end{array} \right] \\ & = \left[\begin{array}{ccc} (1,2) \cdot (-1,0) & (1,2) \cdot (1,1) & (1,2) \cdot (1,0) \\ (3,4) \cdot (-1,0) & (3,4) \cdot (1,1) & (3,4) \cdot (1,0) \end{array} \right] \\ & = \left[\begin{array}{ccc} -1 & 3 & 1 \\ -3 & 7 & 3 \end{array} \right]. \end{aligned}$	
johnston-linear-matrix-algebra-EQ0557	222	0.499	$\begin{aligned} & \operatorname{operatorname{minimize}}: \quad 2x_1-x_2 \\ & \text{subject to: } \quad x_1+x_2 \leq 4 \\ & \quad \quad \quad 2x_1+3x_2 \geq 4 \\ & \quad \quad \quad x_1, \quad x_2 \geq 0 \end{aligned}$	$\begin{aligned} & \operatorname{minimize}: \quad 2x_1-x_2 \\ & \text{subject to: } \quad x_1+x_2 \leq 4 \\ & \quad \quad \quad 2x_1+3x_2 \geq 4 \\ & \quad \quad \quad x_1, \quad x_2 \geq 0 \end{aligned}$	

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johnston-linear-matrix-algebra- EQ1468	462	0.499	$\begin{aligned} & \left\{ \left[\begin{array}{ccc c} 1 & 0 & 2 & x \\ 2 & 1 & 5 & y \\ 1 & -1 & 1 & z \end{array} \right] \right\} \\ & \xrightarrow[R_3-R_1]{R_2-2R_1} \left\{ \left[\begin{array}{ccc c} 1 & 0 & 2 & x \\ 0 & 1 & 1 & y-2x \\ 0 & -1 & -1 & z-x \end{array} \right] \right\} \\ & \xrightarrow{R_3+R_2} \left\{ \left[\begin{array}{ccc c} 1 & 0 & 2 & x \\ 0 & 1 & 1 & y-2x \\ 0 & 0 & 0 & -3x+y+z \end{array} \right] \right\}. \end{aligned}$	$\begin{aligned} & \left[\begin{array}{ccc c} 1 & 0 & 2 & x \\ 2 & 1 & 5 & y \\ 1 & -1 & 1 & z \end{array} \right] \\ & \xrightarrow[R_3-R_1]{R_2-2R_1} \left[\begin{array}{ccc c} 1 & 0 & 2 & x \\ 0 & 1 & 1 & y-2x \\ 0 & -1 & -1 & z-x \end{array} \right] \\ & \xrightarrow{R_3+R_2} \left[\begin{array}{ccc c} 1 & 0 & 2 & x \\ 0 & 1 & 1 & y-2x \\ 0 & 0 & 0 & -3x+y+z \end{array} \right]. \end{aligned}$	
johnston-linear-matrix-algebra- EQ1641	482	0.505	$\begin{gathered} \left\{ \left[\begin{array}{cccc} p_{3,1} & p_{3,2} & p_{3,3} & p_{3,4} \\ p_{4,1} & p_{4,2} & p_{4,3} & p_{4,4} \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{array} \right] \right\} \\ = \left\{ \left[\begin{array}{cccc} 0 & 0 & p_{1,2} & p_{1,3} \\ 0 & 0 & p_{2,2} & p_{2,3} \\ 0 & 0 & p_{3,2} & p_{3,3} \\ 0 & 0 & p_{4,2} & p_{4,3} \end{array} \right] \right\}. \end{gathered}$	$\begin{aligned} & \left[\begin{array}{cccc} p_{3,1} & p_{3,2} & p_{3,3} & p_{3,4} \\ p_{4,1} & p_{4,2} & p_{4,3} & p_{4,4} \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{array} \right] \\ = & \left[\begin{array}{cccc} 0 & 0 & p_{1,2} & p_{1,3} \\ 0 & 0 & p_{2,2} & p_{2,3} \\ 0 & 0 & p_{3,2} & p_{3,3} \\ 0 & 0 & p_{4,2} & p_{4,3} \end{array} \right]. \end{aligned}$	$\begin{aligned} & \begin{bmatrix} p_{3,1} & p_{3,2} & p_{3,3} & p_{3,4} \\ p_{4,1} & p_{4,2} & p_{4,3} & p_{4,4} \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix} \\ = & \begin{bmatrix} 0 & 0 & p_{1,2} & p_{1,3} \\ 0 & 0 & p_{2,2} & p_{2,3} \\ 0 & 0 & p_{3,2} & p_{3,3} \\ 0 & 0 & p_{4,2} & p_{4,3} \end{bmatrix}. \end{aligned}$
johnston-linear-matrix-algebra- EQ9945	336	0.508	$D=\left[\begin{array}{ccc} 2 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & -1 \end{array}\right], P=\left[\begin{array}{ccc} 3 & 0 & 1 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{array}\right], \text{ and } P^{-1}=\frac{1}{3}\left[\begin{array}{ccc} 1 & 0 & -1 \\ -1 & 3 & 1 \\ 0 & 0 & 3 \end{array}\right].$	$D=\begin{bmatrix} 2 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & -1 \end{bmatrix}, P=\begin{bmatrix} 3 & 0 & 1 \\ 1 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}, \text{ and } P^{-1}=\frac{1}{3}\begin{bmatrix} 1 & 0 & -1 \\ -1 & 3 & 1 \\ 0 & 0 & 3 \end{bmatrix}.$	
johnston-linear-matrix-algebra- EQ9635 (0)	243	0.508	$E=\left[\begin{array}{ccccc} 0 & 0 & \star & 0 & 0 \\ 0 & 0 & \star & \star & 0 \\ \star & 0 & \star & \star & 0 \\ \star & 0 & \star & \star & \star \\ \star & \star & \star & \star & \star \end{array}\right] \text{ then } P^T=\left[\begin{array}{ccccc} 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 \\ 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 \\ 0 & 1 & 0 & 0 & 0 \end{array}\right].$	$E=\begin{bmatrix} 0 & 0 & \star & 0 & 0 \\ 0 & 0 & \star & \star & 0 \\ \star & 0 & \star & \star & 0 \\ \star & 0 & \star & \star & \star \\ \star & \star & \star & \star & \star \end{bmatrix} \text{ then } P^T=\begin{bmatrix} 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 \\ 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 \\ 0 & 1 & 0 & 0 & 0 \end{bmatrix}.$	

Identifier	Page	Conf.	LaTeX source	Rendered	Image
johnston-linear-matrix-algebra-EQ0799	295	0.514	$\begin{aligned} & \operatorname{det}\left(\begin{bmatrix} a & b & c \\ d & e & f \\ g & h & i \end{bmatrix}\right) = a \operatorname{det}\left(\begin{bmatrix} e & f \\ h & i \end{bmatrix}\right) - b \operatorname{det}\left(\begin{bmatrix} d & f \\ g & i \end{bmatrix}\right) + c \operatorname{det}\left(\begin{bmatrix} d & e \\ g & h \end{bmatrix}\right) \\ & = a(ei - fh) - b(di - fg) + c(dh - eg) \\ & = aei + bfg + cdh - afh - bdi - ceg \end{aligned}$	$\det \begin{pmatrix} \begin{bmatrix} a & b & c \\ d & e & f \\ g & h & i \end{bmatrix} \end{pmatrix} = a \det \left(\begin{bmatrix} e & f \\ h & i \end{bmatrix} \right) - b \det \left(\begin{bmatrix} d & f \\ g & i \end{bmatrix} \right) + c \det \left(\begin{bmatrix} d & e \\ g & h \end{bmatrix} \right)$ $= a(ei - fh) - b(di - fg) + c(dh - eg)$ $= aei + bfg + cdh - afh - bdi - ceg$	$\det \begin{pmatrix} \begin{bmatrix} a & b & c \\ d & e & f \\ g & h & i \end{bmatrix} \end{pmatrix} = a \det \left(\begin{bmatrix} e & f \\ h & i \end{bmatrix} \right) - b \det \left(\begin{bmatrix} d & f \\ g & i \end{bmatrix} \right) + c \det \left(\begin{bmatrix} d & e \\ g & h \end{bmatrix} \right)$ $= a(ei - fh) - b(di - fg) + c(dh - eg)$ $= aei + bfg + cdh - afh - bdi - ceg,$
johnston-linear-matrix-algebra-EQ0168	84	0.514	$\begin{aligned} A^3 &= \begin{bmatrix} 0 & 1 & 0 & 1 \\ 1 & 0 & 0 & 1 \\ 0 & 0 & 0 & 1 \\ 1 & 1 & 1 & 0 \end{bmatrix} \begin{bmatrix} 0 & 1 & 0 & 1 \\ 1 & 0 & 0 & 1 \\ 0 & 0 & 0 & 1 \\ 1 & 1 & 1 & 0 \end{bmatrix} \begin{bmatrix} 0 & 1 & 0 & 1 \\ 1 & 0 & 0 & 1 \\ 0 & 0 & 0 & 1 \\ 1 & 1 & 1 & 0 \end{bmatrix} \\ &= \begin{bmatrix} 2 & 1 & 1 & 1 \\ 1 & 2 & 1 & 1 \\ 1 & 1 & 1 & 0 \\ 1 & 1 & 0 & 3 \end{bmatrix} \begin{bmatrix} 0 & 1 & 0 & 1 \\ 1 & 0 & 0 & 1 \\ 0 & 0 & 0 & 1 \\ 1 & 1 & 1 & 0 \end{bmatrix} = \begin{bmatrix} 2 & 3 & 1 & 4 \\ 3 & 2 & 1 & 4 \\ 1 & 1 & 0 & 3 \\ 4 & 4 & 3 & 2 \end{bmatrix} \end{aligned}$	$A^3 = \begin{bmatrix} 0 & 1 & 0 & 1 \\ 1 & 0 & 0 & 1 \\ 0 & 0 & 0 & 1 \\ 1 & 1 & 1 & 0 \end{bmatrix} \begin{bmatrix} 0 & 1 & 0 & 1 \\ 1 & 0 & 0 & 1 \\ 0 & 0 & 0 & 1 \\ 1 & 1 & 1 & 0 \end{bmatrix} \begin{bmatrix} 0 & 1 & 0 & 1 \\ 1 & 0 & 0 & 1 \\ 0 & 0 & 0 & 1 \\ 1 & 1 & 1 & 0 \end{bmatrix}$ $= \begin{bmatrix} 2 & 1 & 1 & 1 \\ 1 & 2 & 1 & 1 \\ 1 & 1 & 1 & 0 \\ 1 & 1 & 0 & 3 \end{bmatrix} \begin{bmatrix} 0 & 1 & 0 & 1 \\ 1 & 0 & 0 & 1 \\ 0 & 0 & 0 & 1 \\ 1 & 1 & 1 & 0 \end{bmatrix} = \begin{bmatrix} 2 & 3 & 1 & 4 \\ 3 & 2 & 1 & 4 \\ 1 & 1 & 0 & 3 \\ 4 & 4 & 3 & 2 \end{bmatrix}.$	
johnston-linear-matrix-algebra-EQ0196 (5)	95	0.516	$\begin{bmatrix} 3 & -2 & 1 \\ 2 & 3 & -2 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} -3 \\ 5 \end{bmatrix}$	$\begin{bmatrix} 3 & -2 & 1 \\ 2 & 3 & -2 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} -3 \\ 5 \end{bmatrix}.$	
johnston-linear-matrix-algebra-EQ1378	452	0.520	$(S_{i,j}^c)^2 = (I + cE_{i,j})^2 = I + cE_{i,j} + cE_{i,j} + c^2E_{i,j}^2 = I + 2cE_{i,j} = S_{i,j}^{2c}.$	$(S_{i,j}^c)^2 = (I + cE_{i,j})^2 = I + cE_{i,j} + cE_{i,j} + c^2E_{i,j}^2 = I + 2cE_{i,j} = S_{i,j}^{2c}.$	$(S_{i,j}^c)^2 = (I + cE_{i,j})^2 = I + cE_{i,j} + cE_{i,j} + c^2E_{i,j}^2 = I + 2cE_{i,j} = S_{i,j}^{2c}.$

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johnston-linear-matrix-algebra-EQ0833	304	0.520	$\begin{aligned} & \left[\begin{array}{ccc c} 3 & 3 & 3 & 0 \\ 3 & 3 & -1 & 0 \\ 0 & 0 & 4 & 0 \end{array} \right] \xrightarrow{R_2-R_1} \left[\begin{array}{ccc c} 3 & 3 & 3 & 0 \\ 0 & 0 & -4 & 0 \\ 0 & 0 & 4 & 0 \end{array} \right] \\ & \xrightarrow{R_3+R_2} \left[\begin{array}{ccc c} 3 & 3 & 3 & 0 \\ 0 & 0 & -4 & 0 \\ 0 & 0 & 0 & 0 \end{array} \right]. \end{aligned}$		
johnston-linear-matrix-algebra-EQ0569	222	0.520	$\begin{aligned} & \text{maximize: } x_1 + 2x_2 + x_3 + 2x_4 \\ & \text{subject to: } 2x_1 + 4x_2 - x_3 \leq 3 \\ & \quad 3x_2 + x_3 + 2x_4 = 1 \\ & \quad 3x_1 - x_2 + 2x_3 + 2x_4 \leq 3 \\ & \quad x_1, x_3 \geq 0 \end{aligned}$	$\begin{aligned} & \text{maximize: } x_1 + 2x_2 + x_3 + 2x_4 \\ & \text{subject to: } 2x_1 + 4x_2 - x_3 \leq 3 \\ & \quad 3x_2 + x_3 + 2x_4 = 1 \\ & \quad 3x_1 - x_2 + 2x_3 + 2x_4 \leq 3 \\ & \quad x_1, x_3 \geq 0 \end{aligned}$	
johnston-linear-matrix-algebra-EQ0202	98	0.523	$\begin{array}{l} \text{(equation 2)} \\ -(\text{equation 1}) \\ \hline \text{(new equation 2)} \end{array}$	$\begin{array}{l} \text{(equation 2)} \\ -(\text{equation 1}) \\ \hline \text{(new equation 2)} \end{array}$	
johnston-linear-matrix-algebra-EQ0330 (3)	130	0.523	$\begin{aligned} & \left[\begin{array}{ccc ccc} 1 & 2 & 3 & 1 & 0 & 0 \\ 4 & 5 & 6 & 0 & 1 & 0 \\ 7 & 8 & 9 & 0 & 0 & 1 \end{array} \right] \xrightarrow{\substack{R_2-4R_1 \\ R_3-7R_1}} \left[\begin{array}{ccc ccc} 1 & 2 & 3 & 1 & 0 & 0 \\ 0 & -3 & -6 & -4 & 1 & 0 \\ 0 & -6 & -12 & -7 & 0 & 1 \end{array} \right] \\ & \xrightarrow{R_3-2R_2} \left[\begin{array}{ccc ccc} 1 & 2 & 3 & 1 & 0 & 0 \\ 0 & -3 & -6 & -4 & 1 & 0 \\ 0 & 0 & 0 & 1 & -2 & 1 \end{array} \right]. \end{aligned}$		
johnston-linear-matrix-algebra-EQ1423	457	0.525	$\begin{array}{l} 2x - y - 2z = 0 \\ w - y = 0 \\ w - z = 0. \end{array}$	$\begin{array}{l} 2x - y - 2z = 0 \\ w - y = 0 \\ w - z = 0. \end{array}$	$\begin{array}{l} 2x - y - 2z = 0 \\ w - y = 0 \\ w - z = 0. \end{array}$

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johnston-linear-matrix-algebra-EQ1326	445	0.526	$\begin{aligned} \lim_{x \rightarrow \infty} q(x) c^x &= \lim_{x \rightarrow \infty} \frac{c_k(x+1)^k + \dots + c_1(x+1) + c_0}{b^x} \\ &\stackrel{\text{L'Hôp}}{=} \lim_{x \rightarrow \infty} \frac{c_k k(x+1)^{k-1} + \dots + c_1}{b^x \ln(b)} \\ &\vdots \\ &\stackrel{\text{L'Hôp}}{=} \lim_{x \rightarrow \infty} \frac{c_k k!(x+1) + c_{k-1}(k-1)!}{b^x (\ln(b))^{k-1}} \\ &\stackrel{\text{L'Hôp}}{=} \lim_{x \rightarrow \infty} \frac{c_k k!}{b^x (\ln(b))^k} \\ &= 0, \end{aligned}$	$\lim_{x \rightarrow \infty} q(x) c^x = \lim_{x \rightarrow \infty} \frac{c_k(x+1)^k + \dots + c_1(x+1) + c_0}{b^x}$ $\stackrel{\text{L'Hôp}}{=} \lim_{x \rightarrow \infty} \frac{c_k k(x+1)^{k-1} + \dots + c_1}{b^x \ln(b)}$ $\vdots$ $\stackrel{\text{L'Hôp}}{=} \lim_{x \rightarrow \infty} \frac{c_k k!(x+1) + c_{k-1}(k-1)!}{b^x (\ln(b))^{k-1}}$ $\stackrel{\text{L'Hôp}}{=} \lim_{x \rightarrow \infty} \frac{c_k k!}{b^x (\ln(b))^k}$ $= 0$	$\lim_{x \rightarrow \infty} q(x) c^x = \lim_{x \rightarrow \infty} \frac{c_k(x+1)^k + \dots + c_1(x+1) + c_0}{b^x}$ $\stackrel{\text{L'Hôp}}{=} \lim_{x \rightarrow \infty} \frac{c_k k(x+1)^{k-1} + \dots + c_1}{b^x \ln(b)}$ $\vdots$ $\stackrel{\text{L'Hôp}}{=} \lim_{x \rightarrow \infty} \frac{c_k k!(x+1) + c_{k-1}(k-1)!}{b^x (\ln(b))^{k-1}}$ $\stackrel{\text{L'Hôp}}{=} \lim_{x \rightarrow \infty} \frac{c_k k!}{b^x (\ln(b))^k}$ $= 0,$
johnston-linear-matrix-algebra-EQ1192 (1)	406	0.530	$V = \begin{bmatrix} 1 & 1 & \dots & 1 \\ \lambda_0 & \lambda_1 & \dots & \lambda_{k-1} \\ \lambda_0^2 & \lambda_1^2 & \dots & \lambda_{k-1}^2 \\ \vdots & \vdots & \ddots & \vdots \\ \lambda_0^{k-1} & \lambda_1^{k-1} & \dots & \lambda_{k-1}^{k-1} \end{bmatrix} \text{ and } D = \begin{bmatrix} \lambda_0 & 0 & \dots & 0 \\ 0 & \lambda_1 & \dots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \dots & \lambda_{k-1} \end{bmatrix}.$	$V = \begin{bmatrix} 1 & 1 & \dots & 1 \\ \lambda_0 & \lambda_1 & \dots & \lambda_{k-1} \\ \lambda_0^2 & \lambda_1^2 & \dots & \lambda_{k-1}^2 \\ \vdots & \vdots & \ddots & \vdots \\ \lambda_0^{k-1} & \lambda_1^{k-1} & \dots & \lambda_{k-1}^{k-1} \end{bmatrix} \text{ and } D = \begin{bmatrix} \lambda_0 & 0 & \dots & 0 \\ 0 & \lambda_1 & \dots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \dots & \lambda_{k-1} \end{bmatrix}.$	$V = \begin{bmatrix} 1 & 1 & \dots & 1 \\ \lambda_0 & \lambda_1 & \dots & \lambda_{k-1} \\ \lambda_0^2 & \lambda_1^2 & \dots & \lambda_{k-1}^2 \\ \vdots & \vdots & \ddots & \vdots \\ \lambda_0^{k-1} & \lambda_1^{k-1} & \dots & \lambda_{k-1}^{k-1} \end{bmatrix} \text{ and } D = \begin{bmatrix} \lambda_0 & 0 & \dots & 0 \\ 0 & \lambda_1 & \dots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \dots & \lambda_{k-1} \end{bmatrix}.$
johnston-linear-matrix-algebra-EQ0717	270	0.531	$A = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}, \quad B = \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}, \quad \text{and} \quad C = \begin{bmatrix} 0 & 0 \\ 1 & 0 \end{bmatrix},$	$A = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}, \quad B = \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}, \quad \text{and} \quad C = \begin{bmatrix} 0 & 0 \\ 1 & 0 \end{bmatrix},$	$A = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}, \quad B = \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}, \quad \text{and} \quad C = \begin{bmatrix} 0 & 0 \\ 1 & 0 \end{bmatrix},$
johnston-linear-matrix-algebra-EQ0214	101	0.533	$\begin{bmatrix} \star & \star & \star & \star \\ 0 & 0 & \star & \star \\ 0 & 0 & 0 & \star \\ 0 & 0 & 0 & 0 \end{bmatrix} \text{ and } \begin{bmatrix} 0 & 0 & \star & \star & \star & \star & \star & \star & \star & \star \\ 0 & 0 & 0 & \star & \star & \star & \star & \star & \star & \star \\ 0 & 0 & 0 & 0 & 0 & \star & \star & \star & \star & \star \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \star & \star \end{bmatrix}.$	$\begin{bmatrix} \star & \star & \star & \star \\ 0 & 0 & \star & \star \\ 0 & 0 & 0 & \star \\ 0 & 0 & 0 & 0 \end{bmatrix} \text{ and } \begin{bmatrix} 0 & 0 & \star & \star & \star & \star & \star & \star & \star & \star \\ 0 & 0 & 0 & \star & \star & \star & \star & \star & \star & \star \\ 0 & 0 & 0 & 0 & 0 & \star & \star & \star & \star & \star \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \star & \star \end{bmatrix}.$	$\begin{bmatrix} \star & \star & \star & \star \\ 0 & 0 & \star & \star \\ 0 & 0 & 0 & \star \\ 0 & 0 & 0 & 0 \end{bmatrix} \text{ and } \begin{bmatrix} 0 & 0 & \star & \star & \star & \star & \star & \star & \star & \star \\ 0 & 0 & 0 & \star & \star & \star & \star & \star & \star & \star \\ 0 & 0 & 0 & 0 & 0 & \star & \star & \star & \star & \star \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \star & \star \end{bmatrix}.$
johnston-linear-matrix-algebra-EQ1195	406	0.535	$x_n = 2x_{n-1} - x_{n-2} + 2x_{n-3} \quad \text{for all } n \geq 3,$	$x_n = 2x_{n-1} - x_{n-2} + 2x_{n-3} \quad \text{for all } n \geq 3,$	$x_n = 2x_{n-1} - x_{n-2} + 2x_{n-3} \quad \text{for all } n \geq 3,$

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johnston-linear-matrix-algebra-EQ1467	462	0.538	$\begin{aligned} & \left[\begin{array}{cc c} 1 & 2 & x \\ 2 & 1 & y \end{array} \right] \\ & \xrightarrow{R_2-2R_1} \left[\begin{array}{cc c} 1 & 2 & x \\ 0 & -3 & y-2x \end{array} \right] \end{aligned}$	$\begin{bmatrix} 1 & 2 & x \\ 2 & 1 & y \end{bmatrix} \xrightarrow{R_2-2R_1} \begin{bmatrix} 1 & 2 & x \\ 0 & -3 & y-2x \end{bmatrix}$	$\begin{bmatrix} 1 & 2 & x \\ 2 & 1 & y \end{bmatrix} \xrightarrow{R_2-2R_1} \begin{bmatrix} 1 & 2 & x \\ 0 & -3 & y-2x \end{bmatrix}$
johnston-linear-matrix-algebra-EQ0975	48	0.539	$\mathbf{A} \mathbf{b}_1 = \begin{bmatrix} -1 \\ -3 \end{bmatrix}, \quad \mathbf{A} \mathbf{b}_2 = \begin{bmatrix} 3 \\ 7 \end{bmatrix}, \quad \text{and} \quad \mathbf{A} \mathbf{b}_3 = \begin{bmatrix} 1 \\ 3 \end{bmatrix}.$	$\mathbf{A} \mathbf{b}_1 = \begin{bmatrix} -1 \\ -3 \end{bmatrix}, \quad \mathbf{A} \mathbf{b}_2 = \begin{bmatrix} 3 \\ 7 \end{bmatrix}, \quad \text{and} \quad \mathbf{A} \mathbf{b}_3 = \begin{bmatrix} 1 \\ 3 \end{bmatrix}.$	$\mathbf{A} \mathbf{b}_1 = \begin{bmatrix} -1 \\ -3 \end{bmatrix}, \quad \mathbf{A} \mathbf{b}_2 = \begin{bmatrix} 3 \\ 7 \end{bmatrix}, \quad \text{and} \quad \mathbf{A} \mathbf{b}_3 = \begin{bmatrix} 1 \\ 3 \end{bmatrix}.$
johnston-linear-matrix-algebra-EQ0837	305	0.539	$\begin{aligned} & \operatorname{det}(A - \lambda I) = \det \begin{pmatrix} 1-\lambda & 2 & 3 \\ 1 & -2-\lambda & 1 \\ 3 & 2 & 1-\lambda \end{pmatrix} \\ & = (1-\lambda)(-2-\lambda)(1-\lambda) + 6 + 6 - 2(1-\lambda) - 2(1-\lambda) - 9(-2-\lambda) \\ & = -\lambda^3 + 16\lambda + 24 \end{aligned}$	$\det(A - \lambda I) = \det \left(\begin{bmatrix} 1-\lambda & 2 & 3 \\ 1 & -2-\lambda & 1 \\ 3 & 2 & 1-\lambda \end{bmatrix} \right) \\ = (1-\lambda)(-2-\lambda)(1-\lambda) + 6 + 6 - 2(1-\lambda) - 2(1-\lambda) - 9(-2-\lambda) \\ = -\lambda^3 + 16\lambda + 24$	$\det(A - \lambda I) = \det \left(\begin{bmatrix} 1-\lambda & 2 & 3 \\ 1 & -2-\lambda & 1 \\ 3 & 2 & 1-\lambda \end{bmatrix} \right) \\ = (1-\lambda)(-2-\lambda)(1-\lambda) + 6 + 6 - 2(1-\lambda) - 2(1-\lambda) - 9(-2-\lambda) \\ = -\lambda^3 + 16\lambda + 24.$
johnston-linear-matrix-algebra-EQ0571	223	0.541	$\begin{array}{l} \operatorname{maximize}: \\ \text{subject to:} \end{array} \quad \begin{array}{l} x_2 \\ cx_1 + x_2 \leq c \\ -cx_1 + x_2 \leq 0 \\ x_1, x_2 \geq 0 \end{array}$	$\begin{array}{l} \text{maximize:} \\ \text{subject to:} \end{array} \quad \begin{array}{l} x_2 \\ cx_1 + x_2 \leq c \\ -cx_1 + x_2 \leq 0 \\ x_1, x_2 \geq 0 \end{array}$	$\begin{array}{l} \text{maximize:} \\ \text{subject to:} \end{array} \quad \begin{array}{l} x_2 \\ cx_1 + x_2 \leq c \\ -cx_1 + x_2 \leq 0 \\ x_1, x_2 \geq 0 \end{array}$
johnston-linear-matrix-algebra-EQ0191	94	0.542	$\begin{array}{l} x+2y=4 \\ 2x+4y=8 \end{array} \quad \begin{array}{l} x+2y=4 \\ x+2y=3 \end{array}$	$\begin{array}{l} x+2y=4 \\ 2x+4y=8 \end{array} \quad \begin{array}{l} x+2y=4 \\ x+2y=3 \end{array}$	$\begin{array}{l} x+2y=4 \\ 2x+4y=8 \end{array} \quad \begin{array}{l} x+2y=4 \\ x+2y=3 \end{array}$
johnston-linear-matrix-algebra-EQ1287	430	0.543	$\begin{aligned} & \left(x + \frac{b}{2a} \right)^2 = \frac{b^2 - 4ac}{4a^2} \\ & \Rightarrow x + \frac{b}{2a} = \pm \frac{\sqrt{b^2 - 4ac}}{2a} \\ & \Rightarrow x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a} \end{aligned}$	$\begin{aligned} & \left(x + \frac{b}{2a} \right)^2 = \frac{b^2 - 4ac}{4a^2} \\ & \Rightarrow x + \frac{b}{2a} = \pm \frac{\sqrt{b^2 - 4ac}}{2a} \\ & \Rightarrow x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a} \end{aligned}$	$\begin{aligned} & \left(x + \frac{b}{2a} \right)^2 = \frac{b^2 - 4ac}{4a^2} \\ & \Rightarrow x + \frac{b}{2a} = \pm \frac{\sqrt{b^2 - 4ac}}{2a} \\ & \Rightarrow x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a} \end{aligned}$

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johnston-linear-matrix-algebra-EQ1460	461	0.543	$\begin{aligned} & \left[\begin{array}{cccc} A_1 & 0 & \cdots & 0 \\ 0 & A_2 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & A_n \end{array} \right] \left[\begin{array}{cccc} A_1^{-1} & 0 & \cdots & 0 \\ 0 & A_2^{-1} & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & A_n^{-1} \end{array} \right] \\ & = \left[\begin{array}{cccc} I & 0 & \cdots & 0 \\ 0 & I & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & I \end{array} \right], \end{aligned}$	$\begin{bmatrix} A_1 & 0 & \cdots & 0 \\ 0 & A_2 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & A_n \end{bmatrix} \begin{bmatrix} A_1^{-1} & 0 & \cdots & 0 \\ 0 & A_2^{-1} & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & A_n^{-1} \end{bmatrix} \\ = \begin{bmatrix} I & 0 & \cdots & 0 \\ 0 & I & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & I \end{bmatrix},$	
johnston-linear-matrix-algebra-EQ0110 (0)	61	0.545	$\begin{aligned} & \left[F_{\mathbf{u}} \right] = 2 \mathbf{u} \mathbf{u}^T - I = 2 \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} \begin{bmatrix} 1 & 1 & 1 \end{bmatrix} / 3 - \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \\ & = \frac{1}{3} \begin{bmatrix} -1 & 2 & 2 \\ 2 & -1 & 2 \\ 2 & 2 & -1 \end{bmatrix}. \end{aligned}$	$[F_{\mathbf{u}}] = 2\mathbf{u}\mathbf{u}^T - I = 2 \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} \begin{bmatrix} 1 & 1 & 1 \end{bmatrix} / 3 - \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \\ = \frac{1}{3} \begin{bmatrix} -1 & 2 & 2 \\ 2 & -1 & 2 \\ 2 & 2 & -1 \end{bmatrix}.$	$[F_{\mathbf{u}}] = 2\mathbf{u}\mathbf{u}^T - I = 2 \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} \begin{bmatrix} 1 & 1 & 1 \end{bmatrix} / 3 - \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \\ = \frac{1}{3} \begin{bmatrix} -1 & 2 & 2 \\ 2 & -1 & 2 \\ 2 & 2 & -1 \end{bmatrix}.$
johnston-linear-matrix-algebra-EQ1036	357	0.545	$A \left(\frac{1}{\det(A)} \operatorname{cof}(A)^T \right) = I,$	$A \left(\frac{1}{\det(A)} \operatorname{cof}(A)^T \right) = I,$	$A \left(\frac{1}{\det(A)} \operatorname{cof}(A)^T \right) = I,$
johnston-linear-matrix-algebra-EQ0049	41	0.546	$A^4 = (A^2)^2 = \begin{bmatrix} 3 & 5 \\ -5 & 8 \end{bmatrix} \begin{bmatrix} 3 & 5 \\ -5 & 8 \end{bmatrix} = \begin{bmatrix} -16 & 55 \\ -55 & 39 \end{bmatrix}.$	$A^4 = (A^2)^2 = \begin{bmatrix} 3 & 5 \\ -5 & 8 \end{bmatrix} \begin{bmatrix} 3 & 5 \\ -5 & 8 \end{bmatrix} = \begin{bmatrix} -16 & 55 \\ -55 & 39 \end{bmatrix}.$	$A^4 = (A^2)^2 = \begin{bmatrix} 3 & 5 \\ -5 & 8 \end{bmatrix} \begin{bmatrix} 3 & 5 \\ -5 & 8 \end{bmatrix} = \begin{bmatrix} -16 & 55 \\ -55 & 39 \end{bmatrix}.$
johnston-linear-matrix-algebra-EQ0064	313	0.546	$\begin{aligned} & \operatorname{det}(A - \lambda I) = \operatorname{det} \left(\begin{bmatrix} 3 - \lambda & 1 & 1 \\ 0 & 1 - \lambda & 0 \\ -2 & -1 & -\lambda \end{bmatrix} \right) \\ & = (3 - \lambda)(1 - \lambda)(-\lambda) + 2(1 - \lambda) \\ & = (1 - \lambda)(\lambda^2 - 3\lambda + 2) \\ & = (1 - \lambda)^2(2 - \lambda). \end{aligned}$	$\begin{aligned} \det(A - \lambda I) &= \det \left(\begin{bmatrix} 3 - \lambda & 1 & 1 \\ 0 & 1 - \lambda & 0 \\ -2 & -1 & -\lambda \end{bmatrix} \right) \\ &= (3 - \lambda)(1 - \lambda)(-\lambda) + 2(1 - \lambda) \\ &= (1 - \lambda)(\lambda^2 - 3\lambda + 2) \\ &= (1 - \lambda)^2(2 - \lambda). \end{aligned}$	$\begin{aligned} \det(A - \lambda I) &= \det \left(\begin{bmatrix} 3 - \lambda & 1 & 1 \\ 0 & 1 - \lambda & 0 \\ -2 & -1 & -\lambda \end{bmatrix} \right) \\ &= (3 - \lambda)(1 - \lambda)(-\lambda) + 2(1 - \lambda) \\ &= (1 - \lambda)(\lambda^2 - 3\lambda + 2) \\ &= (1 - \lambda)^2(2 - \lambda). \end{aligned}$

Identifier	Page	Conf.	LaTeX source	Rendered	Image
johnston-linear-matrix-algebra-EQ1567	474	0.547	$\begin{aligned} & f(A \ C \ B) = f\left(A \ \mathbf{e}_{\{j\}} \ \mathbf{e}_{\{k\}}^T \ \mathbf{e}_{\{i\}} \ \mathbf{e}_{\{j\}}^T\right) = 0 \ f\left(A \ \mathbf{e}_{\{j\}} \ \mathbf{e}_{\{j\}}^T\right) = 0, \\ & f(A \ B \ C) = f\left(A \ \mathbf{e}_{\{i\}} \ \mathbf{e}_{\{j\}}^T \ \mathbf{e}_{\{k\}}^T\right) = f\left(A \ \mathbf{e}_{\{i\}} \ \mathbf{e}_{\{k\}}^T\right) \end{aligned}$	$f(ACB) = f(Ae_j e_k^T e_i e_j^T) = 0 f(Ae_j e_j^T) = 0,$ $f(ABC) = f(Ae_i e_j^T e_j e_k^T) = f(Ae_i e_k^T).$	$f(ACB) = f(Ae_j e_k^T e_i e_j^T) = 0 f(Ae_j e_j^T) = 0,$ $f(ABC) = f(Ae_i e_j^T e_j e_k^T) = f(Ae_i e_k^T).$
johnston-linear-matrix-algebra-EQ1592	476	0.551	$\begin{gathered} \left[\begin{array}{cc} I_n & \mathbf{0} \\ \mathbf{w}^T & 1 \end{array} \right] \left[\begin{array}{cc} I_n + \mathbf{v} \mathbf{w}^T & \mathbf{v} \\ \mathbf{0}^T & 1 \end{array} \right] \left[\begin{array}{cc} I_n & \mathbf{0} \\ -\mathbf{w}^T & 1 \end{array} \right] \\ = \left[\begin{array}{cc} I_n & \mathbf{v} \\ \mathbf{0}^T & \mathbf{w}^T \mathbf{v} + 1 \end{array} \right] \end{gathered}$	$\left[\begin{array}{cc} I_n & \mathbf{0} \\ \mathbf{w}^T & 1 \end{array} \right] \left[\begin{array}{cc} I_n + \mathbf{v} \mathbf{w}^T & \mathbf{v} \\ \mathbf{0}^T & 1 \end{array} \right] \left[\begin{array}{cc} I_n & \mathbf{0} \\ -\mathbf{w}^T & 1 \end{array} \right] \\ = \left[\begin{array}{cc} I_n & \mathbf{v} \\ \mathbf{0}^T & \mathbf{w}^T \mathbf{v} + 1 \end{array} \right]$	$\left[\begin{array}{cc} I_n & \mathbf{0} \\ \mathbf{w}^T & 1 \end{array} \right] \left[\begin{array}{cc} I_n + \mathbf{v} \mathbf{w}^T & \mathbf{v} \\ \mathbf{0}^T & 1 \end{array} \right] \left[\begin{array}{cc} I_n & \mathbf{0} \\ -\mathbf{w}^T & 1 \end{array} \right] \\ = \left[\begin{array}{cc} I_n & \mathbf{v} \\ \mathbf{0}^T & \mathbf{w}^T \mathbf{v} + 1 \end{array} \right]$
johnston-linear-matrix-algebra-EQ0305	122	0.553	$\begin{aligned} & \left[\begin{array}{cc} 1 & 2 \\ 3 & 4 \end{array} \right] \left(\frac{1}{2} \left[\begin{array}{cc} -4 & 2 \\ 3 & -1 \end{array} \right] \right) = \left[\begin{array}{cc} 1 & 0 \\ 0 & 1 \end{array} \right] \quad \text{and} \\ & \left(\frac{1}{2} \left[\begin{array}{cc} -4 & 2 \\ 3 & -1 \end{array} \right] \right) \left[\begin{array}{cc} 1 & 2 \\ 3 & 4 \end{array} \right] = \left[\begin{array}{cc} 1 & 0 \\ 0 & 1 \end{array} \right]. \end{aligned}$	$\left[\begin{array}{cc} 1 & 2 \\ 3 & 4 \end{array} \right] \left(\frac{1}{2} \left[\begin{array}{cc} -4 & 2 \\ 3 & -1 \end{array} \right] \right) = \left[\begin{array}{cc} 1 & 0 \\ 0 & 1 \end{array} \right] \quad \text{and}$ $\left(\frac{1}{2} \left[\begin{array}{cc} -4 & 2 \\ 3 & -1 \end{array} \right] \right) \left[\begin{array}{cc} 1 & 2 \\ 3 & 4 \end{array} \right] = \left[\begin{array}{cc} 1 & 0 \\ 0 & 1 \end{array} \right].$	$\left[\begin{array}{cc} 1 & 2 \\ 3 & 4 \end{array} \right] \left(\frac{1}{2} \left[\begin{array}{cc} -4 & 2 \\ 3 & -1 \end{array} \right] \right) = \left[\begin{array}{cc} 1 & 0 \\ 0 & 1 \end{array} \right] \quad \text{and}$ $\left(\frac{1}{2} \left[\begin{array}{cc} -4 & 2 \\ 3 & -1 \end{array} \right] \right) \left[\begin{array}{cc} 1 & 2 \\ 3 & 4 \end{array} \right] = \left[\begin{array}{cc} 1 & 0 \\ 0 & 1 \end{array} \right].$
johnston-linear-matrix-algebra-EQ0604	233	0.557	$\begin{aligned} & \left[\begin{array}{cccc ccc} 2 & 1 & -2 & 1 & 1 & 0 & 0 \\ -4 & -4 & 3 & 0 & 0 & 1 & 0 \\ 2 & -5 & -2 & 8 & 0 & 0 & 1 \end{array} \right] \\ & \xrightarrow[R_3 - R_1]{R_2 + 2R_1} \left[\begin{array}{cccc ccc} 2 & 1 & -2 & 1 & 1 & 0 & 0 \\ 0 & -2 & -1 & 2 & 2 & 1 & 0 \\ 0 & -6 & 0 & 7 & -1 & 0 & 1 \end{array} \right] \\ & \xrightarrow{R_3 - 3R_2} \left[\begin{array}{cccc ccc} 2 & 1 & -2 & 1 & 1 & 0 & 0 \\ 0 & -2 & -1 & 2 & 2 & 1 & 0 \\ 0 & 0 & 3 & 1 & -7 & -3 & 1 \end{array} \right] \end{aligned}$	$\left[\begin{array}{cccc ccc} 2 & 1 & -2 & 1 & 1 & 0 & 0 \\ -4 & -4 & 3 & 0 & 0 & 1 & 0 \\ 2 & -5 & -2 & 8 & 0 & 0 & 1 \end{array} \right] \\ \xrightarrow[R_3 - R_1]{R_2 + 2R_1} \left[\begin{array}{cccc ccc} 2 & 1 & -2 & 1 & 1 & 0 & 0 \\ 0 & -2 & -1 & 2 & 2 & 1 & 0 \\ 0 & -6 & 0 & 7 & -1 & 0 & 1 \end{array} \right] \\ \xrightarrow{R_3 - 3R_2} \left[\begin{array}{cccc ccc} 2 & 1 & -2 & 1 & 1 & 0 & 0 \\ 0 & -2 & -1 & 2 & 2 & 1 & 0 \\ 0 & 0 & 3 & 1 & -7 & -3 & 1 \end{array} \right]$	$\left[\begin{array}{cccc ccc} 2 & 1 & -2 & 1 & 1 & 0 & 0 \\ -4 & -4 & 3 & 0 & 0 & 1 & 0 \\ 2 & -5 & -2 & 8 & 0 & 0 & 1 \end{array} \right] \\ \xrightarrow[R_3 - R_1]{R_2 + 2R_1} \left[\begin{array}{cccc ccc} 2 & 1 & -2 & 1 & 1 & 0 & 0 \\ 0 & -2 & -1 & 2 & 2 & 1 & 0 \\ 0 & -6 & 0 & 7 & -1 & 0 & 1 \end{array} \right] \\ \xrightarrow{R_3 - 3R_2} \left[\begin{array}{cccc ccc} 2 & 1 & -2 & 1 & 1 & 0 & 0 \\ 0 & -2 & -1 & 2 & 2 & 1 & 0 \\ 0 & 0 & 3 & 1 & -7 & -3 & 1 \end{array} \right]$
johnston-linear-matrix-algebra-EQ1238	417	0.558	$1 = \varepsilon \times 0 = \varepsilon \times (0 + 0) = (\varepsilon \times 0) + (\varepsilon \times 0) = 1 + 1 = 2.$	$1 = \varepsilon \times 0 = \varepsilon \times (0 + 0) = (\varepsilon \times 0) + (\varepsilon \times 0) = 1 + 1 = 2.$	$1 = \varepsilon \times 0 = \varepsilon \times (0 + 0) = (\varepsilon \times 0) + (\varepsilon \times 0) = 1 + 1 = 2.$

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johnston-linear-matrix-algebra-E01228	413	0.559	$\begin{aligned} & \text{\texttt{\backslash begin{aligned}}} \ \& \text{\texttt{x_{n}}=x_{n-1}+2 \ x_{n-2}-2} \\ & \text{\texttt{x_{n-4}-4 \ x_{n-5}+x_{n-6}+4 \ x_{n-7}+2 \ x_{n-8} \ \backslash}} \\ & \text{\texttt{\ \& \ -x_{n-9}-2 \ x_{n-10}-3 \ x_{n-13}-3 \ x_{n-14}+5}} \\ & \text{\texttt{x_{n-15}+8 \ x_{n-16}+6 \ x_{n-17} \ \backslash\ \& \ -12 \ x_{n-18}-9}} \\ & \text{\texttt{x_{n-19}-x_{n-20}+x_{n-21}+13 \ x_{n-22}-8 \ x_{n-23}+7}} \\ & \text{\texttt{x_{n-24} \ \backslash\ \& \ +12 \ x_{n-25}-11 \ x_{n-26}-4 \ x_{n-27}-19}} \\ & \text{\texttt{x_{n-28}+4 \ x_{n-29}+16 \ x_{n-30}+9 \ x_{n-31} \ \backslash\ \& \ +9}} \\ & \text{\texttt{x_{n-32}-32 \ x_{n-33}+7 \ x_{n-34}+5 \ x_{n-35}-8 \ x_{n-36}+18}} \\ & \text{\texttt{x_{n-37}-19 \ x_{n-38} \ \backslash\ \& \ +5 \ x_{n-39}+20 \ x_{n-40}-13}} \\ & \text{\texttt{x_{n-41}+x_{n-42}-3 \ x_{n-43}-13 \ x_{n-44}+17 \ x_{n-46} \ \backslash}} \\ & \text{\texttt{\ \& \ -10 \ x_{n-47}+10 \ x_{n-48}+9 \ x_{n-49}-15 \ x_{n-50}-8}} \\ & \text{\texttt{x_{n-51}-x_{n-52}+23 \ x_{n-53} \ \backslash\ \& \ -10 \ x_{n-55}-25}} \\ & \text{\texttt{x_{n-56}+8 \ x_{n-57}+24 \ x_{n-58}+9 \ x_{n-59}-13 \ x_{n-60}-16}} \\ & \text{\texttt{x_{n-61} \ \backslash\ \& \ +x_{n-62}+7 \ x_{n-63}+6 \ x_{n-64}+3}} \\ & \text{\texttt{x_{n-65}-13 \ x_{n-66}+3 \ x_{n-67}-2 \ x_{n-68} \ \backslash\ \& \ +7}} \\ & \text{\texttt{x_{n-69}+7 \ x_{n-70}-9 \ x_{n-71}+3 \ x_{n-72}-9 \ x_{n-73}+6}} \\ & \text{\texttt{x_{n-74} \ \end{aligned}}$	$\begin{aligned} x_n = & x_{n-1} + 2x_{n-2} - 2x_{n-4} - 4x_{n-5} + x_{n-6} + 4x_{n-7} + 2x_{n-8} \\ & - x_{n-9} - 2x_{n-10} - 3x_{n-13} - 3x_{n-14} + 5x_{n-15} + 8x_{n-16} + 6x_{n-17} \\ & - 12x_{n-18} - 9x_{n-19} - x_{n-20} + x_{n-21} + 13x_{n-22} - 8x_{n-23} + 7x_{n-24} \\ & + 12x_{n-25} - 11x_{n-26} - 4x_{n-27} - 19x_{n-28} + 4x_{n-29} + 16x_{n-30} + 9x_{n-31} \\ & + 9x_{n-32} - 32x_{n-33} + 7x_{n-34} + 5x_{n-35} - 8x_{n-36} + 18x_{n-37} - 19x_{n-38} \\ & + 5x_{n-39} + 20x_{n-40} - 13x_{n-41} + x_{n-42} - 3x_{n-43} - 13x_{n-44} + 17x_{n-46} \\ & - 10x_{n-47} + 10x_{n-48} + 9x_{n-49} - 15x_{n-50} - 8x_{n-51} - x_{n-52} + 23x_{n-53} \\ & - 10x_{n-55} - 25x_{n-56} + 8x_{n-57} + 24x_{n-58} + 9x_{n-59} - 13x_{n-60} - 16x_{n-61} \\ & + x_{n-62} + 7x_{n-63} + 6x_{n-64} + 3x_{n-65} - 13x_{n-66} + 3x_{n-67} - 2x_{n-68} \\ & + 7x_{n-69} + 7x_{n-70} - 9x_{n-71} + 3x_{n-72} - 9x_{n-73} + 6x_{n-74} \end{aligned}$	$\begin{aligned} x_n = & x_{n-1} + 2x_{n-2} - 2x_{n-4} - 4x_{n-5} + x_{n-6} + 4x_{n-7} + 2x_{n-8} \\ & - x_{n-9} - 2x_{n-10} - 3x_{n-13} - 3x_{n-14} + 5x_{n-15} + 8x_{n-16} + 6x_{n-17} \\ & - 12x_{n-18} - 9x_{n-19} - x_{n-20} + x_{n-21} + 13x_{n-22} - 8x_{n-23} + 7x_{n-24} \\ & + 12x_{n-25} - 11x_{n-26} - 4x_{n-27} - 19x_{n-28} + 4x_{n-29} + 16x_{n-30} + 9x_{n-31} \\ & + 9x_{n-32} - 32x_{n-33} + 7x_{n-34} + 5x_{n-35} - 8x_{n-36} + 18x_{n-37} - 19x_{n-38} \\ & + 5x_{n-39} + 20x_{n-40} - 13x_{n-41} + x_{n-42} - 3x_{n-43} - 13x_{n-44} + 17x_{n-46} \\ & - 10x_{n-47} + 10x_{n-48} + 9x_{n-49} - 15x_{n-50} - 8x_{n-51} - x_{n-52} + 23x_{n-53} \\ & - 10x_{n-55} - 25x_{n-56} + 8x_{n-57} + 24x_{n-58} + 9x_{n-59} - 13x_{n-60} - 16x_{n-61} \\ & + x_{n-62} + 7x_{n-63} + 6x_{n-64} + 3x_{n-65} - 13x_{n-66} + 3x_{n-67} - 2x_{n-68} \\ & + 7x_{n-69} + 7x_{n-70} - 9x_{n-71} + 3x_{n-72} - 9x_{n-73} + 6x_{n-74} \end{aligned}$
johnston-linear-matrix-algebra-E00098	57	0.560	$\begin{array}{l} \left[\begin{array}{lll} 1 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 3 \end{array} \right] \left[\begin{array}{lll} 1 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{array} \right] = \left[\begin{array}{lll} 1 & 1 & 1 \\ 2 & 2 & 2 \\ 3 & 3 & 3 \end{array} \right] \text{ and} \\ \left[\begin{array}{lll} 1 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{array} \right] \left[\begin{array}{lll} 1 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 3 \end{array} \right] = \left[\begin{array}{lll} 1 & 2 & 3 \\ 1 & 2 & 3 \\ 1 & 2 & 3 \end{array} \right] \end{array}$	$\begin{aligned} & \left[\begin{array}{lll} 1 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 3 \end{array} \right] \left[\begin{array}{lll} 1 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{array} \right] = \left[\begin{array}{lll} 1 & 1 & 1 \\ 2 & 2 & 2 \\ 3 & 3 & 3 \end{array} \right] \text{ and} \\ & \left[\begin{array}{lll} 1 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{array} \right] \left[\begin{array}{lll} 1 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 3 \end{array} \right] = \left[\begin{array}{lll} 1 & 2 & 3 \\ 1 & 2 & 3 \\ 1 & 2 & 3 \end{array} \right] \end{aligned}$	$\begin{aligned} & \left[\begin{array}{lll} 1 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 3 \end{array} \right] \left[\begin{array}{lll} 1 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{array} \right] = \left[\begin{array}{lll} 1 & 1 & 1 \\ 2 & 2 & 2 \\ 3 & 3 & 3 \end{array} \right] \text{ and} \\ & \left[\begin{array}{lll} 1 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{array} \right] \left[\begin{array}{lll} 1 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 3 \end{array} \right] = \left[\begin{array}{lll} 1 & 2 & 3 \\ 1 & 2 & 3 \\ 1 & 2 & 3 \end{array} \right] \end{aligned}$

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johnston-linear-matrix-algebra EQ0620	238	0.563	$\begin{aligned} & \left[\begin{array}{cccc} 1 & 0 & 0 & 0 \\ 1 & 1 & 1 & 1 \\ 1 & 2 & 4 & 8 \\ 1 & 3 & 9 & 27 \end{array} \right] \xrightarrow{\begin{smallmatrix} R_2-R_1 \\ R_3-R_1 \\ R_4-R_1 \end{smallmatrix}} \left[\begin{array}{cccc} 1 & 0 & 0 & 0 \\ 0 & 1 & 1 & 1 \\ 0 & 2 & 4 & 8 \\ 0 & 3 & 9 & 27 \end{array} \right] \\ & \xrightarrow{\begin{smallmatrix} R_3-2R_2 \\ R_4-3R_2 \end{smallmatrix}} \left[\begin{array}{cccc} 1 & 0 & 0 & 0 \\ 0 & 1 & 1 & 1 \\ 0 & 0 & 2 & 6 \\ 0 & 0 & 6 & 24 \end{array} \right] \xrightarrow{R_4-3R_3} \left[\begin{array}{cccc} 1 & 0 & 0 & 0 \\ 0 & 1 & 1 & 1 \\ 0 & 0 & 2 & 6 \\ 0 & 0 & 0 & 6 \end{array} \right]. \end{aligned}$		
johnston-linear-matrix-algebra EQ0331 (0)	130	0.564	$\begin{aligned} & \left[\begin{array}{ccc ccc} 1 & 1 & 1 & 1 & 0 & 0 \\ 1 & 2 & 4 & 0 & 1 & 0 \\ 1 & 3 & 9 & 0 & 0 & 1 \end{array} \right] \xrightarrow{\begin{smallmatrix} R_2-R_1 \\ R_3-R_1 \end{smallmatrix}} \left[\begin{array}{ccc ccc} 1 & 1 & 1 & 1 & 0 & 0 \\ 0 & 1 & 3 & -1 & 1 & 0 \\ 0 & 2 & 8 & -1 & 0 & 1 \end{array} \right] \\ & \xrightarrow{R_1-R_2} \left[\begin{array}{ccc ccc} 1 & 0 & -2 & 2 & -1 & 0 \\ 0 & 1 & 3 & -1 & 1 & 0 \\ 0 & 0 & 2 & 1 & -2 & 1 \end{array} \right] \\ & \xrightarrow{\frac{1}{2}R_3} \left[\begin{array}{ccc ccc} 1 & 0 & -2 & 2 & -1 & 0 \\ 0 & 1 & 3 & -1 & 1 & 0 \\ 0 & 0 & 1 & 1/2 & -1 & 1/2 \end{array} \right] \\ & \xrightarrow{R_1+2R_3} \left[\begin{array}{ccc ccc} 1 & 0 & 0 & 3 & -3 & 1 \\ 0 & 1 & 0 & -5/2 & 4 & -3/2 \\ 0 & 0 & 1 & 1/2 & -1 & 1/2 \end{array} \right]. \end{aligned}$		
johnston-linear-matrix-algebra EQ0647	247	0.564	$A = \left[\begin{array}{ccc} 0 & 1 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{array} \right] \quad \text{and} \quad B = \left[\begin{array}{ccc} 0 & 1 & 0 \\ 0 & 1 & 0 \\ 1 & 0 & 0 \end{array} \right]$	$A = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \quad \text{and} \quad B = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 1 & 0 \\ 1 & 0 & 0 \end{bmatrix}$	

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johnston-linear-matrix-algebra-EQ0221	103	0.570	$\left[\begin{array}{ccccc} 0 & 2 & -2 & 6 & -3 \\ 0 & 0 & -1 & 1 & -1 \\ 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 \end{array}\right] \xrightarrow[\substack{R_1+3R_3 \\ R_2+R_3}]{} \left[\begin{array}{ccccc} 0 & 2 & -2 & 6 & 0 \\ 0 & 0 & -1 & 1 & 0 \\ 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 \end{array}\right] \text{ new zeros}$		
johnston-linear-matrix-algebra-EQ0661 (5)	253	0.570	$\left[\begin{array}{c c} 2 & 1 & 5 \\ 1 & 2 & 1 \end{array}\right] \xrightarrow{R_2-\frac{1}{2}R_1} \left[\begin{array}{c c} 2 & 1 & 5 \\ 0 & 3/2 & -3/2 \end{array}\right].$		
johnston-linear-matrix-algebra-EQ0088	53	0.570	$\text{if } A=\begin{bmatrix} 1 & -1 \\ 1 & 1 \end{bmatrix} \text{ then } A\mathbf{v}=\begin{bmatrix} 1 & -1 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} v_1 \\ v_2 \end{bmatrix}=\begin{bmatrix} v_1-v_2 \\ v_1+v_2 \end{bmatrix},$		
johnston-linear-matrix-algebra-EQ1120	386	0.570	$\begin{aligned} \frac{1}{3}r_B+r_C+\frac{1}{2}r_D+\frac{1}{4}r_E &=r_A, \\ \frac{1}{2}r_A+\frac{1}{4}r_E &=r_B, \\ \frac{1}{2}r_A+\frac{1}{4}r_E &=r_C, \\ \frac{1}{3}r_B+\frac{1}{4}r_E &=r_D, \\ \frac{1}{3}r_B+\frac{1}{2}r_D &=r_E. \end{aligned}$		
johnston-linear-matrix-algebra-EQ0454	181	0.573	$\begin{array}{llll} 0+0=0 & 0+1=1 & 1+0=1 & 1+1=0, \\ 0\times 0=0 & 0\times 1=0 & 1\times 0=0 & 1\times 1=1. \end{array} \quad \text{and}$		
johnston-linear-matrix-algebra-EQ1147	395	0.580	$B=\text{diag}\left(r_1\left[R^{\theta_1}\right], \ldots, r_\ell\left[R^{\theta_\ell}\right], \lambda_1, \ldots, \lambda_m\right) \quad \text{and} \\ Q=\left[\operatorname{Re}\left(\mathbf{v}_1\right) \mid-\operatorname{Im}\left(\mathbf{v}_1\right) \mid \cdots \mid \operatorname{Re}\left(\mathbf{v}_\ell\right) \mid-\operatorname{Im}\left(\mathbf{v}_\ell\right) \mid \mathbf{w}_1 \mid \cdots \mid \mathbf{w}_m\right].$		

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johnston-linear-matrix-algebra EQ0985	345	0.580	$\begin{aligned} A^{\{2\}} &= \left[\begin{array}{cc} 0 & 4 \\ -1 & 5 \end{array} \right] \left[\begin{array}{cc} 0 & 4 \\ -1 & 5 \end{array} \right] = \left[\begin{array}{cc} -4 & 20 \\ -5 & 21 \end{array} \right] \text{ and} \\ A^3 &= A^2 A = \left[\begin{array}{cc} -4 & 20 \\ -5 & 21 \end{array} \right] \left[\begin{array}{cc} 0 & 4 \\ -1 & 5 \end{array} \right] = \left[\begin{array}{cc} -20 & 84 \\ -21 & 85 \end{array} \right]. \end{aligned}$	$A^2 = \begin{bmatrix} 0 & 4 \\ -1 & 5 \end{bmatrix} \begin{bmatrix} 0 & 4 \\ -1 & 5 \end{bmatrix} = \begin{bmatrix} -4 & 20 \\ -5 & 21 \end{bmatrix} \text{ and}$ $A^3 = A^2 A = \begin{bmatrix} -4 & 20 \\ -5 & 21 \end{bmatrix} \begin{bmatrix} 0 & 4 \\ -1 & 5 \end{bmatrix} = \begin{bmatrix} -20 & 84 \\ -21 & 85 \end{bmatrix}.$	$A^2 = \begin{bmatrix} 0 & 4 \\ -1 & 5 \end{bmatrix} \begin{bmatrix} 0 & 4 \\ -1 & 5 \end{bmatrix} = \begin{bmatrix} -4 & 20 \\ -5 & 21 \end{bmatrix} \text{ and}$ $A^3 = A^2 A = \begin{bmatrix} -4 & 20 \\ -5 & 21 \end{bmatrix} \begin{bmatrix} 0 & 4 \\ -1 & 5 \end{bmatrix} = \begin{bmatrix} -20 & 84 \\ -21 & 85 \end{bmatrix}.$
johnston-linear-matrix-algebra EQ1193	406	0.585	$\mathbf{x}_n = C^n \mathbf{x}_0 = V D^n V^{-1} \mathbf{x}_0 = V D^n \mathbf{c} \quad \text{for all } n \geq 0.$	$\mathbf{x}_n = C^n \mathbf{x}_0 = V D^n V^{-1} \mathbf{x}_0 = V D^n \mathbf{c} \quad \text{for all } n \geq 0.$	$\mathbf{x}_n = C^n \mathbf{x}_0 = V D^n V^{-1} \mathbf{x}_0 = V D^n \mathbf{c} \quad \text{for all } n \geq 0.$
johnston-linear-matrix-algebra EQ1160 (2)	399	0.591	$\begin{aligned} A^{\{40\}} &= \left(Q B Q^{-1} \right)^{40} = Q B^{40} Q^{-1} \\ &= \frac{1}{2} \begin{bmatrix} 1 & 0 & 0 \\ 0 & -1 & 1 \\ 1 & 0 & 2 \end{bmatrix} \left[\begin{array}{c c} \frac{1}{2^{20}} [R^{40\pi/4}] & \mathbf{0} \\ \hline \mathbf{0}^T & 2^{40} \end{array} \right] \begin{bmatrix} 2 & 0 & 0 \\ -1 & -2 & 1 \\ -1 & 0 & 1 \end{bmatrix} \\ &= \frac{1}{2} \begin{bmatrix} 1 & 0 & 0 \\ 0 & -1 & 1 \\ 1 & 0 & 2 \end{bmatrix} \begin{bmatrix} 1/2^{20} & 0 & 0 \\ 0 & 1/2^{20} & 0 \\ 0 & 0 & 2^{40} \end{bmatrix} \begin{bmatrix} 2 & 0 & 0 \\ -1 & -2 & 1 \\ -1 & 0 & 1 \end{bmatrix} \\ &= \frac{1}{2^{21}} \begin{bmatrix} 2 & 0 & 0 \\ 1 - 2^{60} & 2 & 2^{60} - 1 \\ 2 - 2^{61} & 0 & 2^{61} \end{bmatrix}. \end{aligned}$	$A^{40} = (QBQ^{-1})^{40} = QB^{40}Q^{-1}$ $= \frac{1}{2} \begin{bmatrix} 1 & 0 & 0 \\ 0 & -1 & 1 \\ 1 & 0 & 2 \end{bmatrix} \left[\begin{array}{c c} \frac{1}{2^{20}} [R^{40\pi/4}] & \mathbf{0} \\ \hline \mathbf{0}^T & 2^{40} \end{array} \right] \begin{bmatrix} 2 & 0 & 0 \\ -1 & -2 & 1 \\ -1 & 0 & 1 \end{bmatrix}$ $= \frac{1}{2} \begin{bmatrix} 1 & 0 & 0 \\ 0 & -1 & 1 \\ 1 & 0 & 2 \end{bmatrix} \begin{bmatrix} 1/2^{20} & 0 & 0 \\ 0 & 1/2^{20} & 0 \\ 0 & 0 & 2^{40} \end{bmatrix} \begin{bmatrix} 2 & 0 & 0 \\ -1 & -2 & 1 \\ -1 & 0 & 1 \end{bmatrix}$ $= \frac{1}{2^{21}} \begin{bmatrix} 2 & 0 & 0 \\ 1 - 2^{60} & 2 & 2^{60} - 1 \\ 2 - 2^{61} & 0 & 2^{61} \end{bmatrix}.$	$A^{40} = (QBQ^{-1})^{40} = QB^{40}Q^{-1}$ $= \frac{1}{2} \begin{bmatrix} 1 & 0 & 0 \\ 0 & -1 & 1 \\ 1 & 0 & 2 \end{bmatrix} \left[\begin{array}{c c} \frac{1}{2^{20}} [R^{40\pi/4}] & \mathbf{0} \\ \hline \mathbf{0}^T & 2^{40} \end{array} \right] \begin{bmatrix} 2 & 0 & 0 \\ -1 & -2 & 1 \\ -1 & 0 & 1 \end{bmatrix}$ $= \frac{1}{2} \begin{bmatrix} 1 & 0 & 0 \\ 0 & -1 & 1 \\ 1 & 0 & 2 \end{bmatrix} \begin{bmatrix} 1/2^{20} & 0 & 0 \\ 0 & 1/2^{20} & 0 \\ 0 & 0 & 2^{40} \end{bmatrix} \begin{bmatrix} 2 & 0 & 0 \\ -1 & -2 & 1 \\ -1 & 0 & 1 \end{bmatrix}$ $= \frac{1}{2^{21}} \begin{bmatrix} 2 & 0 & 0 \\ 1 - 2^{60} & 2 & 2^{60} - 1 \\ 2 - 2^{61} & 0 & 2^{61} \end{bmatrix}.$
johnston-linear-matrix-algebra EQ0616	237	0.602	$\left[\begin{array}{cccc} 2 & 4 & -1 & -1 \\ 4 & 9 & 0 & -1 \\ -6 & -9 & 7 & 6 \\ -2 & -2 & 9 & 0 \end{array} \right] \left[\begin{array}{c} w \\ x \\ y \\ z \end{array} \right] = \left[\begin{array}{c} 0 \\ 2 \\ 0 \\ 1 \end{array} \right]$	$\begin{bmatrix} 2 & 4 & -1 & -1 \\ 4 & 9 & 0 & -1 \\ -6 & -9 & 7 & 6 \\ -2 & -2 & 9 & 0 \end{bmatrix} \begin{bmatrix} w \\ x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 0 \\ 2 \\ 0 \\ 1 \end{bmatrix}$	$\begin{bmatrix} 2 & 4 & -1 & -1 \\ 4 & 9 & 0 & -1 \\ -6 & -9 & 7 & 6 \\ -2 & -2 & 9 & 0 \end{bmatrix} \begin{bmatrix} w \\ x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 0 \\ 2 \\ 0 \\ 1 \end{bmatrix}.$

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johnston-linear-matrix-algebra-EQ1051	360	0.606	$\begin{aligned} A &= \left[\begin{array}{ccc} 3 & -2 & 1 \\ 2 & 3 & -2 \\ 0 & 1 & 1 \end{array} \right], \quad A_1 = \left[\begin{array}{ccc} -3 & -2 & 1 \\ 5 & 3 & -2 \\ 4 & 1 & 1 \end{array} \right], \\ A_2 &= \left[\begin{array}{ccc} 3 & -3 & 1 \\ 2 & 5 & -2 \\ 0 & 4 & 1 \end{array} \right], \quad A_3 = \left[\begin{array}{ccc} -3 & -2 & 1 \\ 5 & 3 & -2 \\ 4 & 1 & 1 \end{array} \right], \end{aligned}$	$A = \begin{bmatrix} 3 & -2 & 1 \\ 2 & 3 & -2 \\ 0 & 1 & 1 \end{bmatrix}, \quad A_1 = \begin{bmatrix} -3 & -2 & 1 \\ 5 & 3 & -2 \\ 4 & 1 & 1 \end{bmatrix}, \\ A_2 = \begin{bmatrix} 3 & -3 & 1 \\ 2 & 5 & -2 \\ 0 & 4 & 1 \end{bmatrix}, \quad A_3 = \begin{bmatrix} 3 & -2 & -3 \\ 2 & 3 & 5 \\ 0 & 1 & 4 \end{bmatrix}.$	
johnston-linear-matrix-algebra-EQ1152	396	0.608	$\begin{aligned} \operatorname{\tilde{H}} D &= \operatorname{diag}(H, H, \ldots, H, 1, 1, \ldots, 1) \operatorname{diag}(D_1, D_2, \ldots, D_\ell, \lambda_1, \ldots, \lambda_m) \\ &= \operatorname{diag}(HD_1, HD_2, \ldots, HD_\ell, \lambda_1, \ldots, \lambda_m) \\ &= \operatorname{diag}(r_1[R^{\theta_1}]H, r_2[R^{\theta_2}]H, \ldots, r_\ell[R^{\theta_\ell}]H, \lambda_1, \ldots, \lambda_m) \\ &= B\operatorname{\tilde{H}}. \end{aligned}$	$\begin{aligned} \tilde{H} D &= \operatorname{diag}(H, H, \ldots, H, 1, 1, \ldots, 1) \operatorname{diag}(D_1, D_2, \ldots, D_\ell, \lambda_1, \ldots, \lambda_m) \\ &= \operatorname{diag}(HD_1, HD_2, \ldots, HD_\ell, \lambda_1, \ldots, \lambda_m) \\ &= \operatorname{diag}(r_1[R^{\theta_1}]H, r_2[R^{\theta_2}]H, \ldots, r_\ell[R^{\theta_\ell}]H, \lambda_1, \ldots, \lambda_m) \\ &= B\tilde{H}. \end{aligned}$	$\begin{aligned} \tilde{H} D &= \operatorname{diag}(H, H, \ldots, H, 1, 1, \ldots, 1) \operatorname{diag}(D_1, D_2, \ldots, D_\ell, \lambda_1, \ldots, \lambda_m) \\ &= \operatorname{diag}(HD_1, HD_2, \ldots, HD_\ell, \lambda_1, \ldots, \lambda_m) \\ &= \operatorname{diag}(r_1[R^{\theta_1}]H, r_2[R^{\theta_2}]H, \ldots, r_\ell[R^{\theta_\ell}]H, \lambda_1, \ldots, \lambda_m) \\ &= B\tilde{H}. \end{aligned}$
johnston-linear-matrix-algebra-EQ0364	142	0.611	$\begin{array}{r} c_1 + 2c_2 = x \\ 2c_1 + c_2 = y \end{array}$	$\begin{array}{l} c_1 + 2c_2 = x \\ 2c_1 + c_2 = y \end{array}$	$\begin{array}{l} c_1 + 2c_2 = x \\ 2c_1 + c_2 = y \end{array}$
johnston-linear-matrix-algebra-EQ0897	322	0.612	$\begin{aligned} &\left[\begin{array}{ccc c} 0 & 2 & 3 & 0 \\ 0 & 3 & 5 & 0 \\ 0 & 0 & 5 & 0 \end{array} \right] \xrightarrow{R_2 - \frac{3}{2}R_1} \left[\begin{array}{ccc c} 0 & 2 & 3 & 0 \\ 0 & 0 & 1/2 & 0 \\ 0 & 0 & 5 & 0 \end{array} \right] \\ &\xrightarrow{R_3 - 10R_2} \left[\begin{array}{ccc c} 0 & 2 & 3 & 0 \\ 0 & 0 & 1/2 & 0 \\ 0 & 0 & 0 & 0 \end{array} \right]. \end{aligned}$	$\begin{aligned} &\left[\begin{array}{ccc c} 0 & 2 & 3 & 0 \\ 0 & 3 & 5 & 0 \\ 0 & 0 & 5 & 0 \end{array} \right] \xrightarrow{R_2 - \frac{3}{2}R_1} \left[\begin{array}{ccc c} 0 & 2 & 3 & 0 \\ 0 & 0 & 1/2 & 0 \\ 0 & 0 & 5 & 0 \end{array} \right] \\ &\xrightarrow{R_3 - 10R_2} \left[\begin{array}{ccc c} 0 & 2 & 3 & 0 \\ 0 & 0 & 1/2 & 0 \\ 0 & 0 & 0 & 0 \end{array} \right]. \end{aligned}$	
johnston-linear-matrix-algebra-EQ0206	98	0.614	$\begin{array}{r} \text{(equation 3)} \\ + \text{(equation 2)} \\ \hline \text{(new equation 3)} \end{array} \quad \begin{array}{r} -2y + 9z = 2 \\ 2y - 6z = 4 \\ \hline 3z = 6 \end{array}$	$\begin{array}{r} \text{(equation 3)} \\ + \text{(equation 2)} \\ \hline \text{(new equation 3)} \end{array} \quad \begin{array}{r} -2y + 9z = 2 \\ 2y - 6z = 4 \\ \hline 3z = 6 \end{array}$	$\begin{array}{r} \text{(equation 3)} \\ + \text{(equation 2)} \\ \hline \text{(new equation 3)} \end{array} \quad \begin{array}{r} -2y + 9z = 2 \\ 2y - 6z = 4 \\ \hline 3z = 6 \end{array}$
johnston-linear-matrix-algebra-EQ0451	177	0.615	$A_2 = \left[\begin{array}{cc} 1 & 2 \\ 3 & 4 \end{array} \right] \quad \text{and} \quad A_3 = \left[\begin{array}{ccc} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{array} \right].$	$A_2 = \left[\begin{array}{cc} 1 & 2 \\ 3 & 4 \end{array} \right] \quad \text{and} \quad A_3 = \left[\begin{array}{ccc} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{array} \right].$	$A_2 = \left[\begin{array}{cc} 1 & 2 \\ 3 & 4 \end{array} \right] \quad \text{and} \quad A_3 = \left[\begin{array}{ccc} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{array} \right].$

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johnston-linear-matrix-algebra-E09538	216	0.615	$\begin{array}{cc} \operatorname{maximize}: & x_1 + 2x_2 \\ \text{subject to:} & x_1 + x_2 \leq 3 \\ & -x_1 + x_2 \leq 1 \\ & x_1, \quad x_2 \geq 0 \end{array}$	$\begin{array}{ll} \text{maximize :} & x_1 + 2x_2 \\ \text{subject to:} & x_1 + x_2 \leq 3 \\ & -x_1 + x_2 \leq 1 \\ & x_1, \quad x_2 \geq 0 \end{array}$	$\begin{array}{ll} \text{maximize:} & x_1 + 2x_2 \\ \text{subject to:} & x_1 + x_2 \leq 3 \\ & -x_1 + x_2 \leq 1 \\ & x_1, \quad x_2 \geq 0 \end{array}$
johnston-linear-matrix-algebra-E09567	222	0.618	$\begin{aligned} & \operatorname{minimize}: & x_1 + 2x_2 - x_3 \\ & \text{subject to:} & x_2 - x_3 \leq 3 \\ & & 2x_1 + 2x_2 - x_3 = 1 \\ & & x_1 + 2x_2 + 2x_3 \leq 3 \\ & & x_1, \quad x_2, \quad x_3 \geq 0 \end{aligned}$	$\begin{array}{ll} \text{minimize :} & x_1 + 2x_2 - x_3 \\ \text{subject to:} & x_2 - x_3 \leq 3 \\ & 2x_1 + 2x_2 - x_3 = 1 \\ & x_1 + 2x_2 + 2x_3 \leq 3 \\ & x_1, \quad x_2, \quad x_3 \geq 0 \end{array}$	$\begin{array}{ll} \text{minimize:} & x_1 + 2x_2 - x_3 \\ \text{subject to:} & x_2 - x_3 \leq 3 \\ & 2x_1 + 2x_2 - x_3 = 1 \\ & x_1 + 2x_2 + 2x_3 \leq 3 \\ & x_1, \quad x_2, \quad x_3 \geq 0 \end{array}$
johnston-linear-matrix-algebra-E09663 (3)	253	0.623	$\left[\begin{array}{ccc} 1 & 2 & 5 \\ 2 & 1 & 4 \\ 1 & 1 & 3 \end{array} \right] \xrightarrow[R_3 - R_1]{R_2 - 2R_1} \left[\begin{array}{ccc} 1 & 2 & 5 \\ 0 & -3 & -6 \\ 0 & -1 & -2 \end{array} \right] \xrightarrow[R_3 + R_2]{-\frac{1}{3}R_2} \left[\begin{array}{ccc} 1 & 2 & 5 \\ 0 & 1 & 2 \\ 0 & 0 & 0 \end{array} \right].$	$\left[\begin{array}{ccc c} 1 & 2 & 5 & \\ 2 & 1 & 4 & \\ 1 & 1 & 3 & \end{array} \right] \xrightarrow[R_3 - R_1]{R_2 - 2R_1} \left[\begin{array}{ccc c} 1 & 2 & 5 & \\ 0 & -3 & -6 & \\ 0 & -1 & -2 & \end{array} \right] \xrightarrow[R_3 + R_2]{-\frac{1}{3}R_2} \left[\begin{array}{ccc c} 1 & 2 & 5 & \\ 0 & 1 & 2 & \\ 0 & 0 & 0 & \end{array} \right].$	$\left[\begin{array}{ccc c} 1 & 2 & 5 & \\ 2 & 1 & 4 & \\ 1 & 1 & 3 & \end{array} \right] \xrightarrow[R_3 - R_1]{R_2 - 2R_1} \left[\begin{array}{ccc c} 1 & 2 & 5 & \\ 0 & -3 & -6 & \\ 0 & -1 & -2 & \end{array} \right] \xrightarrow[R_3 + R_2]{-\frac{1}{3}R_2} \left[\begin{array}{ccc c} 1 & 2 & 5 & \\ 0 & 1 & 2 & \\ 0 & 0 & 0 & \end{array} \right].$
johnston-linear-matrix-algebra-E01187	405	0.628	$\begin{aligned} & x_k = \sum_{n=0}^{k-1} a_n x_n \quad (\text{recurrence that defines } x_k) \\ & = \sum_{n=0}^{k-1} a_n \left(\sum_{j=0}^{k-1} c_j \lambda_j^n \right) \quad \left(\text{since } x_n = \sum_{j=0}^{k-1} c_j \lambda_j^n \text{ when } i \leq k-1 \right) \\ & = \sum_{j=0}^{k-1} c_j \left(\sum_{n=0}^{k-1} a_n \lambda_j^n \right) \quad (\text{swap the order of the sums}) \\ & = \sum_{j=0}^{k-1} c_j \lambda_j^k, \quad (\text{characteristic polynomial says } \lambda_j^k = \sum_{n=0}^{k-1} a_n \lambda_j^n) \end{aligned}$	$\begin{aligned} x_k &= \sum_{n=0}^{k-1} a_n x_n && (\text{recurrence that defines } x_k) \\ &= \sum_{n=0}^{k-1} a_n \left(\sum_{j=0}^{k-1} c_j \lambda_j^n \right) && \left(\text{since } x_n = \sum_{j=0}^{k-1} c_j \lambda_j^n \text{ when } i \leq k-1 \right) \\ &= \sum_{j=0}^{k-1} c_j \left(\sum_{n=0}^{k-1} a_n \lambda_j^n \right) && (\text{swap the order of the sums}) \\ &= \sum_{j=0}^{k-1} c_j \lambda_j^k, && (\text{characteristic polynomial says } \lambda_j^k = \sum_{n=0}^{k-1} a_n \lambda_j^n) \end{aligned}$	$\begin{aligned} x_k &= \sum_{n=0}^{k-1} a_n x_n && (\text{recurrence that defines } x_k) \\ &= \sum_{n=0}^{k-1} a_n \left(\sum_{j=0}^{k-1} c_j \lambda_j^n \right) && (\text{since } x_n = \sum_{j=0}^{k-1} c_j \lambda_j^n \text{ when } i \leq k-1) \\ &= \sum_{j=0}^{k-1} c_j \left(\sum_{n=0}^{k-1} a_n \lambda_j^n \right) && (\text{swap the order of the sums}) \\ &= \sum_{j=0}^{k-1} c_j \lambda_j^k, && (\text{characteristic polynomial says } \lambda_j^k = \sum_{n=0}^{k-1} a_n \lambda_j^n) \end{aligned}$
johnston-linear-matrix-algebra-E09556	221	0.628	$\begin{aligned} & \operatorname{maximize}: & x_1 + x_2 \\ & \text{subject to:} & 2x_1 + x_2 \geq 1 \\ & & x_1 + 2x_2 \leq 3 \\ & & x_1, \quad x_2 \geq 0 \end{aligned}$	$\begin{array}{ll} \text{maximize :} & x_1 + x_2 \\ \text{subject to:} & 2x_1 + x_2 \geq 1 \\ & x_1 + 2x_2 \leq 3 \\ & x_1, \quad x_2 \geq 0 \end{array}$	$\begin{array}{ll} \text{maximize:} & x_1 + x_2 \\ \text{subject to:} & 2x_1 + x_2 \geq 1 \\ & x_1 + 2x_2 \leq 3 \\ & x_1, \quad x_2 \geq 0 \end{array}$
johnston-linear-matrix-algebra-E01488	464	0.630	$\begin{aligned} & \left[\begin{array}{ccc c} 1 & 1 & -1 & 0 \\ 2 & -1 & 2 & 0 \end{array} \right] \\ & \xrightarrow{R_2 - 2R_1} \left[\begin{array}{ccc c} 1 & 1 & -1 & 0 \\ 0 & -3 & 4 & 0 \end{array} \right] \end{aligned}$	$\left[\begin{array}{ccc c} 1 & 1 & -1 & 0 \\ 2 & -1 & 2 & 0 \end{array} \right] \xrightarrow{R_2 - 2R_1} \left[\begin{array}{ccc c} 1 & 1 & -1 & 0 \\ 0 & -3 & 4 & 0 \end{array} \right]$	$\left[\begin{array}{ccc c} 1 & 1 & -1 & 0 \\ 2 & -1 & 2 & 0 \end{array} \right] \xrightarrow{R_2 - 2R_1} \left[\begin{array}{ccc c} 1 & 1 & -1 & 0 \\ 0 & -3 & 4 & 0 \end{array} \right]$

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johnston-linear-matrix-algebra-EQ1142	394	0.630	$D=\left[\begin{array}{cc} -1+2i & 0 \\ i & -1-2i \end{array}\right] \text{ and } P=\left[\begin{array}{cc} 1 & 1 \\ -1-i & -1+i \end{array}\right] .$	$D=\left[\begin{array}{cc} -1+2i & 0 \\ 0 & -1-2i \end{array}\right] \text{ and } P=\left[\begin{array}{cc} 1 & 1 \\ -1-i & -1+i \end{array}\right] .$	$D=\left[\begin{array}{cc} -1+2i & 0 \\ 0 & -1-2i \end{array}\right] \text{ and } P=\left[\begin{array}{cc} 1 & 1 \\ -1-i & -1+i \end{array}\right] .$
johnston-linear-matrix-algebra-EQ1032	356	0.633	$\begin{aligned} & \left[A \operatorname{cof}(A)^T\right]_{i,i} = a_{i,1} [\operatorname{cof}(A)^T]_{1,i} + a_{i,2} [\operatorname{cof}(A)^T]_{2,i} + \cdots + a_{i,n} [\operatorname{cof}(A)^T]_{n,i} \\ & = a_{i,1}c_{i,1} + a_{i,2}c_{i,2} + \cdots + a_{i,n}c_{i,n} \\ & = \det(A). \end{aligned}$	$\begin{aligned} & \left[A \operatorname{cof}(A)^T\right]_{i,i} = a_{i,1} [\operatorname{cof}(A)^T]_{1,i} + a_{i,2} [\operatorname{cof}(A)^T]_{2,i} + \cdots + a_{i,n} [\operatorname{cof}(A)^T]_{n,i} \\ & = a_{i,1}c_{i,1} + a_{i,2}c_{i,2} + \cdots + a_{i,n}c_{i,n} \\ & = \det(A). \end{aligned}$	$\begin{aligned} & \left[A \operatorname{cof}(A)^T\right]_{i,i} = a_{i,1} [\operatorname{cof}(A)^T]_{1,i} + a_{i,2} [\operatorname{cof}(A)^T]_{2,i} + \cdots + a_{i,n} [\operatorname{cof}(A)^T]_{n,i} \\ & = a_{i,1}c_{i,1} + a_{i,2}c_{i,2} + \cdots + a_{i,n}c_{i,n} \\ & = \det(A). \end{aligned}$
johnston-linear-matrix-algebra-EQ0070	47	0.634	$A \mathbf{v}=\left[\begin{array}{c} v_1 \\ v_2 \\ \vdots \\ v_n \end{array}\right] \text{ and } \mathbf{a}_1, \mathbf{a}_2, \ldots, \mathbf{a}_n \text{ are vectors in } \mathbb{R}^n \text{ such that } \mathbf{a}_1, \mathbf{a}_2, \ldots, \mathbf{a}_n \text{ are linearly independent.}$	$A \mathbf{v}=\left[\begin{array}{c} v_1 \\ v_2 \\ \vdots \\ v_n \end{array}\right] \text{ and } \mathbf{a}_1, \mathbf{a}_2, \ldots, \mathbf{a}_n \text{ are vectors in } \mathbb{R}^n \text{ such that } \mathbf{a}_1, \mathbf{a}_2, \ldots, \mathbf{a}_n \text{ are linearly independent.}$	$A \mathbf{v}=\left[\begin{array}{c} v_1 \\ v_2 \\ \vdots \\ v_n \end{array}\right] \text{ and } \mathbf{a}_1, \mathbf{a}_2, \ldots, \mathbf{a}_n \text{ are vectors in } \mathbb{R}^n \text{ such that } \mathbf{a}_1, \mathbf{a}_2, \ldots, \mathbf{a}_n \text{ are linearly independent.}$
johnston-linear-matrix-algebra-EQ0527 (2.B.11)	212	0.635	$\begin{aligned} & \text{maximize: } -y \\ & \text{subject to: } x_1+x_2 \leq 2 \\ & \quad -x_1-x_2-y \leq -1 \\ & \quad -2x_1+x_2-y \leq -2 \\ & \quad x_1, x_2, y \geq 0 \end{aligned}$	$\begin{aligned} & \text{maximize: } -y \\ & \text{subject to: } x_1+x_2 \leq 2 \\ & \quad -x_1-x_2-y \leq -1 \\ & \quad -2x_1+x_2-y \leq -2 \\ & \quad x_1, x_2, y \geq 0 \end{aligned}$	$\begin{aligned} & \text{maximize: } -y \\ & \text{subject to: } x_1+x_2 \leq 2 \\ & \quad -x_1-x_2-y \leq -1 \\ & \quad -2x_1+x_2-y \leq -2 \\ & \quad x_1, x_2, y \geq 0 \end{aligned}$
johnston-linear-matrix-algebra-EQ1639 (0)	482	0.641	$P=\left[\begin{array}{ccc} 1 & 0 & 1 \\ 1 & 1 & 1 \\ 0 & 1 & 1 \end{array}\right] .$	$P=\left[\begin{array}{ccc} 1 & 0 & 1 \\ 1 & 1 & 1 \\ 0 & 1 & 1 \end{array}\right] .$	$P=\left[\begin{array}{ccc} 1 & 0 & 1 \\ 1 & 1 & 1 \\ 0 & 1 & 1 \end{array}\right] .$
johnston-linear-matrix-algebra-EQ0517	206	0.643	$\xrightarrow{\begin{array}{l} R_1+2R_3 \\ R_2+3R_3 \\ R_4+R_3 \end{array}} \left[\begin{array}{c ccc ccc c} 1 & 0 & 2 & 0 & 0 & -3 & 3 & 4 \\ 0 & 1 & 3 & 0 & 0 & -5 & 5 & 7 \\ 0 & 0 & 1 & 0 & 1 & -1 & 1 & 2 \\ 0 & 0 & 1 & 1 & 0 \uparrow & 2 & -2 & 5 \end{array}\right] .$	$\xrightarrow{\begin{array}{l} R_1+2R_3 \\ R_2+3R_3 \\ R_4+R_3 \end{array}} \left[\begin{array}{c ccc ccc c} 1 & 0 & 2 & 0 & 0 & -3 & 3 & 4 \\ 0 & 1 & 3 & 0 & 0 & -5 & 5 & 7 \\ 0 & 0 & 1 & 0 & 1 & -1 & 1 & 2 \\ 0 & 0 & 1 & 1 & 0 \uparrow & 2 & -2 & 5 \end{array}\right] .$	$\xrightarrow{\begin{array}{l} R_1+2R_3 \\ R_2+3R_3 \\ R_4+R_3 \end{array}} \left[\begin{array}{c ccc ccc c} 1 & 0 & 2 & 0 & 0 & -3 & 3 & 4 \\ 0 & 1 & 3 & 0 & 0 & -5 & 5 & 7 \\ 0 & 0 & 1 & 0 & 1 & -1 & 1 & 2 \\ 0 & 0 & 1 & 1 & 0 \uparrow & 2 & -2 & 5 \end{array}\right] .$
johnston-linear-matrix-algebra-EQ1369	451	0.644	$[S]=\left[\begin{array}{ccc} 1 & 0 & 0 \\ 1 & 1 & 0 \\ 1 & 1 & 1 \end{array}\right], \quad [T]=\left[\begin{array}{cc} 1 & 1 \\ 2 & -1 \\ -1 & 3 \end{array}\right],$	$[S]=\left[\begin{array}{ccc} 1 & 0 & 0 \\ 1 & 1 & 0 \\ 1 & 1 & 1 \end{array}\right], \quad [T]=\left[\begin{array}{cc} 1 & 1 \\ 2 & -1 \\ -1 & 3 \end{array}\right],$	$[S]=\left[\begin{array}{ccc} 1 & 0 & 0 \\ 1 & 1 & 0 \\ 1 & 1 & 1 \end{array}\right], \quad [T]=\left[\begin{array}{cc} 1 & 1 \\ 2 & -1 \\ -1 & 3 \end{array}\right],$

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johnston-linear-matrix-algebra-EQ0347	135	0.651	$\left[\begin{array}{ll} A & O \\ O & D \end{array}\right]^{-1}=\left[\begin{array}{cc} A^{-1} & O \\ O & D^{-1} \end{array}\right],$	$\left[\begin{array}{ll} A & O \\ O & D \end{array}\right]^{-1}=\left[\begin{array}{cc} A^{-1} & O \\ O & D^{-1} \end{array}\right],$	$\begin{bmatrix} A & O \\ O & D \end{bmatrix}^{-1}=\begin{bmatrix} A^{-1} & O \\ O & D^{-1} \end{bmatrix},$
johnston-linear-matrix-algebra-EQ1001	349	0.651	$\begin{aligned} e^{A+B} &= \frac{1}{2}\left[\begin{array}{cc} e^2+1 & e^2-1 \\ e^2-1 & e^2+1 \end{array}\right] \quad \text{but} \\ e^Ae^B &= \left[\begin{array}{cc} e & e-1 \\ 0 & 1 \end{array}\right]\left[\begin{array}{cc} 1 & 0 \\ e-1 & e \end{array}\right]=\left[\begin{array}{cc} e^2-e+1 & e^2-e \\ e-1 & e \end{array}\right]. \end{aligned}$	$e^{A+B}=\frac{1}{2}\left[\begin{array}{cc} e^2+1 & e^2-1 \\ e^2-1 & e^2+1 \end{array}\right], \quad \text{but}$ $e^Ae^B=\left[\begin{array}{cc} e & e-1 \\ 0 & 1 \end{array}\right]\left[\begin{array}{cc} 1 & 0 \\ e-1 & e \end{array}\right]=\left[\begin{array}{cc} e^2-e+1 & e^2-e \\ e-1 & e \end{array}\right].$	$e^{A+B}=\frac{1}{2}\left[\begin{array}{cc} e^2+1 & e^2-1 \\ e^2-1 & e^2+1 \end{array}\right], \quad \text{but}$ $e^Ae^B=\left[\begin{array}{cc} e & e-1 \\ 0 & 1 \end{array}\right]\left[\begin{array}{cc} 1 & 0 \\ e-1 & e \end{array}\right]=\left[\begin{array}{cc} e^2-e+1 & e^2-e \\ e-1 & e \end{array}\right].$
johnston-linear-matrix-algebra-EQ0690	261	0.654	$\begin{aligned} P_{C \leftarrow B} &= [[\mathbf{v}_1]_C [\mathbf{v}_2]_C \cdots [\mathbf{v}_k]_C] \\ [T] &= [T(\mathbf{e}_1) T(\mathbf{e}_2) \cdots T(\mathbf{e}_k)]. \end{aligned}$	$P_{C \leftarrow B} = [[\mathbf{v}_1]_C [\mathbf{v}_2]_C \cdots [\mathbf{v}_k]_C]$ $[T] = [T(\mathbf{e}_1) T(\mathbf{e}_2) \cdots T(\mathbf{e}_k)].$	$P_{C \leftarrow B} = \left[\begin{array}{c} [\mathbf{v}_1]_C \mid [\mathbf{v}_2]_C \mid \cdots \mid [\mathbf{v}_k]_C \end{array} \right] \quad \text{and}$ $[T] = \left[\begin{array}{c} T(\mathbf{e}_1) \mid T(\mathbf{e}_2) \mid \cdots \mid T(\mathbf{e}_k) \end{array} \right].$
johnston-linear-matrix-algebra-EQ0957 (1)	338	0.654	$\begin{aligned} \left[\begin{array}{c} F_{n+1} \\ F_n \end{array}\right] &= \left[\begin{array}{cc} 1 & 1 \\ 1 & 0 \end{array}\right]^n \left[\begin{array}{c} 1 \\ 0 \end{array}\right] \\ &= \frac{1}{\sqrt{5}} \left[\begin{array}{cc} \phi & 1-\phi \\ 1 & 1 \end{array}\right] \left[\begin{array}{cc} \phi^n & 0 \\ 0 & (1-\phi)^n \end{array}\right] \left[\begin{array}{cc} 1 & \phi-1 \\ -1 & \phi \end{array}\right] \left[\begin{array}{c} 1 \\ 0 \end{array}\right] \\ &= \frac{1}{\sqrt{5}} \left[\begin{array}{cc} \phi & 1-\phi \\ 1 & 1 \end{array}\right] \left[\begin{array}{cc} \phi^n & 0 \\ 0 & (1-\phi)^n \end{array}\right] \left[\begin{array}{c} 1 \\ -1 \end{array}\right] \\ &= \frac{1}{\sqrt{5}} \left[\begin{array}{cc} \phi & 1-\phi \\ 1 & 1 \end{array}\right] \left[\begin{array}{c} \phi^n \\ -(1-\phi)^n \end{array}\right] \\ &= \frac{1}{\sqrt{5}} \left[\begin{array}{cc} \phi^{n+1} - (1-\phi)^{n+1} \\ \phi^n - (1-\phi)^n \end{array}\right]. \end{aligned}$	$\left[\begin{array}{c} F_{n+1} \\ F_n \end{array}\right] = \left[\begin{array}{cc} 1 & 1 \\ 1 & 0 \end{array}\right]^n \left[\begin{array}{c} 1 \\ 0 \end{array}\right]$ $= \frac{1}{\sqrt{5}} \left[\begin{array}{cc} \phi & 1-\phi \\ 1 & 1 \end{array}\right] \left[\begin{array}{cc} \phi^n & 0 \\ 0 & (1-\phi)^n \end{array}\right] \left[\begin{array}{cc} 1 & \phi-1 \\ -1 & \phi \end{array}\right] \left[\begin{array}{c} 1 \\ 0 \end{array}\right]$ $= \frac{1}{\sqrt{5}} \left[\begin{array}{cc} \phi & 1-\phi \\ 1 & 1 \end{array}\right] \left[\begin{array}{cc} \phi^n & 0 \\ 0 & (1-\phi)^n \end{array}\right] \left[\begin{array}{c} 1 \\ -1 \end{array}\right]$ $= \frac{1}{\sqrt{5}} \left[\begin{array}{cc} \phi & 1-\phi \\ 1 & 1 \end{array}\right] \left[\begin{array}{c} \phi^n \\ -(1-\phi)^n \end{array}\right]$ $= \frac{1}{\sqrt{5}} \left[\begin{array}{cc} \phi^{n+1} - (1-\phi)^{n+1} \\ \phi^n - (1-\phi)^n \end{array}\right].$	$\left[\begin{array}{c} F_{n+1} \\ F_n \end{array}\right] = \left[\begin{array}{cc} 1 & 1 \\ 1 & 0 \end{array}\right]^n \left[\begin{array}{c} 1 \\ 0 \end{array}\right]$ $= \frac{1}{\sqrt{5}} \left[\begin{array}{cc} \phi & 1-\phi \\ 1 & 1 \end{array}\right] \left[\begin{array}{cc} \phi^n & 0 \\ 0 & (1-\phi)^n \end{array}\right] \left[\begin{array}{cc} 1 & \phi-1 \\ -1 & \phi \end{array}\right] \left[\begin{array}{c} 1 \\ 0 \end{array}\right]$ $= \frac{1}{\sqrt{5}} \left[\begin{array}{cc} \phi & 1-\phi \\ 1 & 1 \end{array}\right] \left[\begin{array}{cc} \phi^n & 0 \\ 0 & (1-\phi)^n \end{array}\right] \left[\begin{array}{c} 1 \\ -1 \end{array}\right]$ $= \frac{1}{\sqrt{5}} \left[\begin{array}{cc} \phi & 1-\phi \\ 1 & 1 \end{array}\right] \left[\begin{array}{c} \phi^n \\ -(1-\phi)^n \end{array}\right]$ $= \frac{1}{\sqrt{5}} \left[\begin{array}{cc} \phi^{n+1} - (1-\phi)^{n+1} \\ \phi^n - (1-\phi)^n \end{array}\right].$

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johnston-linear-matrix-algebra-E09106	60	0.657	$\begin{aligned} F_{\mathbf{u}}(\mathbf{v}) &= \mathbf{v} + 2(P_{\mathbf{u}}(\mathbf{v}) - \mathbf{v}) && \text{(by Figure 1.17)} \\ &= 2P_{\mathbf{u}}(\mathbf{v}) - \mathbf{v} && \text{(expand parentheses)} \\ &= 2(\mathbf{u}\mathbf{u}^T)\mathbf{v} - \mathbf{v} && \text{(since } P_{\mathbf{u}}(\mathbf{v}) = (\mathbf{u}\mathbf{u}^T)\mathbf{v}\text{)} \\ &= (2\mathbf{u}\mathbf{u}^T - I)\mathbf{v}. && \text{(factor carefully)} \end{aligned}$	$\begin{aligned} F_{\mathbf{u}}(\mathbf{v}) &= \mathbf{v} + 2(P_{\mathbf{u}}(\mathbf{v}) - \mathbf{v}) && \text{(by Figure 1.17)} \\ &= 2P_{\mathbf{u}}(\mathbf{v}) - \mathbf{v} && \text{(expand parentheses)} \\ &= 2(\mathbf{u}\mathbf{u}^T)\mathbf{v} - \mathbf{v} && \text{(since } P_{\mathbf{u}}(\mathbf{v}) = (\mathbf{u}\mathbf{u}^T)\mathbf{v}\text{)} \\ &= (2\mathbf{u}\mathbf{u}^T - I)\mathbf{v}. && \text{(factor carefully)} \end{aligned}$	$\begin{aligned} F_{\mathbf{u}}(\mathbf{v}) &= \mathbf{v} + 2(P_{\mathbf{u}}(\mathbf{v}) - \mathbf{v}) && \text{(by Figure 1.17)} \\ &= 2P_{\mathbf{u}}(\mathbf{v}) - \mathbf{v} && \text{(expand parentheses)} \\ &= 2(\mathbf{u}\mathbf{u}^T)\mathbf{v} - \mathbf{v} && \text{(since } P_{\mathbf{u}}(\mathbf{v}) = (\mathbf{u}\mathbf{u}^T)\mathbf{v}\text{)} \\ &= (2\mathbf{u}\mathbf{u}^T - I)\mathbf{v}. && \text{(factor carefully)} \end{aligned}$
johnston-linear-matrix-algebra-E09334	131	0.658	$\left(\frac{1}{ad-bc} \begin{bmatrix} d & -b \\ -c & a \end{bmatrix}\right) \begin{bmatrix} a & b \\ c & d \end{bmatrix} = \frac{1}{ad-bc} \begin{bmatrix} ad-bc & 0 \\ 0 & ad-bc \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix},$	$\left(\frac{1}{ad-bc} \begin{bmatrix} d & -b \\ -c & a \end{bmatrix}\right) \begin{bmatrix} a & b \\ c & d \end{bmatrix} = \frac{1}{ad-bc} \begin{bmatrix} ad-bc & 0 \\ 0 & ad-bc \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix},$	$\left(\frac{1}{ad-bc} \begin{bmatrix} d & -b \\ -c & a \end{bmatrix}\right) \begin{bmatrix} a & b \\ c & d \end{bmatrix} = \frac{1}{ad-bc} \begin{bmatrix} ad-bc & 0 \\ 0 & ad-bc \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix},$
johnston-linear-matrix-algebra-E09638	244	0.658	$\begin{aligned} L &= P^{-1}E^{-1} = P^TE^{-1} \\ &= \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 1 & 0 & 0 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 & 0 \\ 1 & 0 & -1/2 & 1 \\ 1 & 1 & 0 & 0 \\ 0 & 1 & 1 & 0 \end{bmatrix} \\ &= \begin{bmatrix} 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 0 \\ 0 & 1 & 1 & 0 \\ 1 & 0 & -1/2 & 1 \end{bmatrix}. \end{aligned}$	$L = P^{-1}E^{-1} = P^TE^{-1} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 1 & 0 & 0 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 & 0 \\ 1 & 0 & -1/2 & 1 \\ 1 & 1 & 0 & 0 \\ 0 & 1 & 1 & 0 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 0 \\ 0 & 1 & 1 & 0 \\ 1 & 0 & -1/2 & 1 \end{bmatrix}.$	$L = P^{-1}E^{-1} = P^TE^{-1} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 1 & 0 & 0 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 & 0 \\ 1 & 0 & -1/2 & 1 \\ 1 & 1 & 0 & 0 \\ 0 & 1 & 1 & 0 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 0 \\ 0 & 1 & 1 & 0 \\ 1 & 0 & -1/2 & 1 \end{bmatrix}.$
johnston-linear-matrix-algebra-E09415	160	0.662	$\begin{aligned} &\left(\begin{array}{ccc c} 1 & 1 & 3 & 0 \\ 1 & 2 & 2 & 0 \\ 1 & 3 & 1 & 0 \end{array}\right) \xrightarrow[R_3-R_1]{R_2-R_1} \left(\begin{array}{ccc c} 1 & 1 & 3 & 0 \\ 0 & 1 & -1 & 0 \\ 0 & 2 & -2 & 0 \end{array}\right) \\ &\xrightarrow{R_3-2R_2} \left(\begin{array}{ccc c} 1 & 1 & 3 & 0 \\ 0 & 1 & -1 & 0 \\ 0 & 0 & 0 & 0 \end{array}\right). \end{aligned}$	$\begin{aligned} &\left(\begin{array}{ccc c} 1 & 1 & 3 & 0 \\ 1 & 2 & 2 & 0 \\ 1 & 3 & 1 & 0 \end{array}\right) \xrightarrow[R_3-R_1]{R_2-R_1} \left(\begin{array}{ccc c} 1 & 1 & 3 & 0 \\ 0 & 1 & -1 & 0 \\ 0 & 2 & -2 & 0 \end{array}\right) \\ &\xrightarrow{R_3-2R_2} \left(\begin{array}{ccc c} 1 & 1 & 3 & 0 \\ 0 & 1 & -1 & 0 \\ 0 & 0 & 0 & 0 \end{array}\right). \end{aligned}$	$\begin{aligned} &\left(\begin{array}{ccc c} 1 & 1 & 3 & 0 \\ 1 & 2 & 2 & 0 \\ 1 & 3 & 1 & 0 \end{array}\right) \xrightarrow[R_3-R_1]{R_2-R_1} \left(\begin{array}{ccc c} 1 & 1 & 3 & 0 \\ 0 & 1 & -1 & 0 \\ 0 & 2 & -2 & 0 \end{array}\right) \\ &\xrightarrow{R_3-2R_2} \left(\begin{array}{ccc c} 1 & 1 & 3 & 0 \\ 0 & 1 & -1 & 0 \\ 0 & 0 & 0 & 0 \end{array}\right). \end{aligned}$

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johnston-linear-matrix-algebra EQ0443	173	0.665	$\begin{array}{r} \left[\begin{array}{ccccc} 0 & 0 & -2 & 2 & -2 \\ 2 & -2 & -1 & 3 & 3 \\ -1 & 1 & -1 & 0 & -3 \end{array} \right] \xrightarrow{R_1 \leftrightarrow R_3} \left[\begin{array}{ccccc} -1 & 1 & -1 & 0 & -3 \\ 2 & -2 & -1 & 3 & 3 \\ 0 & 0 & -2 & 2 & -2 \end{array} \right] \\ \xrightarrow{R_2+2R_1} \left[\begin{array}{ccccc} -1 & 1 & -1 & 0 & -3 \\ 0 & 0 & -3 & 3 & -3 \\ 0 & 0 & -2 & 2 & -2 \end{array} \right] \xrightarrow{R_3-\frac{2}{3}R_2} \left[\begin{array}{ccccc} -1 & 1 & -1 & 0 & -3 \\ 0 & 0 & -3 & 3 & -3 \\ 0 & 0 & 0 & 0 & 0 \end{array} \right] \end{array}$		
johnston-linear-matrix-algebra EQ0892	320	0.670	$\begin{array}{l} \begin{array}{c} \left[\begin{array}{cccc c} -3 & 0 & -2 & 1 & 0 \\ 0 & 0 & -2 & 1 & 0 \\ 3 & 0 & 4 & 1 & 0 \\ -3 & 0 & 2 & 5 & 0 \end{array} \right] \xrightarrow{\substack{R_3+R_1 \\ R_4-R_1}} \left[\begin{array}{cccc c} -3 & 0 & -2 & 1 & 0 \\ 0 & 0 & -2 & 1 & 0 \\ 0 & 0 & 2 & 2 & 0 \\ 0 & 0 & 4 & 4 & 0 \end{array} \right] \\ \xrightarrow{\substack{R_1+R_3 \\ R_2+R_3}} \left[\begin{array}{cccc c} -3 & 0 & 0 & 3 & 0 \\ 0 & 0 & 0 & 3 & 0 \\ 0 & 0 & 2 & 2 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{array} \right] \xrightarrow{R_1-R_2} \left[\begin{array}{cccc c} -3 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 3 & 0 \\ 0 & 0 & 2 & 2 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{array} \right] \end{array} \\ \begin{array}{c} \left[\begin{array}{cccc c} -3 & 0 & -2 & 1 & 0 \\ 0 & 0 & -2 & 1 & 0 \\ 3 & 0 & 4 & 1 & 0 \\ -3 & 0 & 2 & 5 & 0 \end{array} \right] \xrightarrow{\substack{R_3+R_1 \\ R_4-R_1}} \left[\begin{array}{cccc c} -3 & 0 & -2 & 1 & 0 \\ 0 & 0 & -2 & 1 & 0 \\ 0 & 0 & 2 & 2 & 0 \\ 0 & 0 & 4 & 4 & 0 \end{array} \right] \\ \xrightarrow{\substack{R_1+R_3 \\ R_2+R_3}} \left[\begin{array}{cccc c} -3 & 0 & 0 & 3 & 0 \\ 0 & 0 & 0 & 3 & 0 \\ 0 & 0 & 2 & 2 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{array} \right] \xrightarrow{R_1-R_2} \left[\begin{array}{cccc c} -3 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 3 & 0 \\ 0 & 0 & 2 & 2 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{array} \right] \end{array} \end{array}$	$\left[\begin{array}{cccc c} -3 & 0 & -2 & 1 & 0 \\ 0 & 0 & -2 & 1 & 0 \\ 3 & 0 & 4 & 1 & 0 \\ -3 & 0 & 2 & 5 & 0 \end{array} \right] \xrightarrow{\substack{R_3+R_1 \\ R_4-R_1}} \left[\begin{array}{cccc c} -3 & 0 & -2 & 1 & 0 \\ 0 & 0 & -2 & 1 & 0 \\ 0 & 0 & 2 & 2 & 0 \\ 0 & 0 & 4 & 4 & 0 \end{array} \right] \\ \xrightarrow{\substack{R_1+R_3 \\ R_2+R_3 \\ R_4-2R_3}} \left[\begin{array}{cccc c} -3 & 0 & 0 & 3 & 0 \\ 0 & 0 & 0 & 3 & 0 \\ 0 & 0 & 2 & 2 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{array} \right] \xrightarrow{R_1-R_2} \left[\begin{array}{cccc c} -3 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 3 & 0 \\ 0 & 0 & 2 & 2 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{array} \right].$	
johnston-linear-matrix-algebra EQ0359	140	0.671	$\left[\begin{array}{ll l} 2 & -2 & 0 \\ 1 & -1 & 0 \end{array} \right] \xrightarrow{R_1 \leftrightarrow R_2} \left[\begin{array}{ll l} 1 & -1 & 0 \\ 2 & -2 & 0 \end{array} \right] \xrightarrow{R_2-2R_1} \left[\begin{array}{ll l} 1 & -1 & 0 \\ 0 & 0 & 0 \end{array} \right],$		
johnston-linear-matrix-algebra EQ0962 (2)	339	0.676	$P^{-1}=\frac{1}{30}\left[\begin{array}{ccccc} 6 & 6 & -6 & -6 & -6 \\ 15 & -15 & 0 & 0 & 0 \\ 0 & 0 & 20 & -10 & -10 \\ 0 & 0 & 10 & 10 & -20 \\ 3 & 3 & 2 & 2 & 2 \end{array} \right],$	$P^{-1}=\frac{1}{30}\left[\begin{array}{ccccc} 6 & 6 & -6 & -6 & -6 \\ 15 & -15 & 0 & 0 & 0 \\ 0 & 0 & 20 & -10 & -10 \\ 0 & 0 & 10 & 10 & -20 \\ 3 & 3 & 2 & 2 & 2 \end{array} \right],$	

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johnston-linear-matrix-algebra-EQ0521	207	0.682	$\begin{aligned} & \begin{array}{lllllll} z & s_1 & s_2 & s_3 & x_1 & x_2 & x_3 \end{array} \\ & \left[\begin{array}{c ccc ccc} 1 & 0 & 2 & 0 & 0 & -3 & 3 & 4 \\ 0 & 1 & 3 & 0 & 0 & -5 & 5 & 7 \\ 0 & 0 & 1 & 0 & 1 & -1 & 1 & 2 \\ 0 & 0 & 1 & 1 & 0 & 2 & -2 & 5 \end{array} \right] \end{aligned}$ new "z" value ↓ $\begin{array}{c} (1/2)R_4 \\ R_1+3R_4 \\ R_2+5R_4 \\ R_3+R_4 \end{array} \rightarrow \left[\begin{array}{c ccc ccc} 1 & 0 & 7/2 & 3/2 & 0 & 0 & 0 & 23/2 \\ 0 & 1 & 11/2 & 5/2 & 0 & 0 & 0 & 39/2 \\ 0 & 0 & 3/2 & 1/2 & 1 & 0 & 0 & 9/2 \\ 0 & 0 & 1/2 & 1/2 & 0 & 1 & -1 & 5/2 \end{array} \right]$ ↑ new "leading" entry	$\begin{array}{c} z \quad s_1 \quad s_2 \quad s_3 \quad x_1 \quad x_2 \quad x_3 \\ \left[\begin{array}{c ccc ccc} 1 & 0 & 2 & 0 & 0 & -3 & 3 & 4 \\ 0 & 1 & 3 & 0 & 0 & -5 & 5 & 7 \\ 0 & 0 & 1 & 0 & 1 & -1 & 1 & 2 \\ 0 & 0 & 1 & 1 & 0 & 2 & -2 & 5 \end{array} \right] \end{array}$ new "z" value ↓ $\begin{array}{c} (1/2)R_4 \\ R_1+3R_4 \\ R_2+5R_4 \\ R_3+R_4 \end{array} \rightarrow \left[\begin{array}{c ccc ccc} 1 & 0 & 7/2 & 3/2 & 0 & 0 & 0 & 23/2 \\ 0 & 1 & 11/2 & 5/2 & 0 & 0 & 0 & 39/2 \\ 0 & 0 & 3/2 & 1/2 & 1 & 0 & 0 & 9/2 \\ 0 & 0 & 1/2 & 1/2 & 0 & 1 & -1 & 5/2 \end{array} \right]$ ↑ new "leading" entry	
johnston-linear-matrix-algebra-EQ1329	446	0.685	$\begin{aligned} & \text{(a) We just use the definition of vector addition and scalar multiplication, together with properties of real numbers:} \\ & (c+d)\mathbf{v} = ((c+d)v_1, \dots, (c+d)v_n) \\ & = (cv_1 + dv_1, \dots, cv_n + dv_n) \\ & = (cv_1, \dots, cv_n) + (dv_1, \dots, dv_n) \\ & = c\mathbf{v} + d\mathbf{v}. \end{aligned}$	(a) We just use the definition of vector addition and scalar multiplication, together with properties of real numbers: $\begin{aligned} (c+d)\mathbf{v} &= ((c+d)v_1, \dots, (c+d)v_n) \\ &= (cv_1 + dv_1, \dots, cv_n + dv_n) \\ &= (cv_1, \dots, cv_n) + (dv_1, \dots, dv_n) \\ &= c\mathbf{v} + d\mathbf{v}. \end{aligned}$	(a) We just use the definition of vector addition and scalar multiplication, together with properties of real numbers: $\begin{aligned} (c+d)\mathbf{v} &= ((c+d)v_1, \dots, (c+d)v_n) \\ &= (cv_1 + dv_1, \dots, cv_n + dv_n) \\ &= (cv_1, \dots, cv_n) + (dv_1, \dots, dv_n) \\ &= c\mathbf{v} + d\mathbf{v}. \end{aligned}$
johnston-linear-matrix-algebra-EQ0760	286	0.686	$\operatorname{det}\left(\left[\begin{array}{cc} a & b \\ 0 & d \end{array}\right]\right)=a d$	$\det\left(\left[\begin{array}{cc} a & b \\ 0 & d \end{array}\right]\right)=a d$	$\det\left(\left[\begin{array}{cc} a & b \\ 0 & d \end{array}\right]\right)=a d$
johnston-linear-matrix-algebra-EQ1230	414	0.686	$\lim _{n \rightarrow \infty} \frac{x_{n+1}}{x_n} \approx 1.3036 .$	$\lim _{n \rightarrow \infty} \frac{x_{n+1}}{x_n} \approx 1.3036 .$	$\lim _{n \rightarrow \infty} \frac{x_{n+1}}{x_n} \approx 1.3036 .$

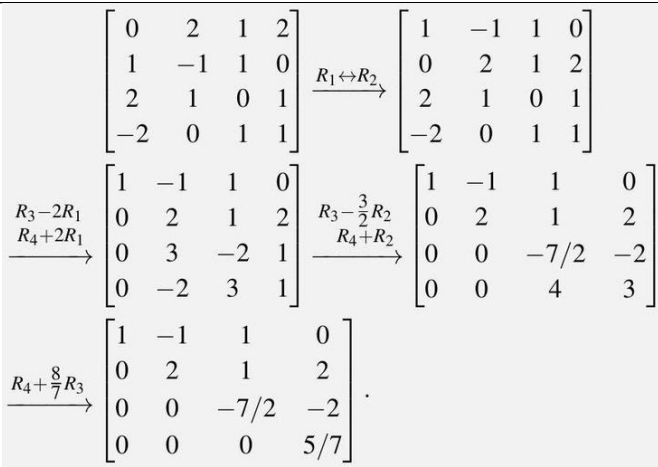
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johnston-linear-matrix-algebra EQ0472	195	0.687	<pre>\begin{aligned} & \left[\begin{array}{llll l} 2 & 0 & 4 & 0 & 1 \\ 0 & 1 & 1 & 0 & 1 \\ 0 & 4 & 4 & 1 & 1 \\ 0 & 3 & 3 & 2 & 2 \end{array}\right] \xrightarrow{3R_1} \left[\begin{array}{llll l} 1 & 0 & 2 & 0 & 3 \\ 0 & 1 & 1 & 0 & 1 \\ 0 & 4 & 4 & 1 & 1 \\ 0 & 3 & 3 & 2 & 2 \end{array}\right] \\ & \xrightarrow{R_2-3R_1, R_4-3R_1} \left[\begin{array}{llll l} 1 & 0 & 2 & 0 & 3 \\ 0 & 1 & 1 & 0 & 1 \\ 0 & 4 & 4 & 1 & 1 \\ 0 & 0 & 0 & 2 & 4 \end{array}\right] \\ & \xrightarrow{R_3-4R_2, R_4-3R_2} \left[\begin{array}{llll l} 1 & 0 & 2 & 0 & 3 \\ 0 & 1 & 1 & 0 & 1 \\ 0 & 0 & 0 & 1 & 2 \\ 0 & 0 & 0 & 0 & 0 \end{array}\right] \\ & \xrightarrow{R_4-2R_3} \left[\begin{array}{llll l} 1 & 0 & 2 & 0 & 3 \\ 0 & 1 & 1 & 0 & 1 \\ 0 & 0 & 0 & 1 & 2 \\ 0 & 0 & 0 & 0 & 0 \end{array}\right]. \end{aligned}</pre>		
johnston-linear-matrix-algebra EQ0582	225	0.689	<pre>\begin{aligned} C^T &= \left[\begin{array}{cccccccccc} 1 & 0 & 1 & 0 & 1 & 0 & 1 & 0 & 1 & 0 & 1 \\ 0 & 1 & 0 & 2 & 0 & -1 & 0 & 2 & 0 & 1 & 0 \end{array}\right] \text{ and} \\ R &= \left[\begin{array}{cccccccccc} 1 & 0 & -1 & 0 & 2 & 0 & 2 & 0 & -1 & 0 & 1 \\ 0 & 2 & 0 & 1 & 0 & 3 & 0 & 1 & 0 & 2 & 0 \end{array}\right], \end{aligned}</pre>	$C^T = \begin{bmatrix} 1 & 0 & 1 & 0 & 1 & 0 & 1 & 0 & 1 & 0 & 1 \\ 0 & 1 & 0 & 2 & 0 & -1 & 0 & 2 & 0 & 1 & 0 \end{bmatrix}$ and $R = \begin{bmatrix} 1 & 0 & -1 & 0 & 2 & 0 & 2 & 0 & -1 & 0 & 1 \\ 0 & 2 & 0 & 1 & 0 & 3 & 0 & 1 & 0 & 2 & 0 \end{bmatrix},$	
johnston-linear-matrix-algebra EQ1335	447	0.690	<pre>\begin{aligned} & \overline{\mathbf{w}} \cdot \mathbf{v} = \overline{w_1} \overline{v_1} + \cdots + \overline{w_n} \overline{v_n} \\ & = \overline{w_1 v_1} + \cdots + \overline{w_n v_n} \\ & = \overline{v_1 w_1} + \cdots + \overline{v_n w_n} \\ & = \mathbf{v} \cdot \mathbf{w}, \end{aligned}</pre>	$\overline{\mathbf{w}} \cdot \mathbf{v} = \overline{w_1} \overline{v_1} + \cdots + \overline{w_n} \overline{v_n}$ $= \overline{w_1 v_1} + \cdots + \overline{w_n v_n}$ $= \overline{v_1 w_1} + \cdots + \overline{v_n w_n}$ $= \mathbf{v} \cdot \mathbf{w},$	$\overline{\mathbf{w}} \cdot \mathbf{v} = \overline{w_1} \overline{v_1} + \cdots + \overline{w_n} \overline{v_n}$ $= \overline{w_1 v_1} + \cdots + \overline{w_n v_n}$ $= \overline{v_1 w_1} + \cdots + \overline{v_n w_n}$ $= \mathbf{v} \cdot \mathbf{w},$
johnston-linear-matrix-algebra EQ0699	263	0.690	<pre>\begin{aligned} (3,3) &= 3(1,1) + 0(1,-1) \text{ so } [(3,3)]_B = (3,0), \text{ and} \\ (1,-1) &= 0(1,1) + 1(1,-1) \text{ so } [(1,-1)]_B = (0,1). \end{aligned}</pre>	$(3,3) = 3(1,1) + 0(1,-1)$ so $[(3,3)]_B = (3,0),$ and $(1,-1) = 0(1,1) + 1(1,-1)$ so $[(1,-1)]_B = (0,1).$	$(3,3) = 3(1,1) + 0(1,-1)$ so $[(3,3)]_B = (3,0),$ and $(1,-1) = 0(1,1) + 1(1,-1)$ so $[(1,-1)]_B = (0,1).$

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johnston-linear-matrix-algebra-EQ0213	101	0.691	$\left[\begin{array}{ccc c} 1 & 3 & -2 & 5 \\ 0 & 2 & -6 & 4 \\ 0 & 0 & 3 & 6 \end{array}\right] \quad \text{and} \quad \left[\begin{array}{ccc c} 1 & 0 & 0 & -15 \\ 0 & 1 & 0 & 8 \\ 0 & 0 & 1 & 2 \end{array}\right]$		$\left[\begin{array}{ccc c} 1 & 3 & -2 & 5 \\ 0 & 2 & -6 & 4 \\ 0 & 0 & 3 & 6 \end{array}\right] \quad \text{and} \quad \left[\begin{array}{ccc c} 1 & 0 & 0 & -15 \\ 0 & 1 & 0 & 8 \\ 0 & 0 & 1 & 2 \end{array}\right]$
johnston-linear-matrix-algebra-EQ1293	437	0.693	$A=\left[\begin{array}{cccc} A_{1,1} & A_{1,2} & \cdots & A_{1,n} \\ A_{2,1} & A_{2,2} & \cdots & A_{2,n} \\ \vdots & \vdots & \ddots & \vdots \\ A_{m,1} & A_{m,2} & \cdots & A_{m,n} \end{array}\right] \quad \text{and} \quad B=\left[\begin{array}{cccc} B_{1,1} & B_{1,2} & \cdots & B_{1,p} \\ B_{2,1} & B_{2,2} & \cdots & B_{2,p} \\ \vdots & \vdots & \ddots & \vdots \\ B_{n,1} & B_{n,2} & \cdots & B_{n,p} \end{array}\right]$		$A=\left[\begin{array}{cccc} A_{1,1} & A_{1,2} & \cdots & A_{1,n} \\ A_{2,1} & A_{2,2} & \cdots & A_{2,n} \\ \vdots & \vdots & \ddots & \vdots \\ A_{m,1} & A_{m,2} & \cdots & A_{m,n} \end{array}\right] \quad \text{and} \quad B=\left[\begin{array}{cccc} B_{1,1} & B_{1,2} & \cdots & B_{1,p} \\ B_{2,1} & B_{2,2} & \cdots & B_{2,p} \\ \vdots & \vdots & \ddots & \vdots \\ B_{n,1} & B_{n,2} & \cdots & B_{n,p} \end{array}\right]$
johnston-linear-matrix-algebra-EQ1630	481	0.693	$A=\left[\begin{array}{ll} 0 & 1 \\ 0 & 0 \end{array}\right] \quad \text{and} \quad B=\left[\begin{array}{cc} 0 & 2 \\ 0 & 0 \end{array}\right]$		$A=\left[\begin{array}{cc} 0 & 1 \\ 0 & 0 \end{array}\right] \quad \text{and} \quad B=\left[\begin{array}{cc} 0 & 2 \\ 0 & 0 \end{array}\right]$
johnston-linear-matrix-algebra-EQ0585	226	0.694	$A=C \quad R=\left[\begin{array}{c} \mathbf{v}_1 \\ \mathbf{v}_2 \\ \vdots \\ \mathbf{v}_r \end{array}\right] \quad \left[\begin{array}{c} \frac{\mathbf{w}_1^T}{\mathbf{w}_2^T} \\ \vdots \\ \frac{\mathbf{w}_n^T}{\mathbf{w}_n^T} \end{array}\right] = \sum_{j=1}^r \mathbf{v}_j \mathbf{w}_j^T,$		$A=CR=\left[\mathbf{v}_1 \mid \mathbf{v}_2 \mid \cdots \mid \mathbf{v}_r\right] \left[\begin{array}{c} \frac{\mathbf{w}_1^T}{\mathbf{w}_2^T} \\ \vdots \\ \frac{\mathbf{w}_n^T}{\mathbf{w}_n^T} \end{array}\right] = \sum_{j=1}^r \mathbf{v}_j \mathbf{w}_j^T,$
johnston-linear-matrix-algebra-EQ0927	330	0.696	$\left[\begin{array}{cc c} 1 & -1 & 0 \\ 1 & -1 & 0 \end{array}\right] \xrightarrow{R_2-R_1} \left[\begin{array}{cc c} 1 & -1 & 0 \\ 0 & 0 & 0 \end{array}\right].$		$\left[\begin{array}{cc c} 1 & -1 & 0 \\ 1 & -1 & 0 \end{array}\right] \xrightarrow{R_2-R_1} \left[\begin{array}{cc c} 1 & -1 & 0 \\ 0 & 0 & 0 \end{array}\right].$
johnston-linear-matrix-algebra-EQ1357	450	0.696	$\begin{aligned} [A(BC)]_{i,j} &= \sum_{k=1}^n a_{i,k} \left(\sum_{\ell=1}^n b_{k,\ell} c_{\ell,j} \right) \\ &= \sum_{\ell=1}^n \left(\sum_{k=1}^n a_{i,k} b_{k,\ell} \right) c_{\ell,j} \\ &= [(AB)C]_{i,j} \end{aligned}$		$\begin{aligned} [A(BC)]_{i,j} &= \sum_{k=1}^n a_{i,k} \left(\sum_{\ell=1}^n b_{k,\ell} c_{\ell,j} \right) \\ &= \sum_{\ell=1}^n \left(\sum_{k=1}^n a_{i,k} b_{k,\ell} \right) c_{\ell,j} \\ &= [(AB)C]_{i,j}. \end{aligned}$
johnston-linear-matrix-algebra-EQ0143 (0)	71	0.697	$\begin{aligned} \left[R_{\mathbf{u}}^\theta\right] &= \cos(\theta)I + (1 - \cos(\theta))\mathbf{u}\mathbf{u}^T \\ &+ \sin(\theta) \begin{bmatrix} 0 & -u_3 & u_2 \\ u_3 & 0 & -u_1 \\ -u_2 & u_1 & 0 \end{bmatrix}. \end{aligned}$		$\begin{aligned} [R_{\mathbf{u}}^\theta] &= \cos(\theta)I + (1 - \cos(\theta))\mathbf{u}\mathbf{u}^T \\ &+ \sin(\theta) \begin{bmatrix} 0 & -u_3 & u_2 \\ u_3 & 0 & -u_1 \\ -u_2 & u_1 & 0 \end{bmatrix}. \end{aligned}$

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johnston-linear-matrix-algebra EQ1080	368	0.697	$\begin{array}{l} \operatorname{det}(A_{1,2})=1, \text{ \& } \operatorname{det}(A_{1,3})=-5, \\ \text{ \& } \operatorname{det}(A_{2,3})=-2, \text{ \& } \\ \operatorname{det}(B_{1,2})=11, \text{ \& } \\ \operatorname{det}(B_{1,3})=7, \text{ \& } \text{and } \\ \operatorname{det}(B_{2,3})=-1. \end{array}$	$\det(A_{1,2}) = 1, \quad \det(A_{1,3}) = -5, \quad \det(A_{2,3}) = -2, \\ \det(B_{1,2}) = 11, \quad \det(B_{1,3}) = 7, \quad \text{and} \\ \det(B_{2,3}) = -1.$	$\det(A_{1,2}) = 1, \quad \det(A_{1,3}) = -5, \quad \det(A_{2,3}) = -2, \\ \det(B_{1,2}) = 11, \quad \det(B_{1,3}) = 7, \quad \text{and} \quad \det(B_{2,3}) = -1.$
johnston-linear-matrix-algebra EQ0560	222	0.697	$\begin{array}{l} \operatorname{maximize}: \text{ \& } 4 \\ x_1+x_2 \text{ \& } \text{subject to: } \text{ \& } 2x_1-2x_2 \\ \leq 5 \text{ \& } x_1+3x_2 \leq 3 \text{ \& } x_1, \\ \quad x_2 \geq 0 \end{array}$	$\begin{array}{l} \text{maximize :} \\ \text{subject to:} \end{array} \quad \begin{array}{l} 4x_1 + x_2 \\ 2x_1 - 2x_2 \leq 5 \\ x_1 + 3x_2 \leq 3 \\ x_1, \quad x_2 \geq 0 \end{array}$	$\begin{array}{l} \text{maximize:} \\ \text{subject to:} \end{array} \quad \begin{array}{l} 4x_1 + x_2 \\ 2x_1 - 2x_2 \leq 5 \\ x_1 + 3x_2 \leq 3 \\ x_1, \quad x_2 \geq 0 \end{array}$
johnston-linear-matrix-algebra EQ0561	222	0.699	$\begin{array}{l} \operatorname{maximize}: \text{ \& } \\ x_1+x_2+x_3 \text{ \& } \text{subject to: } \text{ \& } x_1+2 \\ x_2+3x_3 \leq 2 \text{ \& } 3x_1+2x_2+x_3 \leq 1 \text{ \& } \\ \text{ \& } x_1, \quad x_2, \quad x_3 \geq 0 \end{array}$	$\begin{array}{l} \text{maximize :} \\ \text{subject to:} \end{array} \quad \begin{array}{l} x_1 + x_2 + x_3 \\ x_1 + 2x_2 + 3x_3 \leq 2 \\ 3x_1 + 2x_2 + x_3 \leq 1 \\ x_1, \quad x_2, \quad x_3 \geq 0 \end{array}$	$\begin{array}{l} \text{maximize:} \\ \text{subject to:} \end{array} \quad \begin{array}{l} x_1 + x_2 + x_3 \\ x_1 + 2x_2 + 3x_3 \leq 2 \\ 3x_1 + 2x_2 + x_3 \leq 1 \\ x_1, \quad x_2, \quad x_3 \geq 0 \end{array}$
johnston-linear-matrix-algebra EQ0759	286	0.699	$\operatorname{det}(\left[\begin{array}{cc} a & b \\ c & d \end{array}\right]) = \operatorname{det}(\left[\begin{array}{cc} a & b \\ 0 & d \end{array}\right]) + \operatorname{det}(\left[\begin{array}{cc} 0 & b \\ c & d \end{array}\right]),$	$\det\left(\left[\begin{array}{cc} a & b \\ c & d \end{array}\right]\right) = \det\left(\left[\begin{array}{cc} a & b \\ 0 & d \end{array}\right]\right) + \det\left(\left[\begin{array}{cc} 0 & b \\ c & d \end{array}\right]\right),$	$\det\left(\left[\begin{array}{cc} a & b \\ c & d \end{array}\right]\right) = \det\left(\left[\begin{array}{cc} a & b \\ 0 & d \end{array}\right]\right) + \det\left(\left[\begin{array}{cc} 0 & b \\ c & d \end{array}\right]\right),$
johnston-linear-matrix-algebra EQ0939 (3.4.5)	335	0.699	$A^k = \frac{1}{7} \left((-1)^k \left[\begin{array}{cc} 5 & -2 \\ -5 & 2 \end{array}\right] + 6^k \left[\begin{array}{cc} 2 & 2 \\ 5 & 5 \end{array}\right] \right).$	$A^k = \frac{1}{7} \left((-1)^k \left[\begin{array}{cc} 5 & -2 \\ -5 & 2 \end{array}\right] + 6^k \left[\begin{array}{cc} 2 & 2 \\ 5 & 5 \end{array}\right] \right).$	$A^k = \frac{1}{7} \left((-1)^k \left[\begin{array}{cc} 5 & -2 \\ -5 & 2 \end{array}\right] + 6^k \left[\begin{array}{cc} 2 & 2 \\ 5 & 5 \end{array}\right] \right).$
johnston-linear-matrix-algebra EQ1043	358	0.699	$\begin{array}{l} A^{-1} = \frac{1}{\operatorname{det}(A)} \\ \operatorname{cof}(A)^T \text{ \& } = \frac{1}{aei + bfg + cdh - afh - bdi - ceg} \\ \left[\begin{array}{ccc} ei - fh & ch - bi & bf - ce \\ fg - di & ai - cg & cd - af \\ dh - eg & bg - ah & ae - bd \end{array}\right]. \end{array}$	$\begin{array}{l} A^{-1} = \frac{1}{\det(A)} \operatorname{cof}(A)^T \\ \\ = \frac{1}{aei + bfg + cdh - afh - bdi - ceg} \left[\begin{array}{ccc} ei - fh & ch - bi & bf - ce \\ fg - di & ai - cg & cd - af \\ dh - eg & bg - ah & ae - bd \end{array}\right]. \end{array}$	$\begin{array}{l} A^{-1} = \frac{1}{\det(A)} \operatorname{cof}(A)^T \\ \\ = \frac{1}{aei + bfg + cdh - afh - bdi - ceg} \left[\begin{array}{ccc} ei - fh & ch - bi & bf - ce \\ fg - di & ai - cg & cd - af \\ dh - eg & bg - ah & ae - bd \end{array}\right]. \end{array}$
johnston-linear-matrix-algebra EQ0909	325	0.700	$A^k = \underbrace{(PD \overbrace{P^{-1}}^{\text{left. } P^{-1}}})_{k \text{ times}} (P \overbrace{P^{-1}}^{\text{left. } P^{-1}}) \dots (PD \overbrace{P^{-1}}^{\text{left. } P^{-1}}) = PD^k P^{-1}.$	$A^k = \underbrace{(PD \overbrace{P^{-1}}^{\text{left. } P^{-1}}) (P \overbrace{P^{-1}}^{\text{left. } P^{-1}}) (P \overbrace{P^{-1}}^{\text{left. } P^{-1}}) \dots (PD \overbrace{P^{-1}}^{\text{left. } P^{-1}})}_{k \text{ times}} = PD^k P^{-1}.$	$A^k = \underbrace{(PD \overbrace{P^{-1}}^{\text{left. } P^{-1}}) (P \overbrace{P^{-1}}^{\text{left. } P^{-1}}) (P \overbrace{P^{-1}}^{\text{left. } P^{-1}}) \dots (PD \overbrace{P^{-1}}^{\text{left. } P^{-1}})}_{k \text{ times}} = PD^k P^{-1}.$

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johnston-linear-matrix-algebra-EQ0327	129	0.706	$\begin{aligned} & \left[\begin{array}{ll ll} 2 & 2 & 1 & 0 \\ 4 & 5 & 0 & 1 \end{array} \right] \xrightarrow{R_2-2R_1} \left[\begin{array}{ll ll} 2 & 2 & 1 & 0 \\ 0 & 1 & -2 & 1 \end{array} \right] \\ & \xrightarrow{R_1-2R_2} \left[\begin{array}{ll ll} 2 & 0 & 5 & -2 \\ 0 & 1 & -2 & 1 \end{array} \right] \xrightarrow{\frac{1}{2}R_1} \left[\begin{array}{ll ll} 1 & 0 & 5/2 & -1 \\ 0 & 1 & -2 & 1 \end{array} \right]. \end{aligned}$	$\begin{aligned} & \left[\begin{array}{ll ll} 2 & 2 & 1 & 0 \\ 4 & 5 & 0 & 1 \end{array} \right] \xrightarrow{R_2-2R_1} \left[\begin{array}{ll ll} 2 & 2 & 1 & 0 \\ 0 & 1 & -2 & 1 \end{array} \right] \\ & \xrightarrow{R_1-2R_2} \left[\begin{array}{ll ll} 2 & 0 & 5 & -2 \\ 0 & 1 & -2 & 1 \end{array} \right] \xrightarrow{\frac{1}{2}R_1} \left[\begin{array}{ll ll} 1 & 0 & 5/2 & -1 \\ 0 & 1 & -2 & 1 \end{array} \right]. \end{aligned}$	$\begin{aligned} & \left[\begin{array}{ll ll} 2 & 2 & 1 & 0 \\ 4 & 5 & 0 & 1 \end{array} \right] \xrightarrow{R_2-2R_1} \left[\begin{array}{ll ll} 2 & 2 & 1 & 0 \\ 0 & 1 & -2 & 1 \end{array} \right] \\ & \xrightarrow{R_1-2R_2} \left[\begin{array}{ll ll} 2 & 0 & 5 & -2 \\ 0 & 1 & -2 & 1 \end{array} \right] \xrightarrow{\frac{1}{2}R_1} \left[\begin{array}{ll ll} 1 & 0 & 5/2 & -1 \\ 0 & 1 & -2 & 1 \end{array} \right]. \end{aligned}$
johnston-linear-matrix-algebra-EQ1219	411	0.709	$\lim_{n \rightarrow \infty} \frac{x_{n+1}}{x_n} = \lambda_0.$	$\lim_{n \rightarrow \infty} \frac{x_{n+1}}{x_n} = \lambda_0.$	$\lim_{n \rightarrow \infty} \frac{x_{n+1}}{x_n} = \lambda_0.$
johnston-linear-matrix-algebra-EQ1185	404	0.712	$\begin{aligned} & \left[\begin{array}{cccc} 1 & 1 & \dots & 1 \\ \lambda_0 & \lambda_1 & \dots & \lambda_{k-1} \\ \lambda_0^2 & \lambda_1^2 & \dots & \lambda_{k-1}^2 \\ \vdots & \vdots & \ddots & \vdots \\ \lambda_0^{k-1} & \lambda_1^{k-1} & \dots & \lambda_{k-1}^{k-1} \end{array} \right] \left[\begin{array}{c} c_0 \\ c_1 \\ c_2 \\ \vdots \\ c_{k-1} \end{array} \right] = \left[\begin{array}{c} x_0 \\ x_1 \\ x_2 \\ \vdots \\ x_{k-1} \end{array} \right]. \end{aligned}$	$\begin{aligned} & \left[\begin{array}{cccc} 1 & 1 & \dots & 1 \\ \lambda_0 & \lambda_1 & \dots & \lambda_{k-1} \\ \lambda_0^2 & \lambda_1^2 & \dots & \lambda_{k-1}^2 \\ \vdots & \vdots & \ddots & \vdots \\ \lambda_0^{k-1} & \lambda_1^{k-1} & \dots & \lambda_{k-1}^{k-1} \end{array} \right] \left[\begin{array}{c} c_0 \\ c_1 \\ c_2 \\ \vdots \\ c_{k-1} \end{array} \right] = \left[\begin{array}{c} x_0 \\ x_1 \\ x_2 \\ \vdots \\ x_{k-1} \end{array} \right]. \end{aligned}$	$\begin{aligned} & \left[\begin{array}{cccc} 1 & 1 & \dots & 1 \\ \lambda_0 & \lambda_1 & \dots & \lambda_{k-1} \\ \lambda_0^2 & \lambda_1^2 & \dots & \lambda_{k-1}^2 \\ \vdots & \vdots & \ddots & \vdots \\ \lambda_0^{k-1} & \lambda_1^{k-1} & \dots & \lambda_{k-1}^{k-1} \end{array} \right] \left[\begin{array}{c} c_0 \\ c_1 \\ c_2 \\ \vdots \\ c_{k-1} \end{array} \right] = \left[\begin{array}{c} x_0 \\ x_1 \\ x_2 \\ \vdots \\ x_{k-1} \end{array} \right]. \end{aligned}$
johnston-linear-matrix-algebra-EQ0771	288	0.712	$\begin{aligned} & (3 \cdot (-2) \cdot (-1)) + (6 \cdot 3 \cdot 2) + ((-3) \cdot 0 \cdot 4) \\ & - (3 \cdot 3 \cdot 4) - (6 \cdot 0 \cdot (-1)) - ((-3) \cdot (-2) \cdot 2) \\ & = 6 + 36 + 0 - 36 - 0 - 12 = -6. \end{aligned}$	$\begin{aligned} & (3 \cdot (-2) \cdot (-1)) + (6 \cdot 3 \cdot 2) + ((-3) \cdot 0 \cdot 4) \\ & - (3 \cdot 3 \cdot 4) - (6 \cdot 0 \cdot (-1)) - ((-3) \cdot (-2) \cdot 2) \\ & = 6 + 36 + 0 - 36 - 0 - 12 = -6. \end{aligned}$	$\begin{aligned} & (3 \cdot (-2) \cdot (-1)) + (6 \cdot 3 \cdot 2) + ((-3) \cdot 0 \cdot 4) \\ & - (3 \cdot 3 \cdot 4) - (6 \cdot 0 \cdot (-1)) - ((-3) \cdot (-2) \cdot 2) \\ & = 6 + 36 + 0 - 36 - 0 - 12 = -6. \end{aligned}$
johnston-linear-matrix-algebra-EQ0768	287	0.712	$\operatorname{det} \left(\left[\begin{array}{lll} 0 & b & c \\ d & e & f \\ 0 & h & i \end{array} \right] \right) = cdh - bdi, \quad \text{and} \quad \operatorname{det} \left(\left[\begin{array}{lll} 0 & b & c \\ 0 & e & f \\ g & h & i \end{array} \right] \right) = bfg - ceg,$	$\det \left(\left[\begin{array}{lll} 0 & b & c \\ d & e & f \\ 0 & h & i \end{array} \right] \right) = cdh - bdi, \quad \text{and} \quad \det \left(\left[\begin{array}{lll} 0 & b & c \\ 0 & e & f \\ g & h & i \end{array} \right] \right) = bfg - ceg,$	$\det \left(\left[\begin{array}{lll} 0 & b & c \\ d & e & f \\ 0 & h & i \end{array} \right] \right) = cdh - bdi, \quad \text{and} \quad \det \left(\left[\begin{array}{lll} 0 & b & c \\ 0 & e & f \\ g & h & i \end{array} \right] \right) = bfg - ceg,$
johnston-linear-matrix-algebra-EQ0386	148	0.715	$\mathbf{A} \mathbf{x} = \left[\begin{array}{l} \mathbf{a}_1 \\ \mathbf{a}_2 \\ \vdots \\ \mathbf{a}_n \end{array} \right] \quad \text{where} \quad \mathbf{a}_1, \mathbf{a}_2, \dots, \mathbf{a}_n \text{ are column vectors.}$	$\mathbf{A} \mathbf{x} = \left[\begin{array}{l} \mathbf{a}_1 \\ \mathbf{a}_2 \\ \vdots \\ \mathbf{a}_n \end{array} \right] \quad \text{where} \quad \mathbf{a}_1, \mathbf{a}_2, \dots, \mathbf{a}_n \text{ are column vectors.}$	$\mathbf{A} \mathbf{x} = \left[\begin{array}{l} \mathbf{a}_1 \\ \mathbf{a}_2 \\ \vdots \\ \mathbf{a}_n \end{array} \right] \quad \text{where} \quad \mathbf{a}_1, \mathbf{a}_2, \dots, \mathbf{a}_n \text{ are column vectors.}$

Identifier	Page	Conf.	LaTeX source	Rendered	Image
johnston-linear-matrix-algebra-EQ1252	420	0.718	$\begin{aligned} e^x &= 1 + x + \frac{x^2}{2} + \frac{x^3}{3!} + \frac{x^4}{4!} + \frac{x^5}{5!} + \cdots, \\ \cos(x) &= 1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \cdots, \quad \text{and} \\ \sin(x) &= x - \frac{x^3}{3!} + \frac{x^5}{5!} - \cdots, \end{aligned}$	$e^x = 1 + x + \frac{x^2}{2} + \frac{x^3}{3!} + \frac{x^4}{4!} + \frac{x^5}{5!} + \cdots,$ $\cos(x) = 1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \cdots, \quad \text{and}$ $\sin(x) = x - \frac{x^3}{3!} + \frac{x^5}{5!} - \cdots,$	$e^x = 1 + x + \frac{x^2}{2} + \frac{x^3}{3!} + \frac{x^4}{4!} + \frac{x^5}{5!} + \cdots,$ $\cos(x) = 1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \cdots, \quad \text{and}$ $\sin(x) = x - \frac{x^3}{3!} + \frac{x^5}{5!} - \cdots,$
johnston-linear-matrix-algebra-EQ0519	207	0.720	$\begin{gathered} \left[\begin{array}{cccc cccc} 1 & 0 & 2 & 0 & 0 & -3 & 3 & 4 \\ 0 & 1 & 3 & 0 & 0 & -3 & 5 & 7 \\ 0 & 0 & 1 & 0 & 1 & -1 & 1 & 2 \\ 0 & 0 & 1 & 1 & 0 & 2 & -2 & 5 \end{array} \right] \\ 5/2 \text{ is smallest (and only) positive ratio} \end{gathered}$	$\left[\begin{array}{cccc cccc} 1 & 0 & 2 & 0 & 0 & -3 & 3 & 4 \\ 0 & 1 & 3 & 0 & 0 & -3 & 5 & 7 \\ 0 & 0 & 1 & 0 & 1 & -1 & 1 & 2 \\ 0 & 0 & 1 & 1 & 0 & 2 & -2 & 5 \end{array} \right]$ 5/2 is smallest (and only) positive ratio	$\left[\begin{array}{cccc cccc} 1 & 0 & 2 & 0 & 0 & -3 & 3 & 4 \\ 0 & 1 & 3 & 0 & 0 & -3 & 5 & 7 \\ 0 & 0 & 1 & 0 & 1 & -1 & 1 & 2 \\ 0 & 0 & 1 & 1 & 0 & 2 & -2 & 5 \end{array} \right]$ 5/2 is smallest (and only) positive ratio
johnston-linear-matrix-algebra-EQ0714 (0)	269	0.721	$\begin{aligned} \operatorname{tr}(A B) &= \operatorname{tr} \left(\left[\begin{array}{cc} 1 & 2 \\ 3 & 4 \end{array} \right] \left[\begin{array}{cc} 1 & -1 \\ -2 & 1 \end{array} \right] \right) = \operatorname{tr} \left(\left[\begin{array}{cc} -3 & 1 \\ -5 & 1 \end{array} \right] \right) = -2 \quad \text{and} \\ \operatorname{tr}(B A) &= \operatorname{tr} \left(\left[\begin{array}{cc} 1 & -1 \\ -2 & 1 \end{array} \right] \left[\begin{array}{cc} 1 & 2 \\ 3 & 4 \end{array} \right] \right) = \operatorname{tr} \left(\left[\begin{array}{cc} -2 & -2 \\ 1 & 0 \end{array} \right] \right) = -2, \end{aligned}$	$\operatorname{tr}(AB) = \operatorname{tr} \left(\left[\begin{array}{cc} 1 & 2 \\ 3 & 4 \end{array} \right] \left[\begin{array}{cc} 1 & -1 \\ -2 & 1 \end{array} \right] \right) = \operatorname{tr} \left(\left[\begin{array}{cc} -3 & 1 \\ -5 & 1 \end{array} \right] \right) = -2 \quad \text{and}$ $\operatorname{tr}(BA) = \operatorname{tr} \left(\left[\begin{array}{cc} 1 & -1 \\ -2 & 1 \end{array} \right] \left[\begin{array}{cc} 1 & 2 \\ 3 & 4 \end{array} \right] \right) = \operatorname{tr} \left(\left[\begin{array}{cc} -2 & -2 \\ 1 & 0 \end{array} \right] \right) = -2,$	$\operatorname{tr}(AB) = \operatorname{tr} \left(\left[\begin{array}{cc} 1 & 2 \\ 3 & 4 \end{array} \right] \left[\begin{array}{cc} 1 & -1 \\ -2 & 1 \end{array} \right] \right) = \operatorname{tr} \left(\left[\begin{array}{cc} -3 & 1 \\ -5 & 1 \end{array} \right] \right) = -2 \quad \text{and}$ $\operatorname{tr}(BA) = \operatorname{tr} \left(\left[\begin{array}{cc} 1 & -1 \\ -2 & 1 \end{array} \right] \left[\begin{array}{cc} 1 & 2 \\ 3 & 4 \end{array} \right] \right) = \operatorname{tr} \left(\left[\begin{array}{cc} -2 & -2 \\ 1 & 0 \end{array} \right] \right) = -2,$

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johnston-linear-matrix-algebra-EQ0757 (1)	285	0.721	$\begin{aligned} & \left[\begin{array}{cccc} 0 & 2 & 1 & 2 \\ 1 & -1 & 1 & 0 \\ 2 & 1 & 0 & 1 \\ -2 & 0 & 1 & 1 \end{array} \right] \xrightarrow{R_1 \leftrightarrow R_2} \left[\begin{array}{cccc} 1 & -1 & 1 & 0 \\ 0 & 2 & 1 & 2 \\ 2 & 1 & 0 & 1 \\ -2 & 0 & 1 & 1 \end{array} \right] \\ & \xrightarrow{\begin{smallmatrix} R_3-2R_1 \\ R_4+2R_1 \end{smallmatrix}} \left[\begin{array}{cccc} 1 & -1 & 1 & 0 \\ 0 & 2 & 1 & 2 \\ 0 & 3 & -2 & 1 \\ 0 & -2 & 3 & 1 \end{array} \right] \xrightarrow{\begin{smallmatrix} R_3-\frac{3}{2}R_2 \\ R_4+R_2 \end{smallmatrix}} \left[\begin{array}{cccc} 1 & -1 & 1 & 0 \\ 0 & 2 & 1 & 2 \\ 0 & 0 & -7/2 & -2 \\ 0 & 0 & 4 & 3 \end{array} \right] \\ & \xrightarrow{R_4+\frac{8}{7}R_3} \left[\begin{array}{cccc} 1 & -1 & 1 & 0 \\ 0 & 2 & 1 & 2 \\ 0 & 0 & -7/2 & -2 \\ 0 & 0 & 0 & 5/7 \end{array} \right]. \end{aligned}$		
johnston-linear-matrix-algebra-EQ1176	402	0.721	$x_n=4x_{n-1}-x_{n-2}-6x_{n-3} \quad \text{for all } n \geq 3,$	$x_n = 4x_{n-1} - x_{n-2} - 6x_{n-3} \quad \text{for all } n \geq 3,$	$x_n = 4x_{n-1} - x_{n-2} - 6x_{n-3} \quad \text{for all } n \geq 3,$
johnston-linear-matrix-algebra-EQ0935 (4)	36	0.723	$\left[\begin{array}{ccc} 1 & 2 & 3 \\ 4 & 5 & \end{array} \right] \text{ and } \left[\begin{array}{ccc} 2 & -3 & 4 \\ & 1 & 2 & 0 \end{array} \right].$	$\left[\begin{array}{ccc} 1 & 2 & 3 \\ 4 & 5 & \end{array} \right] \text{ and } \left[\begin{array}{ccc} 2 & -3 & 4 \\ & 1 & 2 & 0 \end{array} \right].$	$\left[\begin{array}{ccc} 1 & 2 & 3 \\ 4 & 5 & \end{array} \right] \text{ and } \left[\begin{array}{ccc} 2 & -3 & 4 \\ & 1 & 2 & 0 \end{array} \right].$
johnston-linear-matrix-algebra-EQ1458	461	0.724	$\begin{aligned} & \left[\begin{array}{cc} A & 0 \\ 0 & D \end{array} \right] \left[\begin{array}{cc} A^{-1} & 0 \\ 0 & D^{-1} \end{array} \right] = \left[\begin{array}{cc} AA^{-1} & 0 \\ 0 & DD^{-1} \end{array} \right] \\ & = \left[\begin{array}{cc} I & 0 \\ 0 & I \end{array} \right]. \end{aligned}$	$\left[\begin{array}{cc} A & 0 \\ 0 & D \end{array} \right] \left[\begin{array}{cc} A^{-1} & 0 \\ 0 & D^{-1} \end{array} \right] = \left[\begin{array}{cc} AA^{-1} & 0 \\ 0 & DD^{-1} \end{array} \right] \\ = \left[\begin{array}{cc} I & 0 \\ 0 & I \end{array} \right].$	$\left[\begin{array}{cc} A & 0 \\ 0 & D \end{array} \right] \left[\begin{array}{cc} A^{-1} & 0 \\ 0 & D^{-1} \end{array} \right] = \left[\begin{array}{cc} AA^{-1} & 0 \\ 0 & DD^{-1} \end{array} \right] \\ = \left[\begin{array}{cc} I & 0 \\ 0 & I \end{array} \right].$
johnston-linear-matrix-algebra-EQ0494	199	0.727	$\begin{aligned} & \text{maximize : } x_1 + 2x_2 \\ & \text{subject to: } 3x_1 - 4x_2 \geq 1 \\ & \quad 2x_1 + x_2 = 5 \\ & \quad x_1, x_2 \geq 0 \end{aligned}$	$\begin{aligned} & \text{maximize : } x_1 + 2x_2 \\ & \text{subject to: } 3x_1 - 4x_2 \geq 1 \\ & \quad 2x_1 + x_2 = 5 \\ & \quad x_1, x_2 \geq 0 \end{aligned}$	maximize: $x_1 + 2x_2$ subject to: $3x_1 - 4x_2 \geq 1$ $2x_1 + x_2 = 5$ $x_1, x_2 \geq 0$
johnston-linear-matrix-algebra-EQ0500	201	0.730	$\begin{aligned} & \text{minimize : } 3x_1 - 2x_2 \\ & \text{subject to: } x_1 + 3x_2 = 4 \\ & \quad x_1 + 2x_2 \geq 3 \\ & \quad x_2 \geq 0 \end{aligned}$	$\begin{aligned} & \text{minimize : } 3x_1 - 2x_2 \\ & \text{subject to: } x_1 + 3x_2 = 4 \\ & \quad x_1 + 2x_2 \geq 3 \\ & \quad x_2 \geq 0 \end{aligned}$	minimize: $3x_1 - 2x_2$ subject to: $x_1 + 3x_2 = 4$ $x_1 + 2x_2 \geq 3$ $x_2 \geq 0$

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johnston-linear-matrix-algebra-EQ0682	259	0.730	$\begin{aligned} P_{C \leftarrow B}[\mathbf{v}]_B &= \left[\begin{array}{c} \mathbf{v}_1 \\ \mathbf{v}_2 \\ \vdots \\ \mathbf{v}_k \end{array} \right]_C & \text{(definition of } P_{C \leftarrow B} \text{)} \\ &= c_1 [\mathbf{v}_1]_C + c_2 [\mathbf{v}_2]_C + \cdots + c_k [\mathbf{v}_k]_C & \text{(block matrix mult.)} \\ &= [c_1 \mathbf{v}_1 + c_2 \mathbf{v}_2 + \cdots + c_k \mathbf{v}_k]_C & \text{(by Theorem 3.1.2)} \\ &= [\mathbf{v}]_C. & \text{(since } [\mathbf{v}]_B = (c_1, c_2, \dots, c_k) \text{)} \end{aligned}$	$P_{C \leftarrow B}[\mathbf{v}]_B = \left[\begin{array}{c} [\mathbf{v}_1]_C \\ [\mathbf{v}_2]_C \\ \vdots \\ [\mathbf{v}_k]_C \end{array} \right] \quad \text{(definition of } P_{C \leftarrow B} \text{)}$ $= c_1 [\mathbf{v}_1]_C + c_2 [\mathbf{v}_2]_C + \cdots + c_k [\mathbf{v}_k]_C \quad \text{(block matrix mult.)}$ $= [c_1 \mathbf{v}_1 + c_2 \mathbf{v}_2 + \cdots + c_k \mathbf{v}_k]_C \quad \text{(by Theorem 3.1.2)}$ $= [\mathbf{v}]_C \quad \text{(since } [\mathbf{v}]_B = (c_1, c_2, \dots, c_k) \text{)}$	$P_{C \leftarrow B}[\mathbf{v}]_B = \left[\begin{array}{c} [\mathbf{v}_1]_C \\ [\mathbf{v}_2]_C \\ \vdots \\ [\mathbf{v}_k]_C \end{array} \right] \quad \text{(definition of } P_{C \leftarrow B} \text{)}$ $= c_1 [\mathbf{v}_1]_C + c_2 [\mathbf{v}_2]_C + \cdots + c_k [\mathbf{v}_k]_C \quad \text{(block matrix mult.)}$ $= [c_1 \mathbf{v}_1 + c_2 \mathbf{v}_2 + \cdots + c_k \mathbf{v}_k]_C \quad \text{(by Theorem 3.1.2)}$ $= [\mathbf{v}]_C. \quad \text{(since } [\mathbf{v}]_B = (c_1, c_2, \dots, c_k) \text{)}$
johnston-linear-matrix-algebra-EQ1164 (3.D.1)	400	0.731	$x_n = a_{k-1} x_{n-1} + a_{k-2} x_{n-2} + \cdots + a_0 x_{n-k} \quad \text{for all } n \geq k.$	$x_n = a_{k-1} x_{n-1} + a_{k-2} x_{n-2} + \cdots + a_0 x_{n-k} \quad \text{for all } n \geq k.$	$x_n = a_{k-1} x_{n-1} + a_{k-2} x_{n-2} + \cdots + a_0 x_{n-k} \quad \text{for all } n \geq k.$
johnston-linear-matrix-algebra-EQ1199	407	0.731	$\begin{aligned} x_n &= 2^n + (1-3i)i^n + (1+3i)(-i)^n \\ &= 2^n + (1-3i)e^{i\pi n/2} + (1+3i)e^{-i\pi n/2} \\ &= 2^n + (e^{i\pi n/2} + e^{-i\pi n/2}) - 3i(e^{i\pi n/2} - e^{-i\pi n/2}) \\ &= 2^n + 2\cos(\pi n/2) + 6\sin(\pi n/2) \quad \text{for all } n \geq 0, \end{aligned}$	$x_n = 2^n + (1-3i)i^n + (1+3i)(-i)^n$ $= 2^n + (1-3i)e^{i\pi n/2} + (1+3i)e^{-i\pi n/2}$ $= 2^n + (e^{i\pi n/2} + e^{-i\pi n/2}) - 3i(e^{i\pi n/2} - e^{-i\pi n/2})$ $= 2^n + 2\cos(\pi n/2) + 6\sin(\pi n/2) \quad \text{for all } n \geq 0,$	$x_n = 2^n + (1-3i)i^n + (1+3i)(-i)^n$ $= 2^n + (1-3i)e^{i\pi n/2} + (1+3i)e^{-i\pi n/2}$ $= 2^n + (e^{i\pi n/2} + e^{-i\pi n/2}) - 3i(e^{i\pi n/2} - e^{-i\pi n/2})$ $= 2^n + 2\cos(\pi n/2) + 6\sin(\pi n/2) \quad \text{for all } n \geq 0,$
johnston-linear-matrix-algebra-EQ1090	370	0.734	$\operatorname{per}(A) = \sum_{\sigma \in S_n} a_{\sigma(1),1} a_{\sigma(2),2} \cdots a_{\sigma(n),n}.$	$\operatorname{per}(A) = \sum_{\sigma \in S_n} a_{\sigma(1),1} a_{\sigma(2),2} \cdots a_{\sigma(n),n}.$	$\operatorname{per}(A) = \sum_{\sigma \in S_n} a_{\sigma(1),1} a_{\sigma(2),2} \cdots a_{\sigma(n),n}.$
johnston-linear-matrix-algebra-EQ1601	477	0.734	$\overline{A\mathbf{v}} = \overline{A} \overline{\mathbf{v}} = \overline{\lambda \mathbf{v}} = \overline{\lambda} \overline{\mathbf{v}},$	$\overline{A\mathbf{v}} = \overline{A} \overline{\mathbf{v}} = \overline{\lambda \mathbf{v}} = \overline{\lambda} \overline{\mathbf{v}},$	$\overline{A\mathbf{v}} = \overline{A} \overline{\mathbf{v}} = \overline{\lambda \mathbf{v}} = \overline{\lambda} \overline{\mathbf{v}},$
johnston-linear-matrix-algebra-EQ0788	292	0.737	$\left[\begin{array}{cccc} 2 & -1 & -1 & 0 \\ 0 & +2 & 1 & 3 \\ 0 & -0 & 1 & 0 \\ -1 & +3 & 0 & 2 \end{array} \right]$	$\left[\begin{array}{cccc} 2 & -1 & -1 & 0 \\ 0 & +2 & 1 & 3 \\ 0 & -0 & 1 & 0 \\ -1 & +3 & 0 & 2 \end{array} \right]$	$\left[\begin{array}{cccc} 2 & -1 & -1 & 0 \\ 0 & +2 & 1 & 3 \\ 0 & -0 & 1 & 0 \\ -1 & +3 & 0 & 2 \end{array} \right]$
johnston-linear-matrix-algebra-EQ0703	266	0.737	$[T] = \left[\begin{array}{ccc} 2 & 1 & -3 \\ 1 & 0 & 1 \\ 0 & 1 & 2 \end{array} \right].$	$[T] = \left[\begin{array}{ccc} 2 & 1 & -3 \\ 1 & 0 & 1 \\ 0 & 1 & 2 \end{array} \right].$	$[T] = \left[\begin{array}{ccc} 2 & 1 & -3 \\ 1 & 0 & 1 \\ 0 & 1 & 2 \end{array} \right].$
johnston-linear-matrix-algebra-EQ1012	351	0.743	$L_0 = 2, \quad L_1 = 1, \quad L_{n+1} = L_n + L_{n-1} \quad \text{for all } n \geq 1.$	$L_0 = 2, \quad L_1 = 1, \quad L_{n+1} = L_n + L_{n-1} \quad \text{for all } n \geq 1.$	$L_0 = 2, \quad L_1 = 1, \quad L_{n+1} = L_n + L_{n-1} \quad \text{for all } n \geq 1.$

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johnston-linear-matrix-algebra-EQ0832	304	0.743	$\begin{aligned} \operatorname{det}(A-\lambda I) &= \operatorname{det}\left(\begin{bmatrix} 1-\lambda & 3 & 3 \\ 3 & 1-\lambda & -1 \\ 0 & 0 & 2-\lambda \end{bmatrix}\right) \\ &= (1-\lambda)(1-\lambda)(2-\lambda) + 0 + 0 - 0 - 0 - 9(2-\lambda) \\ &= (2-\lambda)((1-\lambda)^2 - 9) \\ &= (2-\lambda)(\lambda^2 - 2\lambda - 8) \\ &= (2-\lambda)(\lambda+2)(\lambda-4) \end{aligned}$	$\det(A - \lambda I) = \det \left(\begin{bmatrix} 1-\lambda & 3 & 3 \\ 3 & 1-\lambda & -1 \\ 0 & 0 & 2-\lambda \end{bmatrix} \right)$ $= (1-\lambda)(1-\lambda)(2-\lambda) + 0 + 0 - 0 - 0 - 9(2-\lambda)$ $= (2-\lambda)((1-\lambda)^2 - 9)$ $= (2-\lambda)(\lambda^2 - 2\lambda - 8)$ $= (2-\lambda)(\lambda+2)(\lambda-4).$	$\det(A - \lambda I) = \det \left(\begin{bmatrix} 1-\lambda & 3 & 3 \\ 3 & 1-\lambda & -1 \\ 0 & 0 & 2-\lambda \end{bmatrix} \right)$ $= (1-\lambda)(1-\lambda)(2-\lambda) + 0 + 0 - 0 - 0 - 9(2-\lambda)$ $= (2-\lambda)((1-\lambda)^2 - 9)$ $= (2-\lambda)(\lambda^2 - 2\lambda - 8)$ $= (2-\lambda)(\lambda+2)(\lambda-4).$
johnston-linear-matrix-algebra-EQ0931	333	0.745	$\begin{aligned} p_A(\lambda) &= \operatorname{det}(A-\lambda I) = \operatorname{det}\left(\begin{bmatrix} 1-\lambda & 1 \\ 1 & -\lambda \end{bmatrix}\right) \\ &= (1-\lambda)(-\lambda) - 1 \\ &= \lambda^2 - \lambda - 1. \end{aligned}$	$p_A(\lambda) = \det(A - \lambda I) = \det \left(\begin{bmatrix} 1-\lambda & 1 \\ 1 & -\lambda \end{bmatrix} \right) = (1-\lambda)(-\lambda) - 1$ $= \lambda^2 - \lambda - 1.$	$p_A(\lambda) = \det(A - \lambda I) = \det \left(\begin{bmatrix} 1-\lambda & 1 \\ 1 & -\lambda \end{bmatrix} \right) = (1-\lambda)(-\lambda) - 1$ $= \lambda^2 - \lambda - 1.$
johnston-linear-matrix-algebra-EQ0972	342	0.748	$\begin{aligned} \left(A^{1/2}\right)^2 &= \left(P \begin{bmatrix} \lambda_1^{1/2} & \cdots & 0 \\ \vdots & \ddots & \vdots \\ 0 & \cdots & \lambda_n^{1/2} \end{bmatrix} P^{-1}\right) \left(P \begin{bmatrix} \lambda_1^{1/2} & \cdots & 0 \\ \vdots & \ddots & \vdots \\ 0 & \cdots & \lambda_n^{1/2} \end{bmatrix} P^{-1}\right) \\ &= P \begin{bmatrix} \lambda_1^{1/2} \lambda_1^{1/2} & \cdots & 0 \\ \vdots & \ddots & \vdots \\ 0 & \cdots & \lambda_n^{1/2} \lambda_n^{1/2} \end{bmatrix} P^{-1} = P \begin{bmatrix} \lambda_1 & \cdots & 0 \\ \vdots & \ddots & \vdots \\ 0 & \cdots & \lambda_n \end{bmatrix} P^{-1} = A. \end{aligned}$	$\left(A^{1/2}\right)^2 = \left(P \begin{bmatrix} \lambda_1^{1/2} & \cdots & 0 \\ \vdots & \ddots & \vdots \\ 0 & \cdots & \lambda_n^{1/2} \end{bmatrix} P^{-1}\right) \left(P \begin{bmatrix} \lambda_1^{1/2} & \cdots & 0 \\ \vdots & \ddots & \vdots \\ 0 & \cdots & \lambda_n^{1/2} \end{bmatrix} P^{-1}\right)$ $= P \begin{bmatrix} \lambda_1^{1/2} \lambda_1^{1/2} & \cdots & 0 \\ \vdots & \ddots & \vdots \\ 0 & \cdots & \lambda_n^{1/2} \lambda_n^{1/2} \end{bmatrix} P^{-1} = P \begin{bmatrix} \lambda_1 & \cdots & 0 \\ \vdots & \ddots & \vdots \\ 0 & \cdots & \lambda_n \end{bmatrix} P^{-1} = A.$	$\left(A^{1/2}\right)^2 = \left(P \begin{bmatrix} \lambda_1^{1/2} & \cdots & 0 \\ \vdots & \ddots & \vdots \\ 0 & \cdots & \lambda_n^{1/2} \end{bmatrix} P^{-1}\right) \left(P \begin{bmatrix} \lambda_1^{1/2} & \cdots & 0 \\ \vdots & \ddots & \vdots \\ 0 & \cdots & \lambda_n^{1/2} \end{bmatrix} P^{-1}\right)$ $= P \begin{bmatrix} \lambda_1^{1/2} \lambda_1^{1/2} & \cdots & 0 \\ \vdots & \ddots & \vdots \\ 0 & \cdots & \lambda_n^{1/2} \lambda_n^{1/2} \end{bmatrix} P^{-1} = P \begin{bmatrix} \lambda_1 & \cdots & 0 \\ \vdots & \ddots & \vdots \\ 0 & \cdots & \lambda_n \end{bmatrix} P^{-1} = A.$
johnston-linear-matrix-algebra-EQ0182	88	0.751	$\frac{1}{n} \sum_{j=1}^n w_j. \quad \text{(average number of friendships = total friendships divided by number of people)}$	$\frac{1}{n} \sum_{j=1}^n w_j.$ (average number of friendships = total friendships divided by number of people)	$\frac{1}{n} \sum_{j=1}^n w_j.$ (average number of friendships = total friendships divided by number of people)

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johnston-linear-matrix-algebra- EQ0715	269	0.752	$\begin{gathered} \operatorname{tr}(A)=\operatorname{tr}\left(P B P^{-1}\right)=\operatorname{tr}\left(P\left(B P^{-1}\right)\right)=\underset{\uparrow}{\operatorname{tr}\left(\left(B P^{-1}\right) P\right)}=\operatorname{tr}\left(B\left(P^{-1} P\right)\right)=\operatorname{tr}(B) . \\ \text { by Theorem 3.1.6(c) } \end{gathered}$	$\operatorname{tr}(A)=\operatorname{tr}\left(P B P^{-1}\right)=\operatorname{tr}\left(P\left(B P^{-1}\right)\right)=\underset{\uparrow}{\operatorname{tr}\left(\left(B P^{-1}\right) P\right)}=\operatorname{tr}\left(B\left(P^{-1} P\right)\right)=\operatorname{tr}(B) .$ by Theorem 3.1.6(c)	$\operatorname{tr}(A)=\operatorname{tr}\left(P B P^{-1}\right)=\operatorname{tr}\left(P\left(B P^{-1}\right)\right)=\underset{\uparrow}{\operatorname{tr}\left(\left(B P^{-1}\right) P\right)}=\operatorname{tr}\left(B\left(P^{-1} P\right)\right)=\operatorname{tr}(B) .$ ↑ by Theorem 3.1.6(c)
johnston-linear-matrix-algebra- EQ1328	446	0.755	$\begin{aligned} (\mathbf{v}+\mathbf{w})+\mathbf{x} &= \left(\left(v_1, \ldots, v_n\right)+\left(w_1, \ldots, w_n\right)\right)+\left(x_1, \ldots, x_n\right) \\ &= \left(v_1+w_1, \ldots, v_n+w_n\right)+\left(x_1, \ldots, x_n\right) \\ &= \left(v_1+w_1+x_1, \ldots, v_n+w_n+x_n\right) \\ &= \left(v_1, \ldots, v_n\right)+\left(w_1+x_1, \ldots, w_n+x_n\right) \\ &= \left(v_1, \ldots, v_n\right)+\left(\left(w_1, \ldots, w_n\right)+\left(x_1, \ldots, x_n\right)\right) \\ &= \mathbf{v}+(\mathbf{w}+\mathbf{x}) . \end{aligned}$	$\begin{aligned} (\mathbf{v}+\mathbf{w})+\mathbf{x} &= \left(\left(v_1, \ldots, v_n\right)+\left(w_1, \ldots, w_n\right)\right)+\left(x_1, \ldots, x_n\right) \\ &= \left(v_1+w_1, \ldots, v_n+w_n\right)+\left(x_1, \ldots, x_n\right) \\ &= \left(v_1+w_1+x_1, \ldots, v_n+w_n+x_n\right) \\ &= \left(v_1, \ldots, v_n\right)+\left(w_1+x_1, \ldots, w_n+x_n\right) \\ &= \left(v_1, \ldots, v_n\right)+\left(\left(w_1, \ldots, w_n\right)+\left(x_1, \ldots, x_n\right)\right) \\ &= \mathbf{v}+(\mathbf{w}+\mathbf{x}) \end{aligned}$	$\begin{aligned} (\mathbf{v}+\mathbf{w})+\mathbf{x} &= \left(\left(v_1, \ldots, v_n\right)+\left(w_1, \ldots, w_n\right)\right)+\left(x_1, \ldots, x_n\right) \\ &= \left(v_1+w_1, \ldots, v_n+w_n\right)+\left(x_1, \ldots, x_n\right) \\ &= \left(v_1+w_1+x_1, \ldots, v_n+w_n+x_n\right) \\ &= \left(v_1, \ldots, v_n\right)+\left(w_1+x_1, \ldots, w_n+x_n\right) \\ &= \left(v_1, \ldots, v_n\right)+\left(\left(w_1, \ldots, w_n\right)+\left(x_1, \ldots, x_n\right)\right) \\ &= \mathbf{v}+(\mathbf{w}+\mathbf{x}) . \end{aligned}$
johnston-linear-matrix-algebra- EQ1590	476	0.755	$\begin{aligned} \operatorname{det}\left(\left[\begin{array}{cc} I_m & O \\ B & I_n+B A \end{array}\right]\right) &= \operatorname{det}\left(I_m\right) \operatorname{det}\left(I_n+B A\right) \\ &= \operatorname{det}\left(I_n+B A\right) . \end{aligned}$	$\operatorname{det}\left(\left[\begin{array}{cc} I_m & O \\ B & I_n+B A \end{array}\right]\right)=\operatorname{det}\left(I_m\right) \operatorname{det}\left(I_n+B A\right)=\operatorname{det}\left(I_n+B A\right) .$	$\operatorname{det}\left(\left[\begin{array}{cc} I_m & O \\ B & I_n+B A \end{array}\right]\right)=\operatorname{det}\left(I_m\right) \operatorname{det}\left(I_n+B A\right)=\operatorname{det}\left(I_n+B A\right) .$
johnston-linear-matrix-algebra- EQ0479	196	0.757	$\begin{aligned} r+s+w+x &=1 \\ r+v+x+y &=0 \\ s+v+w+z &=1 \\ r+w+y+z &=1 \\ s+w+x+z &=0 \end{aligned}$	$\begin{aligned} r+s+w+x &=1 \\ r+v+x+y &=0 \\ s+v+w+z &=1 \\ r+w+y+z &=1 \\ s+w+x+z &=0 \end{aligned}$	$\begin{aligned} r+s &+w+x &=1 \\ r &+v &+x+y &=0 \\ &s+v+w &+z &=1 \\ r &+w &+y+z &=1 \\ &s &+w+x &+z &=0 \end{aligned}$
johnston-linear-matrix-algebra- EQ1608	478	0.759	$\begin{aligned} B^* \mathbf{w}_j &= \left(A-\lambda_1 \mathbf{v}_1 \mathbf{v}_1^*\right)^* \mathbf{w}_j \\ &= \left(A^*-\overline{\lambda_1} \mathbf{v}_1 \mathbf{v}_1^*\right) \mathbf{w}_j=A^* \mathbf{w}_j=\overline{\lambda_j} \mathbf{w}_j, \end{aligned}$	$B^* \mathbf{w}_j=\left(A-\lambda_1 \mathbf{v}_1 \mathbf{v}_1^*\right)^* \mathbf{w}_j=\left(A^*-\overline{\lambda_1} \mathbf{v}_1 \mathbf{v}_1^*\right) \mathbf{w}_j=A^* \mathbf{w}_j=\overline{\lambda_j} \mathbf{w}_j,$	$B^* \mathbf{w}_j=\left(A-\lambda_1 \mathbf{v}_1 \mathbf{v}_1^*\right)^* \mathbf{w}_j=\left(A^*-\overline{\lambda_1} \mathbf{v}_1 \mathbf{v}_1^*\right) \mathbf{w}_j=A^* \mathbf{w}_j=\overline{\lambda_j} \mathbf{w}_j,$
johnston-linear-matrix-algebra- EQ0475	196	0.760	$\begin{aligned} &x+y+z=0 \\ &x+y=1 \end{aligned}$	$\begin{aligned} x+y+z &=0 \\ x+y &=1 \end{aligned}$	$\begin{aligned} x+y+z &=0 \\ x+y &=1 \end{aligned}$
johnston-linear-matrix-algebra- EQ0310	124	0.765	$A^{-k} \stackrel{\text { def }}{=} \left(A^{-1}\right)^k \quad \text { for all integers } \quad k \geq 1,$	$A^{-k} \stackrel{\text { def }}{=} \left(A^{-1}\right)^k \quad \text { for all integers } \quad k \geq 1,$	$A^{-k} \stackrel{\text { def }}{=} \left(A^{-1}\right)^k \quad \text { for all integers } \quad k \geq 1,$

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johnston-linear-matrix-algebra-EQ1330	446	0.769	$\begin{aligned} c(d\mathbf{v}) &= c(dv_1, \dots, dv_n) \\ &= (cdv_1, \dots, cdv_n) \\ &= ((cd)v_1, \dots, (cd)v_n) \\ &= (cd)\mathbf{v}. \end{aligned}$	$\begin{aligned} c(d\mathbf{v}) &= c(dv_1, \dots, dv_n) \\ &= (cdv_1, \dots, cdv_n) \\ &= ((cd)v_1, \dots, (cd)v_n) \\ &= (cd)\mathbf{v} \end{aligned}$	$\begin{aligned} c(d\mathbf{v}) &= c(dv_1, \dots, dv_n) \\ &= (cdv_1, \dots, cdv_n) \\ &= ((cd)v_1, \dots, (cd)v_n) \\ &= (cd)\mathbf{v}. \end{aligned}$
johnston-linear-matrix-algebra-EQ0431	167	0.771	$\begin{aligned} &\{(2, -1, 0, 0, 0, 0), (3, 0, 6, -1, 0, 0), \\ &\quad (4, 0, 7, 0, -1, 0), (5, 0, 8, 0, 0, 9, -1)\} \end{aligned}$	$\{(2, -1, 0, 0, 0, 0), (3, 0, 6, -1, 0, 0, 0), \\ (4, 0, 7, 0, -1, 0, 0), (5, 0, 8, 0, 0, 9, -1)\}$	$\{(2, -1, 0, 0, 0, 0), (3, 0, 6, -1, 0, 0, 0), \\ (4, 0, 7, 0, -1, 0, 0), (5, 0, 8, 0, 0, 9, -1)\}.$
johnston-linear-matrix-algebra-EQ0241	110	0.774	$\begin{aligned} &\left[\begin{array}{cc c} -1 & 3 & 1 \\ 3 & 1 & 3 \\ 2 & -1 & 1 \end{array} \right] \xrightarrow{R_2+3R_1} \left[\begin{array}{cc c} -1 & 3 & 1 \\ R_3+2R_1 & 10 & 6 \\ 0 & 5 & 3 \end{array} \right] \\ &\xrightarrow{R_3-\frac{1}{2}R_2} \left[\begin{array}{cc c} -1 & 3 & 1 \\ 0 & 10 & 6 \\ 0 & 0 & 0 \end{array} \right]. \end{aligned}$	$\begin{aligned} &\left[\begin{array}{cc c} -1 & 3 & 1 \\ 3 & 1 & 3 \\ 2 & -1 & 1 \end{array} \right] \xrightarrow{R_2+3R_1} \left[\begin{array}{cc c} -1 & 3 & 1 \\ R_3+2R_1 & 10 & 6 \\ 0 & 5 & 3 \end{array} \right] \\ &\xrightarrow{R_3-\frac{1}{2}R_2} \left[\begin{array}{cc c} -1 & 3 & 1 \\ 0 & 10 & 6 \\ 0 & 0 & 0 \end{array} \right]. \end{aligned}$	$\begin{aligned} &\left[\begin{array}{cc c} -1 & 3 & 1 \\ 3 & 1 & 3 \\ 2 & -1 & 1 \end{array} \right] \xrightarrow{R_2+3R_1} \left[\begin{array}{cc c} -1 & 3 & 1 \\ 0 & 10 & 6 \\ 0 & 5 & 3 \end{array} \right] \\ &\xrightarrow{R_3-\frac{1}{2}R_2} \left[\begin{array}{cc c} -1 & 3 & 1 \\ 0 & 10 & 6 \\ 0 & 0 & 0 \end{array} \right]. \end{aligned}$
johnston-linear-matrix-algebra-EQ0473	195	0.775	$(3, 1, 0, 2), (1, 0, 1, 2), (4, 4, 2, 2), (2, 3, 3, 2), \text{ and } (0, 2, 4, 2).$	$(3, 1, 0, 2), (1, 0, 1, 2), (4, 4, 2, 2), (2, 3, 3, 2), \text{ and } (0, 2, 4, 2).$	$(3, 1, 0, 2), (1, 0, 1, 2), (4, 4, 2, 2), (2, 3, 3, 2), \text{ and } (0, 2, 4, 2).$
johnston-linear-matrix-algebra-EQ0026	32	0.777	$\ \mathbf{v}\ ^2 + \ \mathbf{w}\ ^2 - 2\ \mathbf{v}\ \ \mathbf{w}\ \cos(\theta) = \ \mathbf{v}\ ^2 - 2(\mathbf{v} \cdot \mathbf{w}) + \ \mathbf{w}\ ^2.$	$\ \mathbf{v}\ ^2 + \ \mathbf{w}\ ^2 - 2\ \mathbf{v}\ \ \mathbf{w}\ \cos(\theta) = \ \mathbf{v}\ ^2 - 2(\mathbf{v} \cdot \mathbf{w}) + \ \mathbf{w}\ ^2.$	$\ \mathbf{v}\ ^2 + \ \mathbf{w}\ ^2 - 2\ \mathbf{v}\ \ \mathbf{w}\ \cos(\theta) = \ \mathbf{v}\ ^2 - 2(\mathbf{v} \cdot \mathbf{w}) + \ \mathbf{w}\ ^2.$
johnston-linear-matrix-algebra-EQ0946	336	0.780	$P = [\mathbf{p}_1 \mathbf{p}_2 \mathbf{p}_3] = \begin{bmatrix} 3 & 0 & 1 \\ 1 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}, \quad P^{-1} = \begin{bmatrix} \mathbf{q}_1^* \\ \mathbf{q}_2^* \\ \mathbf{q}_3^* \end{bmatrix} = \frac{1}{3} \begin{bmatrix} 1 & 0 & -1 \\ -1 & 3 & 1 \\ 0 & 0 & 3 \end{bmatrix},$	$P = [\mathbf{p}_1 \mathbf{p}_2 \mathbf{p}_3] = \begin{bmatrix} 3 & 0 & 1 \\ 1 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}, \quad P^{-1} = \begin{bmatrix} \mathbf{q}_1^* \\ \mathbf{q}_2^* \\ \mathbf{q}_3^* \end{bmatrix} = \frac{1}{3} \begin{bmatrix} 1 & 0 & -1 \\ -1 & 3 & 1 \\ 0 & 0 & 3 \end{bmatrix},$	$P = [\mathbf{p}_1 \mathbf{p}_2 \mathbf{p}_3] = \begin{bmatrix} 3 & 0 & 1 \\ 1 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}, \quad P^{-1} = \begin{bmatrix} \mathbf{q}_1^* \\ \mathbf{q}_2^* \\ \mathbf{q}_3^* \end{bmatrix} = \frac{1}{3} \begin{bmatrix} 1 & 0 & -1 \\ -1 & 3 & 1 \\ 0 & 0 & 3 \end{bmatrix},$
johnston-linear-matrix-algebra-EQ1061	363	0.780	$P_{\sigma} = [\mathbf{e}_3 \mathbf{e}_4 \mathbf{e}_2 \mathbf{e}_1] = \begin{bmatrix} 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \end{bmatrix}.$	$P_{\sigma} = [\mathbf{e}_3 \mathbf{e}_4 \mathbf{e}_2 \mathbf{e}_1] = \begin{bmatrix} 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \end{bmatrix}.$	$P_{\sigma} = [\mathbf{e}_3 \mathbf{e}_4 \mathbf{e}_2 \mathbf{e}_1] = \begin{bmatrix} 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \end{bmatrix}.$

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johnston-linear-matrix-algebra-EQ00155	77	0.781	$\begin{aligned} \sqrt{\ \mathbf{v}\ ^2\ \mathbf{w}\ ^2 - (\mathbf{v} \cdot \mathbf{w})^2} &= \sqrt{\ \mathbf{v}\ ^2\ \mathbf{w}\ ^2 - \ \mathbf{v}\ ^2\ \mathbf{w}\ ^2 \cos^2(\theta)} \\ &= \sqrt{\ \mathbf{v}\ ^2\ \mathbf{w}\ ^2 (1 - \cos^2(\theta))} \\ &= \sqrt{\ \mathbf{v}\ ^2\ \mathbf{w}\ ^2 \sin^2(\theta)} \\ &= \ \mathbf{v}\ \ \mathbf{w}\ \sin(\theta), \end{aligned}$	$\sqrt{\ \mathbf{v}\ ^2\ \mathbf{w}\ ^2 - (\mathbf{v} \cdot \mathbf{w})^2} = \sqrt{\ \mathbf{v}\ ^2\ \mathbf{w}\ ^2 - \ \mathbf{v}\ ^2\ \mathbf{w}\ ^2 \cos^2(\theta)}$ $= \sqrt{\ \mathbf{v}\ ^2\ \mathbf{w}\ ^2 (1 - \cos^2(\theta))}$ $= \sqrt{\ \mathbf{v}\ ^2\ \mathbf{w}\ ^2 \sin^2(\theta)}$ $= \ \mathbf{v}\ \ \mathbf{w}\ \sin(\theta),$	$\sqrt{\ \mathbf{v}\ ^2\ \mathbf{w}\ ^2 - (\mathbf{v} \cdot \mathbf{w})^2} = \sqrt{\ \mathbf{v}\ ^2\ \mathbf{w}\ ^2 - \ \mathbf{v}\ ^2\ \mathbf{w}\ ^2 \cos^2(\theta)}$ $= \sqrt{\ \mathbf{v}\ ^2\ \mathbf{w}\ ^2 (1 - \cos^2(\theta))}$ $= \sqrt{\ \mathbf{v}\ ^2\ \mathbf{w}\ ^2 \sin^2(\theta)}$ $= \ \mathbf{v}\ \ \mathbf{w}\ \sin(\theta),$
johnston-linear-matrix-algebra-EQ00012	26	0.783	$\begin{aligned} \left(v_1, v_2, \dots, v_n \right) \cdot \mathbf{e}_j &= 0v_1 + \dots + 0v_{j-1} + 1v_j + 0v_{j+1} + \dots + 0v_n \\ &= v_j \end{aligned}$	$(v_1, v_2, \dots, v_n) \cdot \mathbf{e}_j = 0v_1 + \dots + 0v_{j-1} + 1v_j + 0v_{j+1} + \dots + 0v_n$ $= v_j$	$(v_1, v_2, \dots, v_n) \cdot \mathbf{e}_j = 0v_1 + \dots + 0v_{j-1} + 1v_j + 0v_{j+1} + \dots + 0v_n$ $= v_j.$
johnston-linear-matrix-algebra-EQ00051	310	0.783	$\begin{aligned} p_B(\lambda) &= \det(B - \lambda I) = \det \begin{pmatrix} 8 - \lambda & -5 & -5 \\ 5 & -2 - \lambda & -5 \\ -5 & 5 & 8 - \lambda \end{pmatrix} \\ &= (8 - \lambda)(-2 - \lambda)(8 - \lambda) - 125 - 125 \\ &\quad + 25(8 - \lambda) - 25(-2 - \lambda) + 25(8 - \lambda) \\ &= -\lambda^3 + 14\lambda^2 - 57\lambda + 72 \end{aligned}$	$p_B(\lambda) = \det(B - \lambda I) = \det \left(\begin{bmatrix} 8 - \lambda & -5 & -5 \\ 5 & -2 - \lambda & -5 \\ -5 & 5 & 8 - \lambda \end{bmatrix} \right)$ $= (8 - \lambda)(-2 - \lambda)(8 - \lambda) - 125 - 125$ $+ 25(8 - \lambda) - 25(-2 - \lambda) + 25(8 - \lambda)$ $= -\lambda^3 + 14\lambda^2 - 57\lambda + 72$	$p_B(\lambda) = \det(B - \lambda I) = \det \left(\begin{bmatrix} 8 - \lambda & -5 & -5 \\ 5 & -2 - \lambda & -5 \\ -5 & 5 & 8 - \lambda \end{bmatrix} \right)$ $= (8 - \lambda)(-2 - \lambda)(8 - \lambda) - 125 - 125$ $+ 25(8 - \lambda) - 25(-2 - \lambda) + 25(8 - \lambda)$ $= -\lambda^3 + 14\lambda^2 - 57\lambda + 72.$
johnston-linear-matrix-algebra-EQ1040	358	0.783	$\begin{aligned} \det(A) &= 2 \quad c_{1,1} = 6 \quad c_{1,2} = -5 \quad c_{1,3} = 1 \\ c_{2,1} &= -6 \quad c_{2,2} = 8 \quad c_{2,3} = -2 \\ c_{3,1} &= 2 \quad c_{3,2} = -3 \quad c_{3,3} = 1 \end{aligned}$	$\det(A) = 2 \quad c_{1,1} = 6 \quad c_{1,2} = -5 \quad c_{1,3} = 1$ $c_{2,1} = -6 \quad c_{2,2} = 8 \quad c_{2,3} = -2$ $c_{3,1} = 2 \quad c_{3,2} = -3 \quad c_{3,3} = 1$	$\det(A) = 2 \quad c_{1,1} = 6 \quad c_{1,2} = -5 \quad c_{1,3} = 1$ $c_{2,1} = -6 \quad c_{2,2} = 8 \quad c_{2,3} = -2$ $c_{3,1} = 2 \quad c_{3,2} = -3 \quad c_{3,3} = 1.$
johnston-linear-matrix-algebra-EQ00066	46	0.787	$\left[\begin{array}{ll} A & B \\ B & A \end{array} \right] = \left[\begin{array}{ccc ccc} 1 & 2 & -1 & 1 & 1 & \\ -1 & 1 & 1 & 1 & 2 & \\ 0 & 1 & 0 & 3 & 4 & \end{array} \right],$	$\left[\begin{array}{cc ccc} A & B & 1 & 2 & -1 & 1 & 1 \\ B & A & -1 & 1 & 1 & 1 & 2 \\ & & 0 & 1 & 0 & 3 & 4 \end{array} \right],$	$\left[\begin{array}{cc ccc} A & B & 1 & 2 & -1 & 1 & 1 \\ B & A & -1 & 1 & 1 & 1 & 2 \\ & & 0 & 1 & 0 & 3 & 4 \end{array} \right],$
johnston-linear-matrix-algebra-EQ00022	31	0.788	$\begin{aligned} c^2 &= a^2 + b^2 \\ &\quad - 2ab \cos(\theta). \end{aligned}$	$c^2 = a^2 + b^2$ $- 2ab \cos(\theta).$	$c^2 = a^2 + b^2$ $- 2ab \cos(\theta).$
johnston-linear-matrix-algebra-EQ00540	217	0.788	$\begin{aligned} \text{minimize:} & \quad 3y_1 + y_2 \\ \text{subject to:} & \quad y_1 - y_2 \geq 1 \\ & \quad y_1 + y_2 \geq 2 \\ & \quad y_1, \quad y_2 \geq 0 \end{aligned}$	$\text{minimize :} \quad 3y_1 + y_2$ $\text{subject to:} \quad y_1 - y_2 \geq 1$ $y_1 + y_2 \geq 2$ $y_1, \quad y_2 \geq 0$	$\text{minimize:} \quad 3y_1 + y_2$ $\text{subject to:} \quad y_1 - y_2 \geq 1$ $y_1 + y_2 \geq 2$ $y_1, \quad y_2 \geq 0$

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johnston-linear-matrix-algebra-EQ0054	43	0.791	A=\left[\begin{array}{ccc} 1 & 2 & 3 \\ \end{array}\right] \quad \Longrightarrow \quad A^T=\begin{array}{cc} 7 & 4 \\ 2 & 5 \\ 3 & 6 \end{array} \\ \left[\begin{array}{cc} 7 & 4 \\ 2 & 5 \\ 3 & 6 \end{array}\right] .	$A = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{bmatrix} \implies A^T = \begin{bmatrix} 7 & 4 \\ 2 & 5 \\ 3 & 6 \end{bmatrix}.$	
johnston-linear-matrix-algebra-EQ1596	477	0.793	\begin{aligned} & \operatorname{det}\left(\begin{array}{cc} 1-\lambda & 2 \\ -1 & -2-\lambda \end{array}\right) \\ & \quad		

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johnston-linear-matrix-algebra-EQ1532	471	0.799	$\begin{aligned} L_1 &= \left[\begin{array}{ll} 1 & 0 \\ 0 & 1 \end{array} \right], & U_1 &= \left[\begin{array}{ll} 0 & 0 \\ 0 & 1 \end{array} \right], \text{ and } \\ L_1 &= \left[\begin{array}{ll} 1 & 0 \\ 1 & 1 \end{array} \right], & U_1 &= \left[\begin{array}{ll} 0 & 0 \\ 0 & 1 \end{array} \right]. \end{aligned}$	$L_1 = \left[\begin{array}{ll} 1 & 0 \\ 0 & 1 \end{array} \right], \quad U_1 = \left[\begin{array}{ll} 0 & 0 \\ 0 & 1 \end{array} \right], \text{ and}$ $L_1 = \left[\begin{array}{ll} 1 & 0 \\ 1 & 1 \end{array} \right], \quad U_1 = \left[\begin{array}{ll} 0 & 0 \\ 0 & 1 \end{array} \right].$	$L_1 = \left[\begin{array}{ll} 1 & 0 \\ 0 & 1 \end{array} \right], \quad U_1 = \left[\begin{array}{ll} 0 & 0 \\ 0 & 1 \end{array} \right], \quad \text{and}$ $L_1 = \left[\begin{array}{ll} 1 & 0 \\ 1 & 1 \end{array} \right], \quad U_1 = \left[\begin{array}{ll} 0 & 0 \\ 0 & 1 \end{array} \right].$
johnston-linear-matrix-algebra-EQ0488 (2.B.1)	197	0.799	$\begin{array}{l} \operatorname{maximize} : \quad \mathbf{c} \cdot \mathbf{x} \\ \text{subject to: } \quad \mathbf{Ax} \leq \mathbf{b} \\ \quad \mathbf{x} \geq \mathbf{0} \end{array}$	$\begin{array}{l} \text{maximize :} \quad \mathbf{c} \cdot \mathbf{x} \\ \text{subject to:} \quad \mathbf{Ax} \leq \mathbf{b} \\ \quad \mathbf{x} \geq \mathbf{0} \end{array}$	$\begin{array}{l} \text{maximize:} \quad \mathbf{c} \cdot \mathbf{x} \\ \text{subject to:} \quad \mathbf{Ax} \leq \mathbf{b} \\ \quad \mathbf{x} \geq \mathbf{0} \end{array}$
johnston-linear-matrix-algebra-EQ0419	163	0.804	$\begin{aligned} & \left[\begin{array}{ccc c} 1 & 2 & 1 & 0 \\ -1 & 2 & 1 & 0 \\ -1 & -1 & 0 & 0 \end{array} \right] \xrightarrow[R_3+R_1]{R_2+R_1} \left[\begin{array}{ccc c} 1 & 2 & 1 & 0 \\ 0 & 4 & 2 & 0 \\ 0 & 1 & 1 & 0 \end{array} \right] \\ & \xrightarrow{R_3-\frac{1}{4}R_2} \left[\begin{array}{ccc c} 1 & 2 & 1 & 0 \\ 0 & 4 & 2 & 0 \\ 0 & 0 & 1/2 & 0 \end{array} \right]. \end{aligned}$	$\left[\begin{array}{ccc c} 1 & 2 & 1 & 0 \\ -1 & 2 & 1 & 0 \\ -1 & -1 & 0 & 0 \end{array} \right] \xrightarrow[R_3+R_1]{R_2+R_1} \left[\begin{array}{ccc c} 1 & 2 & 1 & 0 \\ 0 & 4 & 2 & 0 \\ 0 & 1 & 1 & 0 \end{array} \right]$ $\xrightarrow{R_3-\frac{1}{4}R_2} \left[\begin{array}{ccc c} 1 & 2 & 1 & 0 \\ 0 & 4 & 2 & 0 \\ 0 & 0 & 1/2 & 0 \end{array} \right].$	$\left[\begin{array}{ccc c} 1 & 2 & 1 & 0 \\ -1 & 2 & 1 & 0 \\ -1 & -1 & 0 & 0 \end{array} \right] \xrightarrow[R_3+R_1]{R_2+R_1} \left[\begin{array}{ccc c} 1 & 2 & 1 & 0 \\ 0 & 4 & 2 & 0 \\ 0 & 1 & 1 & 0 \end{array} \right]$ $\xrightarrow{R_3-\frac{1}{4}R_2} \left[\begin{array}{ccc c} 1 & 2 & 1 & 0 \\ 0 & 4 & 2 & 0 \\ 0 & 0 & 1/2 & 0 \end{array} \right].$
johnston-linear-matrix-algebra-EQ1523	469	0.805	$\begin{aligned} C &= \left[\begin{array}{ccc} 2 & -1 & 0 \\ 1 & 0 & 0 \\ -1 & 0 & 2 \\ 2 & 0 & 1 \end{array} \right] \text{ and } \\ R &= \left[\begin{array}{cccccc} 2 & 2 & 1 & 2 & 1 & 1 \\ 0 & -1 & 2 & 0 & 1 & -1 \\ -1 & 1 & 2 & 0 & -1 & 1 \end{array} \right]. \end{aligned}$	$C = \left[\begin{array}{ccc} 2 & -1 & 0 \\ 1 & 0 & 0 \\ -1 & 0 & 2 \\ 2 & 0 & 1 \end{array} \right] \text{ and}$ $R = \left[\begin{array}{cccccc} 2 & 2 & 1 & 2 & 1 & 1 \\ 0 & -1 & 2 & 0 & 1 & -1 \\ -1 & 1 & 2 & 0 & -1 & 1 \end{array} \right].$	$C = \left[\begin{array}{ccc} 2 & -1 & 0 \\ 1 & 0 & 0 \\ -1 & 0 & 2 \\ 2 & 0 & 1 \end{array} \right] \text{ and}$ $R = \left[\begin{array}{cccccc} 2 & 2 & 1 & 2 & 1 & 1 \\ 0 & -1 & 2 & 0 & 1 & -1 \\ -1 & 1 & 2 & 0 & -1 & 1 \end{array} \right].$
johnston-linear-matrix-algebra-EQ0723	271	0.810	$\begin{aligned} B &= \{ (-1, -1, 1, 0, 2), (-1, -1, 0, 1, 0), \\ C &= \{ (2, 2, 2, -1, -1), (1, 1, 1, 2, 0), (1, 2, 0, 1, 2) \}, \\ & \quad (2, -1, 3, 0, 1), (0, 1, 0, 1, 0), (3, -1, 0), (1, 1, 0, 1, 0), (3, -1, 2, 3, 3) \} \end{aligned}$	$B = \{(-1, -1, 1, 0, 2), (-1, -1, 0, 1, 0),$ $C = \{(2, 2, 2, -1, -1), (1, 1, 1, 2, 0), (1, 2, 0, 1, 2)\},$ $(2, -1, 3, 0, 1), (0, 1, 0, 1, 0), (3, -1, 0), (1, 1, 0, 1, 0), (3, -1, 2, 3, 3)\}$	$B = \{(-1, -1, 1, 0, 2), (-1, -1, 0, 1, 0),$ $(2, 2, 2, -1, -1), (1, 1, 1, 2, 0), (1, 2, 0, 1, 2)\},$ $C = \{(2, -1, 3, 0, 1), (0, 1, 0, 1, 0),$ $(2, -1, 2, -1, 0), (1, 1, 0, 1, 0), (3, -1, 2, 3, 3)\}$
johnston-linear-matrix-algebra-EQ1138	394	0.810	$H D H^{-1} = H (P^{-1} A P) H^{-1} = (H P^{-1}) A (H P^{-1})^{-1} = Q^{-1} A Q,$	$H D H^{-1} = H (P^{-1} A P) H^{-1} = (H P^{-1}) A (H P^{-1})^{-1} = Q^{-1} A Q,$	$H D H^{-1} = H (P^{-1} A P) H^{-1} = (H P^{-1}) A (H P^{-1})^{-1} = Q^{-1} A Q,$
johnston-linear-matrix-algebra-EQ0105	60	0.810	$\begin{aligned} & 2(\mathbf{u}\mathbf{u}^T)\mathbf{v} - \mathbf{v} \\ &= (2\mathbf{u}\mathbf{u}^T - 1)\mathbf{v}, \end{aligned}$	$2(\mathbf{u}\mathbf{u}^T)\mathbf{v} - \mathbf{v}$ $= (2\mathbf{u}\mathbf{u}^T - 1)\mathbf{v},$	$2(\mathbf{u}\mathbf{u}^T)\mathbf{v} - \mathbf{v}$ $= (2\mathbf{u}\mathbf{u}^T - 1)\mathbf{v},$
johnston-linear-matrix-algebra-EQ1268	425	0.813	$x - 2 \overline{3x^3 + 0x^2 - 14x + 4}$	$x - 2 \overline{3x^3 + 0x^2 - 14x + 4}$	$x - 2 \overline{3x^3 + 0x^2 - 14x + 4}$

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johnston-linear-matrix-algebra-EQ00004	21	0.813	$\begin{aligned} c(\mathbf{v}+\mathbf{w}) &= c\left(v_1+w_1, v_2+w_2, \ldots, v_n+w_n\right) \\ &= (c(v_1+w_1), c(v_2+w_2), \ldots, c(v_n+w_n)) \\ &= (cv_1+cw_1, cv_2+cw_2, \ldots, cv_n+cw_n) \\ &= (cv_1, cv_2, \ldots, cv_n) + (cw_1, cw_2, \ldots, cw_n) \\ &= c(v_1, v_2, \ldots, v_n) + c(w_1, w_2, \ldots, w_n) \\ &= c\mathbf{v} + c\mathbf{w}. \end{aligned}$	$\begin{aligned} c(\mathbf{v}+\mathbf{w}) &= c\left(v_1+w_1, v_2+w_2, \ldots, v_n+w_n\right) && \text{(vector addition)} \\ &= (c(v_1+w_1), c(v_2+w_2), \ldots, c(v_n+w_n)) && \text{(scalar mult.)} \\ &= (cv_1+cw_1, cv_2+cw_2, \ldots, cv_n+cw_n) && \text{(property of } \mathbb{R} \text{)} \\ &= (cv_1, cv_2, \ldots, cv_n) + (cw_1, cw_2, \ldots, cw_n) && \text{(vector addition)} \\ &= c(v_1, v_2, \ldots, v_n) + c(w_1, w_2, \ldots, w_n) && \text{(scalar mult.)} \\ &= c\mathbf{v} + c\mathbf{w}. \end{aligned}$	$\begin{aligned} c(\mathbf{v}+\mathbf{w}) &= c\left(v_1+w_1, v_2+w_2, \ldots, v_n+w_n\right) && \text{(vector addition)} \\ &= (c(v_1+w_1), c(v_2+w_2), \ldots, c(v_n+w_n)) && \text{(scalar mult.)} \\ &= (cv_1+cw_1, cv_2+cw_2, \ldots, cv_n+cw_n) && \text{(property of } \mathbb{R} \text{)} \\ &= (cv_1, cv_2, \ldots, cv_n) + (cw_1, cw_2, \ldots, cw_n) && \text{(vector addition)} \\ &= c(v_1, v_2, \ldots, v_n) + c(w_1, w_2, \ldots, w_n) && \text{(scalar mult.)} \\ &= c\mathbf{v} + c\mathbf{w}. \end{aligned}$
johnston-linear-matrix-algebra-EQ0702 (a)	265	0.815	$\begin{aligned} P_{C \leftarrow B}[T]_B P_{B \leftarrow C}[\mathbf{v}]_C &= P_{C \leftarrow B}[T]_B[\mathbf{v}]_B && (P_{B \leftarrow C}[\mathbf{v}]_C = [\mathbf{v}]_B \text{ by Theorem 3.1.3} \\ &= P_{C \leftarrow B}[T(\mathbf{v})]_B && ([T]_B[\mathbf{v}]_B = [T(\mathbf{v})]_B \text{ by Theorem 3.1.3(a) again}) \\ &= [T(\mathbf{v})]_C. \end{aligned}$	$\begin{aligned} P_{C \leftarrow B}[T]_B P_{B \leftarrow C}[\mathbf{v}]_C &= P_{C \leftarrow B}[T]_B[\mathbf{v}]_B && (P_{B \leftarrow C}[\mathbf{v}]_C = [\mathbf{v}]_B \text{ by Theorem 3.1.3} \\ &= P_{C \leftarrow B}[T(\mathbf{v})]_B && ([T]_B[\mathbf{v}]_B = [T(\mathbf{v})]_B \text{ by Theorem 3.1.3(a) again}) \\ &= [T(\mathbf{v})]_C. \end{aligned}$	$\begin{aligned} P_{C \leftarrow B}[T]_B P_{B \leftarrow C}[\mathbf{v}]_C &= P_{C \leftarrow B}[T]_B[\mathbf{v}]_B && (P_{B \leftarrow C}[\mathbf{v}]_C = [\mathbf{v}]_B \text{ by Theorem 3.1.3} \\ &= P_{C \leftarrow B}[T(\mathbf{v})]_B && ([T]_B[\mathbf{v}]_B = [T(\mathbf{v})]_B \text{ by Theorem 3.1.3(a) again}) \\ &= [T(\mathbf{v})]_C. \end{aligned}$
johnston-linear-matrix-algebra-EQ1511	468	0.815	$\begin{aligned} &\text{minimize: } 2y_1 + 3y_2 \\ &\text{subject to: } 2y_1 + y_2 \geq 0 \\ &\quad -2y_1 + 3y_2 \geq 1 \\ &\quad -y_1, \quad y_2 \geq 0 \end{aligned}$	$\begin{aligned} &\text{minimize: } 2y_1 + 3y_2 \\ &\text{subject to: } 2y_1 + y_2 \geq 0 \\ &\quad -2y_1 + 3y_2 \geq 1 \\ &\quad -y_1, \quad y_2 \geq 0 \end{aligned}$	$\begin{aligned} &\text{minimize: } 2y_1 + 3y_2 \\ &\text{subject to: } 2y_1 + y_2 \geq 0 \\ &\quad -2y_1 + 3y_2 \geq 1 \\ &\quad -y_1, \quad y_2 \geq 0 \end{aligned}$
johnston-linear-matrix-algebra-EQ0963	340	0.817	$\begin{aligned} [A^k]_{1,4} &= [PD^k P^{-1}]_{1,4} = \frac{1}{30} \overbrace{((-2)^k, (-1)^k, 0, 0, 3^{k+1})}^{\text{1 st row of } PD^k} \cdot \overbrace{(-6, 0, -10, 10, 2)}^{30 \times (\text{4th column of } P^{-1})} \\ &= \frac{1}{5} (3^k - (-2)^k) \quad \text{for all integers } k \geq 0. \end{aligned}$	$\begin{aligned} [A^k]_{1,4} &= [PD^k P^{-1}]_{1,4} = \frac{1}{30} \overbrace{((-2)^k, (-1)^k, 0, 0, 3^{k+1})}^{\text{1 st row of } PD^k} \cdot \overbrace{(-6, 0, -10, 10, 2)}^{30 \times (\text{4th column of } P^{-1})} \\ &= \frac{1}{5} (3^k - (-2)^k) \quad \text{for all integers } k \geq 0. \end{aligned}$	$\begin{aligned} [A^k]_{1,4} &= [PD^k P^{-1}]_{1,4} = \frac{1}{30} \overbrace{((-2)^k, (-1)^k, 0, 0, 3^{k+1})}^{\text{1st row of } PD^k} \cdot \overbrace{(-6, 0, -10, 10, 2)}^{30 \times (\text{4th column of } P^{-1})} \\ &= \frac{1}{5} (3^k - (-2)^k) \quad \text{for all integers } k \geq 0. \end{aligned}$
johnston-linear-matrix-algebra-EQ09563	222	0.818	$\begin{aligned} &\text{maximize: } 2x_1 + x_2 + x_3 + x_4 \\ &\text{subject to: } x_1 + 2x_2 - 3x_3 + 2x_4 \leq 1 \\ &\quad 2x_1 - x_2 + x_3 + x_4 \leq 0 \\ &\quad 3x_1 + x_2 - 2x_3 - x_4 \leq 1 \\ &\quad x_1, \quad x_2, \quad x_3, \quad x_4 \geq 0 \end{aligned}$	$\begin{aligned} &\text{maximize: } 2x_1 + x_2 + x_3 + x_4 \\ &\text{subject to: } x_1 + 2x_2 - 3x_3 + 2x_4 \leq 1 \\ &\quad 2x_1 - x_2 + x_3 + x_4 \leq 0 \\ &\quad 3x_1 + x_2 - 2x_3 - x_4 \leq 1 \\ &\quad x_1, \quad x_2, \quad x_3, \quad x_4 \geq 0 \end{aligned}$	$\begin{aligned} &\text{maximize: } 2x_1 + x_2 + x_3 + x_4 \\ &\text{subject to: } x_1 + 2x_2 - 3x_3 + 2x_4 \leq 1 \\ &\quad 2x_1 - x_2 + x_3 + x_4 \leq 0 \\ &\quad 3x_1 + x_2 - 2x_3 - x_4 \leq 1 \\ &\quad x_1, \quad x_2, \quad x_3, \quad x_4 \geq 0 \end{aligned}$

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johnston-linear-matrix-algebra-EQ1068	364	0.820	$\operatorname{sgn}(\sigma)=\operatorname{det}\left(P_{\sigma}\right)=\operatorname{det}\left(\begin{bmatrix} 0 & 0 & 1 & 0 & 0 \\ 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 1 \\ 0 & 1 & 0 & 0 & 0 \end{bmatrix}\right)=1$.	$\operatorname{sgn}(\sigma)=\det(P_{\sigma})=\det\left(\begin{bmatrix} 0 & 0 & 1 & 0 & 0 \\ 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 1 \\ 0 & 1 & 0 & 0 & 0 \end{bmatrix}\right)=1.$	$\operatorname{sgn}(\sigma)=\det(P_{\sigma})=\det\left(\begin{bmatrix} 0 & 0 & 1 & 0 & 0 \\ 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 1 \\ 0 & 1 & 0 & 0 & 0 \end{bmatrix}\right)=1.$
johnston-linear-matrix-algebra-EQ0746	382	0.821	$\left[\begin{array}{ll} 0 & 2 \\ 3 & 4 \end{array}\right] \xrightarrow{R_1 \leftrightarrow R_2} \left[\begin{array}{ll} 3 & 4 \\ 0 & 2 \end{array}\right] \xrightarrow{R_1-2R_2} \left[\begin{array}{ll} 3 & 0 \\ 0 & 2 \end{array}\right] \xrightarrow{(1/3)R_1} \left[\begin{array}{ll} 1 & 0 \\ (1/2)R_2 & 1 \end{array}\right]$.	$\left[\begin{array}{ll} 0 & 2 \\ 3 & 4 \end{array}\right] \xrightarrow{R_1 \leftrightarrow R_2} \left[\begin{array}{ll} 3 & 4 \\ 0 & 2 \end{array}\right] \xrightarrow{R_1-2R_2} \left[\begin{array}{ll} 3 & 0 \\ 0 & 2 \end{array}\right] \xrightarrow{(1/3)R_1} \left[\begin{array}{ll} 1 & 0 \\ (1/2)R_2 & 1 \end{array}\right]$.	$\left[\begin{array}{ll} 0 & 2 \\ 3 & 4 \end{array}\right] \xrightarrow{R_1 \leftrightarrow R_2} \left[\begin{array}{ll} 3 & 4 \\ 0 & 2 \end{array}\right] \xrightarrow{R_1-2R_2} \left[\begin{array}{ll} 3 & 0 \\ 0 & 2 \end{array}\right] \xrightarrow{(1/3)R_1} \left[\begin{array}{ll} 1 & 0 \\ (1/2)R_2 & 1 \end{array}\right]$.
johnston-linear-matrix-algebra-EQ1463	461	0.822	$A=\left[\begin{array}{ll} 1 & 2 \\ 3 & 4 \end{array}\right]$.	$A=\left[\begin{array}{ll} 1 & 2 \\ 3 & 4 \end{array}\right]$.	$A=\left[\begin{array}{ll} 1 & 2 \\ 3 & 4 \end{array}\right]$.
johnston-linear-matrix-algebra-EQ0024	31	0.826	$\ \mathbf{v}-\mathbf{w}\ ^2=\ \mathbf{v}\ ^2+\ \mathbf{w}\ ^2-2\ \mathbf{v}\ \ \mathbf{w}\ \cos(\theta)$.	$\ \mathbf{v}-\mathbf{w}\ ^2=\ \mathbf{v}\ ^2+\ \mathbf{w}\ ^2-2\ \mathbf{v}\ \ \mathbf{w}\ \cos(\theta).$	$\ \mathbf{v}-\mathbf{w}\ ^2=\ \mathbf{v}\ ^2+\ \mathbf{w}\ ^2-2\ \mathbf{v}\ \ \mathbf{w}\ \cos(\theta).$
johnston-linear-matrix-algebra-EQ0101	58	0.827	$P_{\mathbf{u}}=\mathbf{u}\mathbf{u}^T=\left[\begin{array}{l} 1 \\ 0 \end{array}\right]\left[\begin{array}{ll} 1 & 0 \end{array}\right]=\left[\begin{array}{ll} 1 & 0 \\ 0 & 0 \end{array}\right]$.	$[P_{\mathbf{u}}]=\mathbf{u}\mathbf{u}^T=\left[\begin{array}{l} 1 \\ 0 \end{array}\right]\left[\begin{array}{ll} 1 & 0 \end{array}\right]=\left[\begin{array}{ll} 1 & 0 \\ 0 & 0 \end{array}\right]$.	$[P_{\mathbf{u}}]=\mathbf{u}\mathbf{u}^T=\left[\begin{array}{l} 1 \\ 0 \end{array}\right]\left[\begin{array}{ll} 1 & 0 \end{array}\right]=\left[\begin{array}{ll} 1 & 0 \\ 0 & 0 \end{array}\right]$.
johnston-linear-matrix-algebra-EQ0210	99	0.827	$\begin{gathered} \left[\begin{array}{ccc c} 1 & 3 & -2 & 5 \\ 1 & 5 & -8 & 9 \\ 2 & 4 & 5 & 12 \end{array}\right] \xrightarrow{R_2-R_1} \left[\begin{array}{ccc c} 1 & 3 & -2 & 5 \\ 0 & 2 & -6 & 4 \\ 2 & 4 & 5 & 12 \end{array}\right] \xrightarrow{R_3-2R_1} \left[\begin{array}{ccc c} 1 & 3 & -2 & 5 \\ 0 & 2 & -6 & 4 \\ 0 & -2 & 9 & 2 \end{array}\right] \xrightarrow{R_3+R_2} \left[\begin{array}{ccc c} 1 & 3 & -2 & 5 \\ 0 & 2 & -6 & 4 \\ 0 & 0 & 3 & 6 \end{array}\right] \end{gathered}$.	$\begin{gathered} \left[\begin{array}{ccc c} 1 & 3 & -2 & 5 \\ 1 & 5 & -8 & 9 \\ 2 & 4 & 5 & 12 \end{array}\right] \xrightarrow{R_2-R_1} \left[\begin{array}{ccc c} 1 & 3 & -2 & 5 \\ 0 & 2 & -6 & 4 \\ 2 & 4 & 5 & 12 \end{array}\right] \xrightarrow{R_3-2R_1} \left[\begin{array}{ccc c} 1 & 3 & -2 & 5 \\ 0 & 2 & -6 & 4 \\ 0 & -2 & 9 & 2 \end{array}\right] \xrightarrow{R_3+R_2} \left[\begin{array}{ccc c} 1 & 3 & -2 & 5 \\ 0 & 2 & -6 & 4 \\ 0 & 0 & 3 & 6 \end{array}\right] \end{gathered}$.	$\begin{gathered} \left[\begin{array}{ccc c} 1 & 3 & -2 & 5 \\ 1 & 5 & -8 & 9 \\ 2 & 4 & 5 & 12 \end{array}\right] \xrightarrow{R_2-R_1} \left[\begin{array}{ccc c} 1 & 3 & -2 & 5 \\ 0 & 2 & -6 & 4 \\ 2 & 4 & 5 & 12 \end{array}\right] \xrightarrow{R_3-2R_1} \left[\begin{array}{ccc c} 1 & 3 & -2 & 5 \\ 0 & 2 & -6 & 4 \\ 0 & -2 & 9 & 2 \end{array}\right] \xrightarrow{R_3+R_2} \left[\begin{array}{ccc c} 1 & 3 & -2 & 5 \\ 0 & 2 & -6 & 4 \\ 0 & 0 & 3 & 6 \end{array}\right] \end{gathered}$.
johnston-linear-matrix-algebra-EQ1536 (0)	471	0.831	$L=\left[\begin{array}{ccc} 1 & 0 & 0 \\ -1 & 1 & 0 \\ 3 & 1 & 1 \end{array}\right], D=\left[\begin{array}{ccc} 1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & -2 \end{array}\right], U=\left[\begin{array}{ccc} 1 & 2 & -1 \\ 0 & 1 & 3 \\ 0 & 0 & 1 \end{array}\right]$.	$L=\left[\begin{array}{ccc} 1 & 0 & 0 \\ -1 & 1 & 0 \\ 3 & 1 & 1 \end{array}\right], D=\left[\begin{array}{ccc} 1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & -2 \end{array}\right], U=\left[\begin{array}{ccc} 1 & 2 & -1 \\ 0 & 1 & 3 \\ 0 & 0 & 1 \end{array}\right]$.	$L=\left[\begin{array}{ccc} 1 & 0 & 0 \\ -1 & 1 & 0 \\ 3 & 1 & 1 \end{array}\right], D=\left[\begin{array}{ccc} 1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & -2 \end{array}\right], U=\left[\begin{array}{ccc} 1 & 2 & -1 \\ 0 & 1 & 3 \\ 0 & 0 & 1 \end{array}\right]$.

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johnston-linear-matrix-algebra- EQ0784	291	0.831	$\left[\begin{array}{cccc} {}^+2 & -1 & \pm 1 & 0 \\ 0 & -2 & 1 & 3 \\ 0 & 0 & 1 & 0 \\ -1 & 3 & 0 & 2 \end{array}\right]$	$\begin{bmatrix} +2 & -1 & \pm 1 & 0 \\ 0 & -2 & 1 & 3 \\ 0 & 0 & 1 & 0 \\ -1 & 3 & 0 & 2 \end{bmatrix}$	
johnston-linear-matrix-algebra- EQ0570	222	0.838	$\begin{array}{l} \operatorname{maximize}: \quad x_1 + x_2 \\ \text{subject to: } \quad 2x_1 + x_2 \leq 2 \\ \quad \quad \quad 2x_1 + 3x_2 \geq 3 \\ \quad \quad \quad x_1 - 2x_2 \geq -1 \\ \quad \quad \quad x_1, \quad x_2 \geq 0 \end{array}$	$\begin{array}{ll} \text{maximize :} & x_1 + x_2 \\ \text{subject to:} & 2x_1 + x_2 \leq 2 \\ & 2x_1 + 3x_2 \geq 3 \\ & x_1 - 2x_2 \geq -1 \\ & x_1, \quad x_2 \geq 0 \end{array}$	$\begin{array}{ll} \text{maximize:} & x_1 + x_2 \\ \text{subject to:} & 2x_1 + x_2 \leq 2 \\ & 2x_1 + 3x_2 \geq 3 \\ & x_1 - 2x_2 \geq -1 \\ & x_1, \quad x_2 \geq 0 \end{array}$
johnston-linear-matrix-algebra- EQ0506 (2.B.5)	203	0.840	$\begin{array}{l} \operatorname{maximize}: \quad x_1 + 2x_2 \\ \text{subject to: } \quad x_1 - x_2 \leq -1 \\ \quad \quad \quad -x_1 + x_2 \leq -1 \\ \quad \quad \quad x_1, x_2 \geq 0 \end{array}$	$\begin{array}{ll} \text{maximize :} & x_1 + 2x_2 \\ \text{subject to:} & x_1 - x_2 \leq -1 \\ & -x_1 + x_2 \leq -1 \\ & x_1, x_2 \geq 0 \end{array}$	$\begin{array}{ll} \text{maximize:} & x_1 + 2x_2 \\ \text{subject to:} & x_1 - x_2 \leq -1 \\ & -x_1 + x_2 \leq -1 \\ & x_1, \quad x_2 \geq 0 \end{array}$
johnston-linear-matrix-algebra- EQ1519	469	0.843	$\begin{array}{l} \operatorname{minimize}: \quad (-\mathbf{c}) \cdot \mathbf{x} \\ \text{subject to: } \quad (-A^T)^T \mathbf{x} \geq -\mathbf{b} \\ \quad \quad \quad \mathbf{x} \geq \mathbf{0} \end{array}$	$\begin{array}{ll} -\text{minimize :} & (-\mathbf{c}) \cdot \mathbf{x} \\ \text{subject to:} & (-A^T)^T \mathbf{x} \geq -\mathbf{b} \\ & \mathbf{x} \geq \mathbf{0} \end{array}$	$\begin{array}{ll} -\text{minimize:} & (-\mathbf{c}) \cdot \mathbf{x} \\ \text{subject to:} & (-A^T)^T \mathbf{x} \geq -\mathbf{b} \\ & \mathbf{x} \geq \mathbf{0} \end{array}$
johnston-linear-matrix-algebra- EQ0536	215	0.846	$\begin{array}{l} \operatorname{minimize}: \quad \mathbf{b} \cdot \mathbf{y} \\ \text{subject to: } \quad A^T \mathbf{y} \geq \mathbf{c} \\ \quad \quad \quad \mathbf{y} \geq \mathbf{0} \end{array}$	$\begin{array}{ll} \text{minimize :} & \mathbf{b} \cdot \mathbf{y} \\ \text{subject to:} & A^T \mathbf{y} \geq \mathbf{c} \\ & \mathbf{y} \geq \mathbf{0} \end{array}$	$\begin{array}{ll} \text{minimize:} & \mathbf{b} \cdot \mathbf{y} \\ \text{subject to:} & A^T \mathbf{y} \geq \mathbf{c} \\ & \mathbf{y} \geq \mathbf{0} \end{array}$

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johnston-linear-matrix-algebra EQ0618 (3)	237	0.848	$\begin{gathered} \left[\begin{array}{cccc c} 2 & 4 & -1 & -1 & 0 \\ 0 & 1 & 2 & 1 & 2 \\ 0 & 0 & -2 & 0 & -6 \\ 0 & 0 & 0 & -3 & -15 \end{array} \right] \xrightarrow{\begin{smallmatrix} R_1/2 \\ -R_3/2 \\ -R_4/3 \end{smallmatrix}} \left[\begin{array}{cccc c} 1 & 2 & -1/2 & -1/2 & 0 \\ 0 & 1 & 2 & 1 & 2 \\ 0 & 0 & 1 & 0 & 3 \\ 0 & 0 & 0 & 1 & 5 \end{array} \right] \\ \xrightarrow{\begin{smallmatrix} R_1+\frac{1}{2}R_4 \\ R_2-R_4 \end{smallmatrix}} \left[\begin{array}{cccc c} 1 & 2 & -1/2 & 0 & 5/2 \\ 0 & 1 & 2 & 0 & -3 \\ 0 & 0 & 1 & 0 & 3 \\ 0 & 0 & 0 & 1 & 5 \end{array} \right] \xrightarrow{\begin{smallmatrix} R_1+\frac{1}{2}R_3 \\ R_2-2R_3 \end{smallmatrix}} \left[\begin{array}{cccc c} 1 & 2 & 0 & 0 & 4 \\ 0 & 1 & 0 & 0 & -9 \\ 0 & 0 & 1 & 0 & 3 \\ 0 & 0 & 0 & 1 & 5 \end{array} \right] \\ \xrightarrow{R_1-2R_2} \left[\begin{array}{cccc c} 1 & 0 & 0 & 0 & 22 \\ 0 & 1 & 0 & 0 & -9 \\ 0 & 0 & 1 & 0 & 3 \\ 0 & 0 & 0 & 1 & 5 \end{array} \right]. \end{gathered}$	$\begin{gathered} \left[\begin{array}{cccc c} 2 & 4 & -1 & -1 & 0 \\ 0 & 1 & 2 & 1 & 2 \\ 0 & 0 & -2 & 0 & -6 \\ 0 & 0 & 0 & -3 & -15 \end{array} \right] \xrightarrow{\begin{smallmatrix} R_1/2 \\ -R_3/2 \\ -R_4/3 \end{smallmatrix}} \left[\begin{array}{cccc c} 1 & 2 & -1/2 & -1/2 & 0 \\ 0 & 1 & 2 & 1 & 2 \\ 0 & 0 & 1 & 0 & 3 \\ 0 & 0 & 0 & 1 & 5 \end{array} \right] \\ \xrightarrow{\begin{smallmatrix} R_1+\frac{1}{2}R_4 \\ R_2-R_4 \end{smallmatrix}} \left[\begin{array}{cccc c} 1 & 2 & -1/2 & 0 & 5/2 \\ 0 & 1 & 2 & 0 & -3 \\ 0 & 0 & 1 & 0 & 3 \\ 0 & 0 & 0 & 1 & 5 \end{array} \right] \xrightarrow{\begin{smallmatrix} R_1+\frac{1}{2}R_3 \\ R_2-2R_3 \end{smallmatrix}} \left[\begin{array}{cccc c} 1 & 2 & 0 & 0 & 4 \\ 0 & 1 & 0 & 0 & -9 \\ 0 & 0 & 1 & 0 & 3 \\ 0 & 0 & 0 & 1 & 5 \end{array} \right] \\ \xrightarrow{R_1-2R_2} \left[\begin{array}{cccc c} 1 & 0 & 0 & 0 & 22 \\ 0 & 1 & 0 & 0 & -9 \\ 0 & 0 & 1 & 0 & 3 \\ 0 & 0 & 0 & 1 & 5 \end{array} \right]. \end{gathered}$	
johnston-linear-matrix-algebra EQ0273	115	0.848	$\mathrm{ZnS} + \mathrm{O}_2 \rightarrow \mathrm{ZnO} + \mathrm{SO}_2.$	$\mathrm{ZnS} + \mathrm{O}_2 \rightarrow \mathrm{ZnO} + \mathrm{SO}_2.$	$\mathrm{ZnS} + \mathrm{O}_2 \rightarrow \mathrm{ZnO} + \mathrm{SO}_2.$
johnston-linear-matrix-algebra EQ0824	301	0.848	$\begin{aligned} \operatorname{det}(A - \lambda I) &= \operatorname{det} \left(\begin{bmatrix} 1 - \lambda & 2 \\ 5 & 4 - \lambda \end{bmatrix} \right) \\ &= (1 - \lambda)(4 - \lambda) - 10 = \lambda^2 - 5\lambda - 6 \end{aligned}$	$\begin{aligned} \det(A - \lambda I) &= \det \left(\begin{bmatrix} 1 - \lambda & 2 \\ 5 & 4 - \lambda \end{bmatrix} \right) \\ &= (1 - \lambda)(4 - \lambda) - 10 = \lambda^2 - 5\lambda - 6 \end{aligned}$	$\begin{aligned} \det(A - \lambda I) &= \det \left(\begin{bmatrix} 1 - \lambda & 2 \\ 5 & 4 - \lambda \end{bmatrix} \right) \\ &= (1 - \lambda)(4 - \lambda) - 10 = \lambda^2 - 5\lambda - 6. \end{aligned}$
johnston-linear-matrix-algebra EQ1634	481	0.850	$\begin{aligned} \left(A^r \right)^s &= \left(P D^r P^{-1} \right)^s = P \left(D^r \right)^s P^{-1} \\ &= P D^{rs} P^{-1} = A^{rs}, \end{aligned}$	$\begin{aligned} \left(A^r \right)^s &= \left(P D^r P^{-1} \right)^s = P \left(D^r \right)^s P^{-1} \\ &= P D^{rs} P^{-1} = A^{rs} \end{aligned}$	$\begin{aligned} \left(A^r \right)^s &= \left(P D^r P^{-1} \right)^s = P \left(D^r \right)^s P^{-1} \\ &= P D^{rs} P^{-1} = A^{rs}, \end{aligned}$
johnston-linear-matrix-algebra EQ0524	209	0.853	$\begin{aligned} \operatorname{maximize} : & \quad x_p + 0.8x_c + 1.5x_{pc} \\ \operatorname{subject} \text{ to: } & \quad x_p + x_c + x_{pc} \leq 100 \\ & \quad x_p, x_c, x_{pc} \geq 0 \end{aligned}$	$\begin{aligned} \operatorname{maximize} : & \quad x_p + 0.8x_c + 1.5x_{pc} \\ \operatorname{subject} \text{ to: } & \quad x_p + x_c + x_{pc} \leq 100 \\ & \quad x_p, x_c, x_{pc} \geq 0 \end{aligned}$	$\begin{aligned} \operatorname{maximize} : & \quad x_p + 0.8x_c + 1.5x_{pc} \\ \operatorname{subject} \text{ to: } & \quad x_p + x_c + x_{pc} \leq 100 \\ & \quad x_p, x_c, x_{pc} \geq 0 \end{aligned}$

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johnston-linear-matrix-algebra-E00734	277	0.853	$\begin{aligned} & \operatorname{det}\left(2 A^{-3} B^{-2}\left(C^T B\right)^4(C / 5)^2\right) \\ & \quad = 2^3 \operatorname{det}(A)^{-3} \operatorname{det}(B)^{-2} \operatorname{det}\left(C^T\right)^4 \operatorname{det}(B)^4 \operatorname{det}(C / 5)^2 \\ & \quad = 8 \cdot(1 / 8) \cdot(1 / 9) \cdot 5^4 \cdot 3^4 \cdot(1 / 25)^2=9 . \end{aligned}$	$\begin{aligned} & \operatorname{det}\left(2 A^{-3} B^{-2}\left(C^T B\right)^4(C / 5)^2\right) \\ & \quad = 2^3 \operatorname{det}(A)^{-3} \operatorname{det}(B)^{-2} \operatorname{det}\left(C^T\right)^4 \operatorname{det}(B)^4 \operatorname{det}(C / 5)^2 \\ & \quad = 8 \cdot(1 / 8) \cdot(1 / 9) \cdot 5^4 \cdot 3^4 \cdot(1 / 25)^2=9 . \end{aligned}$	$\begin{aligned} \operatorname{det}\left(2 A^{-3} B^{-2}\left(C^T B\right)^4(C / 5)^2\right) \\ & \quad = 2^3 \operatorname{det}(A)^{-3} \operatorname{det}(B)^{-2} \operatorname{det}\left(C^T\right)^4 \operatorname{det}(B)^4 \operatorname{det}(C / 5)^2 \\ & \quad = 8 \cdot(1 / 8) \cdot(1 / 9) \cdot 5^4 \cdot 3^4 \cdot(1 / 25)^2=9 . \end{aligned}$
johnston-linear-matrix-algebra-E00845	308	0.856	$A=\left[\begin{array}{cc} -3 & -2 \\ 4 & 1 \end{array}\right] .$	$A=\left[\begin{array}{cc} -3 & -2 \\ 4 & 1 \end{array}\right] .$	$A=\left[\begin{array}{cc} -3 & -2 \\ 4 & 1 \end{array}\right] .$
johnston-linear-matrix-algebra-E00410	157	0.857	$\begin{aligned} & \left[\begin{array}{ccc c} 1 & 1 & 2 & 0 \\ 1 & 2 & 1 & 0 \\ 2 & 1 & 1 & 0 \end{array}\right] \xrightarrow{R_2-R_1}\left[\begin{array}{ccc c} 1 & 1 & 2 & 0 \\ R_3-2 R_1 & 1 & -1 & 0 \\ 0 & -1 & -3 & 0 \end{array}\right] \\ & \xrightarrow{R_3+R_2}\left[\begin{array}{ccc c} 1 & 1 & 2 & 0 \\ 0 & 1 & -1 & 0 \\ 0 & 0 & -4 & 0 \end{array}\right] . \end{aligned}$	$\begin{aligned} & \left[\begin{array}{ccc c} 1 & 1 & 2 & 0 \\ 1 & 2 & 1 & 0 \\ 2 & 1 & 1 & 0 \end{array}\right] \xrightarrow{R_2-R_1}\left[\begin{array}{ccc c} 1 & 1 & 2 & 0 \\ R_3-2 R_1 & 1 & -1 & 0 \\ 0 & -1 & -3 & 0 \end{array}\right] \\ & \xrightarrow{R_3+R_2}\left[\begin{array}{ccc c} 1 & 1 & 2 & 0 \\ 0 & 1 & -1 & 0 \\ 0 & 0 & -4 & 0 \end{array}\right] . \end{aligned}$	$\begin{aligned} & \left[\begin{array}{ccc c} 1 & 1 & 2 & 0 \\ 1 & 2 & 1 & 0 \\ 2 & 1 & 1 & 0 \end{array}\right] \xrightarrow{R_2-R_1}\left[\begin{array}{ccc c} 1 & 1 & 2 & 0 \\ 0 & 1 & -1 & 0 \\ 0 & -1 & -3 & 0 \end{array}\right] \\ & \xrightarrow{R_3+R_2}\left[\begin{array}{ccc c} 1 & 1 & 2 & 0 \\ 0 & 1 & -1 & 0 \\ 0 & 0 & -4 & 0 \end{array}\right] . \end{aligned}$
johnston-linear-matrix-algebra-E01079	368	0.858	$\begin{aligned} & A_{1,2}=\left[\begin{array}{cc} 1 & 0 \\ 2 & 1 \end{array}\right], \quad A_{1,3}=\left[\begin{array}{cc} 1 & 2 \\ 2 & -1 \end{array}\right], \quad A_{2,3}=\left[\begin{array}{cc} 0 & 2 \\ 1 & -1 \end{array}\right], \\ & B_{1,2}=\left[\begin{array}{cc} 3 & 1 \\ -2 & 3 \end{array}\right], \quad B_{1,3}=\left[\begin{array}{cc} 3 & 1 \\ -1 & 2 \end{array}\right], \quad \text { and } \quad B_{2,3}=\left[\begin{array}{cc} -2 & 3 \\ -1 & 2 \end{array}\right] . \end{aligned}$	$\begin{aligned} & A_{1,2}=\left[\begin{array}{cc} 1 & 0 \\ 2 & 1 \end{array}\right], \quad A_{1,3}=\left[\begin{array}{cc} 1 & 2 \\ 2 & -1 \end{array}\right], \quad A_{2,3}=\left[\begin{array}{cc} 0 & 2 \\ 1 & -1 \end{array}\right], \\ & B_{1,2}=\left[\begin{array}{cc} 3 & 1 \\ -2 & 3 \end{array}\right], \quad B_{1,3}=\left[\begin{array}{cc} 3 & 1 \\ -1 & 2 \end{array}\right], \quad \text { and } \quad B_{2,3}=\left[\begin{array}{cc} -2 & 3 \\ -1 & 2 \end{array}\right] . \end{aligned}$	$\begin{aligned} & A_{1,2}=\left[\begin{array}{cc} 1 & 0 \\ 2 & 1 \end{array}\right], \quad A_{1,3}=\left[\begin{array}{cc} 1 & 2 \\ 2 & -1 \end{array}\right], \quad A_{2,3}=\left[\begin{array}{cc} 0 & 2 \\ 1 & -1 \end{array}\right], \\ & B_{1,2}=\left[\begin{array}{cc} 3 & 1 \\ -2 & 3 \end{array}\right], \quad B_{1,3}=\left[\begin{array}{cc} 3 & 1 \\ -1 & 2 \end{array}\right], \quad \text { and } \quad B_{2,3}=\left[\begin{array}{cc} -2 & 3 \\ -1 & 2 \end{array}\right] . \end{aligned}$
johnston-linear-matrix-algebra-E00526 (2.B.10)	211	0.860	$\begin{aligned} & \operatorname{maximize}: x_1+2 x_2 \\ & \text { subject to: } x_1+x_2 \leq 2 \\ & \quad -x_1-x_2 \leq-1 \\ & \quad -2 x_1+x_2 \leq-2 \\ & \quad x_1, \quad x_2 \geq 0 \end{aligned}$	$\begin{aligned} & \operatorname{maximize}: x_1+2 x_2 \\ & \text { subject to: } x_1+x_2 \leq 2 \\ & \quad -x_1-x_2 \leq-1 \\ & \quad -2 x_1+x_2 \leq-2 \\ & \quad x_1, \quad x_2 \geq 0 \end{aligned}$	$\begin{aligned} & \operatorname{maximize}: x_1+2 x_2 \\ & \text { subject to: } x_1+x_2 \leq 2 \\ & \quad -x_1-x_2 \leq-1 \\ & \quad -2 x_1+x_2 \leq-2 \\ & \quad x_1, \quad x_2 \geq 0 \end{aligned}$
johnston-linear-matrix-algebra-E00761	286	0.862	$\operatorname{det}\left(\left[\begin{array}{cc} 0 & b \\ c & d \end{array}\right]\right)=-\operatorname{det}\left(\left[\begin{array}{cc} c & d \\ 0 & b \end{array}\right]\right)=-b c,$	$\operatorname{det}\left(\left[\begin{array}{cc} 0 & b \\ c & d \end{array}\right]\right)=-\operatorname{det}\left(\left[\begin{array}{cc} c & d \\ 0 & b \end{array}\right]\right)=-b c,$	$\operatorname{det}\left(\left[\begin{array}{cc} 0 & b \\ c & d \end{array}\right]\right)=-\operatorname{det}\left(\left[\begin{array}{cc} c & d \\ 0 & b \end{array}\right]\right)=-b c,$

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johnston-linear-matrix-algebra-EQ09967	341	0.863	$\begin{aligned} A^{\{r\}} &= \frac{1}{3} \left[\begin{array}{cc} 4 & 1 \\ 1 & 1 \end{array} \right] \left[\begin{array}{cc} 1^r & 0 \\ 0 & 4^r \end{array} \right] \left[\begin{array}{cc} 1 & -1 \\ -1 & 4 \end{array} \right] \\ &= \frac{1}{3} \left[\begin{array}{cc} 4 & 1 \\ 1 & 1 \end{array} \right] \left[\begin{array}{cc} 1 & -1 \\ -4^r & 4^{r+1} \end{array} \right] = \frac{1}{3} \left[\begin{array}{cc} 4-4^r & 4^{r+1}-4 \\ 1-4^r & 4^{r+1}-1 \end{array} \right]. \end{aligned}$	$A^r = \frac{1}{3} \begin{bmatrix} 4 & 1 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} 1^r & 0 \\ 0 & 4^r \end{bmatrix} \begin{bmatrix} 1 & -1 \\ -1 & 4 \end{bmatrix} = \frac{1}{3} \begin{bmatrix} 4 & 1 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} 1 & -1 \\ -4^r & 4^{r+1} \end{bmatrix} = \frac{1}{3} \begin{bmatrix} 4-4^r & 4^{r+1}-4 \\ 1-4^r & 4^{r+1}-1 \end{bmatrix}.$	$A^r = \frac{1}{3} \begin{bmatrix} 4 & 1 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} 1^r & 0 \\ 0 & 4^r \end{bmatrix} \begin{bmatrix} 1 & -1 \\ -1 & 4 \end{bmatrix} = \frac{1}{3} \begin{bmatrix} 4 & 1 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} 1 & -1 \\ -4^r & 4^{r+1} \end{bmatrix} = \frac{1}{3} \begin{bmatrix} 4-4^r & 4^{r+1}-4 \\ 1-4^r & 4^{r+1}-1 \end{bmatrix}.$
johnston-linear-matrix-algebra-EQ1165	400	0.863	$x_{\{n\}}=4 \ x_{\{n-1\}}-x_{\{n-2\}}-6 \ x_{\{n-3\}} \ \text{for all } n \geq 3,$	$x_n = 4x_{n-1} - x_{n-2} - 6x_{n-3} \text{ for all } n \geq 3,$	$x_n = 4x_{n-1} - x_{n-2} - 6x_{n-3} \text{ for all } n \geq 3,$
johnston-linear-matrix-algebra-EQ0025	31	0.863	$\begin{aligned} \ \mathbf{v}-\mathbf{w}\ ^2 &= (\mathbf{v}-\mathbf{w}) \cdot (\mathbf{v}-\mathbf{w}) \\ &= \mathbf{v} \cdot \mathbf{v} - \mathbf{v} \cdot \mathbf{w} - \mathbf{w} \cdot \mathbf{v} + \mathbf{w} \cdot \mathbf{w} = \ \mathbf{v}\ ^2 - 2(\mathbf{v} \cdot \mathbf{w}) + \ \mathbf{w}\ ^2. \end{aligned}$	$\ \mathbf{v}-\mathbf{w}\ ^2 = (\mathbf{v}-\mathbf{w}) \cdot (\mathbf{v}-\mathbf{w}) = \mathbf{v} \cdot \mathbf{v} - \mathbf{v} \cdot \mathbf{w} - \mathbf{w} \cdot \mathbf{v} + \mathbf{w} \cdot \mathbf{w} = \ \mathbf{v}\ ^2 - 2(\mathbf{v} \cdot \mathbf{w}) + \ \mathbf{w}\ ^2.$	$\ \mathbf{v}-\mathbf{w}\ ^2 = (\mathbf{v}-\mathbf{w}) \cdot (\mathbf{v}-\mathbf{w}) = \mathbf{v} \cdot \mathbf{v} - \mathbf{v} \cdot \mathbf{w} - \mathbf{w} \cdot \mathbf{v} + \mathbf{w} \cdot \mathbf{w} = \ \mathbf{v}\ ^2 - 2(\mathbf{v} \cdot \mathbf{w}) + \ \mathbf{w}\ ^2.$
johnston-linear-matrix-algebra-EQ1615	479	0.863	$\begin{aligned} D &= \left[\begin{array}{ccc} 3 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & 2 \end{array} \right], P = \left[\begin{array}{ccc} 1 & 0 & 3 \\ 0 & 1 & -2 \\ 0 & 0 & -3 \end{array} \right], \\ P^{-1} &= \frac{1}{3} \left[\begin{array}{ccc} 3 & 0 & 3 \\ 0 & 3 & -2 \\ 0 & 0 & -1 \end{array} \right]. \end{aligned}$	$D = \begin{bmatrix} 3 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & 2 \end{bmatrix}, P = \begin{bmatrix} 1 & 0 & 3 \\ 0 & 1 & -2 \\ 0 & 0 & -3 \end{bmatrix},$ $P^{-1} = \frac{1}{3} \begin{bmatrix} 3 & 0 & 3 \\ 0 & 3 & -2 \\ 0 & 0 & -1 \end{bmatrix}.$	$D = \begin{bmatrix} 3 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & 2 \end{bmatrix}, P = \begin{bmatrix} 1 & 0 & 3 \\ 0 & 1 & -2 \\ 0 & 0 & -3 \end{bmatrix},$ $P^{-1} = \frac{1}{3} \begin{bmatrix} 3 & 0 & 3 \\ 0 & 3 & -2 \\ 0 & 0 & -1 \end{bmatrix}.$
johnston-linear-matrix-algebra-EQ0485	196	0.864	$\begin{aligned} &v \ 3w+x+y+z \ \text{is even} \\ &-2v-w+2x+y \ \text{is odd} \\ &v+w+y+z \ \text{is odd} \\ &w-x+2y-z \ \text{is even, and} \\ &v+w-3x+y+2z \ \text{is even.} \end{aligned}$	$\begin{aligned} &v3w+x+y+z \text{ is even} \\ &-2v-w+2x+y \text{ is odd} \\ &v+w+y+z \text{ is odd} \\ &w-x+2y-z \text{ is even, and} \\ &v+w-3x+y+2z \text{ is even.} \end{aligned}$	$\begin{aligned} &v \ 3w+x+y+z \text{ is even} \\ &-2v-w+2x+y \text{ is odd} \\ &v+w+y+z \text{ is odd} \\ &w-x+2y-z \text{ is even, and} \\ &v+w-3x+y+2z \text{ is even.} \end{aligned}$

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johnston-linear-matrix-algebra-EQ1337	447	0.867	$(c \, \mathbf{v}) \cdot \mathbf{w} = \overline{(c \, \mathbf{v}) \cdot \mathbf{w}} = \overline{c(\mathbf{w} \cdot \mathbf{v})} = \overline{c} \overline{\mathbf{w} \cdot \mathbf{v}} = \overline{c} \mathbf{v} \cdot \overline{\mathbf{w}}.$	$(c\mathbf{v}) \cdot \mathbf{w} = \mathbf{w} \cdot (c\mathbf{v}) = \overline{c}(\mathbf{w} \cdot \mathbf{v}) = \overline{c}\mathbf{v} \cdot \overline{\mathbf{w}}.$	$(c\mathbf{v}) \cdot \mathbf{w} = \mathbf{w} \cdot (c\mathbf{v}) = \overline{c}(\mathbf{w} \cdot \mathbf{v}) = \overline{c}\mathbf{v} \cdot \overline{\mathbf{w}}.$
johnston-linear-matrix-algebra-EQ1505	467	0.871	$\begin{aligned} A \, \mathbf{v} &= A(c_1 \mathbf{v}_1 + c_2 \mathbf{v}_2 + \cdots + c_n \mathbf{v}_n) \\ &= c_1 A \mathbf{v}_1 + c_2 A \mathbf{v}_2 + \cdots + c_n A \mathbf{v}_n \\ &= c_1 B \mathbf{v}_1 + c_2 B \mathbf{v}_2 + \cdots + c_n B \mathbf{v}_n \\ &= B(c_1 \mathbf{v}_1 + c_2 \mathbf{v}_2 + \cdots + c_n \mathbf{v}_n) = B \mathbf{v} \end{aligned}$	$\begin{aligned} A\mathbf{v} &= A(c_1\mathbf{v}_1 + c_2\mathbf{v}_2 + \cdots + c_n\mathbf{v}_n) \\ &= c_1A\mathbf{v}_1 + c_2A\mathbf{v}_2 + \cdots + c_nA\mathbf{v}_n \\ &= c_1B\mathbf{v}_1 + c_2B\mathbf{v}_2 + \cdots + c_nB\mathbf{v}_n \\ &= B(c_1\mathbf{v}_1 + c_2\mathbf{v}_2 + \cdots + c_n\mathbf{v}_n) = B\mathbf{v} \end{aligned}$	$\begin{aligned} A\mathbf{v} &= A(c_1\mathbf{v}_1 + c_2\mathbf{v}_2 + \cdots + c_n\mathbf{v}_n) \\ &= c_1A\mathbf{v}_1 + c_2A\mathbf{v}_2 + \cdots + c_nA\mathbf{v}_n \\ &= c_1B\mathbf{v}_1 + c_2B\mathbf{v}_2 + \cdots + c_nB\mathbf{v}_n \\ &= B(c_1\mathbf{v}_1 + c_2\mathbf{v}_2 + \cdots + c_n\mathbf{v}_n) = B\mathbf{v}. \end{aligned}$
johnston-linear-matrix-algebra-EQ0468	191	0.871	$\begin{aligned} w + y + 2z &= 2 \\ x + y + z &= 1 \\ 2w + x + y + z &= 0 \\ w + y &= 1 \end{aligned}$	$\begin{aligned} w + y + 2z &= 2 \\ x + y + z &= 1 \\ 2w + x + y + z &= 0 \\ w + y &= 1 \end{aligned}$	$\begin{aligned} w + y + 2z &= 2 \\ x + y + z &= 1 \\ 2w + x + y + z &= 0 \\ w + y &= 1 \end{aligned}$
johnston-linear-matrix-algebra-EQ0487	197	0.872	$\begin{aligned} \operatorname{maximize} : & x_1 + 3x_2 \\ \text{subject to: } & 2x_1 + x_2 \leq 4 \\ & x_1 + 3x_2 \leq 6 \\ & x_1, x_2 \geq 0 \end{aligned}$	$\begin{aligned} \operatorname{maximize} : & x_1 + 3x_2 \\ \text{subject to: } & 2x_1 + x_2 \leq 4 \\ & x_1 + 3x_2 \leq 6 \\ & x_1, x_2 \geq 0 \end{aligned}$	$\begin{aligned} \operatorname{maximize} : & x_1 + 3x_2 \\ \text{subject to: } & 2x_1 + x_2 \leq 4 \\ & x_1 + 3x_2 \leq 6 \\ & x_1, x_2 \geq 0 \end{aligned}$
johnston-linear-matrix-algebra-EQ0300 (0)	121	0.873	$\begin{aligned} & \left[\begin{array}{cccc ccc} 1 & 0 & -2 & -1 & -1/2 & 1 & 0 \\ 0 & 1 & 2 & 0 & 1/2 & 0 & 0 \\ 0 & 0 & 0 & 4 & -1/2 & -3 & 1 \end{array} \right] \\ & \xrightarrow{\frac{1}{4}R_3} \left[\begin{array}{cccc ccc} 1 & 0 & -2 & -1 & -1/2 & 1 & 0 \\ 0 & 1 & 2 & 0 & 1/2 & 0 & 0 \\ 0 & 0 & 0 & 1 & -1/8 & -3/4 & 1/4 \end{array} \right] \\ & \xrightarrow{R_1+R_3} \left[\begin{array}{cccc ccc} 1 & 0 & -2 & 0 & -5/8 & 1/4 & 1/4 \\ 0 & 1 & 2 & 0 & 1/2 & 0 & 0 \\ 0 & 0 & 0 & 1 & -1/8 & -3/4 & 1/4 \end{array} \right] \end{aligned}$	$\begin{aligned} & \left[\begin{array}{cccc ccc} 1 & 0 & -2 & -1 & -1/2 & 1 & 0 \\ 0 & 1 & 2 & 0 & 1/2 & 0 & 0 \\ 0 & 0 & 0 & 4 & -1/2 & -3 & 1 \end{array} \right] \\ & \xrightarrow{\frac{1}{4}R_3} \left[\begin{array}{cccc ccc} 1 & 0 & -2 & -1 & -1/2 & 1 & 0 \\ 0 & 1 & 2 & 0 & 1/2 & 0 & 0 \\ 0 & 0 & 0 & 1 & -1/8 & -3/4 & 1/4 \end{array} \right] \\ & \xrightarrow{R_1+R_3} \left[\begin{array}{cccc ccc} 1 & 0 & -2 & 0 & -5/8 & 1/4 & 1/4 \\ 0 & 1 & 2 & 0 & 1/2 & 0 & 0 \\ 0 & 0 & 0 & 1 & -1/8 & -3/4 & 1/4 \end{array} \right] \end{aligned}$	$\begin{aligned} & \left[\begin{array}{cccc ccc} 1 & 0 & -2 & -1 & -1/2 & 1 & 0 \\ 0 & 1 & 2 & 0 & 1/2 & 0 & 0 \\ 0 & 0 & 0 & 4 & -1/2 & -3 & 1 \end{array} \right] \\ & \xrightarrow{\frac{1}{4}R_3} \left[\begin{array}{cccc ccc} 1 & 0 & -2 & -1 & -1/2 & 1 & 0 \\ 0 & 1 & 2 & 0 & 1/2 & 0 & 0 \\ 0 & 0 & 0 & 1 & -1/8 & -3/4 & 1/4 \end{array} \right] \\ & \xrightarrow{R_1+R_3} \left[\begin{array}{cccc ccc} 1 & 0 & -2 & 0 & -5/8 & 1/4 & 1/4 \\ 0 & 1 & 2 & 0 & 1/2 & 0 & 0 \\ 0 & 0 & 0 & 1 & -1/8 & -3/4 & 1/4 \end{array} \right] \end{aligned}$
johnston-linear-matrix-algebra-EQ0511	205	0.873	$\begin{aligned} \operatorname{maximize} : & \mathbf{c} \cdot \mathbf{x} \\ \text{subject to: } & A\mathbf{x} \leq \mathbf{b} \\ & \mathbf{x} \geq \mathbf{0} \end{aligned}$	$\begin{aligned} \operatorname{maximize} : & \mathbf{c} \cdot \mathbf{x} \\ \text{subject to: } & A\mathbf{x} \leq \mathbf{b} \\ & \mathbf{x} \geq \mathbf{0} \end{aligned}$	$\begin{aligned} \operatorname{maximize} : & \mathbf{c} \cdot \mathbf{x} \\ \text{subject to: } & A\mathbf{x} \leq \mathbf{b} \\ & \mathbf{x} \geq \mathbf{0} \end{aligned}$
johnston-linear-matrix-algebra-EQ0100	58	0.874	$[P_{\mathbf{u}}] = \mathbf{u} \mathbf{u}^T.$	$[P_{\mathbf{u}}] = \mathbf{u} \mathbf{u}^T.$	$[P_{\mathbf{u}}] = \mathbf{u} \mathbf{u}^T.$

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johnston-linear-matrix-algebra-EQ1317	443	0.875	$\begin{aligned} & p^{(k+1)}(x) = (m-k)(x-r)^{m-(k+1)}(c_k q(x) + s_k(x)) \\ & + (x-r)^{m-k}(c_k q'(x) + s'_k(x)) \\ & = (x-r)^{m-(k+1)}((m-k)c_k q(x) + (m-k)s_k(x) \\ & + (x-r)(c_k q'(x) + s'_k(x))), \end{aligned}$	$p^{(k+1)}(x) = (m-k)(x-r)^{m-(k+1)}(c_k q(x) + s_k(x)) + (x-r)^{m-k}(c_k q'(x) + s'_k(x)) = (x-r)^{m-(k+1)}((m-k)c_k q(x) + (m-k)s_k(x) + (x-r)(c_k q'(x) + s'_k(x)))$	$p^{(k+1)}(x) = (m-k)(x-r)^{m-(k+1)}(c_k q(x) + s_k(x)) + (x-r)^{m-k}(c_k q'(x) + s'_k(x)) = (x-r)^{m-(k+1)}((m-k)c_k q(x) + (m-k)s_k(x) + (x-r)(c_k q'(x) + s'_k(x))),$
johnston-linear-matrix-algebra-EQ1013	351	0.875	$P_0=0, \quad P_1=1, \quad P_{n+1}=2P_n+P_{n-1} \\ \quad \text{for all } n \geq 1.$	$P_0 = 0, \quad P_1 = 1, \quad P_{n+1} = 2P_n + P_{n-1} \quad \text{for all } n \geq 1.$	$P_0 = 0, \quad P_1 = 1, \quad P_{n+1} = 2P_n + P_{n-1} \quad \text{for all } n \geq 1.$
johnston-linear-matrix-algebra-EQ1656	483	0.877	$A = \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix} \quad \text{and} \quad B = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$	$A = \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix} \quad \text{and} \quad B = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$	$A = \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix} \quad \text{and} \quad B = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$
johnston-linear-matrix-algebra-EQ0103 (14)	59	0.877	$[P_u] = \mathbf{u} \mathbf{u}^T = \frac{1}{14} \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix} \begin{bmatrix} 1 & 2 & 3 \end{bmatrix} = \frac{1}{14} \begin{bmatrix} 1 & 2 & 3 \\ 2 & 4 & 6 \\ 3 & 6 & 9 \end{bmatrix}.$	$[P_u] = \mathbf{u} \mathbf{u}^T = \frac{1}{14} \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix} \begin{bmatrix} 1 & 2 & 3 \end{bmatrix} = \frac{1}{14} \begin{bmatrix} 1 & 2 & 3 \\ 2 & 4 & 6 \\ 3 & 6 & 9 \end{bmatrix}.$	$[P_u] = \mathbf{u} \mathbf{u}^T = \frac{1}{14} \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix} \begin{bmatrix} 1 & 2 & 3 \end{bmatrix} = \frac{1}{14} \begin{bmatrix} 1 & 2 & 3 \\ 2 & 4 & 6 \\ 3 & 6 & 9 \end{bmatrix}.$
johnston-linear-matrix-algebra-EQ0344	134	0.879	$\begin{bmatrix} a & 1 & 1 \\ 0 & a & 1 \\ 0 & 0 & a \end{bmatrix}.$	$\begin{bmatrix} a & 1 & 1 \\ 0 & a & 1 \\ 0 & 0 & a \end{bmatrix}.$	$\begin{bmatrix} a & 1 & 1 \\ 0 & a & 1 \\ 0 & 0 & a \end{bmatrix}.$
johnston-linear-matrix-algebra-EQ0735	278	0.881	$R_i + cR_j \begin{bmatrix} 1 & & & & \\ & \ddots & & & \\ & & 1 & & \\ & & \vdots & \ddots & \\ & & & & \ddots & \\ & & & & & 1 \end{bmatrix},$	$R_i + cR_j \begin{bmatrix} 1 & & & & \\ & \ddots & & & \\ & & 1 & & \\ & & \vdots & \ddots & \\ & & c & \cdots & 1 & \\ & & & & & \ddots & \\ & & & & & & 1 \end{bmatrix},$	$R_i + cR_j \begin{bmatrix} 1 & & & & \\ & \ddots & & & \\ & & 1 & & \\ & & \vdots & \ddots & \\ & & c & \cdots & 1 & \\ & & & & & \ddots & \\ & & & & & & 1 \end{bmatrix},$
johnston-linear-matrix-algebra-EQ0382	147	0.884	$\begin{aligned} & \begin{bmatrix} 1 & -2 & 1 \\ -1 & 1 & 1 \\ 0 & 2 & -4 \end{bmatrix} \xrightarrow{R_2+R_1} \begin{bmatrix} 1 & -2 & 1 \\ 0 & -1 & 2 \\ 0 & 2 & -4 \end{bmatrix} \\ & \xrightarrow{R_3+2R_2} \begin{bmatrix} 1 & -2 & 1 \\ 0 & -1 & 2 \\ 0 & 0 & 0 \end{bmatrix}. \end{aligned}$	$\begin{bmatrix} 1 & -2 & 1 \\ -1 & 1 & 1 \\ 0 & 2 & -4 \end{bmatrix} \xrightarrow{R_2+R_1} \begin{bmatrix} 1 & -2 & 1 \\ 0 & -1 & 2 \\ 0 & 2 & -4 \end{bmatrix} \xrightarrow{R_3+2R_2} \begin{bmatrix} 1 & -2 & 1 \\ 0 & -1 & 2 \\ 0 & 0 & 0 \end{bmatrix}.$	$\begin{bmatrix} 1 & -2 & 1 \\ -1 & 1 & 1 \\ 0 & 2 & -4 \end{bmatrix} \xrightarrow{R_2+R_1} \begin{bmatrix} 1 & -2 & 1 \\ 0 & -1 & 2 \\ 0 & 2 & -4 \end{bmatrix} \xrightarrow{R_3+2R_2} \begin{bmatrix} 1 & -2 & 1 \\ 0 & -1 & 2 \\ 0 & 0 & 0 \end{bmatrix}.$
johnston-linear-matrix-algebra-EQ0769	288	0.887	$2 \cdot 5 - 2 \cdot 4 = 10 - 8 = 2.$	$2 \cdot 5 - 2 \cdot 4 = 10 - 8 = 2.$	$2 \cdot 5 - 2 \cdot 4 = 10 - 8 = 2.$

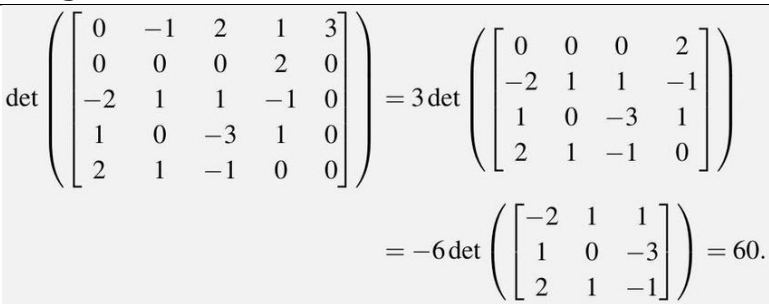
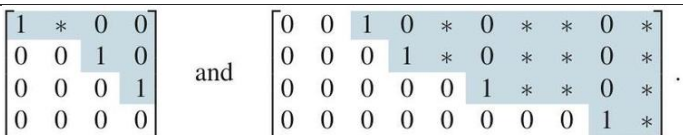
Identifier	Page	Conf.	LaTeX source	Rendered	Image
johnston-linear-matrix-algebra-EQ0767	287	0.888	$\operatorname{det}\left(\left[\begin{array}{ccc} a & b & c \\ 0 & e & f \\ 0 & h & i \end{array}\right]\right)=aei-\operatorname{det}\left(\left[\begin{array}{ccc} a & 0 & c \\ 0 & h & i \\ 0 & 0 & f \end{array}\right]\right)=aei-afh.$	$\det\left(\left[\begin{array}{ccc} a & b & c \\ 0 & e & f \\ 0 & h & i \end{array}\right]\right)=aei-\det\left(\left[\begin{array}{ccc} a & 0 & c \\ 0 & h & i \\ 0 & 0 & f \end{array}\right]\right)=aei-afh.$	$\det\left(\left[\begin{array}{ccc} a & b & c \\ 0 & e & f \\ 0 & h & i \end{array}\right]\right)=aei-\det\left(\left[\begin{array}{ccc} a & 0 & c \\ 0 & h & i \\ 0 & 0 & f \end{array}\right]\right)=aei-afh.$
johnston-linear-matrix-algebra-EQ1320	444	0.890	$\begin{aligned} p_0(x) &= c_0 + c_1x + c_2x^2 + \cdots + c_nx^n \\ p_1(x) &= c_1x + 2c_2x^2 + \cdots + nc_nx^n \\ p_2(x) &= c_1x + 2^2c_2x^2 + \cdots + n^2c_nx^n \\ &\vdots \\ p_m(x) &= c_1x + 2^mc_2x^2 + \cdots + n^mc_nx^n. \end{aligned}$	$\begin{aligned} p_0(x) &= c_0 + c_1x + c_2x^2 + \cdots + c_nx^n \\ p_1(x) &= c_1x + 2c_2x^2 + \cdots + nc_nx^n \\ p_2(x) &= c_1x + 2^2c_2x^2 + \cdots + n^2c_nx^n \\ &\vdots \\ p_m(x) &= c_1x + 2^mc_2x^2 + \cdots + n^mc_nx^n. \end{aligned}$	$\begin{aligned} p_0(x) &= c_0 + c_1x + c_2x^2 + \cdots + c_nx^n \\ p_1(x) &= c_1x + 2c_2x^2 + \cdots + nc_nx^n \\ p_2(x) &= c_1x + 2^2c_2x^2 + \cdots + n^2c_nx^n \\ &\vdots \\ p_m(x) &= c_1x + 2^mc_2x^2 + \cdots + n^mc_nx^n. \end{aligned}$
johnston-linear-matrix-algebra-EQ0509 (2.B.8)	204	0.891	$\begin{array}{l} \operatorname{maximize}: \quad \& \\ \& 2x_1 + x_2 - x_3 \\ \text{subject to: } \& \\ s_1 \quad -3x_1 - 2x_2 + 2x_3 = 1 \\ \& \\ s_2 \quad +x_1 - x_2 + x_3 = 2 \\ \& \\ s_3 - x_1 + 3x_2 - 3x_3 = 3 \\ \& \\ s_1, \quad s_2, \quad s_3, \quad x_1, \quad x_2, \quad x_3 \geq 0 \end{array}$	$\begin{array}{l} \text{maximize :} \\ \text{subject to:} \end{array} \quad \begin{array}{rcl} & 2x_1 + x_2 - x_3 & \\ s_1 & -3x_1 - 2x_2 + 2x_3 & = 1 \\ & +x_1 - x_2 + x_3 & = 2 \\ & s_3 - x_1 + 3x_2 - 3x_3 & = 3 \\ s_1, \quad s_2, \quad s_3, \quad x_1, \quad x_2, \quad x_3 & \geq 0 \end{array}$	$\begin{array}{l} \text{maximize:} \\ \text{subject to:} \end{array} \quad \begin{array}{rcl} & 2x_1 + x_2 - x_3 & \\ s_1 & -3x_1 - 2x_2 + 2x_3 & = 1 \\ & +x_1 - x_2 + x_3 & = 2 \\ & s_3 - x_1 + 3x_2 - 3x_3 & = 3 \\ s_1, \quad s_2, \quad s_3, \quad x_1, \quad x_2, \quad x_3 & \geq 0 \end{array}$
johnston-linear-matrix-algebra-EQ1064	364	0.894	$\begin{aligned} & \& P_{\sigma} \mathbf{e}_j = \mathbf{e}_{\sigma(j)} \text{ for all } j, \text{ so} \\ & (P_{\sigma} P_{\tau}) \mathbf{e}_j = P_{\sigma} (P_{\tau} \mathbf{e}_j) = P_{\sigma} \mathbf{e}_{\tau(j)} = \mathbf{e}_{\sigma(\tau(j))} = \mathbf{e}_{(\sigma \circ \tau)(j)}. \end{aligned}$	$P_{\sigma} \mathbf{e}_j = \mathbf{e}_{\sigma(j)} \text{ for all } j, \text{ so}$ $(P_{\sigma} P_{\tau}) \mathbf{e}_j = P_{\sigma} (P_{\tau} \mathbf{e}_j) = P_{\sigma} \mathbf{e}_{\tau(j)} = \mathbf{e}_{\sigma(\tau(j))} = \mathbf{e}_{(\sigma \circ \tau)(j)}.$	$P_{\sigma} \mathbf{e}_j = \mathbf{e}_{\sigma(j)} \text{ for all } j, \text{ so}$ $(P_{\sigma} P_{\tau}) \mathbf{e}_j = P_{\sigma} (P_{\tau} \mathbf{e}_j) = P_{\sigma} \mathbf{e}_{\tau(j)} = \mathbf{e}_{\sigma(\tau(j))} = \mathbf{e}_{(\sigma \circ \tau)(j)}.$
johnston-linear-matrix-algebra-EQ1155	397	0.894	$D = \left[\begin{array}{ccc} (1+i)/2 & 0 & 0 \\ 0 & (1-i)/2 & 0 \\ 0 & 0 & 2 \end{array}\right] \text{ and } P = \left[\begin{array}{ccc} 1 & 1 & 0 \\ i & -i & 1 \\ 1 & 1 & 2 \end{array}\right].$	$D = \left[\begin{array}{ccc} (1+i)/2 & 0 & 0 \\ 0 & (1-i)/2 & 0 \\ 0 & 0 & 2 \end{array}\right] \text{ and } P = \left[\begin{array}{ccc} 1 & 1 & 0 \\ i & -i & 1 \\ 1 & 1 & 2 \end{array}\right].$	$D = \left[\begin{array}{ccc} (1+i)/2 & 0 & 0 \\ 0 & (1-i)/2 & 0 \\ 0 & 0 & 2 \end{array}\right] \text{ and } P = \left[\begin{array}{ccc} 1 & 1 & 0 \\ i & -i & 1 \\ 1 & 1 & 2 \end{array}\right].$
johnston-linear-matrix-algebra-EQ1058	362	0.899	$\begin{aligned} & \& \sigma(\tau(1)) = \sigma(5) = 4 \quad \sigma(\tau(2)) = \sigma(1) = 3 \quad \sigma(\tau(3)) = \sigma(4) = 1 \\ & \& \sigma(\tau(4)) = \sigma(2) = 2 \quad \sigma(\tau(5)) = \sigma(3) = 5. \end{aligned}$	$\begin{aligned} \sigma(\tau(1)) &= \sigma(5) = 4 & \sigma(\tau(2)) &= \sigma(1) = 3 & \sigma(\tau(3)) &= \sigma(4) = 1 \\ \sigma(\tau(4)) &= \sigma(2) = 2 & \sigma(\tau(5)) &= \sigma(3) = 5. \end{aligned}$	$\begin{aligned} \sigma(\tau(1)) &= \sigma(5) = 4 & \sigma(\tau(2)) &= \sigma(1) = 3 & \sigma(\tau(3)) &= \sigma(4) = 1 \\ \sigma(\tau(4)) &= \sigma(2) = 2 & \sigma(\tau(5)) &= \sigma(3) = 5. \end{aligned}$
johnston-linear-matrix-algebra-EQ0802	297	0.899	$\operatorname{det}\left(\left[\begin{array}{ccc} a & b & c \\ d & e & f \\ g & h & i \end{array}\right]\right)=4.$	$\det\left(\left[\begin{array}{ccc} a & b & c \\ d & e & f \\ g & h & i \end{array}\right]\right)=4.$	$\det\left(\left[\begin{array}{ccc} a & b & c \\ d & e & f \\ g & h & i \end{array}\right]\right)=4.$

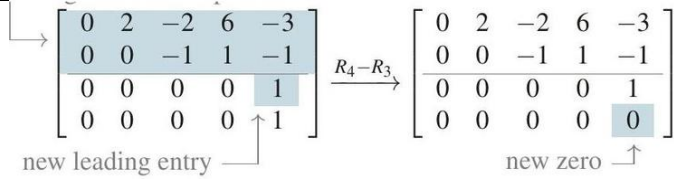
Identifier	Page	Conf.	LaTeX source	Rendered	Image
johnston-linear-matrix-algebra-EQ1344	449	0.899	$A^{\ell+1}=A\ A^{\ell}=\left[\begin{array}{ll} 1 & 1 \\ 0 & 1 \end{array}\right]^{\ell}\left[\begin{array}{ll} 1 & \ell \\ 0 & 1 \end{array}\right]=\left[\begin{array}{ll} 1 & \ell+1 \\ 0 & 1 \end{array}\right],$	$A^{\ell+1}=AA^{\ell}=\left[\begin{array}{ll} 1 & 1 \\ 0 & 1 \end{array}\right]\left[\begin{array}{ll} 1 & \ell \\ 0 & 1 \end{array}\right]=\left[\begin{array}{ll} 1 & \ell+1 \\ 0 & 1 \end{array}\right],$	
johnston-linear-matrix-algebra-EQ1166	400	0.902	$\begin{aligned} & x_3=4x_2-x_1-6x_0=4(8)-(-2)-6(3)=16, \\ & x_4=4x_3-x_2-6x_1=4(16)-8-6(-2)=68, \text{ and } \\ & x_5=4x_4-x_3-6x_2=4(68)-16-6(8)=208. \end{aligned}$	$\begin{aligned} x_3 &= 4x_2 - x_1 - 6x_0 = 4(8) - (-2) - 6(3) = 16, \\ x_4 &= 4x_3 - x_2 - 6x_1 = 4(16) - 8 - 6(-2) = 68, \text{ and } \\ x_5 &= 4x_4 - x_3 - 6x_2 = 4(68) - 16 - 6(8) = 208. \end{aligned}$	$\begin{aligned} x_3 &= 4x_2 - x_1 - 6x_0 = 4(8) - (-2) - 6(3) = 16, \\ x_4 &= 4x_3 - x_2 - 6x_1 = 4(16) - 8 - 6(-2) = 68, \text{ and } \\ x_5 &= 4x_4 - x_3 - 6x_2 = 4(68) - 16 - 6(8) = 208. \end{aligned}$
johnston-linear-matrix-algebra-EQ0127	67	0.904	$\left[F_{\mathbf{u}}\right]=2\left[\begin{array}{l} 3 / 5 \\ 4 / 5 \end{array}\right]\left[\begin{array}{ll} 3 / 5 & 4 / 5 \end{array}\right]-\left[\begin{array}{ll} 1 & 0 \\ 0 & 1 \end{array}\right]=\frac{1}{25}\left[\begin{array}{ll} -7 & 24 \\ 24 & 7 \end{array}\right].$	$\left[F_{\mathbf{u}}\right]=2\left[\begin{array}{l} 3/5 \\ 4/5 \end{array}\right]\left[\begin{array}{ll} 3/5 & 4/5 \end{array}\right]-\left[\begin{array}{ll} 1 & 0 \\ 0 & 1 \end{array}\right]=\frac{1}{25}\left[\begin{array}{ll} -7 & 24 \\ 24 & 7 \end{array}\right].$	
johnston-linear-matrix-algebra-EQ1014	351	0.905	$\left[\begin{array}{cc} F_{n+1} & F_n \\ F_n & F_{n-1} \end{array}\right]=\left[\begin{array}{ll} 1 & 1 \\ 1 & 0 \end{array}\right]^n \quad \text{for all } n \geq 1.$	$\left[\begin{array}{cc} F_{n+1} & F_n \\ F_n & F_{n-1} \end{array}\right]=\left[\begin{array}{ll} 1 & 1 \\ 1 & 0 \end{array}\right]^n \quad \text{for all } n \geq 1.$	$\left[\begin{array}{cc} F_{n+1} & F_n \\ F_n & F_{n-1} \end{array}\right]=\left[\begin{array}{ll} 1 & 1 \\ 1 & 0 \end{array}\right]^n \quad \text{for all } n \geq 1.$
johnston-linear-matrix-algebra-EQ0112	62	0.905	$\left[F_{\mathbf{u}}\right]=2\mathbf{u}\mathbf{u}^T-I=\frac{1}{2}\left[\begin{array}{c} 1 \\ \sqrt{3} \end{array}\right]\left[\begin{array}{cc} 1 & \sqrt{3} \end{array}\right]-\left[\begin{array}{ll} 1 & 0 \\ 0 & 1 \end{array}\right]=\frac{1}{2}\left[\begin{array}{ll} -1 & \sqrt{3} \\ \sqrt{3} & 1 \end{array}\right].$	$\left[F_{\mathbf{u}}\right]=2\mathbf{u}\mathbf{u}^T-I=\frac{1}{2}\left[\begin{array}{c} 1 \\ \sqrt{3} \end{array}\right]\left[\begin{array}{cc} 1 & \sqrt{3} \end{array}\right]-\left[\begin{array}{ll} 1 & 0 \\ 0 & 1 \end{array}\right]=\frac{1}{2}\left[\begin{array}{ll} -1 & \sqrt{3} \\ \sqrt{3} & 1 \end{array}\right].$	$\left[F_{\mathbf{u}}\right]=2\mathbf{u}\mathbf{u}^T-I=\frac{1}{2}\left[\begin{array}{c} 1 \\ \sqrt{3} \end{array}\right]\left[\begin{array}{cc} 1 & \sqrt{3} \end{array}\right]-\left[\begin{array}{ll} 1 & 0 \\ 0 & 1 \end{array}\right]=\frac{1}{2}\left[\begin{array}{ll} -1 & \sqrt{3} \\ \sqrt{3} & 1 \end{array}\right].$
johnston-linear-matrix-algebra-EQ0565	222	0.906	$\begin{array}{l} \text{maximize :} \\ \text{subject to: } x_1+3x_2\leq 3 \\ x_1, \quad x_2\geq 0 \end{array}$	$\begin{array}{l} \text{maximize :} \\ \text{subject to: } 2x_1-2x_2\geq 2 \\ x_1+3x_2\leq 3 \\ x_1, \quad x_2\geq 0 \end{array}$	$\begin{array}{l} \text{maximize:} \\ \text{subject to:} \end{array} \begin{array}{l} x_2 \\ 2x_1-2x_2\geq 2 \\ x_1+3x_2\leq 3 \\ x_1, \quad x_2\geq 0 \end{array}$
johnston-linear-matrix-algebra-EQ1105	380	0.908	$\begin{aligned} B &= A - \lambda_1 \mathbf{w}_1 \mathbf{w}_1^T \approx \left[\begin{array}{lll} 1 & 2 & 3 \\ 4 & 5 & 4 \\ 3 & 2 & 1 \end{array}\right] - 8.53 \left[\begin{array}{lll} 0.14 & 0.32 & 0.14 \\ 0.32 & 0.72 & 0.32 \\ 0.14 & 0.32 & 0.14 \end{array}\right] \\ &= \left[\begin{array}{lll} -0.21 & -0.73 & 1.80 \\ 1.27 & -1.14 & 1.31 \\ 1.80 & -0.69 & -0.18 \end{array}\right] \end{aligned}$	$\begin{aligned} B &= A - \lambda_1 \mathbf{w}_1 \mathbf{w}_1^T \approx \left[\begin{array}{lll} 1 & 2 & 3 \\ 4 & 5 & 4 \\ 3 & 2 & 1 \end{array}\right] - 8.53 \left[\begin{array}{lll} 0.14 & 0.32 & 0.14 \\ 0.32 & 0.72 & 0.32 \\ 0.14 & 0.32 & 0.14 \end{array}\right] \\ &= \left[\begin{array}{lll} -0.21 & -0.73 & 1.80 \\ 1.27 & -1.14 & 1.31 \\ 1.80 & -0.69 & -0.18 \end{array}\right] \end{aligned}$	$\begin{aligned} B &= A - \lambda_1 \mathbf{w}_1 \mathbf{w}_1^T \approx \left[\begin{array}{lll} 1 & 2 & 3 \\ 4 & 5 & 4 \\ 3 & 2 & 1 \end{array}\right] - 8.53 \left[\begin{array}{lll} 0.14 & 0.32 & 0.14 \\ 0.32 & 0.72 & 0.32 \\ 0.14 & 0.32 & 0.14 \end{array}\right] \\ &= \left[\begin{array}{lll} -0.21 & -0.73 & 1.80 \\ 1.27 & -1.14 & 1.31 \\ 1.80 & -0.69 & -0.18 \end{array}\right] \end{aligned}$

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johnston-linear-matrix-algebra- EQ1156 (2)	397	■ 0.909	$\begin{aligned} & B = \left[\begin{array}{c c} \frac{1}{\sqrt{2}} [R^{\pi/4}] & \mathbf{0} \\ \hline \mathbf{0}^T & 2 \end{array} \right] = \begin{bmatrix} 1/2 & -1/2 & 0 \\ 1/2 & 1/2 & 0 \\ 0 & 0 & 2 \end{bmatrix} \text{ and} \\ & Q = [\operatorname{Re}(\mathbf{v}) \mid -\operatorname{Im}(\mathbf{v}) \mid \mathbf{w}] = \begin{bmatrix} 1 & 0 & 0 \\ 0 & -1 & 1 \\ 1 & 0 & 2 \end{bmatrix}. \end{aligned}$	$B = \left[\begin{array}{c c} \frac{1}{\sqrt{2}} [R^{\pi/4}] & \mathbf{0} \\ \hline \mathbf{0}^T & 2 \end{array} \right] = \begin{bmatrix} 1/2 & -1/2 & 0 \\ 1/2 & 1/2 & 0 \\ 0 & 0 & 2 \end{bmatrix} \text{ and}$ $Q = [\operatorname{Re}(\mathbf{v}) \mid -\operatorname{Im}(\mathbf{v}) \mid \mathbf{w}] = \begin{bmatrix} 1 & 0 & 0 \\ 0 & -1 & 1 \\ 1 & 0 & 2 \end{bmatrix}.$	$B = \left[\begin{array}{c c} \frac{1}{\sqrt{2}} [R^{\pi/4}] & \mathbf{0} \\ \hline \mathbf{0}^T & 2 \end{array} \right] = \begin{bmatrix} 1/2 & -1/2 & 0 \\ 1/2 & 1/2 & 0 \\ 0 & 0 & 2 \end{bmatrix} \text{ and}$ $Q = [\operatorname{Re}(\mathbf{v}) \mid -\operatorname{Im}(\mathbf{v}) \mid \mathbf{w}] = \begin{bmatrix} 1 & 0 & 0 \\ 0 & -1 & 1 \\ 1 & 0 & 2 \end{bmatrix}.$
johnston-linear-matrix-algebra- EQ0349	135	■ 0.909	$\begin{aligned} & \left[\begin{array}{cc} A & B \\ C & D \end{array} \right]^{-1} = \\ & \left[\begin{array}{cc} A^{-1} + A^{-1}BS^{-1}CA^{-1} & -A^{-1}BS^{-1} \\ -S^{-1}CA^{-1} & S^{-1} \end{array} \right]. \end{aligned}$	$\left[\begin{array}{cc} A & B \\ C & D \end{array} \right]^{-1} =$ $\left[\begin{array}{cc} A^{-1} + A^{-1}BS^{-1}CA^{-1} & -A^{-1}BS^{-1} \\ -S^{-1}CA^{-1} & S^{-1} \end{array} \right].$	$\left[\begin{array}{cc} A & B \\ C & D \end{array} \right]^{-1} =$ $\left[\begin{array}{cc} A^{-1} + A^{-1}BS^{-1}CA^{-1} & -A^{-1}BS^{-1} \\ -S^{-1}CA^{-1} & S^{-1} \end{array} \right].$
johnston-linear-matrix-algebra- EQ0326	129	■ 0.909	$\begin{aligned} & \left[\begin{array}{ccc} 1 & 2 & -2 \\ 0 & 1 & -2 \\ 1 & 1 & 1 \end{array} \right] \xrightarrow{R_3 - R_1} \left[\begin{array}{ccc} 1 & 2 & -2 \\ 0 & 1 & -2 \\ 0 & -1 & 3 \end{array} \right] \\ & \xrightarrow{\substack{R_1 - 2R_2 \\ R_3 + R_2}} \left[\begin{array}{ccc} 1 & 0 & 2 \\ 0 & 1 & -2 \\ 0 & 0 & 1 \end{array} \right] \xrightarrow{\substack{R_1 - 2R_3 \\ R_2 + 2R_3}} \left[\begin{array}{ccc} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{array} \right]. \end{aligned}$	$\left[\begin{array}{ccc} 1 & 2 & -2 \\ 0 & 1 & -2 \\ 1 & 1 & 1 \end{array} \right] \xrightarrow{R_3 - R_1} \left[\begin{array}{ccc} 1 & 2 & -2 \\ 0 & 1 & -2 \\ 0 & -1 & 3 \end{array} \right]$ $\xrightarrow{\substack{R_1 - 2R_2 \\ R_3 + R_2}} \left[\begin{array}{ccc} 1 & 0 & 2 \\ 0 & 1 & -2 \\ 0 & 0 & 1 \end{array} \right] \xrightarrow{\substack{R_1 - 2R_3 \\ R_2 + 2R_3}} \left[\begin{array}{ccc} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{array} \right].$	$\left[\begin{array}{ccc} 1 & 2 & -2 \\ 0 & 1 & -2 \\ 1 & 1 & 1 \end{array} \right] \xrightarrow{R_3 - R_1} \left[\begin{array}{ccc} 1 & 2 & -2 \\ 0 & 1 & -2 \\ 0 & -1 & 3 \end{array} \right]$ $\xrightarrow{\substack{R_1 - 2R_2 \\ R_3 + R_2}} \left[\begin{array}{ccc} 1 & 0 & 2 \\ 0 & 1 & -2 \\ 0 & 0 & 1 \end{array} \right] \xrightarrow{\substack{R_1 - 2R_3 \\ R_2 + 2R_3}} \left[\begin{array}{ccc} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{array} \right].$
johnston-linear-matrix-algebra- EQ1485	464	■ 0.910	$A = \left[\begin{array}{cc} 0 & 1 \\ 0 & 0 \end{array} \right] \text{ and } B = \left[\begin{array}{cc} 0 & 0 \\ 1 & 0 \end{array} \right].$	$A = \left[\begin{array}{cc} 0 & 1 \\ 0 & 0 \end{array} \right] \text{ and } B = \left[\begin{array}{cc} 0 & 0 \\ 1 & 0 \end{array} \right].$	$A = \left[\begin{array}{cc} 0 & 1 \\ 0 & 0 \end{array} \right] \text{ and } B = \left[\begin{array}{cc} 0 & 0 \\ 1 & 0 \end{array} \right].$

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johnston-linear-matrix-algebra- EQ1325	445	0.912	$\begin{aligned} \lim_{x \rightarrow \infty} \frac{q(x+1)}{q(x)} &= \lim_{x \rightarrow \infty} \frac{c_k(x+1)^k + \cdots + c_1(x+1) + c_0}{c_kx^k + \cdots + c_1x + c_0} \\ &\stackrel{\text{L'Hôp}}{=} \lim_{x \rightarrow \infty} \frac{c_kk(x+1)^{k-1} + \cdots + c_1}{c_kkx^{k-1} + \cdots + c_1} \\ &\vdots \\ &\stackrel{\text{L'Hôp}}{=} \lim_{x \rightarrow \infty} \frac{c_kk!(x+1) + c_{k-1}(k-1)!}{c_kk!x + c_{k-1}(k-1)!} \\ &\stackrel{\text{L'Hôp}}{=} \lim_{x \rightarrow \infty} \frac{c_kk!}{c_kk!} \\ &= 1, \end{aligned}$	$\lim_{x \rightarrow \infty} \frac{q(x+1)}{q(x)} = \lim_{x \rightarrow \infty} \frac{c_k(x+1)^k + \cdots + c_1(x+1) + c_0}{c_kx^k + \cdots + c_1x + c_0}$ $\stackrel{\text{L'Hôp}}{=} \lim_{x \rightarrow \infty} \frac{c_kk(x+1)^{k-1} + \cdots + c_1}{c_kkx^{k-1} + \cdots + c_1}$ $\vdots$ $\stackrel{\text{L'Hôp}}{=} \lim_{x \rightarrow \infty} \frac{c_kk!(x+1) + c_{k-1}(k-1)!}{c_kk!x + c_{k-1}(k-1)!}$ $\stackrel{\text{L'Hôp}}{=} \lim_{x \rightarrow \infty} \frac{c_kk!}{c_kk!}$ $= 1,$	
johnston-linear-matrix-algebra- EQ1563	473	0.913	$\begin{aligned} [T]_B[\mathbf{v}]_B &= [[T(\mathbf{v}_1)]_B \mid \cdots \mid [T(\mathbf{v}_n)]_B] \begin{bmatrix} c_1 \\ \vdots \\ c_n \end{bmatrix} \\ &= c_1[T(\mathbf{v}_1)]_B + \cdots + c_n[T(\mathbf{v}_n)]_B \\ &= [T(c_1\mathbf{v}_1 + \cdots + c_n\mathbf{v}_n)]_B \\ &= [T(\mathbf{v})]_B. \end{aligned}$	$[T]_B[\mathbf{v}]_B = [[T(\mathbf{v}_1)]_B \mid \cdots \mid [T(\mathbf{v}_n)]_B] \begin{bmatrix} c_1 \\ \vdots \\ c_n \end{bmatrix}$ $= c_1[T(\mathbf{v}_1)]_B + \cdots + c_n[T(\mathbf{v}_n)]_B$ $= [T(c_1\mathbf{v}_1 + \cdots + c_n\mathbf{v}_n)]_B$ $= [T(\mathbf{v})]_B.$	$[T]_B[\mathbf{v}]_B = [[T(\mathbf{v}_1)]_B \mid \cdots \mid [T(\mathbf{v}_n)]_B] \begin{bmatrix} c_1 \\ \vdots \\ c_n \end{bmatrix}$ $= c_1[T(\mathbf{v}_1)]_B + \cdots + c_n[T(\mathbf{v}_n)]_B$ $= [T(c_1\mathbf{v}_1 + \cdots + c_n\mathbf{v}_n)]_B$ $= [T(\mathbf{v})]_B.$
johnston-linear-matrix-algebra- EQ1070	365	0.914	$(25134) \rightarrow (15234) \rightarrow (12534) \rightarrow (12354) \rightarrow (12345).$	$(25134) \rightarrow (15234) \rightarrow (12534) \rightarrow (12354) \rightarrow (12345).$	$(2 \ 5 \ 1 \ 3 \ 4) \rightarrow (1 \ 5 \ 2 \ 3 \ 4) \rightarrow (1 \ 2 \ 5 \ 3 \ 4) \rightarrow (1 \ 2 \ 3 \ 5 \ 4) \rightarrow (1 \ 2 \ 3 \ 4 \ 5).$
johnston-linear-matrix-algebra- EQ0461	184	0.915	$\begin{aligned} \mathbf{s} &= (0,0,0,0,0,0,0,0,0,0,0,0,0,0,0) \\ \mathbf{e} &= (1,1,1,1,1,1,1,1,1,1,1,1,1,1,1) \end{aligned}$	$\mathbf{v}_s = (0,0,0,0,0,0,0,0,0,0,0,0,0,0,0) \quad \text{and}$ $\mathbf{v}_e = (1,1,1,1,1,1,1,1,1,1,1,1,1,1,1).$	$\mathbf{v}_s = (0,0,0,0,0,0,0,0,0,0,0,0,0,0,0) \quad \text{and}$ $\mathbf{v}_e = (1,1,1,1,1,1,1,1,1,1,1,1,1,1,1).$
johnston-linear-matrix-algebra- EQ1361	450	0.915	$A B = \begin{bmatrix} \mathbf{a}_1^T \\ \mathbf{a}_2^T \\ \vdots \\ \mathbf{a}_m^T \end{bmatrix} B = \begin{bmatrix} \mathbf{a}_1^T B \\ \mathbf{a}_2^T B \\ \vdots \\ \mathbf{a}_m^T B \end{bmatrix},$	$AB = \begin{bmatrix} \mathbf{a}_1^T \\ \mathbf{a}_2^T \\ \vdots \\ \mathbf{a}_m^T \end{bmatrix} B = \begin{bmatrix} \mathbf{a}_1^T B \\ \mathbf{a}_2^T B \\ \vdots \\ \mathbf{a}_m^T B \end{bmatrix},$	

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johnston-linear-matrix-algebra-EQ1614	479	0.916	$\begin{aligned} & D=\left[\begin{array}{ccc} 2 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & -1 \end{array}\right], P=\left[\begin{array}{ccc} 1 & 0 & 0 \\ 0 & 1 & 1 \\ 0 & 1 & -1 \end{array}\right], \\ & P^{-1}=\frac{1}{2}\left[\begin{array}{ccc} 2 & 0 & 0 \\ 0 & 1 & 1 \\ 0 & 1 & -1 \end{array}\right]. \end{aligned}$	$D=\left[\begin{array}{ccc} 2 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & -1 \end{array}\right], P=\left[\begin{array}{ccc} 1 & 0 & 0 \\ 0 & 1 & 1 \\ 0 & 1 & -1 \end{array}\right],$ $P^{-1}=\frac{1}{2}\left[\begin{array}{ccc} 2 & 0 & 0 \\ 0 & 1 & 1 \\ 0 & 1 & -1 \end{array}\right].$	$D=\left[\begin{array}{ccc} 2 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & -1 \end{array}\right], P=\left[\begin{array}{ccc} 1 & 0 & 0 \\ 0 & 1 & 1 \\ 0 & 1 & -1 \end{array}\right],$ $P^{-1}=\frac{1}{2}\left[\begin{array}{ccc} 2 & 0 & 0 \\ 0 & 1 & 1 \\ 0 & 1 & -1 \end{array}\right].$
johnston-linear-matrix-algebra-EQ1626	480	0.917	$\begin{aligned} B &= P D^{1/3} P^{-1} \\ &= \frac{1}{3} \left[\begin{array}{cc} 1 & 2 \\ 1 & -1 \end{array}\right] \left[\begin{array}{cc} -1 & 0 \\ 0 & 2 \end{array}\right] \left[\begin{array}{cc} 1 & 2 \\ 1 & -1 \end{array}\right] \\ &= \left[\begin{array}{cc} 1 & -2 \\ -1 & 0 \end{array}\right]. \end{aligned}$	$B = PD^{1/3}P^{-1}$ $= \frac{1}{3} \begin{bmatrix} 1 & 2 \\ 1 & -1 \end{bmatrix} \begin{bmatrix} -1 & 0 \\ 0 & 2 \end{bmatrix} \begin{bmatrix} 1 & 2 \\ 1 & -1 \end{bmatrix}$ $= \begin{bmatrix} 1 & -2 \\ -1 & 0 \end{bmatrix}.$	$B = PD^{1/3}P^{-1}$ $= \frac{1}{3} \begin{bmatrix} 1 & 2 \\ 1 & -1 \end{bmatrix} \begin{bmatrix} -1 & 0 \\ 0 & 2 \end{bmatrix} \begin{bmatrix} 1 & 2 \\ 1 & -1 \end{bmatrix}$ $= \begin{bmatrix} 1 & -2 \\ -1 & 0 \end{bmatrix}.$
johnston-linear-matrix-algebra-EQ1359	450	0.917	$\begin{aligned} [c(A\ B)]_{i,j} &= c \sum_{k=1}^n a_{i,k} b_{k,j} \\ &= \sum_{k=1}^n (ca_{i,k}) b_{k,j} = [(cA)B]_{i,j}. \end{aligned}$	$[c(AB)]_{i,j} = c \sum_{k=1}^n a_{i,k} b_{k,j}$ $= \sum_{k=1}^n (ca_{i,k}) b_{k,j} = [(cA)B]_{i,j}$	$[c(AB)]_{i,j} = c \sum_{k=1}^n a_{i,k} b_{k,j}$ $= \sum_{k=1}^n (ca_{i,k}) b_{k,j} = [(cA)B]_{i,j}.$
johnston-linear-matrix-algebra-EQ1509	468	0.917	$\left[\begin{array}{lllll l} 1 & 1 & 1 & 1 & 1 & 0 \\ 0 & 1 & 0 & 1 & 0 & 1 \\ 1 & 1 & 0 & 1 & 1 & 1 \\ 0 & 1 & 1 & 0 & 1 & 0 \\ 1 & 1 & 1 & 1 & 0 & 0 \end{array}\right].$	$\left[\begin{array}{lllll l} 1 & 1 & 1 & 1 & 1 & 0 \\ 0 & 1 & 0 & 1 & 0 & 1 \\ 1 & 1 & 0 & 1 & 1 & 1 \\ 0 & 1 & 1 & 0 & 1 & 0 \\ 1 & 1 & 1 & 1 & 0 & 0 \end{array}\right].$	$\left[\begin{array}{lllll l} 1 & 1 & 1 & 1 & 1 & 0 \\ 0 & 1 & 0 & 1 & 0 & 1 \\ 1 & 1 & 0 & 1 & 1 & 1 \\ 0 & 1 & 1 & 0 & 1 & 0 \\ 1 & 1 & 1 & 1 & 0 & 0 \end{array}\right].$
johnston-linear-matrix-algebra-EQ0360	141	0.918	$\operatorname{span}(B) \quad \text{or} \quad \operatorname{span}(\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_k).$	$\operatorname{span}(B) \quad \text{or} \quad \operatorname{span}(\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_k).$	$\operatorname{span}(B) \quad \text{or} \quad \operatorname{span}(\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_k).$
johnston-linear-matrix-algebra-EQ0490 (2.B.2)	198	0.919	$\begin{aligned} & \operatorname{maximize}: \mathbf{c} \cdot \mathbf{x} \\ & \text{subject to: } \mathbf{A} \mathbf{x} \leq \mathbf{b} \\ & \mathbf{x} \geq \mathbf{0} \end{aligned}$	$\begin{aligned} & \operatorname{maximize}: \mathbf{c} \cdot \mathbf{x} \\ & \text{subject to: } \mathbf{A} \mathbf{x} \leq \mathbf{b} \\ & \mathbf{x} \geq \mathbf{0} \end{aligned}$	$\begin{aligned} & \operatorname{maximize}: \mathbf{c} \cdot \mathbf{x} \\ & \text{subject to: } \mathbf{A} \mathbf{x} \leq \mathbf{b} \\ & \mathbf{x} \geq \mathbf{0} \end{aligned}$
johnston-linear-matrix-algebra-EQ0492	199	0.919	$\begin{aligned} & \operatorname{maximize}: x_1 + 2x_2 \\ & \text{subject to: } x_1 + x_2 \leq 3 \\ & -x_1 + x_2 \leq 1 \\ & x_1, x_2 \geq 0 \end{aligned}$	$\begin{aligned} & \operatorname{maximize}: x_1 + 2x_2 \\ & \text{subject to: } x_1 + x_2 \leq 3 \\ & -x_1 + x_2 \leq 1 \\ & x_1, x_2 \geq 0 \end{aligned}$	$\begin{aligned} & \operatorname{maximize}: x_1 + 2x_2 \\ & \text{subject to: } x_1 + x_2 \leq 3 \\ & -x_1 + x_2 \leq 1 \\ & x_1, x_2 \geq 0 \end{aligned}$

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johnston-linear-matrix-algebra-EQ0798	295	0.919	$\begin{aligned} & \operatorname{det}\left(\left[\begin{array}{ccccc} 0 & -1 & 2 & 1 & 3 \\ 0 & 0 & 0 & 2 & 0 \\ -2 & 1 & 1 & -1 & 0 \\ 1 & 0 & -3 & 1 & 0 \\ 2 & 1 & -1 & 0 & 0 \end{array}\right]\right) = 3 \operatorname{det}\left(\left[\begin{array}{cccc} 0 & 0 & 0 & 2 \\ -2 & 1 & 1 & -1 \\ 1 & 0 & -3 & 1 \\ 2 & 1 & -1 & 0 \end{array}\right]\right) \\ & = -6 \operatorname{det}\left(\left[\begin{array}{ccc} -2 & 1 & 1 \\ 1 & 0 & -3 \\ 2 & 1 & -1 \end{array}\right]\right) = 60. \end{aligned}$		
johnston-linear-matrix-algebra-EQ1370	451	0.919	$[S \circ T] = [S][T] = \left[\begin{array}{cc} 1 & 1 \\ 3 & 0 \\ 2 & 3 \end{array}\right].$	$[S \circ T] = [S][T] = \left[\begin{array}{cc} 1 & 1 \\ 3 & 0 \\ 2 & 3 \end{array}\right].$	$[S \circ T] = [S][T] = \left[\begin{array}{cc} 1 & 1 \\ 3 & 0 \\ 2 & 3 \end{array}\right].$
johnston-linear-matrix-algebra-EQ0215	101	0.921	$\left[\begin{array}{cccc} 1 & * & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \end{array}\right] \quad \text{and} \quad \left[\begin{array}{ccccccccc} 0 & 0 & 1 & 0 & * & 0 & * & * & 0 & * \\ 0 & 0 & 0 & 1 & * & 0 & * & * & 0 & * \\ 0 & 0 & 0 & 0 & 0 & 1 & * & * & 0 & * \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & * \end{array}\right].$	$\left[\begin{array}{cccc} 1 & * & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \end{array}\right] \quad \text{and} \quad \left[\begin{array}{ccccccccc} 0 & 0 & 1 & 0 & * & 0 & * & * & 0 & * \\ 0 & 0 & 0 & 1 & * & 0 & * & * & 0 & * \\ 0 & 0 & 0 & 0 & 0 & 1 & * & * & 0 & * \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & * \end{array}\right].$	
johnston-linear-matrix-algebra-EQ0104	59	0.922	$\left[P_{\mathbf{u}}\right] \mathbf{w} = \frac{1}{14} \left[\begin{array}{ccc} 1 & 2 & 3 \\ 2 & 4 & 6 \\ 3 & 6 & 9 \end{array}\right] \left[\begin{array}{c} 1 \\ 2 \\ 3 \end{array}\right] = \left[\begin{array}{c} 1 \\ 2 \\ 3 \end{array}\right] = \mathbf{w}.$	$[P_{\mathbf{u}}] \mathbf{w} = \frac{1}{14} \left[\begin{array}{ccc} 1 & 2 & 3 \\ 2 & 4 & 6 \\ 3 & 6 & 9 \end{array}\right] \left[\begin{array}{c} 1 \\ 2 \\ 3 \end{array}\right] = \left[\begin{array}{c} 1 \\ 2 \\ 3 \end{array}\right] = \mathbf{w}.$	$[P_{\mathbf{u}}] \mathbf{w} = \frac{1}{14} \left[\begin{array}{ccc} 1 & 2 & 3 \\ 2 & 4 & 6 \\ 3 & 6 & 9 \end{array}\right] \left[\begin{array}{c} 1 \\ 2 \\ 3 \end{array}\right] = \left[\begin{array}{c} 1 \\ 2 \\ 3 \end{array}\right] = \mathbf{w}.$
johnston-linear-matrix-algebra-EQ0226	104	0.923	$\begin{aligned} & x+2z=0 \\ & y-z=0 \\ & 0=1 \\ & 0=0 \end{aligned}$	$\begin{aligned} & x+2z=0 \\ & y-z=0 \\ & 0=1 \\ & 0=0 \end{aligned}$	$\begin{aligned} & x+2z=0 \\ & y-z=0 \\ & 0=1 \\ & 0=0 \end{aligned}$
johnston-linear-matrix-algebra-EQ0728	273	0.924	$\begin{aligned} & \operatorname{det}\left(\left[\mathbf{a}_1 \mid \cdots \mid \mathbf{v}+\mathbf{w} \mid \cdots \mid \mathbf{a}_n\right]\right) \\ & = \operatorname{det}\left(\left[\mathbf{a}_1 \mid \cdots \mid \mathbf{v} \mid \cdots \mid \mathbf{a}_n\right]\right) + \operatorname{det}\left(\left[\mathbf{a}_1 \mid \cdots \mid \mathbf{w} \mid \cdots \mid \mathbf{a}_n\right]\right). \end{aligned}$	$\begin{aligned} & \operatorname{det}\left(\left[\mathbf{a}_1 \mid \cdots \mid \mathbf{v}+\mathbf{w} \mid \cdots \mid \mathbf{a}_n\right]\right) \\ & = \operatorname{det}\left(\left[\mathbf{a}_1 \mid \cdots \mid \mathbf{v} \mid \cdots \mid \mathbf{a}_n\right]\right) + \operatorname{det}\left(\left[\mathbf{a}_1 \mid \cdots \mid \mathbf{w} \mid \cdots \mid \mathbf{a}_n\right]\right). \end{aligned}$	$\begin{aligned} & \operatorname{det}\left(\left[\mathbf{a}_1 \mid \cdots \mid \mathbf{v}+\mathbf{w} \mid \cdots \mid \mathbf{a}_n\right]\right) \\ & = \operatorname{det}\left(\left[\mathbf{a}_1 \mid \cdots \mid \mathbf{v} \mid \cdots \mid \mathbf{a}_n\right]\right) + \operatorname{det}\left(\left[\mathbf{a}_1 \mid \cdots \mid \mathbf{w} \mid \cdots \mid \mathbf{a}_n\right]\right). \end{aligned}$

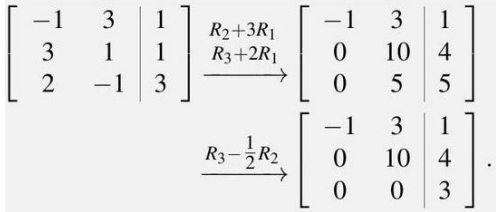
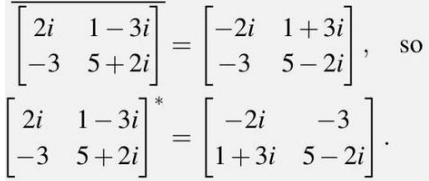
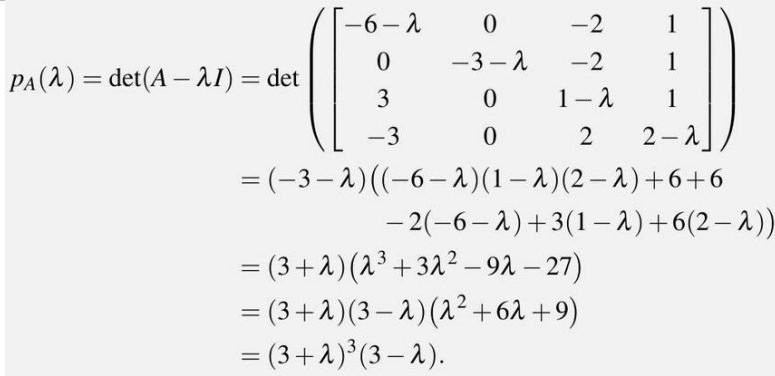
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johnston-linear-matrix-algebra-EQ1607	478	0.925	$\begin{aligned} B \, \mathbf{v}_{_1} &= \left(A - \lambda_1 \mathbf{v}_{_1} \mathbf{v}_{_1}^* \right) \mathbf{v}_{_1} \\ &= A \mathbf{v}_{_1} - \lambda_1 \mathbf{v}_{_1} = \lambda_1 \mathbf{v}_{_1} - \lambda_1 \mathbf{v}_{_1} = \mathbf{0}. \end{aligned}$	$B \mathbf{v}_1 = (A - \lambda_1 \mathbf{v}_1 \mathbf{v}_1^*) \mathbf{v}_1$ $= A \mathbf{v}_1 - \lambda_1 \mathbf{v}_1 = \lambda_1 \mathbf{v}_1 - \lambda_1 \mathbf{v}_1 = \mathbf{0}.$	
johnston-linear-matrix-algebra-EQ1270	426	0.926	$\begin{array}{r} 3 \, x^{\{2\}} + 6 \, x \, \, \, x - 2 \, \, \, \text{longdiv} \, \, \, 3 \, x^{\{3\}} + 0 \, x^{\{2\}} - 14 \, x + 4 \, \, \, \frac{3 \, x^{\{3\}} - 6 \, x^2}{6 \, x^2} - 14 \, x + 4 \\ x^{\{2\}} \{ 6 \, x^{\{2\}} - 14 \, x + 4 \, \, \, \frac{6 \, x^{\{2\}} - 12 \, x}{-2 \, x} + 4 \end{array}$	$x-2 \overline{) 3x^3 + 0x^2 - 14x + 4}$ $\underline{3x^3 - 6x^2} $ $6x^2 - 14x + 4$ $\underline{6x^2 - 12x} $ $-2x + 4$	$x-2 \overline{) 3x^3 + 0x^2 - 14x + 4}$ $\underline{3x^3 - 6x^2} $ $6x^2 - 14x + 4$ $\underline{6x^2 - 12x} $ $-2x + 4$
johnston-linear-matrix-algebra-EQ1253	420	0.926	$\begin{aligned} e^{i \, \theta} &= 1 + i \, \theta - \frac{\theta^2}{2} - i \, \frac{\theta^3}{3!} + \frac{\theta^4}{4!} + i \, \frac{\theta^5}{5!} - \cdots \\ &= \left(1 - \frac{\theta^2}{2} + \frac{\theta^4}{4!} - \cdots \right) + i \left(\theta - \frac{\theta^3}{3!} + \frac{\theta^5}{5!} + \cdots \right), \end{aligned}$	$e^{i \theta} = 1 + i \theta - \frac{\theta^2}{2} - i \frac{\theta^3}{3!} + \frac{\theta^4}{4!} + i \frac{\theta^5}{5!} - \cdots$ $= \left(1 - \frac{\theta^2}{2} + \frac{\theta^4}{4!} - \cdots \right) + i \left(\theta - \frac{\theta^3}{3!} + \frac{\theta^5}{5!} + \cdots \right),$	$e^{i \theta} = 1 + i \theta - \frac{\theta^2}{2} - i \frac{\theta^3}{3!} + \frac{\theta^4}{4!} + i \frac{\theta^5}{5!} - \cdots$ $= \left(1 - \frac{\theta^2}{2} + \frac{\theta^4}{4!} - \cdots \right) + i \left(\theta - \frac{\theta^3}{3!} + \frac{\theta^5}{5!} + \cdots \right),$
johnston-linear-matrix-algebra-EQ1616	479	0.926	$\sum_{j=1}^n a_{i,j} \, v_{_j} = \lambda v_{_i} = \lambda v_{_i} \, .$	$\sum_{j=1}^n a_{i,j} v_j = \lambda v_i = \lambda.$	$\sum_{j=1}^n a_{i,j} v_j = \lambda v_i = \lambda.$
johnston-linear-matrix-algebra-EQ0220	103	0.927	$\begin{gathered} \left\{ \left[\begin{array}{ccccc} 0 & 2 & -2 & 6 & -3 \\ 0 & 0 & -1 & 1 & -1 \\ 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & \uparrow \end{array} \right] \xrightarrow{R_4 - R_3} \left[\begin{array}{ccccc} 0 & 2 & -2 & 6 & -3 \\ 0 & 0 & -1 & 1 & -1 \\ 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 \end{array} \right] \right. \\ \left. \xrightarrow{\text{new leading entry}} \left[\begin{array}{ccccc} 0 & 2 & -2 & 6 & -3 \\ 0 & 0 & -1 & 1 & -1 \\ 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 \end{array} \right] \right\} \end{gathered}$		
johnston-linear-matrix-algebra-EQ0244	111	0.928	$0.035 \, x + 0.01 \, y = 10 \, .$	$0.035x + 0.01y = 10.$	$0.035x + 0.01y = 10.$
johnston-linear-matrix-algebra-EQ1578	475	0.929	$A = \left[\begin{array}{ll} 1 & 0 \\ 0 & 1 \\ 0 & 0 \end{array} \right] \, \text{and} \, B = \left[\begin{array}{lll} 1 & 0 & 0 \\ 0 & 1 & 0 \end{array} \right] .$	$A = \begin{bmatrix} 1 & 0 \\ 0 & 1 \\ 0 & 0 \end{bmatrix} \, \text{and} \, B = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \end{bmatrix} .$	$A = \begin{bmatrix} 1 & 0 \\ 0 & 1 \\ 0 & 0 \end{bmatrix} \, \text{and} \, B = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \end{bmatrix} .$
johnston-linear-matrix-algebra-EQ0234	108	0.931	$\begin{aligned} v_{_1} + 2 \, v_{_2} + 3 \, v_{_3} &= 0 \\ v_{_2} - v_{_3} &= 0 \end{aligned}$	$v_1 + 2v_2 + 3v_3 = 0$ $v_2 - v_3 = 0$	$v_1 + 2v_2 + 3v_3 = 0$ $v_2 - v_3 = 0$
johnston-linear-matrix-algebra-EQ0970	341	0.932	$A^{\pi} = \frac{1}{3} \left[\begin{array}{ll} 4 - 4^{\pi} & 4^{\pi+1} - 4 \\ 1 - 4^{\pi} & 4^{\pi+1} - 1 \end{array} \right] ,$	$A^{\pi} = \frac{1}{3} \left[\begin{array}{ll} 4 - 4^{\pi} & 4^{\pi+1} - 4 \\ 1 - 4^{\pi} & 4^{\pi+1} - 1 \end{array} \right] ,$	$A^{\pi} = \frac{1}{3} \left[\begin{array}{ll} 4 - 4^{\pi} & 4^{\pi+1} - 4 \\ 1 - 4^{\pi} & 4^{\pi+1} - 1 \end{array} \right] ,$

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johnston-linear-matrix-algebra-EQ1492	465	0.932	$\begin{aligned} \operatorname{nullity}(A) &= n - \operatorname{rank}(A) \quad \text{and} \\ \operatorname{nullity}(A^T) &= m - \operatorname{rank}(A^T) = m - \operatorname{rank}(A). \end{aligned}$	$\operatorname{nullity}(A) = n - \operatorname{rank}(A) \quad \text{and}$ $\operatorname{nullity}(A^T) = m - \operatorname{rank}(A^T) = m - \operatorname{rank}(A).$	$\operatorname{nullity}(A) = n - \operatorname{rank}(A) \quad \text{and}$ $\operatorname{nullity}(A^T) = m - \operatorname{rank}(A^T) = m - \operatorname{rank}(A).$
johnston-linear-matrix-algebra-EQ0497	200	0.932	$\begin{array}{l} \operatorname{maximize}: \quad 2x_1 - x_2 \\ \text{subject to:} \quad x_1 + 3x_2 \leq 4 \\ \quad \quad \quad 3x_1 - 2x_2 \leq 5 \\ \quad \quad \quad x_1 \geq 0 \end{array}$	$\begin{array}{l} \operatorname{maximize}: \quad 2x_1 - x_2 \\ \text{subject to:} \quad x_1 + 3x_2 \leq 4 \\ \quad \quad \quad 3x_1 - 2x_2 \leq 5 \\ \quad \quad \quad x_1 \geq 0 \end{array}$	$\begin{array}{l} \operatorname{maximize}: \quad 2x_1 - x_2 \\ \text{subject to:} \quad x_1 + 3x_2 \leq 4 \\ \quad \quad \quad 3x_1 - 2x_2 \leq 5 \\ \quad \quad \quad x_1 \geq 0 \end{array}$
johnston-linear-matrix-algebra-EQ0617	237	0.932	$\begin{aligned} &\left[\begin{array}{cccc c} 1 & 0 & 0 & 0 & 0 \\ 2 & 1 & 0 & 0 & 2 \\ -3 & 3 & 1 & 0 & 0 \\ -1 & 2 & -2 & 1 & 1 \end{array} \right] \xrightarrow{\substack{R_2-2R_1 \\ R_3+3R_1 \\ R_4+R_1}} \left[\begin{array}{cccc c} 1 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 2 \\ 0 & 3 & 1 & 0 & 0 \\ 0 & 2 & -2 & 1 & 1 \end{array} \right] \\ &\xrightarrow{\substack{R_3-3R_2 \\ R_4-2R_2}} \left[\begin{array}{cccc c} 1 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 2 \\ 0 & 0 & 1 & 0 & -6 \\ 0 & 0 & -2 & 1 & -3 \end{array} \right] \xrightarrow{R_4+2R_3} \left[\begin{array}{cccc c} 1 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 2 \\ 0 & 0 & 1 & 0 & -6 \\ 0 & 0 & 0 & 1 & -15 \end{array} \right], \end{aligned}$	$\begin{aligned} &\left[\begin{array}{cccc c} 1 & 0 & 0 & 0 & 0 \\ 2 & 1 & 0 & 0 & 2 \\ -3 & 3 & 1 & 0 & 0 \\ -1 & 2 & -2 & 1 & 1 \end{array} \right] \xrightarrow{\substack{R_2-2R_1 \\ R_3+3R_1 \\ R_4+R_1}} \left[\begin{array}{cccc c} 1 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 2 \\ 0 & 3 & 1 & 0 & 0 \\ 0 & 2 & -2 & 1 & 1 \end{array} \right] \\ &\xrightarrow{\substack{R_3-3R_2 \\ R_4-2R_2}} \left[\begin{array}{cccc c} 1 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 2 \\ 0 & 0 & 1 & 0 & -6 \\ 0 & 0 & -2 & 1 & -3 \end{array} \right] \xrightarrow{R_4+2R_3} \left[\begin{array}{cccc c} 1 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 2 \\ 0 & 0 & 1 & 0 & -6 \\ 0 & 0 & 0 & 1 & -15 \end{array} \right], \end{aligned}$	
johnston-linear-matrix-algebra-EQ0304	122	0.933	$B=I \quad B=(C \ A) \quad B=C(A \ B)=C \quad I=C,$	$B = IB = (CA)B = C(AB) = CI = C,$	$B = IB = (CA)B = C(AB) = CI = C,$
johnston-linear-matrix-algebra-EQ1218	411	0.933	$\left \lambda_0 \right > \left \lambda_j \right \quad \text{for all } 0 < j < m.$	$ \lambda_0 > \lambda_j \quad \text{for all } 0 < j < m.$	$ \lambda_0 > \lambda_j \quad \text{for all } 0 < j < m.$
johnston-linear-matrix-algebra-EQ1611	478	0.933	$\begin{aligned} \det(C - \lambda I) &= \det \left(\begin{bmatrix} -\lambda & 1 & 0 \\ 0 & -\lambda & 1 \\ -a_0 & -a_1 & -a_2 - \lambda \end{bmatrix} \right) \\ &= -\lambda^2(a_2 + \lambda) - a_0 - a_1\lambda \\ &= -(\lambda^3 + a^2\lambda^2 + a_1\lambda + a_0), \end{aligned}$	$\det(C - \lambda I) = \det \left(\begin{bmatrix} -\lambda & 1 & 0 \\ 0 & -\lambda & 1 \\ -a_0 & -a_1 & -a_2 - \lambda \end{bmatrix} \right)$ $= -\lambda^2(a_2 + \lambda) - a_0 - a_1\lambda$ $= -(\lambda^3 + a^2\lambda^2 + a_1\lambda + a_0),$	$\det(C - \lambda I) = \det \left(\begin{bmatrix} -\lambda & 1 & 0 \\ 0 & -\lambda & 1 \\ -a_0 & -a_1 & -a_2 - \lambda \end{bmatrix} \right)$ $= -\lambda^2(a_2 + \lambda) - a_0 - a_1\lambda$ $= -(\lambda^3 + a^2\lambda^2 + a_1\lambda + a_0),$

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johnston-linear-matrix-algebra-EQ1146	395	0.933	$\begin{aligned} D &= \operatorname{diag}(r_1 e^{i\theta_1}, r_1 e^{-i\theta_1}, \dots, r_\ell e^{i\theta_\ell}, r_\ell e^{-i\theta_\ell}, \lambda_1, \dots, \lambda_m) \quad \text{and} \\ P &= [\mathbf{v}_1 \overline{\mathbf{v}_1} \dots \mathbf{v}_\ell \overline{\mathbf{v}_\ell} \mathbf{w}_1 \dots \mathbf{w}_m], \end{aligned}$	$D = \operatorname{diag}(r_1 e^{i\theta_1}, r_1 e^{-i\theta_1}, \dots, r_\ell e^{i\theta_\ell}, r_\ell e^{-i\theta_\ell}, \lambda_1, \dots, \lambda_m) \quad \text{and}$ $P = [\mathbf{v}_1 \overline{\mathbf{v}_1} \dots \mathbf{v}_\ell \overline{\mathbf{v}_\ell} \mathbf{w}_1 \dots \mathbf{w}_m],$	$D = \operatorname{diag}(r_1 e^{i\theta_1}, r_1 e^{-i\theta_1}, \dots, r_\ell e^{i\theta_\ell}, r_\ell e^{-i\theta_\ell}, \lambda_1, \dots, \lambda_m) \quad \text{and}$ $P = [\mathbf{v}_1 \overline{\mathbf{v}_1} \dots \mathbf{v}_\ell \overline{\mathbf{v}_\ell} \mathbf{w}_1 \dots \mathbf{w}_m],$
johnston-linear-matrix-algebra-EQ9915	327	0.935	$D = \begin{bmatrix} \lambda_1 & 0 & 0 \\ 0 & \lambda_2 & 0 \\ 0 & 0 & \lambda_3 \end{bmatrix} = \begin{bmatrix} 2 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & -1 \end{bmatrix}, P = [\mathbf{v}_1 \mathbf{v}_2 \mathbf{v}_3] = \begin{bmatrix} 3 & 0 & 1 \\ 1 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}.$	$D = \begin{bmatrix} \lambda_1 & 0 & 0 \\ 0 & \lambda_2 & 0 \\ 0 & 0 & \lambda_3 \end{bmatrix} = \begin{bmatrix} 2 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & -1 \end{bmatrix}, P = [\mathbf{v}_1 \mathbf{v}_2 \mathbf{v}_3] = \begin{bmatrix} 3 & 0 & 1 \\ 1 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}.$	$D = \begin{bmatrix} \lambda_1 & 0 & 0 \\ 0 & \lambda_2 & 0 \\ 0 & 0 & \lambda_3 \end{bmatrix} = \begin{bmatrix} 2 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & -1 \end{bmatrix}, P = [\mathbf{v}_1 \mathbf{v}_2 \mathbf{v}_3] = \begin{bmatrix} 3 & 0 & 1 \\ 1 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}.$
johnston-linear-matrix-algebra-EQ9942	335	0.935	$\begin{aligned} P D Q^* &= \begin{bmatrix} \mathbf{p}_1 & \mathbf{p}_2 & \dots & \mathbf{p}_n \end{bmatrix} \begin{bmatrix} d_1 & 0 & \dots & 0 \\ 0 & d_2 & \dots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \dots & d_n \end{bmatrix} \begin{bmatrix} \mathbf{q}_1^* \\ \mathbf{q}_2^* \\ \vdots \\ \mathbf{q}_n^* \end{bmatrix} \\ &= \begin{bmatrix} \mathbf{p}_1 & \mathbf{p}_2 & \dots & \mathbf{p}_n \end{bmatrix} \begin{bmatrix} \frac{d_1 \mathbf{q}_1^*}{d_2 \mathbf{q}_2^*} \\ \vdots \\ \frac{d_n \mathbf{q}_n^*}{d_n \mathbf{q}_n^*} \end{bmatrix} = \sum_{j=1}^n d_j \mathbf{p}_j \mathbf{q}_j^*, \end{aligned}$	$P D Q^* = [\mathbf{p}_1 \mathbf{p}_2 \dots \mathbf{p}_n] \begin{bmatrix} d_1 & 0 & \dots & 0 \\ 0 & d_2 & \dots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \dots & d_n \end{bmatrix} \begin{bmatrix} \mathbf{q}_1^* \\ \mathbf{q}_2^* \\ \vdots \\ \mathbf{q}_n^* \end{bmatrix}$ $= [\mathbf{p}_1 \mathbf{p}_2 \dots \mathbf{p}_n] \begin{bmatrix} \frac{d_1 \mathbf{q}_1^*}{d_2 \mathbf{q}_2^*} \\ \vdots \\ \frac{d_n \mathbf{q}_n^*}{d_n \mathbf{q}_n^*} \end{bmatrix} = \sum_{j=1}^n d_j \mathbf{p}_j \mathbf{q}_j^*,$	$P D Q^* = [\mathbf{p}_1 \mathbf{p}_2 \dots \mathbf{p}_n] \begin{bmatrix} d_1 & 0 & \dots & 0 \\ 0 & d_2 & \dots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \dots & d_n \end{bmatrix} \begin{bmatrix} \mathbf{q}_1^* \\ \mathbf{q}_2^* \\ \vdots \\ \mathbf{q}_n^* \end{bmatrix}$ $= [\mathbf{p}_1 \mathbf{p}_2 \dots \mathbf{p}_n] \begin{bmatrix} \frac{d_1 \mathbf{q}_1^*}{d_2 \mathbf{q}_2^*} \\ \vdots \\ \frac{d_n \mathbf{q}_n^*}{d_n \mathbf{q}_n^*} \end{bmatrix} = \sum_{j=1}^n d_j \mathbf{p}_j \mathbf{q}_j^*,$
johnston-linear-matrix-algebra-EQ9798 (3)	267	0.936	$\begin{aligned} P B P^{-1} &= \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix} \begin{bmatrix} 5 & -1 \\ -2 & 0 \end{bmatrix} \begin{bmatrix} 1/2 & 1/2 \\ 1/2 & -1/2 \end{bmatrix} \\ &= \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix} \begin{bmatrix} 2 & 3 \\ -1 & -1 \end{bmatrix} = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix} = A. \end{aligned}$	$P B P^{-1} = \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix} \begin{bmatrix} 5 & -1 \\ -2 & 0 \end{bmatrix} \begin{bmatrix} 1/2 & 1/2 \\ 1/2 & -1/2 \end{bmatrix}$ $= \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix} \begin{bmatrix} 2 & 3 \\ -1 & -1 \end{bmatrix} = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix} = A$	$P B P^{-1} = \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix} \begin{bmatrix} 5 & -1 \\ -2 & 0 \end{bmatrix} \begin{bmatrix} 1/2 & 1/2 \\ 1/2 & -1/2 \end{bmatrix}$ $= \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix} \begin{bmatrix} 2 & 3 \\ -1 & -1 \end{bmatrix} = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix} = A.$
johnston-linear-matrix-algebra-EQ0277	115	0.936	$\begin{aligned} 1/x - 2/z &= -4 \\ 1/x + 1/y + 1/z &= 4 \\ 2/x - 6/y - 2/z &= 4 \end{aligned}$	$\begin{aligned} 1/x - 2/z &= -4 \\ 1/x + 1/y + 1/z &= 4 \\ 2/x - 6/y - 2/z &= 4 \end{aligned}$	$\begin{aligned} 1/x - 2/z &= -4 \\ 1/x + 1/y + 1/z &= 4 \\ 2/x - 6/y - 2/z &= 4 \end{aligned}$

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johnston-linear-matrix-algebra- EQ0257	113	■ 0.937	$\begin{array}{r} x+2y=3 \\ 2x+y=3 \end{array}$	$\begin{array}{l} x+2y=3 \\ 2x+y=3 \end{array}$	$\begin{array}{l} x+2y=3 \\ 2x+y=3 \end{array}$
johnston-linear-matrix-algebra- EQ1373	451	■ 0.938	$D A=\left[\begin{array}{cccc} d_1 & 0 & \cdots & 0 \\ 0 & d_2 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & d_n \end{array}\right]\left[\begin{array}{c} \frac{\mathbf{a}_1}{\mathbf{a}_2} \\ \vdots \\ \mathbf{a}_n \end{array}\right]=\left[\begin{array}{c} \frac{d_1 \mathbf{a}_1}{d_2 \mathbf{a}_2} \\ \vdots \\ d_n \mathbf{a}_n \end{array}\right].$	$DA=\left[\begin{array}{cccc} d_1 & 0 & \cdots & 0 \\ 0 & d_2 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & d_n \end{array}\right]\left[\begin{array}{c} \frac{\mathbf{a}_1}{\mathbf{a}_2} \\ \vdots \\ \mathbf{a}_n \end{array}\right]=\left[\begin{array}{c} \frac{d_1 \mathbf{a}_1}{d_2 \mathbf{a}_2} \\ \vdots \\ d_n \mathbf{a}_n \end{array}\right].$	$DA=\left[\begin{array}{cccc} d_1 & 0 & \cdots & 0 \\ 0 & d_2 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & d_n \end{array}\right]\left[\begin{array}{c} \mathbf{a}_1 \\ \mathbf{a}_2 \\ \vdots \\ \mathbf{a}_n \end{array}\right]=\left[\begin{array}{c} \frac{d_1 \mathbf{a}_1}{d_2 \mathbf{a}_2} \\ \vdots \\ d_n \mathbf{a}_n \end{array}\right].$
johnston-linear-matrix-algebra- EQ1637	482	■ 0.938	$\begin{aligned} C \mathbf{v} &= \left[\begin{array}{cccc} 0 & 1 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & 1 \\ -a_0 & -a_1 & \cdots & -a_{n-1} \end{array}\right] \left[\begin{array}{c} v_1 \\ v_2 \\ \vdots \\ v_{n-1} \\ v_n \end{array}\right] \\ &= \left[\begin{array}{c} v_2 \\ v_3 \\ \vdots \\ v_n \\ -a_0 v_1 - \cdots - a_{n-1} v_n \end{array}\right] = \left[\begin{array}{c} \lambda v_1 \\ \lambda v_2 \\ \vdots \\ \lambda v_{n-1} \\ \lambda v_n \end{array}\right]. \end{aligned}$	$C \mathbf{v} = \left[\begin{array}{cccc} 0 & 1 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & 1 \\ -a_0 & -a_1 & \cdots & -a_{n-1} \end{array}\right] \left[\begin{array}{c} v_1 \\ v_2 \\ \vdots \\ v_{n-1} \\ v_n \end{array}\right] \\ = \left[\begin{array}{c} v_2 \\ v_3 \\ \vdots \\ v_n \\ -a_0 v_1 - \cdots - a_{n-1} v_n \end{array}\right] = \left[\begin{array}{c} \lambda v_1 \\ \lambda v_2 \\ \vdots \\ \lambda v_{n-1} \\ \lambda v_n \end{array}\right].$	$C \mathbf{v} = \left[\begin{array}{cccc} 0 & 1 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & 1 \\ -a_0 & -a_1 & \cdots & -a_{n-1} \end{array}\right] \left[\begin{array}{c} v_1 \\ v_2 \\ \vdots \\ v_{n-1} \\ v_n \end{array}\right] \\ = \left[\begin{array}{c} v_2 \\ v_3 \\ \vdots \\ v_n \\ -a_0 v_1 - \cdots - a_{n-1} v_n \end{array}\right] = \left[\begin{array}{c} \lambda v_1 \\ \lambda v_2 \\ \vdots \\ \lambda v_{n-1} \\ \lambda v_n \end{array}\right].$
johnston-linear-matrix-algebra- EQ0568	222	■ 0.939	$\begin{aligned} \operatorname{maximize}: \quad & 3x_1 + 2x_2 - x_3 + 2x_4 \\ \text{subject to:} \quad & 2x_1 - 2x_2 - x_3 + 2x_4 \geq 2 \\ & x_1 + 3x_2 - 2x_3 + x_4 = 2 \\ & 3x_1 + x_2 - 3x_3 + 3x_4 \leq 3 \\ & x_1, \quad x_2, \quad x_4 \geq 0 \end{aligned}$	$\begin{aligned} \operatorname{maximize}: \quad & 3x_1 + 2x_2 - x_3 + 2x_4 \\ \text{subject to:} \quad & 2x_1 - 2x_2 - x_3 + 2x_4 \geq 2 \\ & x_1 + 3x_2 - 2x_3 + x_4 = 2 \\ & 3x_1 + x_2 - 3x_3 + 3x_4 \leq 3 \\ & x_1, \quad x_2, \quad x_4 \geq 0 \end{aligned}$	$\begin{aligned} \operatorname{maximize}: \quad & 3x_1 + 2x_2 - x_3 + 2x_4 \\ \text{subject to:} \quad & 2x_1 - 2x_2 - x_3 + 2x_4 \geq 2 \\ & x_1 + 3x_2 - 2x_3 + x_4 = 2 \\ & 3x_1 + x_2 - 3x_3 + 3x_4 \leq 3 \\ & x_1, \quad x_2, \quad x_4 \geq 0 \end{aligned}$
johnston-linear-matrix-algebra- EQ0646	247	■ 0.940	$\left[3\right], \quad \left[\begin{array}{ll} 3 & -2 \\ 3 & -1 \end{array}\right], \quad \text{and} \quad \left[\begin{array}{lll} 3 & -2 & -1 \\ 3 & -1 & 4 \\ 1 & 5 & 2 \end{array}\right].$	$\left[3\right], \quad \left[\begin{array}{ll} 3 & -2 \\ 3 & -1 \end{array}\right], \quad \text{and} \quad \left[\begin{array}{lll} 3 & -2 & -1 \\ 3 & -1 & 4 \\ 1 & 5 & 2 \end{array}\right].$	$\left[3\right], \quad \left[\begin{array}{ll} 3 & -2 \\ 3 & -1 \end{array}\right], \quad \text{and} \quad \left[\begin{array}{lll} 3 & -2 & -1 \\ 3 & -1 & 4 \\ 1 & 5 & 2 \end{array}\right].$

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johnston-linear-matrix-algebra-E00894	321	0.943	$\begin{aligned} p_A(\lambda) &= \det(A - \lambda I) = \det \left(\begin{bmatrix} 1-\lambda & 2 & 3 \\ 0 & 4-\lambda & 5 \\ 0 & 0 & 6-\lambda \end{bmatrix} \right) \\ &= (1-\lambda)(4-\lambda)(6-\lambda). \end{aligned}$	$p_A(\lambda) = \det(A - \lambda I) = \det \left(\begin{bmatrix} 1-\lambda & 2 & 3 \\ 0 & 4-\lambda & 5 \\ 0 & 0 & 6-\lambda \end{bmatrix} \right) = (1-\lambda)(4-\lambda)(6-\lambda).$	$p_A(\lambda) = \det(A - \lambda I) = \det \left(\begin{bmatrix} 1-\lambda & 2 & 3 \\ 0 & 4-\lambda & 5 \\ 0 & 0 & 6-\lambda \end{bmatrix} \right) = (1-\lambda)(4-\lambda)(6-\lambda).$
johnston-linear-matrix-algebra-E00762	286	0.944	$\det \left(\begin{bmatrix} a & b \\ c & d \end{bmatrix} \right) = ad - bc.$	$\det \left(\begin{bmatrix} a & b \\ c & d \end{bmatrix} \right) = ad - bc.$	$\det \left(\begin{bmatrix} a & b \\ c & d \end{bmatrix} \right) = ad - bc.$
johnston-linear-matrix-algebra-E00350	136	0.945	$\left(A + \mathbf{vw}^T \right)^{-1} = A^{-1} - \frac{A^{-1} \mathbf{vw}^T A^{-1}}{1 + \mathbf{w}^T A^{-1} \mathbf{v}},$	$(A + \mathbf{vw}^T)^{-1} = A^{-1} - \frac{A^{-1} \mathbf{vw}^T A^{-1}}{1 + \mathbf{w}^T A^{-1} \mathbf{v}}$	$(A + \mathbf{vw}^T)^{-1} = A^{-1} - \frac{A^{-1} \mathbf{vw}^T A^{-1}}{1 + \mathbf{w}^T A^{-1} \mathbf{v}},$
johnston-linear-matrix-algebra-E01222	411	0.945	$\lim_{n \rightarrow \infty} \frac{q(n+1)}{q(n)} = 1 \quad \text{and} \quad \lim_{n \rightarrow \infty} q(n)c^n = 0.$	$\lim_{n \rightarrow \infty} \frac{q(n+1)}{q(n)} = 1 \quad \text{and} \quad \lim_{n \rightarrow \infty} q(n)c^n = 0.$	$\lim_{n \rightarrow \infty} \frac{q(n+1)}{q(n)} = 1 \quad \text{and} \quad \lim_{n \rightarrow \infty} q(n)c^n = 0.$
johnston-linear-matrix-algebra-E01431	457	0.945	$\begin{aligned} (1,0) &= c_1(1,1) + c_2(1,-1) \\ (0,1) &= c_3(1,1) + c_4(1,-1). \end{aligned}$	$\begin{aligned} (1,0) &= c_1(1,1) + c_2(1,-1) \\ (0,1) &= c_3(1,1) + c_4(1,-1). \end{aligned}$	$\begin{aligned} (1,0) &= c_1(1,1) + c_2(1,-1) \\ (0,1) &= c_3(1,1) + c_4(1,-1). \end{aligned}$
johnston-linear-matrix-algebra-E00189	93	0.945	$\begin{aligned} \sqrt{x} + 3y &= 4, & 2x - 7y^2 &= 3, & \text{and} & \ln(x) - y/z = 2. \\ 2^x - 2^y &= 3, & \cos(x) + \sin(y) &= 0, & & \end{aligned}$	$\begin{aligned} \sqrt{x} + 3y &= 4, & 2x - 7y^2 &= 3, & & 4y + 2xz = 1, \\ 2^x - 2^y &= 3, & \cos(x) + \sin(y) &= 0, & \text{and} & \ln(x) - y/z = 2. \end{aligned}$	$\begin{aligned} \sqrt{x} + 3y &= 4, & 2x - 7y^2 &= 3, & & 4y + 2xz = 1, \\ 2^x - 2^y &= 3, & \cos(x) + \sin(y) &= 0, & \text{and} & \ln(x) - y/z = 2. \end{aligned}$
johnston-linear-matrix-algebra-E01312	442	0.945	$\begin{aligned} p_A'(\lambda) &= \frac{d}{d\lambda} f(\lambda, \lambda, \dots, \lambda) \\ &= \frac{\partial}{\partial \lambda_1} f(\lambda_1, \lambda, \dots, \lambda) + \frac{\partial}{\partial \lambda_2} f(\lambda, \lambda_2, \dots, \lambda) + \dots + \frac{\partial}{\partial \lambda_n} f(\lambda, \lambda, \dots, \lambda_n) \\ &= -\det(A_1 - \lambda I) - \det(A_2 - \lambda I) - \dots - \det(A_n - \lambda I) \\ &= -p_{A_1}(\lambda) - p_{A_2}(\lambda) - \dots - p_{A_n}(\lambda), \end{aligned}$	$\begin{aligned} p_A'(\lambda) &= \frac{d}{d\lambda} f(\lambda, \lambda, \dots, \lambda) \\ &= \frac{\partial}{\partial \lambda_1} f(\lambda_1, \lambda, \dots, \lambda) + \frac{\partial}{\partial \lambda_2} f(\lambda, \lambda_2, \dots, \lambda) + \dots + \frac{\partial}{\partial \lambda_n} f(\lambda, \lambda, \dots, \lambda_n) \\ &= -\det(A_1 - \lambda I) - \det(A_2 - \lambda I) - \dots - \det(A_n - \lambda I) \\ &= -p_{A_1}(\lambda) - p_{A_2}(\lambda) - \dots - p_{A_n}(\lambda), \end{aligned}$	$\begin{aligned} p_A'(\lambda) &= \frac{d}{d\lambda} f(\lambda, \lambda, \dots, \lambda) \\ &= \frac{\partial}{\partial \lambda_1} f(\lambda_1, \lambda, \dots, \lambda) + \frac{\partial}{\partial \lambda_2} f(\lambda, \lambda_2, \dots, \lambda) + \dots + \frac{\partial}{\partial \lambda_n} f(\lambda, \lambda, \dots, \lambda_n) \\ &= -\det(A_1 - \lambda I) - \det(A_2 - \lambda I) - \dots - \det(A_n - \lambda I) \\ &= -p_{A_1}(\lambda) - p_{A_2}(\lambda) - \dots - p_{A_n}(\lambda), \end{aligned}$
johnston-linear-matrix-algebra-E01088	369	0.946	$\begin{aligned} 2w - x - 2y - z &= 1 \\ w - 2x - y - 3z &= 2 \\ 4w + y - 2z &= 3 \\ 2x - y + 2z &= 2 \end{aligned}$	$\begin{aligned} 2w - x - 2y - z &= 1 \\ w - 2x - y - 3z &= 2 \\ 4w + y - 2z &= 3 \\ 2x - y + 2z &= 2 \end{aligned}$	$\begin{aligned} 2w - x - 2y - z &= 1 \\ w - 2x - y - 3z &= 2 \\ 4w + y - 2z &= 3 \\ 2x - y + 2z &= 2 \end{aligned}$

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johnston-linear-matrix-algebra- EQ0239	110	■0.947	<pre>\begin{aligned} & \left[\begin{array}{cc c} -1 & 3 & 1 \\ 3 & 1 & 1 \\ 2 & -1 & 3 \end{array}\right] & \xrightarrow{R_2+3R_1} & \left[\begin{array}{cc c} -1 & 3 & 1 \\ R_3+2R_1 & 10 & 4 \\ 0 & 5 & 5 \end{array}\right] \\ & & \xrightarrow{R_3-\frac{1}{2}R_2} & \left[\begin{array}{cc c} -1 & 3 & 1 \\ 0 & 10 & 4 \\ 0 & 0 & 3 \end{array}\right]. \end{aligned}</pre>		
johnston-linear-matrix-algebra- EQ0722	271	■0.949	<pre>\begin{aligned} B=&\{(3,0,0,3),(3,0,1,1), \\ &\{(2,-1,0,1),(2,-1,3,-1)\}, \\ C=&\{(0,1,3,2),(-1,1,2,-1), \\ &\{(1,-1,-1,-1),(0,2,2,0)\} \end{aligned}</pre>	$B = \{(3,0,0,3), (3,0,1,1), (2,-1,0,1), (2,-1,3,-1)\},$ $C = \{(0,1,3,2), (-1,1,2,-1), (1,-1,-1,-1), (0,2,2,0)\}$	$B = \{(3,0,0,3), (3,0,1,1), (2,-1,0,1), (2,-1,3,-1)\},$ $C = \{(0,1,3,2), (-1,1,2,-1), (1,-1,-1,-1), (0,2,2,0)\}$
johnston-linear-matrix-algebra- EQ0867	314	■0.949	<pre>\begin{aligned} & \overline{\left[\begin{array}{cc} 2i & 1-3i \\ -3 & 5+2i \end{array}\right]} = \left[\begin{array}{cc} -2i & 1+3i \\ -3 & 5-2i \end{array}\right], \quad \text{so} \\ & \left[\begin{array}{cc} 2i & 1-3i \\ -3 & 5+2i \end{array}\right]^* = \left[\begin{array}{cc} -2i & -3 \\ 1+3i & 5-2i \end{array}\right]. \end{aligned}</pre>		$\overline{\begin{bmatrix} 2i & 1-3i \\ -3 & 5+2i \end{bmatrix}} = \begin{bmatrix} -2i & 1+3i \\ -3 & 5-2i \end{bmatrix}, \quad \text{so}$ $\begin{bmatrix} 2i & 1-3i \\ -3 & 5+2i \end{bmatrix}^* = \begin{bmatrix} -2i & -3 \\ 1+3i & 5-2i \end{bmatrix}.$
johnston-linear-matrix-algebra- EQ0891	320	■0.949	<pre>\begin{aligned} p_A(\lambda)=&\operatorname{det}(A-\lambda I) = \operatorname{det}\left(\begin{array}{cccc} -6-\lambda & 0 & -2 & 1 \\ 0 & -3-\lambda & -2 & 1 \\ 3 & 0 & 1-\lambda & 1 \\ -3 & 0 & 2 & 2-\lambda \end{array}\right) \\ &= (-3-\lambda)((-6-\lambda)(1-\lambda)(2-\lambda) + 6 + 6 - 2(-6-\lambda) + 3(1-\lambda) + 6(2-\lambda)) \\ &= (3+\lambda)(\lambda^3+3\lambda^2-9\lambda-27) \\ &= (3+\lambda)(3-\lambda)(\lambda^2+6\lambda+9) \\ &= (3+\lambda)^3(3-\lambda). \end{aligned}</pre>		$p_A(\lambda) = \det(A - \lambda I) = \det \left(\begin{bmatrix} -6-\lambda & 0 & -2 & 1 \\ 0 & -3-\lambda & -2 & 1 \\ 3 & 0 & 1-\lambda & 1 \\ -3 & 0 & 2 & 2-\lambda \end{bmatrix} \right)$ $= (-3-\lambda)((-6-\lambda)(1-\lambda)(2-\lambda) + 6 + 6 - 2(-6-\lambda) + 3(1-\lambda) + 6(2-\lambda))$ $= (3+\lambda)(\lambda^3+3\lambda^2-9\lambda-27)$ $= (3+\lambda)(3-\lambda)(\lambda^2+6\lambda+9)$ $= (3+\lambda)^3(3-\lambda).$

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johnston-linear-matrix-algebra-EQ0992	347	0.950	$\begin{aligned} \sin(A) &= \frac{1}{3} \left[\begin{array}{ll} 4 & 1 \\ 1 & 1 \end{array} \right] \left[\begin{array}{cc} \sin(1) & 0 \\ 0 & \sin(4) \end{array} \right] \left[\begin{array}{rr} 1 & -1 \\ -1 & 4 \end{array} \right] \\ &= \frac{1}{3} \left[\begin{array}{ll} 4 & 1 \\ 1 & 1 \end{array} \right] \left[\begin{array}{cc} \sin(1) & -\sin(1) \\ -\sin(4) & 4\sin(4) \end{array} \right] \\ &= \frac{1}{3} \left[\begin{array}{cc} 4\sin(1) - \sin(4) & 4\sin(4) - 4\sin(1) \\ \sin(1) - \sin(4) & 4\sin(4) - \sin(1) \end{array} \right]. \end{aligned}$	$\sin(A) = \frac{1}{3} \begin{bmatrix} 4 & 1 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} \sin(1) & 0 \\ 0 & \sin(4) \end{bmatrix} \begin{bmatrix} 1 & -1 \\ -1 & 4 \end{bmatrix}$ $= \frac{1}{3} \begin{bmatrix} 4 & 1 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} \sin(1) & -\sin(1) \\ -\sin(4) & 4\sin(4) \end{bmatrix}$ $= \frac{1}{3} \begin{bmatrix} 4\sin(1) - \sin(4) & 4\sin(4) - 4\sin(1) \\ \sin(1) - \sin(4) & 4\sin(4) - \sin(1) \end{bmatrix}.$	$\sin(A) = \frac{1}{3} \begin{bmatrix} 4 & 1 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} \sin(1) & 0 \\ 0 & \sin(4) \end{bmatrix} \begin{bmatrix} 1 & -1 \\ -1 & 4 \end{bmatrix}$ $= \frac{1}{3} \begin{bmatrix} 4 & 1 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} \sin(1) & -\sin(1) \\ -\sin(4) & 4\sin(4) \end{bmatrix}$ $= \frac{1}{3} \begin{bmatrix} 4\sin(1) - \sin(4) & 4\sin(4) - 4\sin(1) \\ \sin(1) - \sin(4) & 4\sin(4) - \sin(1) \end{bmatrix}.$
johnston-linear-matrix-algebra-EQ1353	449	0.950	$\begin{aligned} \left[(A+B)+C \right]_{i,j} &= (a_{i,j} + b_{i,j}) + c_{i,j} \\ &= a_{i,j} + (b_{i,j} + c_{i,j}) \\ &= [A + (B+C)]_{i,j}. \end{aligned}$	$[(A+B)+C]_{i,j} = (a_{i,j} + b_{i,j}) + c_{i,j}$ $= a_{i,j} + (b_{i,j} + c_{i,j})$ $= [A + (B+C)]_{i,j}.$	$[(A+B)+C]_{i,j} = (a_{i,j} + b_{i,j}) + c_{i,j}$ $= a_{i,j} + (b_{i,j} + c_{i,j})$ $= [A + (B+C)]_{i,j}.$
johnston-linear-matrix-algebra-EQ1619	479	0.950	$\begin{aligned} D &= \left[\begin{array}{ccc} 2 & 0 & 0 \\ 0 & i & 0 \\ 0 & 0 & -i \end{array} \right], P = \left[\begin{array}{ccc} 1 & 0 & 0 \\ 0 & 1 & i \\ 0 & 1 & -i \end{array} \right], \\ P^{-1} &= \frac{1}{2} \left[\begin{array}{ccc} 2 & 0 & 0 \\ 0 & 1 & -i \\ 0 & 1 & i \end{array} \right]. \end{aligned}$	$D = \begin{bmatrix} 2 & 0 & 0 \\ 0 & i & 0 \\ 0 & 0 & -i \end{bmatrix}, P = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & i \\ 0 & 1 & -i \end{bmatrix},$ $P^{-1} = \frac{1}{2} \begin{bmatrix} 2 & 0 & 0 \\ 0 & 1 & -i \\ 0 & 1 & i \end{bmatrix}.$	$D = \begin{bmatrix} 2 & 0 & 0 \\ 0 & i & 0 \\ 0 & 0 & -i \end{bmatrix}, P = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & i \\ 0 & 1 & -i \end{bmatrix},$ $P^{-1} = \frac{1}{2} \begin{bmatrix} 2 & 0 & 0 \\ 0 & 1 & -i \\ 0 & 1 & i \end{bmatrix}.$
johnston-linear-matrix-algebra-EQ0319	126	0.950	$\left[\begin{array}{ll} 1 & 2 \\ 0 & 1 \end{array} \right]^{-1} = \left[\begin{array}{cc} 1 & -2 \\ 0 & 1 \end{array} \right] \quad \text{and} \quad \left[\begin{array}{ll} 1 & 0 \\ 0 & 3 \end{array} \right]^{-1} = \left[\begin{array}{cc} 1 & 0 \\ 0 & 1/3 \end{array} \right].$	$\begin{bmatrix} 1 & 2 \\ 0 & 1 \end{bmatrix}^{-1} = \begin{bmatrix} 1 & -2 \\ 0 & 1 \end{bmatrix} \quad \text{and} \quad \begin{bmatrix} 1 & 0 \\ 0 & 3 \end{bmatrix}^{-1} = \begin{bmatrix} 1 & 0 \\ 0 & 1/3 \end{bmatrix}.$	$\begin{bmatrix} 1 & 2 \\ 0 & 1 \end{bmatrix}^{-1} = \begin{bmatrix} 1 & -2 \\ 0 & 1 \end{bmatrix} \quad \text{and} \quad \begin{bmatrix} 1 & 0 \\ 0 & 3 \end{bmatrix}^{-1} = \begin{bmatrix} 1 & 0 \\ 0 & 1/3 \end{bmatrix}.$
johnston-linear-matrix-algebra-EQ1023	355	0.950	$\begin{aligned} A &= \left[\begin{array}{llll} 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{array} \right] \quad \text{and} \quad B = \left[\begin{array}{cccc} 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \end{array} \right]. \end{aligned}$	$A = \begin{bmatrix} 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix} \quad \text{and} \quad B = \begin{bmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \end{bmatrix}.$	$A = \begin{bmatrix} 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix} \quad \text{and} \quad B = \begin{bmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \end{bmatrix}.$

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johnston-linear-matrix-algebra EQ0424	165	0.951	$\begin{aligned} & \left[\begin{array}{ccc c} 1 & 1 & 2 & 0 \\ 2 & -1 & 1 & 0 \\ 1 & 0 & 1 & 0 \end{array} \right] \xrightarrow[\frac{-1}{3}R_2]{R_2-2R_1 \atop R_3-R_1} \left[\begin{array}{ccc c} 1 & 1 & 2 & 0 \\ 0 & -3 & -3 & 0 \\ 0 & -1 & -1 & 0 \end{array} \right] \\ & \xrightarrow[\frac{-1}{3}R_2]{R_1-R_2 \atop R_3+R_2} \left[\begin{array}{ccc c} 1 & 0 & 1 & 0 \\ 0 & -1 & -1 & 0 \\ 0 & 0 & 0 & 0 \end{array} \right] . \end{aligned}$	$\begin{aligned} & \left[\begin{array}{ccc c} 1 & 1 & 2 & 0 \\ 2 & -1 & 1 & 0 \\ 1 & 0 & 1 & 0 \end{array} \right] \xrightarrow[\frac{-1}{3}R_2]{R_2-2R_1 \atop R_3-R_1} \left[\begin{array}{ccc c} 1 & 1 & 2 & 0 \\ 0 & -3 & -3 & 0 \\ 0 & -1 & -1 & 0 \end{array} \right] \\ & \xrightarrow[\frac{-1}{3}R_2]{R_1-R_2 \atop R_3+R_2} \left[\begin{array}{ccc c} 1 & 0 & 1 & 0 \\ 0 & -1 & -1 & 0 \\ 0 & 0 & 0 & 0 \end{array} \right] . \end{aligned}$	
johnston-linear-matrix-algebra EQ1529	470	0.953	$A=P_1\left[\begin{array}{ll} I_r & 0 \\ 0 & 0 \end{array}\right]Q_1 \text{ and } B=P_2\left[\begin{array}{cc} I_r & 0 \\ 0 & 0 \end{array}\right]Q_2 .$	$A=P_1\left[\begin{array}{cc} I_r & O \\ O & O \end{array}\right]Q_1 \text{ and } B=P_2\left[\begin{array}{cc} I_r & O \\ O & O \end{array}\right]Q_2 .$	$A=P_1\left[\begin{array}{cc} I_r & O \\ O & O \end{array}\right]Q_1 \text{ and } B=P_2\left[\begin{array}{cc} I_r & O \\ O & O \end{array}\right]Q_2 .$
johnston-linear-matrix-algebra EQ0628	241	0.954	$\begin{aligned} & \left[\begin{array}{cccc cccc} 0 & 2 & 4 & 2 & 1 & 0 & 0 & 0 \\ 2 & 1 & 1 & 2 & 0 & 1 & 0 & 0 \\ 1 & 1 & 2 & 3 & 0 & 0 & 1 & 0 \\ 1 & 2 & 3 & 4 & 0 & 0 & 0 & 1 \end{array} \right] \xrightarrow{R_1 \leftrightarrow R_3} \left[\begin{array}{cccc cccc} 1 & 1 & 2 & 3 & 0 & 0 & 1 & 0 \\ 2 & 1 & 1 & 2 & 0 & 1 & 0 & 0 \\ 0 & 2 & 4 & 2 & 1 & 0 & 0 & 0 \\ 1 & 2 & 3 & 4 & 0 & 0 & 0 & 1 \end{array} \right] \\ & \xrightarrow{R_2-2R_1} \left[\begin{array}{cccc cccc} 1 & 1 & 4 & 3 & 0 & 0 & 1 & 0 \\ 0 & -1 & -3 & -4 & 0 & 1 & -2 & 0 \\ 0 & 2 & 4 & 2 & 1 & 0 & 0 & 0 \\ 0 & 1 & 1 & 1 & 0 & 0 & -1 & 1 \end{array} \right] \\ & \xrightarrow{R_4-R_1+2R_2} \left[\begin{array}{cccc cccc} 1 & 1 & 4 & 3 & 0 & 0 & 1 & 0 \\ 0 & -1 & -3 & -4 & 0 & 1 & -2 & 0 \\ 0 & 0 & -2 & -6 & 1 & 2 & -4 & 0 \\ 0 & 0 & -2 & -3 & 0 & 1 & -3 & 1 \end{array} \right] \\ & \xrightarrow{R_4-R_3} \left[\begin{array}{cccc cccc} 1 & 1 & 4 & 3 & 0 & 0 & 1 & 0 \\ 0 & -1 & -3 & -4 & 0 & 1 & -2 & 0 \\ 0 & 0 & -2 & -6 & 1 & 2 & -4 & 0 \\ 0 & 0 & 0 & 3 & -1 & -1 & 1 & 1 \end{array} \right] . \end{aligned}$	$\begin{aligned} & \left[\begin{array}{cccc cccc} 0 & 2 & 4 & 2 & 1 & 0 & 0 & 0 \\ 2 & 1 & 1 & 2 & 0 & 1 & 0 & 0 \\ 1 & 1 & 2 & 3 & 0 & 0 & 1 & 0 \\ 1 & 2 & 3 & 4 & 0 & 0 & 0 & 1 \end{array} \right] \xrightarrow{R_1 \leftrightarrow R_3} \left[\begin{array}{cccc cccc} 1 & 1 & 2 & 3 & 0 & 0 & 1 & 0 \\ 2 & 1 & 1 & 2 & 0 & 1 & 0 & 0 \\ 0 & 2 & 4 & 2 & 1 & 0 & 0 & 0 \\ 1 & 2 & 3 & 4 & 0 & 0 & 0 & 1 \end{array} \right] \\ & \xrightarrow[\begin{smallmatrix} R_2-2R_1 \\ R_4-R_1 \end{smallmatrix}]{R_1 \leftrightarrow R_3} \left[\begin{array}{cccc cccc} 1 & 1 & 4 & 3 & 0 & 0 & 1 & 0 \\ 0 & -1 & -3 & -4 & 0 & 1 & -2 & 0 \\ 0 & 2 & 4 & 2 & 1 & 0 & 0 & 0 \\ 0 & 1 & 1 & 1 & 0 & 0 & -1 & 1 \end{array} \right] \\ & \xrightarrow[\begin{smallmatrix} R_3+2R_2 \\ R_4+R_2 \end{smallmatrix}]{R_1 \leftrightarrow R_3} \left[\begin{array}{cccc cccc} 1 & 1 & 4 & 3 & 0 & 0 & 1 & 0 \\ 0 & -1 & -3 & -4 & 0 & 1 & -2 & 0 \\ 0 & 0 & -2 & -6 & 1 & 2 & -4 & 0 \\ 0 & 0 & -2 & -3 & 0 & 1 & -3 & 1 \end{array} \right] \\ & \xrightarrow{R_4-R_3} \left[\begin{array}{cccc cccc} 1 & 1 & 4 & 3 & 0 & 0 & 1 & 0 \\ 0 & -1 & -3 & -4 & 0 & 1 & -2 & 0 \\ 0 & 0 & -2 & -6 & 1 & 2 & -4 & 0 \\ 0 & 0 & 0 & 3 & -1 & -1 & 1 & 1 \end{array} \right] . \end{aligned}$	

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johnston-linear-matrix-algebra-EQ0169	84	0.954	$A^6=\left(A^3\right)^2=\left[\begin{array}{llll} 2 & 3 & 1 & 4 \\ 3 & 2 & 1 & 4 \\ 1 & 1 & 0 & 3 \\ 4 & 4 & 3 & 2 \end{array}\right]^2=\left[\begin{array}{llll} 30 & 29 & 17 & 31 \\ 29 & 30 & 17 & 31 \\ 17 & 17 & 11 & 14 \\ 31 & 31 & 14 & 45 \end{array}\right] .$		$A^6=(A^3)^2=\begin{bmatrix} 2 & 3 & 1 & 4 \\ 3 & 2 & 1 & 4 \\ 1 & 1 & 0 & 3 \\ 4 & 4 & 3 & 2 \end{bmatrix}^2=\begin{bmatrix} 30 & 29 & 17 & 31 \\ 29 & 30 & 17 & 31 \\ 17 & 17 & 11 & 14 \\ 31 & 31 & 14 & 45 \end{bmatrix}.$
johnston-linear-matrix-algebra-EQ1087	369	0.955	$\begin{aligned} x+y+z &= 1 \\ -x+y+z &= 2 \\ 2x+y &= 0 \end{aligned}$	$\begin{aligned} x+y+z &= 1 \\ -x+y+z &= 2 \\ 2x+y &= 0 \end{aligned}$	$\begin{aligned} x+y+z &= 1 \\ -x+y+z &= 2 \\ 2x+y &= 0 \end{aligned}$
johnston-linear-matrix-algebra-EQ1197	406	0.955	$\left[\begin{array}{ccc} 1 & 1 & 1 \\ 2 & i & -i \\ -1 & -1 & \end{array}\right]\left[\begin{array}{l} c_0 \\ c_1 \\ c_2 \end{array}\right]=\left[\begin{array}{l} 3 \\ 8 \\ 2 \end{array}\right] .$	$\left[\begin{array}{ccc} 1 & 1 & 1 \\ 2 & i & -i \\ 4 & -1 & -1 \end{array}\right]\left[\begin{array}{l} c_0 \\ c_1 \\ c_2 \end{array}\right]=\left[\begin{array}{l} 3 \\ 8 \\ 2 \end{array}\right] .$	$\left[\begin{array}{ccc} 1 & 1 & 1 \\ 2 & i & -i \\ 4 & -1 & -1 \end{array}\right]\left[\begin{array}{l} c_0 \\ c_1 \\ c_2 \end{array}\right]=\left[\begin{array}{l} 3 \\ 8 \\ 2 \end{array}\right] .$
johnston-linear-matrix-algebra-EQ0562	222	0.955	$\begin{array}{l} \operatorname{maximize}: \\ \text{subject to: } \\ x_1+x_2+x_3 \text{ \& \& \text{subject to:} } \\ x_1-x_2+x_3 \leq 1 \text{ \& } x_1+3 x_2 \leq 2 \\ 2-x_1+x_2+2 x_3 \leq 3 \text{ \& } x_1, \\ x_2, \quad x_3 \geq 0 \end{array}$	$\begin{array}{l} \text{maximize:} \\ \text{subject to:} \end{array} \quad \begin{array}{l} x_1+x_2+x_3 \\ 2x_1-x_2+x_3 \leq 1 \\ x_1+3x_2 \leq 2 \\ -x_1+x_2+2x_3 \leq 3 \\ x_1, \quad x_2, \quad x_3 \geq 0 \end{array}$	$\begin{array}{l} \text{maximize:} \\ \text{subject to:} \end{array} \quad \begin{array}{l} x_1+x_2+x_3 \\ 2x_1-x_2+x_3 \leq 1 \\ x_1+3x_2 \leq 2 \\ -x_1+x_2+2x_3 \leq 3 \\ x_1, \quad x_2, \quad x_3 \geq 0 \end{array}$
johnston-linear-matrix-algebra-EQ0057	44	0.957	$\begin{aligned} &\left[B^T A^T\right]_{i, j}=\left[B^T\right]_{i, 1}\left[A^T\right]_{1, j}+\left[B^T\right]_{i, 2}\left[A^T\right]_{2, j}+\cdots+\left[B^T\right]_{i, n}\left[A^T\right]_{n, j} \\ &=b_{1, i} a_{j, 1}+b_{2, i} a_{j, 2}+\cdots+b_{n, i} a_{j, n} . \end{aligned}$	$\begin{aligned} \left[B^T A^T\right]_{i, j} &= \left[B^T\right]_{i, 1} \left[A^T\right]_{1, j} + \left[B^T\right]_{i, 2} \left[A^T\right]_{2, j} + \cdots + \left[B^T\right]_{i, n} \left[A^T\right]_{n, j} \\ &= b_{1, i} a_{j, 1} + b_{2, i} a_{j, 2} + \cdots + b_{n, i} a_{j, n} . \end{aligned}$	$\begin{aligned} \left[B^T A^T\right]_{i, j} &= \left[B^T\right]_{i, 1} \left[A^T\right]_{1, j} + \left[B^T\right]_{i, 2} \left[A^T\right]_{2, j} + \cdots + \left[B^T\right]_{i, n} \left[A^T\right]_{n, j} \\ &= b_{1, i} a_{j, 1} + b_{2, i} a_{j, 2} + \cdots + b_{n, i} a_{j, n} . \end{aligned}$
johnston-linear-matrix-algebra-EQ0950	337	0.957	$\left[\begin{array}{c} F_{n+1} \\ F_n \end{array}\right]=\left[\begin{array}{cc} 1 & 1 \\ 1 & 0 \end{array}\right]\left[\begin{array}{c} F_n \\ F_{n-1} \end{array}\right] \quad \text { for all } \quad n \geq 1 .$	$\left[\begin{array}{c} F_{n+1} \\ F_n \end{array}\right]=\left[\begin{array}{cc} 1 & 1 \\ 1 & 0 \end{array}\right]\left[\begin{array}{c} F_n \\ F_{n-1} \end{array}\right] \quad \text { for all } \quad n \geq 1 .$	$\left[\begin{array}{c} F_{n+1} \\ F_n \end{array}\right]=\left[\begin{array}{cc} 1 & 1 \\ 1 & 0 \end{array}\right]\left[\begin{array}{c} F_n \\ F_{n-1} \end{array}\right] \quad \text { for all } \quad n \geq 1 .$
johnston-linear-matrix-algebra-EQ1606	478	0.957	$\begin{aligned} &\mathbf{w} \cdot(A \mathbf{v})=\mathbf{w} \cdot(\lambda \mathbf{v})=\bar{\lambda}(\mathbf{w} \cdot \mathbf{v}) \quad \text { and } \\ &\mathbf{w} \cdot(A \mathbf{v})=\left(A^* \mathbf{w}\right) \cdot \mathbf{v}=(\mu \mathbf{w}) \cdot \mathbf{v}=\mu(\mathbf{w} \cdot \mathbf{v}) . \end{aligned}$	$\begin{aligned} \mathbf{w} \cdot(A \mathbf{v}) &= \mathbf{w} \cdot(\lambda \mathbf{v}) = \bar{\lambda}(\mathbf{w} \cdot \mathbf{v}) \quad \text { and } \\ \mathbf{w} \cdot(A \mathbf{v}) &= (A^* \mathbf{w}) \cdot \mathbf{v} = (\mu \mathbf{w}) \cdot \mathbf{v} = \mu(\mathbf{w} \cdot \mathbf{v}) . \end{aligned}$	$\begin{aligned} \mathbf{w} \cdot(A \mathbf{v}) &= \mathbf{w} \cdot(\lambda \mathbf{v}) = \bar{\lambda}(\mathbf{w} \cdot \mathbf{v}) \quad \text { and } \\ \mathbf{w} \cdot(A \mathbf{v}) &= (A^* \mathbf{w}) \cdot \mathbf{v} = (\mu \mathbf{w}) \cdot \mathbf{v} = \mu(\mathbf{w} \cdot \mathbf{v}) . \end{aligned}$
johnston-linear-matrix-algebra-EQ0290 (0)	119	0.958	$E_1=\left[\begin{array}{lll} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{array}\right], \quad E_2=\left[\begin{array}{lll} 1 & 0 & 0 \\ 0 & 1 & 0 \\ -3 & 0 & 1 \end{array}\right], \quad \text { and } \quad E_3=\left[\begin{array}{lll} 1 & 0 & 0 \\ 0 & 1 / 2 & 0 \\ 0 & 0 & 1 \end{array}\right],$	$E_1=\left[\begin{array}{lll} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{array}\right], \quad E_2=\left[\begin{array}{lll} 1 & 0 & 0 \\ 0 & 1 & 0 \\ -3 & 0 & 1 \end{array}\right], \quad \text { and } \quad E_3=\left[\begin{array}{lll} 1 & 0 & 0 \\ 0 & 1 / 2 & 0 \\ 0 & 0 & 1 \end{array}\right],$	$E_1=\left[\begin{array}{lll} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{array}\right], \quad E_2=\left[\begin{array}{lll} 1 & 0 & 0 \\ 0 & 1 & 0 \\ -3 & 0 & 1 \end{array}\right], \quad \text { and } \quad E_3=\left[\begin{array}{lll} 1 & 0 & 0 \\ 0 & 1 / 2 & 0 \\ 0 & 0 & 1 \end{array}\right],$

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johnston-linear-matrix-algebra-EQ0275	115	0.958	$\begin{aligned} 1 / x+2 / y \&=3 / x y \backslash \backslash x+y \&=2 \\ \end{aligned}$	$\frac{1}{x}+2 / y=3 / x y$ $x+y=2$	$\frac{1}{x}+2 / y=3 / x y$ $x+y=2$
johnston-linear-matrix-algebra-EQ0113	62	0.959	$\left[F_{\mathbf{u}}\right] \mathbf{v}=\frac{1}{2}\left[\begin{array}{cc}-1 & \sqrt{3} \\ \sqrt{3} & 1\end{array}\right]\left[\begin{array}{c}-1 \\ 3\end{array}\right]=\frac{1}{2}\left[\begin{array}{c}3 \sqrt{3}+1 \\ 3-\sqrt{3}\end{array}\right] \approx\left[\begin{array}{c}3.0981 \\ 0.6340\end{array}\right] .$	$\left[F_{\mathbf{u}}\right] \mathbf{v}=\frac{1}{2}\left[\begin{array}{cc}-1 & \sqrt{3} \\ \sqrt{3} & 1\end{array}\right]\left[\begin{array}{c}-1 \\ 3\end{array}\right]=\frac{1}{2}\left[\begin{array}{c}3 \sqrt{3}+1 \\ 3-\sqrt{3}\end{array}\right] \approx\left[\begin{array}{c}3.0981 \\ 0.6340\end{array}\right] .$	$\left[F_{\mathbf{u}}\right] \mathbf{v}=\frac{1}{2}\left[\begin{array}{cc}-1 & \sqrt{3} \\ \sqrt{3} & 1\end{array}\right]\left[\begin{array}{c}-1 \\ 3\end{array}\right]=\frac{1}{2}\left[\begin{array}{c}3 \sqrt{3}+1 \\ 3-\sqrt{3}\end{array}\right] \approx\left[\begin{array}{c}3.0981 \\ 0.6340\end{array}\right] .$
johnston-linear-matrix-algebra-EQ1091	370	0.959	$\mathbf{v}_k=\frac{A \mathbf{v}_{k-1}}{\left\ A \mathbf{v}_{k-1}\right\ } \quad \text { for all } \quad k \geq 1 .$	$\mathbf{v}_k=\frac{A \mathbf{v}_{k-1}}{\left\ A \mathbf{v}_{k-1}\right\ } \quad \text { for all } \quad k \geq 1 .$	$\mathbf{v}_k=\frac{A \mathbf{v}_{k-1}}{\left\ A \mathbf{v}_{k-1}\right\ } \quad \text { for all } \quad k \geq 1 .$
johnston-linear-matrix-algebra-EQ0530 (2.B.12)	214	0.962	$\begin{array}{l} \text { maximize: } \quad x_1+2 x_2 \\ \text { subject to: } \quad x_1+x_2 \leq 3 \\ \quad \quad \quad -x_1+x_2 \leq 1 \\ \quad \quad \quad x_1, \quad x_2 \geq 0 \end{array}$	$\begin{array}{l} \text { maximize : } \quad x_1+2 x_2 \\ \text { subject to: } \quad x_1+x_2 \leq 3 \\ \quad \quad \quad -x_1+x_2 \leq 1 \\ \quad \quad \quad x_1, \quad x_2 \geq 0 \end{array}$	$\begin{array}{l} \text { maximize: } \quad x_1+2 x_2 \\ \text { subject to: } \quad x_1+x_2 \leq 3 \\ \quad \quad \quad -x_1+x_2 \leq 1 \\ \quad \quad \quad x_1, \quad x_2 \geq 0 \end{array}$
johnston-linear-matrix-algebra-EQ0601	231	0.962	$\begin{aligned} &\left[\begin{array}{ccc ccc} 1 & -2 & -1 & 1 & 0 & 0 \\ 2 & -1 & -1 & 0 & 1 & 0 \\ 3 & 6 & 2 & 0 & 0 & 1 \end{array}\right] \xrightarrow[R_3-3 R_1]{R_2-2 R_1}\left[\begin{array}{ccc ccc} 1 & -2 & -1 & 1 & 0 & 0 \\ 0 & 3 & 1 & -2 & 1 & 0 \\ 0 & 12 & 5 & -3 & 0 & 1 \end{array}\right] \\ &\xrightarrow{R_3-4 R_2}\left[\begin{array}{ccc ccc} 1 & -2 & -1 & 1 & 0 & 0 \\ 0 & 3 & 1 & -2 & 1 & 0 \\ 0 & 0 & 1 & 5 & -4 & 1 \end{array}\right] . \end{aligned}$	$\left[\begin{array}{ccc ccc} 1 & -2 & -1 & 1 & 0 & 0 \\ 2 & -1 & -1 & 0 & 1 & 0 \\ 3 & 6 & 2 & 0 & 0 & 1 \end{array}\right] \xrightarrow[R_3-3 R_1]{R_2-2 R_1}\left[\begin{array}{ccc ccc} 1 & -2 & -1 & 1 & 0 & 0 \\ 0 & 3 & 1 & -2 & 1 & 0 \\ 0 & 12 & 5 & -3 & 0 & 1 \end{array}\right] \xrightarrow{R_3-4 R_2}\left[\begin{array}{ccc ccc} 1 & -2 & -1 & 1 & 0 & 0 \\ 0 & 3 & 1 & -2 & 1 & 0 \\ 0 & 0 & 1 & 5 & -4 & 1 \end{array}\right] .$	
johnston-linear-matrix-algebra-EQ1465	461	0.963	$\begin{aligned} A &=\left[\begin{array}{ll} 1 & 1 \\ 1 & 2 \end{array}\right], B=\left[\begin{array}{ccc} 2 & -1 & 0 \\ 2 & 0 & 1 \end{array}\right], \text { and } \\ D &=\left[\begin{array}{ccc} 1 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 3 \end{array}\right] . \end{aligned}$	$A=\left[\begin{array}{cc} 1 & 1 \\ 1 & 2 \end{array}\right], B=\left[\begin{array}{ccc} 2 & -1 & 0 \\ 2 & 0 & 1 \end{array}\right], \text { and }$ $D=\left[\begin{array}{ccc} 1 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 3 \end{array}\right] .$	$A=\left[\begin{array}{cc} 1 & 1 \\ 1 & 2 \end{array}\right], B=\left[\begin{array}{ccc} 2 & -1 & 0 \\ 2 & 0 & 1 \end{array}\right], \text { and }$ $D=\left[\begin{array}{ccc} 1 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 3 \end{array}\right] .$
johnston-linear-matrix-algebra-EQ1332	447	0.964	$\begin{aligned} &(-1,-2,-3) \cdot(2,-2,-4)=-2+4+12=14 \\ &(-1,-2,-3) \cdot(-3,0,1)=3+0-3=0 \\ &(-3,0,1) \cdot(2,-2,-4)=-6+0-4=-10 . \end{aligned}$	$\begin{aligned} (-1,-2,-3) \cdot(2,-2,-4) &=-2+4+12=14 \\ (-1,-2,-3) \cdot(-3,0,1) &=3+0-3=0 \\ (-3,0,1) \cdot(2,-2,-4) &=-6+0-4=-10 . \end{aligned}$	$\begin{aligned} (-1,-2,-3) \cdot(2,-2,-4) &=-2+4+12=14 \\ (-1,-2,-3) \cdot(-3,0,1) &=3+0-3=0 \\ (-3,0,1) \cdot(2,-2,-4) &=-6+0-4=-10 . \end{aligned}$

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johnston-linear-matrix-algebra-EQ0579	225	0.964	$\hat{R}=\left[\begin{array}{cccc} 1 & 0 & 0 & -1 \\ 1 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{array}\right], \quad E=\left[\begin{array}{ccc} 1 & -1 & 0 \\ 0 & 1 & 0 \\ -2 & 1 & 0 \end{array}\right], \quad \text{and } P=\left[\begin{array}{ccc} 1 & 1 & 0 \\ 0 & 1 & 0 \\ 1 & 0 & 0 \end{array}\right].$	$\hat{R} = \begin{bmatrix} 1 & 0 & 0 & -1 \\ 0 & 1 & 1 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}, \quad E = \begin{bmatrix} 1 & -1 & 0 \\ 0 & 1 & 0 \\ 1 & -2 & 1 \end{bmatrix}, \quad \text{and } P = \begin{bmatrix} 1 & 1 & 0 \\ 0 & 1 & 0 \\ -1 & 1 & 1 \end{bmatrix}.$	$\hat{R} = \begin{bmatrix} 1 & 0 & 0 & -1 \\ 0 & 1 & 1 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}, \quad E = \begin{bmatrix} 1 & -1 & 0 \\ 0 & 1 & 0 \\ 1 & -2 & 1 \end{bmatrix}, \quad \text{and } P = \begin{bmatrix} 1 & 1 & 0 \\ 0 & 1 & 0 \\ -1 & 1 & 1 \end{bmatrix}.$
johnston-linear-matrix-algebra-EQ0525	211	0.965	$\begin{array}{lr} \operatorname{maximize}: & x_1+3x_2 \\ \text{subject to:} & x_1+2x_2 \leq 4 \\ & -2x_1+x_2 \leq 0 \\ & x_1, \quad x_2 \geq 0 \end{array}$	$\begin{array}{ll} \text{maximize :} & x_1 + 3x_2 \\ \text{subject to:} & x_1 + 2x_2 \leq 4 \\ & -2x_1 + x_2 \leq 0 \\ & x_1, \quad x_2 \geq 0 \end{array}$	$\begin{array}{ll} \text{maximize:} & x_1 + 3x_2 \\ \text{subject to:} & x_1 + 2x_2 \leq 4 \\ & -2x_1 + x_2 \leq 0 \\ & x_1, \quad x_2 \geq 0 \end{array}$
johnston-linear-matrix-algebra-EQ0318	126	0.965	$\begin{aligned} & (A \, B) \left(B^{-1} A^{-1} \right) = A \left(B B^{-1} \right) A^{-1} = A I A^{-1} = A A^{-1} = I, \quad \text{and} \\ & \left(B^{-1} A^{-1} \right) (A B) = B^{-1} \left(A^{-1} A \right) B = B^{-1} I B = B^{-1} B = I. \end{aligned}$	$(AB) \left(B^{-1} A^{-1} \right) = A \left(B B^{-1} \right) A^{-1} = A I A^{-1} = A A^{-1} = I, \quad \text{and} \\ (B^{-1} A^{-1}) (AB) = B^{-1} \left(A^{-1} A \right) B = B^{-1} I B = B^{-1} B = I.$	$(AB)(B^{-1}A^{-1}) = A(BB^{-1})A^{-1} = AIA^{-1} = AA^{-1} = I, \quad \text{and} \\ (B^{-1}A^{-1})(AB) = B^{-1}(A^{-1}A)B = B^{-1}IB = B^{-1}B = I.$
johnston-linear-matrix-algebra-EQ0535	215	0.965	$\begin{aligned} & \operatorname{maximize}: \mathbf{c} \cdot \mathbf{x} \\ & \text{subject to: } A \mathbf{x} \leq \mathbf{b} \\ & \mathbf{x} \geq \mathbf{0} \end{aligned}$	$\begin{array}{ll} \text{maximize :} & \mathbf{c} \cdot \mathbf{x} \\ \text{subject to:} & A \mathbf{x} \leq \mathbf{b} \\ & \mathbf{x} \geq \mathbf{0} \end{array}$	$\begin{array}{ll} \text{maximize:} & \mathbf{c} \cdot \mathbf{x} \\ \text{subject to:} & A \mathbf{x} \leq \mathbf{b} \\ & \mathbf{x} \geq \mathbf{0} \end{array}$
johnston-linear-matrix-algebra-EQ1083	369	0.965	$\begin{array}{r} x+2y=3 \\ 2x+y=3 \end{array}$	$\begin{array}{l} x+2y=3 \\ 2x+y=3 \end{array}$	$\begin{array}{l} x+2y=3 \\ 2x+y=3 \end{array}$
johnston-linear-matrix-algebra-EQ0639	244	0.965	$A \, \mathbf{x} = P \, L \, U \, \mathbf{x} = P \, L \, (U \, \mathbf{x}) = P \, L \, \mathbf{y} = P \, \mathbf{c} = \mathbf{b},$	$A\mathbf{x} = PLU\mathbf{x} = PL(U\mathbf{x}) = PL\mathbf{y} = P(L\mathbf{y}) = P\mathbf{c} = \mathbf{b},$	$A\mathbf{x} = PLU\mathbf{x} = PL(U\mathbf{x}) = PL\mathbf{y} = P(L\mathbf{y}) = P\mathbf{c} = \mathbf{b},$
johnston-linear-matrix-algebra-EQ0457	182	0.965	$\begin{aligned} & x_1+x_3+x_4 = 1 \\ & x_2+x_3 = 1 \\ & x_1+x_3+x_4+x_5 = 0 \\ & x_1+x_2+x_5 = 0 \\ & x_2+x_3+x_5 = 0 \end{aligned}$	$\begin{array}{l} x_1 + x_3 + x_4 = 1 \\ x_2 + x_3 = 1 \\ x_1 + x_3 + x_4 + x_5 = 0 \\ x_1 + x_2 + x_5 = 0 \\ x_2 + x_3 + x_5 = 0 \end{array}$	$\begin{array}{l} x_1 + x_3 + x_4 = 1 \\ x_2 + x_3 = 1 \\ x_1 + x_3 + x_4 + x_5 = 0 \\ x_1 + x_2 + x_5 = 0 \\ x_2 + x_3 + x_5 = 0 \end{array}$
johnston-linear-matrix-algebra-EQ0772	288	0.966	$\begin{aligned} & (0 \cdot (-1) \cdot 0) + (2 \cdot 1 \cdot 2) + (1 \cdot 1 \cdot 1) \\ & - (0 \cdot 1 \cdot 1) - (2 \cdot 1 \cdot 0) - (1 \cdot (-1) \cdot 2) \\ & = 0 + 4 + 1 - 0 - 0 - (-2) = 7. \end{aligned}$	$\begin{array}{l} (0 \cdot (-1) \cdot 0) + (2 \cdot 1 \cdot 2) + (1 \cdot 1 \cdot 1) \\ - (0 \cdot 1 \cdot 1) - (2 \cdot 1 \cdot 0) - (1 \cdot (-1) \cdot 2) \\ = 0 + 4 + 1 - 0 - 0 - (-2) = 7. \end{array}$	$\begin{array}{l} (0 \cdot (-1) \cdot 0) + (2 \cdot 1 \cdot 2) + (1 \cdot 1 \cdot 1) \\ - (0 \cdot 1 \cdot 1) - (2 \cdot 1 \cdot 0) - (1 \cdot (-1) \cdot 2) \\ = 0 + 4 + 1 - 0 - 0 - (-2) = 7. \end{array}$
johnston-linear-matrix-algebra-EQ0700 (0)	263	0.966	$[T]_B=\left[\begin{array}{l} [T(1,1)]_B \\ [T(1,-1)]_B \end{array}\right]=\left[\begin{array}{cc} 3 & 0 \\ 0 & 1 \end{array}\right].$	$[T]_B = \left[\begin{array}{l} [T(1,1)]_B \\ [T(1,-1)]_B \end{array}\right] = \left[\begin{array}{cc} 3 & 0 \\ 0 & 1 \end{array}\right].$	$[T]_B = \left[\begin{array}{l} [T(1,1)]_B \\ [T(1,-1)]_B \end{array}\right] = \left[\begin{array}{cc} 3 & 0 \\ 0 & 1 \end{array}\right].$
johnston-linear-matrix-algebra-EQ0732	275	0.967	$\left(P^{-1} A \right) \mathbf{e}_1 = \left(P^{-1} A \right) \left(P \mathbf{e}_1 \right) = \left(P^{-1} A \right) \mathbf{x} = P^{-1} (A \mathbf{x}) = P^{-1} \mathbf{0} = \mathbf{0}.$	$(P^{-1}AP) \mathbf{e}_1 = (P^{-1}A) (P\mathbf{e}_1) = (P^{-1}A) \mathbf{x} = P^{-1}(A\mathbf{x}) = P^{-1}\mathbf{0} = \mathbf{0}.$	$(P^{-1}AP)\mathbf{e}_1 = (P^{-1}A)(P\mathbf{e}_1) = (P^{-1}A)\mathbf{x} = P^{-1}(A\mathbf{x}) = P^{-1}\mathbf{0} = \mathbf{0}.$

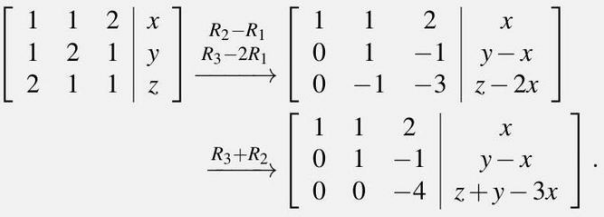
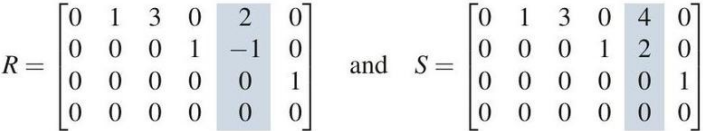
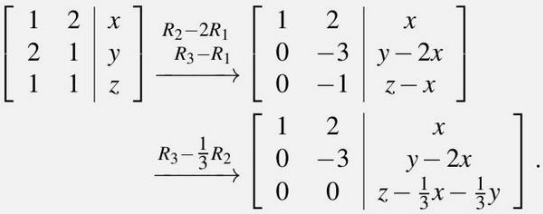
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johnston-linear-matrix-algebra- EQ1269	426	0.967	$\begin{array}{r} 3x^2 \\ x-2 \overline{) 3x^3+0x^2-14x+4} \\ \underline{3x^3-6x^2} \\ 6x^2-14x+4 \end{array}$	$\begin{array}{r} 3x^2 \\ x-2 \overline{) 3x^3+0x^2-14x+4} \\ \underline{3x^3-6x^2} \\ 6x^2-14x+4 \end{array}$	$\begin{array}{r} 3x^2 \\ x-2 \overline{) 3x^3+0x^2-14x+4} \\ \underline{3x^3-6x^2} \\ 6x^2-14x+4 \end{array}$
johnston-linear-matrix-algebra- EQ9504 (2.B.3)	201	0.967	$\begin{array}{l} \text{\texttt{\textbackslash begin\{array\}\{lc\} \operatorname{maximize}\: \& x_{\{1\}}+2} \\ \text{\texttt{\textbackslash x_{\{2\}} \\\ \texttt{\textbackslash text\{ subject to: \} \& x_{\{1\}}+x_{\{2\}} \leq 3 \}} \\ \text{\texttt{\textbackslash \& -x_{\{1\}}+x_{\{2\}} \leq 1 \\\ \& x_{\{1\}}, \quad x_{\{2\}} \geq 0} \\ \text{\texttt{\textbackslash end\{array\}}} \end{array}$	$\begin{array}{ll} \text{maximize :} & x_1 + 2x_2 \\ \text{subject to:} & x_1 + x_2 \leq 3 \\ & -x_1 + x_2 \leq 1 \\ & x_1, \quad x_2 \geq 0 \end{array}$	$\begin{array}{ll} \text{maximize:} & x_1 + 2x_2 \\ \text{subject to:} & x_1 + x_2 \leq 3 \\ & -x_1 + x_2 \leq 1 \\ & x_1, \quad x_2 \geq 0 \end{array}$
johnston-linear-matrix-algebra- EQ1194	406	0.967	$x_{\{n\}}=c_{\{0\}} \lambda_{\{0\}}^n+c_{\{1\}} \lambda_{\{1\}}^n+\dots+c_{\{k-1\}} \lambda_{\{k-1\}}^n \quad \text{\texttt{\textbackslash quad \texttt{\textbackslash text\{ for all \}}} \\ \quad n \geq 0,$	$x_n = c_0 \lambda_0^n + c_1 \lambda_1^n + \cdots + c_{k-1} \lambda_{k-1}^n \quad \text{for all } n \geq 0,$	$x_n = c_0 \lambda_0^n + c_1 \lambda_1^n + \cdots + c_{k-1} \lambda_{k-1}^n \quad \text{for all } n \geq 0,$
johnston-linear-matrix-algebra- EQ0713	269	0.970	$\begin{aligned} \operatorname{tr}(A B) &= \sum_{i=1}^m [A B]_{i,i} = \sum_{i=1}^m \sum_{j=1}^n a_{i,j} b_{j,i} \\ &= \sum_{j=1}^n \sum_{i=1}^m b_{j,i} a_{i,j} = \sum_{j=1}^n [B A]_{j,j} = \operatorname{tr}(B A), \end{aligned}$	$\operatorname{tr}(A B) = \sum_{i=1}^m [A B]_{i,i} = \sum_{i=1}^m \sum_{j=1}^n a_{i,j} b_{j,i} \\ = \sum_{j=1}^n \sum_{i=1}^m b_{j,i} a_{i,j} = \sum_{j=1}^n [B A]_{j,j} = \operatorname{tr}(B A),$	$\operatorname{tr}(A B) = \sum_{i=1}^m [A B]_{i,i} = \sum_{i=1}^m \sum_{j=1}^n a_{i,j} b_{j,i} \\ = \sum_{j=1}^n \sum_{i=1}^m b_{j,i} a_{i,j} = \sum_{j=1}^n [B A]_{j,j} = \operatorname{tr}(B A),$
johnston-linear-matrix-algebra- EQ0781	290	0.970	$\begin{array}{l} c_{\{1,1\}}=-1 \& c_{\{1,2\}}=2 \& c_{\{1,3\}}=2 \\ \text{\texttt{\textbackslash \& c_{\{2,1\}}=1 \& c_{\{2,2\}}=-2 \& c_{\{2,3\}}=-5 \\\ \& c_{\{3,1\}}=3 \& c_{\{3,2\}}=1 \& c_{\{3,3\}}=-2 .} \end{array}$	$\begin{array}{lll} c_{1,1} = -1 & c_{1,2} = 2 & c_{1,3} = 2 \\ c_{2,1} = 1 & c_{2,2} = -2 & c_{2,3} = -5 \\ c_{3,1} = 3 & c_{3,2} = 1 & c_{3,3} = -2. \end{array}$	$\begin{array}{lll} c_{1,1} = -1 & c_{1,2} = 2 & c_{1,3} = 2 \\ c_{2,1} = 1 & c_{2,2} = -2 & c_{2,3} = -5 \\ c_{3,1} = 3 & c_{3,2} = 1 & c_{3,3} = -2. \end{array}$
johnston-linear-matrix-algebra- EQ0566	222	0.970	$\begin{array}{l} \text{\texttt{\textbackslash begin\{array\}\{lr\} \operatorname{minimize}\: \& 2} \\ \text{\texttt{\textbackslash x_{\{1\}}-x_{\{2\}}-2 x_{\{3\}} \\\ \texttt{\textbackslash text\{ subject to: \} \& 2} \\ \text{\texttt{\textbackslash x_{\{1\}}-x_{\{2\}}+4 x_{\{3\}} \geq 2 \\\ \& x_{\{1\}}+2 x_{\{2\}}-3 x_{\{3\}} \\ \text{\texttt{\textbackslash \leq 1 \\\ \& x_{\{1\}}, \quad x_{\{2\}}, \quad x_{\{3\}} \geq 0} \\ \text{\texttt{\textbackslash end\{array\}}} \end{array}$	$\begin{array}{ll} \text{minimize :} & 2x_1 - x_2 - 2x_3 \\ \text{subject to:} & 2x_1 - x_2 + 4x_3 \geq 2 \\ & x_1 + 2x_2 - 3x_3 \leq 1 \\ & x_1, \quad x_2, \quad x_3 \geq 0 \end{array}$	$\begin{array}{ll} \text{minimize:} & 2x_1 - x_2 - 2x_3 \\ \text{subject to:} & 2x_1 - x_2 + 4x_3 \geq 2 \\ & x_1 + 2x_2 - 3x_3 \leq 1 \\ & x_1, \quad x_2, \quad x_3 \geq 0 \end{array}$
johnston-linear-matrix-algebra- EQ0365	142	0.971	$\left[\begin{array}{cc c} 1 & 2 & x \\ 2 & 1 & y \end{array} \right] \xrightarrow{R_2-2R_1} \left[\begin{array}{cc c} 1 & 2 & x \\ 0 & -3 & y-2x \end{array} \right].$	$\left[\begin{array}{cc c} 1 & 2 & x \\ 2 & 1 & y \end{array} \right] \xrightarrow{R_2-2R_1} \left[\begin{array}{cc c} 1 & 2 & x \\ 0 & -3 & y-2x \end{array} \right].$	$\left[\begin{array}{cc c} 1 & 2 & x \\ 2 & 1 & y \end{array} \right] \xrightarrow{R_2-2R_1} \left[\begin{array}{cc c} 1 & 2 & x \\ 0 & -3 & y-2x \end{array} \right].$
johnston-linear-matrix-algebra- EQ1448 (1)	459	0.971	$X=\left[\begin{array}{cc} 0 & 1 \\ 1 & 0 \end{array} \right], A=\left[\begin{array}{cc} 1 & 0 \\ 0 & -1 \end{array} \right], B=\left[\begin{array}{cc} 0 & -1 \\ 1 & 0 \end{array} \right]$	$X = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}, A = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}, B = \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix}$	$X = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}, A = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}, B = \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix}$
johnston-linear-matrix-algebra- EQ0834	305	0.971	$(A-2I)\mathbf{v}=\mathbf{0} \text{\texttt{\textbackslash text\{ . Since the reduced row echelon form of \}} A-2I \text{\texttt{\textbackslash text\{ is \}}$	$(A-2I)\mathbf{v} = \mathbf{0}. \text{ Since the reduced row echelon form of } A-2I \text{ is}$	$(A-2I)\mathbf{v} = \mathbf{0}. \text{ Since the reduced row echelon form of } A-2I \text{ is}$
johnston-linear-matrix-algebra- EQ0389	148	0.972	$\begin{aligned} & (1,-1,0)=-\frac{2}{3}(-2,1,2)-\frac{1}{3}(1,1,-4), \\ & (-2,1,2)=-\frac{3}{2}(1,-1,0)-\frac{1}{2}(1,1,-4), \quad \text{and} \\ & (1,1,-4)=-3(1,-1,0)-2(-2,1,2). \end{aligned}$	$\begin{aligned} (1,-1,0) &= -\frac{2}{3}(-2,1,2) - \frac{1}{3}(1,1,-4), \\ (-2,1,2) &= -\frac{3}{2}(1,-1,0) - \frac{1}{2}(1,1,-4), \quad \text{and} \\ (1,1,-4) &= -3(1,-1,0) - 2(-2,1,2). \end{aligned}$	$\begin{aligned} (1,-1,0) &= -\frac{2}{3}(-2,1,2) - \frac{1}{3}(1,1,-4), \\ (-2,1,2) &= -\frac{3}{2}(1,-1,0) - \frac{1}{2}(1,1,-4), \quad \text{and} \\ (1,1,-4) &= -3(1,-1,0) - 2(-2,1,2). \end{aligned}$

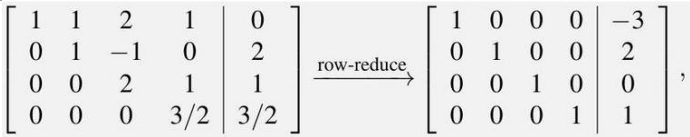
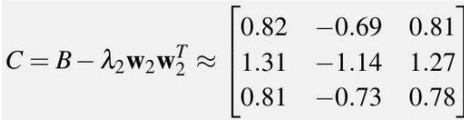
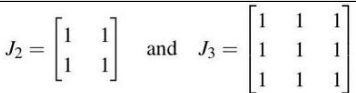
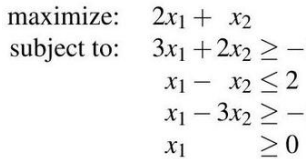
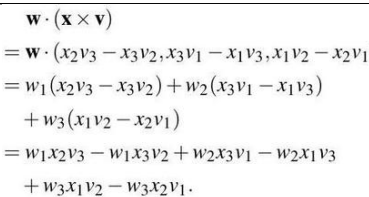
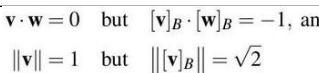
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johnston-linear-matrix-algebra-EQ1149	396	■ 0.972	$D_j = \begin{bmatrix} r_j e^{i\theta_j} & 0 \\ 0 & r_j e^{-i\theta_j} \end{bmatrix}, \quad P_j = [\mathbf{v}_j \mid \overline{\mathbf{v}_j}], \quad \text{and} \quad Q_j = [\operatorname{Re}(\mathbf{v}_j) \mid -\operatorname{Im}(\mathbf{v}_j)]$	$D_j = \begin{bmatrix} r_j e^{i\theta_j} & 0 \\ 0 & r_j e^{-i\theta_j} \end{bmatrix}, \quad P_j = [\mathbf{v}_j \mid \overline{\mathbf{v}_j}], \quad \text{and} \quad Q_j = [\operatorname{Re}(\mathbf{v}_j) \mid -\operatorname{Im}(\mathbf{v}_j)]$	$D_j = \begin{bmatrix} r_j e^{i\theta_j} & 0 \\ 0 & r_j e^{-i\theta_j} \end{bmatrix}, \quad P_j = [\mathbf{v}_j \mid \overline{\mathbf{v}_j}], \quad \text{and} \quad Q_j = [\operatorname{Re}(\mathbf{v}_j) \mid -\operatorname{Im}(\mathbf{v}_j)]$
johnston-linear-matrix-algebra-EQ1585	476	■ 0.972	$\det(\begin{bmatrix} I_k & B \\ O & C \end{bmatrix}) = \det(C),$	$\det \left(\begin{bmatrix} I_k & B \\ O & C \end{bmatrix} \right) = \det(C),$	$\det \left(\begin{bmatrix} I_k & B \\ O & C \end{bmatrix} \right) = \det(C),$
johnston-linear-matrix-algebra-EQ9914	327	■ 0.972	$\begin{aligned} P D P^{-1} &= \begin{bmatrix} -1 & 2 \\ 1 & 5 \end{bmatrix} \begin{bmatrix} -1 & 0 \\ 0 & 6 \end{bmatrix} \begin{bmatrix} \frac{1}{7} & -5 \\ 1 & 1 \end{bmatrix} \\ &= \frac{1}{7} \begin{bmatrix} -1 & 2 \\ 1 & 5 \end{bmatrix} \begin{bmatrix} 5 & -2 \\ 6 & 6 \end{bmatrix} = \frac{1}{7} \begin{bmatrix} 7 & 14 \\ 35 & 28 \end{bmatrix} = \begin{bmatrix} 1 & 2 \\ 5 & 4 \end{bmatrix} = A. \end{aligned}$	$\begin{aligned} P D P^{-1} &= \begin{bmatrix} -1 & 2 \\ 1 & 5 \end{bmatrix} \begin{bmatrix} -1 & 0 \\ 0 & 6 \end{bmatrix} \left(\frac{1}{7} \begin{bmatrix} -5 & 2 \\ 1 & 1 \end{bmatrix} \right) \\ &= \frac{1}{7} \begin{bmatrix} -1 & 2 \\ 1 & 5 \end{bmatrix} \begin{bmatrix} 5 & -2 \\ 6 & 6 \end{bmatrix} = \frac{1}{7} \begin{bmatrix} 7 & 14 \\ 35 & 28 \end{bmatrix} = \begin{bmatrix} 1 & 2 \\ 5 & 4 \end{bmatrix} = A. \end{aligned}$	$\begin{aligned} P D P^{-1} &= \begin{bmatrix} -1 & 2 \\ 1 & 5 \end{bmatrix} \begin{bmatrix} -1 & 0 \\ 0 & 6 \end{bmatrix} \left(\frac{1}{7} \begin{bmatrix} -5 & 2 \\ 1 & 1 \end{bmatrix} \right) \\ &= \frac{1}{7} \begin{bmatrix} -1 & 2 \\ 1 & 5 \end{bmatrix} \begin{bmatrix} 5 & -2 \\ 6 & 6 \end{bmatrix} = \frac{1}{7} \begin{bmatrix} 7 & 14 \\ 35 & 28 \end{bmatrix} = \begin{bmatrix} 1 & 2 \\ 5 & 4 \end{bmatrix} = A. \end{aligned}$
johnston-linear-matrix-algebra-EQ9654	251	■ 0.972	$\begin{aligned} \mathbf{v} &= c_1 \mathbf{v}_1 + c_2 \mathbf{v}_2 + \cdots + c_k \mathbf{v}_k \quad \text{and} \\ \mathbf{v} &= d_1 \mathbf{v}_1 + d_2 \mathbf{v}_2 + \cdots + d_k \mathbf{v}_k \end{aligned}$	$\mathbf{v} = c_1 \mathbf{v}_1 + c_2 \mathbf{v}_2 + \cdots + c_k \mathbf{v}_k \quad \text{and} \\ \mathbf{v} = d_1 \mathbf{v}_1 + d_2 \mathbf{v}_2 + \cdots + d_k \mathbf{v}_k$	$\mathbf{v} = c_1 \mathbf{v}_1 + c_2 \mathbf{v}_2 + \cdots + c_k \mathbf{v}_k \quad \text{and} \\ \mathbf{v} = d_1 \mathbf{v}_1 + d_2 \mathbf{v}_2 + \cdots + d_k \mathbf{v}_k$
johnston-linear-matrix-algebra-EQ9951	337	■ 0.972	$\begin{aligned} \begin{bmatrix} F_{n+1} \\ F_n \end{bmatrix} &= \begin{bmatrix} 1 & 1 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} F_n \\ F_{n-1} \end{bmatrix} \\ &= \begin{bmatrix} 1 & 1 \\ 1 & 0 \end{bmatrix}^2 \begin{bmatrix} F_{n-1} \\ F_{n-2} \end{bmatrix} = \cdots = \begin{bmatrix} 1 & 1 \\ 1 & 0 \end{bmatrix}^n \begin{bmatrix} F_1 \\ F_0 \end{bmatrix} = \begin{bmatrix} 1 & 1 \\ 1 & 0 \end{bmatrix}^n \begin{bmatrix} 1 \\ 0 \end{bmatrix}. \end{aligned}$	$\begin{aligned} \begin{bmatrix} F_{n+1} \\ F_n \end{bmatrix} &= \begin{bmatrix} 1 & 1 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} F_n \\ F_{n-1} \end{bmatrix} \\ &= \begin{bmatrix} 1 & 1 \\ 1 & 0 \end{bmatrix}^2 \begin{bmatrix} F_{n-1} \\ F_{n-2} \end{bmatrix} = \cdots = \begin{bmatrix} 1 & 1 \\ 1 & 0 \end{bmatrix}^n \begin{bmatrix} F_1 \\ F_0 \end{bmatrix} = \begin{bmatrix} 1 & 1 \\ 1 & 0 \end{bmatrix}^n \begin{bmatrix} 1 \\ 0 \end{bmatrix}. \end{aligned}$	$\begin{aligned} \begin{bmatrix} F_{n+1} \\ F_n \end{bmatrix} &= \begin{bmatrix} 1 & 1 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} F_n \\ F_{n-1} \end{bmatrix} \\ &= \begin{bmatrix} 1 & 1 \\ 1 & 0 \end{bmatrix}^2 \begin{bmatrix} F_{n-1} \\ F_{n-2} \end{bmatrix} = \cdots = \begin{bmatrix} 1 & 1 \\ 1 & 0 \end{bmatrix}^n \begin{bmatrix} F_1 \\ F_0 \end{bmatrix} = \begin{bmatrix} 1 & 1 \\ 1 & 0 \end{bmatrix}^n \begin{bmatrix} 1 \\ 0 \end{bmatrix}. \end{aligned}$
johnston-linear-matrix-algebra-EQ1425	457	■ 0.972	$\begin{bmatrix} P_{\mathbf{u}} \end{bmatrix} = \mathbf{u} \mathbf{u}^T = \frac{1}{9} \begin{bmatrix} 1 & 2 & 2 \\ 2 & 4 & 4 \\ 2 & 4 & 4 \end{bmatrix}.$	$[P_{\mathbf{u}}] = \mathbf{u} \mathbf{u}^T = \frac{1}{9} \begin{bmatrix} 1 & 2 & 2 \\ 2 & 4 & 4 \\ 2 & 4 & 4 \end{bmatrix}.$	$[P_{\mathbf{u}}] = \mathbf{u} \mathbf{u}^T = \frac{1}{9} \begin{bmatrix} 1 & 2 & 2 \\ 2 & 4 & 4 \\ 2 & 4 & 4 \end{bmatrix}.$

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johnston-linear-matrix-algebra-EQ0455	181	■ 0.973	$\begin{aligned} x_1+x_3+x_5 &= 1 \\ x_1+x_2+x_5 &= 0 \\ x_3+x_4 &= 1 \\ x_1+x_3+x_4+x_5 &= 0 \\ x_2+x_3+x_5 &= 1 \end{aligned}$	$\begin{aligned} x_1+x_3+x_5 &= 1 \\ x_1+x_2+x_5 &= 0 \\ x_3+x_4 &= 1 \\ x_1+x_3+x_4+x_5 &= 0 \\ x_2+x_3+x_5 &= 1 \end{aligned}$	$\begin{aligned} x_1+x_3+x_5 &= 1 \\ x_1+x_2+x_5 &= 0 \\ x_3+x_4 &= 1 \\ x_1+x_3+x_4+x_5 &= 0 \\ x_2+x_3+x_5 &= 1 \end{aligned}$
johnston-linear-matrix-algebra-EQ1475	463	■ 0.973	$\left[\begin{array}{ccc} 1 & 1 & 1 \\ 1 & 2 & 4 \\ 1 & 4 & 16 \end{array}\right] \left[\begin{array}{c} c_0 \\ c_1 \\ c_2 \end{array}\right] = \left[\begin{array}{c} 1 \\ 2 \\ 10 \end{array}\right]$		$\begin{bmatrix} 1 & 1 & 1 \\ 1 & 2 & 4 \\ 1 & 4 & 16 \end{bmatrix} \begin{bmatrix} c_0 \\ c_1 \\ c_2 \end{bmatrix} = \begin{bmatrix} 1 \\ 2 \\ 10 \end{bmatrix}.$
johnston-linear-matrix-algebra-EQ0803 (1)	297	■ 0.973	$B_2 = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} \quad \text{and} \quad B_3 = \begin{bmatrix} 0 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 0 & 0 \end{bmatrix}$		$B_2 = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} \quad \text{and} \quad B_3 = \begin{bmatrix} 0 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 0 & 0 \end{bmatrix}.$
johnston-linear-matrix-algebra-EQ1291	435	■ 0.973	$P(n): "1+3+5+\cdots+(2n-1)=n^2"$	$P(n): "1+3+5+\cdots+(2n-1)=n^2"$	$P(n): "1+3+5+\cdots+(2n-1)=n^2"$
johnston-linear-matrix-algebra-EQ0665	254	■ 0.974	$A = \begin{bmatrix} 1 & 0 & 1 \\ 2 & 1 & 1 \\ 0 & 1 & -1 \\ 2 & 1 & 1 \\ 1 & 0 & 1 \end{bmatrix}$		$A = \begin{bmatrix} 1 & 0 & 1 \\ 2 & 1 & 1 \\ 0 & 1 & -1 \\ 2 & 1 & 1 \\ 1 & 0 & 1 \end{bmatrix}.$
johnston-linear-matrix-algebra-EQ0513	205	■ 0.975	$z - \mathbf{c} \cdot \mathbf{x} = 0 \left[\begin{array}{c ccc ccc} z & s_1 & s_2 & s_3 & x_1 & x_2 & x_3 & \\ \hline 1 & 0 & 0 & 0 & -2 & -1 & 1 & 0 \\ 0 & 1 & 0 & 0 & -3 & -2 & 2 & 1 \\ 0 & 0 & 1 & 0 & 1 & -1 & 1 & 2 \\ 0 & 0 & 0 & 1 & -1 & 3 & -3 & 3 \end{array} \right],$		$z - \mathbf{c} \cdot \mathbf{x} = 0 \left[\begin{array}{c ccc ccc} z & s_1 & s_2 & s_3 & x_1 & x_2 & x_3 & \\ \hline 1 & 0 & 0 & 0 & -2 & -1 & 1 & 0 \\ 0 & 1 & 0 & 0 & -3 & -2 & 2 & 1 \\ 0 & 0 & 1 & 0 & 1 & -1 & 1 & 2 \\ 0 & 0 & 0 & 1 & -1 & 3 & -3 & 3 \end{array} \right],$

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johnston-linear-matrix-algebra-EQ1456	460	0.975	$\begin{aligned} & A A^{-1} = \left[\begin{array}{cccc} a_1 & * & \cdots & * \\ 0 & a_2 & \cdots & * \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & a_n \end{array} \right] \left[\begin{array}{cccc} b_1 & * & \cdots & * \\ 0 & b_2 & \cdots & * \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & b_n \end{array} \right] \\ & = \left[\begin{array}{cccc} a_1 b_1 & * & \cdots & * \\ 0 & a_2 b_2 & \cdots & * \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & a_n b_n \end{array} \right], \end{aligned}$	$A A^{-1} = \begin{bmatrix} a_1 & * & \cdots & * \\ 0 & a_2 & \cdots & * \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & a_n \end{bmatrix} \begin{bmatrix} b_1 & * & \cdots & * \\ 0 & b_2 & \cdots & * \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & b_n \end{bmatrix} = \begin{bmatrix} a_1 b_1 & * & \cdots & * \\ 0 & a_2 b_2 & \cdots & * \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & a_n b_n \end{bmatrix},$	
johnston-linear-matrix-algebra-EQ1059	362	0.976	$\begin{array}{l} \tau(\sigma(1))=\tau(3)=4 \ \& \ \tau(\sigma(2))=\tau(2)=1 \ \& \ \tau(\sigma(3))=\tau(5)=3 \\ \tau(\sigma(4))=\tau(1)=5 \ \& \ \tau(\sigma(5))=\tau(4)=2 \ . \end{array}$	$\tau(\sigma(1))=\tau(3)=4 \quad \tau(\sigma(2))=\tau(2)=1 \quad \tau(\sigma(3))=\tau(5)=3$ $\tau(\sigma(4))=\tau(1)=5 \quad \tau(\sigma(5))=\tau(4)=2.$	$\tau(\sigma(1))=\tau(3)=4 \quad \tau(\sigma(2))=\tau(2)=1 \quad \tau(\sigma(3))=\tau(5)=3$ $\tau(\sigma(4))=\tau(1)=5 \quad \tau(\sigma(5))=\tau(4)=2.$
johnston-linear-matrix-algebra-EQ1572	474	0.976	$\left\ [\mathbf{v}]_B \right\ = \sqrt{[\mathbf{v}]_B \cdot [\mathbf{v}]_B} = \sqrt{\mathbf{v} \cdot \mathbf{v}} = \ \mathbf{v}\ .$	$\ [\mathbf{v}]_B\ = \sqrt{[\mathbf{v}]_B \cdot [\mathbf{v}]_B} = \sqrt{\mathbf{v} \cdot \mathbf{v}} = \ \mathbf{v}\ .$	$\ [\mathbf{v}]_B\ = \sqrt{[\mathbf{v}]_B \cdot [\mathbf{v}]_B} = \sqrt{\mathbf{v} \cdot \mathbf{v}} = \ \mathbf{v}\ .$

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johnston-linear-matrix-algebra-EQ1072	366	0.976	$\begin{aligned} \operatorname{det}(A) &= \operatorname{det}(\left[\mathbf{a}_1 \mid \mathbf{a}_2 \mid \cdots \mid \mathbf{a}_n\right] \\ &= \operatorname{det}\left(\left[\sum_{i_1=1}^n a_{i_1,1} \mathbf{e}_{i_1} \mid \mathbf{a}_2 \mid \cdots \mid \mathbf{a}_n\right]\right) \\ &= \sum_{i_1=1}^n a_{i_1,1} \operatorname{det}(\left[\mathbf{e}_{i_1} \mid \mathbf{a}_2 \mid \cdots \mid \mathbf{a}_n\right]) \\ &= \sum_{i_1=1}^n a_{i_1,1} \operatorname{det}\left(\left[\mathbf{e}_{i_1} \mid \sum_{i_2=1}^n a_{i_2,2} \mathbf{e}_{i_2} \mid \cdots \mid \mathbf{a}_n\right]\right) \\ &= \sum_{i_1, i_2=1}^n (a_{i_1,1} a_{i_2,2}) \operatorname{det}(\left[\mathbf{e}_{i_1} \mid \mathbf{e}_{i_2} \mid \cdots \mid \mathbf{a}_n\right]) \\ &\vdots \\ &= \sum_{i_1, i_2, \dots, i_n=1}^n (a_{i_1,1} a_{i_2,2} \cdots a_{i_n,n}) \operatorname{det}(\left[\mathbf{e}_{i_1} \mid \mathbf{e}_{i_2} \mid \cdots \mid \mathbf{e}_{i_n}\right]). \end{aligned}$	$\det(A) = \det([\mathbf{a}_1 \mid \mathbf{a}_2 \mid \cdots \mid \mathbf{a}_n])$ $= \det\left(\left[\sum_{i_1=1}^n a_{i_1,1} \mathbf{e}_{i_1} \mid \mathbf{a}_2 \mid \cdots \mid \mathbf{a}_n\right]\right)$ $= \sum_{i_1=1}^n a_{i_1,1} \det([\mathbf{e}_{i_1} \mid \mathbf{a}_2 \mid \cdots \mid \mathbf{a}_n])$ $= \sum_{i_1=1}^n a_{i_1,1} \det\left(\left[\mathbf{e}_{i_1} \mid \sum_{i_2=1}^n a_{i_2,2} \mathbf{e}_{i_2} \mid \cdots \mid \mathbf{a}_n\right]\right)$ $= \sum_{i_1, i_2=1}^n (a_{i_1,1} a_{i_2,2}) \det([\mathbf{e}_{i_1} \mid \mathbf{e}_{i_2} \mid \cdots \mid \mathbf{a}_n])$ $\vdots$ $= \sum_{i_1, i_2, \dots, i_n=1}^n (a_{i_1,1} a_{i_2,2} \cdots a_{i_n,n}) \det([\mathbf{e}_{i_1} \mid \mathbf{e}_{i_2} \mid \cdots \mid \mathbf{e}_{i_n}]).$	$\det(A) = \det([\mathbf{a}_1 \mid \mathbf{a}_2 \mid \cdots \mid \mathbf{a}_n])$ $= \det\left(\left[\sum_{i_1=1}^n a_{i_1,1} \mathbf{e}_{i_1} \mid \mathbf{a}_2 \mid \cdots \mid \mathbf{a}_n\right]\right)$ $= \sum_{i_1=1}^n a_{i_1,1} \det([\mathbf{e}_{i_1} \mid \mathbf{a}_2 \mid \cdots \mid \mathbf{a}_n])$ $= \sum_{i_1=1}^n a_{i_1,1} \det\left(\left[\mathbf{e}_{i_1} \mid \sum_{i_2=1}^n a_{i_2,2} \mathbf{e}_{i_2} \mid \cdots \mid \mathbf{a}_n\right]\right)$ $= \sum_{i_1, i_2=1}^n (a_{i_1,1} a_{i_2,2}) \det([\mathbf{e}_{i_1} \mid \mathbf{e}_{i_2} \mid \cdots \mid \mathbf{a}_n])$ $\vdots$ $= \sum_{i_1, i_2, \dots, i_n=1}^n (a_{i_1,1} a_{i_2,2} \cdots a_{i_n,n}) \det([\mathbf{e}_{i_1} \mid \mathbf{e}_{i_2} \mid \cdots \mid \mathbf{e}_{i_n}]).$
johnston-linear-matrix-algebra-EQ0162	82	0.977	$a_{i,j} = \begin{cases} 1, & \text{if there is an edge between the } i\text{-th and } j\text{-th vertices} \\ 0, & \text{otherwise.} \end{cases}$	$a_{i,j} = \begin{cases} 1, & \text{if there is an edge between the } i\text{-th and } j\text{-th vertices} \\ 0, & \text{otherwise.} \end{cases}$	$a_{i,j} = \begin{cases} 1, & \text{if there is an edge between the } i\text{-th and } j\text{-th vertices} \\ 0, & \text{otherwise.} \end{cases}$
johnston-linear-matrix-algebra-EQ0263	113	0.977	$\begin{aligned} v-2w-x-2y-z &= 1 \\ 2v-2x-6y-4z &= 2 \\ 4w+y-2z &= 3 \end{aligned}$	$\begin{aligned} v-2w-x-2y-z &= 1 \\ 2v-2x-6y-4z &= 2 \\ 4w+y-2z &= 3 \end{aligned}$	$\begin{aligned} v-2w-x-2y-z &= 1 \\ 2v-2x-6y-4z &= 2 \\ 4w+y-2z &= 3 \end{aligned}$
johnston-linear-matrix-algebra-EQ0602 (2.D.1)	231	0.977	$R = \begin{bmatrix} 1 & -2 & -1 \\ 0 & 3 & 1 \\ 0 & 0 & 1 \end{bmatrix} \quad \text{and} \quad P = \begin{bmatrix} 1 & 0 & 0 \\ -2 & 1 & 0 \\ 5 & -4 & 1 \end{bmatrix}^{-1} = \begin{bmatrix} 1 & 0 & 0 \\ 2 & 1 & 0 \\ 3 & 4 & 1 \end{bmatrix}.$	$R = \begin{bmatrix} 1 & -2 & -1 \\ 0 & 3 & 1 \\ 0 & 0 & 1 \end{bmatrix} \quad \text{and} \quad P = \begin{bmatrix} 1 & 0 & 0 \\ -2 & 1 & 0 \\ 5 & -4 & 1 \end{bmatrix}^{-1} = \begin{bmatrix} 1 & 0 & 0 \\ 2 & 1 & 0 \\ 3 & 4 & 1 \end{bmatrix}.$	$R = \begin{bmatrix} 1 & -2 & -1 \\ 0 & 3 & 1 \\ 0 & 0 & 1 \end{bmatrix} \quad \text{and} \quad P = \begin{bmatrix} 1 & 0 & 0 \\ -2 & 1 & 0 \\ 5 & -4 & 1 \end{bmatrix}^{-1} = \begin{bmatrix} 1 & 0 & 0 \\ 2 & 1 & 0 \\ 3 & 4 & 1 \end{bmatrix}.$

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johnston-linear-matrix-algebra EQ0409	157	0.978	$\begin{aligned} & \left[\begin{array}{lll l} 1 & 1 & 2 & x \\ 1 & 2 & 1 & y \\ 2 & 1 & 1 & z \end{array} \right] \xrightarrow[R_3-2R_1]{R_2-R_1} \left[\begin{array}{lll l} 1 & 1 & 2 & x \\ 0 & 1 & -1 & y-x \\ 0 & -1 & -3 & z-2x \end{array} \right] \\ & \xrightarrow{R_3+R_2} \left[\begin{array}{lll l} 1 & 1 & 2 & x \\ 0 & 1 & -1 & y-x \\ 0 & 0 & -4 & z+y-3x \end{array} \right]. \end{aligned}$		
johnston-linear-matrix-algebra EQ1298	439	0.978	$R=\left[\begin{array}{cccccc} 0 & 1 & 3 & 0 & 2 & 0 \\ 0 & 0 & 0 & 1 & -1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{array}\right] \quad \text{and} \quad S=\left[\begin{array}{cccccc} 0 & 1 & 3 & 0 & 4 & 0 \\ 0 & 0 & 0 & 1 & 2 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{array}\right]$		
johnston-linear-matrix-algebra EQ0367	143	0.979	$\begin{aligned} & \left[\begin{array}{ll l} 1 & 2 & x \\ 2 & 1 & y \\ 1 & 1 & z \end{array} \right] \xrightarrow[R_3-R_1]{R_2-2R_1} \left[\begin{array}{ll l} 1 & 2 & x \\ 0 & -3 & y-2x \\ 0 & -1 & z-x \end{array} \right] \\ & \xrightarrow{R_3-\frac{1}{3}R_2} \left[\begin{array}{ll l} 1 & 2 & x \\ 0 & -3 & y-2x \\ 0 & 0 & z-\frac{1}{3}x-\frac{1}{3}y \end{array} \right]. \end{aligned}$		
johnston-linear-matrix-algebra EQ0522	209	0.979	$x_{\mathrm{p}}+0.8\; x_{\mathrm{c}}+1.5\; x_{\mathrm{pc}},$	$x_{\mathrm{p}}+0.8x_{\mathrm{c}}+1.5x_{\mathrm{pc}},$	$x_{\mathrm{p}}+0.8x_{\mathrm{c}}+1.5x_{\mathrm{pc}},$
johnston-linear-matrix-algebra EQ0478	196	0.979	$\begin{aligned} & v+w+x = 0 \\ & w+x+y = 1 \\ & x+y+z = 0 \\ & v+y+z = 1 \\ & v+w+z = 0 \end{aligned}$	$\begin{aligned} & v+w+x = 0 \\ & w+x+y = 1 \\ & x+y+z = 0 \\ & v+y+z = 1 \\ & v+w+z = 0 \end{aligned}$	$\begin{aligned} & v+w+x = 0 \\ & w+x+y = 1 \\ & x+y+z = 0 \\ & v+y+z = 1 \\ & v+w+z = 0 \end{aligned}$
johnston-linear-matrix-algebra EQ0611	235	0.979	$A=\left[\begin{array}{cccc} 2 & 4 & -1 & -1 \\ 4 & 9 & 0 & -1 \\ -6 & -9 & 7 & 6 \\ -2 & -2 & 9 & 0 \end{array}\right].$	$A=\left[\begin{array}{cccc} 2 & 4 & -1 & -1 \\ 4 & 9 & 0 & -1 \\ -6 & -9 & 7 & 6 \\ -2 & -2 & 9 & 0 \end{array}\right].$	$A=\left[\begin{array}{cccc} 2 & 4 & -1 & -1 \\ 4 & 9 & 0 & -1 \\ -6 & -9 & 7 & 6 \\ -2 & -2 & 9 & 0 \end{array}\right].$

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johnston-linear-matrix-algebra-EQ0642	245	0.979	$\left[\begin{array}{cccc c} 1 & 1 & 2 & 1 & 0 \\ 0 & 1 & -1 & 0 & 2 \\ 0 & 0 & 2 & 1 & 1 \\ 0 & 0 & 0 & 3/2 & 3/2 \end{array}\right] \xrightarrow{\text{row-reduce}} \left[\begin{array}{cccc c} 1 & 0 & 0 & 0 & -3 \\ 0 & 1 & 0 & 0 & 2 \\ 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 & 1 \end{array}\right],$		
johnston-linear-matrix-algebra-EQ1106	380	0.979	$C=B-\lambda_2\mathbf{w}_2\mathbf{w}_2^T\approx\left[\begin{array}{ccc} 0.82 & -0.69 & 0.81 \\ 1.31 & -1.14 & 1.27 \\ 0.81 & -0.73 & 0.78 \end{array}\right]$	$C=B-\lambda_2\mathbf{w}_2\mathbf{w}_2^T\approx\left[\begin{array}{ccc} 0.82 & -0.69 & 0.81 \\ 1.31 & -1.14 & 1.27 \\ 0.81 & -0.73 & 0.78 \end{array}\right]$	
johnston-linear-matrix-algebra-EQ1555	473	0.979	$\begin{aligned} & c_1[\mathbf{w}_1]_B+\cdots+c_m[\mathbf{w}_m]_B \\ & \quad = [c_1\mathbf{w}_1+\cdots+c_m\mathbf{w}_m]_B = [\mathbf{0}]_B = \mathbf{0}. \end{aligned}$	$c_1[\mathbf{w}_1]_B+\cdots+c_m[\mathbf{w}_m]_B \\ = [c_1\mathbf{w}_1+\cdots+c_m\mathbf{w}_m]_B = [\mathbf{0}]_B = \mathbf{0}.$	$c_1[\mathbf{w}_1]_B+\cdots+c_m[\mathbf{w}_m]_B \\ = [c_1\mathbf{w}_1+\cdots+c_m\mathbf{w}_m]_B = [\mathbf{0}]_B = \mathbf{0}.$
johnston-linear-matrix-algebra-EQ0531	215	0.980	$(x_1+x_2)+(-x_1+x_2)=2x_2\leq 3+1=4,\quad\text{so}\quad x_2\leq 2.$	$(x_1+x_2)+(-x_1+x_2)=2x_2\leq 3+1=4,\quad\text{so}\quad x_2\leq 2.$	$(x_1+x_2)+(-x_1+x_2)=2x_2\leq 3+1=4,\quad\text{so}\quad x_2\leq 2.$
johnston-linear-matrix-algebra-EQ0080	49	0.980	$J_2=\left[\begin{array}{cc} 1 & 1 \\ 1 & 1 \end{array}\right]\quad\text{and}\quad J_3=\left[\begin{array}{ccc} 1 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{array}\right].$	$J_2=\left[\begin{array}{cc} 1 & 1 \\ 1 & 1 \end{array}\right]\quad\text{and}\quad J_3=\left[\begin{array}{ccc} 1 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{array}\right].$	
johnston-linear-matrix-algebra-EQ0507 (2.B.6)	204	0.980	$\begin{aligned} & \text{maximize: } 2x_1+x_2 \\ & \text{subject to: } 3x_1+2x_2\geq-1 \\ & \quad \quad \quad x_1-x_2\leq 2 \\ & \quad \quad \quad x_1-3x_2\geq-3 \\ & \quad \quad \quad x_1\geq 0 \end{aligned}$	$\begin{aligned} & \text{maximize: } 2x_1+x_2 \\ & \text{subject to: } 3x_1+2x_2\geq-1 \\ & \quad \quad \quad x_1-x_2\leq 2 \\ & \quad \quad \quad x_1-3x_2\geq-3 \\ & \quad \quad \quad x_1\geq 0 \end{aligned}$	
johnston-linear-matrix-algebra-EQ1391	454	0.981	$\begin{aligned} & \mathbf{w}\cdot(\mathbf{x}\times\mathbf{v}) \\ & =\mathbf{w}\cdot(x_2v_3-x_3v_2,x_3v_1-x_1v_3,x_1v_2-x_2v_1) \\ & =w_1(x_2v_3-x_3v_2)+w_2(x_3v_1-x_1v_3) \\ & \quad +w_3(x_1v_2-x_2v_1) \\ & =w_1x_2v_3-w_1x_3v_2+w_2x_3v_1-w_2x_1v_3 \\ & \quad +w_3x_1v_2-w_3x_2v_1. \end{aligned}$	$\begin{aligned} & \mathbf{w}\cdot(\mathbf{x}\times\mathbf{v}) \\ & =\mathbf{w}\cdot(x_2v_3-x_3v_2,x_3v_1-x_1v_3,x_1v_2-x_2v_1) \\ & =w_1(x_2v_3-x_3v_2)+w_2(x_3v_1-x_1v_3) \\ & \quad +w_3(x_1v_2-x_2v_1) \\ & =w_1x_2v_3-w_1x_3v_2+w_2x_3v_1-w_2x_1v_3 \\ & \quad +w_3x_1v_2-w_3x_2v_1. \end{aligned}$	
johnston-linear-matrix-algebra-EQ1573	474	0.981	$\begin{aligned} & \mathbf{v}\cdot\mathbf{w}=0\quad\text{but}\quad[\mathbf{v}]_B\cdot[\mathbf{w}]_B=-1,\text{ and} \\ & \ \mathbf{v}\ =1\quad\text{but}\quad\ [\mathbf{v}]_B\ =\sqrt{2} \end{aligned}$	$\mathbf{v}\cdot\mathbf{w}=0\quad\text{but}\quad[\mathbf{v}]_B\cdot[\mathbf{w}]_B=-1,\text{ and} \\ \ \mathbf{v}\ =1\quad\text{but}\quad\ [\mathbf{v}]_B\ =\sqrt{2}$	

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johnston-linear-matrix-algebra-EQ1215	410	0.982	$\left[\begin{array}{cccc} 1 & 0 & 0 & 1 \\ 1 & 1 & 1 & 3 \\ 1 & 2 & 4 & 9 \\ 1 & 3 & 9 & 27 \end{array}\right] \left[\begin{array}{c} c_0 \\ c_1 \\ c_2 \\ c_3 \end{array}\right] = \left[\begin{array}{c} 0 \\ 3 \\ 8 \\ 7 \end{array}\right],$	$\left[\begin{array}{cccc} 1 & 0 & 0 & 1 \\ 1 & 1 & 1 & 3 \\ 1 & 2 & 4 & 9 \\ 1 & 3 & 9 & 27 \end{array}\right] \left[\begin{array}{c} c_0 \\ c_1 \\ c_2 \\ c_3 \end{array}\right] = \left[\begin{array}{c} 0 \\ 3 \\ 8 \\ 7 \end{array}\right],$	$\left[\begin{array}{cccc} 1 & 0 & 0 & 1 \\ 1 & 1 & 1 & 3 \\ 1 & 2 & 4 & 9 \\ 1 & 3 & 9 & 27 \end{array}\right] \left[\begin{array}{c} c_0 \\ c_1 \\ c_2 \\ c_3 \end{array}\right] = \left[\begin{array}{c} 0 \\ 3 \\ 8 \\ 7 \end{array}\right],$
johnston-linear-matrix-algebra-EQ1342	448	0.982	$A^{1000}=A^{996}=A^{992}=\cdots=A^4=I$.	$A^{1000} = A^{996} = A^{992} = \dots = A^4 = I.$	$A^{1000} = A^{996} = A^{992} = \dots = A^4 = I.$
johnston-linear-matrix-algebra-EQ0322	127	0.982	$A\left(A^{-1}\mathbf{b}\right)=\left(AA^{-1}\right)\mathbf{b}=\mathbf{b}.$	$A\left(A^{-1}\mathbf{b}\right)=\left(AA^{-1}\right)\mathbf{b}=\mathbf{b}.$	$A\left(A^{-1}\mathbf{b}\right)=\left(AA^{-1}\right)\mathbf{b}=\mathbf{b}.$
johnston-linear-matrix-algebra-EQ0572	223	0.982	$\ \mathbf{v}\ _1 \stackrel{\text{def}}{=} \sum_{j=1}^n v_j $ and $\ \mathbf{v}\ _\infty \stackrel{\text{def}}{=} \max_{1 \leq j \leq n} \{ v_j \},$	$\ \mathbf{v}\ _1 \stackrel{\text{def}}{=} \sum_{j=1}^n v_j $ and $\ \mathbf{v}\ _\infty \stackrel{\text{def}}{=} \max_{1 \leq j \leq n} \{ v_j \},$	$\ \mathbf{v}\ _1 \stackrel{\text{def}}{=} \sum_{j=1}^n v_j $ and $\ \mathbf{v}\ _\infty \stackrel{\text{def}}{=} \max_{1 \leq j \leq n} \{ v_j \},$
johnston-linear-matrix-algebra-EQ0576	224	0.982	$P=[C \mid D]$ and $\hat{R}=\left[\begin{array}{c} R \\ O \end{array}\right],$	$P=[C \mid D]$ and $\hat{R}=\left[\begin{array}{c} R \\ O \end{array}\right],$	$P=[C \mid D]$ and $\hat{R}=\left[\begin{array}{c} R \\ O \end{array}\right],$
johnston-linear-matrix-algebra-EQ0612	235	0.982	$\begin{array}{l} \left[\begin{array}{cccc} 2 & 4 & -1 & -1 \\ 4 & 9 & 0 & -1 \\ -6 & -9 & 7 & 6 \\ -2 & -2 & 9 & 0 \end{array}\right] \xrightarrow[R_4+R_1]{R_2-2R_1, R_3+3R_1} \left[\begin{array}{cccc} 2 & 4 & -1 & -1 \\ 0 & 1 & 2 & 1 \\ 0 & 3 & 4 & 3 \\ 0 & 2 & 8 & -1 \end{array}\right] \\ \xrightarrow[R_3-3R_2]{R_4-2R_2} \left[\begin{array}{cccc} 2 & 4 & -1 & -1 \\ 0 & 1 & 2 & 1 \\ 0 & 0 & -2 & 0 \\ 0 & 0 & 4 & -3 \end{array}\right] \xrightarrow{R_4+2R_3} \left[\begin{array}{cccc} 2 & 4 & -1 & -1 \\ 0 & 1 & 2 & 1 \\ 0 & 0 & -2 & 0 \\ 0 & 0 & 0 & -3 \end{array}\right]. \end{array}$	$\begin{array}{l} \left[\begin{array}{cccc} 2 & 4 & -1 & -1 \\ 4 & 9 & 0 & -1 \\ -6 & -9 & 7 & 6 \\ -2 & -2 & 9 & 0 \end{array}\right] \xrightarrow[R_4+R_1]{R_2-2R_1, R_3+3R_1} \left[\begin{array}{cccc} 2 & 4 & -1 & -1 \\ 0 & 1 & 2 & 1 \\ 0 & 3 & 4 & 3 \\ 0 & 2 & 8 & -1 \end{array}\right] \\ \xrightarrow[R_3-3R_2]{R_4-2R_2} \left[\begin{array}{cccc} 2 & 4 & -1 & -1 \\ 0 & 1 & 2 & 1 \\ 0 & 0 & -2 & 0 \\ 0 & 0 & 4 & -3 \end{array}\right] \xrightarrow{R_4+2R_3} \left[\begin{array}{cccc} 2 & 4 & -1 & -1 \\ 0 & 1 & 2 & 1 \\ 0 & 0 & -2 & 0 \\ 0 & 0 & 0 & -3 \end{array}\right]. \end{array}$	$\begin{array}{l} \left[\begin{array}{cccc} 2 & 4 & -1 & -1 \\ 4 & 9 & 0 & -1 \\ -6 & -9 & 7 & 6 \\ -2 & -2 & 9 & 0 \end{array}\right] \xrightarrow[R_4+R_1]{R_2-2R_1, R_3+3R_1} \left[\begin{array}{cccc} 2 & 4 & -1 & -1 \\ 0 & 1 & 2 & 1 \\ 0 & 3 & 4 & 3 \\ 0 & 2 & 8 & -1 \end{array}\right] \\ \xrightarrow[R_3-3R_2]{R_4-2R_2} \left[\begin{array}{cccc} 2 & 4 & -1 & -1 \\ 0 & 1 & 2 & 1 \\ 0 & 0 & -2 & 0 \\ 0 & 0 & 4 & -3 \end{array}\right] \xrightarrow{R_4+2R_3} \left[\begin{array}{cccc} 2 & 4 & -1 & -1 \\ 0 & 1 & 2 & 1 \\ 0 & 0 & -2 & 0 \\ 0 & 0 & 0 & -3 \end{array}\right]. \end{array}$
johnston-linear-matrix-algebra-EQ0786	292	0.982	$\left[\begin{array}{cccc} 2 & 1 & -1 & 0 \\ 0 & -2 & 1 & 3 \\ +0 & 0 & 1 & 0 \\ -1 & 3 & 0 & 2 \end{array}\right]$	$\left[\begin{array}{cccc} 2 & 1 & -1 & 0 \\ 0 & -2 & 1 & 3 \\ +0 & 0 & 1 & 0 \\ -1 & 3 & 0 & 2 \end{array}\right]$	$\left[\begin{array}{cccc} 2 & 1 & -1 & 0 \\ 0 & -2 & 1 & 3 \\ +0 & 0 & 1 & 0 \\ -1 & 3 & 0 & 2 \end{array}\right]$

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johnston-linear-matrix-algebra EQ1054	360	0.983	$\begin{aligned} & x_1 = \frac{\operatorname{det}\left(\operatorname{det}\left(\begin{bmatrix} b_1 & a_{1,2} & a_{1,3} \\ b_2 & a_{2,2} & a_{2,3} \\ b_3 & a_{3,2} & a_{3,3} \end{bmatrix}\right)\right)}{\operatorname{det}\left(\operatorname{det}\left(\begin{bmatrix} a_{1,1} & a_{1,2} & a_{1,3} \\ a_{2,1} & a_{2,2} & a_{2,3} \\ a_{3,1} & a_{3,2} & a_{3,3} \end{bmatrix}\right)\right)} = \\ & \frac{b_1 a_{2,2} a_{3,3} + a_{1,2} a_{2,3} b_3 + a_{1,3} b_2 a_{3,2} - b_1 a_{2,3} a_{3,2} - a_{1,2} b_2 a_{3,3} - a_{1,3} a_{2,2} b_3}{a_{1,1} a_{2,2} a_{3,3} + a_{1,2} a_{2,3} a_{3,1} + a_{1,3} a_{2,1} a_{3,2} - a_{1,1} a_{2,3} a_{3,2} - a_{1,2} a_{2,1} a_{3,3} - a_{1,3} a_{2,2} a_{3,1}}, \end{aligned}$	$x_1 = \frac{\det\left(\begin{bmatrix} b_1 & a_{1,2} & a_{1,3} \\ b_2 & a_{2,2} & a_{2,3} \\ b_3 & a_{3,2} & a_{3,3} \end{bmatrix}\right)}{\det\left(\begin{bmatrix} a_{1,1} & a_{1,2} & a_{1,3} \\ a_{2,1} & a_{2,2} & a_{2,3} \\ a_{3,1} & a_{3,2} & a_{3,3} \end{bmatrix}\right)} = \frac{b_1 a_{2,2} a_{3,3} + a_{1,2} a_{2,3} b_3 + a_{1,3} b_2 a_{3,2} - b_1 a_{2,3} a_{3,2} - a_{1,2} b_2 a_{3,3} - a_{1,3} a_{2,2} b_3}{a_{1,1} a_{2,2} a_{3,3} + a_{1,2} a_{2,3} a_{3,1} + a_{1,3} a_{2,1} a_{3,2} - a_{1,1} a_{2,3} a_{3,2} - a_{1,2} a_{2,1} a_{3,3} - a_{1,3} a_{2,2} a_{3,1}},$	$x_1 = \frac{\det\left(\begin{bmatrix} b_1 & a_{1,2} & a_{1,3} \\ b_2 & a_{2,2} & a_{2,3} \\ b_3 & a_{3,2} & a_{3,3} \end{bmatrix}\right)}{\det\left(\begin{bmatrix} a_{1,1} & a_{1,2} & a_{1,3} \\ a_{2,1} & a_{2,2} & a_{2,3} \\ a_{3,1} & a_{3,2} & a_{3,3} \end{bmatrix}\right)} = \frac{b_1 a_{2,2} a_{3,3} + a_{1,2} a_{2,3} b_3 + a_{1,3} b_2 a_{3,2} - b_1 a_{2,3} a_{3,2} - a_{1,2} b_2 a_{3,3} - a_{1,3} a_{2,2} b_3}{a_{1,1} a_{2,2} a_{3,3} + a_{1,2} a_{2,3} a_{3,1} + a_{1,3} a_{2,1} a_{3,2} - a_{1,1} a_{2,3} a_{3,2} - a_{1,2} a_{2,1} a_{3,3} - a_{1,3} a_{2,2} a_{3,1}},$
johnston-linear-matrix-algebra EQ0229	105	0.983	$\mid$		—
johnston-linear-matrix-algebra EQ1130	391	0.983	$\overline{B \, \overline{\mathbf{v}}} = \overline{\bar{B} \, \mathbf{v}} = \overline{\overline{B \, \mathbf{v}}} = \overline{\bar{\lambda} \, \mathbf{v}} = \bar{\lambda} \, \overline{\mathbf{v}}.$	$B\overline{\mathbf{v}} = \overline{B\mathbf{v}} = \overline{\overline{B\mathbf{v}}} = \overline{\bar{\lambda}\mathbf{v}} = \bar{\lambda}\overline{\mathbf{v}}.$	$B\overline{\mathbf{v}} = \overline{B\mathbf{v}} = \overline{\overline{B\mathbf{v}}} = \overline{\bar{\lambda}\mathbf{v}} = \bar{\lambda}\overline{\mathbf{v}}.$
johnston-linear-matrix-algebra EQ1318	443	0.983	$c_{k+1} = (m-k) c_k \quad \text{and} \quad s_{k+1}(x) = (m-k) s_k(x) + (x-r) (c_k q'(x) + s'_k(x)).$	$c_{k+1} = (m-k) c_k \quad \text{and} \quad s_{k+1}(x) = (m-k) s_k(x) + (x-r) (c_k q'(x) + s'_k(x)).$	$c_{k+1} = (m-k) c_k \quad \text{and} \quad s_{k+1}(x) = (m-k) s_k(x) + (x-r) (c_k q'(x) + s'_k(x)).$
johnston-linear-matrix-algebra EQ0737	279	0.983	$\begin{aligned} & 0 = \operatorname{det}\left(\operatorname{det}\left(\begin{bmatrix} \mathbf{e}_1 & \cdots & \mathbf{e}_i + \mathbf{e}_j & \cdots & \mathbf{e}_i + \mathbf{e}_j & \cdots & \mathbf{e}_n \end{bmatrix}\right)\right) \\ & = \operatorname{det}\left(\operatorname{det}\left(\begin{bmatrix} \mathbf{e}_1 & \cdots & \mathbf{e}_i & \cdots & \mathbf{e}_i + \mathbf{e}_j & \cdots & \mathbf{e}_n \end{bmatrix}\right)\right) \\ & \quad + \operatorname{det}\left(\operatorname{det}\left(\begin{bmatrix} \mathbf{e}_1 & \cdots & \mathbf{e}_j & \cdots & \mathbf{e}_i + \mathbf{e}_j & \cdots & \mathbf{e}_n \end{bmatrix}\right)\right) \\ & = \operatorname{det}\left(\operatorname{det}\left(\begin{bmatrix} \mathbf{e}_1 & \cdots & \mathbf{e}_i & \cdots & \mathbf{e}_i & \cdots & \mathbf{e}_n \end{bmatrix}\right)\right) \quad (\text{not invertible, det} = 0) \\ & \quad + \operatorname{det}\left(\operatorname{det}\left(\begin{bmatrix} \mathbf{e}_1 & \cdots & \mathbf{e}_i & \cdots & \mathbf{e}_j & \cdots & \mathbf{e}_n \end{bmatrix}\right)\right) \quad (\text{identity matrix, det} = 1) \\ & \quad + \operatorname{det}\left(\operatorname{det}\left(\begin{bmatrix} \mathbf{e}_1 & \cdots & \mathbf{e}_j & \cdots & \mathbf{e}_i & \cdots & \mathbf{e}_n \end{bmatrix}\right)\right) \quad (\text{elementary swap matrix}) \\ & \quad + \operatorname{det}\left(\operatorname{det}\left(\begin{bmatrix} \mathbf{e}_1 & \cdots & \mathbf{e}_j & \cdots & \mathbf{e}_j & \cdots & \mathbf{e}_n \end{bmatrix}\right)\right). \quad (\text{not invertible, det} = 0) \end{aligned}$	$\begin{aligned} 0 &= \det\left(\begin{bmatrix} \mathbf{e}_1 & \cdots & \mathbf{e}_i + \mathbf{e}_j & \cdots & \mathbf{e}_i + \mathbf{e}_j & \cdots & \mathbf{e}_n \end{bmatrix}\right) \\ &= \det\left(\begin{bmatrix} \mathbf{e}_1 & \cdots & \mathbf{e}_i & \cdots & \mathbf{e}_i + \mathbf{e}_j & \cdots & \mathbf{e}_n \end{bmatrix}\right) \\ &\quad + \det\left(\begin{bmatrix} \mathbf{e}_1 & \cdots & \mathbf{e}_j & \cdots & \mathbf{e}_i + \mathbf{e}_j & \cdots & \mathbf{e}_n \end{bmatrix}\right) \\ &= \det\left(\begin{bmatrix} \mathbf{e}_1 & \cdots & \mathbf{e}_i & \cdots & \mathbf{e}_i & \cdots & \mathbf{e}_n \end{bmatrix}\right) \quad (\text{not invertible, det} = 0) \\ &\quad + \det\left(\begin{bmatrix} \mathbf{e}_1 & \cdots & \mathbf{e}_i & \cdots & \mathbf{e}_j & \cdots & \mathbf{e}_n \end{bmatrix}\right) \quad (\text{identity matrix, det} = 1) \\ &\quad + \det\left(\begin{bmatrix} \mathbf{e}_1 & \cdots & \mathbf{e}_j & \cdots & \mathbf{e}_i & \cdots & \mathbf{e}_n \end{bmatrix}\right) \quad (\text{elementary swap matrix}) \\ &\quad + \det\left(\begin{bmatrix} \mathbf{e}_1 & \cdots & \mathbf{e}_j & \cdots & \mathbf{e}_j & \cdots & \mathbf{e}_n \end{bmatrix}\right). \quad (\text{not invertible, det} = 0) \end{aligned}$	$\begin{aligned} 0 &= \det\left(\begin{bmatrix} \mathbf{e}_1 & \cdots & \mathbf{e}_i + \mathbf{e}_j & \cdots & \mathbf{e}_i + \mathbf{e}_j & \cdots & \mathbf{e}_n \end{bmatrix}\right) \\ &= \det\left(\begin{bmatrix} \mathbf{e}_1 & \cdots & \mathbf{e}_i & \cdots & \mathbf{e}_i + \mathbf{e}_j & \cdots & \mathbf{e}_n \end{bmatrix}\right) \\ &\quad + \det\left(\begin{bmatrix} \mathbf{e}_1 & \cdots & \mathbf{e}_j & \cdots & \mathbf{e}_i + \mathbf{e}_j & \cdots & \mathbf{e}_n \end{bmatrix}\right) \\ &= \det\left(\begin{bmatrix} \mathbf{e}_1 & \cdots & \mathbf{e}_i & \cdots & \mathbf{e}_i & \cdots & \mathbf{e}_n \end{bmatrix}\right) \quad (\text{not invertible, det} = 0) \\ &\quad + \det\left(\begin{bmatrix} \mathbf{e}_1 & \cdots & \mathbf{e}_i & \cdots & \mathbf{e}_j & \cdots & \mathbf{e}_n \end{bmatrix}\right) \quad (\text{identity matrix, det} = 1) \\ &\quad + \det\left(\begin{bmatrix} \mathbf{e}_1 & \cdots & \mathbf{e}_j & \cdots & \mathbf{e}_i & \cdots & \mathbf{e}_n \end{bmatrix}\right) \quad (\text{elementary swap matrix}) \\ &\quad + \det\left(\begin{bmatrix} \mathbf{e}_1 & \cdots & \mathbf{e}_j & \cdots & \mathbf{e}_j & \cdots & \mathbf{e}_n \end{bmatrix}\right). \quad (\text{not invertible, det} = 0) \end{aligned}$

Identifier	Page	Conf.	LaTeX source	Rendered	Image
johnston-linear-matrix-algebra-EQ1057	361	0.984	$\sigma \circ \sigma^{-1} = \text{id}$ and $\sigma^{-1} \circ \sigma = \text{id}$.	$\sigma \circ \sigma^{-1} = \text{id}$ and $\sigma^{-1} \circ \sigma = \text{id}$.	$\sigma \circ \sigma^{-1} = \text{id}$ and $\sigma^{-1} \circ \sigma = \text{id}$.
johnston-linear-matrix-algebra-EQ0021	360	0.985	$\begin{aligned} 0 &\leq \ \mathbf{v}\ ^2 \ \mathbf{w}\ ^2 - 2\ \mathbf{w}\ (\mathbf{v} \cdot \mathbf{w})^2 / \ \mathbf{w}\ + (\mathbf{v} \cdot \mathbf{w})^2 \ \mathbf{w}\ ^2 / \ \mathbf{w}\ ^2 \\ &= \ \mathbf{v}\ ^2 \ \mathbf{w}\ ^2 - (\mathbf{v} \cdot \mathbf{w})^2. \end{aligned}$	$0 \leq \ \mathbf{v}\ ^2 \ \mathbf{w}\ ^2 - 2\ \mathbf{w}\ (\mathbf{v} \cdot \mathbf{w})^2 / \ \mathbf{w}\ + (\mathbf{v} \cdot \mathbf{w})^2 \ \mathbf{w}\ ^2 / \ \mathbf{w}\ ^2 \\ = \ \mathbf{v}\ ^2 \ \mathbf{w}\ ^2 - (\mathbf{v} \cdot \mathbf{w})^2.$	$0 \leq \ \mathbf{v}\ ^2 \ \mathbf{w}\ ^2 - 2\ \mathbf{w}\ (\mathbf{v} \cdot \mathbf{w})^2 / \ \mathbf{w}\ + (\mathbf{v} \cdot \mathbf{w})^2 \ \mathbf{w}\ ^2 / \ \mathbf{w}\ ^2 \\ = \ \mathbf{v}\ ^2 \ \mathbf{w}\ ^2 - (\mathbf{v} \cdot \mathbf{w})^2.$
johnston-linear-matrix-algebra-EQ0345	134	0.985	$A = \begin{bmatrix} a & b & b & \cdots & b \\ b & a & b & \cdots & b \\ b & b & a & \cdots & b \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ b & b & b & \cdots & a \end{bmatrix} \in \mathcal{M}_n.$	$A = \begin{bmatrix} a & b & b & \cdots & b \\ b & a & b & \cdots & b \\ b & b & a & \cdots & b \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ b & b & b & \cdots & a \end{bmatrix} \in \mathcal{M}_n.$	$A = \begin{bmatrix} a & b & b & \cdots & b \\ b & a & b & \cdots & b \\ b & b & a & \cdots & b \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ b & b & b & \cdots & a \end{bmatrix} \in \mathcal{M}_n.$
johnston-linear-matrix-algebra-EQ0053	250	0.985	$\begin{aligned} (2,1) &= 2(1,0) + (0,1) \\ &= -(0,1) + 2(1,1) \\ &= (1,0) + (1,1), \end{aligned}$	$\begin{aligned} (2,1) &= 2(1,0) + (0,1) \\ &= -(0,1) + 2(1,1) \\ &= (1,0) + (1,1), \end{aligned}$	$\begin{aligned} (2,1) &= 2(1,0) + (0,1) \\ &= -(0,1) + 2(1,1) \\ &= (1,0) + (1,1), \end{aligned}$
johnston-linear-matrix-algebra-EQ1343	448	0.985	$A^k = \begin{bmatrix} 1 & k \\ 0 & 1 \end{bmatrix} \text{ for all integers } k \geq 0.$	$A^k = \begin{bmatrix} 1 & k \\ 0 & 1 \end{bmatrix} \text{ for all integers } k \geq 0.$	$A^k = \begin{bmatrix} 1 & k \\ 0 & 1 \end{bmatrix} \text{ for all integers } k \geq 0.$
johnston-linear-matrix-algebra-EQ1632	481	0.985	$\begin{aligned} p(A) \mathbf{v} &= (c_k A^k + \cdots + c_1 A + c_0 I) \mathbf{v} \\ &= c_k A^k \mathbf{v} + \cdots + c_1 A \mathbf{v} + c_0 \mathbf{v} \\ &= c_k \lambda^k \mathbf{v} + \cdots + c_1 \lambda \mathbf{v} + c_0 \mathbf{v} \\ &= (c_k \lambda^k + \cdots + c_1 \lambda + c_0) \mathbf{v} = p(\lambda) \mathbf{v}. \end{aligned}$	$\begin{aligned} p(A) \mathbf{v} &= (c_k A^k + \cdots + c_1 A + c_0 I) \mathbf{v} \\ &= c_k A^k \mathbf{v} + \cdots + c_1 A \mathbf{v} + c_0 \mathbf{v} \\ &= c_k \lambda^k \mathbf{v} + \cdots + c_1 \lambda \mathbf{v} + c_0 \mathbf{v} \\ &= (c_k \lambda^k + \cdots + c_1 \lambda + c_0) \mathbf{v} = p(\lambda) \mathbf{v}. \end{aligned}$	$\begin{aligned} p(A) \mathbf{v} &= (c_k A^k + \cdots + c_1 A + c_0 I) \mathbf{v} \\ &= c_k A^k \mathbf{v} + \cdots + c_1 A \mathbf{v} + c_0 \mathbf{v} \\ &= c_k \lambda^k \mathbf{v} + \cdots + c_1 \lambda \mathbf{v} + c_0 \mathbf{v} \\ &= (c_k \lambda^k + \cdots + c_1 \lambda + c_0) \mathbf{v} = p(\lambda) \mathbf{v}. \end{aligned}$
johnston-linear-matrix-algebra-EQ0546	220	0.986	$\begin{aligned} \text{maximize: } & x_1 + 2x_2 \\ \text{subject to: } & x_1 + x_2 \leq 3 \\ & -x_1 - x_2 \leq -3 \\ & -x_1 + x_2 \leq 1 \\ & x_1, x_2 \geq 0 \end{aligned}$	$\begin{aligned} \text{maximize: } & x_1 + 2x_2 \\ \text{subject to: } & x_1 + x_2 \leq 3 \\ & -x_1 - x_2 \leq -3 \\ & -x_1 + x_2 \leq 1 \\ & x_1, x_2 \geq 0 \end{aligned}$	$\begin{aligned} \text{maximize: } & x_1 + 2x_2 \\ \text{subject to: } & x_1 + x_2 \leq 3 \\ & -x_1 - x_2 \leq -3 \\ & -x_1 + x_2 \leq 1 \\ & x_1, x_2 \geq 0 \end{aligned}$
johnston-linear-matrix-algebra-EQ0420	163	0.986	$3(1) - (-1) + 4(-1) = 0 \text{ and } 3(2) - (2) + 4(-1) = 0.$	$3(1) - (-1) + 4(-1) = 0 \text{ and } 3(2) - (2) + 4(-1) = 0.$	$3(1) - (-1) + 4(-1) = 0 \text{ and } 3(2) - (2) + 4(-1) = 0.$
johnston-linear-matrix-algebra-EQ0688	260	0.986	$[\mathbf{v}]_C = P_{C \leftarrow B} [\mathbf{v}]_B = \begin{bmatrix} 1 & 0 & 0 \\ -1 & -1 & -2 \\ 1 & 1 & 1 \end{bmatrix} \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix} = \begin{bmatrix} 1 \\ -9 \\ 6 \end{bmatrix}.$	$[\mathbf{v}]_C = P_{C \leftarrow B} [\mathbf{v}]_B = \begin{bmatrix} 1 & 0 & 0 \\ -1 & -1 & -2 \\ 1 & 1 & 1 \end{bmatrix} \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix} = \begin{bmatrix} 1 \\ -9 \\ 6 \end{bmatrix}.$	$[\mathbf{v}]_C = P_{C \leftarrow B} [\mathbf{v}]_B = \begin{bmatrix} 1 & 0 & 0 \\ -1 & -1 & -2 \\ 1 & 1 & 1 \end{bmatrix} \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix} = \begin{bmatrix} 1 \\ -9 \\ 6 \end{bmatrix}.$

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johnston-linear-matrix-algebra-EQ0084	52	0.987	$\begin{aligned} T(\mathbf{v}+\mathbf{w}) &= T(v_1+w_1, v_2+w_2) \\ &= ((v_1+w_1)-(v_2+w_2), (v_1+w_1)+(v_2+w_2)) \\ &= (v_1-v_2, v_1+v_2) + (w_1-w_2, w_1+w_2) = T(\mathbf{v}) + T(\mathbf{w}), \end{aligned}$	$\begin{aligned} T(\mathbf{v}+\mathbf{w}) &= T(v_1+w_1, v_2+w_2) \\ &= ((v_1+w_1)-(v_2+w_2), (v_1+w_1)+(v_2+w_2)) \\ &= (v_1-v_2, v_1+v_2) + (w_1-w_2, w_1+w_2) = T(\mathbf{v}) + T(\mathbf{w}), \end{aligned}$	$\begin{aligned} T(\mathbf{v}+\mathbf{w}) &= T(v_1+w_1, v_2+w_2) \\ &= ((v_1+w_1)-(v_2+w_2), (v_1+w_1)+(v_2+w_2)) \\ &= (v_1-v_2, v_1+v_2) + (w_1-w_2, w_1+w_2) = T(\mathbf{v}) + T(\mathbf{w}), \end{aligned}$
johnston-linear-matrix-algebra-EQ0729	274	0.987	$\begin{aligned} &\operatorname{det}(\left[\begin{array}{c} \mathbf{a}_1 \\ \vdots \\ \mathbf{a}_n \end{array} + c \begin{array}{c} \mathbf{v} \\ \vdots \\ \mathbf{w} \end{array}\right]) \\ &= \operatorname{det}(\left[\begin{array}{c} \mathbf{a}_1 \\ \vdots \\ \mathbf{a}_n \end{array}\right] + c \begin{array}{c} \mathbf{v} \\ \vdots \\ \mathbf{w} \end{array}) = \operatorname{det}(\left[\begin{array}{c} \mathbf{a}_1 \\ \vdots \\ \mathbf{a}_n \end{array}\right]) + c \cdot \operatorname{det}(\left[\begin{array}{c} \mathbf{v} \\ \vdots \\ \mathbf{w} \end{array}\right]). \end{aligned}$	$\begin{aligned} \det([\mathbf{a}_1 \cdots \mathbf{v} + c\mathbf{w} \cdots \mathbf{a}_n]) \\ = \det([\mathbf{a}_1 \cdots \mathbf{v} \cdots \mathbf{a}_n]) + c \cdot \det([\mathbf{a}_1 \cdots \mathbf{w} \cdots \mathbf{a}_n]). \end{aligned}$	$\begin{aligned} \det([\mathbf{a}_1 \cdots \mathbf{v} + c\mathbf{w} \cdots \mathbf{a}_n]) \\ = \det([\mathbf{a}_1 \cdots \mathbf{v} \cdots \mathbf{a}_n]) + c \cdot \det([\mathbf{a}_1 \cdots \mathbf{w} \cdots \mathbf{a}_n]). \end{aligned}$
johnston-linear-matrix-algebra-EQ0920 (3.4.2)	329	0.987	$\begin{aligned} \lambda_k \mathbf{v}_k &= A \mathbf{v}_k = c_1 A \mathbf{v}_1 + c_2 A \mathbf{v}_2 + \cdots + c_{k-1} A \mathbf{v}_{k-1} \\ &= c_1 \lambda_1 \mathbf{v}_1 + c_2 \lambda_2 \mathbf{v}_2 + \cdots + c_{k-1} \lambda_{k-1} \mathbf{v}_{k-1}. \end{aligned}$	$\begin{aligned} \lambda_k \mathbf{v}_k &= A \mathbf{v}_k = c_1 A \mathbf{v}_1 + c_2 A \mathbf{v}_2 + \cdots + c_{k-1} A \mathbf{v}_{k-1} \\ &= c_1 \lambda_1 \mathbf{v}_1 + c_2 \lambda_2 \mathbf{v}_2 + \cdots + c_{k-1} \lambda_{k-1} \mathbf{v}_{k-1}. \end{aligned}$	$\begin{aligned} \lambda_k \mathbf{v}_k &= A \mathbf{v}_k = c_1 A \mathbf{v}_1 + c_2 A \mathbf{v}_2 + \cdots + c_{k-1} A \mathbf{v}_{k-1} \\ &= c_1 \lambda_1 \mathbf{v}_1 + c_2 \lambda_2 \mathbf{v}_2 + \cdots + c_{k-1} \lambda_{k-1} \mathbf{v}_{k-1}. \end{aligned}$
johnston-linear-matrix-algebra-EQ1290	435	0.987	$\begin{aligned} 1 &= 1 \\ 1+3 &= 4 \\ 1+3+5 &= 9 \\ 1+3+5+7 &= 16 \\ 1+3+5+7+9 &= 25, \end{aligned}$	$\begin{aligned} 1 &= 1 \\ 1+3 &= 4 \\ 1+3+5 &= 9 \\ 1+3+5+7 &= 16 \\ 1+3+5+7+9 &= 25, \end{aligned}$	$\begin{aligned} 1 &= 1 \\ 1+3 &= 4 \\ 1+3+5 &= 9 \\ 1+3+5+7 &= 16 \\ 1+3+5+7+9 &= 25, \end{aligned}$
johnston-linear-matrix-algebra-EQ1322 (B.7.1)	444	0.987	$\lim_{x \rightarrow \infty} \frac{q(x+1)}{q(x)} = 1 \quad \text{and} \quad \lim_{x \rightarrow \infty} q(x)c^x = 0.$	$\lim_{x \rightarrow \infty} \frac{q(x+1)}{q(x)} = 1 \quad \text{and} \quad \lim_{x \rightarrow \infty} q(x)c^x = 0.$	$\lim_{x \rightarrow \infty} \frac{q(x+1)}{q(x)} = 1 \quad \text{and} \quad \lim_{x \rightarrow \infty} q(x)c^x = 0.$
johnston-linear-matrix-algebra-EQ1351	449	0.987	$\begin{aligned} A \mathbf{I}_n &= \left[A \begin{array}{c} \mathbf{e}_1 \\ \vdots \\ \mathbf{e}_n \end{array} \right] \\ &= \left[\begin{array}{c} A \mathbf{e}_1 \\ \vdots \\ A \mathbf{e}_n \end{array} \right] \\ &= A, \end{aligned}$	$\begin{aligned} A \mathbf{I}_n &= [A \mathbf{e}_1 A \mathbf{e}_2 \cdots A \mathbf{e}_n] \\ &= [\mathbf{a}_1 \mathbf{a}_2 \cdots \mathbf{a}_n] \\ &= A, \end{aligned}$	$\begin{aligned} A \mathbf{I}_n &= [A \mathbf{e}_1 A \mathbf{e}_2 \cdots A \mathbf{e}_n] \\ &= [\mathbf{a}_1 \mathbf{a}_2 \cdots \mathbf{a}_n] \\ &= A, \end{aligned}$
johnston-linear-matrix-algebra-EQ0766	287	0.987	$\begin{aligned} \det(\left[\begin{array}{ccc} a & b & c \\ 0 & e & f \\ 0 & h & i \end{array}\right]) &= \det(\left[\begin{array}{ccc} a & b & c \\ 0 & e & f \\ 0 & 0 & i \end{array}\right]) + \det(\left[\begin{array}{ccc} a & 0 & c \\ 0 & 0 & f \\ 0 & h & i \end{array}\right]), \end{aligned}$	$\det\left(\left[\begin{array}{ccc} a & b & c \\ 0 & e & f \\ 0 & h & i \end{array}\right]\right) = \det\left(\left[\begin{array}{ccc} a & b & c \\ 0 & e & f \\ 0 & 0 & i \end{array}\right]\right) + \det\left(\left[\begin{array}{ccc} a & 0 & c \\ 0 & 0 & f \\ 0 & h & i \end{array}\right]\right),$	$\det\left(\left[\begin{array}{ccc} a & b & c \\ 0 & e & f \\ 0 & h & i \end{array}\right]\right) = \det\left(\left[\begin{array}{ccc} a & b & c \\ 0 & e & f \\ 0 & 0 & i \end{array}\right]\right) + \det\left(\left[\begin{array}{ccc} a & 0 & c \\ 0 & 0 & f \\ 0 & h & i \end{array}\right]\right),$

Identifier	Page	Conf.	LaTeX source	Rendered	Image
johnston-linear-matrix-algebra-EQ0118	64	■ 0.987	$\begin{aligned} \left\{ \left[R^{\pi/4} \right] \right\} &= \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & -1 \\ 1 & 1 \end{bmatrix} \\ \left[R^{\pi/4} \right] \mathbf{v} &= \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & -1 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} 1 \\ 3 \end{bmatrix} = \begin{bmatrix} -\sqrt{2} \\ 2\sqrt{2} \end{bmatrix} \end{aligned}$	$\left[R^{\pi/4} \right] = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & -1 \\ 1 & 1 \end{bmatrix}, \quad \text{so}$ $\left[R^{\pi/4} \right] \mathbf{v} = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & -1 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} 1 \\ 3 \end{bmatrix} = \begin{bmatrix} -\sqrt{2} \\ 2\sqrt{2} \end{bmatrix}.$	$\left[R^{\pi/4} \right] = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & -1 \\ 1 & 1 \end{bmatrix}, \quad \text{so}$ $\left[R^{\pi/4} \right] \mathbf{v} = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & -1 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} 1 \\ 3 \end{bmatrix} = \begin{bmatrix} -\sqrt{2} \\ 2\sqrt{2} \end{bmatrix}.$
johnston-linear-matrix-algebra-EQ1188	405	■ 0.987	$x_{\{n\}}=8\; x_{\{n-1\}}-17\; x_{\{n-2\}}+10\; x_{\{n-3\}} \; \text{for all } n \geq 3,$	$x_n = 8x_{n-1} - 17x_{n-2} + 10x_{n-3} \; \text{for all } n \geq 3,$	$x_n = 8x_{n-1} - 17x_{n-2} + 10x_{n-3} \; \text{for all } n \geq 3,$
johnston-linear-matrix-algebra-EQ0085	52	■ 0.988	$\begin{aligned} T(c\; \mathbf{v}) &= T\left(c\; v_{\{1\}},\; c\; v_{\{2\}} \right) \\ &= \left(c\; v_{\{1\}}-c\; v_{\{2\}},\; c\; v_{\{1\}}+c\; v_{\{2\}} \right) \\ &= c\left(v_{\{1\}}-v_{\{2\}},\; v_{\{1\}}+v_{\{2\}} \right) = c\; T(\mathbf{v}). \end{aligned}$	$T(c\mathbf{v}) = T(cv_1, cv_2) = (cv_1 - cv_2, cv_1 + cv_2) = c(v_1 - v_2, v_1 + v_2) = cT(\mathbf{v}).$	$T(c\mathbf{v}) = T(cv_1, cv_2) = (cv_1 - cv_2, cv_1 + cv_2) = c(v_1 - v_2, v_1 + v_2) = cT(\mathbf{v}).$
johnston-linear-matrix-algebra-EQ0782	291	■ 0.988	$\begin{aligned} \det(A) &= a_{i,1}c_{i,1} + a_{i,2}c_{i,2} + \cdots + a_{i,n}c_{i,n} \quad \text{for all } 1 \leq i \leq n \quad \text{and} \\ \det(A) &= a_{1,j}c_{1,j} + a_{2,j}c_{2,j} + \cdots + a_{n,j}c_{n,j} \quad \text{for all } 1 \leq j \leq n. \end{aligned}$	$\det(A) = a_{i,1}c_{i,1} + a_{i,2}c_{i,2} + \cdots + a_{i,n}c_{i,n} \quad \text{for all } 1 \leq i \leq n \quad \text{and}$ $\det(A) = a_{1,j}c_{1,j} + a_{2,j}c_{2,j} + \cdots + a_{n,j}c_{n,j} \quad \text{for all } 1 \leq j \leq n.$	$\det(A) = a_{i,1}c_{i,1} + a_{i,2}c_{i,2} + \cdots + a_{i,n}c_{i,n} \quad \text{for all } 1 \leq i \leq n \quad \text{and}$ $\det(A) = a_{1,j}c_{1,j} + a_{2,j}c_{2,j} + \cdots + a_{n,j}c_{n,j} \quad \text{for all } 1 \leq j \leq n.$
johnston-linear-matrix-algebra-EQ0984	345	■ 0.988	$\begin{aligned} p(A) &= p\left(P\; D\; P^{-1}\right) = c_k\left(P\; D\; P^{-1}\right)^k + \cdots + c_2\left(P\; D\; P^{-1}\right)^2 + c_1\left(P\; D\; P^{-1}\right) + c_0I \\ &= c_k\left(P\; D^k\; P^{-1}\right) + \cdots + c_2\left(P\; D^2\; P^{-1}\right) + c_1\left(P\; D\; P^{-1}\right) + c_0I \\ &= P\left(c_kD^k + \cdots + c_2D^2 + c_1D + c_0I\right)P^{-1} \\ &= Pp(D)P^{-1}, \end{aligned}$	$p(A) = p\left(PDP^{-1}\right) = c_k\left(PDP^{-1}\right)^k + \cdots + c_2\left(PDP^{-1}\right)^2 + c_1\left(PDP^{-1}\right) + c_0I$ $= c_k\left(PD^kP^{-1}\right) + \cdots + c_2\left(PD^2P^{-1}\right) + c_1\left(PDP^{-1}\right) + c_0I$ $= P\left(c_kD^k + \cdots + c_2D^2 + c_1D + c_0I\right)P^{-1}$ $= Pp(D)P^{-1},$	$p(A) = p\left(PDP^{-1}\right) = c_k\left(PDP^{-1}\right)^k + \cdots + c_2\left(PDP^{-1}\right)^2 + c_1\left(PDP^{-1}\right) + c_0I$ $= c_k\left(PD^kP^{-1}\right) + \cdots + c_2\left(PD^2P^{-1}\right) + c_1\left(PDP^{-1}\right) + c_0I$ $= P\left(c_kD^k + \cdots + c_2D^2 + c_1D + c_0I\right)P^{-1}$ $= Pp(D)P^{-1},$
johnston-linear-matrix-algebra-EQ0259	113	■ 0.988	$\begin{aligned} x-y &= 2 \\ x+2y &= 4 \\ 2x-y &= 5 \end{aligned}$	$\begin{aligned} x-y &= 2 \\ x+2y &= 4 \\ 2x-y &= 5 \end{aligned}$	$\begin{aligned} x-y &= 2 \\ x+2y &= 4 \\ 2x-y &= 5 \end{aligned}$
johnston-linear-matrix-algebra-EQ0872	314	■ 0.988	$\left(\mathbf{v}^* A \mathbf{v}\right)^* = \mathbf{v}^* A^* \mathbf{v} = \mathbf{v}^* A \mathbf{v}.$	$\left(\mathbf{v}^* A \mathbf{v}\right)^* = \mathbf{v}^* A^* \mathbf{v} = \mathbf{v}^* A \mathbf{v}.$	$\left(\mathbf{v}^* A \mathbf{v}\right)^* = \mathbf{v}^* A^* \mathbf{v} = \mathbf{v}^* A \mathbf{v}.$

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johnston-linear-matrix-algebra-EQ1046	359	0.988	$A=\left[\begin{array}{cccc} a_{1,1} & \cdots & a_{1,j} & \cdots & a_{1,n} \\ a_{2,1} & \cdots & a_{2,j} & \cdots & a_{2,n} \\ \vdots & \ddots & \vdots & \ddots & \vdots \\ a_{n,1} & \cdots & a_{n,j} & \cdots & a_{n,n} \end{array}\right], A_j=\left[\begin{array}{cccc} a_{1,1} & \cdots & b_1 & \cdots & a_{1,n} \\ a_{2,1} & \cdots & b_2 & \cdots & a_{2,n} \\ \vdots & \ddots & \vdots & \ddots & \vdots \\ a_{n,1} & \cdots & b_n & \cdots & a_{n,n} \end{array}\right]$	$A=\begin{bmatrix} a_{1,1} & \cdots & a_{1,j} & \cdots & a_{1,n} \\ a_{2,1} & \cdots & a_{2,j} & \cdots & a_{2,n} \\ \vdots & \ddots & \vdots & \ddots & \vdots \\ a_{n,1} & \cdots & a_{n,j} & \cdots & a_{n,n} \end{bmatrix}, A_j=\begin{bmatrix} a_{1,1} & \cdots & b_1 & \cdots & a_{1,n} \\ a_{2,1} & \cdots & b_2 & \cdots & a_{2,n} \\ \vdots & \ddots & \vdots & \ddots & \vdots \\ a_{n,1} & \cdots & b_n & \cdots & a_{n,n} \end{bmatrix}$	$A=\begin{bmatrix} a_{1,1} & \cdots & a_{1,j} & \cdots & a_{1,n} \\ a_{2,1} & \cdots & a_{2,j} & \cdots & a_{2,n} \\ \vdots & \ddots & \vdots & \ddots & \vdots \\ a_{n,1} & \cdots & a_{n,j} & \cdots & a_{n,n} \end{bmatrix}, A_j=\begin{bmatrix} a_{1,1} & \cdots & b_1 & \cdots & a_{1,n} \\ a_{2,1} & \cdots & b_2 & \cdots & a_{2,n} \\ \vdots & \ddots & \vdots & \ddots & \vdots \\ a_{n,1} & \cdots & b_n & \cdots & a_{n,n} \end{bmatrix}$
johnston-linear-matrix-algebra-EQ1145 (0)	395	0.988	$\operatorname{diag}\left(I_2,5,\left[R^{\pi/7}\right]\right)=\left[\begin{array}{ccc ccc} 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 5 & 0 & 0 & 0 \\ \hline 0 & 0 & 0 & \cos(\pi/7) & -\sin(\pi/7) \\ 0 & 0 & 0 & \sin(\pi/7) & \cos(\pi/7) \end{array}\right].$	$\operatorname{diag}\left(I_2,5,\left[R^{\pi/7}\right]\right)=\left[\begin{array}{ccc ccc} 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 5 & 0 & 0 & 0 \\ \hline 0 & 0 & 0 & \cos(\pi/7) & -\sin(\pi/7) \\ 0 & 0 & 0 & \sin(\pi/7) & \cos(\pi/7) \end{array}\right].$	$\operatorname{diag}\left(I_2,5,\left[R^{\pi/7}\right]\right)=\left[\begin{array}{ccc ccc} 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 5 & 0 & 0 & 0 \\ \hline 0 & 0 & 0 & \cos(\pi/7) & -\sin(\pi/7) \\ 0 & 0 & 0 & \sin(\pi/7) & \cos(\pi/7) \end{array}\right].$
johnston-linear-matrix-algebra-EQ1244	418	0.988	$\overline{\overline{a+b\,i}}=\overline{a-b\,i}=a+b\,i.$	$\overline{\overline{a+bi}}=\overline{a-bi}=a+bi.$	$\overline{\overline{a+bi}}=\overline{a-bi}=a+bi.$
johnston-linear-matrix-algebra-EQ1021	354	0.988	$A=P\,D\,P^{-1}\quad\text{\rm and}\quad B=Q\,D\,Q^{-1}\quad\text{\rm for some invertible }P,Q\in\mathcal{M}_n(\mathbb{C}).$	$A=PDP^{-1}\quad\text{and}\quad B=QDQ^{-1}\quad\text{for some invertible }P,Q\in\mathcal{M}_n(\mathbb{C}).$	$A=PDP^{-1}\quad\text{and}\quad B=QDQ^{-1}\quad\text{for some invertible }P,Q\in\mathcal{M}_n(\mathbb{C}).$
johnston-linear-matrix-algebra-EQ0966	341	0.989	$A^{1/2}=\frac{1}{3}\begin{bmatrix}4&1\\1&1\end{bmatrix}\begin{bmatrix}1^{1/2}&0\\0&4^{1/2}\end{bmatrix}\begin{bmatrix}1&-1\\-1&4\end{bmatrix}$	$A^{1/2}=\frac{1}{3}\begin{bmatrix}4&1\\1&1\end{bmatrix}\begin{bmatrix}1^{1/2}&0\\0&4^{1/2}\end{bmatrix}\begin{bmatrix}1&-1\\-1&4\end{bmatrix}$	$A^{1/2}=\frac{1}{3}\begin{bmatrix}4&1\\1&1\end{bmatrix}\begin{bmatrix}1^{1/2}&0\\0&4^{1/2}\end{bmatrix}\begin{bmatrix}1&-1\\-1&4\end{bmatrix}$
johnston-linear-matrix-algebra-EQ1410	456	0.989	$\left[\begin{array}{cc c} a & b & 1 \\ c & d & 1 \end{array}\right]\xrightarrow{aR_2-cR_1}\left[\begin{array}{cc c} a & b & 1 \\ 0 & ad-bc & a-c \end{array}\right].$	$\left[\begin{array}{cc c} a & b & 1 \\ c & d & 1 \end{array}\right]\xrightarrow{aR_2-cR_1}\left[\begin{array}{cc c} a & b & 1 \\ 0 & ad-bc & a-c \end{array}\right].$	$\left[\begin{array}{cc c} a & b & 1 \\ c & d & 1 \end{array}\right]\xrightarrow{aR_2-cR_1}\left[\begin{array}{cc c} a & b & 1 \\ 0 & ad-bc & a-c \end{array}\right].$
johnston-linear-matrix-algebra-EQ1489	464	0.989	$d_1(c_1\mathbf{v}_1)+d_2(c_2\mathbf{v}_2)+\cdots+d_n(c_n\mathbf{v}_n)=\mathbf{0}.$	$d_1(c_1\mathbf{v}_1)+d_2(c_2\mathbf{v}_2)+\cdots+d_n(c_n\mathbf{v}_n)=\mathbf{0}.$	$d_1(c_1\mathbf{v}_1)+d_2(c_2\mathbf{v}_2)+\cdots+d_n(c_n\mathbf{v}_n)=\mathbf{0}.$

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johnston-linear-matrix-algebra-EQ0108	60	0.989	$\left[F_{\mathbf{u}}\right]=2\mathbf{u}^T\mathbf{u}-I=2\left[\begin{array}{l}0\\1\end{array}\right]\left[\begin{array}{ll}0&1\end{array}\right]-\left[\begin{array}{ll}1&0\\0&1\end{array}\right]=\left[\begin{array}{ll}-1&0\\0&1\end{array}\right].$	$[F_{\mathbf{u}}] = 2\mathbf{u}\mathbf{u}^T - I = 2 \begin{bmatrix} 0 & \\ 1 & \end{bmatrix} \begin{bmatrix} 0 & 1 \end{bmatrix} - \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} -1 & 0 \\ 0 & 1 \end{bmatrix}.$	$[F_{\mathbf{u}}] = 2\mathbf{u}\mathbf{u}^T - I = 2 \begin{bmatrix} 0 \\ 1 \end{bmatrix} \begin{bmatrix} 0 & 1 \end{bmatrix} - \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} -1 & 0 \\ 0 & 1 \end{bmatrix}.$
johnston-linear-matrix-algebra-EQ0430	167	0.989	$\left[\begin{array}{l}2\\0\\0\\0\\0\end{array}\right]\rightarrow\left[\begin{array}{l}2\\-1\\0\\0\\0\end{array}\right],\quad\left[\begin{array}{l}3\\6\\0\end{array}\right]\rightarrow\left[\begin{array}{l}3\\0\\6\\-1\\0\\0\end{array}\right],\quad\left[\begin{array}{l}4\\7\\0\end{array}\right]\rightarrow\left[\begin{array}{l}4\\0\\7\\-1\\0\\0\end{array}\right],\quad\left[\begin{array}{l}5\\8\\9\end{array}\right]\rightarrow\left[\begin{array}{l}5\\0\\8\\0\\9\\-1\end{array}\right].$	$\begin{bmatrix} 2 \\ 0 \\ 0 \\ 0 \\ 0 \end{bmatrix} \rightarrow \begin{bmatrix} 2 \\ -1 \\ 0 \\ 0 \\ 0 \end{bmatrix}, \quad \begin{bmatrix} 3 \\ 6 \\ 0 \end{bmatrix} \rightarrow \begin{bmatrix} 3 \\ 0 \\ 6 \\ -1 \\ 0 \\ 0 \end{bmatrix}, \quad \begin{bmatrix} 4 \\ 7 \\ 0 \end{bmatrix} \rightarrow \begin{bmatrix} 4 \\ 0 \\ 7 \\ -1 \\ 0 \\ 0 \end{bmatrix}, \quad \begin{bmatrix} 5 \\ 8 \\ 9 \end{bmatrix} \rightarrow \begin{bmatrix} 5 \\ 0 \\ 8 \\ 0 \\ 9 \\ -1 \end{bmatrix}$	
johnston-linear-matrix-algebra-EQ1250	419	0.989	$\begin{aligned} &\frac{z+\bar{z}}{2}=\frac{(a+bi)+(a-bi)}{2}=\frac{2a}{2}=a=\operatorname{Re}(z) \quad \text{and} \\ &\frac{z-\bar{z}}{2i}=\frac{(a+bi)-(a-bi)}{2i}=\frac{2bi}{2i}=b=\operatorname{Im}(z). \end{aligned}$	$\frac{z+\bar{z}}{2}=\frac{(a+bi)+(a-bi)}{2}=\frac{2a}{2}=a=\operatorname{Re}(z) \quad \text{and} \\ \frac{z-\bar{z}}{2i}=\frac{(a+bi)-(a-bi)}{2i}=\frac{2bi}{2i}=b=\operatorname{Im}(z).$	$\frac{z+\bar{z}}{2}=\frac{(a+bi)+(a-bi)}{2}=\frac{2a}{2}=a=\operatorname{Re}(z) \quad \text{and} \\ \frac{z-\bar{z}}{2i}=\frac{(a+bi)-(a-bi)}{2i}=\frac{2bi}{2i}=b=\operatorname{Im}(z).$
johnston-linear-matrix-algebra-EQ1517	469	0.989	$\begin{array}{l} \operatorname{minimize}: \mathbf{b}^T\mathbf{y} \\ \text{subject to: } \mathbf{A}^T\mathbf{y}\geq\mathbf{c} \\ \mathbf{y}\geq\mathbf{0} \end{array}$	$\begin{array}{l} \text{minimize :} \quad \mathbf{b} \cdot \mathbf{y} \\ \text{subject to: } \quad A^T \mathbf{y} \geq \mathbf{c} \\ \quad \mathbf{y} \geq \mathbf{0} \end{array}$	$\begin{array}{l} \text{minimize:} \quad \mathbf{b} \cdot \mathbf{y} \\ \text{subject to:} \quad A^T \mathbf{y} \geq \mathbf{c} \\ \quad \mathbf{y} \geq \mathbf{0} \end{array}$
johnston-linear-matrix-algebra-EQ0640 (1)	245	0.989	$\mathbf{c}=P^T\mathbf{b}=\left[\begin{array}{llll}1&0&0&0\\0&0&1&0\\0&0&0&1\\0&1&0&0\end{array}\right]\left[\begin{array}{l}0\\1\\2\\3\end{array}\right]=\left[\begin{array}{l}0\\2\\3\\1\end{array}\right].$	$\mathbf{c} = P^T \mathbf{b} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 1 & 0 & 0 \end{bmatrix} \begin{bmatrix} 0 \\ 1 \\ 2 \\ 3 \end{bmatrix} = \begin{bmatrix} 0 \\ 2 \\ 3 \\ 1 \end{bmatrix}.$	$\mathbf{c} = P^T \mathbf{b} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 1 & 0 & 0 \end{bmatrix} \begin{bmatrix} 0 \\ 1 \\ 2 \\ 3 \end{bmatrix} = \begin{bmatrix} 0 \\ 2 \\ 3 \\ 1 \end{bmatrix}.$
johnston-linear-matrix-algebra-EQ0046	40	0.989	$I_2=\left[\begin{array}{ll}1&0\\0&1\end{array}\right] \quad \text{and} \quad I_3=\left[\begin{array}{lll}1&0&0\\0&1&0\\0&0&1\end{array}\right].$	$I_2 = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \quad \text{and} \quad I_3 = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}.$	$I_2 = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \quad \text{and} \quad I_3 = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}.$

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johnston-linear-matrix-algebra_E00637	244	0.990	$P^T=\left[\begin{array}{cccc} 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 1 & 0 & 0 \end{array}\right] \quad \text{and} \quad E^{-1}=\left[\begin{array}{cccc} 1 & 0 & 0 & 0 \\ 1 & 0 & -1/2 & 1 \\ 1 & 1 & 0 & 0 \\ 0 & 1 & 1 & 0 \end{array}\right].$		$P^T=\left[\begin{array}{cccc} 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 1 & 0 & 0 \end{array}\right] \quad \text{and} \quad E^{-1}=\left[\begin{array}{cccc} 1 & 0 & 0 & 0 \\ 1 & 0 & -1/2 & 1 \\ 1 & 1 & 0 & 0 \\ 0 & 1 & 1 & 0 \end{array}\right].$
johnston-linear-matrix-algebra_E01154	396	0.990	$A=\frac{1}{4}\left[\begin{array}{ccc} 3 & 2 & -1 \\ -5 & 2 & 3 \\ -5 & 2 & 7 \end{array}\right].$		$A=\frac{1}{4}\left[\begin{array}{ccc} 3 & 2 & -1 \\ -5 & 2 & 3 \\ -5 & 2 & 7 \end{array}\right].$
johnston-linear-matrix-algebra_E01274	427	0.990	$\begin{aligned} & (x-2)\left(x+\frac{1}{3}\right) \\ &=x^2-\frac{5}{3}x-\frac{2}{3}, \end{aligned}$	$(x-2)\left(x+\frac{1}{3}\right) \\ =x^2-\frac{5}{3}x-\frac{2}{3},$	$(x-2)\left(x+\frac{1}{3}\right) \\ =x^2-\frac{5}{3}x-\frac{2}{3},$
johnston-linear-matrix-algebra_E01521	469	0.990	$\begin{array}{l} \operatorname{maximize}: \quad 0 \\ \text{subject to: } \quad (A-I)\mathbf{x}=\mathbf{0} \\ \quad \quad \quad \mathbf{x} \geq \mathbf{0} \end{array}$	$\begin{array}{l} \operatorname{maximize}: \quad 0 \\ \text{subject to: } \quad (A-I)\mathbf{x}=\mathbf{0} \\ \quad \quad \quad \mathbf{x} \geq \mathbf{0} \end{array}$	$\begin{array}{l} \operatorname{maximize}: \quad 0 \\ \text{subject to: } \quad (A-I)\mathbf{x}=\mathbf{0} \\ \quad \quad \quad \mathbf{x} \geq \mathbf{0} \end{array}$
johnston-linear-matrix-algebra_E00644	246	0.990	$A=\left[\begin{array}{c c} B & \mathbf{v} \\ \hline \mathbf{w}^T & c \end{array}\right]=\underbrace{\left[\begin{array}{c c} L & \mathbf{0} \\ \hline \mathbf{x}^T & 1 \end{array}\right]}_{\text{"new" } L}\underbrace{\left[\begin{array}{c c} U & \mathbf{y} \\ \hline \mathbf{0}^T & d \end{array}\right]}_{\text{"new" } U}=\left[\begin{array}{c c} LU & Ly \\ \hline \mathbf{x}^TU & d+\mathbf{x}^T\mathbf{y} \end{array}\right].$	$A=\left[\begin{array}{c c} B & \mathbf{v} \\ \hline \mathbf{w}^T & c \end{array}\right]=\underbrace{\left[\begin{array}{c c} L & \mathbf{0} \\ \hline \mathbf{x}^T & 1 \end{array}\right]}_{\text{"new" } L}\underbrace{\left[\begin{array}{c c} U & \mathbf{y} \\ \hline \mathbf{0}^T & d \end{array}\right]}_{\text{"new" } U}=\left[\begin{array}{c c} LU & Ly \\ \hline \mathbf{x}^TU & d+\mathbf{x}^T\mathbf{y} \end{array}\right].$	$A=\left[\begin{array}{c c} B & \mathbf{v} \\ \hline \mathbf{w}^T & c \end{array}\right]=\underbrace{\left[\begin{array}{c c} L & \mathbf{0} \\ \hline \mathbf{x}^T & 1 \end{array}\right]}_{\text{"new" } L}\underbrace{\left[\begin{array}{c c} U & \mathbf{y} \\ \hline \mathbf{0}^T & d \end{array}\right]}_{\text{"new" } U}=\left[\begin{array}{c c} LU & Ly \\ \hline \mathbf{x}^TU & d+\mathbf{x}^T\mathbf{y} \end{array}\right].$
johnston-linear-matrix-algebra_E00647	41	0.990	$A^k \stackrel{\text{def}}{=} \underbrace{A A \cdots A}_k.$	$A^k \stackrel{\text{def}}{=} \underbrace{A A \cdots A}_k.$	$A^k \stackrel{\text{def}}{=} \underbrace{A A \cdots A}_k.$
johnston-linear-matrix-algebra_E00058	44	0.990	$\mathbf{v}^T\mathbf{w}=\left[\begin{array}{cccc} v_1 & v_2 & \cdots & v_n \end{array}\right]\left[\begin{array}{c} w_1 \\ w_2 \\ \vdots \\ w_n \end{array}\right]=v_1w_1+v_2w_2+\cdots+v_nw_n=\mathbf{v}\cdot\mathbf{w}.$	$\mathbf{v}^T\mathbf{w}=\left[\begin{array}{cccc} v_1 & v_2 & \cdots & v_n \end{array}\right]\left[\begin{array}{c} w_1 \\ w_2 \\ \vdots \\ w_n \end{array}\right]=v_1w_1+v_2w_2+\cdots+v_nw_n=\mathbf{v}\cdot\mathbf{w}.$	$\mathbf{v}^T\mathbf{w}=\left[\begin{array}{cccc} v_1 & v_2 & \cdots & v_n \end{array}\right]\left[\begin{array}{c} w_1 \\ w_2 \\ \vdots \\ w_n \end{array}\right]=v_1w_1+v_2w_2+\cdots+v_nw_n=\mathbf{v}\cdot\mathbf{w}.$
johnston-linear-matrix-algebra_E00960	339	0.990	$\begin{array}{l} \lambda_1=-2 \quad \mathbf{v}_1=(1,1,-1,-1,-1) \\ \lambda_2=-1 \quad \mathbf{v}_2=(1,-1,0,0,0) \\ \lambda_3=0 \quad \mathbf{v}_3=(0,0,1,-1,0) \\ \lambda_4=0 \quad \mathbf{v}_4=(0,0,0,1,-1) \\ \lambda_5=3 \quad \mathbf{v}_5=(3,3,2,2,2). \end{array}$	$\begin{array}{l} \lambda_1=-2 \quad \mathbf{v}_1=(1,1,-1,-1,-1) \\ \lambda_2=-1 \quad \mathbf{v}_2=(1,-1,0,0,0) \\ \lambda_3=0 \quad \mathbf{v}_3=(0,0,1,-1,0) \\ \lambda_4=0 \quad \mathbf{v}_4=(0,0,0,1,-1) \\ \lambda_5=3 \quad \mathbf{v}_5=(3,3,2,2,2). \end{array}$	$\begin{array}{l} \lambda_1=-2 \quad \mathbf{v}_1=(1,1,-1,-1,-1) \\ \lambda_2=-1 \quad \mathbf{v}_2=(1,-1,0,0,0) \\ \lambda_3=0 \quad \mathbf{v}_3=(0,0,1,-1,0) \\ \lambda_4=0 \quad \mathbf{v}_4=(0,0,0,1,-1) \\ \lambda_5=3 \quad \mathbf{v}_5=(3,3,2,2,2). \end{array}$

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johnston-linear-matrix-algebra-EQ1336	447	0.990	$\begin{aligned} \mathbf{v} \cdot (c\mathbf{w}) &= \overline{v_1}(cw_1) + \cdots + \overline{v_n}(cw_n) \\ &= c(\overline{v_1}w_1 + \cdots + \overline{v_n}w_n) \\ &= c(\mathbf{v} \cdot \mathbf{w}). \end{aligned}$	$\mathbf{v} \cdot (c\mathbf{w}) = \overline{v_1}(cw_1) + \cdots + \overline{v_n}(cw_n) \\ = c(\overline{v_1}w_1 + \cdots + \overline{v_n}w_n) \\ = c(\mathbf{v} \cdot \mathbf{w}).$	$\mathbf{v} \cdot (c\mathbf{w}) = \overline{v_1}(cw_1) + \cdots + \overline{v_n}(cw_n) \\ = c(\overline{v_1}w_1 + \cdots + \overline{v_n}w_n) \\ = c(\mathbf{v} \cdot \mathbf{w}).$
johnston-linear-matrix-algebra-EQ1581	475	0.990	$\begin{aligned} &\vdots \\ &= c^n \det([\mathbf{a}_1 \mid \mathbf{a}_2 \mid \cdots \mid \mathbf{a}_n]) = c^n \det(A). \end{aligned}$	$\vdots \\ = c^n \det([\mathbf{a}_1 \mid \mathbf{a}_2 \mid \cdots \mid \mathbf{a}_n]) = c^n \det(A).$	$\vdots \\ = c^n \det([\mathbf{a}_1 \mid \mathbf{a}_2 \mid \cdots \mid \mathbf{a}_n]) = c^n \det(A).$
johnston-linear-matrix-algebra-EQ1484	463	0.991	$c_1 \mathbf{v}_1 + \cdots + c_k \mathbf{v}_k + 0 \mathbf{w}_1 + \cdots + 0 \mathbf{w}_m = \mathbf{0}.$	$c_1 \mathbf{v}_1 + \cdots + c_k \mathbf{v}_k + 0 \mathbf{w}_1 + \cdots + 0 \mathbf{w}_m = \mathbf{0}.$	$c_1 \mathbf{v}_1 + \cdots + c_k \mathbf{v}_k + 0 \mathbf{w}_1 + \cdots + 0 \mathbf{w}_m = \mathbf{0}.$
johnston-linear-matrix-algebra-EQ1549	473	0.991	$\mathbf{v} = (c_1 + d_1) \mathbf{v}_1 + (c_2 + d_2) \mathbf{v}_2 + \cdots + (c_k + d_k) \mathbf{v}_k.$	$\mathbf{v} = (c_1 + d_1) \mathbf{v}_1 + (c_2 + d_2) \mathbf{v}_2 + \cdots + (c_k + d_k) \mathbf{v}_k.$	$\mathbf{v} = (c_1 + d_1) \mathbf{v}_1 + (c_2 + d_2) \mathbf{v}_2 + \cdots + (c_k + d_k) \mathbf{v}_k.$
johnston-linear-matrix-algebra-EQ0853	310	0.991	$\begin{aligned} c_0 = \det(A) &= \lambda_1 \lambda_2 \cdots \lambda_n \quad \text{and} \\ (-1)^{n-1} c_{n-1} &= \text{tr}(A) = \lambda_1 + \lambda_2 + \cdots + \lambda_n. \end{aligned}$	$c_0 = \det(A) = \lambda_1 \lambda_2 \cdots \lambda_n \quad \text{and} \\ (-1)^{n-1} c_{n-1} = \text{tr}(A) = \lambda_1 + \lambda_2 + \cdots + \lambda_n.$	$c_0 = \det(A) = \lambda_1 \lambda_2 \cdots \lambda_n \quad \text{and} \\ (-1)^{n-1} c_{n-1} = \text{tr}(A) = \lambda_1 + \lambda_2 + \cdots + \lambda_n.$
johnston-linear-matrix-algebra-EQ1535	471	0.991	$\begin{aligned} L &= \begin{bmatrix} 1 & 0 & 0 & 0 & 0 & 0 \\ \frac{-1}{2} & 1 & 0 & 0 & 0 & 0 \\ 0 & \frac{-2}{5} & 1 & 0 & 0 & 0 \\ 0 & 0 & \frac{-5}{18} & 1 & 0 & 0 \\ 0 & 0 & 0 & \frac{-18}{67} & 1 & 0 \\ 0 & \frac{2}{5} & \frac{1}{9} & \frac{2}{67} & \frac{-19}{37} & 1 \end{bmatrix}, \text{ and} \\ U &= \begin{bmatrix} 2 & -1 & 0 & 0 & 0 & 0 \\ 0 & \frac{5}{2} & -1 & 0 & 1 & 0 \\ 0 & 0 & \frac{18}{5} & -1 & \frac{2}{5} & 0 \\ 0 & 0 & 0 & \frac{67}{18} & \frac{-8}{9} & 0 \\ 0 & 0 & 0 & 0 & \frac{185}{67} & -1 \\ 0 & 0 & 0 & 0 & 0 & \frac{55}{37} \end{bmatrix}. \end{aligned}$	$L = \begin{bmatrix} 1 & 0 & 0 & 0 & 0 & 0 \\ \frac{-1}{2} & 1 & 0 & 0 & 0 & 0 \\ 0 & \frac{-2}{5} & 1 & 0 & 0 & 0 \\ 0 & 0 & \frac{-5}{18} & 1 & 0 & 0 \\ 0 & 0 & 0 & \frac{-18}{67} & 1 & 0 \\ 0 & \frac{2}{5} & \frac{1}{9} & \frac{2}{67} & \frac{-19}{37} & 1 \end{bmatrix}, \text{ and}$ $U = \begin{bmatrix} 2 & -1 & 0 & 0 & 0 & 0 \\ 0 & \frac{5}{2} & -1 & 0 & 1 & 0 \\ 0 & 0 & \frac{18}{5} & -1 & \frac{2}{5} & 0 \\ 0 & 0 & 0 & \frac{67}{18} & \frac{-8}{9} & 0 \\ 0 & 0 & 0 & 0 & \frac{185}{67} & -1 \\ 0 & 0 & 0 & 0 & 0 & \frac{55}{37} \end{bmatrix}.$	$L = \begin{bmatrix} 1 & 0 & 0 & 0 & 0 & 0 \\ \frac{-1}{2} & 1 & 0 & 0 & 0 & 0 \\ 0 & \frac{-2}{5} & 1 & 0 & 0 & 0 \\ 0 & 0 & \frac{-5}{18} & 1 & 0 & 0 \\ 0 & 0 & 0 & \frac{-18}{67} & 1 & 0 \\ 0 & \frac{2}{5} & \frac{1}{9} & \frac{2}{67} & \frac{-19}{37} & 1 \end{bmatrix}, \text{ and}$ $U = \begin{bmatrix} 2 & -1 & 0 & 0 & 0 & 0 \\ 0 & \frac{5}{2} & -1 & 0 & 1 & 0 \\ 0 & 0 & \frac{18}{5} & -1 & \frac{2}{5} & 0 \\ 0 & 0 & 0 & \frac{67}{18} & \frac{-8}{9} & 0 \\ 0 & 0 & 0 & 0 & \frac{185}{67} & -1 \\ 0 & 0 & 0 & 0 & 0 & \frac{55}{37} \end{bmatrix}.$
johnston-linear-matrix-algebra-EQ0882	56	0.991	$A B = \begin{bmatrix} \frac{\mathbf{a}_1^T B}{\mathbf{a}_2^T B} \\ \vdots \\ \frac{\mathbf{a}_m^T B}{\mathbf{a}_m^T B} \end{bmatrix}.$	$AB = \begin{bmatrix} \frac{\mathbf{a}_1^T B}{\mathbf{a}_2^T B} \\ \vdots \\ \frac{\mathbf{a}_m^T B}{\mathbf{a}_m^T B} \end{bmatrix}.$	$AB = \begin{bmatrix} \frac{\mathbf{a}_1^T B}{\mathbf{a}_2^T B} \\ \vdots \\ \frac{\mathbf{a}_m^T B}{\mathbf{a}_m^T B} \end{bmatrix}.$
johnston-linear-matrix-algebra-EQ0354	139	0.991	$(v_1 + w_1) + (v_2 + w_2) - 3(v_3 + w_3) = 0,$	$(v_1 + w_1) + (v_2 + w_2) - 3(v_3 + w_3) = 0,$	$(v_1 + w_1) + (v_2 + w_2) - 3(v_3 + w_3) = 0,$

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johnston-linear-matrix-algebra-EQ0358	140	0.991	$\mathbf{A} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 2 & -2 \\ 1 & -1 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 2(x-y) \\ x-y \end{bmatrix}.$	$\mathbf{A}\mathbf{x} = \begin{bmatrix} 2 & -2 \\ 1 & -1 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 2(x-y) \\ x-y \end{bmatrix}.$	$\mathbf{A}\mathbf{x} = \begin{bmatrix} 2 & -2 \\ 1 & -1 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 2(x-y) \\ x-y \end{bmatrix}.$
johnston-linear-matrix-algebra-EQ0979	343	0.991	$\mathbf{B} = \frac{1}{3} \begin{bmatrix} 4 & 1 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 0 & -2 \end{bmatrix} \begin{bmatrix} 1 & -1 \\ -1 & 4 \end{bmatrix} = \frac{1}{3} \begin{bmatrix} 4 & 1 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} 1 & -1 \\ 2 & -8 \end{bmatrix} = \begin{bmatrix} 2 & -4 \\ 1 & -3 \end{bmatrix},$	$B = \frac{1}{3} \begin{bmatrix} 4 & 1 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 0 & -2 \end{bmatrix} \begin{bmatrix} 1 & -1 \\ -1 & 4 \end{bmatrix} = \frac{1}{3} \begin{bmatrix} 4 & 1 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} 1 & -1 \\ 2 & -8 \end{bmatrix} = \begin{bmatrix} 2 & -4 \\ 1 & -3 \end{bmatrix},$	$B = \frac{1}{3} \begin{bmatrix} 4 & 1 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 0 & -2 \end{bmatrix} \begin{bmatrix} 1 & -1 \\ -1 & 4 \end{bmatrix} = \frac{1}{3} \begin{bmatrix} 4 & 1 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} 1 & -1 \\ 2 & -8 \end{bmatrix} = \begin{bmatrix} 2 & -4 \\ 1 & -3 \end{bmatrix},$
johnston-linear-matrix-algebra-EQ1254	420	0.991	$\begin{bmatrix} a \\ b \end{bmatrix} = r \begin{bmatrix} \cos(\theta) \\ \sin(\theta) \end{bmatrix} \quad \text{and} \quad r = \sqrt{a^2 + b^2} \quad \text{and} \quad \theta = \operatorname{sign}(b) \arccos\left(\frac{a}{\sqrt{a^2 + b^2}}\right).$	$\begin{aligned} a &= r \cos(\theta) & r &= \sqrt{a^2 + b^2} \\ b &= r \sin(\theta) & \theta &= \operatorname{sign}(b) \arccos\left(\frac{a}{\sqrt{a^2 + b^2}}\right). \end{aligned}$	$\begin{aligned} a &= r \cos(\theta) & r &= \sqrt{a^2 + b^2} \\ b &= r \sin(\theta) & \theta &= \operatorname{sign}(b) \arccos\left(\frac{a}{\sqrt{a^2 + b^2}}\right). \end{aligned}$
johnston-linear-matrix-algebra-EQ1659	484	0.991	$\lim_{k \rightarrow \infty} \ A \mathbf{v}_k\ = \lim_{k \rightarrow \infty} \ \lambda_1 \mathbf{v}_k\ ,$	$\lim_{k \rightarrow \infty} \ A \mathbf{v}_k\ = \lim_{k \rightarrow \infty} \ \lambda_1 \mathbf{v}_k\ ,$	$\lim_{k \rightarrow \infty} \ A \mathbf{v}_k\ = \lim_{k \rightarrow \infty} \ \lambda_1 \mathbf{v}_k\ ,$
johnston-linear-matrix-algebra-EQ1515	469	0.991	$\mathbf{A}^{-1} \mathbf{b} = \begin{bmatrix} -3 & 2 \\ 2 & -1 \end{bmatrix} \begin{bmatrix} 2 \\ 2 \end{bmatrix} = \begin{bmatrix} -2 \\ 2 \end{bmatrix} \neq \begin{bmatrix} 1 \\ 0 \end{bmatrix} = \mathbf{x}.$	$\mathbf{A}^{-1}\mathbf{b} = \begin{bmatrix} -3 & 2 \\ 2 & -1 \end{bmatrix} \begin{bmatrix} 2 \\ 2 \end{bmatrix} = \begin{bmatrix} -2 \\ 2 \end{bmatrix} \neq \begin{bmatrix} 1 \\ 0 \end{bmatrix} = \mathbf{x}.$	$\mathbf{A}^{-1}\mathbf{b} = \begin{bmatrix} -3 & 2 \\ 2 & -1 \end{bmatrix} \begin{bmatrix} 2 \\ 2 \end{bmatrix} = \begin{bmatrix} -2 \\ 2 \end{bmatrix} \neq \begin{bmatrix} 1 \\ 0 \end{bmatrix} = \mathbf{x}.$

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johnston-linear-matrix-algebra-EQ1612	479	0.991	$\begin{aligned} & \operatorname{det}(C-\lambda I) \\ & =\operatorname{det}\left(\left[\begin{array}{ccccc}-\lambda & 1 & 0 & \cdots & 0 \\ 0 & -\lambda & 1 & \cdots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & 0 & \cdots & 1 \\ -a_0 & -a_1 & -a_2 & \cdots & -a_{n-1}-\lambda \end{array}\right]-\lambda& 1 & 0 & \cdots & 0 \\& 0 & -\lambda & 1 & \cdots & 0 \\& \vdots & \vdots & \vdots & \ddots & \vdots \\& 0 & 0 & 0 & \cdots & 1 \\& -a_0 & -a_1 & -a_2 & \cdots & -a_{n-1}-\lambda \end{array}\right)\right) \\& =-\lambda \operatorname{det}\left(\left[\begin{array}{cccc}-\lambda & 1 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & 1 \\ -a_1 & -a_2 & \cdots & -a_{n-1}-\lambda \end{array}\right]\right) \\& +(-1)^na_0 \\& =-\lambda\left((-1)^{n-1}\left(\lambda^{n-1}+a_{n-1} \lambda^{n-2}+\cdots+a_2 \lambda+a_1\right)+(-1)^na_0\right) \\& =(-1)^n\left(\lambda^n+a_{n-1} \lambda^{n-1}+\cdots+a_1 \lambda+a_0\right),\end{aligned}$	$\det(C-\lambda I)$ $= \det \left(\begin{bmatrix} -\lambda & 1 & 0 & \cdots & 0 \\ 0 & -\lambda & 1 & \cdots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & 0 & \cdots & 1 \\ -a_0 & -a_1 & -a_2 & \cdots & -a_{n-1} - \lambda \end{bmatrix} \right)$ $= -\lambda \det \left(\begin{bmatrix} -\lambda & 1 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & 1 \\ -a_1 & -a_2 & \cdots & -a_{n-1} - \lambda \end{bmatrix} \right)$ $+ (-1)^n a_0$ $= -\lambda \left((-1)^{n-1} (\lambda^{n-1} + a_{n-1} \lambda^{n-2} + \cdots + a_2 \lambda + a_1) \right) + (-1)^n a_0$ $= (-1)^n (\lambda^n + a_{n-1} \lambda^{n-1} + \cdots + a_1 \lambda + a_0),$	
johnston-linear-matrix-algebra-EQ0044	40	0.992	$A=\left[\begin{array}{ll} 1 & 1 \\ 0 & 1 \end{array}\right] \quad \text { and } \quad B=\left[\begin{array}{ll} 1 & 0 \\ 1 & 1 \end{array}\right]$	$A = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix} \quad \text{and} \quad B = \begin{bmatrix} 1 & 0 \\ 1 & 1 \end{bmatrix}$	$A = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix} \quad \text{and} \quad B = \begin{bmatrix} 1 & 0 \\ 1 & 1 \end{bmatrix}$
johnston-linear-matrix-algebra-EQ0608	234	0.992	$R_{m}+\ell _{m,\,1}R_{1},\;\;\mathrm{\quad} R_{m}+\ell _{m,\,2}R_{2},\;\;\mathrm{\quad} \ldots ,\;\;\mathrm{\quad} R_{m}+\ell _{m,m-1}R_{m-1}$	$R_m + \ell_{m,1} R_1, \quad R_m + \ell_{m,2} R_2, \quad \dots, \quad R_m + \ell_{m,m-1} R_{m-1}$	$R_m + \ell_{m,1} R_1, \quad R_m + \ell_{m,2} R_2, \quad \dots, \quad R_m + \ell_{m,m-1} R_{m-1}$
johnston-linear-matrix-algebra-EQ0696	262	0.992	$[T]_B\stackrel{\rm def}{=} [[T({\bf v}_1)]_B [T({\bf v}_2)]_B \cdots [T({\bf v}_n)]_B].$	$[T]_B \stackrel{\rm def}{=} [[T(\mathbf{v}_1)]_B [T(\mathbf{v}_2)]_B \cdots [T(\mathbf{v}_n)]_B].$	$[T]_B \stackrel{\rm def}{=} [[T(\mathbf{v}_1)]_B [T(\mathbf{v}_2)]_B \cdots [T(\mathbf{v}_n)]_B].$
johnston-linear-matrix-algebra-EQ1333	447	0.992	$\begin{aligned} & \mathbf{v} \cdot (\mathbf{w} + \mathbf{x}) = v_1(w_1+x_1)+\cdots+v_n(w_n+x_n) \\ & = (v_1 w_1+v_1 x_1)+\cdots+(v_n w_n+v_n x_n) \\ & = \mathbf{v} \cdot \mathbf{w} + \mathbf{v} \cdot \mathbf{x}. \end{aligned}$	$\mathbf{v} \cdot (\mathbf{w} + \mathbf{x}) = v_1 (w_1 + x_1) + \cdots + v_n (w_n + x_n)$ $= (v_1 w_1 + v_1 x_1) + \cdots + (v_n w_n + v_n x_n)$ $= \mathbf{v} \cdot \mathbf{w} + \mathbf{v} \cdot \mathbf{x}.$	$\mathbf{v} \cdot (\mathbf{w} + \mathbf{x}) = v_1 (w_1 + x_1) + \cdots + v_n (w_n + x_n)$ $= (v_1 w_1 + v_1 x_1) + \cdots + (v_n w_n + v_n x_n)$ $= \mathbf{v} \cdot \mathbf{w} + \mathbf{v} \cdot \mathbf{x}.$

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johnston-linear-matrix-algebra-EQ1462	461	■ 0.992	$\begin{aligned} & \left[\begin{array}{cc} A & B \\ O & D \end{array} \right] \left[\begin{array}{cc} A^{-1} & -A^{-1}BD^{-1} \\ O & D^{-1} \end{array} \right] \\ & = \left[\begin{array}{cc} AA^{-1} & -AA^{-1}BD^{-1} + BD^{-1} \\ O & DD^{-1} \end{array} \right] \\ & = \left[\begin{array}{cc} I & -BD^{-1} + BD^{-1} \\ O & I \end{array} \right] = \left[\begin{array}{cc} I & O \\ O & I \end{array} \right] \end{aligned}$	$\begin{aligned} & \left[\begin{array}{cc} A & B \\ O & D \end{array} \right] \left[\begin{array}{cc} A^{-1} & -A^{-1}BD^{-1} \\ O & D^{-1} \end{array} \right] \\ & = \left[\begin{array}{cc} AA^{-1} & -AA^{-1}BD^{-1} + BD^{-1} \\ O & DD^{-1} \end{array} \right] \\ & = \left[\begin{array}{cc} I & -BD^{-1} + BD^{-1} \\ O & I \end{array} \right] = \left[\begin{array}{cc} I & O \\ O & I \end{array} \right] \end{aligned}$	
johnston-linear-matrix-algebra-EQ1380	439	■ 0.992	$\begin{aligned} & \begin{array}{c} \left[\begin{array}{c c} I & \mathbf{b} \\ O & \mathbf{0} \end{array} \right] \quad \text{or} \quad \left[\begin{array}{c c} I & \mathbf{0} \\ O & \mathbf{e}_1 \end{array} \right], \quad \text{and} \\ \left[\begin{array}{c c} I & \mathbf{c} \\ O & \mathbf{0} \end{array} \right] \quad \text{or} \quad \left[\begin{array}{c c} I & \mathbf{0} \\ O & \mathbf{e}_1 \end{array} \right]. \end{array} \end{aligned}$	$\begin{aligned} & \begin{array}{c} \left[\begin{array}{c c} I & \mathbf{b} \\ O & \mathbf{0} \end{array} \right] \quad \text{or} \quad \left[\begin{array}{c c} I & \mathbf{0} \\ O & \mathbf{e}_1 \end{array} \right], \quad \text{and} \\ \left[\begin{array}{c c} I & \mathbf{c} \\ O & \mathbf{0} \end{array} \right] \quad \text{or} \quad \left[\begin{array}{c c} I & \mathbf{0} \\ O & \mathbf{e}_1 \end{array} \right]. \end{array} \end{aligned}$	
johnston-linear-matrix-algebra-EQ0067	46	■ 0.992	$\left[\begin{array}{cc} A & B \\ O & I_3 \end{array} \right] \left[\begin{array}{cc} A & A \\ O & A \end{array} \right] = \left[\begin{array}{cc} A^2 & A^2 + BA \\ O & A \end{array} \right],$	$\left[\begin{array}{cc} A & B \\ O & I_3 \end{array} \right] \left[\begin{array}{cc} A & A \\ O & A \end{array} \right] = \left[\begin{array}{cc} A^2 & A^2 + BA \\ O & A \end{array} \right],$	
johnston-linear-matrix-algebra-EQ1471	462	■ 0.992	$\left[\begin{array}{ccc c} 1 & 1 & 1 & 0 \\ 0 & 1 & 2 & 0 \\ -1 & 1 & -1 & 0 \end{array} \right] \xrightarrow{R_3+R_1} \left[\begin{array}{ccc c} 1 & 1 & 1 & 0 \\ 0 & 1 & 2 & 0 \\ 0 & 2 & 0 & 0 \end{array} \right].$	$\left[\begin{array}{ccc c} 1 & 1 & 1 & 0 \\ 0 & 1 & 2 & 0 \\ -1 & 1 & -1 & 0 \end{array} \right] \xrightarrow{R_3+R_1} \left[\begin{array}{ccc c} 1 & 1 & 1 & 0 \\ 0 & 1 & 2 & 0 \\ 0 & 2 & 0 & 0 \end{array} \right].$	
johnston-linear-matrix-algebra-EQ0323	127	■ 0.993	$\mathbf{x} - \mathbf{y} = (A^{-1}A)(\mathbf{x} - \mathbf{y}) = A^{-1}(A(\mathbf{x} - \mathbf{y})) = A^{-1}\mathbf{0} = \mathbf{0},$	$\mathbf{x} - \mathbf{y} = (A^{-1}A)(\mathbf{x} - \mathbf{y}) = A^{-1}(A(\mathbf{x} - \mathbf{y})) = A^{-1}\mathbf{0} = \mathbf{0},$	
johnston-linear-matrix-algebra-EQ0414	158	■ 0.993	$V\mathbf{x} = (WA^T)\mathbf{x} = W(A^T\mathbf{x}) = W\mathbf{0} = \mathbf{0}.$	$V\mathbf{x} = (WA^T)\mathbf{x} = W(A^T\mathbf{x}) = W\mathbf{0} = \mathbf{0}.$	
johnston-linear-matrix-algebra-EQ0991	347	■ 0.993	$e^A = \frac{1}{3} \begin{bmatrix} 4 & 1 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} e & 0 \\ 0 & e^4 \end{bmatrix} \begin{bmatrix} 1 & -1 \\ -1 & 4 \end{bmatrix} = \frac{1}{3} \begin{bmatrix} 4 & 1 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} e & -e \\ -e^4 & 4e^4 \end{bmatrix} = \frac{1}{3} \begin{bmatrix} 4e - e^4 & 4e^4 - 4e \\ e - e^4 & 4e^4 - e \end{bmatrix}.$	$e^A = \frac{1}{3} \begin{bmatrix} 4 & 1 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} e & 0 \\ 0 & e^4 \end{bmatrix} \begin{bmatrix} 1 & -1 \\ -1 & 4 \end{bmatrix} = \frac{1}{3} \begin{bmatrix} 4 & 1 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} e & -e \\ -e^4 & 4e^4 \end{bmatrix} = \frac{1}{3} \begin{bmatrix} 4e - e^4 & 4e^4 - 4e \\ e - e^4 & 4e^4 - e \end{bmatrix}.$	

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johnston-linear-matrix-algebra-EQ1256	421	0.993	$e^{\pi i / 6}, \quad \text{and} \quad e^{9 \pi i / 6} .$	$e^{\pi i / 6}, \quad e^{5 \pi i / 6}, \quad \text{and} \quad e^{9 \pi i / 6} .$	$e^{\pi i / 6}, \quad e^{5 \pi i / 6}, \quad \text{and} \quad e^{9 \pi i / 6} .$
johnston-linear-matrix-algebra-EQ0041	39	0.993	$\begin{aligned} B C &= \left[\begin{array}{ccc} 5 & 6 & 7 \\ 8 & 9 & 10 \end{array} \right] \left[\begin{array}{c} 1 \\ 0 \\ 2 \end{array} \right] \\ &= \left[\begin{array}{ccc} (5,6,7) \cdot (1,0,2) & (5,6,7) \cdot (0,-1,-1) \\ (8,9,10) \cdot (1,0,2) & (8,9,10) \cdot (0,-1,-1) \end{array} \right] \\ &= \left[\begin{array}{cc} 5 \times 1 + 6 \times 0 + 7 \times 2 & 5 \times 0 + 6 \times -1 + 7 \times -1 \\ 8 \times 1 + 9 \times 0 + 10 \times 2 & 8 \times 0 + 9 \times -1 + 10 \times -1 \end{array} \right] \\ &= \left[\begin{array}{cc} 19 & -13 \\ 28 & -19 \end{array} \right] . \end{aligned}$	$BC = \begin{bmatrix} 5 & 6 & 7 \\ 8 & 9 & 10 \end{bmatrix} \begin{bmatrix} 1 \\ 0 \\ 2 \end{bmatrix}$ $= \begin{bmatrix} (5,6,7) \cdot (1,0,2) & (5,6,7) \cdot (0,-1,-1) \\ (8,9,10) \cdot (1,0,2) & (8,9,10) \cdot (0,-1,-1) \end{bmatrix}$ $= \begin{bmatrix} 5 \times 1 + 6 \times 0 + 7 \times 2 & 5 \times 0 + 6 \times -1 + 7 \times -1 \\ 8 \times 1 + 9 \times 0 + 10 \times 2 & 8 \times 0 + 9 \times -1 + 10 \times -1 \end{bmatrix}$ $= \begin{bmatrix} 19 & -13 \\ 28 & -19 \end{bmatrix} .$	$BC = \begin{bmatrix} 5 & 6 & 7 \\ 8 & 9 & 10 \end{bmatrix} \begin{bmatrix} 1 \\ 0 \\ 2 \end{bmatrix}$ $= \begin{bmatrix} (5,6,7) \cdot (1,0,2) & (5,6,7) \cdot (0,-1,-1) \\ (8,9,10) \cdot (1,0,2) & (8,9,10) \cdot (0,-1,-1) \end{bmatrix}$ $= \begin{bmatrix} 5 \times 1 + 6 \times 0 + 7 \times 2 & 5 \times 0 + 6 \times -1 + 7 \times -1 \\ 8 \times 1 + 9 \times 0 + 10 \times 2 & 8 \times 0 + 9 \times -1 + 10 \times -1 \end{bmatrix}$ $= \begin{bmatrix} 19 & -13 \\ 28 & -19 \end{bmatrix} .$
johnston-linear-matrix-algebra-EQ0096	57	0.993	$[T] = \left[c_1 \mathbf{e}_1 \mid c_2 \mathbf{e}_2 \mid \cdots \mid c_n \mathbf{e}_n \right] = \left[\begin{array}{cccc} c_1 & 0 & \cdots & 0 \\ 0 & c_2 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & c_n \end{array} \right] .$	$[T] = [c_1 \mathbf{e}_1 \mid c_2 \mathbf{e}_2 \mid \cdots \mid c_n \mathbf{e}_n] = \begin{bmatrix} c_1 & 0 & \cdots & 0 \\ 0 & c_2 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & c_n \end{bmatrix} .$	$[T] = [c_1 \mathbf{e}_1 \mid c_2 \mathbf{e}_2 \mid \cdots \mid c_n \mathbf{e}_n] = \begin{bmatrix} c_1 & 0 & \cdots & 0 \\ 0 & c_2 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & c_n \end{bmatrix} .$
johnston-linear-matrix-algebra-EQ0731 (3.2.1)	275	0.993	$\begin{aligned} \det(P^{-1}AP) &= \det(P^{-1}) \det(A) \det(P) \\ &= \frac{1}{\det(P)} \det(A) \det(P) = \det(A) \end{aligned}$	$\det(P^{-1}AP) = \det(P^{-1}) \det(A) \det(P)$ $= \frac{1}{\det(P)} \det(A) \det(P) = \det(A)$	$\det(P^{-1}AP) = \det(P^{-1}) \det(A) \det(P)$ $= \frac{1}{\det(P)} \det(A) \det(P) = \det(A),$
johnston-linear-matrix-algebra-EQ0807	288	0.993	$\det \left(\begin{bmatrix} A & B \\ O & C \end{bmatrix} \right) = \det(A) \det(C) .$	$\det \left(\begin{bmatrix} A & B \\ O & C \end{bmatrix} \right) = \det(A) \det(C) .$	$\det \left(\begin{bmatrix} A & B \\ O & C \end{bmatrix} \right) = \det(A) \det(C) .$
johnston-linear-matrix-algebra-EQ0825	301	0.993	$\lambda^2 - 5\lambda - 6 = 0 \iff (\lambda + 1)(\lambda - 6) = 0$ $\iff \lambda = -1 \quad \text{or} \quad \lambda = 6,$	$\lambda^2 - 5\lambda - 6 = 0 \iff (\lambda + 1)(\lambda - 6) = 0$ $\iff \lambda = -1 \quad \text{or} \quad \lambda = 6,$	$\lambda^2 - 5\lambda - 6 = 0 \iff (\lambda + 1)(\lambda - 6) = 0$ $\iff \lambda = -1 \quad \text{or} \quad \lambda = 6,$
johnston-linear-matrix-algebra-EQ1084	369	0.993	$\begin{aligned} x - y &= 2 \\ 2x - y &= 5 \end{aligned}$	$\begin{aligned} x - y &= 2 \\ 2x - y &= 5 \end{aligned}$	$\begin{aligned} x - y &= 2 \\ 2x - y &= 5 \end{aligned}$
johnston-linear-matrix-algebra-EQ1403	455	0.993	$A^n = \begin{bmatrix} 1 & n \\ 0 & 1 \end{bmatrix} ,$	$A^n = \begin{bmatrix} 1 & n \\ 0 & 1 \end{bmatrix} ,$	$A^n = \begin{bmatrix} 1 & n \\ 0 & 1 \end{bmatrix} ,$

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johnston-linear-matrix-algebra EQ1062	363	0.994	P_{\sigma}=\left[\mathbf{e}_{\{2\}}\left \mathbf{e}_{\{5\}}\right.\right.\left.\left \mathbf{e}_{\{1\}}\left \mathbf{e}_{\{3\}}\right.\right.\right.\left.\left \mathbf{e}_{\{4\}}\right.\right]=\left[\begin{array}{ccccc} 0 & 0 & 1 & 0 & 0 \\ 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 & 0 \end{array}\right].	$P_{\sigma} = [\mathbf{e}_2 \mathbf{e}_5 \mathbf{e}_1 \mathbf{e}_3 \mathbf{e}_4] = \begin{bmatrix} 0 & 0 & 1 & 0 & 0 \\ 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 & 0 \end{bmatrix}.$	$P_{\sigma} = [\mathbf{e}_2 \mathbf{e}_5 \mathbf{e}_1 \mathbf{e}_3 \mathbf{e}_4] = \begin{bmatrix} 0 & 0 & 1 & 0 & 0 \\ 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 & 0 \end{bmatrix}.$
johnston-linear-matrix-algebra EQ0883	318	0.994	\begin{aligned} \operatorname{nullity}(A-\lambda I) &= \operatorname{nullity}\left(P(A-\lambda I)P^{-1}\right) \\ &= \operatorname{nullity}(PAP^{-1}-\lambda PP^{-1}) = \operatorname{nullity}(B-\lambda I), \end{aligned}	$\operatorname{nullity}(A-\lambda I) = \operatorname{nullity}\left(P(A-\lambda I)P^{-1}\right) \\ = \operatorname{nullity}\left(PAP^{-1}-\lambda PP^{-1}\right) = \operatorname{nullity}(B-\lambda I),$	$\operatorname{nullity}(A-\lambda I) = \operatorname{nullity}\left(P(A-\lambda I)P^{-1}\right) \\ = \operatorname{nullity}(PAP^{-1}-\lambda PP^{-1}) = \operatorname{nullity}(B-\lambda I),$
johnston-linear-matrix-algebra EQ1461	461	0.994	\left[\begin{array}{cc} I & 0 \\ 0 & I \end{array}\right]	$\left[\begin{array}{cc} I & O \\ O & I \end{array}\right]$	$\left[\begin{array}{cc} I & O \\ O & I \end{array}\right]$
johnston-linear-matrix-algebra EQ0466	189	0.994	\mathbf{y}=\left[\begin{array}{l} 0 \\ 1 \\ 1 \\ 0 \\ 1 \end{array}\right] \quad \text{and} \quad A=\left[\begin{array}{ccccc} 1 & 0 & 1 & 1 & 0 \\ 0 & 1 & 0 & 1 & 0 \\ 1 & 0 & 1 & 1 & 1 \\ 1 & 1 & 1 & 1 & 0 \\ 0 & 0 & 1 & 0 & 1 \end{array}\right] \quad \text{then} \quad B=\left[\begin{array}{ccc} 1 & 0 & 0 \\ 0 & 1 & 1 \\ 0 & 1 & 1 \end{array}\right].	$\mathbf{y} = \begin{bmatrix} 0 \\ 1 \\ 1 \\ 0 \\ 1 \end{bmatrix} \quad \text{and} \quad A = \begin{bmatrix} 1 & 0 & 1 & 1 & 0 \\ 0 & 1 & 0 & 1 & 0 \\ 1 & 0 & 1 & 1 & 1 \\ 1 & 1 & 1 & 1 & 0 \\ 0 & 0 & 1 & 0 & 1 \end{bmatrix} \quad \text{then} \quad B = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 1 \\ 0 & 1 & 1 \end{bmatrix}.$	$\mathbf{y} = \begin{bmatrix} 0 \\ 1 \\ 1 \\ 0 \\ 1 \end{bmatrix} \quad \text{and} \quad A = \begin{bmatrix} 1 & 0 & 1 & 1 & 0 \\ 0 & 1 & 0 & 1 & 0 \\ 1 & 0 & 1 & 1 & 1 \\ 1 & 1 & 1 & 1 & 0 \\ 0 & 0 & 1 & 0 & 1 \end{bmatrix} \quad \text{then} \quad B = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 1 \\ 0 & 1 & 1 \end{bmatrix}.$
johnston-linear-matrix-algebra EQ0222	103	0.995	\left[\begin{array}{ccccc} 0 & 2 & -2 & 6 & 0 \\ 0 & 0 & -1 & 1 & 0 \\ 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 \end{array}\right] \xrightarrow{-R_2} \left[\begin{array}{ccccc} 0 & 2 & -2 & 6 & 0 \\ 0 & 0 & 1 & -1 & 0 \\ 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 \end{array}\right] \quad \text{new zero} \downarrow	$\left[\begin{array}{ccccc} 0 & 2 & -2 & 6 & 0 \\ 0 & 0 & -1 & 1 & 0 \\ 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 \end{array}\right] \xrightarrow{-R_2} \left[\begin{array}{ccccc} 0 & 2 & -2 & 6 & 0 \\ 0 & 0 & 1 & -1 & 0 \\ 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 \end{array}\right] \quad \text{new zero} \downarrow$	$\left[\begin{array}{ccccc} 0 & 2 & -2 & 6 & 0 \\ 0 & 0 & -1 & 1 & 0 \\ 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 \end{array}\right] \xrightarrow{-R_2} \left[\begin{array}{ccccc} 0 & 2 & -2 & 6 & 0 \\ 0 & 0 & 1 & -1 & 0 \\ 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 \end{array}\right] \quad \text{new zero} \downarrow$
johnston-linear-matrix-algebra EQ0758	286	0.995	\operatorname{det}\left(\left[\begin{array}{cc} a & b \\ c & d \end{array}\right]\right)=ad-bc.	$\det\left(\left[\begin{array}{cc} a & b \\ c & d \end{array}\right]\right) = ad - bc.$	$\det\left(\left[\begin{array}{cc} a & b \\ c & d \end{array}\right]\right) = ad - bc.$
johnston-linear-matrix-algebra EQ1137 (3.C.2)	393	0.995	\begin{aligned} QH &= [\operatorname{Re}(\mathbf{v}) \mid -\operatorname{Im}(\mathbf{v})] \left[\begin{array}{cc} 1 & 1 \\ -i & i \end{array}\right] \\ &= [\operatorname{Re}(\mathbf{v}) + i\operatorname{Im}(\mathbf{v}) \mid \operatorname{Re}(\mathbf{v}) - i\operatorname{Im}(\mathbf{v})] = [\mathbf{v} \mid \overline{\mathbf{v}}] = P, \end{aligned}	$QH = [\operatorname{Re}(\mathbf{v}) \mid -\operatorname{Im}(\mathbf{v})] \left[\begin{array}{cc} 1 & 1 \\ -i & i \end{array}\right] \\ = [\operatorname{Re}(\mathbf{v}) + i\operatorname{Im}(\mathbf{v}) \mid \operatorname{Re}(\mathbf{v}) - i\operatorname{Im}(\mathbf{v})] = [\mathbf{v} \mid \overline{\mathbf{v}}] = P,$	$QH = [\operatorname{Re}(\mathbf{v}) \mid -\operatorname{Im}(\mathbf{v})] \left[\begin{array}{cc} 1 & 1 \\ -i & i \end{array}\right] \\ = [\operatorname{Re}(\mathbf{v}) + i\operatorname{Im}(\mathbf{v}) \mid \operatorname{Re}(\mathbf{v}) - i\operatorname{Im}(\mathbf{v})] = [\mathbf{v} \mid \overline{\mathbf{v}}] = P,$
johnston-linear-matrix-algebra EQ1216	410	0.995	x_n=c_0+c_1n+c_2n^2+c_3n^3-1+2n+3n^2-3^n \quad \text{for all } n \geq 0.	$x_n = c_0 + c_1n + c_2n^2 + c_33^n = 1 + 2n + 3n^2 - 3^n \quad \text{for all } n \geq 0.$	$x_n = c_0 + c_1n + c_2n^2 + c_33^n = 1 + 2n + 3n^2 - 3^n \quad \text{for all } n \geq 0.$

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johnston-linear-matrix-algebra-EQ1272	426	0.995	$p(x)=3x^3-14x+4=(x-2)q(x)=(x-2)\left(3x^2+6x-2\right)$.	$p(x)=3x^3-14x+4=(x-2)q(x)=(x-2)(3x^2+6x-2)$.	$p(x)=3x^3-14x+4=(x-2)q(x)=(x-2)(3x^2+6x-2)$.
johnston-linear-matrix-algebra-EQ1318	442	0.995	$\operatorname{det}(A-\Lambda)=a_{i,1}c_{i,1}+\cdots+(a_{i,i}-\lambda_i)c_{i,i}+\cdots+a_{i,n}c_{i,n}$.	$\det(A-\Lambda)=a_{i,1}c_{i,1}+\cdots+(a_{i,i}-\lambda_i)c_{i,i}+\cdots+a_{i,n}c_{i,n}$.	$\det(A-\Lambda)=a_{i,1}c_{i,1}+\cdots+(a_{i,i}-\lambda_i)c_{i,i}+\cdots+a_{i,n}c_{i,n}$.
johnston-linear-matrix-algebra-EQ1117	385	0.995	$a_{i,j}=\begin{cases}\frac{1}{p_j} & \text{if page } j \text{ links to page } i, \text{ and} \\ 0 & \text{otherwise,}\end{cases}$	$a_{i,j}=\begin{cases}\frac{1}{p_j} & \text{if page } j \text{ links to page } i, \text{ and} \\ 0 & \text{otherwise,}\end{cases}$	$a_{i,j}=\begin{cases}\frac{1}{p_j} & \text{if page } j \text{ links to page } i, \text{ and} \\ 0 & \text{otherwise,}\end{cases}$
johnston-linear-matrix-algebra-EQ0401	154	0.995	$A_2=\left[\begin{array}{ll} 1 & 2 \\ 3 & 4 \end{array}\right]$ and $A_3=\left[\begin{array}{lll} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{array}\right]$.	$A_2=\left[\begin{array}{ll} 1 & 2 \\ 3 & 4 \end{array}\right]$ and $A_3=\left[\begin{array}{lll} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{array}\right]$.	$A_2=\left[\begin{array}{ll} 1 & 2 \\ 3 & 4 \end{array}\right]$ and $A_3=\left[\begin{array}{lll} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{array}\right]$.
johnston-linear-matrix-algebra-EQ1281	408	0.995	$x_n=3x_{n-2}+2x_{n-3}$ for all $n\geq 3$,	$x_n=3x_{n-2}+2x_{n-3}$ for all $n\geq 3$,	$x_n=3x_{n-2}+2x_{n-3}$ for all $n\geq 3$,
johnston-linear-matrix-algebra-EQ1374	451	0.995	$\begin{aligned} A D &= \left[\mathbf{a}_1\left \begin{array}{c} \mathbf{a}_2 \\ \vdots \\ \mathbf{a}_n \end{array}\right.\right] \\ &= \left[d_1\mathbf{a}_1\left d_2\mathbf{a}_2\right \cdots\left d_n\mathbf{a}_n\right.\right]. \end{aligned}$	$AD=\left[\mathbf{a}_1\left \mathbf{a}_2\right \cdots\left \mathbf{a}_n\right.\right]\begin{bmatrix}d_1&0&\cdots&0\\0&d_2&\cdots&0\\\vdots&\vdots&\ddots&\vdots\\0&0&\cdots&d_n\end{bmatrix}$ $=\left[d_1\mathbf{a}_1\left d_2\mathbf{a}_2\right \cdots\left d_n\mathbf{a}_n\right.\right]$.	$AD=\left[\mathbf{a}_1\left \mathbf{a}_2\right \cdots\left \mathbf{a}_n\right.\right]\begin{bmatrix}d_1&0&\cdots&0\\0&d_2&\cdots&0\\\vdots&\vdots&\ddots&\vdots\\0&0&\cdots&d_n\end{bmatrix}$ $=\left[d_1\mathbf{a}_1\left d_2\mathbf{a}_2\right \cdots\left d_n\mathbf{a}_n\right.\right]$.
johnston-linear-matrix-algebra-EQ1446	458	0.995	$y=(2/3-d/6)x^3+(d-2)x^2+(13/3-11/6d)x+d$	$y=(2/3-d/6)x^3+(d-2)x^2+(13/3-11/6d)x+d$	$y=(2/3-d/6)x^3+(d-2)x^2+(13/3-11/6d)x+d$
johnston-linear-matrix-algebra-EQ1543	472	0.995	$\left[\begin{array}{cc} p_{1,1} & p_{1,2} \\ 2p_{2,1} & 2p_{2,2} \end{array}\right]=\left[\begin{array}{cc} 3p_{1,1}+p_{1,2} & -2p_{1,1} \\ 3p_{2,1}+p_{2,2} & -2p_{2,1} \end{array}\right]$.	$\left[\begin{array}{cc} p_{1,1} & p_{1,2} \\ 2p_{2,1} & 2p_{2,2} \end{array}\right]=\left[\begin{array}{cc} 3p_{1,1}+p_{1,2} & -2p_{1,1} \\ 3p_{2,1}+p_{2,2} & -2p_{2,1} \end{array}\right]$.	$\left[\begin{array}{cc} p_{1,1} & p_{1,2} \\ 2p_{2,1} & 2p_{2,2} \end{array}\right]=\left[\begin{array}{cc} 3p_{1,1}+p_{1,2} & -2p_{1,1} \\ 3p_{2,1}+p_{2,2} & -2p_{2,1} \end{array}\right]$.
johnston-linear-matrix-algebra-EQ0102	59	0.995	$\left[P_{\mathbf{u}}\right]\mathbf{v}=\left[\begin{array}{ll} 1 & 0 \\ 0 & 0 \end{array}\right]\left[\begin{array}{l} v_1 \\ v_2 \end{array}\right]=\left[\begin{array}{l} v_1 \\ 0 \end{array}\right]$.	$\left[P_{\mathbf{u}}\right]\mathbf{v}=\left[\begin{array}{ll} 1 & 0 \\ 0 & 0 \end{array}\right]\left[\begin{array}{l} v_1 \\ v_2 \end{array}\right]=\left[\begin{array}{l} v_1 \\ 0 \end{array}\right]$.	$\left[P_{\mathbf{u}}\right]\mathbf{v}=\left[\begin{array}{ll} 1 & 0 \\ 0 & 0 \end{array}\right]\left[\begin{array}{l} v_1 \\ v_2 \end{array}\right]=\left[\begin{array}{l} v_1 \\ 0 \end{array}\right]$.

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johnston-linear-matrix-algebra EQ1229	413	0.995	$\begin{aligned} q(\lambda) = & \lambda^{71} - \lambda^{69} - 2\lambda^{68} - \lambda^{67} + 2\lambda^{66} + 2\lambda^{65} + \lambda^{64} - \lambda^{63} \\ & - \lambda^{62} - \lambda^{61} - \lambda^{60} - \lambda^{59} + 2\lambda^{58} + 5\lambda^{57} + 3\lambda^{56} - 2\lambda^{55} \\ & - 10\lambda^{54} - 3\lambda^{53} - 2\lambda^{52} + 6\lambda^{51} + 6\lambda^{50} + \lambda^{49} + 9\lambda^{48} - 3\lambda^{47} \\ & - 7\lambda^{46} - 8\lambda^{45} - 8\lambda^{44} + 10\lambda^{43} + 6\lambda^{42} + 8\lambda^{41} - 5\lambda^{40} - 12\lambda^{39} \\ & + 7\lambda^{38} - 7\lambda^{37} + 7\lambda^{36} + \lambda^{35} - 3\lambda^{34} + 10\lambda^{33} + \lambda^{32} - 6\lambda^{31} \\ & - 2\lambda^{30} - 10\lambda^{29} - 3\lambda^{28} + 2\lambda^{27} + 9\lambda^{26} - 3\lambda^{25} + 14\lambda^{24} - 8\lambda^{23} \\ & - 7\lambda^{21} + 9\lambda^{20} + 3\lambda^{19} - 4\lambda^{18} - 10\lambda^{17} - 7\lambda^{16} + 12\lambda^{15} + 7\lambda^{14} \\ & + 2\lambda^{13} - 12\lambda^{12} - 4\lambda^{11} - 2\lambda^{10} + 5\lambda^9 + \lambda^7 - 7\lambda^6 + 7\lambda^5 \\ & - 4\lambda^4 + 12\lambda^3 - 6\lambda^2 + 3\lambda - 6 \end{aligned}$	$q(\lambda) = \lambda^{71} - \lambda^{69} - 2\lambda^{68} - \lambda^{67} + 2\lambda^{66} + 2\lambda^{65} + \lambda^{64} - \lambda^{63} - \lambda^{62} - \lambda^{61} - \lambda^{60} - \lambda^{59} + 2\lambda^{58} + 5\lambda^{57} + 3\lambda^{56} - 2\lambda^{55} - 10\lambda^{54} - 3\lambda^{53} - 2\lambda^{52} + 6\lambda^{51} + 6\lambda^{50} + \lambda^{49} + 9\lambda^{48} - 3\lambda^{47} - 7\lambda^{46} - 8\lambda^{45} - 8\lambda^{44} + 10\lambda^{43} + 6\lambda^{42} + 8\lambda^{41} - 5\lambda^{40} - 12\lambda^{39} + 7\lambda^{38} - 7\lambda^{37} + 7\lambda^{36} + \lambda^{35} - 3\lambda^{34} + 10\lambda^{33} + \lambda^{32} - 6\lambda^{31} - 2\lambda^{30} - 10\lambda^{29} - 3\lambda^{28} + 2\lambda^{27} + 9\lambda^{26} - 3\lambda^{25} + 14\lambda^{24} - 8\lambda^{23} - 7\lambda^{21} + 9\lambda^{20} + 3\lambda^{19} - 4\lambda^{18} - 10\lambda^{17} - 7\lambda^{16} + 12\lambda^{15} + 7\lambda^{14} + 2\lambda^{13} - 12\lambda^{12} - 4\lambda^{11} - 2\lambda^{10} + 5\lambda^9 + \lambda^7 - 7\lambda^6 + 7\lambda^5 - 4\lambda^4 + 12\lambda^3 - 6\lambda^2 + 3\lambda - 6$	
johnston-linear-matrix-algebra EQ0385	147	0.995	$\begin{aligned} & \left[\begin{array}{ccc c} 1 & 1 & 0 & 0 \\ 2 & 0 & -1 & 0 \\ 3 & 1 & 2 & 0 \end{array} \right] \xrightarrow[R_3-3R_1]{R_2-2R_1} \left[\begin{array}{ccc c} 1 & 1 & 0 & 0 \\ 0 & -2 & -1 & 0 \\ 0 & -2 & 2 & 0 \end{array} \right] \\ & \xrightarrow{R_3-R_2} \left[\begin{array}{ccc c} 1 & 1 & 0 & 0 \\ 0 & -2 & -1 & 0 \\ 0 & 0 & 3 & 0 \end{array} \right] \end{aligned}$	$\left[\begin{array}{ccc c} 1 & 1 & 0 & 0 \\ 2 & 0 & -1 & 0 \\ 3 & 1 & 2 & 0 \end{array} \right] \xrightarrow[R_3-R_2]{R_2-2R_1, R_3-3R_1} \left[\begin{array}{ccc c} 1 & 1 & 0 & 0 \\ 0 & -2 & -1 & 0 \\ 0 & -2 & 2 & 0 \end{array} \right]$	
johnston-linear-matrix-algebra EQ0233 (8)	107	0.996	$\begin{aligned} & \left[\begin{array}{ccc c} 1 & 2 & -2 & -4 \\ 2 & 4 & 1 & 0 \\ 1 & 2 & 7 & 2 \end{array} \right] \xrightarrow[R_3-\frac{9}{5}R_2]{R_2-2R_1, R_3-R_1} \left[\begin{array}{ccc c} 1 & 2 & -2 & -4 \\ 0 & 0 & 5 & 8 \\ 0 & 0 & 9 & 6 \end{array} \right] \\ & \xrightarrow{R_3-\frac{9}{5}R_2} \left[\begin{array}{ccc c} 1 & 2 & -2 & -4 \\ 0 & 0 & 5 & 8 \\ 0 & 0 & 0 & -42/5 \end{array} \right] \end{aligned}$	$\left[\begin{array}{ccc c} 1 & 2 & -2 & -4 \\ 2 & 4 & 1 & 0 \\ 1 & 2 & 7 & 2 \end{array} \right] \xrightarrow[R_3-\frac{9}{5}R_2]{R_2-2R_1, R_3-R_1} \left[\begin{array}{ccc c} 1 & 2 & -2 & -4 \\ 0 & 0 & 5 & 8 \\ 0 & 0 & 9 & 6 \end{array} \right]$	

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johnston-linear-matrix-algebra-EQ0425	165	0.996	$A=\left[\begin{array}{ccc} 1 & 1 & 2 \\ 2 & -1 & 1 \\ 1 & 0 & 1 \end{array}\right] \quad \text{is} \quad \left[\begin{array}{ccc} 1 & 0 & 1 \\ 0 & 1 & 1 \\ 0 & 0 & 0 \end{array}\right],$	$A = \begin{bmatrix} 1 & 1 & 2 \\ 2 & -1 & 1 \\ 1 & 0 & 1 \end{bmatrix} \quad \text{is} \quad \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \\ 0 & 0 & 0 \end{bmatrix},$	$A = \begin{bmatrix} 1 & 1 & 2 \\ 2 & -1 & 1 \\ 1 & 0 & 1 \end{bmatrix} \quad \text{is} \quad \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \\ 0 & 0 & 0 \end{bmatrix},$
johnston-linear-matrix-algebra-EQ1605	478	0.996	$A^{*}\overline{\mathbf{v}}=\overline{A^{*}\mathbf{v}}=\overline{\lambda\mathbf{v}}=\overline{\lambda}\overline{\mathbf{v}},$	$A^{*}\overline{\mathbf{v}} = \overline{A^{*}\mathbf{v}} = \overline{\lambda\mathbf{v}} = \overline{\lambda}\overline{\mathbf{v}},$	$A^{*}\overline{\mathbf{v}} = \overline{A^{*}\mathbf{v}} = \overline{\lambda\mathbf{v}} = \overline{\lambda}\overline{\mathbf{v}},$
johnston-linear-matrix-algebra-EQ0336	132	0.996	$\frac{1}{2}\left[\begin{array}{cc} 5 & -4 \\ -2 & 2 \end{array}\right].$	$\frac{1}{2}\begin{bmatrix} 5 & -4 \\ -2 & 2 \end{bmatrix}.$	$\frac{1}{2}\begin{bmatrix} 5 & -4 \\ -2 & 2 \end{bmatrix}.$
johnston-linear-matrix-algebra-EQ0512	205	0.996	$\left[\begin{array}{c c c c} 1 & \mathbf{0}^T & -\mathbf{c}^T & 0 \\ \hline \mathbf{0} & I & A & \mathbf{b} \end{array}\right].$	$\left[\begin{array}{c c c c} 1 & \mathbf{0}^T & -\mathbf{c}^T & 0 \\ \hline \mathbf{0} & I & A & \mathbf{b} \end{array}\right].$	$\left[\begin{array}{c c c c} 1 & \mathbf{0}^T & -\mathbf{c}^T & 0 \\ \hline \mathbf{0} & I & A & \mathbf{b} \end{array}\right].$
johnston-linear-matrix-algebra-EQ0809	288	0.996	$\operatorname{det}\left(\left[\begin{array}{cc} A & B \\ C & D \end{array}\right]\right)=\operatorname{det}(A)\operatorname{det}(D)-\operatorname{det}(B)\operatorname{det}(C).$	$\det\left(\begin{bmatrix} A & B \\ C & D \end{bmatrix}\right) = \det(A)\det(D) - \det(B)\det(C).$	$\det\left(\begin{bmatrix} A & B \\ C & D \end{bmatrix}\right) = \det(A)\det(D) - \det(B)\det(C).$
johnston-linear-matrix-algebra-EQ1212	409	0.996	$x_{\{n\}}=6\,x_{\{n-1\}}-12\,x_{\{n-2\}}+10\,x_{\{n-3\}}-3\,x_{\{n-4\}}\,\text{for all}\,n\geq 4,$	$x_n = 6x_{n-1} - 12x_{n-2} + 10x_{n-3} - 3x_{n-4}$ for all $n \geq 4,$	$x_n = 6x_{n-1} - 12x_{n-2} + 10x_{n-3} - 3x_{n-4}$ for all $n \geq 4,$
johnston-linear-matrix-algebra-EQ0010	24	0.997	$\mathbf{vw}\stackrel{\text{def}}{=}(v_1w_1,v_2w_2,\ldots,v_nw_n).$	$\mathbf{vw} \stackrel{\text{def}}{=} (v_1w_1, v_2w_2, \dots, v_nw_n).$	$\mathbf{vw} \stackrel{\text{def}}{=} (v_1w_1, v_2w_2, \dots, v_nw_n).$
johnston-linear-matrix-algebra-EQ0062	45	0.997	$A=\left[\begin{array}{cc} I_3 & I_3 \\ O & C \end{array}\right] \quad \text{and} \quad B=\left[\begin{array}{cc} D & O \\ O & D \end{array}\right].$	$A = \begin{bmatrix} I_3 & I_3 \\ O & C \end{bmatrix} \quad \text{and} \quad B = \begin{bmatrix} D & O \\ O & D \end{bmatrix}.$	$A = \begin{bmatrix} I_3 & I_3 \\ O & C \end{bmatrix} \quad \text{and} \quad B = \begin{bmatrix} D & O \\ O & D \end{bmatrix}.$
johnston-linear-matrix-algebra-EQ0284 (2.2.1)	116	0.997	$\left[\begin{array}{lll} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{array}\right]\left[\begin{array}{cccc} 0 & 2 & 4 & 0 \\ 1 & 1 & 0 & -1 \\ 3 & 4 & 2 & 1 \end{array}\right]=\left[\begin{array}{cccc} 1 & 1 & 0 & -1 \\ 0 & 2 & 4 & 0 \\ 3 & 4 & 2 & 1 \end{array}\right].$	$\begin{bmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 0 & 2 & 4 & 0 \\ 1 & 1 & 0 & -1 \\ 3 & 4 & 2 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 1 & 0 & -1 \\ 0 & 2 & 4 & 0 \\ 3 & 4 & 2 & 1 \end{bmatrix}.$	$\begin{bmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 0 & 2 & 4 & 0 \\ 1 & 1 & 0 & -1 \\ 3 & 4 & 2 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 1 & 0 & -1 \\ 0 & 2 & 4 & 0 \\ 3 & 4 & 2 & 1 \end{bmatrix}.$
johnston-linear-matrix-algebra-EQ0533	215	0.997	$y_{\{1\}}\left(x_{\{1\}}+x_{\{2\}}\right)+y_{\{2\}}\left(-x_{\{1\}}+x_{\{2\}}\right)\leq 3\,y_{\{1\}}+y_{\{2\}}.$	$y_1\left(x_1+x_2\right)+y_2\left(-x_1+x_2\right)\leq 3y_1+y_2.$	$y_1\left(x_1+x_2\right)+y_2\left(-x_1+x_2\right)\leq 3y_1+y_2.$
johnston-linear-matrix-algebra-EQ0621	239	0.997	$L=\left[\begin{array}{llll} 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 0 \\ 1 & 2 & 1 & 0 \\ 1 & 3 & 3 & 1 \end{array}\right] \quad \text{and} \quad U=\left[\begin{array}{cccc} 1 & 0 & 0 & 0 \\ 0 & 1 & 1 & 1 \\ 0 & 0 & 2 & 6 \\ 0 & 0 & 0 & 6 \end{array}\right].$	$L = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 0 \\ 1 & 2 & 1 & 0 \\ 1 & 3 & 3 & 1 \end{bmatrix} \quad \text{and} \quad U = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 1 & 1 \\ 0 & 0 & 2 & 6 \\ 0 & 0 & 0 & 6 \end{bmatrix}.$	$L = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 0 \\ 1 & 2 & 1 & 0 \\ 1 & 3 & 3 & 1 \end{bmatrix} \quad \text{and} \quad U = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 1 & 1 \\ 0 & 0 & 2 & 6 \\ 0 & 0 & 0 & 6 \end{bmatrix}.$
johnston-linear-matrix-algebra-EQ1042	358	0.997	$A=\left[\begin{array}{lll} a & b & c \\ d & e & f \\ g & h & i \end{array}\right] \quad \text{is} \quad \operatorname{cof}(A)=\left[\begin{array}{ccc} e\,i-f\,h & f\,g-d\,i & d\,h-e\,g \\ c\,h-b\,i & a\,i-c\,g & b\,g-a\,h \\ b\,f-c\,e & c\,d-a\,f & a\,e-b\,d \end{array}\right].$	$A = \begin{bmatrix} a & b & c \\ d & e & f \\ g & h & i \end{bmatrix} \quad \text{is} \quad \operatorname{cof}(A) = \begin{bmatrix} ei-fh & fg-di & dh-eg \\ ch-bi & ai-cg & bg-ah \\ bf-ce & cd-af & ae-bd \end{bmatrix}.$	$A = \begin{bmatrix} a & b & c \\ d & e & f \\ g & h & i \end{bmatrix} \quad \text{is} \quad \operatorname{cof}(A) = \begin{bmatrix} ei-fh & fg-di & dh-eg \\ ch-bi & ai-cg & bg-ah \\ bf-ce & cd-af & ae-bd \end{bmatrix}.$

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johnston-linear-matrix-algebra-EQ1265	424	■ 0.997	$\begin{array}{ll} (1)^2-2(1)-3=-4 & (-1)^2-2(-1)-3=0 \\ (3)^2-2(3)-3=0 & (-3)^2-2(-3)-3=12 \end{array}$	$\begin{array}{ll} (1)^2-2(1)-3=-4 & (-1)^2-2(-1)-3=0 \\ (3)^2-2(3)-3=0 & (-3)^2-2(-3)-3=12. \end{array}$	$\begin{array}{ll} (1)^2-2(1)-3=-4 & (-1)^2-2(-1)-3=0 \\ (3)^2-2(3)-3=0 & (-3)^2-2(-3)-3=12. \end{array}$
johnston-linear-matrix-algebra-EQ1531	470	■ 0.997	$L \, \widetilde{U} = \left[L \, U \, \mid L \, L^{-1} \right] C = [B \, \mid C] = A,$	$L\widetilde{U} = [LU \mid LL^{-1}C] = [B \mid C] = A,$	$L\widetilde{U} = [LU \mid LL^{-1}C] = [B \mid C] = A,$
johnston-linear-matrix-algebra-EQ1533 (0)	471	■ 0.997	$[1] \quad \text{and} \quad \left[\begin{array}{cc} 1 & 3 \\ 3 & 2 \end{array} \right],$	$[1] \quad \text{and} \quad \left[\begin{array}{cc} 1 & 3 \\ 3 & 2 \end{array} \right],$	$[1] \quad \text{and} \quad \left[\begin{array}{cc} 1 & 3 \\ 3 & 2 \end{array} \right],$
johnston-linear-matrix-algebra-EQ0954	338	■ 0.997	$\begin{gathered} \left\{ \left[\begin{array}{cc c} 1 & -\phi & 1 & 0 \\ 1 & -\phi & 0 & 0 \end{array} \right] \xrightarrow{\text{R}_1 \leftrightarrow \text{R}_2} \left[\begin{array}{cc c} 1 & -\phi & 1 & 0 \\ 1 & -\phi & 0 & 0 \end{array} \right] \xrightarrow{\text{R}_2 + (\phi-1)\text{R}_1} \left[\begin{array}{cc c} 1 & -\phi & 1 & 0 \\ 0 & 0 & 0 & 0 \end{array} \right] \right\} \\ \left\{ \left[\begin{array}{cc c} 1 & -\phi & 1 & 0 \\ 1 & -\phi & 0 & 0 \end{array} \right] \xrightarrow{\text{R}_2 + (\phi-1)\text{R}_1} \left[\begin{array}{cc c} 1 & -\phi & 1 & 0 \\ 0 & 0 & 0 & 0 \end{array} \right] \right\} \end{gathered}$	$\left[\begin{array}{cc c} 1-\phi & 1 & 0 \\ 1 & -\phi & 0 \end{array} \right] \xrightarrow{R_1 \leftrightarrow R_2} \left[\begin{array}{cc c} 1 & -\phi & 0 \\ 1-\phi & 1 & 0 \end{array} \right] \xrightarrow{R_2 + (\phi-1)R_1} \left[\begin{array}{cc c} 1 & -\phi & 0 \\ 0 & 0 & 0 \end{array} \right].$	$\left[\begin{array}{cc c} 1-\phi & 1 & 0 \\ 1 & -\phi & 0 \end{array} \right] \xrightarrow{R_1 \leftrightarrow R_2} \left[\begin{array}{cc c} 1 & -\phi & 0 \\ 1-\phi & 1 & 0 \end{array} \right] \xrightarrow{R_2 + (\phi-1)R_1} \left[\begin{array}{cc c} 1 & -\phi & 0 \\ 0 & 0 & 0 \end{array} \right].$
johnston-linear-matrix-algebra-EQ1473	462	■ 0.997	$\begin{aligned} & \left\{ \left[\begin{array}{ccc c} 0 & 1 & k & x \\ 1 & 2 & -1 & y \\ -1 & 1 & 4 & z \end{array} \right] \xrightarrow{R_1 \leftrightarrow R_2} \left[\begin{array}{ccc c} 1 & 2 & -1 & y \\ 0 & 1 & k & x \\ -1 & 1 & 4 & z \end{array} \right] \xrightarrow{R_3 + R_1} \left[\begin{array}{ccc c} 1 & 2 & -1 & y \\ 0 & 1 & k & x \\ 0 & 3 & 3 & y+z \end{array} \right] \right. \\ & \left. \xrightarrow{R_3 - 3R_2} \left[\begin{array}{ccc c} 1 & 2 & -1 & y \\ 0 & 1 & k & x \\ 0 & 0 & 3-3k & y+z-3x \end{array} \right] \right\} \end{aligned}$	$\left\{ \left[\begin{array}{ccc c} 0 & 1 & k & x \\ 1 & 2 & -1 & y \\ -1 & 1 & 4 & z \end{array} \right] \xrightarrow{R_1 \leftrightarrow R_2} \left[\begin{array}{ccc c} 1 & 2 & -1 & y \\ 0 & 1 & k & x \\ -1 & 1 & 4 & z \end{array} \right] \xrightarrow{R_3 + R_1} \left[\begin{array}{ccc c} 1 & 2 & -1 & y \\ 0 & 1 & k & x \\ 0 & 3 & 3 & y+z \end{array} \right] \right. \\ & \left. \xrightarrow{R_3 - 3R_2} \left[\begin{array}{ccc c} 1 & 2 & -1 & y \\ 0 & 1 & k & x \\ 0 & 0 & 3-3k & y+z-3x \end{array} \right] \right\}.$	$\left\{ \left[\begin{array}{ccc c} 0 & 1 & k & x \\ 1 & 2 & -1 & y \\ -1 & 1 & 4 & z \end{array} \right] \xrightarrow{R_1 \leftrightarrow R_2} \left[\begin{array}{ccc c} 1 & 2 & -1 & y \\ 0 & 1 & k & x \\ -1 & 1 & 4 & z \end{array} \right] \xrightarrow{R_3 + R_1} \left[\begin{array}{ccc c} 1 & 2 & -1 & y \\ 0 & 1 & k & x \\ 0 & 3 & 3 & y+z \end{array} \right] \right. \\ & \left. \xrightarrow{R_3 - 3R_2} \left[\begin{array}{ccc c} 1 & 2 & -1 & y \\ 0 & 1 & k & x \\ 0 & 0 & 3-3k & y+z-3x \end{array} \right] \right\}.$
johnston-linear-matrix-algebra-EQ0733	276	■ 0.997	$\begin{aligned} & \operatorname{det} \left(\left[\begin{array}{c} \mathbf{0} \\ \mathbf{a}_2 \end{array} \right] \cdots \left[\begin{array}{c} \mathbf{a}_n \end{array} \right] \right) = \operatorname{det} \left(\left[\begin{array}{c} \mathbf{a}_1 - \mathbf{a}_1 \\ \mathbf{a}_2 \end{array} \right] \cdots \left[\begin{array}{c} \mathbf{a}_n \end{array} \right] \right) \\ & = \operatorname{det} \left(\left[\begin{array}{c} \mathbf{a}_1 \\ \mathbf{a}_2 \end{array} \right] \cdots \left[\begin{array}{c} \mathbf{a}_n \end{array} \right] \right) - \operatorname{det} \left(\left[\begin{array}{c} \mathbf{a}_1 \\ \mathbf{a}_2 \end{array} \right] \cdots \left[\begin{array}{c} \mathbf{a}_n \end{array} \right] \right) \\ & = 0, \end{aligned}$	$\det \left(\left[\begin{array}{c} \mathbf{0} \\ \mathbf{a}_2 \end{array} \right] \cdots \left[\begin{array}{c} \mathbf{a}_n \end{array} \right] \right) = \det \left(\left[\begin{array}{c} \mathbf{a}_1 - \mathbf{a}_1 \\ \mathbf{a}_2 \end{array} \right] \cdots \left[\begin{array}{c} \mathbf{a}_n \end{array} \right] \right) \\ = \det \left(\left[\begin{array}{c} \mathbf{a}_1 \\ \mathbf{a}_2 \end{array} \right] \cdots \left[\begin{array}{c} \mathbf{a}_n \end{array} \right] \right) - \det \left(\left[\begin{array}{c} \mathbf{a}_1 \\ \mathbf{a}_2 \end{array} \right] \cdots \left[\begin{array}{c} \mathbf{a}_n \end{array} \right] \right) \\ = 0,$	$\det \left(\left[\begin{array}{c} \mathbf{0} \\ \mathbf{a}_2 \end{array} \right] \cdots \left[\begin{array}{c} \mathbf{a}_n \end{array} \right] \right) = \det \left(\left[\begin{array}{c} \mathbf{a}_1 - \mathbf{a}_1 \\ \mathbf{a}_2 \end{array} \right] \cdots \left[\begin{array}{c} \mathbf{a}_n \end{array} \right] \right) \\ = \det \left(\left[\begin{array}{c} \mathbf{a}_1 \\ \mathbf{a}_2 \end{array} \right] \cdots \left[\begin{array}{c} \mathbf{a}_n \end{array} \right] \right) - \det \left(\left[\begin{array}{c} \mathbf{a}_1 \\ \mathbf{a}_2 \end{array} \right] \cdots \left[\begin{array}{c} \mathbf{a}_n \end{array} \right] \right) \\ = 0,$
johnston-linear-matrix-algebra-EQ0842	307	■ 0.997	$p_A(\lambda) = \lambda^2 - 5\lambda - 6 \quad \text{and} \quad p_B(\lambda) = -\lambda^3 + 16\lambda + 24,$	$p_A(\lambda) = \lambda^2 - 5\lambda - 6 \quad \text{and} \quad p_B(\lambda) = -\lambda^3 + 16\lambda + 24,$	$p_A(\lambda) = \lambda^2 - 5\lambda - 6 \quad \text{and} \quad p_B(\lambda) = -\lambda^3 + 16\lambda + 24,$
johnston-linear-matrix-algebra-EQ1315 (B.6.1)	443	■ 0.997	$p^{(k)}(x) = (x-r)^{m-k} (c_k q(x) + s_k(x)) \quad \text{for all } 0 \leq k \leq m,$	$p^{(k)}(x) = (x-r)^{m-k} (c_k q(x) + s_k(x)) \quad \text{for all } 0 \leq k \leq m,$	$p^{(k)}(x) = (x-r)^{m-k} (c_k q(x) + s_k(x)) \quad \text{for all } 0 \leq k \leq m,$

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johnston-linear-matrix-algebra-EQ1826	356	0.997	$\begin{array}{ll} c_{1,1}=\operatorname{det}([4])=4, \\ \& c_{1,2}=-\operatorname{det}([-1])=1, \\ \& c_{2,1}=-\operatorname{det}([3])=-3, \\ \& c_{2,2}=\operatorname{det}([2])=2. \end{array}$	$\begin{array}{ll} c_{1,1} = \det([4]) = 4, & c_{1,2} = -\det([-1]) = 1, \\ c_{2,1} = -\det([3]) = -3, & \text{and} \\ c_{2,2} = \det([2]) = 2. \end{array}$	$\begin{array}{ll} c_{1,1} = \det([4]) = 4, & c_{1,2} = -\det([-1]) = 1, \\ c_{2,1} = -\det([3]) = -3, & \text{and} \\ c_{2,2} = \det([2]) = 2. \end{array}$
johnston-linear-matrix-algebra-EQ1377	452	0.997	$\begin{aligned} & \left[F_{\mathbf{u}} \right] &= 2 \left[\begin{array}{c} 1 \\ m \end{array} \right] \left[\begin{array}{cc} 1 & m \end{array} \right] / (1+m^2) - \left[\begin{array}{cc} 1 & 0 \\ 0 & 1 \end{array} \right] \\ &= \frac{1}{1+m^2} \left[\begin{array}{cc} 1-m^2 & 2m \\ 2m & m^2-1 \end{array} \right]. \end{aligned}$	$\begin{aligned} [F_{\mathbf{u}}] &= 2\mathbf{u}\mathbf{u}^T/\ \mathbf{u}\ ^2 - I \\ &= 2 \begin{bmatrix} 1 \\ m \end{bmatrix} \begin{bmatrix} 1 & m \end{bmatrix} / (1+m^2) - \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \\ &= \frac{1}{1+m^2} \begin{bmatrix} 1-m^2 & 2m \\ 2m & m^2-1 \end{bmatrix}. \end{aligned}$	$\begin{aligned} [F_{\mathbf{u}}] &= 2\mathbf{u}\mathbf{u}^T/\ \mathbf{u}\ ^2 - I \\ &= 2 \begin{bmatrix} 1 \\ m \end{bmatrix} \begin{bmatrix} 1 & m \end{bmatrix} / (1+m^2) - \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \\ &= \frac{1}{1+m^2} \begin{bmatrix} 1-m^2 & 2m \\ 2m & m^2-1 \end{bmatrix}. \end{aligned}$
johnston-linear-matrix-algebra-EQ8011	25	0.997	$\mathbf{v} \cdot \mathbf{w} \stackrel{\text{def}}{=} v_1w_1 + v_2w_2 + \cdots + v_nw_n.$	$\mathbf{v} \cdot \mathbf{w} \stackrel{\text{def}}{=} v_1w_1 + v_2w_2 + \cdots + v_nw_n.$	$\mathbf{v} \cdot \mathbf{w} \stackrel{\text{def}}{=} v_1w_1 + v_2w_2 + \cdots + v_nw_n.$
johnston-linear-matrix-algebra-EQ8017	28	0.997	$\ \mathbf{v}\ \stackrel{\text{def}}{=} \sqrt{\mathbf{v} \cdot \mathbf{v}} = \sqrt{v_1^2 + v_2^2 + \cdots + v_n^2}.$	$\ \mathbf{v}\ \stackrel{\text{def}}{=} \sqrt{\mathbf{v} \cdot \mathbf{v}} = \sqrt{v_1^2 + v_2^2 + \cdots + v_n^2}.$	$\ \mathbf{v}\ \stackrel{\text{def}}{=} \sqrt{\mathbf{v} \cdot \mathbf{v}} = \sqrt{v_1^2 + v_2^2 + \cdots + v_n^2}.$
johnston-linear-matrix-algebra-EQ8154	76	0.997	$\begin{aligned} & \left(v_1^2 + v_2^2 + v_3^2 \right) \left(w_1^2 + w_2^2 + w_3^2 \right) - \left(v_1w_1 + v_2w_2 + v_3w_3 \right)^2 \\ &= v_1^2w_2^2 + v_1^2w_3^2 + v_2^2w_1^2 + v_2^2w_3^2 + v_3^2w_1^2 + v_3^2w_2^2 \\ &\quad - 2 \left(v_2v_3w_2w_3 + v_1v_3w_1w_3 + v_1v_2w_1w_2 \right), \end{aligned}$	$\begin{aligned} \ \mathbf{v}\ ^2\ \mathbf{w}\ ^2 - (\mathbf{v} \cdot \mathbf{w})^2 &= (v_1^2 + v_2^2 + v_3^2)(w_1^2 + w_2^2 + w_3^2) - (v_1w_1 + v_2w_2 + v_3w_3)^2 \\ &= v_1^2w_2^2 + v_1^2w_3^2 + v_2^2w_1^2 + v_2^2w_3^2 + v_3^2w_1^2 + v_3^2w_2^2 \\ &\quad - 2(v_2v_3w_2w_3 + v_1v_3w_1w_3 + v_1v_2w_1w_2) \end{aligned}$	$\begin{aligned} \ \mathbf{v}\ ^2\ \mathbf{w}\ ^2 - (\mathbf{v} \cdot \mathbf{w})^2 &= (v_1^2 + v_2^2 + v_3^2)(w_1^2 + w_2^2 + w_3^2) - (v_1w_1 + v_2w_2 + v_3w_3)^2 \\ &= v_1^2w_2^2 + v_1^2w_3^2 + v_2^2w_1^2 + v_2^2w_3^2 + v_3^2w_1^2 + v_3^2w_2^2 \\ &\quad - 2(v_2v_3w_2w_3 + v_1v_3w_1w_3 + v_1v_2w_1w_2), \end{aligned}$
johnston-linear-matrix-algebra-EQ8488	196	0.997	$\begin{array}{r} x+2y+z=0 \\ 2x+y+2z=1 \\ x+z=2 \end{array}$	$\begin{array}{l} x+2y+2z=0 \\ 2x+y+2z=1 \\ x+z=2 \end{array}$	$\begin{array}{l} x+2y+2z=0 \\ 2x+y+2z=1 \\ x+z=2 \end{array}$
johnston-linear-matrix-algebra-EQ8633	243	0.997	$E = \begin{bmatrix} 0 & 0 & \star & 0 & 0 \\ 0 & 0 & \star & \star & 0 \\ \star & 0 & \star & \star & 0 \\ \star & 0 & \star & \star & \star \\ \star & \star & \star & \star & \star \end{bmatrix}.$	$E = \begin{bmatrix} 0 & 0 & \star & 0 & 0 \\ 0 & 0 & \star & \star & 0 \\ \star & 0 & \star & \star & 0 \\ \star & 0 & \star & \star & \star \\ \star & \star & \star & \star & \star \end{bmatrix}.$	$E = \begin{bmatrix} 0 & 0 & \star & 0 & 0 \\ 0 & 0 & \star & \star & 0 \\ \star & 0 & \star & \star & 0 \\ \star & 0 & \star & \star & \star \\ \star & \star & \star & \star & \star \end{bmatrix}.$
johnston-linear-matrix-algebra-EQ8747 (0)	282	0.997	$\begin{array}{l} \left[\begin{array}{lll} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{array} \right] \xrightarrow[R_3-7R_1]{R_2-4R_1} \left[\begin{array}{lll} 1 & 2 & 3 \\ 0 & -3 & -6 \\ 0 & -6 & -12 \end{array} \right] \xrightarrow{R_3-2R_2} \left[\begin{array}{lll} 1 & 2 & 3 \\ 0 & -3 & -6 \\ 0 & 0 & 0 \end{array} \right]. \end{array}$	$\begin{array}{l} \left[\begin{array}{lll} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{array} \right] \xrightarrow[R_3-7R_1]{R_2-4R_1} \left[\begin{array}{lll} 1 & 2 & 3 \\ 0 & -3 & -6 \\ 0 & -6 & -12 \end{array} \right] \xrightarrow{R_3-2R_2} \left[\begin{array}{lll} 1 & 2 & 3 \\ 0 & -3 & -6 \\ 0 & 0 & 0 \end{array} \right]. \end{array}$	$\begin{array}{l} \left[\begin{array}{lll} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{array} \right] \xrightarrow[R_3-7R_1]{R_2-4R_1} \left[\begin{array}{lll} 1 & 2 & 3 \\ 0 & -3 & -6 \\ 0 & -6 & -12 \end{array} \right] \xrightarrow{R_3-2R_2} \left[\begin{array}{lll} 1 & 2 & 3 \\ 0 & -3 & -6 \\ 0 & 0 & 0 \end{array} \right]. \end{array}$

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johnston-linear-matrix-algebra-EQ1000	349	0.997	$A=\left[\begin{array}{ll} 1 & 1 \\ 0 & 0 \end{array}\right] \quad \text{and} \quad B=\left[\begin{array}{ll} 0 & 0 \\ 1 & 1 \end{array}\right],$	$A = \begin{bmatrix} 1 & 1 \\ 0 & 0 \end{bmatrix} \quad \text{and} \quad B = \begin{bmatrix} 0 & 0 \\ 1 & 1 \end{bmatrix},$	$A = \begin{bmatrix} 1 & 1 \\ 0 & 0 \end{bmatrix} \quad \text{and} \quad B = \begin{bmatrix} 0 & 0 \\ 1 & 1 \end{bmatrix},$
johnston-linear-matrix-algebra-EQ1251	419	0.997	$e^{i\theta} = \cos(\theta) + i \sin(\theta) \quad \text{for all } \theta \in [0, 2\pi).$	$e^{i\theta} = \cos(\theta) + i \sin(\theta) \quad \text{for all } \theta \in [0, 2\pi).$	$e^{i\theta} = \cos(\theta) + i \sin(\theta) \quad \text{for all } \theta \in [0, 2\pi).$
johnston-linear-matrix-algebra-EQ1604	478	0.997	$p_A(\lambda) = \det(A - \lambda I) = (a_{1,1} - \lambda)(a_{2,2} - \lambda) \cdots (a_{n,n} - \lambda),$	$p_A(\lambda) = \det(A - \lambda I) = (a_{1,1} - \lambda)(a_{2,2} - \lambda) \cdots (a_{n,n} - \lambda),$	$p_A(\lambda) = \det(A - \lambda I) = (a_{1,1} - \lambda)(a_{2,2} - \lambda) \cdots (a_{n,n} - \lambda),$
johnston-linear-matrix-algebra-EQ0838	306	0.997	$\begin{array}{l} \begin{array}{l} \{r\} - \lambda^2 + 2\lambda + 12 \\ \hline -\lambda^3 + 0\lambda^2 + 16\lambda + 24 \\ \hline -\lambda^3 - 2\lambda^2 \\ \hline 2\lambda^2 + 16\lambda + 24 \\ \hline 2\lambda^2 + 4\lambda \\ \hline 12\lambda + 24 \\ \hline 12\lambda + 24 \\ \hline 0 \end{array} \end{array}$	$\begin{array}{l} \begin{array}{l} \{r\} - \lambda^2 + 2\lambda + 12 \\ \hline -\lambda^3 + 0\lambda^2 + 16\lambda + 24 \\ \hline -\lambda^3 - 2\lambda^2 \\ \hline 2\lambda^2 + 16\lambda + 24 \\ \hline 2\lambda^2 + 4\lambda \\ \hline 12\lambda + 24 \\ \hline 12\lambda + 24 \\ \hline 0 \end{array} \end{array}$	$\begin{array}{l} \begin{array}{l} \{r\} - \lambda^2 + 2\lambda + 12 \\ \hline -\lambda^3 + 0\lambda^2 + 16\lambda + 24 \\ \hline -\lambda^3 - 2\lambda^2 \\ \hline 2\lambda^2 + 16\lambda + 24 \\ \hline 2\lambda^2 + 4\lambda \\ \hline 12\lambda + 24 \\ \hline 12\lambda + 24 \\ \hline 0 \end{array} \end{array}$
johnston-linear-matrix-algebra-EQ1139	394	0.997	$HD = \begin{bmatrix} 1 & 1 \\ -i & i \end{bmatrix} \begin{bmatrix} re^{i\theta} & 0 \\ 0 & re^{-i\theta} \end{bmatrix} = r \begin{bmatrix} e^{i\theta} & e^{-i\theta} \\ -ie^{i\theta} & ie^{-i\theta} \end{bmatrix}, \quad \text{and}$	$HD = \begin{bmatrix} 1 & 1 \\ -i & i \end{bmatrix} \begin{bmatrix} re^{i\theta} & 0 \\ 0 & re^{-i\theta} \end{bmatrix} = r \begin{bmatrix} e^{i\theta} & e^{-i\theta} \\ -ie^{i\theta} & ie^{-i\theta} \end{bmatrix}, \quad \text{and}$	$HD = \begin{bmatrix} 1 & 1 \\ -i & i \end{bmatrix} \begin{bmatrix} re^{i\theta} & 0 \\ 0 & re^{-i\theta} \end{bmatrix} = r \begin{bmatrix} e^{i\theta} & e^{-i\theta} \\ -ie^{i\theta} & ie^{-i\theta} \end{bmatrix}, \quad \text{and}$
johnston-linear-matrix-algebra-EQ0545 (2.B.13)	220	0.997	$\begin{array}{l} \text{maximize : } x_1 + 2x_2 \\ \text{subject to : } x_1 + x_2 = 3 \\ -x_1 + x_2 \leq 1 \\ x_1, x_2 \geq 0 \end{array}$	$\begin{array}{l} \text{maximize : } x_1 + 2x_2 \\ \text{subject to : } x_1 + x_2 = 3 \\ -x_1 + x_2 \leq 1 \\ x_1, x_2 \geq 0 \end{array}$	$\begin{array}{l} \text{maximize : } x_1 + 2x_2 \\ \text{subject to : } x_1 + x_2 = 3 \\ -x_1 + x_2 \leq 1 \\ x_1, x_2 \geq 0 \end{array}$

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johnston-linear-matrix-algebra-EQ0446	174	0.997	$\begin{aligned} & \left[\begin{array}{ccccc} 1 & 0 & -1 & 0 & 0 \\ 0 & 0 & 2 & 1 & -1 \\ 2 & 1 & -1 & 2 & 2 \\ 1 & 0 & -1 & 0 & 1 \\ -1 & -1 & 0 & -2 & -2 \end{array} \right] \xrightarrow[\begin{array}{c} R_3 - R_1 \\ R_4 + R_1 \end{array}]{R_2 - 2R_1} \left[\begin{array}{ccccc} 1 & 0 & -1 & 0 & 0 \\ 0 & 1 & 1 & 2 & 2 \\ 0 & 0 & 0 & 0 & 1 \\ 0 & -1 & -1 & -2 & -2 \end{array} \right] \\ & \xrightarrow{R_4 + R_2} \left[\begin{array}{ccccc} 1 & 0 & -1 & 0 & 0 \\ 0 & 1 & 1 & 2 & 2 \\ 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 \end{array} \right]. \end{aligned}$		
johnston-linear-matrix-algebra-EQ0072	47	0.998	$A \mathbf{B} = A \left[\mathbf{b}_1 \mid \mathbf{b}_2 \mid \cdots \mid \mathbf{b}_p \right] = \left[A \mathbf{b}_1 \mid A \mathbf{b}_2 \mid \cdots \mid A \mathbf{b}_p \right],$	$AB = A \left[\mathbf{b}_1 \mid \mathbf{b}_2 \mid \cdots \mid \mathbf{b}_p \right] = \left[A \mathbf{b}_1 \mid A \mathbf{b}_2 \mid \cdots \mid A \mathbf{b}_p \right],$	$AB = A \left[\mathbf{b}_1 \mid \mathbf{b}_2 \mid \cdots \mid \mathbf{b}_p \right] = \left[A \mathbf{b}_1 \mid A \mathbf{b}_2 \mid \cdots \mid A \mathbf{b}_p \right],$
johnston-linear-matrix-algebra-EQ0111	61	0.998	$\left[F_{\mathbf{u}} \right] \mathbf{w} = \frac{1}{3} \left[\begin{array}{ccc} -1 & 2 & 2 \\ 2 & -1 & 2 \\ 2 & 2 & -1 \end{array} \right] \left[\begin{array}{c} 1 \\ 1 \\ 1 \end{array} \right] = \left[\begin{array}{c} 1 \\ 1 \\ 1 \end{array} \right] = \mathbf{w}.$	$\left[F_{\mathbf{u}} \right] \mathbf{w} = \frac{1}{3} \left[\begin{array}{ccc} -1 & 2 & 2 \\ 2 & -1 & 2 \\ 2 & 2 & -1 \end{array} \right] \left[\begin{array}{c} 1 \\ 1 \\ 1 \end{array} \right] = \left[\begin{array}{c} 1 \\ 1 \\ 1 \end{array} \right] = \mathbf{w}.$	$\left[F_{\mathbf{u}} \right] \mathbf{w} = \frac{1}{3} \left[\begin{array}{ccc} -1 & 2 & 2 \\ 2 & -1 & 2 \\ 2 & 2 & -1 \end{array} \right] \left[\begin{array}{c} 1 \\ 1 \\ 1 \end{array} \right] = \left[\begin{array}{c} 1 \\ 1 \\ 1 \end{array} \right] = \mathbf{w}.$
johnston-linear-matrix-algebra-EQ0121 (0)	65	0.998	$\begin{aligned} \left[R_{zx} \right] &= \left[\begin{array}{ccc} \cos(\theta) & 0 & -\sin(\theta) \\ 0 & 1 & 0 \\ \sin(\theta) & 0 & \cos(\theta) \end{array} \right] \quad \text{and} \quad \left[R_{xy} \right] = \left[\begin{array}{ccc} \cos(\theta) & -\sin(\theta) & 0 \\ \sin(\theta) & \cos(\theta) & 0 \\ 0 & 0 & 1 \end{array} \right]. \\ \left[R_{zy} \right] &= \left[\begin{array}{ccc} \cos(\theta) & -\sin(\theta) & 0 \\ 0 & 1 & 0 \\ \sin(\theta) & 0 & \cos(\theta) \end{array} \right] \quad \text{and} \quad \left[R_{yx} \right] = \left[\begin{array}{ccc} \cos(\theta) & \sin(\theta) & 0 \\ -\sin(\theta) & \cos(\theta) & 0 \\ 0 & 0 & 1 \end{array} \right]. \end{aligned}$	$\left[R_{zx}^{\theta} \right] = \left[\begin{array}{ccc} \cos(\theta) & 0 & -\sin(\theta) \\ 0 & 1 & 0 \\ \sin(\theta) & 0 & \cos(\theta) \end{array} \right] \quad \text{and} \quad \left[R_{xy}^{\theta} \right] = \left[\begin{array}{ccc} \cos(\theta) & -\sin(\theta) & 0 \\ \sin(\theta) & \cos(\theta) & 0 \\ 0 & 0 & 1 \end{array} \right].$	$\left[R_{zx}^{\theta} \right] = \left[\begin{array}{ccc} \cos(\theta) & 0 & -\sin(\theta) \\ 0 & 1 & 0 \\ \sin(\theta) & 0 & \cos(\theta) \end{array} \right] \quad \text{and} \quad \left[R_{xy}^{\theta} \right] = \left[\begin{array}{ccc} \cos(\theta) & -\sin(\theta) & 0 \\ \sin(\theta) & \cos(\theta) & 0 \\ 0 & 0 & 1 \end{array} \right].$
johnston-linear-matrix-algebra-EQ0324	128	0.998	$A = E_1^{-1} E_2^{-1} \cdots E_k^{-1}.$	$A = E_1^{-1} E_2^{-1} \cdots E_k^{-1}.$	$A = E_1^{-1} E_2^{-1} \cdots E_k^{-1}.$
johnston-linear-matrix-algebra-EQ0589	227	0.998	$A = \left[\begin{array}{c} 1 \\ 0 \\ -1 \end{array} \right] \left[\begin{array}{cccc} 1 & 0 & 0 & -1 \end{array} \right] + \left[\begin{array}{c} 1 \\ 1 \\ 1 \end{array} \right] \left[\begin{array}{cccc} 0 & 1 & 1 & 0 \end{array} \right].$	$A = \left[\begin{array}{c} 1 \\ 0 \\ -1 \end{array} \right] \left[\begin{array}{cccc} 1 & 0 & 0 & -1 \end{array} \right] + \left[\begin{array}{c} 1 \\ 1 \\ 1 \end{array} \right] \left[\begin{array}{cccc} 0 & 1 & 1 & 0 \end{array} \right].$	$A = \left[\begin{array}{c} 1 \\ 0 \\ -1 \end{array} \right] \left[\begin{array}{cccc} 1 & 0 & 0 & -1 \end{array} \right] + \left[\begin{array}{c} 1 \\ 1 \\ 1 \end{array} \right] \left[\begin{array}{cccc} 0 & 1 & 1 & 0 \end{array} \right].$
johnston-linear-matrix-algebra-EQ0692	261	0.998	$\begin{aligned} \left[\mathbf{v}_1 \right]_E &= (1, 1, -1), \quad \left[\mathbf{v}_2 \right]_E = (2, 2, -1), \quad \text{and} \quad \left[\mathbf{v}_3 \right]_E = (-1, -2, 1). \\ \left[\mathbf{v}_1 \right]_E &= (1, 1, -1), \quad \left[\mathbf{v}_2 \right]_E = (2, 2, -1), \quad \text{and} \quad \left[\mathbf{v}_3 \right]_E = (-1, -2, 1). \end{aligned}$	$\left[\mathbf{v}_1 \right]_E = (1, 1, -1), \quad \left[\mathbf{v}_2 \right]_E = (2, 2, -1), \quad \text{and} \quad \left[\mathbf{v}_3 \right]_E = (-1, -2, 1).$	$\left[\mathbf{v}_1 \right]_E = (1, 1, -1), \quad \left[\mathbf{v}_2 \right]_E = (2, 2, -1), \quad \text{and} \quad \left[\mathbf{v}_3 \right]_E = (-1, -2, 1).$

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johnston-linear-matrix-algebra- EQ0704	266	■ 0.998	$P_{E \leftarrow B} = \left[\begin{array}{ccc} 1 & 2 & -1 \\ 2 & -1 & 1 \\ 3 & 0 & 2 \end{array} \right], \quad \text{so} \quad P_{B \leftarrow E} = P_{E \leftarrow B}^{-1} = \frac{1}{7} \left[\begin{array}{ccc} 2 & 4 & -1 \\ 1 & -5 & 3 \\ -3 & -6 & 5 \end{array} \right].$	$P_{E \leftarrow B} = \left[\begin{array}{ccc} 1 & 2 & -1 \\ 2 & -1 & 1 \\ 3 & 0 & 2 \end{array} \right], \quad \text{so} \quad P_{B \leftarrow E} = P_{E \leftarrow B}^{-1} = \frac{1}{7} \left[\begin{array}{ccc} 2 & 4 & -1 \\ 1 & -5 & 3 \\ -3 & -6 & 5 \end{array} \right].$	
johnston-linear-matrix-algebra- EQ0846	308	■ 0.998	$\begin{aligned} p_A(\lambda) &= \det(A - \lambda I) = \det \left(\begin{bmatrix} -3-\lambda & -2 \\ 4 & 1-\lambda \end{bmatrix} \right) \\ &= (-3-\lambda)(1-\lambda) + 8 \\ &= \lambda^2 + 2\lambda + 5. \end{aligned}$	$p_A(\lambda) = \det(A - \lambda I) = \det \left(\begin{bmatrix} -3-\lambda & -2 \\ 4 & 1-\lambda \end{bmatrix} \right) \\ = (-3-\lambda)(1-\lambda) + 8 \\ = \lambda^2 + 2\lambda + 5.$	$p_A(\lambda) = \det(A - \lambda I) = \det \left(\begin{bmatrix} -3-\lambda & -2 \\ 4 & 1-\lambda \end{bmatrix} \right) \\ = (-3-\lambda)(1-\lambda) + 8 \\ = \lambda^2 + 2\lambda + 5.$
johnston-linear-matrix-algebra- EQ0850	310	■ 0.998	$\begin{aligned} p_A(\lambda) &= \det(A - \lambda I) = \det \left(\begin{bmatrix} 4-\lambda & 0 & 2 \\ -2 & 6-\lambda & 2 \\ 2 & 0 & 4-\lambda \end{bmatrix} \right) \\ &= (4-\lambda)(6-\lambda)(4-\lambda) + 0 + 0 \\ &\quad - 0 - 0 - 4(6-\lambda) \\ &= -\lambda^3 + 14\lambda^2 - 60\lambda + 72, \quad \text{but} \end{aligned}$	$p_A(\lambda) = \det(A - \lambda I) = \det \left(\begin{bmatrix} 4-\lambda & 0 & 2 \\ -2 & 6-\lambda & 2 \\ 2 & 0 & 4-\lambda \end{bmatrix} \right) \\ = (4-\lambda)(6-\lambda)(4-\lambda) + 0 + 0 \\ - 0 - 0 - 4(6-\lambda) \\ = -\lambda^3 + 14\lambda^2 - 60\lambda + 72, \quad \text{but}$	$p_A(\lambda) = \det(A - \lambda I) = \det \left(\begin{bmatrix} 4-\lambda & 0 & 2 \\ -2 & 6-\lambda & 2 \\ 2 & 0 & 4-\lambda \end{bmatrix} \right) \\ = (4-\lambda)(6-\lambda)(4-\lambda) + 0 + 0 \\ - 0 - 0 - 4(6-\lambda) \\ = -\lambda^3 + 14\lambda^2 - 60\lambda + 72, \quad \text{but}$
johnston-linear-matrix-algebra- EQ0961 (1)	339	■ 0.998	$D = \left[\begin{array}{cccccc} -2 & 0 & 0 & 0 & 0 & 0 \\ 0 & -1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 3 & 0 \end{array} \right] \quad \text{and} \quad P = \left[\begin{array}{ccccc} 1 & 1 & 0 & 0 & 3 \\ 1 & -1 & 0 & 0 & 3 \\ -1 & 0 & 1 & 0 & 2 \\ -1 & 0 & -1 & 1 & 2 \\ -1 & 0 & 0 & -1 & 2 \end{array} \right].$	$D = \left[\begin{array}{ccccc} -2 & 0 & 0 & 0 & 0 \\ 0 & -1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 3 \end{array} \right] \quad \text{and} \quad P = \left[\begin{array}{ccccc} 1 & 1 & 0 & 0 & 3 \\ 1 & -1 & 0 & 0 & 3 \\ -1 & 0 & 1 & 0 & 2 \\ -1 & 0 & -1 & 1 & 2 \\ -1 & 0 & 0 & -1 & 2 \end{array} \right].$	$D = \left[\begin{array}{ccccc} -2 & 0 & 0 & 0 & 0 \\ 0 & -1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 3 \end{array} \right] \quad \text{and} \quad P = \left[\begin{array}{ccccc} 1 & 1 & 0 & 0 & 3 \\ 1 & -1 & 0 & 0 & 3 \\ -1 & 0 & 1 & 0 & 2 \\ -1 & 0 & -1 & 1 & 2 \\ -1 & 0 & 0 & -1 & 2 \end{array} \right].$

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johnston-linear-matrix-algebra-EQ1102	378	■ 0.998	$\begin{aligned} & \lim_{k \rightarrow \infty} \mathbf{v}_k^* A \mathbf{v}_k = \lim_{k \rightarrow \infty} \frac{ \lambda_1 ^{2k} (c_1 \mathbf{w}_1 + \mathbf{r}_k)^* A (c_1 \mathbf{w}_1 + \mathbf{r}_k)}{\ \lambda_1^k (c_1 \mathbf{w}_1 + \mathbf{r}_k)\ ^2} \\ & = \frac{(c_1 \mathbf{w}_1 + \lim_{k \rightarrow \infty} \mathbf{r}_k)^* A (c_1 \mathbf{w}_1 + \lim_{k \rightarrow \infty} \mathbf{r}_k)}{\ c_1 \mathbf{w}_1 + \lim_{k \rightarrow \infty} \mathbf{r}_k\ ^2} \\ & = \frac{ c_1 ^2 \mathbf{w}_1^* A \mathbf{w}_1}{\ c_1 \mathbf{w}_1\ ^2} = \mathbf{w}_1^* A \mathbf{w}_1 = \mathbf{w}_1^* (\lambda_1 \mathbf{w}_1) = \lambda_1, \end{aligned}$		
johnston-linear-matrix-algebra-EQ1372	451	■ 0.998	$\begin{aligned} & \left[R^{-\pi/4} \right] [P_{\mathbf{u}}] = \frac{1}{2\sqrt{2}} \begin{bmatrix} 1 & 1 \\ -1 & 1 \end{bmatrix} \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix} \\ & = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 1 \\ 0 & 0 \end{bmatrix}. \end{aligned}$		
johnston-linear-matrix-algebra-EQ1577	474	■ 0.998	$\begin{aligned} & R^\theta(\mathbf{e}_1) \cdot R^\theta(\mathbf{e}_2) \\ & = (\cos(\theta), \sin(\theta)) \cdot (-\sin(\theta), \cos(\theta)) \\ & = -\cos(\theta) \sin(\theta) + \sin(\theta) \cos(\theta) = 0. \end{aligned}$		
johnston-linear-matrix-algebra-EQ1595	476	■ 0.998	$\begin{aligned} & \det(A + \mathbf{v} \mathbf{w}^T) = \det(A) \det(I_n + A^{-1} \mathbf{v} \mathbf{w}^T) \\ & = \det(A) \det(I_n + (A^{-1} \mathbf{v}) \mathbf{w}^T) \\ & = \det(A) (1 + \mathbf{w}^T A^{-1} \mathbf{v}), \end{aligned}$		
johnston-linear-matrix-algebra-EQ1633	481	■ 0.998	$\begin{aligned} & A^r A^s = (PD^r P^{-1})(PD^s P^{-1}) \\ & = PD^r D^s P^{-1} = PD^{r+s} P^{-1} = A^{r+s}, \end{aligned}$		

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johnston-linear-matrix-algebra-EQ1655	483	0.998	$\begin{aligned} \operatorname{per}\left(A^T\right) &= \sum_{\sigma \in S_n} a_{1, \sigma(1)} a_{2, \sigma(2)} \cdots a_{n, \sigma(n)} \\ &= \sum_{\sigma \in S_n} a_{\sigma^{-1}(1), 1} a_{\sigma^{-1}(2), 2} \cdots a_{\sigma^{-1}(n), n} \\ &= \sum_{\sigma \in S_n} a_{\sigma(1), 1} a_{\sigma(2), 2} \cdots a_{\sigma(n), n} \\ &= \operatorname{per}(A), \end{aligned}$	$\begin{aligned} \operatorname{per}\left(A^T\right) &= \sum_{\sigma \in S_n} a_{1, \sigma(1)} a_{2, \sigma(2)} \cdots a_{n, \sigma(n)} \\ &= \sum_{\sigma \in S_n} a_{\sigma^{-1}(1), 1} a_{\sigma^{-1}(2), 2} \cdots a_{\sigma^{-1}(n), n} \\ &= \sum_{\sigma \in S_n} a_{\sigma(1), 1} a_{\sigma(2), 2} \cdots a_{\sigma(n), n} \\ &= \operatorname{per}(A), \end{aligned}$	$\begin{aligned} \operatorname{per}\left(A^T\right) &= \sum_{\sigma \in S_n} a_{1, \sigma(1)} a_{2, \sigma(2)} \cdots a_{n, \sigma(n)} \\ &= \sum_{\sigma \in S_n} a_{\sigma^{-1}(1), 1} a_{\sigma^{-1}(2), 2} \cdots a_{\sigma^{-1}(n), n} \\ &= \sum_{\sigma \in S_n} a_{\sigma(1), 1} a_{\sigma(2), 2} \cdots a_{\sigma(n), n} \\ &= \operatorname{per}(A), \end{aligned}$
johnston-linear-matrix-algebra-EQ1221	411	0.998	$\begin{aligned} \lim _{n \rightarrow \infty} \frac{x_{n+1}}{x_n} &= \lim _{n \rightarrow \infty} \frac{q_0(n+1) \lambda_0^{n+1}+\cdots+q_{m-1}(n+1) \lambda_{m-1}^{n+1}}{q_0(n) \lambda_0^n+\cdots+q_{m-1}(n) \lambda_{m-1}^n} \\ &= \lim _{n \rightarrow \infty} \frac{q_0(n+1) \lambda_0^{n+1}+\cdots+q_{m-1}(n+1) \lambda_{m-1}^{n+1}}{q_0(n) \lambda_0^n+\cdots+q_{m-1}(n) \lambda_{m-1}^n} \cdot \frac{\left(\frac{1}{q_0(n) \lambda_0^n}\right)}{\left(\frac{1}{q_0(n) \lambda_0^n}\right)} \\ &= \lim _{n \rightarrow \infty} \frac{\frac{q_0(n+1)}{q_0(n)} \lambda_0+\cdots+\lambda_{m-1} \frac{q_{m-1}(n+1)}{q_0(n)}\left(\frac{\lambda_{m-1}}{\lambda_0}\right)^n}{1+\cdots+\frac{q_{m-1}(n)}{q_0(n)}\left(\frac{\lambda_{m-1}}{\lambda_0}\right)^n} \\ &= \frac{\lambda_0+0+\cdots+0}{1+0+\cdots+0} \\ &= \lambda_0, \end{aligned}$	$\begin{aligned} \lim _{n \rightarrow \infty} \frac{x_{n+1}}{x_n} &= \lim _{n \rightarrow \infty} \frac{q_0(n+1) \lambda_0^{n+1}+\cdots+q_{m-1}(n+1) \lambda_{m-1}^{n+1}}{q_0(n) \lambda_0^n+\cdots+q_{m-1}(n) \lambda_{m-1}^n} \\ &= \lim _{n \rightarrow \infty} \frac{q_0(n+1) \lambda_0^{n+1}+\cdots+q_{m-1}(n+1) \lambda_{m-1}^{n+1}}{q_0(n) \lambda_0^n+\cdots+q_{m-1}(n) \lambda_{m-1}^n} \cdot \frac{\left(\frac{1}{q_0(n) \lambda_0^n}\right)}{\left(\frac{1}{q_0(n) \lambda_0^n}\right)} \\ &= \lim _{n \rightarrow \infty} \frac{\frac{q_0(n+1)}{q_0(n)} \lambda_0+\cdots+\lambda_{m-1} \frac{q_{m-1}(n+1)}{q_0(n)}\left(\frac{\lambda_{m-1}}{\lambda_0}\right)^n}{1+\cdots+\frac{q_{m-1}(n)}{q_0(n)}\left(\frac{\lambda_{m-1}}{\lambda_0}\right)^n} \\ &= \frac{\lambda_0+0+\cdots+0}{1+0+\cdots+0} \\ &= \lambda_0 \end{aligned}$	$\begin{aligned} \lim _{n \rightarrow \infty} \frac{x_{n+1}}{x_n} &= \lim _{n \rightarrow \infty} \frac{q_0(n+1) \lambda_0^{n+1}+\cdots+q_{m-1}(n+1) \lambda_{m-1}^{n+1}}{q_0(n) \lambda_0^n+\cdots+q_{m-1}(n) \lambda_{m-1}^n} \\ &= \lim _{n \rightarrow \infty} \frac{q_0(n+1) \lambda_0^{n+1}+\cdots+q_{m-1}(n+1) \lambda_{m-1}^{n+1}}{q_0(n) \lambda_0^n+\cdots+q_{m-1}(n) \lambda_{m-1}^n} \cdot \frac{\left(\frac{1}{q_0(n) \lambda_0^n}\right)}{\left(\frac{1}{q_0(n) \lambda_0^n}\right)} \\ &= \lim _{n \rightarrow \infty} \frac{\frac{q_0(n+1)}{q_0(n)} \lambda_0+\cdots+\lambda_{m-1} \frac{q_{m-1}(n+1)}{q_0(n)}\left(\frac{\lambda_{m-1}}{\lambda_0}\right)^n}{1+\cdots+\frac{q_{m-1}(n)}{q_0(n)}\left(\frac{\lambda_{m-1}}{\lambda_0}\right)^n} \\ &= \frac{\lambda_0+0+\cdots+0}{1+0+\cdots+0} \\ &= \lambda_0, \end{aligned}$
johnston-linear-matrix-algebra-EQ0224	104	0.998	$\text { next leading entry }\left[\begin{array}{ccccc} 0 & 2 & 0 & 4 & 0 \\ 0 & 0 & 1 & -1 & 0 \\ 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 \end{array}\right] \xrightarrow{\frac{1}{2} R_1}\left[\begin{array}{ccccc} 0 & 1 & 0 & 2 & 0 \\ 0 & 0 & 1 & -1 & 0 \\ 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 \end{array}\right]$	$\text { next leading entry }\left[\begin{array}{ccccc} 0 & 2 & 0 & 4 & 0 \\ 0 & 0 & 1 & -1 & 0 \\ 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 \end{array}\right] \xrightarrow{\frac{1}{2} R_1}\left[\begin{array}{ccccc} 0 & 1 & 0 & 2 & 0 \\ 0 & 0 & 1 & -1 & 0 \\ 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 \end{array}\right]$	next leading entry $\xrightarrow{\quad}$ $\left[\begin{array}{ccccc} 0 & 2 & 0 & 4 & 0 \\ 0 & 0 & 1 & -1 & 0 \\ 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 \end{array}\right] \xrightarrow{\frac{1}{2} R_1} \left[\begin{array}{ccccc} 0 & 1 & 0 & 2 & 0 \\ 0 & 0 & 1 & -1 & 0 \\ 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 \end{array}\right]$ $\xleftarrow{\quad}$ scaled to 1
johnston-linear-matrix-algebra-EQ0129	68	0.998	$[T]=[D]\left[F_{\mathbf{u}}\right]=\frac{1}{25}\left[\begin{array}{cc} 2 & 0 \\ 0 & 3 \end{array}\right]\left[\begin{array}{cc} -7 & 24 \\ 24 & 7 \end{array}\right]=\frac{1}{25}\left[\begin{array}{cc} -14 & 48 \\ 72 & 21 \end{array}\right].$	$[T]=[D]\left[F_{\mathbf{u}}\right]=\frac{1}{25}\left[\begin{array}{cc} 2 & 0 \\ 0 & 3 \end{array}\right]\left[\begin{array}{cc} -7 & 24 \\ 24 & 7 \end{array}\right]=\frac{1}{25}\left[\begin{array}{cc} -14 & 48 \\ 72 & 21 \end{array}\right].$	$[T]=[D]\left[F_{\mathbf{u}}\right]=\frac{1}{25}\left[\begin{array}{cc} 2 & 0 \\ 0 & 3 \end{array}\right]\left[\begin{array}{cc} -7 & 24 \\ 24 & 7 \end{array}\right]=\frac{1}{25}\left[\begin{array}{cc} -14 & 48 \\ 72 & 21 \end{array}\right].$

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johnston-linear-matrix-algebra-EQ0178	87	■ 0.998	$\left[A^2\right]_{1,4}=(0,1,3,0) \cdot (0,1,2,0)=0+1+6+0=7$.	$[A^2]_{1,4} = (0, 1, 3, 0) \cdot (0, 1, 2, 0) = 0 + 1 + 6 + 0 = 7.$	$[A^2]_{1,4} = (0, 1, 3, 0) \cdot (0, 1, 2, 0) = 0 + 1 + 6 + 0 = 7.$
johnston-linear-matrix-algebra-EQ0302	121	■ 0.998	$E_1=\left[\begin{array}{ccc} 1 & 0 & 0 \\ -3 & 0 & 1 \end{array}\right]$ and $E_2=\left[\begin{array}{ccc} 1 & 0 & 0 \\ 3 & 0 & 1 \end{array}\right]$,	$E_1 = \left[\begin{array}{ccc} 1 & 0 & 0 \\ 0 & 1 & 0 \\ -3 & 0 & 1 \end{array}\right]$ and $E_2 = \left[\begin{array}{ccc} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 3 & 0 & 1 \end{array}\right]$,	$E_1 = \left[\begin{array}{ccc} 1 & 0 & 0 \\ 0 & 1 & 0 \\ -3 & 0 & 1 \end{array}\right]$ and $E_2 = \left[\begin{array}{ccc} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 3 & 0 & 1 \end{array}\right]$,
johnston-linear-matrix-algebra-EQ0337	132	■ 0.998	$A \operatorname{\mathbf{x}}=A\left(A^{-1} \operatorname{\mathbf{b}}\right)=\left(A^{-1} A\right) \operatorname{\mathbf{b}}=I \operatorname{\mathbf{b}}=\operatorname{\mathbf{b}}$.	$A\mathbf{x} = A \left(A^{-1}\mathbf{b}\right) = \left(A^{-1}A\right) \mathbf{b} = I\mathbf{b} = \mathbf{b}.$	$A\mathbf{x} = A \left(A^{-1}\mathbf{b}\right) = \left(A^{-1}A\right) \mathbf{b} = I\mathbf{b} = \mathbf{b}.$
johnston-linear-matrix-algebra-EQ0397	152	■ 0.998	$\left[\begin{array}{ccccc} 1 & x_0 & x_0^2 & \cdots & x_0^n \\ 1 & x_1 & x_1^2 & \cdots & x_1^n \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 1 & x_n & x_n^2 & \cdots & x_n^n \end{array}\right]\left[\begin{array}{c} c_0 \\ c_1 \\ \vdots \\ c_n \end{array}\right]=\left[\begin{array}{c} y_0 \\ y_1 \\ \vdots \\ y_n \end{array}\right]$,	$\left[\begin{array}{ccccc} 1 & x_0 & x_0^2 & \cdots & x_0^n \\ 1 & x_1 & x_1^2 & \cdots & x_1^n \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 1 & x_n & x_n^2 & \cdots & x_n^n \end{array}\right]\left[\begin{array}{c} c_0 \\ c_1 \\ \vdots \\ c_n \end{array}\right] = \left[\begin{array}{c} y_0 \\ y_1 \\ \vdots \\ y_n \end{array}\right]$,	$\left[\begin{array}{ccccc} 1 & x_0 & x_0^2 & \cdots & x_0^n \\ 1 & x_1 & x_1^2 & \cdots & x_1^n \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 1 & x_n & x_n^2 & \cdots & x_n^n \end{array}\right]\left[\begin{array}{c} c_0 \\ c_1 \\ \vdots \\ c_n \end{array}\right] = \left[\begin{array}{c} y_0 \\ y_1 \\ \vdots \\ y_n \end{array}\right]$,
johnston-linear-matrix-algebra-EQ0697	263	■ 0.998	$[T]_B=\left[\left[T\left(\operatorname{\mathbf{e}}_1\right)\right]_B \mid\left[T\left(\operatorname{\mathbf{e}}_2\right)\right]_B\right]=\left[\begin{array}{cc} 2 & 1 \\ 1 & 2 \end{array}\right]$.	$[T]_B = [[T(\mathbf{e}_1)]_B \mid [T(\mathbf{e}_2)]_B] = \left[\begin{array}{cc} 2 & 1 \\ 1 & 2 \end{array}\right]$.	$[T]_B = [[T(\mathbf{e}_1)]_B \mid [T(\mathbf{e}_2)]_B] = \left[\begin{array}{cc} 2 & 1 \\ 1 & 2 \end{array}\right]$.
johnston-linear-matrix-algebra-EQ0814	298	■ 0.998	$\operatorname{det}\left(A+\operatorname{\mathbf{v}} \operatorname{\mathbf{w}}^T\right)=\left(1+\operatorname{\mathbf{w}}^T A^{-1} \operatorname{\mathbf{v}}\right) \operatorname{det}(A)$.	$\det \left(A + \mathbf{vw}^T\right) = \left(1 + \mathbf{w}^T A^{-1} \mathbf{v}\right) \det(A).$	$\det(A + \mathbf{vw}^T) = (1 + \mathbf{w}^T A^{-1} \mathbf{v}) \det(A).$
johnston-linear-matrix-algebra-EQ0854	311	■ 0.998	$\begin{aligned} p_A(\lambda)=\operatorname{det}(A-\lambda I) &= (-1)^n \lambda^n + c_{n-1} \lambda^{n-1} + \cdots + c_1 \lambda + c_0 \\ &= (\lambda_1 - \lambda) (\lambda_2 - \lambda) \cdots (\lambda_n - \lambda), \end{aligned}$	$p_A(\lambda) = \det(A - \lambda I) = (-1)^n \lambda^n + c_{n-1} \lambda^{n-1} + \cdots + c_1 \lambda + c_0$ $= (\lambda_1 - \lambda) (\lambda_2 - \lambda) \cdots (\lambda_n - \lambda),$	$p_A(\lambda) = \det(A - \lambda I) = (-1)^n \lambda^n + c_{n-1} \lambda^{n-1} + \cdots + c_1 \lambda + c_0$ $= (\lambda_1 - \lambda) (\lambda_2 - \lambda) \cdots (\lambda_n - \lambda),$
johnston-linear-matrix-algebra-EQ0875	315	■ 0.998	$\begin{aligned} \operatorname{det}\left(\left[\begin{array}{cc} 2-\lambda & 1+i \\ 1-i & 3-\lambda \end{array}\right]\right) &= (2-\lambda)(3-\lambda) - (1+i)(1-i) \\ &= \lambda^2 - 5\lambda + 6 - 2 \\ &= \lambda^2 - 5\lambda + 4 \\ &= (\lambda - 1)(\lambda - 4). \end{aligned}$	$\det \left(\left[\begin{array}{cc} 2-\lambda & 1+i \\ 1-i & 3-\lambda \end{array}\right]\right) = (2-\lambda)(3-\lambda) - (1+i)(1-i)$ $= \lambda^2 - 5\lambda + 6 - 2$ $= \lambda^2 - 5\lambda + 4$ $= (\lambda - 1)(\lambda - 4).$	$\det \left(\left[\begin{array}{cc} 2-\lambda & 1+i \\ 1-i & 3-\lambda \end{array}\right]\right) = (2-\lambda)(3-\lambda) - (1+i)(1-i)$ $= \lambda^2 - 5\lambda + 6 - 2$ $= \lambda^2 - 5\lambda + 4$ $= (\lambda - 1)(\lambda - 4).$
johnston-linear-matrix-algebra-EQ0907	324	■ 0.998	$\left[\begin{array}{ll} 1 / 2 & 0 \\ 1 / 2 & 1 \end{array}\right]$ and $\left[\begin{array}{ll} 1 / 3 & 2 / 3 \\ 1 / 2 & 1 / 2 \end{array}\right]$	$\left[\begin{array}{cc} 1/2 & 0 \\ 1/2 & 1 \end{array}\right]$ and $\left[\begin{array}{cc} 1/3 & 2/3 \\ 1/2 & 1/2 \end{array}\right]$	$\left[\begin{array}{cc} 1/2 & 0 \\ 1/2 & 1 \end{array}\right]$ and $\left[\begin{array}{cc} 1/3 & 2/3 \\ 1/2 & 1/2 \end{array}\right]$
johnston-linear-matrix-algebra-EQ1065	364	■ 0.998	$\left[P_{\sigma}^T P_{\sigma}\right]_{i, j}=\operatorname{\mathbf{e}}_{\sigma(i)}^T \operatorname{\mathbf{e}}_{\sigma(j)}=\operatorname{\mathbf{e}}_{\sigma(i)} \cdot \operatorname{\mathbf{e}}_{\sigma(j)}$.	$[P_{\sigma}^T P_{\sigma}]_{i,j} = \mathbf{e}_{\sigma(i)}^T \mathbf{e}_{\sigma(j)} = \mathbf{e}_{\sigma(i)} \cdot \mathbf{e}_{\sigma(j)}.$	$[P_{\sigma}^T P_{\sigma}]_{i,j} = \mathbf{e}_{\sigma(i)}^T \mathbf{e}_{\sigma(j)} = \mathbf{e}_{\sigma(i)} \cdot \mathbf{e}_{\sigma(j)}.$

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johnston-linear-matrix-algebra-EQ1390	454	0.998	$\begin{aligned} & \mathbf{v} \cdot \mathbf{w} \times \mathbf{x} \\ &= \mathbf{v} \cdot (w_2x_3 - w_3x_2, w_3x_1 - w_1x_3, w_1x_2 - w_2x_1) \\ &= v_1(w_2x_3 - w_3x_2) + v_2(w_3x_1 - w_1x_3) \\ &\quad + v_3(w_1x_2 - w_2x_1) \\ &= v_1w_2x_3 - v_1w_3x_2 + v_2w_3x_1 - v_2w_1x_3 \\ &\quad + v_3w_1x_2 - v_3w_2x_1. \end{aligned}$	$\mathbf{v} \cdot (\mathbf{w} \times \mathbf{x})$ $= \mathbf{v} \cdot (w_2x_3 - w_3x_2, w_3x_1 - w_1x_3, w_1x_2 - w_2x_1)$ $= v_1(w_2x_3 - w_3x_2) + v_2(w_3x_1 - w_1x_3)$ $+ v_3(w_1x_2 - w_2x_1)$ $= v_1w_2x_3 - v_1w_3x_2 + v_2w_3x_1 - v_2w_1x_3$ $+ v_3w_1x_2 - v_3w_2x_1.$	
johnston-linear-matrix-algebra-EQ1435	458	0.998	$\begin{array}{r} 2x+y=3 \\ x+y=2 \end{array}$	$\begin{array}{l} 2x + y = 3 \\ x + y = 2 \end{array}$	$\begin{array}{l} 2x + y = 3 \\ x + y = 2 \end{array}$
johnston-linear-matrix-algebra-EQ1441	458	0.998	$\begin{aligned} b+m &= 2 \\ b+3m &= 8 \end{aligned}$	$\begin{array}{l} b + m = 2 \\ b + 3m = 8 \end{array}$	$\begin{array}{l} b + m = 2 \\ b + 3m = 8 \end{array}$
johnston-linear-matrix-algebra-EQ1586	476	0.998	$\operatorname{det}(\left(\begin{array}{cc} A & 0 \\ 0 & I \end{array}\right)) = \operatorname{det}(A)$	$\det\left(\begin{bmatrix} A & O \\ O & I \end{bmatrix}\right) = \det(A)$	$\det\left(\begin{bmatrix} A & O \\ O & I \end{bmatrix}\right) = \det(A)$
johnston-linear-matrix-algebra-EQ1670	485	0.998	$\begin{aligned} x_n &= i^n + (-i)^n \\ &= e^{in\pi/2} + e^{-in\pi/2} = 2\cos(n\pi/2). \end{aligned}$	$\begin{aligned} x_n &= i^n + (-i)^n \\ &= e^{in\pi/2} + e^{-in\pi/2} = 2\cos(n\pi/2). \end{aligned}$	$\begin{aligned} x_n &= i^n + (-i)^n \\ &= e^{in\pi/2} + e^{-in\pi/2} = 2\cos(n\pi/2). \end{aligned}$
johnston-linear-matrix-algebra-EQ0090	54	0.998	$[T] \stackrel{\text{def}}{=} [T(\mathbf{e}_1) \mid T(\mathbf{e}_2) \mid \cdots \mid T(\mathbf{e}_n)].$	$[T] \stackrel{\text{def}}{=} [T(\mathbf{e}_1) \mid T(\mathbf{e}_2) \mid \cdots \mid T(\mathbf{e}_n)].$	$[T] \stackrel{\text{def}}{=} [T(\mathbf{e}_1) \mid T(\mathbf{e}_2) \mid \cdots \mid T(\mathbf{e}_n)].$
johnston-linear-matrix-algebra-EQ1190	405	0.998	$\left(\begin{array}{ccc} 1 & 1 & 1 \\ 1 & 2 & 5 \\ 1 & 4 & 25 \end{array}\right) \left(\begin{array}{l} c_0 \\ c_1 \\ c_2 \end{array}\right) = \left(\begin{array}{l} 1 \\ 4 \\ 22 \end{array}\right).$	$\begin{bmatrix} 1 & 1 & 1 \\ 1 & 2 & 5 \\ 1 & 4 & 25 \end{bmatrix} \begin{bmatrix} c_0 \\ c_1 \\ c_2 \end{bmatrix} = \begin{bmatrix} 1 \\ 4 \\ 22 \end{bmatrix}.$	$\begin{bmatrix} 1 & 1 & 1 \\ 1 & 2 & 5 \\ 1 & 4 & 25 \end{bmatrix} \begin{bmatrix} c_0 \\ c_1 \\ c_2 \end{bmatrix} = \begin{bmatrix} 1 \\ 4 \\ 22 \end{bmatrix}.$
johnston-linear-matrix-algebra-EQ0086	53	0.999	$T(\mathbf{v}) = T(v_1\mathbf{e}_1 + v_2\mathbf{e}_2 + \cdots + v_n\mathbf{e}_n) = v_1T(\mathbf{e}_1) + v_2T(\mathbf{e}_2) + \cdots + v_nT(\mathbf{e}_n),$	$T(\mathbf{v}) = T(v_1\mathbf{e}_1 + v_2\mathbf{e}_2 + \cdots + v_n\mathbf{e}_n) = v_1T(\mathbf{e}_1) + v_2T(\mathbf{e}_2) + \cdots + v_nT(\mathbf{e}_n),$	$T(\mathbf{v}) = T(v_1\mathbf{e}_1 + v_2\mathbf{e}_2 + \cdots + v_n\mathbf{e}_n) = v_1T(\mathbf{e}_1) + v_2T(\mathbf{e}_2) + \cdots + v_nT(\mathbf{e}_n),$
johnston-linear-matrix-algebra-EQ0150	74	0.999	$\begin{aligned} & \mathbf{v} \cdot \mathbf{w} \times \mathbf{v} \\ &= v_1(v_2w_3 - v_3w_2) + v_2(v_3w_1 - v_1w_3) + v_3(v_1w_2 - v_2w_1) \\ &= v_1v_2w_3 - v_1v_3w_2 + v_2v_3w_1 - v_2v_1w_3 + v_3v_1w_2 - v_3v_2w_1 \\ &= 0. \end{aligned}$	$\mathbf{v} \cdot (\mathbf{v} \times \mathbf{w}) = v_1(v_2w_3 - v_3w_2) + v_2(v_3w_1 - v_1w_3) + v_3(v_1w_2 - v_2w_1)$ $= v_1v_2w_3 - v_1v_3w_2 + v_2v_3w_1 - v_2v_1w_3 + v_3v_1w_2 - v_3v_2w_1$ $= 0.$	$\mathbf{v} \cdot (\mathbf{v} \times \mathbf{w}) = v_1(v_2w_3 - v_3w_2) + v_2(v_3w_1 - v_1w_3) + v_3(v_1w_2 - v_2w_1)$ $= v_1v_2w_3 - v_1v_3w_2 + v_2v_3w_1 - v_2v_1w_3 + v_3v_1w_2 - v_3v_2w_1$ $= 0.$
johnston-linear-matrix-algebra-EQ0308	148	0.999	$3(1, -1, 0) + 2(-2, 1, 2) + (1, 1, -4) = (0, 0, 0).$	$3(1, -1, 0) + 2(-2, 1, 2) + (1, 1, -4) = (0, 0, 0).$	$3(1, -1, 0) + 2(-2, 1, 2) + (1, 1, -4) = (0, 0, 0).$

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johnston-linear-matrix-algebra-E00712	269	0.999	$\begin{aligned} & \{[A\ B]_{i,\ i}=\sum_{j=1}^n a_{i,\ j} b_{j,\ i} \quad \text{for all } 1 \leq i \leq m, \\ & b_{j,\ i} \quad \text{for all } 1 \leq i \leq m, \\ & \quad \text{and } \} \quad \& \{[B\ A]_{j,\ j}=\sum_{i=1}^m b_{j,\ i} a_{i,\ j} \quad \text{for all } 1 \leq j \leq n. \\ & b_{j,\ i} a_{i,\ j} \quad \text{for all } 1 \leq j \leq n. \end{aligned}$	$[AB]_{i,i} = \sum_{j=1}^n a_{i,j} b_{j,i} \quad \text{for all } 1 \leq i \leq m, \quad \text{and}$ $[BA]_{j,j} = \sum_{i=1}^m b_{j,i} a_{i,j} \quad \text{for all } 1 \leq j \leq n.$	$[AB]_{i,i} = \sum_{j=1}^n a_{i,j} b_{j,i} \quad \text{for all } 1 \leq i \leq m, \quad \text{and}$ $[BA]_{j,j} = \sum_{i=1}^m b_{j,i} a_{i,j} \quad \text{for all } 1 \leq j \leq n.$
johnston-linear-matrix-algebra-E00764	287	0.999	$\operatorname{det}\left(\left[\begin{array}{lll} a & b & c \\ d & e & f \\ g & h & i \end{array}\right]\right)=a e i+b f g+c d h-a f h-b d i-c e g .$	$\det \left(\left[\begin{array}{ccc} a & b & c \\ d & e & f \\ g & h & i \end{array}\right]\right)=a e i+b f g+c d h-a f h-b d i-c e g .$	$\det \left(\left[\begin{array}{ccc} a & b & c \\ d & e & f \\ g & h & i \end{array}\right]\right)=a e i+b f g+c d h-a f h-b d i-c e g .$
johnston-linear-matrix-algebra-E00785	291	0.999	$\begin{aligned} \operatorname{det}(A)= & 2 \\ \operatorname{det}\left(\left[\begin{array}{ccc} -2 & 1 & 3 \\ 0 & 1 & 0 \\ 3 & 0 & 2 \end{array}\right]\right)- & \operatorname{det}\left(\left[\begin{array}{ccc} 0 & 1 & 3 \\ 0 & 1 & 0 \\ -1 & 0 & 2 \end{array}\right]\right) \\ & +(-1) \operatorname{det}\left(\left[\begin{array}{ccc} 0 & -2 & 3 \\ 0 & 0 & 0 \\ -1 & 3 & 2 \end{array}\right]\right)-0 \operatorname{det}\left(\left[\begin{array}{ccc} 0 & -2 & 1 \\ 0 & 0 & 1 \\ -1 & 3 & 0 \end{array}\right]\right) \\ = & 2 \cdot(-13)-3+0-0 \\ = & -29 . \end{aligned}$	$\det(A)=2 \det \left(\left[\begin{array}{ccc} -2 & 1 & 3 \\ 0 & 1 & 0 \\ 3 & 0 & 2 \end{array}\right]\right)-\det \left(\left[\begin{array}{ccc} 0 & 1 & 3 \\ 0 & 1 & 0 \\ -1 & 0 & 2 \end{array}\right]\right)$ $+(-1) \det \left(\left[\begin{array}{ccc} 0 & -2 & 3 \\ 0 & 0 & 0 \\ -1 & 3 & 2 \end{array}\right]\right)-0 \det \left(\left[\begin{array}{ccc} 0 & -2 & 1 \\ 0 & 0 & 1 \\ -1 & 3 & 0 \end{array}\right]\right)$ $=2 \cdot(-13)-3+0-0$ $=-29 .$	$\det(A)=2 \det \left(\left[\begin{array}{ccc} -2 & 1 & 3 \\ 0 & 1 & 0 \\ 3 & 0 & 2 \end{array}\right]\right)-\det \left(\left[\begin{array}{ccc} 0 & 1 & 3 \\ 0 & 1 & 0 \\ -1 & 0 & 2 \end{array}\right]\right)$ $+(-1) \det \left(\left[\begin{array}{ccc} 0 & -2 & 3 \\ 0 & 0 & 0 \\ -1 & 3 & 2 \end{array}\right]\right)-0 \det \left(\left[\begin{array}{ccc} 0 & -2 & 1 \\ 0 & 0 & 1 \\ -1 & 3 & 0 \end{array}\right]\right)$ $=2 \cdot(-13)-3+0-0$ $=-29 .$
johnston-linear-matrix-algebra-E00871	314	0.999	$\mathbf{v}^* A \mathbf{v}=\mathbf{v}^*(A \mathbf{v})=\mathbf{v}^*(\lambda \mathbf{v})=\lambda \mathbf{v}^* \mathbf{v} .$	$\mathbf{v}^* A \mathbf{v}=\mathbf{v}^*(A \mathbf{v})=\mathbf{v}^*(\lambda \mathbf{v})=\lambda \mathbf{v}^* \mathbf{v} .$	$\mathbf{v}^* A \mathbf{v}=\mathbf{v}^*(A \mathbf{v})=\mathbf{v}^*(\lambda \mathbf{v})=\lambda \mathbf{v}^* \mathbf{v} .$
johnston-linear-matrix-algebra-E01075	367	0.999	$\begin{aligned} \operatorname{det}(A\ B) & =\sum_{\sigma \in S_m} \operatorname{sgn}(\sigma)\left(\sum_{j=1}^n a_{\sigma(1), j} b_{j, 1}\right) \cdots\left(\sum_{j=1}^n a_{\sigma(m), j} b_{j, m}\right) \\ & =\sum_{1 \leq j_1, \ldots, j_m \leq n} b_{j_1, 1} \cdots b_{j_m, m}\left(\sum_{\sigma \in S_m} \operatorname{sgn}(\sigma) a_{\sigma(1), j_1} \cdots a_{\sigma(m), j_m}\right) \\ & =\sum_{1 \leq j_1, \ldots, j_m \leq n} b_{j_1, 1} \cdots b_{j_m, m} \operatorname{det}\left(A_{j_1, \ldots, j_m}\right) . \end{aligned}$	$\det(A B)=\sum_{\sigma \in S_m} \operatorname{sgn}(\sigma)[A B]_{\sigma(1), 1} \cdots[A B]_{\sigma(m), m}$ $=\sum_{\sigma \in S_m} \operatorname{sgn}(\sigma)\left(\sum_{j=1}^n a_{\sigma(1), j} b_{j, 1}\right) \cdots\left(\sum_{j=1}^n a_{\sigma(m), j} b_{j, m}\right)$ $=\sum_{1 \leq j_1, \ldots, j_m \leq n} b_{j_1, 1} \cdots b_{j_m, m}\left(\sum_{\sigma \in S_m} \operatorname{sgn}(\sigma) a_{\sigma(1), j_1} \cdots a_{\sigma(m), j_m}\right)$ $=\sum_{1 \leq j_1, \ldots, j_m \leq n} b_{j_1, 1} \cdots b_{j_m, m} \det \left(A_{j_1, \ldots, j_m}\right) .$	$\det(A B)=\sum_{\sigma \in S_m} \operatorname{sgn}(\sigma)[A B]_{\sigma(1), 1} \cdots[A B]_{\sigma(m), m}$ $=\sum_{\sigma \in S_m} \operatorname{sgn}(\sigma)\left(\sum_{j=1}^n a_{\sigma(1), j} b_{j, 1}\right) \cdots\left(\sum_{j=1}^n a_{\sigma(m), j} b_{j, m}\right)$ $=\sum_{1 \leq j_1, \ldots, j_m \leq n} b_{j_1, 1} \cdots b_{j_m, m}\left(\sum_{\sigma \in S_m} \operatorname{sgn}(\sigma) a_{\sigma(1), j_1} \cdots a_{\sigma(m), j_m}\right)$ $=\sum_{1 \leq j_1, \ldots, j_m \leq n} b_{j_1, 1} \cdots b_{j_m, m} \det \left(A_{j_1, \ldots, j_m}\right) .$

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johnston-linear-matrix-atqdw-EQ1227	412	0.999	1,11,21,1211,111221,312211,13112221,1113213211, \ldots	1, 11, 21, 1211, 111221, 312211, 13112221, 1113213211, ...	1, 11, 21, 1211, 111221, 312211, 13112221, 1113213211, ...
johnston-linear-matrix-atqdw-EQ1275	427	0.999	$p(x)$ \text { by } 3(x-2)(x+1/3)=3 x^{\{2\}}-5 x-2 \text { : }	$p(x)$ by $3(x-2)(x+1/3)=3x^2-5x-2$:	$p(x)$ by $3(x-2)(x+1/3)=3x^2-5x-2$:
johnston-linear-matrix-atqdw-EQ1389	442	0.999	$f(\lambda_1,\lambda_2,\ldots,\lambda_n)$ \right)=\operatorname{det}(A-\Lambda),	$f(\lambda_1,\lambda_2,\ldots,\lambda_n)=\det(A-\Lambda),$	$f(\lambda_1,\lambda_2,\ldots,\lambda_n)=\det(A-\Lambda),$
johnston-linear-matrix-atqdw-EQ1345	449	0.999	$\left[\begin{array}{ll} A & A \\ A & A \end{array}\right]^2=\left[\begin{array}{ll} 2 A^{\{2\}} & 2 A^{\{2\}} \\ 2 A^{\{2\}} & 2 A^{\{2\}} \end{array}\right]$.	$\left[\begin{array}{cc} A & A \\ A & A \end{array}\right]^2=\left[\begin{array}{cc} 2 A^2 & 2 A^2 \\ 2 A^2 & 2 A^2 \end{array}\right]$.	$\left[\begin{array}{cc} A & A \\ A & A \end{array}\right]^2=\left[\begin{array}{cc} 2 A^2 & 2 A^2 \\ 2 A^2 & 2 A^2 \end{array}\right]$.
johnston-linear-matrix-atqdw-EQ1388	452	0.999	$\left[R_{\mathbf{u}}^{\pi/4}\right]^{\pi/4}=\frac{1}{9}\left[\begin{array}{ccc} 1+4 \sqrt{2} & 2-4 \sqrt{2} & 2+2 \sqrt{2} \\ 2+2 \sqrt{2} & 4+5 / \sqrt{2} & 4-7 / \sqrt{2} \\ 2-4 \sqrt{2} & 4-1 / \sqrt{2} & 4+5 / \sqrt{2} \end{array}\right]$.	$\left[R_{\mathbf{u}}^{\pi/4}\right]=\frac{1}{9}\left[\begin{array}{ccc} 1+4 \sqrt{2} & 2-4 \sqrt{2} & 2+2 \sqrt{2} \\ 2+2 \sqrt{2} & 4+5 / \sqrt{2} & 4-7 / \sqrt{2} \\ 2-4 \sqrt{2} & 4-1 / \sqrt{2} & 4+5 / \sqrt{2} \end{array}\right]$.	$\left[R_{\mathbf{u}}^{\pi/4}\right]=\frac{1}{9}\left[\begin{array}{ccc} 1+4 \sqrt{2} & 2-4 \sqrt{2} & 2+2 \sqrt{2} \\ 2+2 \sqrt{2} & 4+5 / \sqrt{2} & 4-7 / \sqrt{2} \\ 2-4 \sqrt{2} & 4-1 / \sqrt{2} & 4+5 / \sqrt{2} \end{array}\right]$.
johnston-linear-matrix-atqdw-EQ1486	456	0.999	$(v, w, x, y, z)=(4+x+2 z, 1 / 2+z / 2, x, 1, z)$.	$(v, w, x, y, z)=(4+x+2 z, 1 / 2+z / 2, x, 1, z)$.	$(v, w, x, y, z)=(4+x+2 z, 1 / 2+z / 2, x, 1, z)$.
johnston-linear-matrix-atqdw-EQ1426	457	0.999	$\frac{1}{9}\left[\begin{array}{ccc} 1 & 2 & 2 \\ 2 & 4 & 4 \\ 2 & 4 & 4 \end{array}\right]\left[\begin{array}{c} v_1 \\ v_2 \\ v_3 \end{array}\right]=\left[\begin{array}{c} 2 \\ 4 \\ 4 \end{array}\right]$.	$\frac{1}{9}\left[\begin{array}{ccc} 1 & 2 & 2 \\ 2 & 4 & 4 \\ 2 & 4 & 4 \end{array}\right]\left[\begin{array}{c} v_1 \\ v_2 \\ v_3 \end{array}\right]=\left[\begin{array}{c} 2 \\ 4 \\ 4 \end{array}\right]$.	$\frac{1}{9}\left[\begin{array}{ccc} 1 & 2 & 2 \\ 2 & 4 & 4 \\ 2 & 4 & 4 \end{array}\right]\left[\begin{array}{c} v_1 \\ v_2 \\ v_3 \end{array}\right]=\left[\begin{array}{c} 2 \\ 4 \\ 4 \end{array}\right]$.
johnston-linear-matrix-atqdw-EQ1439	458	0.999	$\left[\begin{array}{ll} 1 & 1 \\ a & b \end{array}\right]\left[\begin{array}{ll} 1 & 1 \\ a & b \end{array}\right]=\left[\begin{array}{ll} 1 & 1 \\ a & b \end{array}\right]\left[\begin{array}{ll} 1 & 1 \\ a & b \end{array}\right]$.	$\left[\begin{array}{cc} 1 & 1 \\ 1 & 0 \end{array}\right]\left[\begin{array}{cc} a & b \\ c & d \end{array}\right]=\left[\begin{array}{cc} a & b \\ c & d \end{array}\right]\left[\begin{array}{cc} 1 & 1 \\ 1 & 0 \end{array}\right]$.	$\left[\begin{array}{cc} 1 & 1 \\ 1 & 0 \end{array}\right]\left[\begin{array}{cc} a & b \\ c & d \end{array}\right]=\left[\begin{array}{cc} a & b \\ c & d \end{array}\right]\left[\begin{array}{cc} 1 & 1 \\ 1 & 0 \end{array}\right]$.
johnston-linear-matrix-atqdw-EQ1447	459	0.999	$\left(A^{-1}\right)^7 A^7=\left(A^{-1}\right)^7 0$,	$\left(A^{-1}\right)^7 A^7=\left(A^{-1}\right)^7 O$,	$\left(A^{-1}\right)^7 A^7=\left(A^{-1}\right)^7 O$,
johnston-linear-matrix-atqdw-EQ1581	467	0.999	$\left[\begin{array}{ll} 1 & 1 \\ 0 & 0 \end{array}\right]$ \text { and } \left[\begin{array}{ll} 1 & 0 \\ 0 & 1 \end{array}\right],	$\left[\begin{array}{cc} 1 & 1 \\ 0 & 0 \end{array}\right]$ and $\left[\begin{array}{cc} 1 & 0 \\ 0 & 1 \end{array}\right]$,	$\left[\begin{array}{cc} 1 & 1 \\ 0 & 0 \end{array}\right]$ and $\left[\begin{array}{cc} 1 & 0 \\ 0 & 1 \end{array}\right]$,
johnston-linear-matrix-atqdw-EQ1586	467	0.999	$[A \mid I]$ \text { to } [R \mid E]	$[A \mid I]$ to $[R \mid E]$	$[A \mid I]$ to $[R \mid E]$
johnston-linear-matrix-atqdw-EQ1513	468	0.999	$\begin{aligned} 3 y_1+y_2+3 y_3 & \geq 1 \\ y_3 & \geq 1 \\ -y_1+y_2+2 y_3 & \geq 1 \\ y_1, y_3 & \geq 0 \end{aligned}$	$\begin{aligned} 3 y_1+y_2+3 y_3 & \geq 1 \\ 2 y_1+3 y_3 & \geq 1 \\ 4 y_1+3 y_2-y_3 & =2 \\ -y_1+y_2+2 y_3 & \geq 1 \\ 2 y_2+2 y_3 & =2 \\ y_1, y_3 & \geq 0 \end{aligned}$	$\begin{aligned} 3 y_1+y_2+3 y_3 & \geq 1 \\ 2 y_1+3 y_3 & \geq 1 \\ 4 y_1+3 y_2-y_3 & =2 \\ -y_1+y_2+2 y_3 & \geq 1 \\ 2 y_2+2 y_3 & =2 \\ y_1, y_3 & \geq 0 \end{aligned}$
johnston-linear-matrix-atqdw-EQ1557	473	0.999	$\begin{aligned} & c_1 \mathbf{w}_1+\cdots+c_m \mathbf{w}_m \\ & =\left[c_1 \mathbf{w}_1+\cdots+c_m \mathbf{w}_m\right]_B=\left[\mathbf{v}\right]_B \end{aligned}$	$\begin{aligned} & c_1 \mathbf{w}_1+\cdots+c_m \mathbf{w}_m \\ & =\left[c_1 \mathbf{w}_1+\cdots+c_m \mathbf{w}_m\right]_B=\left[\mathbf{v}\right]_B \end{aligned}$	$\begin{aligned} & c_1 \mathbf{w}_1+\cdots+c_m \mathbf{w}_m \\ & =\left[c_1 \mathbf{w}_1+\cdots+c_m \mathbf{w}_m\right]_B=\left[\mathbf{v}\right]_B \end{aligned}$

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johnston-linear-matrix-algebra-EQ1579	475	0.999	$A B=\left[\begin{array}{lll} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{array}\right] \text { and } B A=\left[\begin{array}{ll} 1 & 0 \\ 0 & 1 \end{array}\right],$	$AB = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix} \text{ and } BA = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix},$	$AB = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix} \text{ and } BA = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix},$
johnston-linear-matrix-algebra-EQ0986	346	0.999	$\begin{aligned} p(A) &= A^3 - 3A^2 + 2A - 4I \\ &= \begin{bmatrix} -20 & 84 \\ -21 & 85 \end{bmatrix} - 3 \begin{bmatrix} -4 & 20 \\ -5 & 21 \end{bmatrix} + 2 \begin{bmatrix} 0 & 4 \\ -1 & 5 \end{bmatrix} - 4 \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \\ &= \begin{bmatrix} -12 & 32 \\ -8 & 28 \end{bmatrix}. \end{aligned}$	$p(A) = A^3 - 3A^2 + 2A - 4I$ $= \begin{bmatrix} -20 & 84 \\ -21 & 85 \end{bmatrix} - 3 \begin{bmatrix} -4 & 20 \\ -5 & 21 \end{bmatrix} + 2 \begin{bmatrix} 0 & 4 \\ -1 & 5 \end{bmatrix} - 4 \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$ $= \begin{bmatrix} -12 & 32 \\ -8 & 28 \end{bmatrix}.$	$p(A) = A^3 - 3A^2 + 2A - 4I$ $= \begin{bmatrix} -20 & 84 \\ -21 & 85 \end{bmatrix} - 3 \begin{bmatrix} -4 & 20 \\ -5 & 21 \end{bmatrix} + 2 \begin{bmatrix} 0 & 4 \\ -1 & 5 \end{bmatrix} - 4 \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$ $= \begin{bmatrix} -12 & 32 \\ -8 & 28 \end{bmatrix}.$
johnston-linear-matrix-algebra-EQ0997	348	0.999	$e^0=\left[\begin{array}{cccc} e^0 & 0 & \cdots & 0 \\ 0 & e^0 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & e^0 \end{array}\right]=\left[\begin{array}{cccc} 1 & 0 & \cdots & 0 \\ 0 & 1 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & 1 \end{array}\right]=I.$	$e^O = \begin{bmatrix} e^0 & 0 & \cdots & 0 \\ 0 & e^0 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & e^0 \end{bmatrix} = \begin{bmatrix} 1 & 0 & \cdots & 0 \\ 0 & 1 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & 1 \end{bmatrix} = I.$	$e^O = \begin{bmatrix} e^0 & 0 & \cdots & 0 \\ 0 & e^0 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & e^0 \end{bmatrix} = \begin{bmatrix} 1 & 0 & \cdots & 0 \\ 0 & 1 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & 1 \end{bmatrix} = I.$
johnston-linear-matrix-algebra-EQ1116	385	0.999	$A \mathbf{v}=\left[\begin{array}{lll} 2 & 1 & 4 \\ 4 & 1 & 2 \\ 1 & 2 & 4 \end{array}\right]\left[\begin{array}{l} 1 \\ 1 \\ 1 \end{array}\right]=\left[\begin{array}{l} 7 \\ 7 \\ 7 \end{array}\right]=7 \mathbf{v}=\lambda \mathbf{v},$	$A \mathbf{v} = \begin{bmatrix} 2 & 1 & 4 \\ 4 & 1 & 2 \\ 1 & 2 & 4 \end{bmatrix} \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} = \begin{bmatrix} 7 \\ 7 \\ 7 \end{bmatrix} = 7 \mathbf{v} = \lambda \mathbf{v},$	$A \mathbf{v} = \begin{bmatrix} 2 & 1 & 4 \\ 4 & 1 & 2 \\ 1 & 2 & 4 \end{bmatrix} \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} = \begin{bmatrix} 7 \\ 7 \\ 7 \end{bmatrix} = 7 \mathbf{v} = \lambda \mathbf{v},$
johnston-linear-matrix-algebra-EQ0013	27	0.999	$\begin{aligned} \mathbf{v} \cdot \mathbf{w} &= v_1 w_1 + v_2 w_2 + \cdots + v_n w_n \\ &= w_1 v_1 + w_2 v_2 + \cdots + w_n v_n = \mathbf{w} \cdot \mathbf{v}. \end{aligned}$	$\mathbf{v} \cdot \mathbf{w} = v_1 w_1 + v_2 w_2 + \cdots + v_n w_n$ $= w_1 v_1 + w_2 v_2 + \cdots + w_n v_n = \mathbf{w} \cdot \mathbf{v}.$	$\mathbf{v} \cdot \mathbf{w} = v_1 w_1 + v_2 w_2 + \cdots + v_n w_n$ $= w_1 v_1 + w_2 v_2 + \cdots + w_n v_n = \mathbf{w} \cdot \mathbf{v}.$
johnston-linear-matrix-algebra-EQ0145 (9)	73	0.999	$A\left[\begin{array}{l} x \\ y \end{array}\right]=\left[\begin{array}{l} 2 x-y \\ y-x \end{array}\right]$	$A \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 2x - y \\ y - x \end{bmatrix}$	$A \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 2x - y \\ y - x \end{bmatrix}$
johnston-linear-matrix-algebra-EQ0181	88	0.999	$\left[A^3\right]_{2,3}=(1,2,1,3) \cdot(2,1,0,0)=2+2+0+0=4.$	$\left[A^3\right]_{2,3} = (1, 2, 1, 3) \cdot (2, 1, 0, 0) = 2 + 2 + 0 + 0 = 4.$	$\left[A^3\right]_{2,3} = (1, 2, 1, 3) \cdot (2, 1, 0, 0) = 2 + 2 + 0 + 0 = 4.$
johnston-linear-matrix-algebra-EQ0248	112	0.999	$2 x=2 y+z, \quad \text { or equivalently } \quad 2 x-2 y-z=0.$	$2x = 2y + z, \quad \text{or equivalently} \quad 2x - 2y - z = 0.$	$2x = 2y + z, \quad \text{or equivalently} \quad 2x - 2y - z = 0.$
johnston-linear-matrix-algebra-EQ0270	114	0.999	$\left[\begin{array}{ll} a & b \\ c & d \end{array}\right]$	$\begin{bmatrix} a & b \\ c & d \end{bmatrix}$	$\begin{bmatrix} a & b \\ c & d \end{bmatrix}$
johnston-linear-matrix-algebra-EQ0291	119	0.999	$E_3 E_2 E_1 A=\left[\begin{array}{cccc} 1 & 1 & 0 & -1 \\ 0 & 1 & 2 & 0 \\ 0 & 1 & 2 & 4 \end{array}\right],$	$E_3 E_2 E_1 A = \begin{bmatrix} 1 & 1 & 0 & -1 \\ 0 & 1 & 2 & 0 \\ 0 & 1 & 2 & 4 \end{bmatrix},$	$E_3 E_2 E_1 A = \begin{bmatrix} 1 & 1 & 0 & -1 \\ 0 & 1 & 2 & 0 \\ 0 & 1 & 2 & 4 \end{bmatrix},$
johnston-linear-matrix-algebra-EQ0470	193	0.999	$0+0+0=0, \quad 1+1+1=0, \quad 2+2+2=0, \quad \text { and } \quad 0+1+2=0.$	$0 + 0 + 0 = 0, \quad 1 + 1 + 1 = 0, \quad 2 + 2 + 2 = 0, \quad \text{and} \quad 0 + 1 + 2 = 0.$	$0 + 0 + 0 = 0, \quad 1 + 1 + 1 = 0, \quad 2 + 2 + 2 = 0, \quad \text{and} \quad 0 + 1 + 2 = 0.$

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johnston-linear-matrix-algebra-EQ0523	209	■ 0.999	$x_{\mathrm{c}}+x_{\mathrm{pc}}\leq 100, \quad x_{\mathrm{p}}+\mathrm{p}+x_{\mathrm{pc}}\leq 80, \quad \text{and} \quad x_{\mathrm{p}}+x_{\mathrm{c}}+x_{\mathrm{pc}}\leq 150,$	$x_c + x_{\mathrm{pc}} \leq 100, \quad x_{\mathrm{p}} + x_{\mathrm{pc}} \leq 80, \quad \text{and} \quad x_{\mathrm{p}} + x_{\mathrm{c}} + x_{\mathrm{pc}} \leq 150,$	$x_{\mathrm{c}} + x_{\mathrm{pc}} \leq 100, \quad x_{\mathrm{p}} + x_{\mathrm{pc}} \leq 80, \quad \text{and} \quad x_{\mathrm{p}} + x_{\mathrm{c}} + x_{\mathrm{pc}} \leq 150,$
johnston-linear-matrix-algebra-EQ0619	238	■ 0.999	$\left[\begin{array}{cccc} 1 & 0 & 0 & 0 \\ 1 & 1 & 1 & 1 \\ 1 & 2 & 4 & 8 \\ 1 & 3 & 9 & 27 \end{array}\right] \left[\begin{array}{c} c_0 \\ c_1 \\ c_2 \\ c_3 \end{array}\right] = \left[\begin{array}{c} 3 \\ -1 \\ -3 \\ 9 \end{array}\right]$	$\left[\begin{array}{cccc} 1 & 0 & 0 & 0 \\ 1 & 1 & 1 & 1 \\ 1 & 2 & 4 & 8 \\ 1 & 3 & 9 & 27 \end{array}\right] \left[\begin{array}{c} c_0 \\ c_1 \\ c_2 \\ c_3 \end{array}\right] = \left[\begin{array}{c} 3 \\ -1 \\ -3 \\ 9 \end{array}\right]$	$\left[\begin{array}{cccc} 1 & 0 & 0 & 0 \\ 1 & 1 & 1 & 1 \\ 1 & 2 & 4 & 8 \\ 1 & 3 & 9 & 27 \end{array}\right] \left[\begin{array}{c} c_0 \\ c_1 \\ c_2 \\ c_3 \end{array}\right] = \left[\begin{array}{c} 3 \\ -1 \\ -3 \\ 9 \end{array}\right]$
johnston-linear-matrix-algebra-EQ0655	251	■ 0.999	$\mathbf{0}=\mathbf{v}-\mathbf{v}=\left(c_1-d_1\right) \mathbf{v}_1+\left(c_2-d_2\right) \mathbf{v}_2+\cdots+\left(c_k-d_k\right) \mathbf{v}_k$	$\mathbf{0} = \mathbf{v} - \mathbf{v} = (c_1 - d_1) \mathbf{v}_1 + (c_2 - d_2) \mathbf{v}_2 + \cdots + (c_k - d_k) \mathbf{v}_k.$	$\mathbf{0} = \mathbf{v} - \mathbf{v} = (c_1 - d_1) \mathbf{v}_1 + (c_2 - d_2) \mathbf{v}_2 + \cdots + (c_k - d_k) \mathbf{v}_k.$
johnston-linear-matrix-algebra-EQ0716	270	■ 0.999	$\operatorname{tr}(A \ B \ C)=\operatorname{tr}(A(B \ C))=\operatorname{tr}((B \ C) \ A)=\operatorname{tr}(B \ C \ A),$	$\operatorname{tr}(ABC) = \operatorname{tr}(A(BC)) = \operatorname{tr}((BC)A) = \operatorname{tr}(BCA),$	$\operatorname{tr}(ABC) = \operatorname{tr}(A(BC)) = \operatorname{tr}((BC)A) = \operatorname{tr}(BCA),$
johnston-linear-matrix-algebra-EQ0815	298	■ 0.999	$\left[\begin{array}{cc} I_n & \mathbf{0} \\ \mathbf{w}^T & 1 \end{array}\right] \left[\begin{array}{cc} I_n + \mathbf{vw}^T & \mathbf{v} \\ \mathbf{0}^T & 1 \end{array}\right] \left[\begin{array}{cc} I_n & \mathbf{0} \\ -\mathbf{w}^T & 1 \end{array}\right]$	$\left[\begin{array}{cc} I_n & \mathbf{0} \\ \mathbf{w}^T & 1 \end{array}\right] \left[\begin{array}{cc} I_n + \mathbf{vw}^T & \mathbf{v} \\ \mathbf{0}^T & 1 \end{array}\right] \left[\begin{array}{cc} I_n & \mathbf{0} \\ -\mathbf{w}^T & 1 \end{array}\right]$	$\left[\begin{array}{cc} I_n & \mathbf{0} \\ \mathbf{w}^T & 1 \end{array}\right] \left[\begin{array}{cc} I_n + \mathbf{vw}^T & \mathbf{v} \\ \mathbf{0}^T & 1 \end{array}\right] \left[\begin{array}{cc} I_n & \mathbf{0} \\ -\mathbf{w}^T & 1 \end{array}\right]$
johnston-linear-matrix-algebra-EQ0849	309	■ 0.999	$\begin{aligned} p_B(\lambda) &= \det(B - \lambda I) = \det(PAP^{-1} - \lambda I) \\ &= \det(P(A - \lambda I)P^{-1}) = \det(A - \lambda I) = p_A(\lambda), \end{aligned}$	$p_B(\lambda) = \det(B - \lambda I) = \det(PAP^{-1} - \lambda I) \\ = \det(P(A - \lambda I)P^{-1}) = \det(A - \lambda I) = p_A(\lambda),$	$p_B(\lambda) = \det(B - \lambda I) = \det(PAP^{-1} - \lambda I) \\ = \det(P(A - \lambda I)P^{-1}) = \det(A - \lambda I) = p_A(\lambda),$
johnston-linear-matrix-algebra-EQ0877	315	■ 0.999	$\begin{array}{l} \lambda^3 - 2\lambda^2 + 9\lambda - 2 \\ \lambda^3 - 2\lambda^2 + 9\lambda - 2 \\ \hline -4\lambda^2 + 8\lambda \\ \hline \lambda - 2 \\ \hline \lambda - 2 \\ \hline 0 \end{array}$	$\begin{array}{l} -\lambda^2 - 4\lambda + 1 \\ \lambda - 2 \overline{) -\lambda^3 - 2\lambda^2 + 9\lambda - 2} \\ \underline{-\lambda^3 + 2\lambda^2} \\ -4\lambda^2 + 9\lambda - 2 \\ \underline{-4\lambda^2 + 8\lambda} \\ \lambda - 2 \\ \underline{\lambda - 2} \\ 0 \end{array}$	$\begin{array}{l} -\lambda^2 - 4\lambda + 1 \\ \lambda - 2 \overline{) -\lambda^3 - 2\lambda^2 + 9\lambda - 2} \\ \underline{-\lambda^3 + 2\lambda^2} \\ -4\lambda^2 + 9\lambda - 2 \\ \underline{-4\lambda^2 + 8\lambda} \\ \lambda - 2 \\ \underline{\lambda - 2} \\ 0 \end{array}$
johnston-linear-matrix-algebra-EQ0922	329	■ 0.999	$\mathbf{0} = c_1(\lambda_1 - \lambda_k) \mathbf{v}_1 + c_2(\lambda_2 - \lambda_k) \mathbf{v}_2 + \cdots + c_{k-1}(\lambda_{k-1} - \lambda_k) \mathbf{v}_{k-1}.$	$\mathbf{0} = c_1(\lambda_1 - \lambda_k) \mathbf{v}_1 + c_2(\lambda_2 - \lambda_k) \mathbf{v}_2 + \cdots + c_{k-1}(\lambda_{k-1} - \lambda_k) \mathbf{v}_{k-1}.$	$\mathbf{0} = c_1(\lambda_1 - \lambda_k) \mathbf{v}_1 + c_2(\lambda_2 - \lambda_k) \mathbf{v}_2 + \cdots + c_{k-1}(\lambda_{k-1} - \lambda_k) \mathbf{v}_{k-1}.$
johnston-linear-matrix-algebra-EQ0937	334	■ 0.999	$D = \begin{bmatrix} -1 & 0 \\ 0 & 6 \end{bmatrix}, \quad P = \begin{bmatrix} -1 & 2 \\ 1 & 5 \end{bmatrix}, \quad \text{and} \quad P^{-1} = \frac{1}{7} \begin{bmatrix} -5 & 2 \\ 1 & 1 \end{bmatrix}.$	$D = \begin{bmatrix} -1 & 0 \\ 0 & 6 \end{bmatrix}, \quad P = \begin{bmatrix} -1 & 2 \\ 1 & 5 \end{bmatrix}, \quad \text{and} \quad P^{-1} = \frac{1}{7} \begin{bmatrix} -5 & 2 \\ 1 & 1 \end{bmatrix}.$	$D = \begin{bmatrix} -1 & 0 \\ 0 & 6 \end{bmatrix}, \quad P = \begin{bmatrix} -1 & 2 \\ 1 & 5 \end{bmatrix}, \quad \text{and} \quad P^{-1} = \frac{1}{7} \begin{bmatrix} -5 & 2 \\ 1 & 1 \end{bmatrix}.$

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johnston-linear-matrix-algebra-EQ09941	335	0.999	$\mathbf{P} \mathbf{D} \mathbf{Q}^* = \sum_{j=1}^n d_j \mathbf{p}_j \mathbf{q}_j^* .$	$PDQ^* = \sum_{j=1}^n d_j \mathbf{p}_j \mathbf{q}_j^* .$	$PDQ^* = \sum_{j=1}^n d_j \mathbf{p}_j \mathbf{q}_j^* .$
johnston-linear-matrix-algebra-EQ09988	346	0.999	$\begin{aligned} p(A) &= \frac{1}{3} \left[\begin{array}{cc} 4 & 1 \\ 1 & 1 \end{array} \right] \left[\begin{array}{cc} p(1) & 0 \\ 0 & p(4) \end{array} \right] \left[\begin{array}{cc} 1 & -1 \\ -1 & 4 \end{array} \right] \\ &= \frac{1}{3} \left[\begin{array}{cc} 4 & 1 \\ 1 & 1 \end{array} \right] \left[\begin{array}{cc} -4 & 0 \\ 0 & 20 \end{array} \right] \left[\begin{array}{cc} 1 & -1 \\ -1 & 4 \end{array} \right] \\ &= \frac{1}{3} \left[\begin{array}{cc} 4 & 1 \\ 1 & 1 \end{array} \right] \left[\begin{array}{cc} -4 & 4 \\ -20 & 80 \end{array} \right] = \left[\begin{array}{cc} -12 & 32 \\ -8 & 28 \end{array} \right] , \end{aligned}$	$p(A) = \frac{1}{3} \left[\begin{array}{cc} 4 & 1 \\ 1 & 1 \end{array} \right] \left[\begin{array}{cc} p(1) & 0 \\ 0 & p(4) \end{array} \right] \left[\begin{array}{cc} 1 & -1 \\ -1 & 4 \end{array} \right] \\ = \frac{1}{3} \left[\begin{array}{cc} 4 & 1 \\ 1 & 1 \end{array} \right] \left[\begin{array}{cc} -4 & 0 \\ 0 & 20 \end{array} \right] \left[\begin{array}{cc} 1 & -1 \\ -1 & 4 \end{array} \right] \\ = \frac{1}{3} \left[\begin{array}{cc} 4 & 1 \\ 1 & 1 \end{array} \right] \left[\begin{array}{cc} -4 & 4 \\ -20 & 80 \end{array} \right] = \left[\begin{array}{cc} -12 & 32 \\ -8 & 28 \end{array} \right] ,$	$p(A) = \frac{1}{3} \left[\begin{array}{cc} 4 & 1 \\ 1 & 1 \end{array} \right] \left[\begin{array}{cc} p(1) & 0 \\ 0 & p(4) \end{array} \right] \left[\begin{array}{cc} 1 & -1 \\ -1 & 4 \end{array} \right] \\ = \frac{1}{3} \left[\begin{array}{cc} 4 & 1 \\ 1 & 1 \end{array} \right] \left[\begin{array}{cc} -4 & 0 \\ 0 & 20 \end{array} \right] \left[\begin{array}{cc} 1 & -1 \\ -1 & 4 \end{array} \right] \\ = \frac{1}{3} \left[\begin{array}{cc} 4 & 1 \\ 1 & 1 \end{array} \right] \left[\begin{array}{cc} -4 & 4 \\ -20 & 80 \end{array} \right] = \left[\begin{array}{cc} -12 & 32 \\ -8 & 28 \end{array} \right] ,$
johnston-linear-matrix-algebra-EQ1099	377	0.999	$\begin{aligned} A^k \mathbf{v}_0 &= A^k (c_1 \mathbf{w}_1 + c_2 \mathbf{w}_2 + \cdots + c_n \mathbf{w}_n) \\ &= c_1 \lambda_1^k \mathbf{w}_1 + c_2 \lambda_2^k \mathbf{w}_2 + \cdots + c_n \lambda_n^k \mathbf{w}_n \\ &= \lambda_1^k \left(c_1 \mathbf{w}_1 + c_2 \left(\frac{\lambda_2}{\lambda_1} \right)^k \mathbf{w}_2 + \cdots + c_n \left(\frac{\lambda_n}{\lambda_1} \right)^k \mathbf{w}_n \right) . \end{aligned}$	$\begin{aligned} A^k \mathbf{v}_0 &= A^k (c_1 \mathbf{w}_1 + c_2 \mathbf{w}_2 + \cdots + c_n \mathbf{w}_n) \\ &= c_1 \lambda_1^k \mathbf{w}_1 + c_2 \lambda_2^k \mathbf{w}_2 + \cdots + c_n \lambda_n^k \mathbf{w}_n \\ &= \lambda_1^k \left(c_1 \mathbf{w}_1 + c_2 \left(\frac{\lambda_2}{\lambda_1} \right)^k \mathbf{w}_2 + \cdots + c_n \left(\frac{\lambda_n}{\lambda_1} \right)^k \mathbf{w}_n \right) . \end{aligned}$	$\begin{aligned} A^k \mathbf{v}_0 &= A^k (c_1 \mathbf{w}_1 + c_2 \mathbf{w}_2 + \cdots + c_n \mathbf{w}_n) \\ &= c_1 \lambda_1^k \mathbf{w}_1 + c_2 \lambda_2^k \mathbf{w}_2 + \cdots + c_n \lambda_n^k \mathbf{w}_n \\ &= \lambda_1^k \left(c_1 \mathbf{w}_1 + c_2 \left(\frac{\lambda_2}{\lambda_1} \right)^k \mathbf{w}_2 + \cdots + c_n \left(\frac{\lambda_n}{\lambda_1} \right)^k \mathbf{w}_n \right) . \end{aligned}$
johnston-linear-matrix-algebra-EQ1123	388	0.999	$\left[\begin{array}{cc} 0 & 1 \\ 1 & 0 \end{array} \right] \quad \text{and} \quad \left[\begin{array}{ccc} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 1 & 0 & 0 \end{array} \right] .$	$\left[\begin{array}{cc} 0 & 1 \\ 1 & 0 \end{array} \right] \quad \text{and} \quad \left[\begin{array}{ccc} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 1 & 0 & 0 \end{array} \right] .$	$\left[\begin{array}{cc} 0 & 1 \\ 1 & 0 \end{array} \right] \quad \text{and} \quad \left[\begin{array}{ccc} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 1 & 0 & 0 \end{array} \right] .$
johnston-linear-matrix-algebra-EQ1134	392	0.999	$A = \left[\begin{array}{cc} -3 & -2 \\ 4 & 1 \end{array} \right] .$	$A = \left[\begin{array}{cc} -3 & -2 \\ 4 & 1 \end{array} \right] .$	$A = \left[\begin{array}{cc} -3 & -2 \\ 4 & 1 \end{array} \right] .$
johnston-linear-matrix-algebra-EQ1151	396	0.999	$\begin{aligned} Q \widetilde{H} &= [Q_1 Q_2 \cdots Q_\ell \mathbf{w}_1 \mathbf{w}_2 \cdots \mathbf{w}_m] \text{diag}(H, H, \dots, H, 1, 1, \dots, 1) \\ &= [Q_1 H Q_2 H \cdots Q_\ell H \mathbf{w}_1 \mathbf{w}_2 \cdots \mathbf{w}_m] \\ &= [P_1 P_2 \cdots P_\ell \mathbf{w}_1 \mathbf{w}_2 \cdots \mathbf{w}_m] = P . \end{aligned}$	$\begin{aligned} Q \widetilde{H} &= [Q_1 Q_2 \cdots Q_\ell \mathbf{w}_1 \mathbf{w}_2 \cdots \mathbf{w}_m] \text{diag}(H, H, \dots, H, 1, 1, \dots, 1) \\ &= [Q_1 H Q_2 H \cdots Q_\ell H \mathbf{w}_1 \mathbf{w}_2 \cdots \mathbf{w}_m] \\ &= [P_1 P_2 \cdots P_\ell \mathbf{w}_1 \mathbf{w}_2 \cdots \mathbf{w}_m] = P . \end{aligned}$	$\begin{aligned} Q \widetilde{H} &= [Q_1 Q_2 \cdots Q_\ell \mathbf{w}_1 \mathbf{w}_2 \cdots \mathbf{w}_m] \text{diag}(H, H, \dots, H, 1, 1, \dots, 1) \\ &= [Q_1 H Q_2 H \cdots Q_\ell H \mathbf{w}_1 \mathbf{w}_2 \cdots \mathbf{w}_m] \\ &= [P_1 P_2 \cdots P_\ell \mathbf{w}_1 \mathbf{w}_2 \cdots \mathbf{w}_m] = P . \end{aligned}$

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johnston-linear-matrix-atlas-2-EQ1191	405	0.999	$x_{\{n\}}=c_{\{0\}}\,\lambda_{\{0\}}^{\{n\}}+c_{\{1\}}\,\lambda_{\{1\}}^{\{n\}}+c_{\{2\}}\,\lambda_{\{2\}}^{\{n\}}=1-2^{\{n\}}+5^{\{n\}}\quad\text{for all}\quad n\geq 0\,.$	$x_n=c_0\lambda_0^n+c_1\lambda_1^n+c_2\lambda_2^n=1-2^n+5^n\quad\text{for all}\quad n\geq 0.$	$x_n=c_0\lambda_0^n+c_1\lambda_1^n+c_2\lambda_2^n=1-2^n+5^n\quad\text{for all}\quad n\geq 0.$
johnston-linear-matrix-atlas-2-EQ1299	439	0.999	$\widetilde{R}=\left[\begin{array}{cc c} 1&0&2\\ 0&1&-1\\ 0&0&0\\ 0&0&0\end{array}\right]\quad\text{and}\quad\widetilde{S}=\left[\begin{array}{cc c} 1&0&4\\ 0&1&2\\ 0&0&0\\ 0&0&0\end{array}\right].$	$\widetilde{R}=\left[\begin{array}{cc c} 1&0&2\\ 0&1&-1\\ 0&0&0\\ 0&0&0\end{array}\right]\quad\text{and}\quad\widetilde{S}=\left[\begin{array}{cc c} 1&0&4\\ 0&1&2\\ 0&0&0\\ 0&0&0\end{array}\right].$	$\widetilde{R}=\left[\begin{array}{cc c} 1&0&2\\ 0&1&-1\\ 0&0&0\\ 0&0&0\end{array}\right]\quad\text{and}\quad\widetilde{S}=\left[\begin{array}{cc c} 1&0&4\\ 0&1&2\\ 0&0&0\\ 0&0&0\end{array}\right].$
johnston-linear-matrix-atlas-2-EQ1339	448	0.999	$\ (\mathbf{v}-\mathbf{w})+\mathbf{w}\ \leq\ \mathbf{v}-\mathbf{w}\ +\ \mathbf{w}\ .$	$\ (\mathbf{v}-\mathbf{w})+\mathbf{w}\ \leq\ \mathbf{v}-\mathbf{w}\ +\ \mathbf{w}\ .$	$\ (\mathbf{v}-\mathbf{w})+\mathbf{w}\ \leq\ \mathbf{v}-\mathbf{w}\ +\ \mathbf{w}\ .$
johnston-linear-matrix-atlas-2-EQ1348	449	0.999	$A\,\mathbf{e}_{\{i\}}=\left[\begin{array}{c} a_{\{1,\,i\}}\\ a_{\{2,\,i\}}\\ \vdots\\ a_{\{m,\,i\}}\end{array}\right],$	$A\mathbf{e}_i=\left[\begin{array}{c} a_{1,i}\\ a_{2,i}\\ \vdots\\ a_{m,i}\end{array}\right],$	$A\mathbf{e}_i=\left[\begin{array}{c} a_{1,i}\\ a_{2,i}\\ \vdots\\ a_{m,i}\end{array}\right],$
johnston-linear-matrix-atlas-2-EQ1386	453	0.999	$\begin{aligned}\mathbf{w}\cdot\mathbf{v}\times\mathbf{w}&=w_1(v_2w_3-v_3w_2)+w_2(v_3w_1-v_1w_3)\\&\quad+w_3(v_1w_2-v_2w_1)\\&=w_1v_2w_3-w_1v_3w_2+w_2v_3w_1-w_2v_1w_3\\&\quad+w_3v_1w_2-w_3v_2w_1\\&=0\end{aligned}$	$\begin{aligned}\mathbf{w}\cdot(\mathbf{v}\times\mathbf{w})&=w_1(v_2w_3-v_3w_2)+w_2(v_3w_1-v_1w_3)\\&\quad+w_3(v_1w_2-v_2w_1)\\&=w_1v_2w_3-w_1v_3w_2+w_2v_3w_1-w_2v_1w_3\\&\quad+w_3v_1w_2-w_3v_2w_1\\&=0\end{aligned}$	$\begin{aligned}\mathbf{w}\cdot(\mathbf{v}\times\mathbf{w})&=w_1(v_2w_3-v_3w_2)+w_2(v_3w_1-v_1w_3)\\&\quad+w_3(v_1w_2-v_2w_1)\\&=w_1v_2w_3-w_1v_3w_2+w_2v_3w_1-w_2v_1w_3\\&\quad+w_3v_1w_2-w_3v_2w_1\\&=0.\end{aligned}$
johnston-linear-matrix-atlas-2-EQ1404	455	0.999	$\begin{aligned}\left[A^{\{k+1\}}\right]_{\{i,\,j\}}&=\left[A^{\{k\}}A\right]_{\{i,\,j\}}\\&=\left[A^{\{k\}}\right]_{\{i,1\}}a_{\{1,j\}}+\left[A^{\{k\}}\right]_{\{i,2\}}a_{\{2,j\}}+\cdots+\left[A^{\{k\}}\right]_{\{i,n\}}a_{\{n,j\}}.\end{aligned}$	$\begin{aligned}[A^{k+1}]_{i,j}&=[A^kA]_{i,j}\\&=[A^k]_{i,1}a_{1,j}+[A^k]_{i,2}a_{2,j}+\cdots+[A^k]_{i,n}a_{n,j}\end{aligned}$	$\begin{aligned}[A^{k+1}]_{i,j}&=[A^kA]_{i,j}\\&=[A^k]_{i,1}a_{1,j}+[A^k]_{i,2}a_{2,j}+\cdots+[A^k]_{i,n}a_{n,j}.\end{aligned}$
johnston-linear-matrix-atlas-2-EQ1450	459	0.999	$\begin{aligned}AB&=((a-b)I_n+bJ_n)(cI_n+dJ_n)\\&=c(a-b)I_n+(bc+d(a-b)+nbd)J_n.\end{aligned}$	$\begin{aligned}AB&=((a-b)I_n+bJ_n)(cI_n+dJ_n)\\&=c(a-b)I_n+(bc+d(a-b)+nbd)J_n.\end{aligned}$	$\begin{aligned}AB&=((a-b)I_n+bJ_n)(cI_n+dJ_n)\\&=c(a-b)I_n+(bc+d(a-b)+nbd)J_n.\end{aligned}$
johnston-linear-matrix-atlas-2-EQ1455	460	0.999	$\begin{aligned}A\,A^{-1}&=A\left[\mathbf{x}_1\mid\mathbf{x}_2\mid\cdots\mid\mathbf{x}_n\right] \\&=A\mathbf{x}_1\mid A\mathbf{x}_2\mid\cdots\mid A\mathbf{x}_n \\&=\mathbf{e}_1\mid\mathbf{e}_2\mid\cdots\mid\mathbf{e}_n=I,\end{aligned}$	$\begin{aligned}AA^{-1}&=A\left[\mathbf{x}_1\mid\mathbf{x}_2\mid\cdots\mid\mathbf{x}_n\right] \\&=A\mathbf{x}_1\mid A\mathbf{x}_2\mid\cdots\mid A\mathbf{x}_n \\&=\mathbf{e}_1\mid\mathbf{e}_2\mid\cdots\mid\mathbf{e}_n=I,\end{aligned}$	$\begin{aligned}AA^{-1}&=A\left[\mathbf{x}_1\mid\mathbf{x}_2\mid\cdots\mid\mathbf{x}_n\right] \\&=A\mathbf{x}_1\mid A\mathbf{x}_2\mid\cdots\mid A\mathbf{x}_n \\&=\mathbf{e}_1\mid\mathbf{e}_2\mid\cdots\mid\mathbf{e}_n=I,\end{aligned}$
johnston-linear-matrix-atlas-2-EQ1518	469	0.999	$\begin{array}{l} -\operatorname{maximize}: \quad (-\mathbf{b})\cdot\mathbf{y} \\ \text{subject to:} \quad (-A^T)\mathbf{y}\leq-\mathbf{c} \\ \quad\quad\quad\mathbf{y}\geq\mathbf{0} \end{array}$	$\begin{array}{l} -\operatorname{maximize}: \quad (-\mathbf{b})\cdot\mathbf{y} \\ \text{subject to:} \quad (-A^T)\mathbf{y}\leq-\mathbf{c} \\ \quad\quad\quad\mathbf{y}\geq\mathbf{0} \end{array}$	$\begin{array}{l} -\operatorname{maximize}: \quad (-\mathbf{b})\cdot\mathbf{y} \\ \text{subject to:} \quad (-A^T)\mathbf{y}\leq-\mathbf{c} \\ \quad\quad\quad\mathbf{y}\geq\mathbf{0} \end{array}$

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johnston-linear-matrix-algebra-EQ0036	36	1.000	$[A+B]_{i,j}=a_{i,j}+b_{i,j} \quad \text{and} \quad [cA]_{i,j}=ca_{i,j},$	$[A+B]_{i,j}=a_{i,j}+b_{i,j} \quad \text{and} \quad [cA]_{i,j}=ca_{i,j},$	$[A+B]_{i,j}=a_{i,j}+b_{i,j} \quad \text{and} \quad [cA]_{i,j}=ca_{i,j},$
johnston-linear-matrix-algebra-EQ0068	47	1.000	$\left[\begin{array}{ll} A & B \\ 0 & I_3 \end{array}\right] \left[\begin{array}{ll} B & 0 \\ I_3 & I_3 \end{array}\right] = \left[\begin{array}{ll} AB+B & B \\ I_3 & I_3 \end{array}\right] = \left[\begin{array}{ccc ccc} -2 & 4 & 2 & -1 & 1 & 1 \\ -3 & 8 & 3 & 0 & 1 & 0 \\ 1 & 0 & 0 & 1 & 0 & 0 \\ 0 & 1 & 0 & 0 & 1 & 0 \\ 0 & 0 & 1 & 0 & 0 & 1 \end{array}\right].$	$\left[\begin{array}{ll} A & B \\ O & I_3 \end{array}\right] \left[\begin{array}{ll} B & O \\ I_3 & I_3 \end{array}\right] = \left[\begin{array}{ll} AB+B & B \\ I_3 & I_3 \end{array}\right] = \left[\begin{array}{ccc ccc} -2 & 4 & 2 & -1 & 1 & 1 \\ -3 & 8 & 3 & 0 & 1 & 0 \\ 1 & 0 & 0 & 1 & 0 & 0 \\ 0 & 1 & 0 & 0 & 1 & 0 \\ 0 & 0 & 1 & 0 & 0 & 1 \end{array}\right].$	$\left[\begin{array}{ll} A & B \\ O & I_3 \end{array}\right] \left[\begin{array}{ll} B & O \\ I_3 & I_3 \end{array}\right] = \left[\begin{array}{ll} AB+B & B \\ I_3 & I_3 \end{array}\right] = \left[\begin{array}{ccc ccc} -2 & 4 & 2 & -1 & 1 & 1 \\ -3 & 8 & 3 & 0 & 1 & 0 \\ 1 & 0 & 0 & 1 & 0 & 0 \\ 0 & 1 & 0 & 0 & 1 & 0 \\ 0 & 0 & 1 & 0 & 0 & 1 \end{array}\right].$
johnston-linear-matrix-algebra-EQ0087	53	1.000	$\begin{aligned} &T(v_1, v_2) = T(v_1 \mathbf{e}_1 + v_2 \mathbf{e}_2) \\ &= v_1 T(\mathbf{e}_1) + v_2 T(\mathbf{e}_2) \\ &= v_1(1, 1) + v_2(-1, 1) = (v_1 - v_2, v_1 + v_2). \end{aligned}$	$T(v_1, v_2) = T(v_1 \mathbf{e}_1 + v_2 \mathbf{e}_2) \\ = v_1 T(\mathbf{e}_1) + v_2 T(\mathbf{e}_2) \\ = v_1(1, 1) + v_2(-1, 1) = (v_1 - v_2, v_1 + v_2).$	$T(v_1, v_2) = T(v_1 \mathbf{e}_1 + v_2 \mathbf{e}_2) \\ = v_1 T(\mathbf{e}_1) + v_2 T(\mathbf{e}_2) \\ = v_1(1, 1) + v_2(-1, 1) = (v_1 - v_2, v_1 + v_2).$
johnston-linear-matrix-algebra-EQ0107	60	1.000	$[F_{\mathbf{u}}] = 2 \mathbf{u} \mathbf{u}^T - I.$	$[F_{\mathbf{u}}] = 2 \mathbf{u} \mathbf{u}^T - I.$	$[F_{\mathbf{u}}] = 2 \mathbf{u} \mathbf{u}^T - I.$
johnston-linear-matrix-algebra-EQ0131	68	1.000	$\cos(\theta + \phi) = \cos(\theta) \cos(\phi) - \sin(\theta) \sin(\phi).$	$\cos(\theta + \phi) = \cos(\theta) \cos(\phi) - \sin(\theta) \sin(\phi).$	$\cos(\theta + \phi) = \cos(\theta) \cos(\phi) - \sin(\theta) \sin(\phi).$
johnston-linear-matrix-algebra-EQ0132	69	1.000	$\begin{aligned} &\left[R^{\theta} \circ R^{\phi}\right] = \left[R^{\theta}\right] \left[R^{\phi}\right] = \begin{bmatrix} \cos(\theta) & -\sin(\theta) \\ \sin(\theta) & \cos(\theta) \end{bmatrix} \begin{bmatrix} \cos(\phi) & -\sin(\phi) \\ \sin(\phi) & \cos(\phi) \end{bmatrix} \\ &= \begin{bmatrix} \cos(\theta) \cos(\phi) - \sin(\theta) \sin(\phi) & -\cos(\theta) \sin(\phi) - \sin(\theta) \cos(\phi) \\ \sin(\theta) \cos(\phi) + \cos(\theta) \sin(\phi) & -\sin(\theta) \sin(\phi) + \cos(\theta) \cos(\phi) \end{bmatrix}. \end{aligned}$	$\begin{aligned} &\left[R^{\theta} \circ R^{\phi}\right] = \left[R^{\theta}\right] \left[R^{\phi}\right] = \begin{bmatrix} \cos(\theta) & -\sin(\theta) \\ \sin(\theta) & \cos(\theta) \end{bmatrix} \begin{bmatrix} \cos(\phi) & -\sin(\phi) \\ \sin(\phi) & \cos(\phi) \end{bmatrix} \\ &= \begin{bmatrix} \cos(\theta) \cos(\phi) - \sin(\theta) \sin(\phi) & -\cos(\theta) \sin(\phi) - \sin(\theta) \cos(\phi) \\ \sin(\theta) \cos(\phi) + \cos(\theta) \sin(\phi) & -\sin(\theta) \sin(\phi) + \cos(\theta) \cos(\phi) \end{bmatrix}. \end{aligned}$	$\begin{aligned} &\left[R^{\theta} \circ R^{\phi}\right] = \left[R^{\theta}\right] \left[R^{\phi}\right] = \begin{bmatrix} \cos(\theta) & -\sin(\theta) \\ \sin(\theta) & \cos(\theta) \end{bmatrix} \begin{bmatrix} \cos(\phi) & -\sin(\phi) \\ \sin(\phi) & \cos(\phi) \end{bmatrix} \\ &= \begin{bmatrix} \cos(\theta) \cos(\phi) - \sin(\theta) \sin(\phi) & -\cos(\theta) \sin(\phi) - \sin(\theta) \cos(\phi) \\ \sin(\theta) \cos(\phi) + \cos(\theta) \sin(\phi) & -\sin(\theta) \sin(\phi) + \cos(\theta) \cos(\phi) \end{bmatrix}. \end{aligned}$
johnston-linear-matrix-algebra-EQ0140	71	1.000	$\frac{1}{1+m^2} \begin{bmatrix} 1-m^2 & 2m \\ 2m & m^2-1 \end{bmatrix}.$	$\frac{1}{1+m^2} \begin{bmatrix} 1-m^2 & 2m \\ 2m & m^2-1 \end{bmatrix}.$	$\frac{1}{1+m^2} \begin{bmatrix} 1-m^2 & 2m \\ 2m & m^2-1 \end{bmatrix}.$
johnston-linear-matrix-algebra-EQ0159	80	1.000	$\mathbf{v} \times (\mathbf{w} \times \mathbf{x}) + \mathbf{w} \times (\mathbf{x} \times \mathbf{v}) + \mathbf{x} \times (\mathbf{v} \times \mathbf{w}) = \mathbf{0}.$	$\mathbf{v} \times (\mathbf{w} \times \mathbf{x}) + \mathbf{w} \times (\mathbf{x} \times \mathbf{v}) + \mathbf{x} \times (\mathbf{v} \times \mathbf{w}) = \mathbf{0}.$	$\mathbf{v} \times (\mathbf{w} \times \mathbf{x}) + \mathbf{w} \times (\mathbf{x} \times \mathbf{v}) + \mathbf{x} \times (\mathbf{v} \times \mathbf{w}) = \mathbf{0}.$
johnston-linear-matrix-algebra-EQ0194	95	1.000	$\begin{aligned} &3x - 2y + z = -3 \\ &2x + 3y - 2z = 5 \end{aligned}$	$\begin{aligned} &3x - 2y + z = -3 \\ &2x + 3y - 2z = 5 \end{aligned}$	$\begin{aligned} &3x - 2y + z = -3 \\ &2x + 3y - 2z = 5 \end{aligned}$
johnston-linear-matrix-algebra-EQ0199	96	1.000	$\begin{aligned} &x + 3y - 2z = 5 \\ &2y - 6z = 4 \\ &3z = 6 \end{aligned}$	$\begin{aligned} &x + 3y - 2z = 5 \\ &2y - 6z = 4 \\ &3z = 6 \end{aligned}$	$\begin{aligned} &x + 3y - 2z = 5 \\ &2y - 6z = 4 \\ &3z = 6 \end{aligned}$

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johnston-linear-matrix-algebra-EQ0201	98	1.000	$\begin{aligned} x+3y-2z &= 5 \\ x+5y-8z &= 9 \\ 2x+4y+5z &= 12 \end{aligned}$	$\begin{aligned} x+3y-2z &= 5 \\ x+5y-8z &= 9 \\ 2x+4y+5z &= 12 \end{aligned}$	$\begin{aligned} x+3y-2z &= 5 \\ x+5y-8z &= 9 \\ 2x+4y+5z &= 12 \end{aligned}$
johnston-linear-matrix-algebra-EQ0209	99	1.000	$\begin{aligned} x+3y-2z &= 5 \\ x+5y-8z &= 9 \\ y+5z &= 12 \end{aligned}$	$\begin{aligned} x+3y-2z &= 5 \\ x+5y-8z &= 9 \\ 2x+4y+5z &= 12 \end{aligned}$	$\begin{aligned} x+3y-2z &= 5 \\ x+5y-8z &= 9 \\ 2x+4y+5z &= 12 \end{aligned}$
johnston-linear-matrix-algebra-EQ0212	100	1.000	$\left[\begin{array}{lll} 1 & 2 & 3 \\ 3 & 4 & 7 \end{array}\right] \xrightarrow{0R_2} \left[\begin{array}{lll} 1 & 2 & 3 \\ 0 & 0 & 0 \end{array}\right],$		$\left[\begin{array}{lll l} 1 & 2 & 3 & 3 \\ 3 & 4 & 7 & 7 \end{array}\right] \xrightarrow{0R_2} \left[\begin{array}{lll l} 1 & 2 & 3 & 3 \\ 0 & 0 & 0 & 0 \end{array}\right],$
johnston-linear-matrix-algebra-EQ0217 (1)	102	1.000	$\left[\begin{array}{cccccc} 0 & 0 & -1 & 1 & 0 & -2 \\ 1 & -5 & 2 & 0 & 2 & -2 \\ 0 & 2 & -2 & 6 & -3 & 0 \\ 0 & -4 & 2 & -10 & 5 & -10 \end{array}\right] \xrightarrow{R_1 \leftrightarrow R_3} \left[\begin{array}{cccccc} 0 & 2 & -2 & 6 & -3 & 0 \\ 0 & -2 & 1 & -5 & 2 & 0 \\ 0 & 0 & -1 & 1 & 0 & -2 \\ 0 & -4 & 2 & -10 & 5 & -10 \end{array}\right]$		$\begin{aligned} &\begin{array}{ccccc} 0 & 0 & -1 & 1 & 0 \\ 0 & -2 & 1 & -5 & 2 \\ 0 & 2 & -2 & 6 & -3 \\ 0 & -4 & 2 & -10 & 5 \end{array} \xrightarrow{R_1 \leftrightarrow R_3} \begin{array}{ccccc} 0 & 2 & -2 & 6 & -3 \\ 0 & -2 & 1 & -5 & 2 \\ 0 & 0 & -1 & 1 & 0 \\ 0 & -4 & 2 & -10 & 5 \end{array} \end{aligned}$ ↓ leading entry
johnston-linear-matrix-algebra-EQ0227	105	1.000	$\left[\begin{array}{ccc c} 1 & 0 & 0 & 4 \\ 0 & 1 & 0 & -3 \\ 0 & 0 & 1 & 2 \end{array}\right] \quad \left \quad \begin{array}{l} x = 4 \\ y = -3 \\ z = 2 \end{array} \right.$		$\left[\begin{array}{ccc c} 1 & 0 & 0 & 4 \\ 0 & 1 & 0 & -3 \\ 0 & 0 & 1 & 2 \end{array}\right] \quad \left \quad \begin{array}{l} x = 4 \\ y = -3 \\ z = 2 \end{array} \right.$
johnston-linear-matrix-algebra-EQ0230	105	1.000	$\begin{aligned} x+4y &= 3 \\ z &= 2 \\ 0 &= 0 \end{aligned}$	$\begin{aligned} x+4y &= 3 \\ z &= 2 \\ 0 &= 0 \end{aligned}$	$\begin{aligned} x+4y &= 3 \\ z &= 2 \\ 0 &= 0 \end{aligned}$
johnston-linear-matrix-algebra-EQ0231 (0)	105	1.000	$\left(\begin{array}{l} x \\ y \\ z \end{array}\right) = \left(\begin{array}{l} 3-4y \\ y \\ 2 \end{array}\right) = \left(\begin{array}{l} 3 \\ 0 \\ 2 \end{array}\right) + y \left(\begin{array}{l} -4 \\ 1 \\ 0 \end{array}\right).$		$\begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} 3-4y \\ y \\ 2 \end{pmatrix} = \begin{pmatrix} 3 \\ 0 \\ 2 \end{pmatrix} + y \begin{pmatrix} -4 \\ 1 \\ 0 \end{pmatrix}.$
johnston-linear-matrix-algebra-EQ0243	110	1.000	$x+y=500.$	$x+y=500.$	$x+y=500.$
johnston-linear-matrix-algebra-EQ0249	112	1.000	$4w-y=0 \quad \text{and} \quad 10w-2z=0.$	$4w-y=0 \quad \text{and} \quad 10w-2z=0.$	$4w-y=0 \quad \text{and} \quad 10w-2z=0.$
johnston-linear-matrix-algebra-EQ0261	113	1.000	$\begin{aligned} x+y+z &= 1 \\ -x+z &= 2 \\ 2x+y &= 0 \end{aligned}$	$\begin{aligned} x+y+z &= 1 \\ -x+z &= 2 \\ 2x+y &= 0 \end{aligned}$	$\begin{aligned} x+y+z &= 1 \\ -x+z &= 2 \\ 2x+y &= 0 \end{aligned}$
johnston-linear-matrix-algebra-EQ0303	122	1.000	$AA^{-1}=A^{-1}A=I.$	$AA^{-1}=A^{-1}A=I.$	$AA^{-1}=A^{-1}A=I.$
johnston-linear-matrix-algebra-EQ0339	132	1.000	$A^{-1}=\left[\begin{array}{ccc} 3 & -3 & 1 \\ -5/2 & 4 & -3/2 \\ 1/2 & -1 & 1/2 \end{array}\right].$		$A^{-1}=\begin{bmatrix} 3 & -3 & 1 \\ -5/2 & 4 & -3/2 \\ 1/2 & -1 & 1/2 \end{bmatrix}.$

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johnston-linear-matrix-algebra EQ0340	132	1.000	$\mathbf{x} = A^{-1} \mathbf{b} = \left[\begin{array}{ccc} 3 & -3 & 1 \\ -5/2 & 4 & -3/2 \\ 1/2 & -1 & 1/2 \end{array} \right] \left[\begin{array}{c} 2 \\ 1 \\ 2 \end{array} \right] = \left[\begin{array}{c} 5 \\ -4 \\ 1 \end{array} \right].$		$\mathbf{x} = A^{-1} \mathbf{b} = \left[\begin{array}{ccc} 3 & -3 & 1 \\ -5/2 & 4 & -3/2 \\ 1/2 & -1 & 1/2 \end{array} \right] \left[\begin{array}{c} 2 \\ 1 \\ 2 \end{array} \right] = \left[\begin{array}{c} 5 \\ -4 \\ 1 \end{array} \right].$
johnston-linear-matrix-algebra EQ0343	133	1.000	$B A = \left[\begin{array}{ccc} 1 & 0 & 0 \\ 0 & 1/2 & 0 \end{array} \right] \left[\begin{array}{cc} 1 & 0 \\ 0 & 2 \\ 0 & 0 \end{array} \right] = \left[\begin{array}{cc} 1 & 0 \\ 0 & 1 \end{array} \right].$		$B A = \left[\begin{array}{ccc} 1 & 0 & 0 \\ 0 & 1/2 & 0 \end{array} \right] \left[\begin{array}{cc} 1 & 0 \\ 0 & 2 \\ 0 & 0 \end{array} \right] = \left[\begin{array}{cc} 1 & 0 \\ 0 & 1 \end{array} \right].$
johnston-linear-matrix-algebra EQ0368	143	1.000	$\begin{aligned} \mathbf{v} &= c_1 \mathbf{v}_1 + c_2 \mathbf{v}_2 + \cdots + c_k \mathbf{v}_k \\ \mathbf{w} &= d_1 \mathbf{v}_1 + d_2 \mathbf{v}_2 + \cdots + d_k \mathbf{v}_k. \end{aligned}$		$\mathbf{v} = c_1 \mathbf{v}_1 + c_2 \mathbf{v}_2 + \cdots + c_k \mathbf{v}_k$ $\mathbf{w} = d_1 \mathbf{v}_1 + d_2 \mathbf{v}_2 + \cdots + d_k \mathbf{v}_k.$
johnston-linear-matrix-algebra EQ0381	147	1.000	$\begin{aligned} c_1 - 2c_2 + c_3 &= 0 \\ -c_1 + c_2 + c_3 &= 0 \\ 2c_2 - 4c_3 &= 0 \end{aligned}$		$\begin{aligned} c_1 - 2c_2 + c_3 &= 0 \\ -c_1 + c_2 + c_3 &= 0 \\ 2c_2 - 4c_3 &= 0 \end{aligned}$
johnston-linear-matrix-algebra EQ0383	147	1.000	$3(1, -1, 0) + 2(-2, 1, 2) + (1, 1, -4) = (0, 0, 0).$		$3(1, -1, 0) + 2(-2, 1, 2) + (1, 1, -4) = (0, 0, 0).$
johnston-linear-matrix-algebra EQ0398	153	1.000	$\left[\begin{array}{ccc} 1 & -2 & 4 \\ 1 & 1 & 1 \\ 1 & 3 & 9 \end{array} \right] \left[\begin{array}{c} c_0 \\ c_1 \\ c_2 \end{array} \right] = \left[\begin{array}{c} 13 \\ -2 \\ 8 \end{array} \right],$		$\left[\begin{array}{ccc} 1 & -2 & 4 \\ 1 & 1 & 1 \\ 1 & 3 & 9 \end{array} \right] \left[\begin{array}{c} c_0 \\ c_1 \\ c_2 \end{array} \right] = \left[\begin{array}{c} 13 \\ -2 \\ 8 \end{array} \right],$
johnston-linear-matrix-algebra EQ0417	162	1.000	$d \mathbf{w} + c_1 \mathbf{v}_1 + c_2 \mathbf{v}_2 + \cdots + c_k \mathbf{v}_k = \mathbf{0},$		$d \mathbf{w} + c_1 \mathbf{v}_1 + c_2 \mathbf{v}_2 + \cdots + c_k \mathbf{v}_k = \mathbf{0},$
johnston-linear-matrix-algebra EQ0477	196	1.000	$\begin{array}{l} x + z = 0 \\ w + y + z = 1 \\ w + x + y + z = 0 \\ x + y + z = 1 \end{array}$		$\begin{array}{l} x + z = 0 \\ w + y + z = 1 \\ w + x + y + z = 0 \\ x + y + z = 1 \end{array}$
johnston-linear-matrix-algebra EQ0505 (2.B.4)	203	1.000	$\begin{array}{l} \text{maximize:} \\ \text{subject to:} \end{array} \begin{array}{l} x_1 + 2x_2 \\ -x_1 + x_2 \leq 1 \\ x_1, x_2 \geq 0 \end{array}$		$\begin{array}{l} \text{maximize:} \\ \text{subject to:} \end{array} \begin{array}{l} x_1 + 2x_2 \\ -x_1 + x_2 \leq 1 \\ x_1, x_2 \geq 0 \end{array}$
johnston-linear-matrix-algebra EQ0626	241	1.000	$\left[\begin{array}{cc} 0 & 1 \\ 1 & 0 \end{array} \right]$		$\left[\begin{array}{cc} 0 & 1 \\ 1 & 0 \end{array} \right]$
johnston-linear-matrix-algebra EQ0694	262	1.000	$P_{B \leftarrow E} = P_{E \leftarrow B}^{-1} = \left[\begin{array}{ccc} 1 & 2 & -1 \\ 1 & 2 & -2 \\ -1 & -1 & 1 \end{array} \right]^{-1} = \left[\begin{array}{ccc} 0 & -1 & -2 \\ 1 & 0 & 1 \\ 1 & -1 & 0 \end{array} \right].$		$P_{B \leftarrow E} = P_{E \leftarrow B}^{-1} = \left[\begin{array}{ccc} 1 & 2 & -1 \\ 1 & 2 & -2 \\ -1 & -1 & 1 \end{array} \right]^{-1} = \left[\begin{array}{ccc} 0 & -1 & -2 \\ 1 & 0 & 1 \\ 1 & -1 & 0 \end{array} \right].$

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johnston-linear-matrix-algebra-EQ0763	286	1.000	$\operatorname{det}(A)=1 \cdot 4 - 2 \cdot 3 = -2 \quad \text{and} \quad \operatorname{det}(B)=0 \cdot 5 - 2 \cdot 3 = -6 \quad .$	$\det(A) = 1 \cdot 4 - 2 \cdot 3 = -2 \quad \text{and} \quad \det(B) = 0 \cdot 5 - 2 \cdot 3 = -6.$	$\det(A) = 1 \cdot 4 - 2 \cdot 3 = -2 \quad \text{and} \quad \det(B) = 0 \cdot 5 - 2 \cdot 3 = -6.$
johnston-linear-matrix-algebra-EQ0775	289	1.000	$c_{1,2}=(-1)^{1+2} \quad m_{1,2}=(-1)(-6)=6 \quad .$	$c_{1,2} = (-1)^{1+2} m_{1,2} = (-1)(-6) = 6.$	$c_{1,2} = (-1)^{1+2} m_{1,2} = (-1)(-6) = 6.$
johnston-linear-matrix-algebra-EQ0822	301	1.000	$\operatorname{det}\left(\left[\begin{array}{ll} a & b \\ c & d \end{array}\right]\right)=a \, d - b \, c \quad .$	$\det \left(\left[\begin{array}{cc} a & b \\ c & d \end{array} \right] \right) = ad - bc.$	$\det \left(\left[\begin{array}{cc} a & b \\ c & d \end{array} \right] \right) = ad - bc.$
johnston-linear-matrix-algebra-EQ0874	314	1.000	$\mathbf{v}^* \mathbf{v} = \left[\begin{array}{l} \overline{v_1} \\ \overline{v_2} \\ \vdots \\ \overline{v_n} \end{array} \right] \left[\begin{array}{l} v_1 \\ v_2 \\ \vdots \\ v_n \end{array} \right] = v_1 ^2 + v_2 ^2 + \cdots + v_n ^2$	$\mathbf{v}^* \mathbf{v} = \left[\begin{array}{cccc} \overline{v_1} & \overline{v_2} & \cdots & \overline{v_n} \end{array} \right] \left[\begin{array}{c} v_1 \\ v_2 \\ \vdots \\ v_n \end{array} \right] = v_1 ^2 + v_2 ^2 + \cdots + v_n ^2$	$\mathbf{v}^* \mathbf{v} = \left[\overline{v_1} \quad \overline{v_2} \quad \cdots \quad \overline{v_n} \right] \left[\begin{array}{c} v_1 \\ v_2 \\ \vdots \\ v_n \end{array} \right] = v_1 ^2 + v_2 ^2 + \cdots + v_n ^2$
johnston-linear-matrix-algebra-EQ0876	315	1.000	$\begin{aligned} \operatorname{det}\left(\left[\begin{array}{ccc} -\lambda & 1 & -1 \\ 1 & -3-\lambda & 2 \\ -1 & 2 & 1-\lambda \end{array}\right]\right) &= -\lambda(-3-\lambda)(1-\lambda) - 2 - 2 \\ &\quad - 4(-\lambda) - (1-\lambda) - (-3-\lambda) \\ &= -\lambda^3 - 2\lambda^2 + 9\lambda - 2 \end{aligned}$	$\det \left(\left[\begin{array}{ccc} -\lambda & 1 & -1 \\ 1 & -3-\lambda & 2 \\ -1 & 2 & 1-\lambda \end{array} \right] \right) = -\lambda(-3-\lambda)(1-\lambda) - 2 - 2 \\ \qquad \qquad \qquad - 4(-\lambda) - (1-\lambda) - (-3-\lambda) \\ \qquad \qquad \qquad = -\lambda^3 - 2\lambda^2 + 9\lambda - 2$	$\det \left(\left[\begin{array}{ccc} -\lambda & 1 & -1 \\ 1 & -3-\lambda & 2 \\ -1 & 2 & 1-\lambda \end{array} \right] \right) = -\lambda(-3-\lambda)(1-\lambda) - 2 - 2 \\ \qquad \qquad \qquad - 4(-\lambda) - (1-\lambda) - (-3-\lambda) \\ \qquad \qquad \qquad = -\lambda^3 - 2\lambda^2 + 9\lambda - 2.$
johnston-linear-matrix-algebra-EQ0921 (3.4.3)	329	1.000	$\lambda_k \mathbf{v}_k = c_1 \lambda_k \mathbf{v}_1 + c_2 \lambda_k \mathbf{v}_2 + \cdots + c_{k-1} \lambda_k \mathbf{v}_{k-1}.$	$\lambda_k \mathbf{v}_k = c_1 \lambda_k \mathbf{v}_1 + c_2 \lambda_k \mathbf{v}_2 + \cdots + c_{k-1} \lambda_k \mathbf{v}_{k-1}.$	$\lambda_k \mathbf{v}_k = c_1 \lambda_k \mathbf{v}_1 + c_2 \lambda_k \mathbf{v}_2 + \cdots + c_{k-1} \lambda_k \mathbf{v}_{k-1}.$
johnston-linear-matrix-algebra-EQ0940	335	1.000	$P=\left \mathbf{p}_1\right \left \mathbf{p}_2\right \cdots \left \mathbf{p}_n\right \quad \text{and} \quad Q=\left \mathbf{q}_1\right \left \mathbf{q}_2\right \cdots \left \mathbf{q}_n\right \quad .$	$P = \left \mathbf{p}_1 \right \left \mathbf{p}_2 \right \cdots \left \mathbf{p}_n \right \quad \text{and} \quad Q = \left \mathbf{q}_1 \right \left \mathbf{q}_2 \right \cdots \left \mathbf{q}_n \right .$	$P = \left[\mathbf{p}_1 \mid \mathbf{p}_2 \mid \cdots \mid \mathbf{p}_n \right] \quad \text{and} \quad Q = \left[\mathbf{q}_1 \mid \mathbf{q}_2 \mid \cdots \mid \mathbf{q}_n \right].$
johnston-linear-matrix-algebra-EQ0955 (0)	338	1.000	$\left[\begin{array}{cc c} \phi & 1 & 0 \\ 1 & \phi-1 & 0 \end{array}\right] \xrightarrow{R_1 \leftrightarrow R_2} \left[\begin{array}{cc c} 1 & \phi-1 & 0 \\ \phi & 1 & 0 \end{array}\right] \xrightarrow{R_2 - \phi R_1} \left[\begin{array}{cc c} 1 & \phi-1 & 0 \\ 0 & 0 & 0 \end{array}\right].$	$\left[\begin{array}{cc c} \phi & 1 & 0 \\ 1 & \phi-1 & 0 \end{array} \right] \xrightarrow{R_1 \leftrightarrow R_2} \left[\begin{array}{cc c} 1 & \phi-1 & 0 \\ \phi & 1 & 0 \end{array} \right] \xrightarrow{R_2 - \phi R_1} \left[\begin{array}{cc c} 1 & \phi-1 & 0 \\ 0 & 0 & 0 \end{array} \right].$	$\left[\begin{array}{cc c} \phi & 1 & 0 \\ 1 & \phi-1 & 0 \end{array} \right] \xrightarrow{R_1 \leftrightarrow R_2} \left[\begin{array}{cc c} 1 & \phi-1 & 0 \\ \phi & 1 & 0 \end{array} \right] \xrightarrow{R_2 - \phi R_1} \left[\begin{array}{cc c} 1 & \phi-1 & 0 \\ 0 & 0 & 0 \end{array} \right].$
johnston-linear-matrix-algebra-EQ0968	341	1.000	$A^{-1}=\frac{1}{3}\left[\begin{array}{cc} 4-4^{-1} & 4^0-4 \\ 1-4^{-1} & 4^0-1 \end{array}\right]=\frac{1}{3}\left[\begin{array}{cc} 15 / 4 & -3 \\ 3 / 4 & 0 \end{array}\right]=\frac{1}{4}\left[\begin{array}{cc} 5 & -4 \\ 1 & 0 \end{array}\right].$	$A^{-1} = \frac{1}{3} \left[\begin{array}{cc} 4 - 4^{-1} & 4^0 - 4 \\ 1 - 4^{-1} & 4^0 - 1 \end{array} \right] = \frac{1}{3} \left[\begin{array}{cc} 15/4 & -3 \\ 3/4 & 0 \end{array} \right] = \frac{1}{4} \left[\begin{array}{cc} 5 & -4 \\ 1 & 0 \end{array} \right].$	$A^{-1} = \frac{1}{3} \left[\begin{array}{cc} 4 - 4^{-1} & 4^0 - 4 \\ 1 - 4^{-1} & 4^0 - 1 \end{array} \right] = \frac{1}{3} \left[\begin{array}{cc} 15/4 & -3 \\ 3/4 & 0 \end{array} \right] = \frac{1}{4} \left[\begin{array}{cc} 5 & -4 \\ 1 & 0 \end{array} \right].$

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johnston-linear-matrix-algebra-EQ0973	342	1.000	$\left(A^{1/2}\right)^2=\left(\frac{1}{3}\left[\begin{array}{cc} 2 & 4 \\ -1 & 7 \end{array}\right]\right)\left(\frac{1}{3}\left[\begin{array}{cc} 2 & 4 \\ -1 & 7 \end{array}\right]\right)=\frac{1}{9}\left[\begin{array}{cc} 0 & 36 \\ -9 & 45 \end{array}\right]=\left[\begin{array}{cc} 0 & 4 \\ -1 & 5 \end{array}\right]=A,$	$\left(A^{1/2}\right)^2=\left(\frac{1}{3}\left[\begin{array}{cc} 2 & 4 \\ -1 & 7 \end{array}\right]\right)\left(\frac{1}{3}\left[\begin{array}{cc} 2 & 4 \\ -1 & 7 \end{array}\right]\right)=\frac{1}{9}\left[\begin{array}{cc} 0 & 36 \\ -9 & 45 \end{array}\right]=\left[\begin{array}{cc} 0 & 4 \\ -1 & 5 \end{array}\right]=A,$	$\left(A^{1/2}\right)^2=\left(\frac{1}{3}\left[\begin{array}{cc} 2 & 4 \\ -1 & 7 \end{array}\right]\right)\left(\frac{1}{3}\left[\begin{array}{cc} 2 & 4 \\ -1 & 7 \end{array}\right]\right)=\frac{1}{9}\left[\begin{array}{cc} 0 & 36 \\ -9 & 45 \end{array}\right]=\left[\begin{array}{cc} 0 & 4 \\ -1 & 5 \end{array}\right]=A,$
johnston-linear-matrix-algebra-EQ0976	343	1.000	$\begin{aligned} B^3 &= \left[\begin{array}{cc} 0 & 2 \\ 1 & -1 \end{array}\right] \left[\begin{array}{cc} 0 & 2 \\ 1 & -1 \end{array}\right] \left[\begin{array}{cc} 0 & 2 \\ 1 & -1 \end{array}\right] \\ &= \left[\begin{array}{cc} 0 & 2 \\ 1 & -1 \end{array}\right] \left[\begin{array}{cc} 2 & -2 \\ -1 & 3 \end{array}\right] = \left[\begin{array}{cc} -2 & 6 \\ 3 & -5 \end{array}\right] = A. \end{aligned}$	$B^3 = \left[\begin{array}{cc} 0 & 2 \\ 1 & -1 \end{array}\right] \left[\begin{array}{cc} 0 & 2 \\ 1 & -1 \end{array}\right] \left[\begin{array}{cc} 0 & 2 \\ 1 & -1 \end{array}\right] \\ = \left[\begin{array}{cc} 0 & 2 \\ 1 & -1 \end{array}\right] \left[\begin{array}{cc} 2 & -2 \\ -1 & 3 \end{array}\right] = \left[\begin{array}{cc} -2 & 6 \\ 3 & -5 \end{array}\right] = A.$	$B^3 = \left[\begin{array}{cc} 0 & 2 \\ 1 & -1 \end{array}\right] \left[\begin{array}{cc} 0 & 2 \\ 1 & -1 \end{array}\right] \left[\begin{array}{cc} 0 & 2 \\ 1 & -1 \end{array}\right] \\ = \left[\begin{array}{cc} 0 & 2 \\ 1 & -1 \end{array}\right] \left[\begin{array}{cc} 2 & -2 \\ -1 & 3 \end{array}\right] = \left[\begin{array}{cc} -2 & 6 \\ 3 & -5 \end{array}\right] = A.$
johnston-linear-matrix-algebra-EQ0981	344	1.000	$p(x)=c_{\{k\}}\ x^{\{k\}}+\cdots+c_{\{2\}}\ x^{\{2\}}+c_{\{1\}}\ x+c_{\{0\}},$	$p(x)=c_kx^k+\cdots+c_2x^2+c_1x+c_0,$	$p(x)=c_kx^k+\cdots+c_2x^2+c_1x+c_0,$
johnston-linear-matrix-algebra-EQ0995	348	1.000	$e^A=\sum_{k=0}^{\infty}\frac{1}{k!}\ A^k\quad\text{for all}\quad A\in\mathcal{M}_n.$	$e^A=\sum_{k=0}^{\infty}\frac{1}{k!}A^k\quad\text{for all}\quad A\in\mathcal{M}_n.$	$e^A=\sum_{k=0}^{\infty}\frac{1}{k!}A^k\quad\text{for all}\quad A\in\mathcal{M}_n.$
johnston-linear-matrix-algebra-EQ1024	355	1.000	$N=\left[\begin{array}{ll} 0 & 1 \\ 0 & 0 \end{array}\right]$	$N=\left[\begin{array}{cc} 0 & 1 \\ 0 & 0 \end{array}\right]$	$N=\left[\begin{array}{cc} 0 & 1 \\ 0 & 0 \end{array}\right]$
johnston-linear-matrix-algebra-EQ1056	361	1.000	$(\sigma\circ\tau)(j)=\sigma(\tau(j))\quad\text{for all}\quad 1\leq j\leq n.$	$(\sigma\circ\tau)(j)=\sigma(\tau(j))\quad\text{for all}\quad 1\leq j\leq n.$	$(\sigma\circ\tau)(j)=\sigma(\tau(j))\quad\text{for all}\quad 1\leq j\leq n.$
johnston-linear-matrix-algebra-EQ1063	363	1.000	$P_{\{\sigma\}}=\left[\mathbf{e}_1\left \mathbf{e}_2\right \cdots\left \mathbf{e}_6\right.\right]=I_6.$	$P_{\sigma}=[\mathbf{e}_1 \mathbf{e}_2 \cdots \mathbf{e}_6]=I_6.$	$P_{\sigma}=[\mathbf{e}_1 \mathbf{e}_2 \cdots \mathbf{e}_6]=I_6.$
johnston-linear-matrix-algebra-EQ1092	372	1.000	$\lambda\approx\mathbf{v}_{\{5\}}^TA\ \mathbf{v}_{\{5\}}\approx\left[\begin{array}{ll} 0.37 & 0.93 \end{array}\right]\left[\begin{array}{l} 2.23 \\ 5.57 \end{array}\right]\approx 6.00,$	$\lambda\approx\mathbf{v}_5^TA\mathbf{v}_5\approx\left[\begin{array}{cc} 0.37 & 0.93 \end{array}\right]\left[\begin{array}{c} 2.23 \\ 5.57 \end{array}\right]\approx 6.00,$	$\lambda\approx\mathbf{v}_5^TA\mathbf{v}_5\approx\left[\begin{array}{cc} 0.37 & 0.93 \end{array}\right]\left[\begin{array}{c} 2.23 \\ 5.57 \end{array}\right]\approx 6.00,$
johnston-linear-matrix-algebra-EQ1127	389	1.000	$\lim_{k\rightarrow\infty}\ A\mathbf{v}_k\ =\lambda_1.$	$\lim_{k\rightarrow\infty}\ A\mathbf{v}_k\ =\lambda_1.$	$\lim_{k\rightarrow\infty}\ A\mathbf{v}_k\ =\lambda_1.$
johnston-linear-matrix-algebra-EQ1167	400	1.000	$3,-2,8,16,68,208,668,2056,6308,\text{ and }19168.$	$3,-2,8,16,68,208,668,2056,6308,\text{ and }19168.$	$3,-2,8,16,68,208,668,2056,6308,\text{ and }19168.$

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johnston-linear-matrix-algebra EQ1179	403	1.000	$\begin{aligned} & \left[\begin{array}{ccc} 0 & 1 & 0 \\ 0 & 0 & 1 \\ -6 & -1 & 4 \end{array} \right]^n \left[\begin{array}{c} 3 \\ -2 \\ 8 \end{array} \right] = PD^n P^{-1} \left[\begin{array}{c} 3 \\ -2 \\ 8 \end{array} \right] \\ & = \left[\begin{array}{ccc} 1 & 1 & 1 \\ 3 & 2 & -1 \\ 9 & 4 & 1 \end{array} \right] \left[\begin{array}{ccc} 3^n & 0 & 0 \\ 0 & 2^n & 0 \\ 0 & 0 & (-1)^n \end{array} \right] \left(\frac{1}{12} \left[\begin{array}{ccc} -6 & -3 & 3 \\ 12 & 8 & -4 \\ 6 & -5 & 1 \end{array} \right] \right) \left[\begin{array}{c} 3 \\ -2 \\ 8 \end{array} \right] \\ & = \left[\begin{array}{ccc} 1 & 1 & 1 \\ 3 & 2 & -1 \\ 9 & 4 & 1 \end{array} \right] \left[\begin{array}{ccc} 3^n & 0 & 0 \\ 0 & 2^n & 0 \\ 0 & 0 & (-1)^n \end{array} \right] \left[\begin{array}{c} 1 \\ -1 \\ 3 \end{array} \right] \\ & = \left[\begin{array}{ccc} 1 & 1 & 1 \\ 3 & 2 & -1 \\ 9 & 4 & 1 \end{array} \right] \left[\begin{array}{c} 3^n \\ -2^n \\ 3(-1)^n \end{array} \right], \end{aligned}$	$\begin{aligned} & \left[\begin{array}{ccc} 0 & 1 & 0 \\ 0 & 0 & 1 \\ -6 & -1 & 4 \end{array} \right]^n \left[\begin{array}{c} 3 \\ -2 \\ 8 \end{array} \right] = PD^n P^{-1} \left[\begin{array}{c} 3 \\ -2 \\ 8 \end{array} \right] \\ & = \left[\begin{array}{ccc} 1 & 1 & 1 \\ 3 & 2 & -1 \\ 9 & 4 & 1 \end{array} \right] \left[\begin{array}{ccc} 3^n & 0 & 0 \\ 0 & 2^n & 0 \\ 0 & 0 & (-1)^n \end{array} \right] \left(\frac{1}{12} \left[\begin{array}{ccc} -6 & -3 & 3 \\ 12 & 8 & -4 \\ 6 & -5 & 1 \end{array} \right] \right) \left[\begin{array}{c} 3 \\ -2 \\ 8 \end{array} \right] \\ & = \left[\begin{array}{ccc} 1 & 1 & 1 \\ 3 & 2 & -1 \\ 9 & 4 & 1 \end{array} \right] \left[\begin{array}{ccc} 3^n & 0 & 0 \\ 0 & 2^n & 0 \\ 0 & 0 & (-1)^n \end{array} \right] \left[\begin{array}{c} 1 \\ -1 \\ 3 \end{array} \right] \\ & = \left[\begin{array}{ccc} 1 & 1 & 1 \\ 3 & 2 & -1 \\ 9 & 4 & 1 \end{array} \right] \left[\begin{array}{c} 3^n \\ -2^n \\ 3(-1)^n \end{array} \right], \end{aligned}$	
johnston-linear-matrix-algebra EQ1266	425	1.000	$\begin{aligned} & 2(5)^3 - 9(5)^2 - 6(5) + 5 = 0 \\ & 2(-5)^3 - 9(-5)^2 - 6(-5) + 5 = -440 \\ & 2(1)^3 - 9(1)^2 - 6(1) + 5 = -8 \\ & 2(-1)^3 - 9(-1)^2 - 6(-1) + 5 = 0 \\ & 2(5/2)^3 - 9(5/2)^2 - 6(5/2) + 5 = -35 \\ & 2(-5/2)^3 - 9(-5/2)^2 - 6(-5/2) + 5 = -135/2 \\ & 2(1/2)^3 - 9(1/2)^2 - 6(1/2) + 5 = 0 \\ & 2(-1/2)^3 - 9(-1/2)^2 - 6(-1/2) + 5 = 11/2. \end{aligned}$	$\begin{aligned} & 2(5)^3 - 9(5)^2 - 6(5) + 5 = 0 \\ & 2(-5)^3 - 9(-5)^2 - 6(-5) + 5 = -440 \\ & 2(1)^3 - 9(1)^2 - 6(1) + 5 = -8 \\ & 2(-1)^3 - 9(-1)^2 - 6(-1) + 5 = 0 \\ & 2(5/2)^3 - 9(5/2)^2 - 6(5/2) + 5 = -35 \\ & 2(-5/2)^3 - 9(-5/2)^2 - 6(-5/2) + 5 = -135/2 \\ & 2(1/2)^3 - 9(1/2)^2 - 6(1/2) + 5 = 0 \\ & 2(-1/2)^3 - 9(-1/2)^2 - 6(-1/2) + 5 = 11/2. \end{aligned}$	$\begin{aligned} & 2(5)^3 - 9(5)^2 - 6(5) + 5 = 0 \\ & 2(-5)^3 - 9(-5)^2 - 6(-5) + 5 = -440 \\ & 2(1)^3 - 9(1)^2 - 6(1) + 5 = -8 \\ & 2(-1)^3 - 9(-1)^2 - 6(-1) + 5 = 0 \\ & 2(5/2)^3 - 9(5/2)^2 - 6(5/2) + 5 = -35 \\ & 2(-5/2)^3 - 9(-5/2)^2 - 6(-5/2) + 5 = -135/2 \\ & 2(1/2)^3 - 9(1/2)^2 - 6(1/2) + 5 = 0 \\ & 2(-1/2)^3 - 9(-1/2)^2 - 6(-1/2) + 5 = 11/2. \end{aligned}$
johnston-linear-matrix-algebra EQ1279	427	1.000	$p(x)=a \ x^3+b \ x^2+c \ x+d,$	$p(x) = ax^3 + bx^2 + cx + d,$	$p(x) = ax^3 + bx^2 + cx + d,$

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johnston-linear-matrix-algebra-EQ1394	440	1.000	$\begin{aligned} E\mathbf{A} &= \left[\mathbf{e}_1 \cdots \mathbf{e}_j \cdots \mathbf{e}_i \cdots \mathbf{e}_m \right] \begin{bmatrix} \mathbf{a}_1^T \\ \vdots \\ \mathbf{a}_m^T \end{bmatrix} \\ &= \mathbf{e}_1 \mathbf{a}_1^T + \mathbf{e}_2 \mathbf{a}_2^T + \cdots + \mathbf{e}_j \mathbf{a}_j^T + \cdots + \mathbf{e}_i \mathbf{a}_i^T + \cdots + \mathbf{e}_m \mathbf{a}_m^T, \end{aligned}$	$EA = [\mathbf{e}_1 \cdots \mathbf{e}_j \cdots \mathbf{e}_i \cdots \mathbf{e}_m] \begin{bmatrix} \mathbf{a}_1^T \\ \vdots \\ \mathbf{a}_m^T \end{bmatrix} = \mathbf{e}_1 \mathbf{a}_1^T + \mathbf{e}_2 \mathbf{a}_2^T + \cdots + \mathbf{e}_j \mathbf{a}_j^T + \cdots + \mathbf{e}_i \mathbf{a}_i^T + \cdots + \mathbf{e}_m \mathbf{a}_m^T$	$EA = [\mathbf{e}_1 \cdots \mathbf{e}_j \cdots \mathbf{e}_i \cdots \mathbf{e}_m] \begin{bmatrix} \mathbf{a}_1^T \\ \vdots \\ \mathbf{a}_m^T \end{bmatrix} = \mathbf{e}_1 \mathbf{a}_1^T + \mathbf{e}_2 \mathbf{a}_2^T + \cdots + \mathbf{e}_j \mathbf{a}_j^T + \cdots + \mathbf{e}_i \mathbf{a}_i^T + \cdots + \mathbf{e}_m \mathbf{a}_m^T,$
johnston-linear-matrix-algebra-EQ1395	441	1.000	$f(A) = \sum_{\sigma \in S_n} \operatorname{sgn}(\sigma) a_{\sigma(1),1} a_{\sigma(2),2} \cdots a_{\sigma(n),n}$	$f(A) = \sum_{\sigma \in S_n} \operatorname{sgn}(\sigma) a_{\sigma(1),1} a_{\sigma(2),2} \cdots a_{\sigma(n),n}$	$f(A) = \sum_{\sigma \in S_n} \operatorname{sgn}(\sigma) a_{\sigma(1),1} a_{\sigma(2),2} \cdots a_{\sigma(n),n}.$
johnston-linear-matrix-algebra-EQ1388	453	1.000	$\begin{aligned} \mathbf{v} \times \mathbf{w} + \mathbf{v} \times \mathbf{x} &= \begin{bmatrix} v_2 w_3 - v_3 w_2 \\ v_3 w_1 - v_1 w_3 \\ v_1 w_2 - v_2 w_1 \end{bmatrix} + \begin{bmatrix} v_2 x_3 - v_3 x_2 \\ v_3 x_1 - v_1 x_3 \\ v_1 x_2 - v_2 x_1 \end{bmatrix} \\ &= \begin{bmatrix} v_2 w_3 - v_3 w_2 + v_2 x_3 - v_3 x_2 \\ v_3 w_1 - v_1 w_3 + v_3 x_1 - v_1 x_3 \\ v_1 w_2 - v_2 w_1 + v_1 x_2 - v_2 x_1 \end{bmatrix} \end{aligned}$	$\mathbf{v} \times \mathbf{w} + \mathbf{v} \times \mathbf{x} = \begin{bmatrix} v_2 w_3 - v_3 w_2 \\ v_3 w_1 - v_1 w_3 \\ v_1 w_2 - v_2 w_1 \end{bmatrix} + \begin{bmatrix} v_2 x_3 - v_3 x_2 \\ v_3 x_1 - v_1 x_3 \\ v_1 x_2 - v_2 x_1 \end{bmatrix} = \begin{bmatrix} v_2 w_3 - v_3 w_2 + v_2 x_3 - v_3 x_2 \\ v_3 w_1 - v_1 w_3 + v_3 x_1 - v_1 x_3 \\ v_1 w_2 - v_2 w_1 + v_1 x_2 - v_2 x_1 \end{bmatrix}$	$\mathbf{v} \times \mathbf{w} + \mathbf{v} \times \mathbf{x} = \begin{bmatrix} v_2 w_3 - v_3 w_2 \\ v_3 w_1 - v_1 w_3 \\ v_1 w_2 - v_2 w_1 \end{bmatrix} + \begin{bmatrix} v_2 x_3 - v_3 x_2 \\ v_3 x_1 - v_1 x_3 \\ v_1 x_2 - v_2 x_1 \end{bmatrix} = \begin{bmatrix} v_2 w_3 - v_3 w_2 + v_2 x_3 - v_3 x_2 \\ v_3 w_1 - v_1 w_3 + v_3 x_1 - v_1 x_3 \\ v_1 w_2 - v_2 w_1 + v_1 x_2 - v_2 x_1 \end{bmatrix}$
johnston-linear-matrix-algebra-EQ1411	456	1.000	$\begin{array}{r} v_1 + 2v_2 + 3v_3 = 0 \\ 3v_1 + 2v_2 + v_3 = 0 \end{array}$	$\begin{array}{l} v_1 + 2v_2 + 3v_3 = 0 \\ 3v_1 + 2v_2 + v_3 = 0 \end{array}$	$\begin{array}{l} v_1 + 2v_2 + 3v_3 = 0 \\ 3v_1 + 2v_2 + v_3 = 0 \end{array}$
johnston-linear-matrix-algebra-EQ1438	458	1.000	$\left[\begin{array}{ll} c & d \end{array} \right] \& \left[\begin{array}{ll} a & b \end{array} \right] \& \left[\begin{array}{ll} b & a \end{array} \right] \& \left[\begin{array}{ll} d & c \end{array} \right]$	$\left[\begin{array}{ll} c & d \\ a & b \end{array} \right] = \left[\begin{array}{ll} b & a \\ d & c \end{array} \right].$	$\left[\begin{array}{ll} c & d \\ a & b \end{array} \right] = \left[\begin{array}{ll} b & a \\ d & c \end{array} \right].$
johnston-linear-matrix-algebra-EQ1493	465	1.000	$\& (k-1)(n+1, n+2, \ldots, 2n) - (k-2)(1, 2, \ldots, n) = ((k-1)n+1, (k-1)n+2, \ldots, kn).$	$(k-1)(n+1, n+2, \ldots, 2n) - (k-2)(1, 2, \ldots, n) = ((k-1)n+1, (k-1)n+2, \ldots, kn).$	$(k-1)(n+1, n+2, \ldots, 2n) - (k-2)(1, 2, \ldots, n) = ((k-1)n+1, (k-1)n+2, \ldots, kn).$
johnston-linear-matrix-algebra-EQ1538	471	1.000	$\begin{aligned} &\left[\begin{array}{ll} 0 & 0 \\ 1 & 1 \end{array} \right] \& \left[\begin{array}{cc} 1 & 0 \\ \ell_{2,1} & 1 \end{array} \right] \& \left[\begin{array}{cc} u_{1,1} & u_{1,2} \\ 0 & u_{2,2} \end{array} \right] \& \left[\begin{array}{cc} u_{1,1} & u_{1,2} \\ \ell_{2,1} u_{1,1} & \ell_{2,1} u_{1,2} + u_{2,2} \end{array} \right] \end{aligned}$	$\left[\begin{array}{ll} 0 & 0 \\ 1 & 1 \end{array} \right] = \left[\begin{array}{ll} 1 & 0 \\ \ell_{2,1} & 1 \end{array} \right] \left[\begin{array}{ll} u_{1,1} & u_{1,2} \\ 0 & u_{2,2} \end{array} \right] = \left[\begin{array}{ll} u_{1,1} & u_{1,2} \\ \ell_{2,1} u_{1,1} & \ell_{2,1} u_{1,2} + u_{2,2} \end{array} \right].$	$\left[\begin{array}{ll} 0 & 0 \\ 1 & 1 \end{array} \right] = \left[\begin{array}{ll} 1 & 0 \\ \ell_{2,1} & 1 \end{array} \right] \left[\begin{array}{ll} u_{1,1} & u_{1,2} \\ 0 & u_{2,2} \end{array} \right] = \left[\begin{array}{ll} u_{1,1} & u_{1,2} \\ \ell_{2,1} u_{1,1} & \ell_{2,1} u_{1,2} + u_{2,2} \end{array} \right].$
johnston-linear-matrix-algebra-EQ1568	474	1.000	$f(\mathbf{A} \mathbf{e}_i \mathbf{e}_k^T) = f(\mathbf{e}_j \mathbf{e}_i^T \mathbf{e}_k^T) = f(\mathbf{e}_j \mathbf{e}_k^T) = 0$	$f(\mathbf{A} \mathbf{e}_i \mathbf{e}_k^T) = f(\mathbf{e}_j \mathbf{e}_i^T \mathbf{e}_k^T) = f(\mathbf{e}_j \mathbf{e}_k^T) = 0$	$f(\mathbf{A} \mathbf{e}_i \mathbf{e}_k^T) = f(\mathbf{e}_j \mathbf{e}_i^T \mathbf{e}_k^T) = f(\mathbf{e}_j \mathbf{e}_k^T) = 0$

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johnston-linear-matrix-algebra-EQ1576	474	1.000	$\begin{aligned} \mathbf{v} \cdot \mathbf{w} &= (c_1 \mathbf{v}_1 + \cdots + c_k \mathbf{v}_k) \cdot (d_1 \mathbf{v}_1 + \cdots + d_k \mathbf{v}_k) \\ &= c_1 d_1 (\mathbf{v}_1 \cdot \mathbf{v}_1) + c_1 d_2 (\mathbf{v}_1 \cdot \mathbf{v}_2) + \cdots \\ &\quad + c_k d_{k-1} (\mathbf{v}_k \cdot \mathbf{v}_{k-1}) + c_k d_k (\mathbf{v}_k \cdot \mathbf{v}_k) \\ &= c_1 d_1 + c_2 d_2 + \cdots + c_k d_k, \end{aligned}$	$\mathbf{v} \cdot \mathbf{w} = (c_1 \mathbf{v}_1 + \cdots + c_k \mathbf{v}_k) \cdot (d_1 \mathbf{v}_1 + \cdots + d_k \mathbf{v}_k)$ $= c_1 d_1 (\mathbf{v}_1 \cdot \mathbf{v}_1) + c_1 d_2 (\mathbf{v}_1 \cdot \mathbf{v}_2) + \cdots$ $+ c_k d_{k-1} (\mathbf{v}_k \cdot \mathbf{v}_{k-1}) + c_k d_k (\mathbf{v}_k \cdot \mathbf{v}_k)$ $= c_1 d_1 + c_2 d_2 + \cdots + c_k d_k,$	
johnston-linear-matrix-algebra-EQ1600	477	1.000	$\begin{aligned} \operatorname{det}(A - \lambda I) &= \operatorname{det} \left(\begin{bmatrix} k - \lambda & 1 \\ -1 & 1 - \lambda \end{bmatrix} \right) \\ &= (k - \lambda)(1 - \lambda) + 1 \\ &= \lambda^2 - (1 + k)\lambda + (k + 1) \\ &= 0 \end{aligned}$	$\det(A - \lambda I) = \det \left(\begin{bmatrix} k - \lambda & 1 \\ -1 & 1 - \lambda \end{bmatrix} \right)$ $= (k - \lambda)(1 - \lambda) + 1$ $= \lambda^2 - (1 + k)\lambda + (k + 1)$ $= 0$	$\det(A - \lambda I) = \det \left(\begin{bmatrix} k - \lambda & 1 \\ -1 & 1 - \lambda \end{bmatrix} \right)$ $= (k - \lambda)(1 - \lambda) + 1$ $= \lambda^2 - (1 + k)\lambda + (k + 1)$ $= 0.$
johnston-linear-matrix-algebra-EQ1635	481	1.000	$\begin{aligned} \operatorname{det}(e^{A+B}) &= e^{\operatorname{tr}(A+B)} = e^{\operatorname{tr}(A) + \operatorname{tr}(B)} = e^{\operatorname{tr}(A)} e^{\operatorname{tr}(B)} \\ &= \operatorname{det}(e^A) \operatorname{det}(e^B) = \operatorname{det}(e^A e^B). \end{aligned}$	$\det(e^{A+B}) = e^{\operatorname{tr}(A+B)} = e^{\operatorname{tr}(A) + \operatorname{tr}(B)} = e^{\operatorname{tr}(A)} e^{\operatorname{tr}(B)}$ $= \det(e^A) \det(e^B) = \det(e^A e^B)$	$\det(e^{A+B}) = e^{\operatorname{tr}(A+B)} = e^{\operatorname{tr}(A) + \operatorname{tr}(B)} = e^{\operatorname{tr}(A)} e^{\operatorname{tr}(B)}$ $= \det(e^A) \det(e^B) = \det(e^A e^B).$
johnston-linear-matrix-algebra-EQ1660	484	1.000	$\lim_{k \rightarrow \infty} \ \lambda_1 \mathbf{v}_k\ = \lim_{k \rightarrow \infty} \lambda_1 = \lambda_1,$	$\lim_{k \rightarrow \infty} \ \lambda_1 \mathbf{v}_k\ = \lim_{k \rightarrow \infty} \lambda_1 = \lambda_1,$	$\lim_{k \rightarrow \infty} \ \lambda_1 \mathbf{v}_k\ = \lim_{k \rightarrow \infty} \lambda_1 = \lambda_1,$
johnston-linear-matrix-algebra-EQ0001	18	1.000	$\mathbf{v} + \mathbf{w} \stackrel{\text{def}}{=} (v_1 + w_1, v_2 + w_2, \dots, v_n + w_n).$	$\mathbf{v} + \mathbf{w} \stackrel{\text{def}}{=} (v_1 + w_1, v_2 + w_2, \dots, v_n + w_n).$	$\mathbf{v} + \mathbf{w} \stackrel{\text{def}}{=} (v_1 + w_1, v_2 + w_2, \dots, v_n + w_n).$
johnston-linear-matrix-algebra-EQ0002	18	1.000	$\begin{aligned} \mathbf{v} + \mathbf{w} &= (v_1 + w_1, v_2 + w_2, \dots, v_n + w_n) \\ &= (w_1 + v_1, w_2 + v_2, \dots, w_n + v_n) = \mathbf{w} + \mathbf{v}. \end{aligned}$	$\mathbf{v} + \mathbf{w} = (v_1 + w_1, v_2 + w_2, \dots, v_n + w_n)$ $= (w_1 + v_1, w_2 + v_2, \dots, w_n + v_n) = \mathbf{w} + \mathbf{v}.$	$\mathbf{v} + \mathbf{w} = (v_1 + w_1, v_2 + w_2, \dots, v_n + w_n)$ $= (w_1 + v_1, w_2 + v_2, \dots, w_n + v_n) = \mathbf{w} + \mathbf{v}.$
johnston-linear-matrix-algebra-EQ0003	20	1.000	$c \mathbf{v} \stackrel{\text{def}}{=} (cv_1, cv_2, \dots, cv_n).$	$c\mathbf{v} \stackrel{\text{def}}{=} (cv_1, cv_2, \dots, cv_n).$	$c\mathbf{v} \stackrel{\text{def}}{=} (cv_1, cv_2, \dots, cv_n).$
johnston-linear-matrix-algebra-EQ0007	23	1.000	$c_1 \mathbf{v}_1 + c_2 \mathbf{v}_2 + \cdots + c_k \mathbf{v}_k,$	$c_1 \mathbf{v}_1 + c_2 \mathbf{v}_2 + \cdots + c_k \mathbf{v}_k,$	$c_1 \mathbf{v}_1 + c_2 \mathbf{v}_2 + \cdots + c_k \mathbf{v}_k,$
johnston-linear-matrix-algebra-EQ0008	23	1.000	$\mathbf{e}_j \stackrel{\text{def}}{=} (0, 0, \dots, 0, 1, 0, \dots, 0).$	$\mathbf{e}_j \stackrel{\text{def}}{=} (0, 0, \dots, 0, 1, 0, \dots, 0).$	$\mathbf{e}_j \stackrel{\text{def}}{=} (0, 0, \dots, 0, 1, 0, \dots, 0).$ ↑ j-th entry
johnston-linear-matrix-algebra-EQ0009	23	1.000	$\mathbf{v} = v_1 \mathbf{e}_1 + v_2 \mathbf{e}_2 + \cdots + v_n \mathbf{e}_n,$	$\mathbf{v} = v_1 \mathbf{e}_1 + v_2 \mathbf{e}_2 + \cdots + v_n \mathbf{e}_n,$	$\mathbf{v} = v_1 \mathbf{e}_1 + v_2 \mathbf{e}_2 + \cdots + v_n \mathbf{e}_n,$
johnston-linear-matrix-algebra-EQ0015	27	1.000	$\ \mathbf{v}\ = \sqrt{\ (v_1, 0)\ ^2 + \ (0, v_2)\ ^2} = \sqrt{ v_1 ^2 + v_2 ^2} = \sqrt{v_1^2 + v_2^2} = \sqrt{\mathbf{v} \cdot \mathbf{v}}.$	$\ \mathbf{v}\ = \sqrt{\ (v_1, 0)\ ^2 + \ (0, v_2)\ ^2} = \sqrt{ v_1 ^2 + v_2 ^2} = \sqrt{v_1^2 + v_2^2} = \sqrt{\mathbf{v} \cdot \mathbf{v}}.$	$\ \mathbf{v}\ = \sqrt{\ (v_1, 0)\ ^2 + \ (0, v_2)\ ^2} = \sqrt{ v_1 ^2 + v_2 ^2} = \sqrt{v_1^2 + v_2^2} = \sqrt{\mathbf{v} \cdot \mathbf{v}}.$

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johnston-linear-matrix-algebra-EQ0018	29	■ 1.000	$\begin{aligned} & c \, \mathbf{v} \, \&= \sqrt{\left(c \, v_1 \right)^2 + \left(c \, v_2 \right)^2 + \cdots + \left(c \, v_n \right)^2} \\ &\&= \sqrt{c^2 \left(v_1^2 + v_2^2 + \cdots + v_n^2 \right)} \\ &\&= \sqrt{c^2} \sqrt{v_1^2 + v_2^2 + \cdots + v_n^2} = c \, \ \mathbf{v}\ . \end{aligned}$	$\begin{aligned} \ \mathbf{c}\mathbf{v}\ &= \sqrt{(cv_1)^2 + (cv_2)^2 + \cdots + (cv_n)^2} \\ &= \sqrt{c^2(v_1^2 + v_2^2 + \cdots + v_n^2)} \\ &= \sqrt{c^2} \sqrt{v_1^2 + v_2^2 + \cdots + v_n^2} = c \ \mathbf{v}\ . \end{aligned}$	$\begin{aligned} \ \mathbf{c}\mathbf{v}\ &= \sqrt{(cv_1)^2 + (cv_2)^2 + \cdots + (cv_n)^2} \\ &= \sqrt{c^2(v_1^2 + v_2^2 + \cdots + v_n^2)} \\ &= \sqrt{c^2} \sqrt{v_1^2 + v_2^2 + \cdots + v_n^2} = c \ \mathbf{v}\ . \end{aligned}$
johnston-linear-matrix-algebra-EQ0019	29	■ 1.000	$\left \frac{\mathbf{v}}{\ \mathbf{v}\ } \right = \frac{1}{\ \mathbf{v}\ } \ \mathbf{v}\ = 1.$	$\left\ \frac{\mathbf{v}}{\ \mathbf{v}\ } \right\ = \frac{1}{\ \mathbf{v}\ } \ \mathbf{v}\ = 1.$	$\left\ \frac{\mathbf{v}}{\ \mathbf{v}\ } \right\ = \frac{1}{\ \mathbf{v}\ } \ \mathbf{v}\ = 1.$
johnston-linear-matrix-algebra-EQ0020	30	■ 1.000	$\begin{aligned} &0 \leq c \, \mathbf{v} + d \, \mathbf{w} ^2 \\ &\&= (c \, \mathbf{v} + d \, \mathbf{w}) \cdot (c \, \mathbf{v} + d \, \mathbf{w}) \\ &\&= c^2 (\mathbf{v} \cdot \mathbf{v}) + 2cd (\mathbf{v} \cdot \mathbf{w}) + d^2 (\mathbf{w} \cdot \mathbf{w}) \\ &\&= c^2 \ \mathbf{v}\ ^2 + 2cd (\mathbf{v} \cdot \mathbf{w}) + d^2 \ \mathbf{w}\ ^2 \end{aligned}$	$\begin{aligned} 0 \leq \ \mathbf{c}\mathbf{v} + d\mathbf{w}\ ^2 &= (\mathbf{c}\mathbf{v} + d\mathbf{w}) \cdot (\mathbf{c}\mathbf{v} + d\mathbf{w}) \\ &= c^2(\mathbf{v} \cdot \mathbf{v}) + 2cd(\mathbf{v} \cdot \mathbf{w}) + d^2(\mathbf{w} \cdot \mathbf{w}) \\ &= c^2\ \mathbf{v}\ ^2 + 2cd(\mathbf{v} \cdot \mathbf{w}) + d^2\ \mathbf{w}\ ^2 \end{aligned}$	$\begin{aligned} 0 \leq \ \mathbf{c}\mathbf{v} + d\mathbf{w}\ ^2 &= (\mathbf{c}\mathbf{v} + d\mathbf{w}) \cdot (\mathbf{c}\mathbf{v} + d\mathbf{w}) \\ &= c^2(\mathbf{v} \cdot \mathbf{v}) + 2cd(\mathbf{v} \cdot \mathbf{w}) + d^2(\mathbf{w} \cdot \mathbf{w}) \\ &= c^2\ \mathbf{v}\ ^2 + 2cd(\mathbf{v} \cdot \mathbf{w}) + d^2\ \mathbf{w}\ ^2 \end{aligned}$
johnston-linear-matrix-algebra-EQ0027	32	■ 1.000	$\cos(\theta) = \frac{\mathbf{v} \cdot \mathbf{w}}{\ \mathbf{v}\ \ \mathbf{w}\ }, \quad \text{so} \quad \theta = \arccos \left(\frac{\mathbf{v} \cdot \mathbf{w}}{\ \mathbf{v}\ \ \mathbf{w}\ } \right).$	$\cos(\theta) = \frac{\mathbf{v} \cdot \mathbf{w}}{\ \mathbf{v}\ \ \mathbf{w}\ }, \quad \text{so} \quad \theta = \arccos \left(\frac{\mathbf{v} \cdot \mathbf{w}}{\ \mathbf{v}\ \ \mathbf{w}\ } \right).$	$\cos(\theta) = \frac{\mathbf{v} \cdot \mathbf{w}}{\ \mathbf{v}\ \ \mathbf{w}\ }, \quad \text{so} \quad \theta = \arccos \left(\frac{\mathbf{v} \cdot \mathbf{w}}{\ \mathbf{v}\ \ \mathbf{w}\ } \right).$
johnston-linear-matrix-algebra-EQ0028	32	■ 1.000	$\theta = \arccos \left(\frac{\mathbf{v} \cdot \mathbf{w}}{\ \mathbf{v}\ \ \mathbf{w}\ } \right).$	$\theta = \arccos \left(\frac{\mathbf{v} \cdot \mathbf{w}}{\ \mathbf{v}\ \ \mathbf{w}\ } \right).$	$\theta = \arccos \left(\frac{\mathbf{v} \cdot \mathbf{w}}{\ \mathbf{v}\ \ \mathbf{w}\ } \right).$
johnston-linear-matrix-algebra-EQ0029	33	■ 1.000	$\theta = \arccos \left(\frac{11}{5\sqrt{5}} \right) \approx 0.1799 \text{ radians (or } \approx 10.30 \text{ degrees).}$	$\theta = \arccos \left(\frac{11}{5\sqrt{5}} \right) \approx 0.1799 \text{ radians (or } \approx 10.30 \text{ degrees).}$	$\theta = \arccos \left(\frac{11}{5\sqrt{5}} \right) \approx 0.1799 \text{ radians (or } \approx 10.30 \text{ degrees).}$
johnston-linear-matrix-algebra-EQ0030	33	■ 1.000	$\theta = \arccos(0) = \pi/2 \text{ (i.e., } 90 \text{ degrees).}$	$\theta = \arccos(0) = \pi/2 \text{ (i.e., } 90 \text{ degrees).}$	$\theta = \arccos(0) = \pi/2 \text{ (i.e., } 90 \text{ degrees).}$
johnston-linear-matrix-algebra-EQ0031	33	■ 1.000	$\theta = \arccos \left(\frac{1}{\sqrt{2} \cdot \sqrt{2}} \right) = \arccos \left(\frac{1}{2} \right) = \pi/3 \text{ (i.e., } 60 \text{ degrees).}$	$\theta = \arccos \left(\frac{1}{\sqrt{2} \cdot \sqrt{2}} \right) = \arccos \left(\frac{1}{2} \right) = \pi/3 \text{ (i.e., } 60 \text{ degrees).}$	$\theta = \arccos \left(\frac{1}{\sqrt{2} \cdot \sqrt{2}} \right) = \arccos \left(\frac{1}{2} \right) = \pi/3 \text{ (i.e., } 60 \text{ degrees).}$
johnston-linear-matrix-algebra-EQ0032	35	■ 1.000	$\left(\frac{v_1 + v_2 + \cdots + v_n}{n} \right)^2 \leq \frac{1}{n} (v_1^2 + v_2^2 + \cdots + v_n^2).$	$\left(\frac{v_1 + v_2 + \cdots + v_n}{n} \right)^2 \leq \frac{1}{n} (v_1^2 + v_2^2 + \cdots + v_n^2).$	$\left(\frac{v_1 + v_2 + \cdots + v_n}{n} \right)^2 \leq \frac{1}{n} (v_1^2 + v_2^2 + \cdots + v_n^2).$
johnston-linear-matrix-algebra-EQ0033	35	■ 1.000	$\mathbf{v} \cdot \mathbf{w} \stackrel{\text{def}}{=} \overline{v_1} w_1 + \overline{v_2} w_2 + \cdots + \overline{v_n} w_n,$	$\mathbf{v} \cdot \mathbf{w} \stackrel{\text{def}}{=} \overline{v_1} w_1 + \overline{v_2} w_2 + \cdots + \overline{v_n} w_n,$	$\mathbf{v} \cdot \mathbf{w} \stackrel{\text{def}}{=} \overline{v_1} w_1 + \overline{v_2} w_2 + \cdots + \overline{v_n} w_n,$
johnston-linear-matrix-algebra-EQ0034	35	■ 1.000	$A = \begin{bmatrix} 1 & 3 \\ 2 & -1 \end{bmatrix} \quad \text{and} \quad B = \begin{bmatrix} 3 & 0 & 2 \\ 0 & -1 & 1 \end{bmatrix}.$	$A = \begin{bmatrix} 1 & 3 \\ 2 & -1 \end{bmatrix} \quad \text{and} \quad B = \begin{bmatrix} 3 & 0 & 2 \\ 0 & -1 & 1 \end{bmatrix}.$	$A = \begin{bmatrix} 1 & 3 \\ 2 & -1 \end{bmatrix} \quad \text{and} \quad B = \begin{bmatrix} 3 & 0 & 2 \\ 0 & -1 & 1 \end{bmatrix}.$

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johnston-linear-matrix-algebra-EQ0037	37	1.000	$\begin{aligned} A+B &= \left[\begin{array}{cccc} a_{1,1}+b_{1,1} & a_{1,2}+b_{1,2} & \cdots & a_{1,n}+b_{1,n} \\ a_{2,1}+b_{2,1} & a_{2,2}+b_{2,2} & \cdots & a_{2,n}+b_{2,n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{m,1}+b_{m,1} & a_{m,2}+b_{m,2} & \cdots & a_{m,n}+b_{m,n} \end{array} \right] \\ &= \left[\begin{array}{cccc} b_{1,1}+a_{1,1} & b_{1,2}+a_{1,2} & \cdots & b_{1,n}+a_{1,n} \\ b_{2,1}+a_{2,1} & b_{2,2}+a_{2,2} & \cdots & b_{2,n}+a_{2,n} \\ \vdots & \vdots & \ddots & \vdots \\ b_{m,1}+a_{m,1} & b_{m,2}+a_{m,2} & \cdots & b_{m,n}+a_{m,n} \end{array} \right] = B+A, \end{aligned}$	$A+B = \left[\begin{array}{cccc} a_{1,1}+b_{1,1} & a_{1,2}+b_{1,2} & \cdots & a_{1,n}+b_{1,n} \\ a_{2,1}+b_{2,1} & a_{2,2}+b_{2,2} & \cdots & a_{2,n}+b_{2,n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{m,1}+b_{m,1} & a_{m,2}+b_{m,2} & \cdots & a_{m,n}+b_{m,n} \end{array} \right] \\ = \left[\begin{array}{cccc} b_{1,1}+a_{1,1} & b_{1,2}+a_{1,2} & \cdots & b_{1,n}+a_{1,n} \\ b_{2,1}+a_{2,1} & b_{2,2}+a_{2,2} & \cdots & b_{2,n}+a_{2,n} \\ \vdots & \vdots & \ddots & \vdots \\ b_{m,1}+a_{m,1} & b_{m,2}+a_{m,2} & \cdots & b_{m,n}+a_{m,n} \end{array} \right] = B+A,$	$A+B = \left[\begin{array}{cccc} a_{1,1}+b_{1,1} & a_{1,2}+b_{1,2} & \cdots & a_{1,n}+b_{1,n} \\ a_{2,1}+b_{2,1} & a_{2,2}+b_{2,2} & \cdots & a_{2,n}+b_{2,n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{m,1}+b_{m,1} & a_{m,2}+b_{m,2} & \cdots & a_{m,n}+b_{m,n} \end{array} \right] \\ = \left[\begin{array}{cccc} b_{1,1}+a_{1,1} & b_{1,2}+a_{1,2} & \cdots & b_{1,n}+a_{1,n} \\ b_{2,1}+a_{2,1} & b_{2,2}+a_{2,2} & \cdots & b_{2,n}+a_{2,n} \\ \vdots & \vdots & \ddots & \vdots \\ b_{m,1}+a_{m,1} & b_{m,2}+a_{m,2} & \cdots & b_{m,n}+a_{m,n} \end{array} \right] = B+A,$
johnston-linear-matrix-algebra-EQ0038	38	1.000	$[A \ B]_{i,j} \stackrel{\text{def}}{=} a_{i,1}b_{1,j} + a_{i,2}b_{2,j} + \cdots + a_{i,n}b_{n,j}.$	$[AB]_{i,j} \stackrel{\text{def}}{=} a_{i,1}b_{1,j} + a_{i,2}b_{2,j} + \cdots + a_{i,n}b_{n,j}.$	$[AB]_{i,j} \stackrel{\text{def}}{=} a_{i,1}b_{1,j} + a_{i,2}b_{2,j} + \cdots + a_{i,n}b_{n,j}.$
johnston-linear-matrix-algebra-EQ0039	38	1.000	$A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}, \quad B = \begin{bmatrix} 5 & 6 & 7 \\ 8 & 9 & 10 \end{bmatrix}, \quad \text{and} \quad C = \begin{bmatrix} 1 & 0 \\ 0 & -1 \\ 2 & -1 \end{bmatrix}.$	$A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}, \quad B = \begin{bmatrix} 5 & 6 & 7 \\ 8 & 9 & 10 \end{bmatrix}, \quad \text{and} \quad C = \begin{bmatrix} 1 & 0 \\ 0 & -1 \\ 2 & -1 \end{bmatrix}.$	$A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}, \quad B = \begin{bmatrix} 5 & 6 & 7 \\ 8 & 9 & 10 \end{bmatrix}, \quad \text{and} \quad C = \begin{bmatrix} 1 & 0 \\ 0 & -1 \\ 2 & -1 \end{bmatrix}.$
johnston-linear-matrix-algebra-EQ0042	40	1.000	$a_{i,1}(b_{1,j}+c_{1,j}) + a_{i,2}(b_{2,j}+c_{2,j}) + \cdots + a_{i,n}(b_{n,j}+c_{n,j}),$	$a_{i,1}(b_{1,j}+c_{1,j}) + a_{i,2}(b_{2,j}+c_{2,j}) + \cdots + a_{i,n}(b_{n,j}+c_{n,j}),$	$a_{i,1}(b_{1,j}+c_{1,j}) + a_{i,2}(b_{2,j}+c_{2,j}) + \cdots + a_{i,n}(b_{n,j}+c_{n,j}),$
johnston-linear-matrix-algebra-EQ0043	40	1.000	$(a_{i,1}b_{1,j} + a_{i,2}b_{2,j} + \cdots + a_{i,n}b_{n,j}) + (a_{i,1}c_{1,j} + a_{i,2}c_{2,j} + \cdots + a_{i,n}c_{n,j}).$	$(a_{i,1}b_{1,j} + a_{i,2}b_{2,j} + \cdots + a_{i,n}b_{n,j}) + (a_{i,1}c_{1,j} + a_{i,2}c_{2,j} + \cdots + a_{i,n}c_{n,j}).$	$(a_{i,1}b_{1,j} + a_{i,2}b_{2,j} + \cdots + a_{i,n}b_{n,j}) + (a_{i,1}c_{1,j} + a_{i,2}c_{2,j} + \cdots + a_{i,n}c_{n,j}).$
johnston-linear-matrix-algebra-EQ0045	40	1.000	$AB = \begin{bmatrix} 2 & 1 \\ 1 & 1 \end{bmatrix} \quad \text{and} \quad BA = \begin{bmatrix} 1 & 1 \\ 1 & 2 \end{bmatrix}.$	$AB = \begin{bmatrix} 2 & 1 \\ 1 & 1 \end{bmatrix} \quad \text{and} \quad BA = \begin{bmatrix} 1 & 1 \\ 1 & 2 \end{bmatrix}.$	$AB = \begin{bmatrix} 2 & 1 \\ 1 & 1 \end{bmatrix} \quad \text{and} \quad BA = \begin{bmatrix} 1 & 1 \\ 1 & 2 \end{bmatrix}.$
johnston-linear-matrix-algebra-EQ0048	41	1.000	$A^2 = \begin{bmatrix} 2 & 1 \\ -1 & 3 \end{bmatrix} \begin{bmatrix} 2 & 1 \\ -1 & 3 \end{bmatrix} = \begin{bmatrix} 3 & 5 \\ -5 & 8 \end{bmatrix}.$	$A^2 = \begin{bmatrix} 2 & 1 \\ -1 & 3 \end{bmatrix} \begin{bmatrix} 2 & 1 \\ -1 & 3 \end{bmatrix} = \begin{bmatrix} 3 & 5 \\ -5 & 8 \end{bmatrix}.$	$A^2 = \begin{bmatrix} 2 & 1 \\ -1 & 3 \end{bmatrix} \begin{bmatrix} 2 & 1 \\ -1 & 3 \end{bmatrix} = \begin{bmatrix} 3 & 5 \\ -5 & 8 \end{bmatrix}.$
johnston-linear-matrix-algebra-EQ0050	42	1.000	$(A+B)^2 = (A+B)(A+B) = A(A+B) + B(A+B) = AA + AB + BA + BB = A^2 + AB + BA + B^2,$	$(A+B)^2 = (A+B)(A+B) = A(A+B) + B(A+B) = AA + AB + BA + BB = A^2 + AB + BA + B^2,$	$(A+B)^2 = (A+B)(A+B) = A(A+B) + B(A+B) = AA + AB + BA + BB = A^2 + AB + BA + B^2,$

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johnston-linear-matrix-algebra-EQ0051	42	1.000	$\begin{aligned} (A+I)^2(A-I) &= \left(A^2+AI+IA+I\right)(A-I) \\ &= \left(A^2+2A+I\right)(A-I) \\ &= \left(A^3+2A^2+A\right)-\left(A^2+2A+I\right) \\ &= A^3+A^2-A-I . \end{aligned}$	$(A+I)^2(A-I) = \left(A^2+AI+IA+I\right)(A-I)$ $= \left(A^2+2A+I\right)(A-I)$ $= \left(A^3+2A^2+A\right)-\left(A^2+2A+I\right)$ $= A^3+A^2-A-I.$	$(A+I)^2(A-I) = \left(A^2+AI+IA+I\right)(A-I)$ $= \left(A^2+2A+I\right)(A-I)$ $= \left(A^3+2A^2+A\right)-\left(A^2+2A+I\right)$ $= A^3+A^2-A-I.$
johnston-linear-matrix-algebra-EQ0053	43	1.000	$\mathbf{w}=\left[\begin{array}{l} 1 \\ 2 \\ 3 \end{array}\right] \in \mathcal{M}_{3,1}$	$\mathbf{w} = \left[\begin{array}{l} 1 \\ 2 \\ 3 \end{array}\right] \in \mathcal{M}_{3,1}$	$\mathbf{w} = \left[\begin{array}{l} 1 \\ 2 \\ 3 \end{array}\right] \in \mathcal{M}_{3,1}$
johnston-linear-matrix-algebra-EQ0055	43	1.000	$A=\left[\begin{array}{ll} 1 & 2 \\ 3 & 4 \end{array}\right] \quad \text { and } \quad B=\left[\begin{array}{ccc} -1 & 1 & 1 \\ 0 & 1 & 0 \end{array}\right] .$	$A = \left[\begin{array}{cc} 1 & 2 \\ 3 & 4 \end{array}\right] \quad \text { and } \quad B = \left[\begin{array}{ccc} -1 & 1 & 1 \\ 0 & 1 & 0 \end{array}\right] .$	$A = \left[\begin{array}{cc} 1 & 2 \\ 3 & 4 \end{array}\right] \quad \text { and } \quad B = \left[\begin{array}{ccc} -1 & 1 & 1 \\ 0 & 1 & 0 \end{array}\right] .$
johnston-linear-matrix-algebra-EQ0056	44	1.000	$\left[(A B)^T\right]_{i, j}=[A B]_{j, i}=a_{j, 1} b_{1, i}+a_{j, 2} b_{2, i}+\cdots+a_{j, n} b_{n, i},$	$\left[(A B)^T\right]_{i, j}=[A B]_{j, i}=a_{j, 1} b_{1, i}+a_{j, 2} b_{2, i}+\cdots+a_{j, n} b_{n, i},$	$\left[(A B)^T\right]_{i, j}=[A B]_{j, i}=a_{j, 1} b_{1, i}+a_{j, 2} b_{2, i}+\cdots+a_{j, n} b_{n, i},$
johnston-linear-matrix-algebra-EQ0059	45	1.000	$A=\left[\begin{array}{cccccc} 1 & 0 & 0 & 1 & 0 & 0 \\ 0 & 1 & 0 & 0 & 1 & 0 \\ 0 & 0 & 1 & 0 & 0 & 1 \\ 0 & 0 & 0 & 2 & 1 & -1 \\ 0 & 0 & 0 & 0 & -2 & 3 \end{array}\right] \quad \text { and } \quad B=\left[\begin{array}{cccc} 1 & 2 & 0 & 0 \\ 2 & 1 & 0 & 0 \\ -1 & 1 & 0 & 0 \\ 0 & 0 & 1 & 2 \\ 0 & 0 & 2 & 1 \\ 0 & 0 & -1 & 1 \end{array}\right]$	$A = \left[\begin{array}{cccccc} 1 & 0 & 0 & 1 & 0 & 0 \\ 0 & 1 & 0 & 0 & 1 & 0 \\ 0 & 0 & 1 & 0 & 0 & 1 \\ 0 & 0 & 0 & 2 & 1 & -1 \\ 0 & 0 & 0 & 0 & -2 & 3 \end{array}\right] \quad \text { and } \quad B = \left[\begin{array}{cccc} 1 & 2 & 0 & 0 \\ 2 & 1 & 0 & 0 \\ -1 & 1 & 0 & 0 \\ 0 & 0 & 1 & 2 \\ 0 & 0 & 2 & 1 \\ 0 & 0 & -1 & 1 \end{array}\right]$	$A = \left[\begin{array}{cccccc} 1 & 0 & 0 & 1 & 0 & 0 \\ 0 & 1 & 0 & 0 & 1 & 0 \\ 0 & 0 & 1 & 0 & 0 & 1 \\ 0 & 0 & 0 & 2 & 1 & -1 \\ 0 & 0 & 0 & 0 & -2 & 3 \end{array}\right] \quad \text { and } \quad B = \left[\begin{array}{cccc} 1 & 2 & 0 & 0 \\ 2 & 1 & 0 & 0 \\ -1 & 1 & 0 & 0 \\ 0 & 0 & 1 & 2 \\ 0 & 0 & 2 & 1 \\ 0 & 0 & -1 & 1 \end{array}\right]$
johnston-linear-matrix-algebra-EQ0060	45	1.000	$A=\left[\begin{array}{ccc ccc} 1 & 0 & 0 & 1 & 0 & 0 \\ 0 & 1 & 0 & 0 & 1 & 0 \\ 0 & 0 & 1 & 0 & 0 & 1 \\ \hline 0 & 0 & 0 & 2 & 1 & -1 \\ 0 & 0 & 0 & 0 & -2 & 3 \end{array}\right] \quad \text { and } \quad B=\left[\begin{array}{cc cc} 1 & 2 & 0 & 0 \\ 2 & 1 & 0 & 0 \\ \hline 0 & 0 & 1 & 2 \\ 0 & 0 & 2 & 1 \\ 0 & 0 & -1 & 1 \end{array}\right] .$	$A = \left[\begin{array}{ccc ccc} 1 & 0 & 0 & 1 & 0 & 0 \\ 0 & 1 & 0 & 0 & 1 & 0 \\ 0 & 0 & 1 & 0 & 0 & 1 \\ \hline 0 & 0 & 0 & 2 & 1 & -1 \\ 0 & 0 & 0 & 0 & -2 & 3 \end{array}\right] \quad \text { and } \quad B = \left[\begin{array}{cc cc} 1 & 2 & 0 & 0 \\ 2 & 1 & 0 & 0 \\ \hline 0 & 0 & 1 & 2 \\ 0 & 0 & 2 & 1 \\ 0 & 0 & -1 & 1 \end{array}\right] .$	$A = \left[\begin{array}{ccc ccc} 1 & 0 & 0 & 1 & 0 & 0 \\ 0 & 1 & 0 & 0 & 1 & 0 \\ 0 & 0 & 1 & 0 & 0 & 1 \\ \hline 0 & 0 & 0 & 2 & 1 & -1 \\ 0 & 0 & 0 & 0 & -2 & 3 \end{array}\right] \quad \text { and } \quad B = \left[\begin{array}{cc cc} 1 & 2 & 0 & 0 \\ 2 & 1 & 0 & 0 \\ \hline 0 & 0 & 1 & 2 \\ 0 & 0 & 2 & 1 \\ 0 & 0 & -1 & 1 \end{array}\right] .$
johnston-linear-matrix-algebra-EQ0061	45	1.000	$C=\left[\begin{array}{ccc} 2 & 1 & -1 \\ 0 & -2 & 3 \end{array}\right] \quad \text { and } \quad D=\left[\begin{array}{cc} 1 & 2 \\ 2 & 1 \\ -1 & 1 \end{array}\right],$	$C = \left[\begin{array}{ccc} 2 & 1 & -1 \\ 0 & -2 & 3 \end{array}\right] \quad \text { and } \quad D = \left[\begin{array}{cc} 1 & 2 \\ 2 & 1 \\ -1 & 1 \end{array}\right],$	$C = \left[\begin{array}{ccc} 2 & 1 & -1 \\ 0 & -2 & 3 \end{array}\right] \quad \text { and } \quad D = \left[\begin{array}{cc} 1 & 2 \\ 2 & 1 \\ -1 & 1 \end{array}\right],$
johnston-linear-matrix-algebra-EQ0064	46	1.000	$A B=\left[\begin{array}{cc} D & D \\ O & C D \end{array}\right]=\left[\begin{array}{cc cc} 1 & 2 & 1 & 2 \\ 2 & 1 & 2 & 1 \\ \hline 0 & 0 & 5 & 4 \\ 0 & 0 & -7 & 1 \end{array}\right],$	$A B = \left[\begin{array}{cc} D & D \\ O & C D \end{array}\right] = \left[\begin{array}{cc cc} 1 & 2 & 1 & 2 \\ 2 & 1 & 2 & 1 \\ \hline 0 & 0 & 5 & 4 \\ 0 & 0 & -7 & 1 \end{array}\right],$	$A B = \left[\begin{array}{cc} D & D \\ O & C D \end{array}\right] = \left[\begin{array}{cc cc} 1 & 2 & 1 & 2 \\ 2 & 1 & 2 & 1 \\ \hline 0 & 0 & 5 & 4 \\ 0 & 0 & -7 & 1 \end{array}\right],$

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johnston-linear-matrix-algebra-EQ0094	55	1.000	$[T]=\left[\begin{array}{ccc} 3 & -1 & 1 \\ 2 & 4 & -2 \end{array}\right] .$	$[T] = \begin{bmatrix} 3 & -1 & 1 \\ 2 & 4 & -2 \end{bmatrix} .$	$[T] = \begin{bmatrix} 3 & -1 & 1 \\ 2 & 4 & -2 \end{bmatrix} .$
johnston-linear-matrix-algebra-EQ0095	56	1.000	$(S+T)(\mathbf{v})=S(\mathbf{v})+T(\mathbf{v}) \quad \text{and} \quad (cT)(\mathbf{v})=cT(\mathbf{v}) \quad \text{for all } \mathbf{v} \in \mathbb{R}^n,$	$(S+T)(\mathbf{v}) = S(\mathbf{v}) + T(\mathbf{v}) \quad \text{and} \quad (cT)(\mathbf{v}) = cT(\mathbf{v}) \quad \text{for all } \mathbf{v} \in \mathbb{R}^n,$	$(S+T)(\mathbf{v}) = S(\mathbf{v}) + T(\mathbf{v}) \quad \text{and} \quad (cT)(\mathbf{v}) = cT(\mathbf{v}) \quad \text{for all } \mathbf{v} \in \mathbb{R}^n,$
johnston-linear-matrix-algebra-EQ0097	57	1.000	$\left[\begin{array}{cccc} c_1 & 0 & \cdots & 0 \\ 0 & c_2 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & c_n \end{array}\right] \left[\begin{array}{cccc} d_1 & 0 & \cdots & 0 \\ 0 & d_2 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & d_n \end{array}\right] = \left[\begin{array}{cccc} c_1 d_1 & 0 & \cdots & 0 \\ 0 & c_2 d_2 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & c_n d_n \end{array}\right] .$	$\begin{bmatrix} c_1 & 0 & \cdots & 0 \\ 0 & c_2 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & c_n \end{bmatrix} \begin{bmatrix} d_1 & 0 & \cdots & 0 \\ 0 & d_2 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & d_n \end{bmatrix} = \begin{bmatrix} c_1 d_1 & 0 & \cdots & 0 \\ 0 & c_2 d_2 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & c_n d_n \end{bmatrix} .$	$\begin{bmatrix} c_1 & 0 & \cdots & 0 \\ 0 & c_2 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & c_n \end{bmatrix} \begin{bmatrix} d_1 & 0 & \cdots & 0 \\ 0 & d_2 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & d_n \end{bmatrix} = \begin{bmatrix} c_1 d_1 & 0 & \cdots & 0 \\ 0 & c_2 d_2 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & c_n d_n \end{bmatrix} .$
johnston-linear-matrix-algebra-EQ0099	58	1.000	$P_{\mathbf{u}}(\mathbf{v})=\mathbf{u}(\ \mathbf{v}\ (\mathbf{u} \cdot \mathbf{v})/\ \mathbf{v}\)=\mathbf{u}(\mathbf{u} \cdot \mathbf{v}).$	$P_{\mathbf{u}}(\mathbf{v}) = \mathbf{u}(\ \mathbf{v}\ (\mathbf{u} \cdot \mathbf{v})/\ \mathbf{v}\) = \mathbf{u}(\mathbf{u} \cdot \mathbf{v}).$	$P_{\mathbf{u}}(\mathbf{v}) = \mathbf{u}(\ \mathbf{v}\ (\mathbf{u} \cdot \mathbf{v})/\ \mathbf{v}\) = \mathbf{u}(\mathbf{u} \cdot \mathbf{v}).$
johnston-linear-matrix-algebra-EQ0109	61	1.000	$\left[F_{\mathbf{u}}\right] \mathbf{v}=\left[\begin{array}{cc} -1 & 0 \\ 0 & 1 \end{array}\right] \left[\begin{array}{c} v_1 \\ v_2 \end{array}\right]=\left[\begin{array}{c} -v_1 \\ v_2 \end{array}\right] .$	$[F_{\mathbf{u}}] \mathbf{v} = \begin{bmatrix} -1 & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} v_1 \\ v_2 \end{bmatrix} = \begin{bmatrix} -v_1 \\ v_2 \end{bmatrix} .$	$[F_{\mathbf{u}}] \mathbf{v} = \begin{bmatrix} -1 & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} v_1 \\ v_2 \end{bmatrix} = \begin{bmatrix} -v_1 \\ v_2 \end{bmatrix} .$
johnston-linear-matrix-algebra-EQ0115	63	1.000	$\left[R^{\theta}\right]=\left[R^{\theta}\left(\mathbf{e}_1\right) \mid R^{\theta}\left(\mathbf{e}_2\right)\right]=\left[\begin{array}{cc} \cos (\theta) & -\sin (\theta) \\ \sin (\theta) & \cos (\theta) \end{array}\right] .$	$\left[R^{\theta}\right] = \left[R^{\theta}(\mathbf{e}_1) \mid R^{\theta}(\mathbf{e}_2)\right] = \begin{bmatrix} \cos(\theta) & -\sin(\theta) \\ \sin(\theta) & \cos(\theta) \end{bmatrix} .$	$\left[R^{\theta}\right] = \left[R^{\theta}(\mathbf{e}_1) \mid R^{\theta}(\mathbf{e}_2)\right] = \begin{bmatrix} \cos(\theta) & -\sin(\theta) \\ \sin(\theta) & \cos(\theta) \end{bmatrix} .$
johnston-linear-matrix-algebra-EQ0116	63	1.000	$\left[R^{\pi / 4}\right]=\left[\begin{array}{cc} \cos (\pi / 4) & -\sin (\pi / 4) \\ \sin (\pi / 4) & \cos (\pi / 4) \end{array}\right]=\frac{1}{\sqrt{2}}\left[\begin{array}{cc} 1 & -1 \\ 1 & 1 \end{array}\right] .$	$\left[R^{\pi/4}\right] = \begin{bmatrix} \cos(\pi/4) & -\sin(\pi/4) \\ \sin(\pi/4) & \cos(\pi/4) \end{bmatrix} = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & -1 \\ 1 & 1 \end{bmatrix} .$	$\left[R^{\pi/4}\right] = \begin{bmatrix} \cos(\pi/4) & -\sin(\pi/4) \\ \sin(\pi/4) & \cos(\pi/4) \end{bmatrix} = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & -1 \\ 1 & 1 \end{bmatrix} .$
johnston-linear-matrix-algebra-EQ0117 (3)	63	1.000	$\left[R^{-\pi / 6}\right]=\left[\begin{array}{cc} \cos (-\pi / 6) & -\sin (-\pi / 6) \\ \sin (-\pi / 6) & \cos (-\pi / 6) \end{array}\right]=\frac{1}{2}\left[\begin{array}{cc} \sqrt{3} & 1 \\ -1 & \sqrt{3} \end{array}\right] .$	$\left[R^{-\pi/6}\right] = \begin{bmatrix} \cos(-\pi/6) & -\sin(-\pi/6) \\ \sin(-\pi/6) & \cos(-\pi/6) \end{bmatrix} = \frac{1}{2} \begin{bmatrix} \sqrt{3} & 1 \\ -1 & \sqrt{3} \end{bmatrix} .$	$\left[R^{-\pi/6}\right] = \begin{bmatrix} \cos(-\pi/6) & -\sin(-\pi/6) \\ \sin(-\pi/6) & \cos(-\pi/6) \end{bmatrix} = \frac{1}{2} \begin{bmatrix} \sqrt{3} & 1 \\ -1 & \sqrt{3} \end{bmatrix} .$
johnston-linear-matrix-algebra-EQ0119 (3)	64	1.000	$\left[R^{-\pi / 6}\right] \mathbf{w}=\frac{1}{2}\left[\begin{array}{cc} \sqrt{3} & 1 \\ -1 & \sqrt{3} \end{array}\right]\left[\begin{array}{c} \sqrt{3} \\ 3 \end{array}\right]=\left[\begin{array}{c} 3 \\ \sqrt{3} \end{array}\right] .$	$\left[R^{-\pi/6}\right] \mathbf{w} = \frac{1}{2} \begin{bmatrix} \sqrt{3} & 1 \\ -1 & \sqrt{3} \end{bmatrix} \begin{bmatrix} \sqrt{3} \\ 3 \end{bmatrix} = \begin{bmatrix} 3 \\ \sqrt{3} \end{bmatrix} .$	$\left[R^{-\pi/6}\right] \mathbf{w} = \frac{1}{2} \begin{bmatrix} \sqrt{3} & 1 \\ -1 & \sqrt{3} \end{bmatrix} \begin{bmatrix} \sqrt{3} \\ 3 \end{bmatrix} = \begin{bmatrix} 3 \\ \sqrt{3} \end{bmatrix} .$

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johnston-linear-matrix-algebra-EQ0120	65	1.000	$\left[R_{yz}^{\theta}\right]=\left[R_{yz}^{\theta}(\mathbf{e}_1)\left R_{yz}^{\theta}(\mathbf{e}_2)\right R_{yz}^{\theta}(\mathbf{e}_3)\right]=\left[\begin{array}{ccc}1&0&0\\0&\cos(\theta)&-\sin(\theta)\\0&\sin(\theta)&\cos(\theta)\end{array}\right].$		
johnston-linear-matrix-algebra-EQ0122 (0)	65	1.000	$\left[R_{xy}^{2\pi/3}\right]=\left[\begin{array}{ccc}\cos(2\pi/3)&-\sin(2\pi/3)&0\\\sin(2\pi/3)&\cos(2\pi/3)&0\\0&0&1\end{array}\right]=\frac{1}{2}\left[\begin{array}{ccc}-1&-\sqrt{3}&0\\\sqrt{3}&-1&0\\0&0&2\end{array}\right].$		
johnston-linear-matrix-algebra-EQ0123 (2.000)	66	1.000	$\left[R_{xy}^{2\pi/3}\right]\mathbf{v}=\frac{1}{2}\left[\begin{array}{ccc}-1&-\sqrt{3}&0\\\sqrt{3}&-1&0\\0&0&2\end{array}\right]\left[\begin{array}{c}3\\-1\\2\end{array}\right]=\frac{1}{2}\left[\begin{array}{c}\sqrt{3}-3\\3\sqrt{3}+1\\4\end{array}\right]\approx\left[\begin{array}{c}-0.6340\\3.0981\\2.000\end{array}\right].$		
johnston-linear-matrix-algebra-EQ0124	66	1.000	$(S\circ T)(\mathbf{v})=S(T(\mathbf{v}))\quad\text{for all}\quad\mathbf{v}\in\mathbb{R}^n.$		
johnston-linear-matrix-algebra-EQ0125	67	1.000	$[S\circ T]=[S][T].$		
johnston-linear-matrix-algebra-EQ0126	67	1.000	$(S\circ T)(\mathbf{v})=S(T(\mathbf{v}))=S([T]\mathbf{v})=[S]([T]\mathbf{v})=([S][T])\mathbf{v}.$		
johnston-linear-matrix-algebra-EQ0128	67	1.000	$[D]=\left[\begin{array}{cc}2&0\\0&3\end{array}\right].$		
johnston-linear-matrix-algebra-EQ0130	68	1.000	$\sin(\theta+\phi)=\sin(\theta)\cos(\phi)+\cos(\theta)\sin(\phi)$		
johnston-linear-matrix-algebra-EQ0133	69	1.000	$[R^{\theta+\phi}]=\left[\begin{array}{cc}\cos(\theta+\phi)&-\sin(\theta+\phi)\\\sin(\theta+\phi)&\cos(\theta+\phi)\end{array}\right].$		
johnston-linear-matrix-algebra-EQ0134	69	1.000	$\sin(\theta+\phi)=\sin(\theta)\cos(\phi)+\cos(\theta)\sin(\phi),$		
johnston-linear-matrix-algebra-EQ0135	69	1.000	$\cos(\theta+\phi)=\cos(\theta)\cos(\phi)-\sin(\theta)\sin(\phi).$		
johnston-linear-matrix-algebra-EQ0136	69	1.000	$R=\left[\begin{array}{cc}\cos(1)&-\sin(1)\\\sin(1)&\cos(1)\end{array}\right]\approx\left[\begin{array}{cc}0.5403&-0.8415\\0.8415&0.5403\end{array}\right]$		

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johnston-linear-matrix-algebra-EQ0137	70	■ 1.000	$T(\textbf{c}_1 \textbf{v}_1 + \cdots + \textbf{c}_k \textbf{v}_k) = \textbf{c}_1 T(\textbf{v}_1) + \cdots + \textbf{c}_k T(\textbf{v}_k)$	$T(c_1 \mathbf{v}_1 + \cdots + c_k \mathbf{v}_k) = c_1 T(\mathbf{v}_1) + \cdots + c_k T(\mathbf{v}_k)$	$T(c_1 \mathbf{v}_1 + \cdots + c_k \mathbf{v}_k) = c_1 T(\mathbf{v}_1) + \cdots + c_k T(\mathbf{v}_k)$
johnston-linear-matrix-algebra-EQ0138	71	■ 1.000	$A = \begin{bmatrix} \cos(\pi/4) & -\sin(\pi/4) \\ \sin(\pi/4) & \cos(\pi/4) \end{bmatrix}$	$A = \begin{bmatrix} \cos(\pi/4) & -\sin(\pi/4) \\ \sin(\pi/4) & \cos(\pi/4) \end{bmatrix}$	$A = \begin{bmatrix} \cos(\pi/4) & -\sin(\pi/4) \\ \sin(\pi/4) & \cos(\pi/4) \end{bmatrix}$
johnston-linear-matrix-algebra-EQ0139	71	■ 1.000	$A = \begin{bmatrix} \cos(\theta) & -\sin(\theta) \\ \sin(\theta) & \cos(\theta) \end{bmatrix}$	$A = \begin{bmatrix} \cos(\theta) & -\sin(\theta) \\ \sin(\theta) & \cos(\theta) \end{bmatrix}$	$A = \begin{bmatrix} \cos(\theta) & -\sin(\theta) \\ \sin(\theta) & \cos(\theta) \end{bmatrix}$
johnston-linear-matrix-algebra-EQ0141	71	■ 1.000	$\begin{bmatrix} \cos(2\theta) & \sin(2\theta) \\ \sin(2\theta) & -\cos(2\theta) \end{bmatrix}$	$\begin{bmatrix} \cos(2\theta) & \sin(2\theta) \\ \sin(2\theta) & -\cos(2\theta) \end{bmatrix}$	$\begin{bmatrix} \cos(2\theta) & \sin(2\theta) \\ \sin(2\theta) & -\cos(2\theta) \end{bmatrix}$
johnston-linear-matrix-algebra-EQ0142	71	■ 1.000	$\begin{bmatrix} 1 & c \\ 0 & 1 \end{bmatrix} \text{ or } \begin{bmatrix} 1 & 0 \\ c & 1 \end{bmatrix}$	$\begin{bmatrix} 1 & c \\ 0 & 1 \end{bmatrix} \text{ or } \begin{bmatrix} 1 & 0 \\ c & 1 \end{bmatrix}$	$\begin{bmatrix} 1 & c \\ 0 & 1 \end{bmatrix} \text{ or } \begin{bmatrix} 1 & 0 \\ c & 1 \end{bmatrix}$
johnston-linear-matrix-algebra-EQ0144	72	■ 1.000	$[T] = [T(\mathbf{e}_1) T(\mathbf{e}_2) \cdots T(\mathbf{e}_n)]$	$[T] = [T(\mathbf{e}_1) T(\mathbf{e}_2) \cdots T(\mathbf{e}_n)]$	$[T] = [T(\mathbf{e}_1) T(\mathbf{e}_2) \cdots T(\mathbf{e}_n)]$
johnston-linear-matrix-algebra-EQ0146	73	■ 1.000	$A = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & -1 \\ 1 & 1 \end{bmatrix}$	$A = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & -1 \\ 1 & 1 \end{bmatrix}$	$A = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & -1 \\ 1 & 1 \end{bmatrix}$
johnston-linear-matrix-algebra-EQ0147	73	■ 1.000	$A = \begin{bmatrix} -1 & 2 \\ 0 & 3 \end{bmatrix} \text{ and } \mathbf{v} = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$	$A = \begin{bmatrix} -1 & 2 \\ 0 & 3 \end{bmatrix} \text{ and } \mathbf{v} = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$	$A = \begin{bmatrix} -1 & 2 \\ 0 & 3 \end{bmatrix} \text{ and } \mathbf{v} = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$
johnston-linear-matrix-algebra-EQ0148	73	■ 1.000	$\begin{bmatrix} \cos(\theta) & -\sin(\theta) \\ \sin(\theta) & \cos(\theta) \end{bmatrix}^n = \begin{bmatrix} \cos(n\theta) & -\sin(n\theta) \\ \sin(n\theta) & \cos(n\theta) \end{bmatrix}$	$\begin{bmatrix} \cos(\theta) & -\sin(\theta) \\ \sin(\theta) & \cos(\theta) \end{bmatrix}^n = \begin{bmatrix} \cos(n\theta) & -\sin(n\theta) \\ \sin(n\theta) & \cos(n\theta) \end{bmatrix}$	$\begin{bmatrix} \cos(\theta) & -\sin(\theta) \\ \sin(\theta) & \cos(\theta) \end{bmatrix}^n = \begin{bmatrix} \cos(n\theta) & -\sin(n\theta) \\ \sin(n\theta) & \cos(n\theta) \end{bmatrix}$
johnston-linear-matrix-algebra-EQ0149	73	■ 1.000	$\mathbf{v} \times \mathbf{w} \stackrel{\text{def}}{=} \begin{pmatrix} v_2 w_3 - v_3 w_2 \\ v_3 w_1 - v_1 w_3 \\ v_1 w_2 - v_2 w_1 \end{pmatrix}$	$\mathbf{v} \times \mathbf{w} \stackrel{\text{def}}{=} \begin{pmatrix} v_2 w_3 - v_3 w_2 \\ v_3 w_1 - v_1 w_3 \\ v_1 w_2 - v_2 w_1 \end{pmatrix}$	$\mathbf{v} \times \mathbf{w} \stackrel{\text{def}}{=} \begin{pmatrix} v_2 w_3 - v_3 w_2 \\ v_3 w_1 - v_1 w_3 \\ v_1 w_2 - v_2 w_1 \end{pmatrix}$
johnston-linear-matrix-algebra-EQ0151	75	■ 1.000	$\begin{aligned} & \left(\begin{bmatrix} v_1 \\ v_2 \\ v_3 \end{bmatrix} \times \begin{bmatrix} w_1 \\ w_2 \\ w_3 \end{bmatrix} \right) = \begin{bmatrix} v_2 w_3 - v_3 w_2 \\ v_3 w_1 - v_1 w_3 \\ v_1 w_2 - v_2 w_1 \end{bmatrix} \\ & \left(\begin{bmatrix} w_1 \\ w_2 \\ w_3 \end{bmatrix} \times \begin{bmatrix} v_1 \\ v_2 \\ v_3 \end{bmatrix} \right) = \begin{bmatrix} w_2 v_3 - w_3 v_2 \\ w_3 v_1 - w_1 v_3 \\ w_1 v_2 - w_2 v_1 \end{bmatrix} \end{aligned}$	$\begin{aligned} & (v_1, v_2, v_3) \times (w_1, w_2, w_3) = (v_2 w_3 - v_3 w_2, v_3 w_1 - v_1 w_3, v_1 w_2 - v_2 w_1) \\ & (w_1, w_2, w_3) \times (v_1, v_2, v_3) = (w_2 v_3 - w_3 v_2, w_3 v_1 - w_1 v_3, w_1 v_2 - w_2 v_1) \end{aligned}$	$\begin{aligned} & (v_1, v_2, v_3) \times (w_1, w_2, w_3) = (v_2 w_3 - v_3 w_2, v_3 w_1 - v_1 w_3, v_1 w_2 - v_2 w_1) \\ & (w_1, w_2, w_3) \times (v_1, v_2, v_3) = (w_2 v_3 - w_3 v_2, w_3 v_1 - w_1 v_3, w_1 v_2 - w_2 v_1) \end{aligned}$
johnston-linear-matrix-algebra-EQ0152	75	■ 1.000	$\left(\begin{bmatrix} v_1 \\ v_2 \\ v_3 \end{bmatrix} \times \begin{bmatrix} v_1 \\ v_2 \\ v_3 \end{bmatrix} \right) = \begin{bmatrix} v_2 v_3 - v_3 v_2 \\ v_3 v_1 - v_1 v_3 \\ v_1 v_2 - v_2 v_1 \end{bmatrix} = (0, 0, 0)$	$(v_1, v_2, v_3) \times (v_1, v_2, v_3) = (v_2 v_3 - v_3 v_2, v_3 v_1 - v_1 v_3, v_1 v_2 - v_2 v_1) = (0, 0, 0)$	$(v_1, v_2, v_3) \times (v_1, v_2, v_3) = (v_2 v_3 - v_3 v_2, v_3 v_1 - v_1 v_3, v_1 v_2 - v_2 v_1) = (0, 0, 0)$

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johnston-linear-matrix-algebra-EQ0167	83	■ 1.000	$A=\left[\begin{array}{llll} 0 & 1 & 0 & 1 \\ 1 & 0 & 0 & 1 \\ 0 & 0 & 0 & 1 \\ 1 & 1 & 1 & 0 \end{array}\right]$.	$A = \begin{bmatrix} 0 & 1 & 0 & 1 \\ 1 & 0 & 0 & 1 \\ 0 & 0 & 0 & 1 \\ 1 & 1 & 1 & 0 \end{bmatrix}$.	$A = \begin{bmatrix} 0 & 1 & 0 & 1 \\ 1 & 0 & 0 & 1 \\ 0 & 0 & 0 & 1 \\ 1 & 1 & 1 & 0 \end{bmatrix}$.
johnston-linear-matrix-algebra-EQ0170	85	■ 1.000	$A=\left[\begin{array}{llllll} 0 & 1 & 0 & 0 & 1 & 1 \\ & 0 & 1 & 1 & 1 & 1 \\ & 0 & 1 & 0 & 1 & 1 \\ & 0 & 1 & 1 & 0 & 0 \\ & 1 & 1 & 1 & 0 & 0 \\ & 1 & 1 & 1 & 0 & 1 \end{array}\right]$.	$A = \begin{bmatrix} 0 & 1 & 0 & 0 & 1 & 1 \\ 1 & 0 & 1 & 1 & 1 & 1 \\ 0 & 1 & 0 & 1 & 1 & 1 \\ 0 & 1 & 1 & 0 & 0 & 0 \\ 1 & 1 & 1 & 0 & 0 & 1 \\ 1 & 1 & 1 & 0 & 1 & 0 \end{bmatrix}$.	$A = \begin{bmatrix} 0 & 1 & 0 & 0 & 1 & 1 \\ 1 & 0 & 1 & 1 & 1 & 1 \\ 0 & 1 & 0 & 1 & 1 & 1 \\ 0 & 1 & 1 & 0 & 0 & 0 \\ 1 & 1 & 1 & 0 & 0 & 1 \\ 1 & 1 & 1 & 0 & 1 & 0 \end{bmatrix}$.
johnston-linear-matrix-algebra-EQ0171	85	■ 1.000	$A^2=\left[\begin{array}{llllll} 3 & 2 & 3 & 1 & 2 & 2 \\ & 2 & 5 & 3 & 1 & 3 \\ & 3 & 3 & 4 & 1 & 2 \\ & 1 & 1 & 1 & 2 & 2 \\ & 2 & 3 & 2 & 2 & 4 \\ & 2 & 3 & 2 & 2 & 3 \end{array}\right]$.	$A^2 = \begin{bmatrix} 3 & 2 & 3 & 1 & 2 & 2 \\ 2 & 5 & 3 & 1 & 3 & 3 \\ 3 & 3 & 4 & 1 & 2 & 2 \\ 1 & 1 & 1 & 2 & 2 & 2 \\ 2 & 3 & 2 & 2 & 4 & 3 \\ 2 & 3 & 2 & 2 & 3 & 4 \end{bmatrix}$.	$A^2 = \begin{bmatrix} 3 & 2 & 3 & 1 & 2 & 2 \\ 2 & 5 & 3 & 1 & 3 & 3 \\ 3 & 3 & 4 & 1 & 2 & 2 \\ 1 & 1 & 1 & 2 & 2 & 2 \\ 2 & 3 & 2 & 2 & 4 & 3 \\ 2 & 3 & 2 & 2 & 3 & 4 \end{bmatrix}$.
johnston-linear-matrix-algebra-EQ0172	85	■ 1.000	$\left[A^3\right]_{4,6}=(0,1,1,0,0,0) \cdot \left(2,3,2,2,3,4\right)=3+2=5$.	$\left[A^3\right]_{4,6}=(0,1,1,0,0,0) \cdot (2,3,2,2,3,4)=3+2=5$.	$\left[A^3\right]_{4,6}=(0,1,1,0,0,0) \cdot (2,3,2,2,3,4)=3+2=5$.
johnston-linear-matrix-algebra-EQ0173	85	■ 1.000	$\left[A^3\right]_{4,6}+\left[A^2\right]_{4,6}+a_{4,6}=5+2+0=7$.	$\left[A^3\right]_{4,6}+\left[A^2\right]_{4,6}+a_{4,6}=5+2+0=7$.	$\left[A^3\right]_{4,6}+\left[A^2\right]_{4,6}+a_{4,6}=5+2+0=7$.
johnston-linear-matrix-algebra-EQ0174	86	■ 1.000	$A=\left[\begin{array}{lllll} 0 & 0 & 0 & 1 & 1 \\ 0 & 0 & 1 & 1 & 0 \\ 1 & 1 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{array}\right]$.	$A = \begin{bmatrix} 0 & 0 & 0 & 1 & 1 \\ 0 & 0 & 1 & 1 & 0 \\ 1 & 1 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}$.	$A = \begin{bmatrix} 0 & 0 & 0 & 1 & 1 \\ 0 & 0 & 1 & 1 & 0 \\ 1 & 1 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}$.
johnston-linear-matrix-algebra-EQ0175	86	■ 1.000	$A^2=\left[\begin{array}{lllll} 1 & 0 & 0 & 0 & 0 \\ & 2 & 1 & 0 & 0 \\ & 0 & 0 & 1 & 2 \\ & 0 & 0 & 0 & 1 \\ & 0 & 0 & 0 & 0 \end{array}\right]$.	$A^2 = \begin{bmatrix} 1 & 0 & 0 & 0 & 0 \\ 2 & 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 2 & 1 \\ 0 & 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}$.	$A^2 = \begin{bmatrix} 1 & 0 & 0 & 0 & 0 \\ 2 & 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 2 & 1 \\ 0 & 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}$.
johnston-linear-matrix-algebra-EQ0176	86	■ 1.000	$\left[A^4\right]_{3,5}=(0,0,1,2,1) \cdot\left(0,0,1,1,0\right)=0+0+1+2+0=3$.	$\left[A^4\right]_{3,5}=(0,0,1,2,1) \cdot (0,0,1,1,0)=0+0+1+2+0=3$.	$\left[A^4\right]_{3,5}=(0,0,1,2,1) \cdot (0,0,1,1,0)=0+0+1+2+0=3$.
johnston-linear-matrix-algebra-EQ0177	87	■ 1.000	$A=\left[\begin{array}{llll} 0 & 1 & 3 & 0 \\ 1 & 1 & 1 & 1 \\ 3 & 1 & 0 & 2 \\ 0 & 1 & 2 & 0 \end{array}\right]$.	$A = \begin{bmatrix} 0 & 1 & 3 & 0 \\ 1 & 1 & 1 & 1 \\ 3 & 1 & 0 & 2 \\ 0 & 1 & 2 & 0 \end{bmatrix}$.	$A = \begin{bmatrix} 0 & 1 & 3 & 0 \\ 1 & 1 & 1 & 1 \\ 3 & 1 & 0 & 2 \\ 0 & 1 & 2 & 0 \end{bmatrix}$.

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johnston-linear-matrix-algebra-EQ0179	87	1.000	$A=\left[\begin{array}{llll} 0 & 1 & 2 & 0 \\ 0 & 1 & 1 & 1 \\ 1 & 0 & 0 & 2 \\ 0 & 1 & 0 & 0 \end{array}\right]$.	$A = \begin{bmatrix} 0 & 1 & 2 & 0 \\ 0 & 1 & 1 & 1 \\ 1 & 0 & 0 & 2 \\ 0 & 1 & 0 & 0 \end{bmatrix}$.	$A = \begin{bmatrix} 0 & 1 & 2 & 0 \\ 0 & 1 & 1 & 1 \\ 1 & 0 & 0 & 2 \\ 0 & 1 & 0 & 0 \end{bmatrix}$.
johnston-linear-matrix-algebra-EQ0180	88	1.000	$A^2=\left[\begin{array}{llll} 2 & 1 & 1 & 5 \\ & 1 & 3 & \\ 0 & 3 & 2 & 0 \\ 0 & 1 & 1 & 1 \end{array}\right]$.	$A^2 = \begin{bmatrix} 2 & 1 & 1 & 5 \\ 1 & 2 & 1 & 3 \\ 0 & 3 & 2 & 0 \\ 0 & 1 & 1 & 1 \end{bmatrix}$.	$A^2 = \begin{bmatrix} 2 & 1 & 1 & 5 \\ 1 & 2 & 1 & 3 \\ 0 & 3 & 2 & 0 \\ 0 & 1 & 1 & 1 \end{bmatrix}$.
johnston-linear-matrix-algebra-EQ0184	89	1.000	$\sum_{j=1}^n w_j = \mathbf{v} \cdot \mathbf{w} \leq \ \mathbf{v}\ \ \mathbf{w}\ = \sqrt{n} \sqrt{\sum_{j=1}^n w_j^2}$.	$\sum_{j=1}^n w_j = \mathbf{v} \cdot \mathbf{w} \leq \ \mathbf{v}\ \ \mathbf{w}\ = \sqrt{n} \sqrt{\sum_{j=1}^n w_j^2}$.	$\sum_{j=1}^n w_j = \mathbf{v} \cdot \mathbf{w} \leq \ \mathbf{v}\ \ \mathbf{w}\ = \sqrt{n} \sqrt{\sum_{j=1}^n w_j^2}$.
johnston-linear-matrix-algebra-EQ0185	89	1.000	$\frac{1}{n} \sum_{j=1}^n w_j \leq \frac{\sum_{j=1}^n w_j^2}{\sum_{j=1}^n w_j}$,	$\frac{1}{n} \sum_{j=1}^n w_j \leq \frac{\sum_{j=1}^n w_j^2}{\sum_{j=1}^n w_j}$,	$\frac{1}{n} \sum_{j=1}^n w_j \leq \frac{\sum_{j=1}^n w_j^2}{\sum_{j=1}^n w_j}$,
johnston-linear-matrix-algebra-EQ0186	91	1.000	$A=\left[\begin{array}{ll} 1 & 1 \\ 0 & 1 \end{array}\right]$.	$A = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}$.	$A = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}$.
johnston-linear-matrix-algebra-EQ0187	92	1.000	$a_1x_1+a_2x_2+\cdots+a_nx_n=b$,	$a_1x_1+a_2x_2+\cdots+a_nx_n=b$,	$a_1x_1+a_2x_2+\cdots+a_nx_n=b$,
johnston-linear-matrix-algebra-EQ0190	93	1.000	$\begin{aligned} x+2y &= 4 \\ -x+y &= -1 \end{aligned}$	$\begin{aligned} x+2y &= 4 \\ -x+y &= -1 \end{aligned}$	$\begin{aligned} x+2y &= 4 \\ -x+y &= -1 \end{aligned}$
johnston-linear-matrix-algebra-EQ0192	95	1.000	$\begin{aligned} a_{1,1}x_1+a_{1,2}x_2+\cdots+a_{1,n}x_n &= b_1 \\ a_{2,1}x_1+a_{2,2}x_2+\cdots+a_{2,n}x_n &= b_2 \\ &\vdots \\ a_{m,1}x_1+a_{m,2}x_2+\cdots+a_{m,n}x_n &= b_m \end{aligned}$	$\begin{aligned} a_{1,1}x_1+a_{1,2}x_2+\cdots+a_{1,n}x_n &= b_1 \\ a_{2,1}x_1+a_{2,2}x_2+\cdots+a_{2,n}x_n &= b_2 \\ &\vdots \\ a_{m,1}x_1+a_{m,2}x_2+\cdots+a_{m,n}x_n &= b_m \end{aligned}$	$\begin{aligned} a_{1,1}x_1+a_{1,2}x_2+\cdots+a_{1,n}x_n &= b_1 \\ a_{2,1}x_1+a_{2,2}x_2+\cdots+a_{2,n}x_n &= b_2 \\ &\vdots \\ a_{m,1}x_1+a_{m,2}x_2+\cdots+a_{m,n}x_n &= b_m \end{aligned}$
johnston-linear-matrix-algebra-EQ0193	95	1.000	$\begin{array}{r} x+2y=4 \\ 3x+4y=6 \end{array}$	$\begin{array}{r} x+2y=4 \\ 3x+4y=6 \end{array}$	$\begin{array}{r} x+2y=4 \\ 3x+4y=6 \end{array}$
johnston-linear-matrix-algebra-EQ0195	95	1.000	$\left[\begin{array}{ll} 1 & 2 \\ 3 & 4 \end{array}\right] \left[\begin{array}{l} x \\ y \end{array}\right] = \left[\begin{array}{l} 4 \\ 6 \end{array}\right]$	$\begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 4 \\ 6 \end{bmatrix}$	$\begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 4 \\ 6 \end{bmatrix}$.
johnston-linear-matrix-algebra-EQ0198	96	1.000	$A((1-c)\mathbf{x}_1+c\mathbf{x}_2)=(1-c)A\mathbf{x}_1+cA\mathbf{x}_2=(1-c)\mathbf{b}+c\mathbf{b}=\mathbf{b}$	$A((1-c)\mathbf{x}_1+c\mathbf{x}_2)=(1-c)A\mathbf{x}_1+cA\mathbf{x}_2=(1-c)\mathbf{b}+c\mathbf{b}=\mathbf{b}$	$A((1-c)\mathbf{x}_1+c\mathbf{x}_2)=(1-c)A\mathbf{x}_1+cA\mathbf{x}_2=(1-c)\mathbf{b}+c\mathbf{b}=\mathbf{b}$,
johnston-linear-matrix-algebra-EQ0200	96	1.000	$\begin{aligned} x+3y &= 9 \\ 2y &= 16 \end{aligned}$	$\begin{aligned} x+3y &= 9 \\ 2y &= 16 \end{aligned}$	$\begin{aligned} x+3y &= 9 \\ 2y &= 16 \end{aligned}$

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johnston-linear-matrix-algebra-EQ0203	98	1.000	$\begin{aligned} x+5y-8z &= 9 \\ -x-3y+2z &= -5 \end{aligned}$ $\hline 2y-6z = 4$	$\begin{aligned} x+5y-8z &= 9 \\ -x-3y+2z &= -5 \end{aligned}$ $\hline 2y-6z = 4$	
johnston-linear-matrix-algebra-EQ0205	98	1.000	$\begin{array}{r} x+3y-2z=5 \\ 2y-6z=4 \\ -2y+9z=2 \end{array}$	$\begin{aligned} x+3y-2z &= 5 \\ 2y-6z &= 4 \\ -2y+9z &= 2 \end{aligned}$	
johnston-linear-matrix-algebra-EQ0207	98	1.000	$\begin{array}{r} x+3y-2z=5 \\ 2y-6z=4 \\ 3z=6 \end{array}$	$\begin{aligned} x+3y-2z &= 5 \\ 2y-6z &= 4 \\ 3z &= 6 \end{aligned}$	
johnston-linear-matrix-algebra-EQ0208	99	1.000	$\left[\begin{array}{ccc c} 1 & 3 & -2 & 5 \\ 1 & 5 & -8 & 9 \\ 2 & 4 & 5 & 12 \end{array}\right]$	$\left[\begin{array}{ccc c} 1 & 3 & -2 & 5 \\ 1 & 5 & -8 & 9 \\ 2 & 4 & 5 & 12 \end{array}\right]$	
johnston-linear-matrix-algebra-EQ0216	102	1.000	$\left[\begin{array}{ccccc} 0 & 0 & -1 & 1 & 0 \\ 0 & -2 & 1 & -5 & 2 \\ 0 & 2 & -2 & 6 & -3 \\ 0 & -4 & 2 & -10 & 5 \end{array}\right]$	$\left[\begin{array}{ccccc} 0 & 0 & -1 & 1 & 0 \\ 0 & -2 & 1 & -5 & 2 \\ 0 & 2 & -2 & 6 & -3 \\ 0 & -4 & 2 & -10 & 5 \end{array}\right]$	
johnston-linear-matrix-algebra-EQ0223	103	1.000	$\xrightarrow{R_1+2R_2}\left[\begin{array}{ccccc} 0 & 2 & 0 & 4 & 0 \\ 0 & 0 & 1 & -1 & 0 \\ 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 \end{array}\right]$	$\xrightarrow{R_1+2R_2}\left[\begin{array}{ccccc} 0 & 2 & 0 & 4 & 0 \\ 0 & 0 & 1 & -1 & 0 \\ 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 \end{array}\right]$	
johnston-linear-matrix-algebra-EQ0225	104	1.000	$\left[\begin{array}{cccc c} 0 & 1 & 0 & 2 & 0 \\ 0 & 0 & 1 & -1 & 0 \\ 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 \end{array}\right]$	$\left[\begin{array}{cccc c} 0 & 1 & 0 & 2 & 0 \\ 0 & 0 & 1 & -1 & 0 \\ 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 \end{array}\right]$	
johnston-linear-matrix-algebra-EQ0228	105	1.000	$\left[\begin{array}{lll l} 1 & 4 & 0 & 3 \\ 0 & 0 & 1 & 2 \\ 0 & 0 & 0 & 0 \end{array}\right]$	$\left[\begin{array}{lll l} 1 & 4 & 0 & 3 \\ 0 & 0 & 1 & 2 \\ 0 & 0 & 0 & 0 \end{array}\right]$	
johnston-linear-matrix-algebra-EQ0232	107	1.000	$\left[\begin{array}{ccc c} 1 & 2 & -2 & -4 \\ 2 & 4 & 1 & 0 \\ 1 & 2 & 7 & 2 \end{array}\right]$	$\left[\begin{array}{ccc c} 1 & 2 & -2 & -4 \\ 2 & 4 & 1 & 0 \\ 1 & 2 & 7 & 2 \end{array}\right]$	
johnston-linear-matrix-algebra-EQ0235	109	1.000	$\begin{aligned} v_1+2v_2+2v_3+2v_4 &= 0 \\ 2v_1+v_2-v_3 &= 0 \\ v_1+2v_3+v_4 &= 0 \end{aligned}$	$\begin{aligned} v_1+2v_2+2v_3+2v_4 &= 0 \\ 2v_1+v_2-v_3 &= 0 \\ v_1+2v_3+v_4 &= 0 \end{aligned}$	
johnston-linear-matrix-algebra-EQ0236	109	1.000	$\left[\begin{array}{cccc c} 1 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 1/2 & 0 \\ 0 & 0 & 1 & 1/2 & 0 \end{array}\right]$	$\left[\begin{array}{cccc c} 1 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 1/2 & 0 \\ 0 & 0 & 1 & 1/2 & 0 \end{array}\right]$	

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johnston-linear-matrix-algebra EQ0237	109	1.000	$(1,1,3)=c_1(-1,3,2)+c_2(3,1,-1)$.	$(1,1,3) = c_1(-1,3,2) + c_2(3,1,-1).$	$(1,1,3) = c_1(-1,3,2) + c_2(3,1,-1).$
johnston-linear-matrix-algebra EQ0238	109	1.000	$\begin{aligned} -c_1+3c_2 &= 1 \\ 3c_1+c_2 &= 1 \\ 2c_1-c_2 &= 3 \end{aligned}$	$\begin{aligned} -c_1+3c_2 &= 1 \\ 3c_1+c_2 &= 1 \\ 2c_1-c_2 &= 3 \end{aligned}$	$\begin{aligned} -c_1+3c_2 &= 1 \\ 3c_1+c_2 &= 1 \\ 2c_1-c_2 &= 3 \end{aligned}$
johnston-linear-matrix-algebra EQ0240	110	1.000	$(1,3,1)=c_1(-1,3,2)+c_2(3,1,-1)$.	$(1,3,1) = c_1(-1,3,2) + c_2(3,1,-1).$	$(1,3,1) = c_1(-1,3,2) + c_2(3,1,-1).$
johnston-linear-matrix-algebra EQ0242	110	1.000	$(1,3,1)=\frac{4}{5}(-1,3,2)+\frac{3}{5}(3,1,-1)$.	$(1,3,1) = \frac{4}{5}(-1,3,2) + \frac{3}{5}(3,1,-1).$	$(1,3,1) = \frac{4}{5}(-1,3,2) + \frac{3}{5}(3,1,-1).$
johnston-linear-matrix-algebra EQ0246	112	1.000	$\mathrm{C}_4\mathrm{H}_{10}+\mathrm{O}_2\rightarrow\mathrm{CO}_2+\mathrm{H}_2\mathrm{O}$.	$\mathrm{C}_4\mathrm{H}_{10} + \mathrm{O}_2 \rightarrow \mathrm{CO}_2 + \mathrm{H}_2\mathrm{O}.$	$\mathrm{C}_4\mathrm{H}_{10} + \mathrm{O}_2 \rightarrow \mathrm{CO}_2 + \mathrm{H}_2\mathrm{O}.$
johnston-linear-matrix-algebra EQ0247	112	1.000	$w\mathrm{C}_4\mathrm{H}_{10}+x\mathrm{O}_2\rightarrow y\mathrm{CO}_2+z\mathrm{H}_2\mathrm{O}$	$w\mathrm{C}_4\mathrm{H}_{10} + x\mathrm{O}_2 \rightarrow y\mathrm{CO}_2 + z\mathrm{H}_2\mathrm{O}$	$w\mathrm{C}_4\mathrm{H}_{10} + x\mathrm{O}_2 \rightarrow y\mathrm{CO}_2 + z\mathrm{H}_2\mathrm{O}$
johnston-linear-matrix-algebra EQ0250	112	1.000	$\left[\begin{array}{cccc c} 0 & 2 & -2 & -1 & 0 \\ 4 & 0 & -1 & 0 & 0 \\ 10 & 0 & 0 & -2 & 0 \end{array}\right] \xrightarrow{\text{row-reduce}} \left[\begin{array}{cccc c} 1 & 0 & 0 & -1/5 & 0 \\ 0 & 1 & 0 & -13/10 & 0 \\ 0 & 0 & 1 & -4/5 & 0 \end{array}\right] .$	$\left[\begin{array}{cccc c} 0 & 2 & -2 & -1 & 0 \\ 4 & 0 & -1 & 0 & 0 \\ 10 & 0 & 0 & -2 & 0 \end{array}\right] \xrightarrow{\text{row-reduce}} \left[\begin{array}{cccc c} 1 & 0 & 0 & -1/5 & 0 \\ 0 & 1 & 0 & -13/10 & 0 \\ 0 & 0 & 1 & -4/5 & 0 \end{array}\right] .$	
johnston-linear-matrix-algebra EQ0251	112	1.000	$2\mathrm{C}_4\mathrm{H}_{10}+13\mathrm{O}_2\rightarrow 8\mathrm{CO}_2+10\mathrm{H}_2\mathrm{O}$.	$2\mathrm{C}_4\mathrm{H}_{10} + 13\mathrm{O}_2 \rightarrow 8\mathrm{CO}_2 + 10\mathrm{H}_2\mathrm{O}.$	$2\mathrm{C}_4\mathrm{H}_{10} + 13\mathrm{O}_2 \rightarrow 8\mathrm{CO}_2 + 10\mathrm{H}_2\mathrm{O}.$
johnston-linear-matrix-algebra EQ0252	113	1.000	$\left[\begin{array}{cccc} 3 & -2 & 0 & -2 \\ 0 & -2 & -2 & 2 \\ 0 & 2 & 0 & -1 \\ 1 & 0 & 2 & -2 \end{array}\right]$	$\left[\begin{array}{cccc} 3 & -2 & 0 & -2 \\ 0 & -2 & -2 & 2 \\ 0 & 2 & 0 & -1 \\ 1 & 0 & 2 & -2 \end{array}\right]$	$\left[\begin{array}{cccc} 3 & -2 & 0 & -2 \\ 0 & -2 & -2 & 2 \\ 0 & 2 & 0 & -1 \\ 1 & 0 & 2 & -2 \end{array}\right]$
johnston-linear-matrix-algebra EQ0253	113	1.000	$\left[\begin{array}{ccccc} 4 & 2 & -1 & 2 & 1 \\ 1 & 2 & -1 & 0 & 4 \\ 5 & 1 & 2 & 6 & -1 \\ -3 & 4 & 2 & 2 & 5 \end{array}\right]$	$\left[\begin{array}{ccccc} 4 & 2 & -1 & 2 & 1 \\ 1 & 2 & -1 & 0 & 4 \\ 5 & 1 & 2 & 6 & -1 \\ -3 & 4 & 2 & 2 & 5 \end{array}\right]$	$\left[\begin{array}{ccccc} 4 & 2 & -1 & 2 & 1 \\ 1 & 2 & -1 & 0 & 4 \\ 5 & 1 & 2 & 6 & -1 \\ -3 & 4 & 2 & 2 & 5 \end{array}\right]$
johnston-linear-matrix-algebra EQ0254	113	1.000	$\left[\begin{array}{cccccc} 0 & -2 & -2 & 3 & 4 & 1 \\ 1 & 0 & -2 & 5 & 4 & 0 \\ 5 & 0 & 2 & 1 & 0 & 3 \\ -1 & 1 & 0 & 2 & 4 & -1 \end{array}\right]$	$\left[\begin{array}{cccccc} 0 & -2 & -2 & 3 & 4 & 1 \\ 1 & 0 & -2 & 5 & 4 & 0 \\ 5 & 0 & 2 & 1 & 0 & 3 \\ -1 & 1 & 0 & 2 & 4 & -1 \end{array}\right]$	$\left[\begin{array}{cccccc} 0 & -2 & -2 & 3 & 4 & 1 \\ 1 & 0 & -2 & 5 & 4 & 0 \\ 5 & 0 & 2 & 1 & 0 & 3 \\ -1 & 1 & 0 & 2 & 4 & -1 \end{array}\right]$
johnston-linear-matrix-algebra EQ0255	113	1.000	$\left[\begin{array}{cccccc} 1 & -1 & -3 & -1 & -3 & -3 \\ 0 & -1 & -2 & 1 & -2 & 3 \\ 3 & -1 & -5 & 0 & 0 & -5 \\ 2 & -3 & -8 & -2 & -9 & -5 \\ -2 & 3 & 8 & -2 & 5 & -3 \end{array}\right]$	$\left[\begin{array}{cccccc} 1 & -1 & -3 & -1 & -3 & -3 \\ 0 & -1 & -2 & 1 & -2 & 3 \\ 3 & -1 & -5 & 0 & 0 & -5 \\ 2 & -3 & -8 & -2 & -9 & -5 \\ -2 & 3 & 8 & -2 & 5 & -3 \end{array}\right]$	$\left[\begin{array}{cccccc} 1 & -1 & -3 & -1 & -3 & -3 \\ 0 & -1 & -2 & 1 & -2 & 3 \\ 3 & -1 & -5 & 0 & 0 & -5 \\ 2 & -3 & -8 & -2 & -9 & -5 \\ -2 & 3 & 8 & -2 & 5 & -3 \end{array}\right]$
johnston-linear-matrix-algebra EQ0256	113	1.000	$\begin{aligned} 2y+3z &= -1 \\ z &= 1 \end{aligned}$	$\begin{aligned} 2y+3z &= -1 \\ z &= 1 \end{aligned}$	$\begin{aligned} 2y+3z &= -1 \\ z &= 1 \end{aligned}$
johnston-linear-matrix-algebra EQ0258	113	1.000	$\begin{aligned} x+y+z &= 4 \\ x-y+z &= 0 \end{aligned}$	$\begin{aligned} x+y+z &= 4 \\ x-y+z &= 0 \end{aligned}$	$\begin{aligned} x+y+z &= 4 \\ x-y+z &= 0 \end{aligned}$

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johnston-linear-matrix-algebra-EQ0260	113	1.000	$\begin{aligned} 2x+y-z &=1 \\ x-3y+z &=-2 \\ 2x-2y+3z &=7 \end{aligned}$	$\begin{aligned} 2x+y-z &=1 \\ x-3y+z &=-2 \\ 2x-2y+3z &=7 \end{aligned}$	$\begin{aligned} 2x+y-z &=1 \\ x-3y+z &=-2 \\ 2x-2y+3z &=7 \end{aligned}$
johnston-linear-matrix-algebra-EQ0262	113	1.000	$\begin{aligned} w+x+y-z &=0 \\ 2x+3y+z &=-1 \\ y-z &=-3 \\ 3z &=3 \end{aligned}$	$\begin{aligned} w+x+y-z &=0 \\ 2x+3y+z &=-1 \\ y-z &=-3 \\ 3z &=3 \end{aligned}$	$\begin{aligned} w+x+y-z &=0 \\ 2x+3y+z &=-1 \\ y-z &=-3 \\ 3z &=3 \end{aligned}$
johnston-linear-matrix-algebra-EQ0264	113	1.000	$\begin{aligned} &6v+5w+3x-2y-2z=1 \\ &3v-w+x+5y+4z=2 \\ &3v+4w+x+3y+4z=3 \\ &2v+6w+x-2y-z=4 \end{aligned}$	$\begin{aligned} 6v+5w+3x-2y-2z &=1 \\ 3v-w+x+5y+4z &=2 \\ 3v+4w+x+3y+4z &=3 \\ 2v+6w+x-2y-z &=4 \end{aligned}$	$\begin{aligned} 6v+5w+3x-2y-2z &=1 \\ 3v-w+x+5y+4z &=2 \\ 3v+4w+x+3y+4z &=3 \\ 2v+6w+x-2y-z &=4 \end{aligned}$
johnston-linear-matrix-algebra-EQ0265	113	1.000	$\begin{aligned} v-w+2x+6y+6z &=3 \\ 4v+3w-y+4z &=0 \\ 5v+w-2x-2y-2z &=2 \\ w-2x+3y-2z &=-1 \\ v+5w-x+5z &=3 \end{aligned}$	$\begin{aligned} v-w+2x+6y+6z &=3 \\ 4v+3w-y+4z &=0 \\ 5v+w-2x-2y-2z &=2 \\ w-2x+3y-2z &=-1 \\ v+5w-x+5z &=3 \end{aligned}$	$\begin{aligned} v-w+2x+6y+6z &=3 \\ 4v+3w-y+4z &=0 \\ 5v+w-2x-2y-2z &=2 \\ w-2x+3y-2z &=-1 \\ v+5w-x+5z &=3 \end{aligned}$
johnston-linear-matrix-algebra-EQ0266	114	1.000	$\left[\begin{array}{cc c} 1 & -1 & 0 \\ 1 & 1 & 1 \\ 0 & 1 & t \end{array}\right].$	$\left[\begin{array}{cc c} 1 & -1 & 0 \\ 1 & 1 & 1 \\ 0 & 1 & t \end{array}\right].$	$\left[\begin{array}{cc c} 1 & -1 & 0 \\ 1 & 1 & 1 \\ 0 & 1 & t \end{array}\right].$
johnston-linear-matrix-algebra-EQ0267	114	1.000	$\begin{array}{r} x+y+h \\ z=1 \\ y-z=k \\ x-y+2z=3 \end{array}$	$\begin{aligned} x+y+hz &=1 \\ y-z &=k \\ x-y+2z &=3 \end{aligned}$	$\begin{aligned} x+y+hz &=1 \\ y-z &=k \\ x-y+2z &=3 \end{aligned}$
johnston-linear-matrix-algebra-EQ0268	114	1.000	$\begin{array}{r} w+x/2+y/3+z/4=1 \\ w/2+x/3+y/4+z/5=1 \\ w/3+x/4+y/5+z/6=1 \\ w/4+x/5+y/6+z/7=h \end{array}$	$\begin{aligned} w+x/2+y/3+z/4 &=1 \\ w/2+x/3+y/4+z/5 &=1 \\ w/3+x/4+y/5+z/6 &=1 \\ w/4+x/5+y/6+z/7 &=h \end{aligned}$	$\begin{aligned} w+x/2+y/3+z/4 &=1 \\ w/2+x/3+y/4+z/5 &=1 \\ w/3+x/4+y/5+z/6 &=1 \\ w/4+x/5+y/6+z/7 &=h \end{aligned}$
johnston-linear-matrix-algebra-EQ0269	114	1.000	$\left[\begin{array}{ll l} a & b & 1 \\ c & d & 1 \end{array}\right].$	$\left[\begin{array}{ll l} a & b & 1 \\ c & d & 1 \end{array}\right].$	$\left[\begin{array}{ll l} a & b & 1 \\ c & d & 1 \end{array}\right].$
johnston-linear-matrix-algebra-EQ0271	114	1.000	$\left[\begin{array}{ll l} 1 & 1 & 3 \\ 1 & 2 & 5 \end{array}\right].$	$\left[\begin{array}{ll l} 1 & 1 & 3 \\ 1 & 2 & 5 \end{array}\right].$	$\left[\begin{array}{ll l} 1 & 1 & 3 \\ 1 & 2 & 5 \end{array}\right].$
johnston-linear-matrix-algebra-EQ0272	114	1.000	$\left(v_2w_3-v_3w_2, v_3w_1-v_1w_3, v_1w_2-v_2w_1\right)$	$(v_2w_3-v_3w_2, v_3w_1-v_1w_3, v_1w_2-v_2w_1)$	$(v_2w_3-v_3w_2, v_3w_1-v_1w_3, v_1w_2-v_2w_1)$
johnston-linear-matrix-algebra-EQ0274	115	1.000	$\mathrm{C}_2\mathrm{H}_6+\mathrm{O}_2\rightarrow\mathrm{CO}_2+\mathrm{H}_2\mathrm{O}.$	$\mathrm{C}_2\mathrm{H}_6+\mathrm{O}_2\rightarrow\mathrm{CO}_2+\mathrm{H}_2\mathrm{O}.$	$\mathrm{C}_2\mathrm{H}_6+\mathrm{O}_2\rightarrow\mathrm{CO}_2+\mathrm{H}_2\mathrm{O}.$
johnston-linear-matrix-algebra-EQ0276	115	1.000	$\begin{aligned} y/x+2x/y &=6/xy \\ x^2+y^2 &=5 \end{aligned}$	$\begin{aligned} y/x+2x/y &=6/xy \\ x^2+y^2 &=5 \end{aligned}$	$\begin{aligned} y/x+2x/y &=6/xy \\ x^2+y^2 &=5 \end{aligned}$
johnston-linear-matrix-algebra-EQ0278	115	1.000	$\begin{aligned} \sin(x)+2\cos(y)-\cos(z) &=3 \\ 2\sin(x)+\cos(y)+\cos(z) &=0 \\ -\sin(x)-\cos(y)+\cos(z) &=-2 \end{aligned}$	$\begin{aligned} \sin(x)+2\cos(y)-\cos(z) &=3 \\ 2\sin(x)+\cos(y)+\cos(z) &=0 \\ -\sin(x)-\cos(y)+\cos(z) &=-2 \end{aligned}$	$\begin{aligned} \sin(x)+2\cos(y)-\cos(z) &=3 \\ 2\sin(x)+\cos(y)+\cos(z) &=0 \\ -\sin(x)-\cos(y)+\cos(z) &=-2 \end{aligned}$
johnston-linear-matrix-algebra-EQ0279	115	1.000	$A=\left[\begin{array}{ll} 0 & 1 \\ 1 & 0 \end{array}\right].$	$A=\left[\begin{array}{ll} 0 & 1 \\ 1 & 0 \end{array}\right].$	$A=\left[\begin{array}{ll} 0 & 1 \\ 1 & 0 \end{array}\right].$

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johnston-linear-matrix-algebra EQ0280	115	1.000	$B=\left[\begin{array}{ll} a & b \\ c & d \end{array}\right]$	$B = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$	$B = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$
johnston-linear-matrix-algebra EQ0281	115	1.000	$C=\left[\begin{array}{ll} 1 & 1 \\ 1 & 0 \end{array}\right]$.	$C = \begin{bmatrix} 1 & 1 \\ 1 & 0 \end{bmatrix}.$	$C = \begin{bmatrix} 1 & 1 \\ 1 & 0 \end{bmatrix}.$
johnston-linear-matrix-algebra EQ0282	116	1.000	$p(x)=a x^5+b x^4+c x^3+d x^2+e x+f$	$p(x) = ax^5 + bx^4 + cx^3 + dx^2 + ex + f$	$p(x) = ax^5 + bx^4 + cx^3 + dx^2 + ex + f$
johnston-linear-matrix-algebra EQ0283	116	1.000	$\left[\begin{array}{cccc} 0 & 2 & 4 & 0 \\ 1 & 1 & 0 & -1 \\ 3 & 4 & 2 & 1 \end{array}\right]$.	$\begin{bmatrix} 0 & 2 & 4 & 0 \\ 1 & 1 & 0 & -1 \\ 3 & 4 & 2 & 1 \end{bmatrix}.$	$\begin{bmatrix} 0 & 2 & 4 & 0 \\ 1 & 1 & 0 & -1 \\ 3 & 4 & 2 & 1 \end{bmatrix}.$
johnston-linear-matrix-algebra EQ0285 (2.2.2)	116	1.000	$\left[\begin{array}{ccc} 1 & 0 & 0 \\ 0 & 1 & 0 \\ -3 & 0 & 1 \end{array}\right]\left[\begin{array}{cccc} 1 & 1 & 0 & -1 \\ 0 & 2 & 4 & 0 \\ 3 & 4 & 2 & 1 \end{array}\right]=\left[\begin{array}{cccc} 1 & 1 & 0 & -1 \\ 0 & 2 & 4 & 0 \\ 0 & 1 & 2 & 4 \end{array}\right],$	$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ -3 & 0 & 1 \end{bmatrix}\begin{bmatrix} 1 & 1 & 0 & -1 \\ 0 & 2 & 4 & 0 \\ 3 & 4 & 2 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 1 & 0 & -1 \\ 0 & 2 & 4 & 0 \\ 0 & 1 & 2 & 4 \end{bmatrix},$	$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ -3 & 0 & 1 \end{bmatrix}\begin{bmatrix} 1 & 1 & 0 & -1 \\ 0 & 2 & 4 & 0 \\ 3 & 4 & 2 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 1 & 0 & -1 \\ 0 & 2 & 4 & 0 \\ 0 & 1 & 2 & 4 \end{bmatrix},$
johnston-linear-matrix-algebra EQ0286 (2.2.3)	116	1.000	$\left[\begin{array}{ccc} 1 & 0 & 0 \\ 0 & 1/2 & 0 \\ 0 & 0 & 1 \end{array}\right]\left[\begin{array}{cccc} 1 & 1 & 0 & -1 \\ 0 & 2 & 4 & 0 \\ 0 & 1 & 2 & 4 \end{array}\right]=\left[\begin{array}{cccc} 1 & 1 & 0 & -1 \\ 0 & 1 & 2 & 0 \\ 0 & 1 & 2 & 4 \end{array}\right].$	$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1/2 & 0 \\ 0 & 0 & 1 \end{bmatrix}\begin{bmatrix} 1 & 1 & 0 & -1 \\ 0 & 2 & 4 & 0 \\ 0 & 1 & 2 & 4 \end{bmatrix} = \begin{bmatrix} 1 & 1 & 0 & -1 \\ 0 & 1 & 2 & 0 \\ 0 & 1 & 2 & 4 \end{bmatrix}.$	$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1/2 & 0 \\ 0 & 0 & 1 \end{bmatrix}\begin{bmatrix} 1 & 1 & 0 & -1 \\ 0 & 2 & 4 & 0 \\ 0 & 1 & 2 & 4 \end{bmatrix} = \begin{bmatrix} 1 & 1 & 0 & -1 \\ 0 & 1 & 2 & 0 \\ 0 & 1 & 2 & 4 \end{bmatrix}.$
johnston-linear-matrix-algebra EQ0287	117	1.000	$\left[\begin{array}{lll} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{array}\right],$	$\begin{bmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix},$	$\begin{bmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix},$
johnston-linear-matrix-algebra EQ0288 (1)	117	1.000	$\left[\begin{array}{ccc} 1 & 0 & 0 \\ 0 & 1 & 0 \\ -3 & 0 & 1 \end{array}\right]\quad\text{and}\quad\left[\begin{array}{ccc} 1 & 0 & 0 \\ 0 & 1/2 & 0 \\ 0 & 0 & 1 \end{array}\right],$	$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ -3 & 0 & 1 \end{bmatrix}\quad\text{and}\quad\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1/2 & 0 \\ 0 & 0 & 1 \end{bmatrix},$	$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ -3 & 0 & 1 \end{bmatrix}\quad\text{and}\quad\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1/2 & 0 \\ 0 & 0 & 1 \end{bmatrix},$
johnston-linear-matrix-algebra EQ0289	119	1.000	$A=\left[\begin{array}{cccc} 0 & 2 & 4 & 0 \\ 1 & 1 & 0 & -1 \\ 3 & 4 & 2 & 1 \end{array}\right]$.	$A = \begin{bmatrix} 0 & 2 & 4 & 0 \\ 1 & 1 & 0 & -1 \\ 3 & 4 & 2 & 1 \end{bmatrix}.$	$A = \begin{bmatrix} 0 & 2 & 4 & 0 \\ 1 & 1 & 0 & -1 \\ 3 & 4 & 2 & 1 \end{bmatrix}.$
johnston-linear-matrix-algebra EQ0292	119	1.000	$E_4=\left[\begin{array}{ccc} 1 & -1 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{array}\right]\quad\text{and}\quad E_5=\left[\begin{array}{ccc} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & -1 & 1 \end{array}\right],$	$E_4 = \begin{bmatrix} 1 & -1 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}\quad\text{and}\quad E_5 = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & -1 & 1 \end{bmatrix},$	$E_4 = \begin{bmatrix} 1 & -1 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}\quad\text{and}\quad E_5 = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & -1 & 1 \end{bmatrix},$
johnston-linear-matrix-algebra EQ0293	119	1.000	$E_5E_4E_3E_2E_1A=\left[\begin{array}{cccc} 1 & 0 & -2 & -1 \\ 0 & 1 & 2 & 0 \\ 0 & 0 & 0 & 4 \end{array}\right].$	$E_5E_4E_3E_2E_1A = \begin{bmatrix} 1 & 0 & -2 & -1 \\ 0 & 1 & 2 & 0 \\ 0 & 0 & 0 & 4 \end{bmatrix}.$	$E_5E_4E_3E_2E_1A = \begin{bmatrix} 1 & 0 & -2 & -1 \\ 0 & 1 & 2 & 0 \\ 0 & 0 & 0 & 4 \end{bmatrix}.$

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johnston-linear-matrix-algebra-EQ0294	119	1.000	$E_6=\left[\begin{array}{ccc} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1/4 \end{array}\right] \quad \text{and} \quad E_7=\left[\begin{array}{ccc} 1 & 0 & 1 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{array}\right],$		$E_6=\left[\begin{array}{ccc} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1/4 \end{array}\right] \quad \text{and} \quad E_7=\left[\begin{array}{ccc} 1 & 0 & 1 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{array}\right],$
johnston-linear-matrix-algebra-EQ0295	119	1.000	$E_7 E_6 E_5 E_4 E_3 E_2 E_1 A=\left[\begin{array}{cccc} 1 & 0 & -2 & 0 \\ 0 & 1 & 2 & 0 \\ 0 & 0 & 0 & 1 \end{array}\right],$	$E_7 E_6 E_5 E_4 E_3 E_2 E_1 A=\left[\begin{array}{cccc} 1 & 0 & -2 & 0 \\ 0 & 1 & 2 & 0 \\ 0 & 0 & 0 & 1 \end{array}\right],$	$E_7 E_6 E_5 E_4 E_3 E_2 E_1 A=\left[\begin{array}{cccc} 1 & 0 & -2 & 0 \\ 0 & 1 & 2 & 0 \\ 0 & 0 & 0 & 1 \end{array}\right],$
johnston-linear-matrix-algebra-EQ0296	120	1.000	$E=E_7 E_6 E_5 E_4 E_3 E_2 E_1=\frac{1}{8}\left[\begin{array}{ccc} -5 & 2 & 2 \\ 4 & 0 & 0 \\ -1 & -6 & 2 \end{array}\right].$	$E=E_7 E_6 E_5 E_4 E_3 E_2 E_1=\frac{1}{8}\left[\begin{array}{ccc} -5 & 2 & 2 \\ 4 & 0 & 0 \\ -1 & -6 & 2 \end{array}\right].$	$E=E_7 E_6 E_5 E_4 E_3 E_2 E_1=\frac{1}{8}\left[\begin{array}{ccc} -5 & 2 & 2 \\ 4 & 0 & 0 \\ -1 & -6 & 2 \end{array}\right].$
johnston-linear-matrix-algebra-EQ0297	120	1.000	$A=\left[\begin{array}{cccc} 0 & 2 & 4 & 0 \\ 1 & 1 & 0 & -1 \\ 3 & 4 & 2 & 1 \end{array}\right].$	$A=\left[\begin{array}{cccc} 0 & 2 & 4 & 0 \\ 1 & 1 & 0 & -1 \\ 3 & 4 & 2 & 1 \end{array}\right].$	$A=\left[\begin{array}{cccc} 0 & 2 & 4 & 0 \\ 1 & 1 & 0 & -1 \\ 3 & 4 & 2 & 1 \end{array}\right].$
johnston-linear-matrix-algebra-EQ0301	121	1.000	$[R \mid E]=E_k \cdots E_2 E_1[A \mid I]=\left[E_k \cdots E_2 E_1 A \mid E_k \cdots E_2 E_1\right].$	$[R \mid E]=E_k \cdots E_2 E_1[A \mid I]=\left[E_k \cdots E_2 E_1 A \mid E_k \cdots E_2 E_1\right].$	$[R \mid E]=E_k \cdots E_2 E_1[A \mid I]=\left[E_k \cdots E_2 E_1 A \mid E_k \cdots E_2 E_1\right].$
johnston-linear-matrix-algebra-EQ0306	123	1.000	$\left[\begin{array}{ll} 1 & 2 \\ 2 & 4 \end{array}\right]$	$\left[\begin{array}{ll} 1 & 2 \\ 2 & 4 \end{array}\right]$	$\left[\begin{array}{ll} 1 & 2 \\ 2 & 4 \end{array}\right]$
johnston-linear-matrix-algebra-EQ0307	123	1.000	$\left[\begin{array}{ll} 1 & 2 \\ 2 & 4 \end{array}\right]\left[\begin{array}{cc} -4 & 2 \\ 2 & -1 \end{array}\right]=\left[\begin{array}{cc} 0 & 0 \\ 0 & 0 \end{array}\right].$	$\left[\begin{array}{ll} 1 & 2 \\ 2 & 4 \end{array}\right]\left[\begin{array}{cc} -4 & 2 \\ 2 & -1 \end{array}\right]=\left[\begin{array}{cc} 0 & 0 \\ 0 & 0 \end{array}\right].$	$\left[\begin{array}{ll} 1 & 2 \\ 2 & 4 \end{array}\right]\left[\begin{array}{cc} -4 & 2 \\ 2 & -1 \end{array}\right]=\left[\begin{array}{cc} 0 & 0 \\ 0 & 0 \end{array}\right].$
johnston-linear-matrix-algebra-EQ0309	124	1.000	$(c A)\left(\frac{1}{c} A^{-1}\right)=\frac{c}{c} A A^{-1}=I \quad \text{and} \quad\left(\frac{1}{c} A^{-1}\right)(c A)=\frac{c}{c} A^{-1} A=I,$	$(c A)\left(\frac{1}{c} A^{-1}\right)=\frac{c}{c} A A^{-1}=I \quad \text{and} \quad\left(\frac{1}{c} A^{-1}\right)(c A)=\frac{c}{c} A^{-1} A=I$	$(c A)\left(\frac{1}{c} A^{-1}\right)=\frac{c}{c} A A^{-1}=I \quad \text{and} \quad\left(\frac{1}{c} A^{-1}\right)(c A)=\frac{c}{c} A^{-1} A=I,$
johnston-linear-matrix-algebra-EQ0311	124	1.000	$A^{-1}=\frac{1}{2}\left[\begin{array}{cc} -4 & 2 \\ 3 & -1 \end{array}\right].$	$A^{-1}=\frac{1}{2}\left[\begin{array}{cc} -4 & 2 \\ 3 & -1 \end{array}\right].$	$A^{-1}=\frac{1}{2}\left[\begin{array}{cc} -4 & 2 \\ 3 & -1 \end{array}\right].$
johnston-linear-matrix-algebra-EQ0312	125	1.000	$A^{-2}=\left(A^{-1}\right)^2=\left(\frac{1}{2}\left[\begin{array}{cc} -4 & 2 \\ 3 & -1 \end{array}\right]\right)\left(\frac{1}{2}\left[\begin{array}{cc} -4 & 2 \\ 3 & -1 \end{array}\right]\right)=\frac{1}{4}\left[\begin{array}{cc} 22 & -10 \\ -15 & 7 \end{array}\right].$	$A^{-2}=\left(A^{-1}\right)^2=\left(\frac{1}{2}\left[\begin{array}{cc} -4 & 2 \\ 3 & -1 \end{array}\right]\right)\left(\frac{1}{2}\left[\begin{array}{cc} -4 & 2 \\ 3 & -1 \end{array}\right]\right)=\frac{1}{4}\left[\begin{array}{cc} 22 & -10 \\ -15 & 7 \end{array}\right].$	$A^{-2}=\left(A^{-1}\right)^2=\left(\frac{1}{2}\left[\begin{array}{cc} -4 & 2 \\ 3 & -1 \end{array}\right]\right)\left(\frac{1}{2}\left[\begin{array}{cc} -4 & 2 \\ 3 & -1 \end{array}\right]\right)=\frac{1}{4}\left[\begin{array}{cc} 22 & -10 \\ -15 & 7 \end{array}\right].$
johnston-linear-matrix-algebra-EQ0313	125	1.000	$\left[\begin{array}{cc} 1 & 0 \\ 0 & 1/3 \end{array}\right].$	$\left[\begin{array}{cc} 1 & 0 \\ 0 & 1/3 \end{array}\right].$	$\left[\begin{array}{cc} 1 & 0 \\ 0 & 1/3 \end{array}\right].$
johnston-linear-matrix-algebra-EQ0314	125	1.000	$\left[\begin{array}{cc} 1 & -2 \\ 0 & 1 \end{array}\right].$	$\left[\begin{array}{cc} 1 & -2 \\ 0 & 1 \end{array}\right].$	$\left[\begin{array}{cc} 1 & -2 \\ 0 & 1 \end{array}\right].$

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johnston-linear-matrix-algebra-EQ0315	125	■ 1.000	$\left[\begin{array}{lll} 1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{array}\right]$.	$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{bmatrix}$.	$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{bmatrix}$.
johnston-linear-matrix-algebra-EQ0316	126	■ 1.000	$\left[\begin{array}{lll} 1 & 0 & 0 \\ 5 & 1 & 0 \\ 0 & 0 & 1 \end{array}\right]$.	$\begin{bmatrix} 1 & 0 & 0 \\ 5 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$.	$\begin{bmatrix} 1 & 0 & 0 \\ 5 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$.
johnston-linear-matrix-algebra-EQ0317	126	■ 1.000	$(A\ B)^{-1}=B^{-1}\ A^{-1}$.	$(AB)^{-1} = B^{-1}A^{-1}$.	$(AB)^{-1} = B^{-1}A^{-1}$.
johnston-linear-matrix-algebra-EQ0320 (1)	126	■ 1.000	$\begin{aligned} &\left[\begin{array}{ll} 1 & 6 \\ 0 & 3 \end{array}\right]^{-1}=\left(\left[\begin{array}{ll} 1 & 2 \\ 0 & 1 \end{array}\right]\left[\begin{array}{ll} 1 & 0 \\ 0 & 3 \end{array}\right]\right)^{-1}=\left[\begin{array}{ll} 1 & 0 \\ 0 & 3 \end{array}\right]^{-1}\left[\begin{array}{ll} 1 & 2 \\ 0 & 1 \end{array}\right]^{-1} \\ &=\left[\begin{array}{ll} 1 & 0 \\ 0 & 1/3 \end{array}\right]\left[\begin{array}{ll} 1 & -2 \\ 0 & 1 \end{array}\right]=\left[\begin{array}{ll} 1 & -2 \\ 0 & 1/3 \end{array}\right] . \end{aligned}$	$\begin{bmatrix} 1 & 6 \\ 0 & 3 \end{bmatrix}^{-1}=\left(\left[\begin{array}{ll} 1 & 2 \\ 0 & 1 \end{array}\right]\left[\begin{array}{ll} 1 & 0 \\ 0 & 3 \end{array}\right]\right)^{-1}=\left[\begin{array}{ll} 1 & 0 \\ 0 & 3 \end{array}\right]^{-1}\left[\begin{array}{ll} 1 & 2 \\ 0 & 1 \end{array}\right]^{-1}$ $=\left[\begin{array}{ll} 1 & 0 \\ 0 & 1/3 \end{array}\right]\left[\begin{array}{ll} 1 & -2 \\ 0 & 1 \end{array}\right]=\left[\begin{array}{ll} 1 & -2 \\ 0 & 1/3 \end{array}\right]$.	$\begin{bmatrix} 1 & 6 \\ 0 & 3 \end{bmatrix}^{-1}=\left(\left[\begin{array}{ll} 1 & 2 \\ 0 & 1 \end{array}\right]\left[\begin{array}{ll} 1 & 0 \\ 0 & 3 \end{array}\right]\right)^{-1}=\left[\begin{array}{ll} 1 & 0 \\ 0 & 3 \end{array}\right]^{-1}\left[\begin{array}{ll} 1 & 2 \\ 0 & 1 \end{array}\right]^{-1}$ $=\left[\begin{array}{ll} 1 & 0 \\ 0 & 1/3 \end{array}\right]\left[\begin{array}{ll} 1 & -2 \\ 0 & 1 \end{array}\right]=\left[\begin{array}{ll} 1 & -2 \\ 0 & 1/3 \end{array}\right]$.
johnston-linear-matrix-algebra-EQ0321	127	■ 1.000	$(A\ B\ C)^{-1}=((A\ B)\ C)^{-1}=C^{-1}(A\ B)^{-1}=C^{-1}B^{-1}A^{-1}$ $B^{-1}\ A^{-1}$.	$(ABC)^{-1} = ((AB)C)^{-1} = C^{-1}(AB)^{-1} = C^{-1}B^{-1}A^{-1}$.	$(ABC)^{-1} = ((AB)C)^{-1} = C^{-1}(AB)^{-1} = C^{-1}B^{-1}A^{-1}$.
johnston-linear-matrix-algebra-EQ0325	128	■ 1.000	$\left[\begin{array}{ll} 1 & 2 \\ 2 & 4 \end{array}\right]\xrightarrow{R_2-2R_1}\left[\begin{array}{ll} 1 & 2 \\ 0 & 0 \end{array}\right]$.	$\begin{bmatrix} 1 & 2 \\ 2 & 4 \end{bmatrix} \xrightarrow{R_2-2R_1} \begin{bmatrix} 1 & 2 \\ 0 & 0 \end{bmatrix}$.	$\begin{bmatrix} 1 & 2 \\ 2 & 4 \end{bmatrix} \xrightarrow{R_2-2R_1} \begin{bmatrix} 1 & 2 \\ 0 & 0 \end{bmatrix}$.
johnston-linear-matrix-algebra-EQ0328	130	■ 1.000	$\left[\begin{array}{c} 5 / 2 & -1 \\ -2 & 1 \end{array}\right]$	$\begin{bmatrix} 5/2 & -1 \\ -2 & 1 \end{bmatrix}$	$\begin{bmatrix} 5/2 & -1 \\ -2 & 1 \end{bmatrix}$.
johnston-linear-matrix-algebra-EQ0329	130	■ 1.000	$\left[\begin{array}{cc cc} 1 & -2 & 1 & 0 \\ -3 & 6 & 0 & 1 \end{array}\right]\xrightarrow{R_2+3R_1}\left[\begin{array}{cc cc} 1 & -2 & 1 & 0 \\ 0 & 0 & 3 & 1 \end{array}\right]$.	$\left[\begin{array}{cc cc} 1 & -2 & 1 & 0 \\ -3 & 6 & 0 & 1 \end{array}\right]\xrightarrow{R_2+3R_1}\left[\begin{array}{cc cc} 1 & -2 & 1 & 0 \\ 0 & 0 & 3 & 1 \end{array}\right]$.	$\left[\begin{array}{cc cc} 1 & -2 & 1 & 0 \\ -3 & 6 & 0 & 1 \end{array}\right]\xrightarrow{R_2+3R_1}\left[\begin{array}{cc cc} 1 & -2 & 1 & 0 \\ 0 & 0 & 3 & 1 \end{array}\right]$.
johnston-linear-matrix-algebra-EQ0332	130	■ 1.000	$\left[\begin{array}{ccc} 3 & -3 & 1 \\ -5/2 & 4 & -3/2 \\ 1/2 & -1 & 1/2 \end{array}\right]$	$\begin{bmatrix} 3 & -3 & 1 \\ -5/2 & 4 & -3/2 \\ 1/2 & -1 & 1/2 \end{bmatrix}$	$\begin{bmatrix} 3 & -3 & 1 \\ -5/2 & 4 & -3/2 \\ 1/2 & -1 & 1/2 \end{bmatrix}$.
johnston-linear-matrix-algebra-EQ0333	131	■ 1.000	$A^{-1}=\frac{1}{ad-bc}\left[\begin{array}{cc} d & -b \\ -c & a \end{array}\right]$.	$A^{-1} = \frac{1}{ad-bc} \begin{bmatrix} d & -b \\ -c & a \end{bmatrix}$.	$A^{-1} = \frac{1}{ad-bc} \begin{bmatrix} d & -b \\ -c & a \end{bmatrix}$.
johnston-linear-matrix-algebra-EQ0335	131	■ 1.000	$\frac{1}{-2}\left[\begin{array}{cc} 4 & -2 \\ -3 & 1 \end{array}\right]=\frac{1}{2}\left[\begin{array}{cc} -4 & 2 \\ 3 & -1 \end{array}\right]$.	$\frac{1}{-2} \begin{bmatrix} 4 & -2 \\ -3 & 1 \end{bmatrix} = \frac{1}{2} \begin{bmatrix} -4 & 2 \\ 3 & -1 \end{bmatrix}$.	$\frac{1}{-2} \begin{bmatrix} 4 & -2 \\ -3 & 1 \end{bmatrix} = \frac{1}{2} \begin{bmatrix} -4 & 2 \\ 3 & -1 \end{bmatrix}$.

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johnston-linear-matrix-algebra-EQ0338 (2)	132	1.000	$A=\left[\begin{array}{lll} 1 & 1 & 1 \\ 1 & 2 & 4 \\ 1 & 3 & 9 \end{array}\right] \quad \text{and} \quad \mathbf{b}=\left[\begin{array}{l} 2 \\ 1 \\ 2 \end{array}\right]$	$A = \begin{bmatrix} 1 & 1 & 1 \\ 1 & 2 & 4 \\ 1 & 3 & 9 \end{bmatrix} \quad \text{and} \quad \mathbf{b} = \begin{bmatrix} 2 \\ 1 \\ 2 \end{bmatrix}.$	$A = \begin{bmatrix} 1 & 1 & 1 \\ 1 & 2 & 4 \\ 1 & 3 & 9 \end{bmatrix} \quad \text{and} \quad \mathbf{b} = \begin{bmatrix} 2 \\ 1 \\ 2 \end{bmatrix}.$
johnston-linear-matrix-algebra-EQ0341 (0)	132	1.000	$\mathbf{b}=\left[\begin{array}{c} 1 \\ 0 \\ -1 \end{array}\right] \quad \text{then} \quad \mathbf{x}=A^{-1}\mathbf{b}=\left[\begin{array}{ccc} 3 & -3 & 1 \\ -5/2 & 4 & -3/2 \\ 1/2 & -1 & 1/2 \end{array}\right]\left[\begin{array}{c} 1 \\ 0 \\ -1 \end{array}\right]=\left[\begin{array}{c} 2 \\ -1 \\ 0 \end{array}\right]$	$\mathbf{b} = \begin{bmatrix} 1 \\ 0 \\ -1 \end{bmatrix} \quad \text{then} \quad \mathbf{x} = A^{-1}\mathbf{b} = \begin{bmatrix} 3 & -3 & 1 \\ -5/2 & 4 & -3/2 \\ 1/2 & -1 & 1/2 \end{bmatrix} \begin{bmatrix} 1 \\ 0 \\ -1 \end{bmatrix} = \begin{bmatrix} 2 \\ -1 \\ 0 \end{bmatrix}.$	$\mathbf{b} = \begin{bmatrix} 1 \\ 0 \\ -1 \end{bmatrix} \quad \text{then} \quad \mathbf{x} = A^{-1}\mathbf{b} = \begin{bmatrix} 3 & -3 & 1 \\ -5/2 & 4 & -3/2 \\ 1/2 & -1 & 1/2 \end{bmatrix} \begin{bmatrix} 1 \\ 0 \\ -1 \end{bmatrix} = \begin{bmatrix} 2 \\ -1 \\ 0 \end{bmatrix}.$
johnston-linear-matrix-algebra-EQ0342	133	1.000	$A=\left[\begin{array}{ll} 1 & 0 \\ 0 & 2 \\ 0 & 0 \end{array}\right]$	$A = \begin{bmatrix} 1 & 0 \\ 0 & 2 \\ 0 & 0 \end{bmatrix}.$	$A = \begin{bmatrix} 1 & 0 \\ 0 & 2 \\ 0 & 0 \end{bmatrix}.$
johnston-linear-matrix-algebra-EQ0346	134	1.000	$A=\left[\begin{array}{ll} 2 & 1 \\ 3 & 2 \end{array}\right]$	$A = \begin{bmatrix} 2 & 1 \\ 3 & 2 \end{bmatrix}.$	$A = \begin{bmatrix} 2 & 1 \\ 3 & 2 \end{bmatrix}.$
johnston-linear-matrix-algebra-EQ0348	135	1.000	$\left[\begin{array}{cccc} A_1 & 0 & \cdots & 0 \\ 0 & A_2 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & A_n \end{array}\right]$	$\begin{bmatrix} A_1 & O & \cdots & O \\ O & A_2 & \cdots & O \\ \vdots & \vdots & \ddots & \vdots \\ O & O & \cdots & A_n \end{bmatrix}$	$\begin{bmatrix} A_1 & O & \cdots & O \\ O & A_2 & \cdots & O \\ \vdots & \vdots & \ddots & \vdots \\ O & O & \cdots & A_n \end{bmatrix}$
johnston-linear-matrix-algebra-EQ0351	136	1.000	$A_n=\left[\begin{array}{cccccc} 1 & 1 & 1 & \cdots & 1 \\ 1 & 2 & 1 & \cdots & 1 \\ 1 & 1 & 3 & \cdots & 1 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 1 & 1 & 1 & \cdots & n \end{array}\right]$	$A_n = \begin{bmatrix} 1 & 1 & 1 & \cdots & 1 \\ 1 & 2 & 1 & \cdots & 1 \\ 1 & 1 & 3 & \cdots & 1 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 1 & 1 & 1 & \cdots & n \end{bmatrix}.$	$A_n = \begin{bmatrix} 1 & 1 & 1 & \cdots & 1 \\ 1 & 2 & 1 & \cdots & 1 \\ 1 & 1 & 3 & \cdots & 1 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 1 & 1 & 1 & \cdots & n \end{bmatrix}.$
johnston-linear-matrix-algebra-EQ0352	136	1.000	$\mathbf{0}=0\mathbf{v}=c\mathbf{v} \in \mathcal{S}.$	$\mathbf{0} = 0\mathbf{v} = c\mathbf{v} \in \mathcal{S}.$	$\mathbf{0} = 0\mathbf{v} = c\mathbf{v} \in \mathcal{S}.$
johnston-linear-matrix-algebra-EQ0353	138	1.000	$c_{\{1\}}\mathbf{v}_{\{1\}}+c_{\{2\}}\mathbf{v}_{\{2\}}+\cdots+c_{\{k\}}\mathbf{v}_{\{k\}} \in \mathcal{S}$	$c_1\mathbf{v}_1 + c_2\mathbf{v}_2 + \cdots + c_k\mathbf{v}_k \in \mathcal{S}$	$c_1\mathbf{v}_1 + c_2\mathbf{v}_2 + \cdots + c_k\mathbf{v}_k \in \mathcal{S}$
johnston-linear-matrix-algebra-EQ0355	139	1.000	$\left(c\mathbf{v}_{\{1\}}\right)+\left(c\mathbf{v}_{\{2\}}\right)-3\left(c\mathbf{v}_{\{3\}}\right)=0,$	$(c\mathbf{v}_1) + (c\mathbf{v}_2) - 3(c\mathbf{v}_3) = 0,$	$(c\mathbf{v}_1) + (c\mathbf{v}_2) - 3(c\mathbf{v}_3) = 0,$
johnston-linear-matrix-algebra-EQ0356	140	1.000	$A(\mathbf{v}+\mathbf{w})=A\mathbf{v}+A\mathbf{w}=\mathbf{0}+\mathbf{0}=\mathbf{0}$	$A(\mathbf{v} + \mathbf{w}) = A\mathbf{v} + A\mathbf{w} = \mathbf{0} + \mathbf{0} = \mathbf{0}$	$A(\mathbf{v} + \mathbf{w}) = A\mathbf{v} + A\mathbf{w} = \mathbf{0} + \mathbf{0} = \mathbf{0},$
johnston-linear-matrix-algebra-EQ0357	140	1.000	$A(c\mathbf{v})=c(A\mathbf{v})=c\mathbf{0}=\mathbf{0}$	$A(c\mathbf{v}) = c(A\mathbf{v}) = c\mathbf{0} = \mathbf{0}$	$A(c\mathbf{v}) = c(A\mathbf{v}) = c\mathbf{0} = \mathbf{0},$
johnston-linear-matrix-algebra-EQ0361	141	1.000	$(x,y)=x\mathbf{e}_{\{1\}}+y\mathbf{e}_{\{2\}}.$	$(x,y) = x\mathbf{e}_1 + y\mathbf{e}_2.$	$(x,y) = x\mathbf{e}_1 + y\mathbf{e}_2.$
johnston-linear-matrix-algebra-EQ0362	141	1.000	$\mathbf{v}=\left(v_{\{1\}},v_{\{2\}},\ldots,v_{\{n\}}\right)=v_{\{1\}}\mathbf{e}_{\{1\}}+v_{\{2\}}\mathbf{e}_{\{2\}}+\cdots+v_{\{n\}}\mathbf{e}_{\{n\}}.$	$\mathbf{v} = (v_1, v_2, \dots, v_n) = v_1\mathbf{e}_1 + v_2\mathbf{e}_2 + \cdots + v_n\mathbf{e}_n.$	$\mathbf{v} = (v_1, v_2, \dots, v_n) = v_1\mathbf{e}_1 + v_2\mathbf{e}_2 + \cdots + v_n\mathbf{e}_n.$
johnston-linear-matrix-algebra-EQ0363	142	1.000	$(x,y)=c_{\{1\}}(1,2)+c_{\{2\}}(2,1).$	$(x,y) = c_1(1,2) + c_2(2,1).$	$(x,y) = c_1(1,2) + c_2(2,1).$

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johnston-linear-matrix-algebra-EQ0366	143	■ 1.000	$(x, y, z) = c_1(1, 2, 1) + c_2(2, 1, 1) \text{ .}$	$(x, y, z) = c_1(1, 2, 1) + c_2(2, 1, 1).$	$(x, y, z) = c_1(1, 2, 1) + c_2(2, 1, 1).$
johnston-linear-matrix-algebra-EQ0369	143	■ 1.000	$\mathbf{v} + \mathbf{w} = \left(c_1 + d_1\right) \mathbf{v}_1 + \left(c_2 + d_2\right) \mathbf{v}_2 + \cdots + \left(c_k + d_k\right) \mathbf{v}_k,$	$\mathbf{v} + \mathbf{w} = (c_1 + d_1) \mathbf{v}_1 + (c_2 + d_2) \mathbf{v}_2 + \cdots + (c_k + d_k) \mathbf{v}_k,$	$\mathbf{v} + \mathbf{w} = (c_1 + d_1) \mathbf{v}_1 + (c_2 + d_2) \mathbf{v}_2 + \cdots + (c_k + d_k) \mathbf{v}_k,$
johnston-linear-matrix-algebra-EQ0370	144	■ 1.000	$c \mathbf{v} = \left(c c_1\right) \mathbf{v}_1 + \left(c c_2\right) \mathbf{v}_2 + \cdots + \left(c c_k\right) \mathbf{v}_k,$	$c \mathbf{v} = (cc_1) \mathbf{v}_1 + (cc_2) \mathbf{v}_2 + \cdots + (cc_k) \mathbf{v}_k,$	$c \mathbf{v} = (cc_1) \mathbf{v}_1 + (cc_2) \mathbf{v}_2 + \cdots + (cc_k) \mathbf{v}_k,$
johnston-linear-matrix-algebra-EQ0371	144	■ 1.000	$\operatorname{range}(A) = \operatorname{span}\left(\mathbf{a}_1, \mathbf{a}_2, \ldots, \mathbf{a}_n\right) \text{ .}$	$\operatorname{range}(A) = \operatorname{span}\left(\mathbf{a}_1, \mathbf{a}_2, \ldots, \mathbf{a}_n\right).$	$\operatorname{range}(A) = \operatorname{span}\left(\mathbf{a}_1, \mathbf{a}_2, \ldots, \mathbf{a}_n\right).$
johnston-linear-matrix-algebra-EQ0372	144	■ 1.000	$A \mathbf{x} = x_1 \mathbf{a}_1 + x_2 \mathbf{a}_2 + \cdots + x_n \mathbf{a}_n \text{ .}$	$A \mathbf{x} = x_1 \mathbf{a}_1 + x_2 \mathbf{a}_2 + \cdots + x_n \mathbf{a}_n.$	$A \mathbf{x} = x_1 \mathbf{a}_1 + x_2 \mathbf{a}_2 + \cdots + x_n \mathbf{a}_n.$
johnston-linear-matrix-algebra-EQ0373	144	■ 1.000	$\operatorname{range}(A) = \operatorname{span}\left((2, 1), (-2, -1)\right) \text{ .}$	$\operatorname{range}(A) = \operatorname{span}\left((2, 1), (-2, -1)\right).$	$\operatorname{range}(A) = \operatorname{span}\left((2, 1), (-2, -1)\right).$
johnston-linear-matrix-algebra-EQ0374	144	■ 1.000	$\operatorname{range}(A) = \operatorname{span}\left((2, 1)\right) \text{ .}$	$\operatorname{range}(A) = \operatorname{span}\left((2, 1)\right).$	$\operatorname{range}(A) = \operatorname{span}\left((2, 1)\right).$
johnston-linear-matrix-algebra-EQ0375	146	■ 1.000	$\left[\begin{array}{cc c} 1 & -1 & 2 \\ -1 & 1 & -2 \end{array}\right] \quad \text{is} \quad \left[\begin{array}{cc c} 1 & -1 & 2 \\ 0 & 0 & 0 \end{array}\right] \text{ .}$	$\left[\begin{array}{cc c} 1 & -1 & 2 \\ -1 & 1 & -2 \end{array}\right] \quad \text{is} \quad \left[\begin{array}{cc c} 1 & -1 & 2 \\ 0 & 0 & 0 \end{array}\right] \text{ .}$	$\left[\begin{array}{cc c} 1 & -1 & 2 \\ -1 & 1 & -2 \end{array}\right] \quad \text{is} \quad \left[\begin{array}{cc c} 1 & -1 & 2 \\ 0 & 0 & 0 \end{array}\right] \text{ .}$
johnston-linear-matrix-algebra-EQ0376	146	■ 1.000	$c_1 \mathbf{v}_1 + c_2 \mathbf{v}_2 + \cdots + c_k \mathbf{v}_k = \mathbf{0} \text{ .}$	$c_1 \mathbf{v}_1 + c_2 \mathbf{v}_2 + \cdots + c_k \mathbf{v}_k = \mathbf{0}.$	$c_1 \mathbf{v}_1 + c_2 \mathbf{v}_2 + \cdots + c_k \mathbf{v}_k = \mathbf{0}.$
johnston-linear-matrix-algebra-EQ0377	146	■ 1.000	$(2, 3) - 2(1, 0) - 3(0, 1) = (0, 0) \text{ .}$	$(2, 3) - 2(1, 0) - 3(0, 1) = (0, 0).$	$(2, 3) - 2(1, 0) - 3(0, 1) = (0, 0).$
johnston-linear-matrix-algebra-EQ0378	146	■ 1.000	$0(1, 0, 0) + 0(0, 1, 0) + 0(0, 0, 1) = (0, 0, 0) \text{ .}$	$0(1, 0, 0) + 0(0, 1, 0) + 0(0, 0, 1) = (0, 0, 0).$	$0(1, 0, 0) + 0(0, 1, 0) + 0(0, 0, 1) = (0, 0, 0).$
johnston-linear-matrix-algebra-EQ0379	146	■ 1.000	$c_1 \mathbf{v}_1 + c_2 \mathbf{v}_2 + \cdots + c_k \mathbf{v}_k = \mathbf{0}$	$c_1 \mathbf{v}_1 + c_2 \mathbf{v}_2 + \cdots + c_k \mathbf{v}_k = \mathbf{0}$	$c_1 \mathbf{v}_1 + c_2 \mathbf{v}_2 + \cdots + c_k \mathbf{v}_k = \mathbf{0}$
johnston-linear-matrix-algebra-EQ0380	147	■ 1.000	$c_1(1, -1, 0) + c_2(-2, 1, 2) + c_3(1, 1, -4) = (0, 0, 0) \text{ .}$	$c_1(1, -1, 0) + c_2(-2, 1, 2) + c_3(1, 1, -4) = (0, 0, 0).$	$c_1(1, -1, 0) + c_2(-2, 1, 2) + c_3(1, 1, -4) = (0, 0, 0).$
johnston-linear-matrix-algebra-EQ0384	147	■ 1.000	$c_1(1, 2, 3) + c_2(1, 0, 1) + c_3(0, -1, 2) = (0, 0, 0) \text{ .}$	$c_1(1, 2, 3) + c_2(1, 0, 1) + c_3(0, -1, 2) = (0, 0, 0).$	$c_1(1, 2, 3) + c_2(1, 0, 1) + c_3(0, -1, 2) = (0, 0, 0).$
johnston-linear-matrix-algebra-EQ0387	148	■ 1.000	$c_1 \mathbf{v}_1 + c_2 \mathbf{v}_2 + \cdots + c_k \mathbf{v}_k = \mathbf{0}$	$c_1 \mathbf{v}_1 + c_2 \mathbf{v}_2 + \cdots + c_k \mathbf{v}_k = \mathbf{0}$	$c_1 \mathbf{v}_1 + c_2 \mathbf{v}_2 + \cdots + c_k \mathbf{v}_k = \mathbf{0}$
johnston-linear-matrix-algebra-EQ0390	148	■ 1.000	$2(1, 2, 1) - (2, 4, 2) + 0(2, 1, 1) = (0, 0, 0) \text{ .}$	$2(1, 2, 1) - (2, 4, 2) + 0(2, 1, 1) = (0, 0, 0).$	$2(1, 2, 1) - (2, 4, 2) + 0(2, 1, 1) = (0, 0, 0).$
johnston-linear-matrix-algebra-EQ0391	150	■ 1.000	$\left[\begin{array}{ccc} 1 & -2 & 1 \\ -1 & 1 & 1 \\ 2 & -4 & \end{array}\right]$	$\left[\begin{array}{ccc} 1 & -2 & 1 \\ -1 & 1 & 1 \\ 0 & 2 & -4 \end{array}\right]$	$\left[\begin{array}{ccc} 1 & -2 & 1 \\ -1 & 1 & 1 \\ 0 & 2 & -4 \end{array}\right]$
johnston-linear-matrix-algebra-EQ0392	150	■ 1.000	$\left[\begin{array}{ccc} 1 & 1 & 0 \\ 2 & 0 & -1 \\ 3 & 1 & 2 \end{array}\right]$	$\left[\begin{array}{ccc} 1 & 1 & 0 \\ 2 & 0 & -1 \\ 3 & 1 & 2 \end{array}\right]$	$\left[\begin{array}{ccc} 1 & 1 & 0 \\ 2 & 0 & -1 \\ 3 & 1 & 2 \end{array}\right]$
johnston-linear-matrix-algebra-EQ0393	151	■ 1.000	$\left[\begin{array}{lll} 1 & 1 & 1 \\ 1 & 2 & 4 \\ 1 & 3 & 9 \end{array}\right]$	$\left[\begin{array}{lll} 1 & 1 & 1 \\ 1 & 2 & 4 \\ 1 & 3 & 9 \end{array}\right]$	$\left[\begin{array}{lll} 1 & 1 & 1 \\ 1 & 2 & 4 \\ 1 & 3 & 9 \end{array}\right]$

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johnston-linear-matrix-algebra-EQ0394	151	1.000	$V=\left[\begin{array}{ccccc} 1 & a_0 & a_0^2 & \cdots & a_0^n \\ 1 & a_1 & a_1^2 & \cdots & a_1^n \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 1 & a_n & a_n^2 & \cdots & a_n^n \end{array}\right],$		
johnston-linear-matrix-algebra-EQ0395 (2.3.1)	151	1.000	$c_{\{0\}}\left[\begin{array}{c} 1 \\ 1 \\ \vdots \\ 1 \end{array}\right]+c_{\{1\}}\left[\begin{array}{c} a_0 \\ a_1 \\ \vdots \\ a_n \end{array}\right]+c_{\{2\}}\left[\begin{array}{c} a_0^2 \\ a_1^2 \\ \vdots \\ a_n^2 \end{array}\right]+\cdots+c_n\left[\begin{array}{c} a_0^n \\ a_1^n \\ \vdots \\ a_n^n \end{array}\right]=\left[\begin{array}{c} 0 \\ 0 \\ \vdots \\ 0 \end{array}\right]$		
johnston-linear-matrix-algebra-EQ0396	152	1.000	$p(x)=c_{\{n\}}\,x^{\{n\}}+\cdots+c_{\{2\}}\,x^{\{2\}}+c_{\{1\}}\,x+c_{\{0\}}$	$p(x)=c_nx^n+\cdots+c_2x^2+c_1x+c_0$	$p(x)=c_nx^n+\cdots+c_2x^2+c_1x+c_0$
johnston-linear-matrix-algebra-EQ0399	153	1.000	$p(x)=c_{\{2\}}\,x^{\{2\}}+c_{\{1\}}\,x+c_{\{0\}}=2\,x^{\{2\}}-3\,x-1\,.$	$p(x)=c_2x^2+c_1x+c_0=2x^2-3x-1.$	$p(x)=c_2x^2+c_1x+c_0=2x^2-3x-1.$
johnston-linear-matrix-algebra-EQ0400	154	1.000	$\{(1,2,3),(-1,k,1),(1,1,0)\}$	$\{(1,2,3),(-1,k,1),(1,1,0)\}$	$\{(1,2,3),(-1,k,1),(1,1,0)\}$
johnston-linear-matrix-algebra-EQ0402	154	1.000	$c_{\{1\}}\,\mathbf{v}_{\{1\}}+\cdots+c_{\{k\}}\,\mathbf{v}_{\{k\}}\in\mathcal{S}$	$c_1\mathbf{v}_1+\cdots+c_k\mathbf{v}_k\in\mathcal{S}$	$c_1\mathbf{v}_1+\cdots+c_k\mathbf{v}_k\in\mathcal{S}$
johnston-linear-matrix-algebra-EQ0403 (0)	154	1.000	$\left[\begin{array}{ccccc} 3 & -1 & 1 & 0 & -1 \\ 2 & 1 & -1 & 4 & 0 \\ 1 & 1 & -1 & 1 & 2 \\ 2 & 1 & 2 & 0 & 0 \\ -1 & 3 & -1 & 3 & 2 \end{array}\right]$		$\left[\begin{array}{ccccc} 3 & -1 & 1 & 0 & -1 \\ 2 & 1 & -1 & 4 & 0 \\ 1 & 1 & -1 & 1 & 2 \\ 2 & 1 & 2 & 0 & 0 \\ -1 & 3 & -1 & 3 & 2 \end{array}\right]$
johnston-linear-matrix-algebra-EQ0404	154	1.000	$\left[\begin{array}{cccccc} 4 & 1 & 2 & 3 & 2 & 2 & -1 \\ 0 & -1 & 4 & 2 & 0 & 4 & 2 \\ -1 & 2 & 0 & 2 & 4 & 3 & 3 \\ 3 & 1 & -1 & 0 & 2 & 2 & 3 \\ -1 & 0 & -1 & 4 & 3 & 0 & 3 \end{array}\right]$		$\left[\begin{array}{cccccc} 4 & 1 & 2 & 3 & 2 & 2 & -1 \\ 0 & -1 & 4 & 2 & 0 & 4 & 2 \\ -1 & 2 & 0 & 2 & 4 & 3 & 3 \\ 3 & 1 & -1 & 0 & 2 & 2 & 3 \\ -1 & 0 & -1 & 4 & 3 & 0 & 3 \end{array}\right]$
johnston-linear-matrix-algebra-EQ0405	154	1.000	$\operatorname{span}(\mathbf{v},\,\mathbf{w})=\operatorname{span}(\mathbf{v},\,\mathbf{v}+\mathbf{w})\,.$	$\operatorname{span}(\mathbf{v},\mathbf{w})=\operatorname{span}(\mathbf{v},\mathbf{v}+\mathbf{w}).$	$\operatorname{span}(\mathbf{v},\mathbf{w})=\operatorname{span}(\mathbf{v},\mathbf{v}+\mathbf{w}).$
johnston-linear-matrix-algebra-EQ0406	154	1.000	$\operatorname{span}\left(\mathbf{w},\,\mathbf{v}_1,\,\mathbf{v}_2,\,\ldots,\,\mathbf{v}_k\right)=\operatorname{span}\left(\mathbf{v}_1,\,\mathbf{v}_2,\,\ldots,\,\mathbf{v}_k\right)\,.$	$\operatorname{span}\left(\mathbf{w},\mathbf{v}_1,\mathbf{v}_2,\ldots,\mathbf{v}_k\right)=\operatorname{span}\left(\mathbf{v}_1,\mathbf{v}_2,\ldots,\mathbf{v}_k\right).$	$\operatorname{span}(\mathbf{w},\mathbf{v}_1,\mathbf{v}_2,\ldots,\mathbf{v}_k)=\operatorname{span}(\mathbf{v}_1,\mathbf{v}_2,\ldots,\mathbf{v}_k).$

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johnston-linear-matrix-algebra-EQ0407	155	1.000	$V_n=\left[\begin{array}{ccccc} 1 & 1 & 1 & \cdots & 1 \\ 1 & 2 & 4 & \cdots & 2^{n-1} \\ 1 & 3 & 9 & \cdots & 3^{n-1} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 1 & n & n^2 & \cdots & n^{n-1} \end{array}\right] .$	$V_n=\begin{bmatrix} 1 & 1 & 1 & \cdots & 1 \\ 1 & 2 & 4 & \cdots & 2^{n-1} \\ 1 & 3 & 9 & \cdots & 3^{n-1} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 1 & n & n^2 & \cdots & n^{n-1} \end{bmatrix} .$	$V_n=\begin{bmatrix} 1 & 1 & 1 & \cdots & 1 \\ 1 & 2 & 4 & \cdots & 2^{n-1} \\ 1 & 3 & 9 & \cdots & 3^{n-1} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 1 & n & n^2 & \cdots & n^{n-1} \end{bmatrix} .$
johnston-linear-matrix-algebra-EQ0408	156	1.000	$c_{\{1\}}\,\mathbf{e}_{\{1\}}+c_{\{2\}}\,\mathbf{e}_{\{2\}}+\cdots+c_{\{n\}}\,\mathbf{e}_{\{n\}}=\mathbf{e}_{\{n\}}=\mathbf{0} \; .$	$c_1\mathbf{e}_1+c_2\mathbf{e}_2+\cdots+c_n\mathbf{e}_n=\mathbf{0}.$	$c_1\mathbf{e}_1+c_2\mathbf{e}_2+\cdots+c_n\mathbf{e}_n=\mathbf{0}.$
johnston-linear-matrix-algebra-EQ0411	158	1.000	$ B \, \leq C \; .$	$ B \leq C .$	$ B \leq C .$
johnston-linear-matrix-algebra-EQ0412 (2.4.1)	158	1.000	$\mathbf{v}_{\{i\}}=a_{\{i, \, 1\}}\,\mathbf{w}_{\{1\}}+a_{\{i, \, 2\}}\,\mathbf{w}_{\{2\}}+\cdots+a_{\{i, \, \ell\}}\,\mathbf{w}_{\{\ell\}} \\ \quad \quad \quad \text{for } 1 \leq i \leq k,$	$\mathbf{v}_i = a_{i,1}\mathbf{w}_1 + a_{i,2}\mathbf{w}_2 + \cdots + a_{i,\ell}\mathbf{w}_\ell \quad \text{for } 1 \leq i \leq k,$	$\mathbf{v}_i = a_{i,1}\mathbf{w}_1 + a_{i,2}\mathbf{w}_2 + \cdots + a_{i,\ell}\mathbf{w}_\ell \quad \text{for } 1 \leq i \leq k,$
johnston-linear-matrix-algebra-EQ0413	158	1.000	$a_{\{i, \, 1\}}\,\mathbf{w}_{\{1\}}+a_{\{i, \, 2\}}\,\mathbf{w}_{\{2\}}+\cdots+a_{\{i, \, \ell\}}\,\mathbf{w}_{\{\ell\}},$	$a_{i,1}\mathbf{w}_1 + a_{i,2}\mathbf{w}_2 + \cdots + a_{i,\ell}\mathbf{w}_\ell,$	$a_{i,1}\mathbf{w}_1 + a_{i,2}\mathbf{w}_2 + \cdots + a_{i,\ell}\mathbf{w}_\ell,$
johnston-linear-matrix-algebra-EQ0418	162	1.000	$c_{\{1\}}\,\mathbf{v}_{\{1\}}+c_{\{2\}}\,\mathbf{v}_{\{2\}}+\cdots+c_{\{k\}}\,\mathbf{v}_{\{k\}}=\mathbf{0} \; .$	$c_1\mathbf{v}_1+c_2\mathbf{v}_2+\cdots+c_k\mathbf{v}_k=\mathbf{0}.$	$c_1\mathbf{v}_1+c_2\mathbf{v}_2+\cdots+c_k\mathbf{v}_k=\mathbf{0}.$
johnston-linear-matrix-algebra-EQ0421	164	1.000	$d\,\mathbf{w}+c_{\{1\}}\,\mathbf{v}_{\{1\}}+c_{\{2\}}\,\mathbf{v}_{\{2\}}+\cdots+c_{\{m\}}\,\mathbf{v}_{\{m\}}=\mathbf{0} \; .$	$d\mathbf{w}+c_1\mathbf{v}_1+c_2\mathbf{v}_2+\cdots+c_m\mathbf{v}_m=\mathbf{0}.$	$d\mathbf{w}+c_1\mathbf{v}_1+c_2\mathbf{v}_2+\cdots+c_m\mathbf{v}_m=\mathbf{0}.$
johnston-linear-matrix-algebra-EQ0422	164	1.000	$\mathbf{w}=-\frac{c_{\{1\}}}{d}\,\mathbf{v}_{\{1\}}-\frac{c_{\{2\}}}{d}\,\mathbf{v}_{\{2\}}-\cdots-\frac{c_{\{m\}}}{d}\,\mathbf{v}_{\{m\}}.$	$\mathbf{w}=-\frac{c_1}{d}\mathbf{v}_1-\frac{c_2}{d}\mathbf{v}_2-\cdots-\frac{c_m}{d}\mathbf{v}_m.$	$\mathbf{w}=-\frac{c_1}{d}\mathbf{v}_1-\frac{c_2}{d}\mathbf{v}_2-\cdots-\frac{c_m}{d}\mathbf{v}_m.$
johnston-linear-matrix-algebra-EQ0423	164	1.000	$\operatorname{range}(A)=\operatorname{span}((1,2,1),(1,-1,0),(2,1,1)).$	$\operatorname{range}(A)=\operatorname{span}((1,2,1),(1,-1,0),(2,1,1)).$	$\operatorname{range}(A)=\operatorname{span}((1,2,1),(1,-1,0),(2,1,1)).$
johnston-linear-matrix-algebra-EQ0426	166	1.000	$A=\left[\begin{array}{ccccc} 1 & 0 & 1 & 0 & -1 \\ 1 & 1 & 0 & 0 & 1 \\ -1 & 0 & -1 & 1 & 4 \\ 2 & 1 & 1 & -1 & -3 \end{array}\right]$	$A=\begin{bmatrix} 1 & 0 & 1 & 0 & -1 \\ 1 & 1 & 0 & 0 & 1 \\ -1 & 0 & -1 & 1 & 4 \\ 2 & 1 & 1 & -1 & -3 \end{bmatrix}$	$A=\begin{bmatrix} 1 & 0 & 1 & 0 & -1 \\ 1 & 1 & 0 & 0 & 1 \\ -1 & 0 & -1 & 1 & 4 \\ 2 & 1 & 1 & -1 & -3 \end{bmatrix}$
johnston-linear-matrix-algebra-EQ0428	166	1.000	$\left(x_{\{1\}},\,x_{\{2\}},\,x_{\{3\}},\,x_{\{4\}},\,x_{\{5\}}\right)=x_{\{3\}}(-1,1,1,0,0)+x_{\{5\}}(1,-2,0,-3,1) \; .$	$(x_1,x_2,x_3,x_4,x_5)=x_3(-1,1,1,0,0)+x_5(1,-2,0,-3,1).$	$(x_1,x_2,x_3,x_4,x_5)=x_3(-1,1,1,0,0)+x_5(1,-2,0,-3,1).$
johnston-linear-matrix-algebra-EQ0429	167	1.000	$\left[\begin{array}{cccccc} 1 & 2 & 0 & 3 & 4 & 0 & 5 \\ 0 & 0 & 1 & 6 & 7 & 0 & 8 \\ 0 & 0 & 0 & 0 & 0 & 1 & 9 \end{array}\right]$	$\begin{bmatrix} 1 & 2 & 0 & 3 & 4 & 0 & 5 \\ 0 & 0 & 1 & 6 & 7 & 0 & 8 \\ 0 & 0 & 0 & 0 & 0 & 1 & 9 \end{bmatrix}$	$\begin{bmatrix} 1 & 2 & 0 & 3 & 4 & 0 & 5 \\ 0 & 0 & 1 & 6 & 7 & 0 & 8 \\ 0 & 0 & 0 & 0 & 0 & 1 & 9 \end{bmatrix}$
johnston-linear-matrix-algebra-EQ0432	167	1.000	$\operatorname{range}(A),\,\operatorname{null}(A),\,\operatorname{range}(A^T),\,\text{and }\operatorname{null}(A^T)$	$\operatorname{range}(A),\operatorname{null}(A),\operatorname{range}(A^T),\,\text{and }\operatorname{null}(A^T)$	$\operatorname{range}(A),\,\operatorname{null}(A),\,\operatorname{range}(A^T),\,\text{and }\operatorname{null}(A^T)$
johnston-linear-matrix-algebra-EQ0433	168	1.000	$A=\left[\begin{array}{ccccc} 1 & 0 & 1 & 0 & -1 \\ 1 & 1 & 0 & 0 & 1 \\ -1 & 0 & -1 & 1 & 4 \\ 2 & 1 & 1 & -1 & -3 \end{array}\right] .$	$A=\begin{bmatrix} 1 & 0 & 1 & 0 & -1 \\ 1 & 1 & 0 & 0 & 1 \\ -1 & 0 & -1 & 1 & 4 \\ 2 & 1 & 1 & -1 & -3 \end{bmatrix} .$	$A=\begin{bmatrix} 1 & 0 & 1 & 0 & -1 \\ 1 & 1 & 0 & 0 & 1 \\ -1 & 0 & -1 & 1 & 4 \\ 2 & 1 & 1 & -1 & -3 \end{bmatrix} .$
johnston-linear-matrix-algebra-EQ0435	168	1.000	$\{(1,0,1,0,-1),(1,1,0,0,1),(-1,0,-1,1,4)\} \; .$	$\{(1,0,1,0,-1),(1,1,0,0,1),(-1,0,-1,1,4)\}.$	$\{(1,0,1,0,-1),(1,1,0,0,1),(-1,0,-1,1,4)\}.$
johnston-linear-matrix-algebra-EQ0436	168	1.000	$\left(x_{\{1\}},\,x_{\{2\}},\,x_{\{3\}},\,x_{\{4\}}\right)=x_{\{4\}}(0,-1,1,1),$	$(x_1,x_2,x_3,x_4)=x_4(0,-1,1,1),$	$(x_1,x_2,x_3,x_4)=x_4(0,-1,1,1),$

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johnston-linear-matrix-algebra-EQ0437	169	■ 1.000	$\operatorname{range}\left(A^T\right)=\operatorname{span}\left(\mathbf{a}_1, \mathbf{a}_2, \ldots, \mathbf{a}_m\right) .$	$\operatorname{range}\left(A^T\right)=\operatorname{span}\left(\mathbf{a}_1, \mathbf{a}_2, \ldots, \mathbf{a}_m\right) .$	$\operatorname{range}\left(A^T\right)=\operatorname{span}\left(\mathbf{a}_1, \mathbf{a}_2, \ldots, \mathbf{a}_m\right) .$
johnston-linear-matrix-algebra-EQ0438	169	■ 1.000	$A^T E^T=\left[A^T \mathbf{v}_1 \mid A^T \mathbf{v}_2 \mid \cdots \mid A^T \mathbf{v}_m\right]=R^T .$	$A^T E^T=\left[A^T \mathbf{v}_1 \mid A^T \mathbf{v}_2 \mid \cdots \mid A^T \mathbf{v}_m\right]=R^T .$	$A^T E^T=\left[A^T \mathbf{v}_1 \mid A^T \mathbf{v}_2 \mid \cdots \mid A^T \mathbf{v}_m\right]=R^T .$
johnston-linear-matrix-algebra-EQ0441	170	■ 1.000	$\left(x_1, x_2, x_3, x_4\right)=x_3(0,-1,1,0)+x_4(1,0,0,1) .$	$\left(x_1, x_2, x_3, x_4\right)=x_3(0,-1,1,0)+x_4(1,0,0,1) .$	$\left(x_1, x_2, x_3, x_4\right)=x_3(0,-1,1,0)+x_4(1,0,0,1) .$
johnston-linear-matrix-algebra-EQ0442	171	■ 1.000	$A=\left[\begin{array}{ccc} 1 & 1 & 2 \\ 2 & -1 & 1 \\ 1 & 0 & 1 \end{array}\right]$	$A=\left[\begin{array}{ccc} 1 & 1 & 2 \\ 2 & -1 & 1 \\ 1 & 0 & 1 \end{array}\right]$	$A=\left[\begin{array}{ccc} 1 & 1 & 2 \\ 2 & -1 & 1 \\ 1 & 0 & 1 \end{array}\right]$
johnston-linear-matrix-algebra-EQ0444	173	■ 1.000	$\operatorname{rank}(A)+\operatorname{nullity}(A)=r+(n-r)=n,$	$\operatorname{rank}(A)+\operatorname{nullity}(A)=r+(n-r)=n,$	$\operatorname{rank}(A)+\operatorname{nullity}(A)=r+(n-r)=n,$
johnston-linear-matrix-algebra-EQ0445	174	■ 1.000	$A=\left[\begin{array}{ccccc} 1 & 0 & -1 & 0 & 0 \\ 2 & 1 & -1 & 2 & 2 \\ -1 & -1 & 0 & -2 & -2 \\ 1 & 0 & -1 & 0 & 1 \end{array}\right] .$	$A=\left[\begin{array}{ccccc} 1 & 0 & -1 & 0 & 0 \\ 2 & 1 & -1 & 2 & 2 \\ -1 & -1 & 0 & -2 & -2 \\ 1 & 0 & -1 & 0 & 1 \end{array}\right] .$	$A=\left[\begin{array}{ccccc} 1 & 0 & -1 & 0 & 0 \\ 2 & 1 & -1 & 2 & 2 \\ -1 & -1 & 0 & -2 & -2 \\ 1 & 0 & -1 & 0 & 1 \end{array}\right] .$
johnston-linear-matrix-algebra-EQ0447	175	■ 1.000	$\operatorname{span}\left(\mathbf{a}_1+\mathbf{b}_1, \mathbf{a}_2+\mathbf{b}_2, \ldots, \mathbf{a}_n+\mathbf{b}_n\right) \subseteq \operatorname{span}\left(\mathbf{a}_1, \mathbf{b}_1, \mathbf{a}_2, \mathbf{b}_2, \ldots, \mathbf{a}_n, \mathbf{b}_n\right)$	$\operatorname{span}\left(\mathbf{a}_1+\mathbf{b}_1, \mathbf{a}_2+\mathbf{b}_2, \ldots, \mathbf{a}_n+\mathbf{b}_n\right) \subseteq \operatorname{span}\left(\mathbf{a}_1, \mathbf{b}_1, \mathbf{a}_2, \mathbf{b}_2, \ldots, \mathbf{a}_n, \mathbf{b}_n\right)$	$\operatorname{span}\left(\mathbf{a}_1+\mathbf{b}_1, \mathbf{a}_2+\mathbf{b}_2, \ldots, \mathbf{a}_n+\mathbf{b}_n\right) \subseteq \operatorname{span}\left(\mathbf{a}_1, \mathbf{b}_1, \mathbf{a}_2, \mathbf{b}_2, \ldots, \mathbf{a}_n, \mathbf{b}_n\right)$
johnston-linear-matrix-algebra-EQ0448	175	■ 1.000	$A B=\left[A \mathbf{b}_1 \mid A \mathbf{b}_2 \mid \cdots \mid A \mathbf{b}_p\right]=\left[\sum_{j=1}^n b_{j, 1} \mathbf{a}_j \mid \sum_{j=1}^n b_{j, 2} \mathbf{a}_j \mid \cdots \mid \sum_{j=1}^n b_{j, p} \mathbf{a}_j\right] .$	$A B=\left[A \mathbf{b}_1 \mid A \mathbf{b}_2 \mid \cdots \mid A \mathbf{b}_p\right]=\left[\sum_{j=1}^n b_{j, 1} \mathbf{a}_j \mid \sum_{j=1}^n b_{j, 2} \mathbf{a}_j \mid \cdots \mid \sum_{j=1}^n b_{j, p} \mathbf{a}_j\right] .$	$A B=\left[A \mathbf{b}_1 \mid A \mathbf{b}_2 \mid \cdots \mid A \mathbf{b}_p\right]=\left[\sum_{j=1}^n b_{j, 1} \mathbf{a}_j \mid \sum_{j=1}^n b_{j, 2} \mathbf{a}_j \mid \cdots \mid \sum_{j=1}^n b_{j, p} \mathbf{a}_j\right] .$
johnston-linear-matrix-algebra-EQ0449	177	■ 1.000	$\left[\begin{array}{ccccccc} 0 & 0 & 2 & 2 & -1 & -2 & 3 & -1 \\ -6 & -12 & -1 & -7 & 1 & 8 & -7 & 6 \\ 4 & 8 & -2 & 2 & 1 & -2 & 1 & -2 \\ -1 & -2 & 0 & -1 & 0 & 1 & -1 & 1 \end{array}\right]$	$\left[\begin{array}{ccccccc} 0 & 0 & 2 & 2 & -1 & -2 & 3 & -1 \\ -6 & -12 & -1 & -7 & 1 & 8 & -7 & 6 \\ 4 & 8 & -2 & 2 & 1 & -2 & 1 & -2 \\ -1 & -2 & 0 & -1 & 0 & 1 & -1 & 1 \end{array}\right]$	$\left[\begin{array}{ccccccc} 0 & 0 & 2 & 2 & -1 & -2 & 3 & -1 \\ -6 & -12 & -1 & -7 & 1 & 8 & -7 & 6 \\ 4 & 8 & -2 & 2 & 1 & -2 & 1 & -2 \\ -1 & -2 & 0 & -1 & 0 & 1 & -1 & 1 \end{array}\right]$
johnston-linear-matrix-algebra-EQ0450	177	■ 1.000	$\left[\begin{array}{cccccc} 5 & 1 & 9 & 6 & -1 & 1 & 8 \\ 2 & 2 & 2 & 4 & 1 & 0 & 0 \\ 5 & 4 & 6 & 9 & 1 & 1 & 4 \\ 6 & 3 & 9 & 9 & 0 & 1 & 7 \\ -3 & 1 & -7 & -2 & 2 & -1 & -8 \end{array}\right]$	$\left[\begin{array}{cccccc} 5 & 1 & 9 & 6 & -1 & 1 & 8 \\ 2 & 2 & 2 & 4 & 1 & 0 & 0 \\ 5 & 4 & 6 & 9 & 1 & 1 & 4 \\ 6 & 3 & 9 & 9 & 0 & 1 & 7 \\ -3 & 1 & -7 & -2 & 2 & -1 & -8 \end{array}\right]$	$\left[\begin{array}{cccccc} 5 & 1 & 9 & 6 & -1 & 1 & 8 \\ 2 & 2 & 2 & 4 & 1 & 0 & 0 \\ 5 & 4 & 6 & 9 & 1 & 1 & 4 \\ 6 & 3 & 9 & 9 & 0 & 1 & 7 \\ -3 & 1 & -7 & -2 & 2 & -1 & -8 \end{array}\right]$
johnston-linear-matrix-algebra-EQ0452	177	■ 1.000	$A=\left[\begin{array}{cccc} x & 1 & 1 & \cdots & 1 \\ 1 & x & 1 & \cdots & 1 \\ 1 & 1 & x & \cdots & 1 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 1 & 1 & 1 & \cdots & x \end{array}\right] \in \mathcal{M}_n .$	$A=\left[\begin{array}{cccc} x & 1 & 1 & \cdots & 1 \\ 1 & x & 1 & \cdots & 1 \\ 1 & 1 & x & \cdots & 1 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 1 & 1 & 1 & \cdots & x \end{array}\right] \in \mathcal{M}_n .$	$A=\left[\begin{array}{cccc} x & 1 & 1 & \cdots & 1 \\ 1 & x & 1 & \cdots & 1 \\ 1 & 1 & x & \cdots & 1 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 1 & 1 & 1 & \cdots & x \end{array}\right] \in \mathcal{M}_n .$

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johnston-linear-matrix-algebra-EQ0453	178	1.000	$B=\{(1,-1,2),(-1,2,3),(2,-2,4)\}$	$B = \{(1, -1, 2), (-1, 2, 3), (2, -2, 4)\}$	$B = \{(1, -1, 2), (-1, 2, 3), (2, -2, 4)\}$
johnston-linear-matrix-algebra-EQ0459	183	1.000	$\left(x_1, x_2, x_3, x_4, x_5\right)=(0,1,0,1,1) \quad \text { and } \quad \left(x_1, x_2, x_3, x_4, x_5\right)=(1,0,1,1,1) .$	$(x_1, x_2, x_3, x_4, x_5) = (0, 1, 0, 1, 1) \quad \text { and } \quad (x_1, x_2, x_3, x_4, x_5) = (1, 0, 1, 1, 1).$	$(x_1, x_2, x_3, x_4, x_5) = (0, 1, 0, 1, 1) \quad \text { and } \quad (x_1, x_2, x_3, x_4, x_5) = (1, 0, 1, 1, 1).$
johnston-linear-matrix-algebra-EQ0462	185	1.000	$\begin{aligned} \mathbf{v}_{\mathrm{s}}+\left(x_1 \mathbf{a}_1+x_2 \mathbf{a}_2+\cdots+x_{16} \mathbf{a}_{16}\right) &=\mathbf{v}_{\mathrm{e}}, \quad \text { or } \\ x_1 \mathbf{a}_1+x_2 \mathbf{a}_2+\cdots+x_{16} \mathbf{a}_{16} &=\mathbf{v}_{\mathrm{e}}-\mathbf{v}_{\mathrm{s}} . \end{aligned}$	$\mathbf{v}_{\mathrm{s}}+\left(x_1 \mathbf{a}_1+x_2 \mathbf{a}_2+\cdots+x_{16} \mathbf{a}_{16}\right)=\mathbf{v}_{\mathrm{e}}, \quad \text { or } \\ x_1 \mathbf{a}_1+x_2 \mathbf{a}_2+\cdots+x_{16} \mathbf{a}_{16} &=\mathbf{v}_{\mathrm{e}}-\mathbf{v}_{\mathrm{s}} .$	$\mathbf{v}_{\mathrm{s}}+\left(x_1 \mathbf{a}_1+x_2 \mathbf{a}_2+\cdots+x_{16} \mathbf{a}_{16}\right)=\mathbf{v}_{\mathrm{e}}, \quad \text { or } \\ x_1 \mathbf{a}_1+x_2 \mathbf{a}_2+\cdots+x_{16} \mathbf{a}_{16} &=\mathbf{v}_{\mathrm{e}}-\mathbf{v}_{\mathrm{s}} .$
johnston-linear-matrix-algebra-EQ0465	186	1.000	$\mathbf{x}=(0,0,1,0,1,0,0,0,0,0,0,1,0,1,0,0) .$	$\mathbf{x} = (0, 0, 1, 0, 1, 0, 0, 0, 0, 0, 0, 1, 0, 1, 0, 0).$	$\mathbf{x} = (0, 0, 1, 0, 1, 0, 0, 0, 0, 0, 0, 1, 0, 1, 0, 0).$
johnston-linear-matrix-algebra-EQ0467	190	1.000	$R=\left[\begin{array}{llllll} 1 & 0 & 0 & 0 & 0 & 1 \\ 0 & 1 & 1 & 0 & 0 & 1 \\ 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{array}\right] \quad \text { then } \quad \text {null}(A)=\text {span}\left(\left[\begin{array}{l} 0 \\ 1 \\ 0 \\ 0 \\ 0 \\ 0 \end{array}\right],\left[\begin{array}{l} 1 \\ 1 \\ 0 \\ 1 \\ 1 \\ 1 \end{array}\right]\right) .$	$R=\left[\begin{array}{cccccc} 1 & 0 & 0 & 0 & 0 & 1 \\ 0 & 1 & 1 & 0 & 0 & 1 \\ 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{array}\right] \quad \text { then } \quad \text {null}(A)=\text {span}\left(\left[\begin{array}{l} 0 \\ 1 \\ 0 \\ 0 \\ 0 \\ 0 \end{array}\right],\left[\begin{array}{l} 1 \\ 1 \\ 0 \\ 1 \\ 1 \\ 1 \end{array}\right]\right) .$	$R=\left[\begin{array}{cccccc} 1 & 0 & 0 & 0 & 0 & 1 \\ 0 & 1 & 1 & 0 & 0 & 1 \\ 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{array}\right] \quad \text { then } \quad \text {null}(A)=\text {span}\left(\left[\begin{array}{l} 0 \\ 1 \\ 0 \\ 0 \\ 0 \\ 0 \end{array}\right],\left[\begin{array}{l} 1 \\ 1 \\ 0 \\ 1 \\ 1 \\ 1 \end{array}\right]\right) .$
johnston-linear-matrix-algebra-EQ0471	195	1.000	$\begin{aligned} 2 w+4 y &=1 \quad \text { and } \quad 3 w+x+2 y=0 \quad \text { and } \quad 4 x+4 y+z=1 \quad \text { and } \quad 3 w+3 x+4 y+2 z=1 \end{aligned}$	$\begin{aligned} 2 w+4 y &=1 \\ 3 w+x+2 y &=0 \\ 4 x+4 y+z &=1 \\ 3 w+3 x+4 y+2 z &=1 \end{aligned}$	$\begin{aligned} 2 w+4 y &=1 \\ 3 w+x+2 y &=0 \\ 4 x+4 y+z &=1 \\ 3 w+3 x+4 y+2 z &=1 \end{aligned}$
johnston-linear-matrix-algebra-EQ0474	196	1.000	$\begin{aligned} x+y+z &=1 \quad \text { and } \quad x+y=1 \quad \text { and } \quad x+z=0 \end{aligned}$	$\begin{aligned} x+y+z &=1 \\ x+y &=1 \\ x+z &=0 \end{aligned}$	$\begin{aligned} x+y+z &=1 \\ x+y &=1 \\ x+z &=0 \end{aligned}$
johnston-linear-matrix-algebra-EQ0482	196	1.000	$\begin{array}{r} x+2 y+3 z=4 \\ 2 x+3 y+z=1 \\ 4 x+y+2 z=0 \end{array}$	$\begin{array}{r} x+2 y+3 z=4 \\ 2 x+3 y+z=1 \\ 4 x+y+2 z=0 \end{array}$	$\begin{array}{r} x+2 y+3 z=4 \\ 2 x+3 y+z=1 \\ 4 x+y+2 z=0 \end{array}$
johnston-linear-matrix-algebra-EQ0489	198	1.000	$A=\left[\begin{array}{ll} 2 & 1 \\ 1 & 3 \end{array}\right], \quad \mathbf{b}=(4,6), \quad \text { and } \quad \mathbf{c}=(1,3),$	$A=\left[\begin{array}{ll} 2 & 1 \\ 1 & 3 \end{array}\right], \quad \mathbf{b}=(4,6), \quad \text { and } \quad \mathbf{c}=(1,3),$	$A=\left[\begin{array}{ll} 2 & 1 \\ 1 & 3 \end{array}\right], \quad \mathbf{b}=(4,6), \quad \text { and } \quad \mathbf{c}=(1,3),$
johnston-linear-matrix-algebra-EQ0493	199	1.000	$A=\left[\begin{array}{cc} 1 & 1 \\ -1 & 1 \end{array}\right], \quad \mathbf{b}=(3,1), \quad \text { and } \quad \mathbf{c}=(1,2) .$	$A=\left[\begin{array}{cc} 1 & 1 \\ -1 & 1 \end{array}\right], \quad \mathbf{b}=(3,1), \quad \text { and } \quad \mathbf{c}=(1,2) .$	$A=\left[\begin{array}{cc} 1 & 1 \\ -1 & 1 \end{array}\right], \quad \mathbf{b}=(3,1), \quad \text { and } \quad \mathbf{c}=(1,2) .$
johnston-linear-matrix-algebra-EQ0496	200	1.000	$A=\left[\begin{array}{cc} -3 & 4 \\ 2 & 1 \\ -2 & -1 \end{array}\right], \quad \mathbf{b}=(-1,5,-5), \quad \text { and } \quad \mathbf{c}=(1,2) .$	$A=\left[\begin{array}{cc} -3 & 4 \\ 2 & 1 \\ -2 & -1 \end{array}\right], \quad \mathbf{b}=(-1,5,-5), \quad \text { and } \quad \mathbf{c}=(1,2) .$	$A=\left[\begin{array}{cc} -3 & 4 \\ 2 & 1 \\ -2 & -1 \end{array}\right], \quad \mathbf{b}=(-1,5,-5), \quad \text { and } \quad \mathbf{c}=(1,2) .$

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johnston-linear-matrix-algebra-EQ0499	200	1.000	$A=\left[\begin{array}{ccc} 1 & 3 & -3 \\ 3 & -2 & 2 \end{array}\right], \quad \mathbf{b}=(4,5), \quad \text{and} \quad \mathbf{c}=(2,-1,1) .$	$A = \begin{bmatrix} 1 & 3 & -3 \\ 3 & -2 & 2 \end{bmatrix}, \quad \mathbf{b} = (4,5), \quad \text{and} \quad \mathbf{c} = (2,-1,1).$	$A = \begin{bmatrix} 1 & 3 & -3 \\ 3 & -2 & 2 \end{bmatrix}, \quad \mathbf{b} = (4,5), \quad \text{and} \quad \mathbf{c} = (2,-1,1).$
johnston-linear-matrix-algebra-EQ0503	201	1.000	$A=\left[\begin{array}{ccc} 1 & -1 & 3 \\ & -3 & -1 \end{array}\right], \quad \mathbf{b}=(4,-4,-3), \quad \text{and} \quad \mathbf{c}=(-3,3,2) .$	$A = \begin{bmatrix} 1 & -1 & 3 \\ -1 & 1 & -3 \\ -1 & 1 & -2 \end{bmatrix}, \quad \mathbf{b} = (4,-4,-3), \quad \text{and} \quad \mathbf{c} = (-3,3,2).$	$A = \begin{bmatrix} 1 & -1 & 3 \\ -1 & 1 & -3 \\ -1 & 1 & -2 \end{bmatrix}, \quad \mathbf{b} = (4,-4,-3), \quad \text{and} \quad \mathbf{c} = (-3,3,2).$
johnston-linear-matrix-algebra-EQ0514	206	1.000	$x_{\{1\}}=x_{\{2\}}=x_{\{3\}}=0, \quad s_{\{1\}}=1, \quad s_{\{2\}}=2, \quad s_{\{3\}}=3, \quad z=0 .$	$x_1 = x_2 = x_3 = 0, \quad s_1 = 1, s_2 = 2, s_3 = 3, \quad z = 0.$	$x_1 = x_2 = x_3 = 0, \quad s_1 = 1, s_2 = 2, s_3 = 3, \quad z = 0.$
johnston-linear-matrix-algebra-EQ0515	206	1.000	$\begin{aligned} & -3 \, x_{\{1\}} - 2 \, x_{\{2\}} + 2 \, x_{\{3\}} \, \& \; = -3 \, x_{\{1\}} \\ & \leq 1 \, \& \; x_{\{1\}} - x_{\{2\}} + x_{\{3\}} \, \& \; = x_{\{1\}} \, \leq 2 \, \& \; - x_{\{1\}} + 3 \, x_{\{2\}} - 3 \, x_{\{3\}} \, \& \; = -x_{\{1\}} \, \leq 3 \, . \end{aligned}$	$\begin{aligned} -3x_1 - 2x_2 + 2x_3 &= -3x_1 \leq 1 \\ x_1 - x_2 + x_3 &= x_1 \leq 2 \\ -x_1 + 3x_2 - 3x_3 &= -x_1 \leq 3. \end{aligned}$	$\begin{aligned} -3x_1 - 2x_2 + 2x_3 &= -3x_1 \leq 1 \\ x_1 - x_2 + x_3 &= x_1 \leq 2 \\ -x_1 + 3x_2 - 3x_3 &= -x_1 \leq 3. \end{aligned}$
johnston-linear-matrix-algebra-EQ0516	206	1.000	$\left[\begin{array}{c ccc ccc} 1 & 0 & 0 & 0 & -2 & -1 & 1 & 0 \\ 0 & 1 & 0 & 0 & -3 & -2 & 2 & 1 \\ 0 & 0 & 1 & 0 & 1 & -1 & 1 & 2 \\ 0 & 0 & 0 & 1 & -1 & 3 & -3 & 3 \end{array}\right]$	$\left[\begin{array}{c ccc ccc} 1 & 0 & 0 & 0 & -2 & -1 & 1 & 0 \\ 0 & 1 & 0 & 0 & -3 & -2 & 2 & 1 \\ 0 & 0 & 1 & 0 & 1 & -1 & 1 & 2 \\ 0 & 0 & 0 & 1 & -1 & 3 & -3 & 3 \end{array}\right]$	$\left[\begin{array}{c ccc ccc} 1 & 0 & 0 & 0 & -2 & -1 & 1 & 0 \\ 0 & 1 & 0 & 0 & -3 & -2 & 2 & 1 \\ 0 & 0 & 1 & 0 & 1 & -1 & 1 & 2 \\ 0 & 0 & 0 & 1 & -1 & 3 & -3 & 3 \end{array}\right]$
johnston-linear-matrix-algebra-EQ0518	206	1.000	$s_{\{2\}}=x_{\{2\}}=x_{\{3\}}=0, \quad s_{\{1\}}=7, \quad x_{\{1\}}=2, \quad s_{\{3\}}=5, \quad z=4 .$	$s_2 = x_2 = x_3 = 0, \quad s_1 = 7, x_1 = 2, s_3 = 5, \quad z = 4.$	$s_2 = x_2 = x_3 = 0, \quad s_1 = 7, x_1 = 2, s_3 = 5, \quad z = 4.$
johnston-linear-matrix-algebra-EQ0529	214	1.000	$\left[\begin{array}{c ccc cc c} 1 & 1 & 0 & 0 & 0 & -1 & 2 \\ 0 & 1 & 0 & 0 & 1 & 1 & 2 \\ 0 & 1 & 1 & 0 & 0 & 0 & 1 \\ 0 & 2 & 0 & 1 & 0 & 3 & 2 \end{array}\right]$	$\left[\begin{array}{c ccc cc c} 1 & 1 & 0 & 0 & 0 & -1 & 2 \\ 0 & 1 & 0 & 0 & 1 & 1 & 2 \\ 0 & 1 & 1 & 0 & 0 & 0 & 1 \\ 0 & 2 & 0 & 1 & 0 & 3 & 2 \end{array}\right]$	$\left[\begin{array}{c ccc ccc} 1 & 1 & 0 & 0 & 0 & -1 & 2 \\ 0 & 1 & 0 & 0 & 1 & 1 & 2 \\ 0 & 1 & 1 & 0 & 0 & 0 & 1 \\ 0 & 2 & 0 & 1 & 0 & 3 & 2 \end{array}\right]$
johnston-linear-matrix-algebra-EQ0532	215	1.000	$x_{\{2\}}+\left(x_{\{1\}}+x_{\{2\}}\right)=x_{\{1\}}+2 \, x_{\{2\}} \, \leq 2+3=5 .$	$x_2 + (x_1 + x_2) = x_1 + 2x_2 \leq 2 + 3 = 5.$	$x_2 + (x_1 + x_2) = x_1 + 2x_2 \leq 2 + 3 = 5.$
johnston-linear-matrix-algebra-EQ0534	215	1.000	$x_{\{1\}}+2 \, x_{\{2\}} \, \leq y_{\{1\}}\left(x_{\{1\}}+x_{\{2\}}\right)+y_{\{2\}} \left(-x_{\{1\}}+x_{\{2\}}\right),$	$x_1 + 2x_2 \leq y_1 (x_1 + x_2) + y_2 (-x_1 + x_2),$	$x_1 + 2x_2 \leq y_1 (x_1 + x_2) + y_2 (-x_1 + x_2),$
johnston-linear-matrix-algebra-EQ0537	216	1.000	$\mathbf{c} \cdot \mathbf{x} \leq \mathbf{b} \cdot \mathbf{y} .$	$\mathbf{c} \cdot \mathbf{x} \leq \mathbf{b} \cdot \mathbf{y}.$	$\mathbf{c} \cdot \mathbf{x} \leq \mathbf{b} \cdot \mathbf{y}.$
johnston-linear-matrix-algebra-EQ0539	216	1.000	$A=\left[\begin{array}{cc} 1 & 1 \\ -1 & 1 \end{array}\right], \quad \mathbf{b}=(3,1), \quad \text{and} \quad \mathbf{c}=(1,2),$	$A = \begin{bmatrix} 1 & 1 \\ -1 & 1 \end{bmatrix}, \quad \mathbf{b} = (3,1), \quad \text{and} \quad \mathbf{c} = (1,2),$	$A = \begin{bmatrix} 1 & 1 \\ -1 & 1 \end{bmatrix}, \quad \mathbf{b} = (3,1), \quad \text{and} \quad \mathbf{c} = (1,2),$
johnston-linear-matrix-algebra-EQ0542	217	1.000	$x_{\{1\}}+x_{\{2\}}+x_{\{3\}}+x_{\{4\}}+x_{\{5\}}=1+1+1+1+1=5$	$x_1 + x_2 + x_3 + x_4 + x_5 = 1 + 1 + 1 + 1 + 1 = 5$	$x_1 + x_2 + x_3 + x_4 + x_5 = 1 + 1 + 1 + 1 + 1 = 5$
johnston-linear-matrix-algebra-EQ0544	218	1.000	$\mathbf{c} \cdot \mathbf{x}_* = \mathbf{b} \cdot \mathbf{y}_* .$	$\mathbf{c} \cdot \mathbf{x}_* = \mathbf{b} \cdot \mathbf{y}_*.$	$\mathbf{c} \cdot \mathbf{x}_* = \mathbf{b} \cdot \mathbf{y}_*.$
johnston-linear-matrix-algebra-EQ0550	221	1.000	$\begin{aligned} & \leq 0 \\ & \text{unc} \\ & \geq 0 \end{aligned}$	$\begin{aligned} & \leq 0 \\ & \text{unc} \\ & \geq 0 \end{aligned}$	$\begin{aligned} & \leq 0 \\ & \text{unc} \\ & \geq 0 \end{aligned}$
johnston-linear-matrix-algebra-EQ0551	221	1.000	unconstrained	<i>unconstrained</i>	unconstrained

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johnston-linear-matrix-algebra-EQ0552	221	1.000	$\begin{aligned} & \geq 0 \quad \& \text{unconstrained} \\ & \leq 0 \end{aligned}$	≥ 0 unconstrained ≤ 0	≥ 0 unconstrained ≤ 0
johnston-linear-matrix-algebra-EQ0553	221	1.000	$\begin{aligned} & \geq \\ & = \\ & \leq \end{aligned}$	$\geq$ $=$ $\leq$	$\geq$ $=$ $\leq$
johnston-linear-matrix-algebra-EQ0573	223	1.000	$\operatorname{rank}(A \ B) \leq \min \{ \operatorname{rank}(A), \operatorname{rank}(B) \} .$	$\operatorname{rank}(AB) \leq \min \{ \operatorname{rank}(A), \operatorname{rank}(B) \} .$	$\operatorname{rank}(AB) \leq \min \{ \operatorname{rank}(A), \operatorname{rank}(B) \} .$
johnston-linear-matrix-algebra-EQ0574	223	1.000	$\operatorname{rank}(A)=\operatorname{rank}(C \ R) \leq \min \{ \operatorname{rank}(C), \operatorname{rank}(R) \} \leq \min \{ m, r, n \} \leq r .$	$\operatorname{rank}(A) = \operatorname{rank}(CR) \leq \min \{ \operatorname{rank}(C), \operatorname{rank}(R) \} \leq \min \{ m, r, n \} \leq r .$	$\operatorname{rank}(A) = \operatorname{rank}(CR) \leq \min \{ \operatorname{rank}(C), \operatorname{rank}(R) \} \leq \min \{ m, r, n \} \leq r .$
johnston-linear-matrix-algebra-EQ0577	224	1.000	$A=P \ \hat{R}=[C \ \mid D]\left[\frac{R}{0}\right]=C \ R+D \ 0=C \ R,$	$A=P\hat{R}=[C \mid D]\left[\frac{R}{0}\right]=CR+DO=CR,$	$A=P\hat{R}=[C \mid D]\left[\frac{R}{0}\right]=CR+DO=CR,$
johnston-linear-matrix-algebra-EQ0578	224	1.000	$\left[\begin{array}{cccc ccc} 1 & 0 & 0 & -1 & 1 & -1 & 0 & 0 \\ 0 & 1 & 1 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & -2 & 1 & 0 \end{array}\right] .$	$\left[\begin{array}{cccc ccc} 1 & 0 & 0 & -1 & 1 & -1 & 0 & 0 \\ 0 & 1 & 1 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & -2 & 1 & 0 \end{array}\right] .$	$\left[\begin{array}{cccc ccc} 1 & 0 & 0 & -1 & 1 & -1 & 0 & 0 \\ 0 & 1 & 1 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & -2 & 1 & 0 \end{array}\right] .$
johnston-linear-matrix-algebra-EQ0580	225	1.000	$A=C \ R, \quad \text{where} \quad C=\left[\begin{array}{cc} 1 & 1 \\ 0 & 1 \\ -1 & 1 \end{array}\right] \quad \text{and} \quad R=\left[\begin{array}{cccc} 1 & 0 & 0 & -1 \\ 0 & 1 & 1 & 0 \end{array}\right] .$	$A=CR, \quad \text{where} \quad C=\left[\begin{array}{cc} 1 & 1 \\ 0 & 1 \\ -1 & 1 \end{array}\right] \quad \text{and} \quad R=\left[\begin{array}{cccc} 1 & 0 & 0 & -1 \\ 0 & 1 & 1 & 0 \end{array}\right] .$	$A=CR, \quad \text{where} \quad C=\left[\begin{array}{cc} 1 & 1 \\ 0 & 1 \\ -1 & 1 \end{array}\right] \quad \text{and} \quad R=\left[\begin{array}{cccc} 1 & 0 & 0 & -1 \\ 0 & 1 & 1 & 0 \end{array}\right] .$
johnston-linear-matrix-algebra-EQ0581	225	1.000	$A=\left[\begin{array}{cccccccccc} 1 & 0 & -1 & 0 & 2 & 0 & 2 & 0 & -1 & 0 & 1 \\ & 2 & 0 & -1 & 0 & 1 & 0 & 2 & 0 & 1 & 0 & 3 & 0 & 1 & 0 & 0 \\ & 2 & 0 & 1 & 0 & -1 & 0 & 2 & 0 & 0 & 2 & 0 & 0 & -1 & 0 & 0 & 1 \\ & 0 & 4 & 0 & 2 & 0 & 6 & 0 & 2 & 0 & 0 & 4 & 0 & 0 \\ & 1 & 0 & -1 & 0 & 2 & 0 & 2 & 0 & -1 & 0 & 1 \\ & 0 & -2 & 0 & -1 & 0 & -3 & 0 & -1 & 0 & -2 & 0 \\ & 1 & 0 & -1 & 0 & 2 & 0 & 2 & 0 & -1 & 0 & 1 \\ & 0 & 4 & 0 & 2 & 0 & 6 & 0 & 2 & 0 & 0 & 4 & 0 \\ & 1 & 0 & -1 & 0 & 2 & 0 & 2 & 0 & -1 & 0 & 1 \\ & 0 & 2 & 0 & 1 & 0 & 3 & 0 & 1 & 0 & 2 & 0 \\ & 1 & 0 & -1 & 0 & 2 & 0 & 2 & 0 & -1 & 0 & 1 \end{array}\right]$	$A=\left[\begin{array}{cccccccccc} 1 & 0 & -1 & 0 & 2 & 0 & 2 & 0 & -1 & 0 & 1 \\ 0 & 2 & 0 & 1 & 0 & 3 & 0 & 1 & 0 & 2 & 0 \\ 1 & 0 & -1 & 0 & 2 & 0 & 2 & 0 & -1 & 0 & 1 \\ 0 & 4 & 0 & 2 & 0 & 6 & 0 & 2 & 0 & 0 & 4 & 0 \\ 1 & 0 & -1 & 0 & 2 & 0 & 2 & 0 & -1 & 0 & 1 \\ 0 & -2 & 0 & -1 & 0 & -3 & 0 & -1 & 0 & -2 & 0 \\ 1 & 0 & -1 & 0 & 2 & 0 & 2 & 0 & -1 & 0 & 1 \\ 0 & 4 & 0 & 2 & 0 & 6 & 0 & 2 & 0 & 0 & 4 & 0 \\ 1 & 0 & -1 & 0 & 2 & 0 & 2 & 0 & -1 & 0 & 1 \\ 0 & 2 & 0 & 1 & 0 & 3 & 0 & 1 & 0 & 2 & 0 \\ 1 & 0 & -1 & 0 & 2 & 0 & 2 & 0 & -1 & 0 & 1 \end{array}\right]$	$A=\left[\begin{array}{cccccccccc} 1 & 0 & -1 & 0 & 2 & 0 & 2 & 0 & -1 & 0 & 1 \\ 0 & 2 & 0 & 1 & 0 & 3 & 0 & 1 & 0 & 2 & 0 \\ 1 & 0 & -1 & 0 & 2 & 0 & 2 & 0 & -1 & 0 & 1 \\ 0 & 4 & 0 & 2 & 0 & 6 & 0 & 2 & 0 & 0 & 4 & 0 \\ 1 & 0 & -1 & 0 & 2 & 0 & 2 & 0 & -1 & 0 & 1 \\ 0 & -2 & 0 & -1 & 0 & -3 & 0 & -1 & 0 & -2 & 0 \\ 1 & 0 & -1 & 0 & 2 & 0 & 2 & 0 & -1 & 0 & 1 \\ 0 & 4 & 0 & 2 & 0 & 6 & 0 & 2 & 0 & 0 & 4 & 0 \\ 1 & 0 & -1 & 0 & 2 & 0 & 2 & 0 & -1 & 0 & 1 \\ 0 & 2 & 0 & 1 & 0 & 3 & 0 & 1 & 0 & 2 & 0 \\ 1 & 0 & -1 & 0 & 2 & 0 & 2 & 0 & -1 & 0 & 1 \end{array}\right]$
johnston-linear-matrix-algebra-EQ0583	226	1.000	$A=\sum_{j=1}^r \mathbf{v}_j \mathbf{w}_j^T$	$A=\sum_{j=1}^r \mathbf{v}_j \mathbf{w}_j^T$	$A=\sum_{j=1}^r \mathbf{v}_j \mathbf{w}_j^T$
johnston-linear-matrix-algebra-EQ0584	226	1.000	$C=\left[\mathbf{v}_1 \mid \mathbf{v}_2 \mid \cdots \mid \mathbf{v}_r\right] \quad \text{and} \quad R^T=\left[\mathbf{w}_1 \mid \mathbf{w}_2 \mid \cdots \mid \mathbf{w}_r\right] .$	$C=\left[\mathbf{v}_1 \mid \mathbf{v}_2 \mid \cdots \mid \mathbf{v}_r\right] \quad \text{and} \quad R^T=\left[\mathbf{w}_1 \mid \mathbf{w}_2 \mid \cdots \mid \mathbf{w}_r\right] .$	$C=\left[\mathbf{v}_1 \mid \mathbf{v}_2 \mid \cdots \mid \mathbf{v}_r\right] \quad \text{and} \quad R^T=\left[\mathbf{w}_1 \mid \mathbf{w}_2 \mid \cdots \mid \mathbf{w}_r\right] .$

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johnston-linear-matrix-algebra-EQ0586	226	1.000	$\operatorname{rank}(A)=\operatorname{rank}\left(\sum_{j=1}^r\mathbf{v}_j\mathbf{w}_j^T\right)\leq\sum_{j=1}^r\operatorname{rank}\left(\mathbf{v}_j\mathbf{w}_j^T\right)=\sum_{j=1}^r1=r,$		$\operatorname{rank}(A)=\operatorname{rank}\left(\sum_{j=1}^r\mathbf{v}_j\mathbf{w}_j^T\right)\leq\sum_{j=1}^r\operatorname{rank}\left(\mathbf{v}_j\mathbf{w}_j^T\right)=\sum_{j=1}^r1=r,$
johnston-linear-matrix-algebra-EQ0587	226	1.000	$A=\left[\begin{array}{cccc}1&1&1&-1\\0&1&1&0\\-1&1&1&1\end{array}\right].$	$A=\left[\begin{array}{cccc}1&1&1&-1\\0&1&1&0\\-1&1&1&1\end{array}\right].$	$A=\left[\begin{array}{cccc}1&1&1&-1\\0&1&1&0\\-1&1&1&1\end{array}\right].$
johnston-linear-matrix-algebra-EQ0588	227	1.000	$A=CR, \quad \text{where } C=\left[\begin{array}{cc}1&1\\0&1\\-1&1\end{array}\right] \quad \text{and } R=\left[\begin{array}{cccc}1&0&0&-1\\0&1&1&0\end{array}\right].$	$A=CR, \quad \text{where } C=\left[\begin{array}{cc}1&1\\0&1\\-1&1\end{array}\right] \quad \text{and } R=\left[\begin{array}{cccc}1&0&0&-1\\0&1&1&0\end{array}\right].$	$A=CR, \quad \text{where } C=\left[\begin{array}{cc}1&1\\0&1\\-1&1\end{array}\right] \quad \text{and } R=\left[\begin{array}{cccc}1&0&0&-1\\0&1&1&0\end{array}\right].$
johnston-linear-matrix-algebra-EQ0590	227	1.000	$A=\left[\begin{array}{cccccc}1&2&3&4&5\\6&7&8&9&10\\8&9&10&11&12&13&14&15\end{array}\right]$	$A=\left[\begin{array}{ccccc}1&2&3&4&5\\6&7&8&9&10\\11&12&13&14&15\end{array}\right]$	$A=\left[\begin{array}{ccccc}1&2&3&4&5\\6&7&8&9&10\\11&12&13&14&15\end{array}\right]$
johnston-linear-matrix-algebra-EQ0591	227	1.000	$A=\left[\begin{array}{ccc}1&2&2\\2&1&-1\\1&0&0\end{array}\right]$	$A=\left[\begin{array}{ccc}1&2&2\\2&1&-1\\1&1&0\end{array}\right]$	$A=\left[\begin{array}{ccc}1&2&2\\2&1&-1\\1&1&0\end{array}\right]$
johnston-linear-matrix-algebra-EQ0592	228	1.000	$\operatorname{rank}(A)=\operatorname{dim}(\operatorname{range}(A))=\operatorname{dim}(\operatorname{span}(\mathbf{a}_1,\mathbf{a}_2,\dots,\mathbf{a}_n)).$	$\operatorname{rank}(A)=\operatorname{dim}(\operatorname{range}(A))=\operatorname{dim}(\operatorname{span}(\mathbf{a}_1,\mathbf{a}_2,\dots,\mathbf{a}_n)).$	$\operatorname{rank}(A)=\operatorname{dim}(\operatorname{range}(A))=\operatorname{dim}(\operatorname{span}(\mathbf{a}_1,\mathbf{a}_2,\dots,\mathbf{a}_n)).$
johnston-linear-matrix-algebra-EQ0593	229	1.000	$\operatorname{rank}(C)=\operatorname{dim}(\operatorname{range}(C^T))=\operatorname{dim}(\operatorname{span}(\mathbf{c}_1,\mathbf{c}_2,\dots,\mathbf{c}_m)).$	$\operatorname{rank}(C)=\operatorname{dim}(\operatorname{range}(C^T))=\operatorname{dim}(\operatorname{span}(\mathbf{c}_1,\mathbf{c}_2,\dots,\mathbf{c}_m)).$	$\operatorname{rank}(C)=\operatorname{dim}(\operatorname{range}(C^T))=\operatorname{dim}(\operatorname{span}(\mathbf{c}_1,\mathbf{c}_2,\dots,\mathbf{c}_m)).$
johnston-linear-matrix-algebra-EQ0596	230	1.000	$\left[\begin{array}{cccccc}4&5&0&4&1&3\\2&2&1&2&1&1\\-4&0&3&-2&-3&1\\3&5&4&4&1&3\end{array}\right]$	$\left[\begin{array}{cccccc}4&5&0&4&1&3\\2&2&1&2&1&1\\-4&0&3&-2&-3&1\\3&5&4&4&1&3\end{array}\right]$	$\left[\begin{array}{cccccc}4&5&0&4&1&3\\2&2&1&2&1&1\\-4&0&3&-2&-3&1\\3&5&4&4&1&3\end{array}\right]$
johnston-linear-matrix-algebra-EQ0597	230	1.000	$\left[\begin{array}{cccccc} -1&0&3&-1&2&-1&-2\\3&1&-3&2&-2&1&4\\3&2&3&1&2&-1&2\\6&6&2&0&2&0&6\\6&4&6&2&4&-2&4\end{array}\right]$	$\left[\begin{array}{cccccc} -1&0&3&-1&2&-1&-2\\3&1&-3&2&-2&1&4\\3&2&3&1&2&-1&2\\6&6&2&0&2&0&6\\6&4&6&2&4&-2&4\end{array}\right]$	$\left[\begin{array}{cccccc} -1&0&3&-1&2&-1&-2\\3&1&-3&2&-2&1&4\\3&2&3&1&2&-1&2\\6&6&2&0&2&0&6\\6&4&6&2&4&-2&4\end{array}\right]$
johnston-linear-matrix-algebra-EQ0598	230	1.000	$D=\left[\begin{array}{cc}I_{\operatorname{rank}(A)}&O\\O&O\end{array}\right].$	$D=\left[\begin{array}{cc}I_{\operatorname{rank}(A)}&O\\O&O\end{array}\right].$	$D=\left[\begin{array}{cc}I_{\operatorname{rank}(A)}&O\\O&O\end{array}\right].$

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johnston-linear-matrix-algebra- EQ0599	230	1.000	$V=\left[\begin{array}{ccccc} 1 & a_{\{0\}} & a_{\{0\}}^2 & \cdots & a_{\{0\}}^n \\ \cdots & a_{\{0\}}^{\{n\}} & \backslash \ 1 & a_{\{1\}} & a_{\{1\}}^2 & \cdots & a_{\{1\}}^{\{n\}} \\ \backslash \vdots & \vdots & \vdots & \ddots & \vdots & \vdots & \vdots \\ \backslash \ 1 & a_{\{m\}} & a_{\{m\}}^2 & \cdots & a_{\{m\}}^{\{n\}} \end{array}\right] \ .$	$V = \begin{bmatrix} 1 & a_0 & a_0^2 & \cdots & a_0^n \\ 1 & a_1 & a_1^2 & \cdots & a_1^n \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 1 & a_m & a_m^2 & \cdots & a_m^n \end{bmatrix} \ .$	$V = \begin{bmatrix} 1 & a_0 & a_0^2 & \cdots & a_0^n \\ 1 & a_1 & a_1^2 & \cdots & a_1^n \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 1 & a_m & a_m^2 & \cdots & a_m^n \end{bmatrix} \ .$
johnston-linear-matrix-algebra- EQ0600	231	1.000	$A=\left[\begin{array}{ccc} 1 & -2 & -1 \\ 2 & -1 & -1 \\ 3 & 6 & 2 \end{array}\right]\backslash\right]$	$A = \begin{bmatrix} 1 & -2 & -1 \\ 2 & -1 & -1 \\ 3 & 6 & 2 \end{bmatrix}$	$A = \begin{bmatrix} 1 & -2 & -1 \\ 2 & -1 & -1 \\ 3 & 6 & 2 \end{bmatrix}$
johnston-linear-matrix-algebra- EQ0603	232	1.000	$\backslash\left[\begin{array}{llll} 1 & 2 & 3 & 4 \\ 0 & 5 & 6 & 7 \\ 0 & 0 & 8 & 9 \end{array}\right]\backslash\right]$.	$\begin{bmatrix} 1 & 2 & 3 & 4 \\ 0 & 5 & 6 & 7 \\ 0 & 0 & 8 & 9 \end{bmatrix} \ .$	$\begin{bmatrix} 1 & 2 & 3 & 4 \\ 0 & 5 & 6 & 7 \\ 0 & 0 & 8 & 9 \end{bmatrix} \ .$
johnston-linear-matrix-algebra- EQ0605	233	1.000	$U=\left[\begin{array}{cccc} 2 & 1 & -2 & 1 \\ 0 & 0 & 3 & 1 \\ 0 & 0 & 3 & 1 \end{array}\right]\backslash\right]$ \text { and } L=\left[\begin{array}{ccc} 1 & 0 & 0 \\ 2 & 1 & 0 \\ -3 & 1 & 0 \end{array}\right]\backslash\right]^{-1}=\left[\begin{array}{ccc} 1 & 0 & 0 \\ -2 & 1 & 0 \\ 1 & 3 & 1 \end{array}\right] \ .	$U = \begin{bmatrix} 2 & 1 & -2 & 1 \\ 0 & -2 & -1 & 3 \\ 0 & 0 & 3 & 1 \end{bmatrix} \text{ and } L = \begin{bmatrix} 1 & 0 & 0 \\ 2 & 1 & 0 \\ -7 & -3 & 1 \end{bmatrix}^{-1} = \begin{bmatrix} 1 & 0 & 0 \\ -2 & 1 & 0 \\ 1 & 3 & 1 \end{bmatrix} \ .$	$U = \begin{bmatrix} 2 & 1 & -2 & 1 \\ 0 & -2 & -1 & 3 \\ 0 & 0 & 3 & 1 \end{bmatrix} \text{ and } L = \begin{bmatrix} 1 & 0 & 0 \\ 2 & 1 & 0 \\ -7 & -3 & 1 \end{bmatrix}^{-1} = \begin{bmatrix} 1 & 0 & 0 \\ -2 & 1 & 0 \\ 1 & 3 & 1 \end{bmatrix} \ .$
johnston-linear-matrix-algebra- EQ0606	233	1.000	$A=\left[\begin{array}{ll} 2 & 4 \\ 2 & 7 \end{array}\right]\backslash\right]$ \quad \text { then } \ A=\left[\begin{array}{ll} 1 & 0 \\ 1 & 1 \end{array}\right]\left[\begin{array}{ll} 2 & 4 \\ 0 & 3 \end{array}\right]\backslash\right]=\left[\begin{array}{ll} 2 & 0 \\ 2 & 3 \end{array}\right]\left[\begin{array}{ll} 1 & 2 \\ 0 & 1 \end{array}\right]\backslash\right]=\left[\begin{array}{lll} 2 & 0 \\ 2 & 3 \end{array}\right]\backslash\right]=\left[\begin{array}{lll} 1 & 2 \\ 0 & 1 \end{array}\right]\backslash\right]	$A = \begin{bmatrix} 2 & 4 \\ 2 & 7 \end{bmatrix} \quad \text{then} \quad A = \begin{bmatrix} 1 & 0 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} 2 & 4 \\ 0 & 3 \end{bmatrix} = \begin{bmatrix} 2 & 0 \\ 2 & 3 \end{bmatrix} \begin{bmatrix} 1 & 2 \\ 0 & 1 \end{bmatrix}$	$A = \begin{bmatrix} 2 & 4 \\ 2 & 7 \end{bmatrix} \quad \text{then} \quad A = \begin{bmatrix} 1 & 0 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} 2 & 4 \\ 0 & 3 \end{bmatrix} = \begin{bmatrix} 2 & 0 \\ 2 & 3 \end{bmatrix} \begin{bmatrix} 1 & 2 \\ 0 & 1 \end{bmatrix}$
johnston-linear-matrix-algebra- EQ0607 (2.D.2)	234	1.000	$\backslash\mathbf{a}_{\{i\}}^{\{T\}}=\mathbf{u}_{\{i\}}^{\{T\}}+\sum_{j=1}^{\{i-1\}}\backslash\ell_{\{i,\ j\}}\backslash\mathbf{u}_{\{j\}}^{\{T\}}\backslash\quad\backslash\text { for all } \backslash\quad\backslash\quad\backslash\leq i \backslash\leq m \ .$	$\mathbf{a}_i^T = \mathbf{u}_i^T + \sum_{j=1}^{i-1} \ell_{i,j} \mathbf{u}_j^T \quad \text{for all} \quad 1 \leq i \leq m.$	$\mathbf{a}_i^T = \mathbf{u}_i^T + \sum_{j=1}^{i-1} \ell_{i,j} \mathbf{u}_j^T \quad \text{for all} \quad 1 \leq i \leq m.$
johnston-linear-matrix-algebra- EQ0609	235	1.000	$\backslash\mathbf{u}_{\{m\}}^{\{T\}}+\ell_{\{m,\ 1\}}\backslash\mathbf{u}_{\{1\}}^{\{T\}}+\backslash\ell_{\{m,\ 2\}}\backslash\mathbf{u}_{\{2\}}^{\{T\}}+\cdots+\ell_{\{m,m-1\}}\backslash\mathbf{u}_{\{m-1\}}^{\{T\}},$	$\mathbf{u}_m^T + \ell_{m,1} \mathbf{u}_1^T + \ell_{m,2} \mathbf{u}_2^T + \cdots + \ell_{m,m-1} \mathbf{u}_{m-1}^T,$	$\mathbf{u}_m^T + \ell_{m,1} \mathbf{u}_1^T + \ell_{m,2} \mathbf{u}_2^T + \cdots + \ell_{m,m-1} \mathbf{u}_{m-1}^T,$
johnston-linear-matrix-algebra- EQ0613	236	1.000	$L=\left[\begin{array}{cccc} 1 & 0 & 0 & 0 \\ 2 & 1 & 0 & 0 \\ -3 & 3 & 1 & 0 \\ -1 & 2 & -2 & 1 \end{array}\right]\backslash\right]$ \quad \text { and } \quad U=\left[\begin{array}{cccc} 2 & 4 & -1 & -1 \\ 0 & 1 & 2 & 1 \\ 0 & 0 & -2 & 0 \\ 0 & 0 & 0 & -3 \end{array}\right]\backslash\right]	$L = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 2 & 1 & 0 & 0 \\ -3 & 3 & 1 & 0 \\ -1 & 2 & -2 & 1 \end{bmatrix} \quad \text{and} \quad U = \begin{bmatrix} 2 & 4 & -1 & -1 \\ 0 & 1 & 2 & 1 \\ 0 & 0 & -2 & 0 \\ 0 & 0 & 0 & -3 \end{bmatrix} \ .$	$L = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 2 & 1 & 0 & 0 \\ -3 & 3 & 1 & 0 \\ -1 & 2 & -2 & 1 \end{bmatrix} \quad \text{and} \quad U = \begin{bmatrix} 2 & 4 & -1 & -1 \\ 0 & 1 & 2 & 1 \\ 0 & 0 & -2 & 0 \\ 0 & 0 & 0 & -3 \end{bmatrix} \ .$
johnston-linear-matrix-algebra- EQ0614	236	1.000	$A=\left[\begin{array}{ll} 0 & 0 \\ 0 & 1 \end{array}\right]\backslash\right],$	$A = \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix},$	$A = \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix},$
johnston-linear-matrix-algebra- EQ0615	237	1.000	$A \backslash\mathbf{x}=L \ U \ \backslash\mathbf{x}=L(U \ \backslash\mathbf{x})=L \ \backslash\mathbf{y}=\backslash\mathbf{b},$	$A \mathbf{x} = L U \mathbf{x} = L(U \mathbf{x}) = L \mathbf{y} = \mathbf{b},$	$A \mathbf{x} = L U \mathbf{x} = L(U \mathbf{x}) = L \mathbf{y} = \mathbf{b},$
johnston-linear-matrix-algebra- EQ0622	239	1.000	$p(x)=2 \ x^{\{3\}}-5 \ x^{\{2\}}-x+3 \ .$	$p(x) = 2x^3 - 5x^2 - x + 3.$	$p(x) = 2x^3 - 5x^2 - x + 3.$

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johnston-linear-matrix-algebra-EQ0623 (5)	239	1.000	$\left[\begin{array}{cccc} 1 & 0 & 0 & 0 \\ 1 & 1 & 1 & 1 \\ 1 & 2 & 4 & 8 \\ 1 & 3 & 9 & 27 \end{array}\right] \left[\begin{array}{c} c_0 \\ c_1 \\ c_2 \\ c_3 \end{array}\right] = \left[\begin{array}{c} -1 \\ -1 \\ -1 \\ 5 \end{array}\right]$		$\begin{bmatrix} 1 & 0 & 0 & 0 \\ 1 & 1 & 1 & 1 \\ 1 & 2 & 4 & 8 \\ 1 & 3 & 9 & 27 \end{bmatrix} \begin{bmatrix} c_0 \\ c_1 \\ c_2 \\ c_3 \end{bmatrix} = \begin{bmatrix} -1 \\ -1 \\ -1 \\ 5 \end{bmatrix}.$
johnston-linear-matrix-algebra-EQ0624	239	1.000	$p(x)=x^3-3x^2+2x-1.$	$p(x) = x^3 - 3x^2 + 2x - 1.$	$p(x) = x^3 - 3x^2 + 2x - 1.$
johnston-linear-matrix-algebra-EQ0625	240	1.000	$A \mathbf{x} = \mathbf{b}, \quad \text{and} \quad A \mathbf{y} = \mathbf{c}, \quad \text{and} \quad A \mathbf{z} = \mathbf{d},$	$A\mathbf{x} = \mathbf{b}, \quad A\mathbf{y} = \mathbf{c}, \quad \text{and} \quad A\mathbf{z} = \mathbf{d},$	$A\mathbf{x} = \mathbf{b}, \quad A\mathbf{y} = \mathbf{c}, \quad \text{and} \quad A\mathbf{z} = \mathbf{d},$
johnston-linear-matrix-algebra-EQ0627	241	1.000	$A = \begin{bmatrix} 0 & 2 & 4 & 2 \\ 2 & 1 & 1 & 2 \\ 1 & 1 & 2 & 3 \\ 1 & 2 & 3 & 4 \end{bmatrix}$	$A = \begin{bmatrix} 0 & 2 & 4 & 2 \\ 2 & 1 & 1 & 2 \\ 1 & 1 & 2 & 3 \\ 1 & 2 & 3 & 4 \end{bmatrix}$	$A = \begin{bmatrix} 0 & 2 & 4 & 2 \\ 2 & 1 & 1 & 2 \\ 1 & 1 & 2 & 3 \\ 1 & 2 & 3 & 4 \end{bmatrix}$
johnston-linear-matrix-algebra-EQ0629	241	1.000	$\begin{aligned} R &= \begin{bmatrix} 1 & 1 & 4 & 3 \\ 0 & -1 & -3 & -4 \\ 0 & 0 & -2 & -6 \\ 0 & 0 & 0 & 3 \end{bmatrix} \quad \text{and} \\ E^{-1} &= \begin{bmatrix} 0 & 0 & 1 & 0 \\ 0 & 1 & -2 & 0 \\ 1 & 2 & -4 & 0 \\ -1 & -1 & 1 & 1 \end{bmatrix}^{-1} = \begin{bmatrix} 0 & -2 & 1 & 0 \\ 2 & 1 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 1 & -1 & 1 & 1 \end{bmatrix}. \end{aligned}$	$R = \begin{bmatrix} 1 & 1 & 4 & 3 \\ 0 & -1 & -3 & -4 \\ 0 & 0 & -2 & -6 \\ 0 & 0 & 0 & 3 \end{bmatrix} \quad \text{and} \\ E^{-1} = \begin{bmatrix} 0 & 0 & 1 & 0 \\ 0 & 1 & -2 & 0 \\ 1 & 2 & -4 & 0 \\ -1 & -1 & 1 & 1 \end{bmatrix}^{-1} = \begin{bmatrix} 0 & -2 & 1 & 0 \\ 2 & 1 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 1 & -1 & 1 & 1 \end{bmatrix}.$	$R = \begin{bmatrix} 1 & 1 & 4 & 3 \\ 0 & -1 & -3 & -4 \\ 0 & 0 & -2 & -6 \\ 0 & 0 & 0 & 3 \end{bmatrix} \quad \text{and} \\ E^{-1} = \begin{bmatrix} 0 & 0 & 1 & 0 \\ 0 & 1 & -2 & 0 \\ 1 & 2 & -4 & 0 \\ -1 & -1 & 1 & 1 \end{bmatrix}^{-1} = \begin{bmatrix} 0 & -2 & 1 & 0 \\ 2 & 1 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 1 & -1 & 1 & 1 \end{bmatrix}.$
johnston-linear-matrix-algebra-EQ0630	241	1.000	$\begin{aligned} &\left[\begin{array}{cccc} 0 & -2 & 1 & 0 \\ 2 & 1 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 1 & -1 & 1 & 1 \end{array}\right] \xrightarrow{R_1 \leftrightarrow R_3} \left[\begin{array}{cccc} 1 & 0 & 0 & 0 \\ 2 & 1 & 0 & 0 \\ 0 & -2 & 1 & 0 \\ 1 & -1 & 1 & 1 \end{array}\right], \\ &\left[\begin{array}{cccc} 1 & 0 & 0 & 0 \\ 2 & 1 & 0 & 0 \\ 0 & -2 & 1 & 0 \\ 1 & -1 & 1 & 1 \end{array}\right] \xrightarrow{\text{row-reduce}} \left[\begin{array}{cccc} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{array}\right], \end{aligned}$	$\begin{bmatrix} 0 & -2 & 1 & 0 \\ 2 & 1 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 1 & -1 & 1 & 1 \end{bmatrix} \xrightarrow{R_1 \leftrightarrow R_3} \begin{bmatrix} 1 & 0 & 0 & 0 \\ 2 & 1 & 0 & 0 \\ 0 & -2 & 1 & 0 \\ 1 & -1 & 1 & 1 \end{bmatrix},$	$\begin{bmatrix} 0 & -2 & 1 & 0 \\ 2 & 1 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 1 & -1 & 1 & 1 \end{bmatrix} \xrightarrow{R_1 \leftrightarrow R_3} \begin{bmatrix} 1 & 0 & 0 & 0 \\ 2 & 1 & 0 & 0 \\ 0 & -2 & 1 & 0 \\ 1 & -1 & 1 & 1 \end{bmatrix},$
johnston-linear-matrix-algebra-EQ0632	242	1.000	$A = P L U$	$A = PLU$	$A = PLU$
johnston-linear-matrix-algebra-EQ0634	243	1.000	$E^{-1} = P_*^{-1} L_*^{-1}.$	$E^{-1} = P_*^{-1} L_*^{-1}.$	$E^{-1} = P_*^{-1} L_*^{-1}.$
johnston-linear-matrix-algebra-EQ0641	245	1.000	$\begin{aligned} &\left[\begin{array}{cccc c} 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 0 & 2 \\ 0 & 1 & 1 & 0 & 3 \\ 1 & 0 & -1/2 & 1 & 1 \end{array}\right] \xrightarrow{\text{row-reduce}} \left[\begin{array}{cccc c} 1 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 2 \\ 0 & 0 & 1 & 0 & 1 \\ 0 & 0 & 0 & 1 & 3/2 \end{array}\right], \\ &\left[\begin{array}{cccc c} 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 0 & 2 \\ 0 & 1 & 1 & 0 & 3 \\ 1 & 0 & -1/2 & 1 & 1 \end{array}\right] \xrightarrow{\text{row-reduce}} \left[\begin{array}{cccc c} 1 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 2 \\ 0 & 0 & 1 & 0 & 1 \\ 0 & 0 & 0 & 1 & 3/2 \end{array}\right], \end{aligned}$	$\left[\begin{array}{cccc c} 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 0 & 2 \\ 0 & 1 & 1 & 0 & 3 \\ 1 & 0 & -1/2 & 1 & 1 \end{array}\right] \xrightarrow{\text{row-reduce}} \left[\begin{array}{cccc c} 1 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 2 \\ 0 & 0 & 1 & 0 & 1 \\ 0 & 0 & 0 & 1 & 3/2 \end{array}\right],$	$\left[\begin{array}{cccc c} 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 0 & 2 \\ 0 & 1 & 1 & 0 & 3 \\ 1 & 0 & -1/2 & 1 & 1 \end{array}\right] \xrightarrow{\text{row-reduce}} \left[\begin{array}{cccc c} 1 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 2 \\ 0 & 0 & 1 & 0 & 1 \\ 0 & 0 & 0 & 1 & 3/2 \end{array}\right],$

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johnston-linear-matrix-algebra-EQ0643	246	■ 1.000	$A=\left[\begin{array}{c c} B & \mathbf{v} \\ \hline \mathbf{w}^T & c \end{array}\right]$.	$A = \left[\begin{array}{c c} B & \mathbf{v} \\ \hline \mathbf{w}^T & c \end{array} \right]$.	$A = \left[\begin{array}{c c} B & \mathbf{v} \\ \hline \mathbf{w}^T & c \end{array} \right]$.
johnston-linear-matrix-algebra-EQ0645	246	■ 1.000	$\operatorname{rank}\left(\left[\frac{B}{\mathbf{w}^T}\right]\right)>r$,	$\operatorname{rank}\left(\left[\frac{B}{\mathbf{w}^T}\right]\right)>r$,	$\operatorname{rank}\left(\left[\frac{B}{\mathbf{w}^T}\right]\right)>r$,
johnston-linear-matrix-algebra-EQ0648	248	■ 1.000	$A=\left[\begin{array}{ll} 0 & 0 \\ 0 & 1 \end{array}\right]$.	$A = \left[\begin{array}{cc} 0 & 0 \\ 0 & 1 \end{array} \right]$.	$A = \left[\begin{array}{cc} 0 & 0 \\ 0 & 1 \end{array} \right]$.
johnston-linear-matrix-algebra-EQ0649	248	■ 1.000	$A=\left[\begin{array}{cccccc} 2 & -1 & 0 & 0 & 0 & 0 \\ -1 & 3 & -1 & 0 & 1 & 0 \\ 0 & -1 & 4 & -1 & 0 & 0 \\ 0 & 0 & -1 & 4 & -1 & 0 \\ 0 & 0 & 0 & -1 & 3 & -1 \\ 0 & 1 & 0 & 0 & -1 & 2 \end{array}\right]$.	$A = \left[\begin{array}{cccccc} 2 & -1 & 0 & 0 & 0 & 0 \\ -1 & 3 & -1 & 0 & 1 & 0 \\ 0 & -1 & 4 & -1 & 0 & 0 \\ 0 & 0 & -1 & 4 & -1 & 0 \\ 0 & 0 & 0 & -1 & 3 & -1 \\ 0 & 1 & 0 & 0 & -1 & 2 \end{array} \right]$.	$A = \left[\begin{array}{cccccc} 2 & -1 & 0 & 0 & 0 & 0 \\ -1 & 3 & -1 & 0 & 1 & 0 \\ 0 & -1 & 4 & -1 & 0 & 0 \\ 0 & 0 & -1 & 4 & -1 & 0 \\ 0 & 0 & 0 & -1 & 3 & -1 \\ 0 & 1 & 0 & 0 & -1 & 2 \end{array} \right]$.
johnston-linear-matrix-algebra-EQ0651	248	■ 1.000	$V=\left[\begin{array}{cccccc} 1 & a_0 & a_0^2 & \cdots & a_0^n \\ 1 & a_1 & a_1^2 & \cdots & a_1^n \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 1 & a_n & a_n^2 & \cdots & a_n^n \end{array}\right]$,	$V = \left[\begin{array}{ccccc} 1 & a_0 & a_0^2 & \cdots & a_0^n \\ 1 & a_1 & a_1^2 & \cdots & a_1^n \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 1 & a_n & a_n^2 & \cdots & a_n^n \end{array} \right]$,	$V = \left[\begin{array}{ccccc} 1 & a_0 & a_0^2 & \cdots & a_0^n \\ 1 & a_1 & a_1^2 & \cdots & a_1^n \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 1 & a_n & a_n^2 & \cdots & a_n^n \end{array} \right]$,
johnston-linear-matrix-algebra-EQ0652	249	■ 1.000	$A=\left[\begin{array}{ll} 2 & 1 \\ 1 & 2 \end{array}\right]$,	$A = \left[\begin{array}{cc} 2 & 1 \\ 1 & 2 \end{array} \right]$,	$A = \left[\begin{array}{cc} 2 & 1 \\ 1 & 2 \end{array} \right]$,
johnston-linear-matrix-algebra-EQ0657	252	■ 1.000	$\mathbf{v}=c_1\mathbf{v}_1+c_2\mathbf{v}_2+\cdots+c_k\mathbf{v}_k$	$\mathbf{v} = c_1\mathbf{v}_1 + c_2\mathbf{v}_2 + \cdots + c_k\mathbf{v}_k$	$\mathbf{v} = c_1\mathbf{v}_1 + c_2\mathbf{v}_2 + \cdots + c_k\mathbf{v}_k$
johnston-linear-matrix-algebra-EQ0658	252	■ 1.000	$[\mathbf{v}]_B \stackrel{\text{def}}{=} (c_1, c_2, \dots, c_k)$	$[\mathbf{v}]_B \stackrel{\text{def}}{=} (c_1, c_2, \dots, c_k)$	$[\mathbf{v}]_B \stackrel{\text{def}}{=} (c_1, c_2, \dots, c_k)$
johnston-linear-matrix-algebra-EQ0660	252	■ 1.000	$(5,1)=c_1(2,1)+c_2(1,2)$.	$(5,1) = c_1(2,1) + c_2(1,2)$.	$(5,1) = c_1(2,1) + c_2(1,2)$.
johnston-linear-matrix-algebra-EQ0662	253	■ 1.000	$(5,4,3)=c_1(1,2,1)+c_2(2,1,1)$.	$(5,4,3) = c_1(1,2,1) + c_2(2,1,1)$.	$(5,4,3) = c_1(1,2,1) + c_2(2,1,1)$.
johnston-linear-matrix-algebra-EQ0664	253	■ 1.000	$\mathbf{v}_j=0\mathbf{v}_1+0\mathbf{v}_2+\cdots+0\mathbf{v}_{j-1}+1\mathbf{v}_j+0\mathbf{v}_{j+1}+\cdots+0\mathbf{v}_k$.	$\mathbf{v}_j = 0\mathbf{v}_1 + 0\mathbf{v}_2 + \cdots + 0\mathbf{v}_{j-1} + 1\mathbf{v}_j + 0\mathbf{v}_{j+1} + \cdots + 0\mathbf{v}_k$.	$\mathbf{v}_j = 0\mathbf{v}_1 + 0\mathbf{v}_2 + \cdots + 0\mathbf{v}_{j-1} + 1\mathbf{v}_j + 0\mathbf{v}_{j+1} + \cdots + 0\mathbf{v}_k$.
johnston-linear-matrix-algebra-EQ0666	254	■ 1.000	$\left[\begin{array}{ccc} 1 & 0 & 1 \\ 2 & 1 & 1 \\ 0 & 1 & -1 \end{array}\right] \xrightarrow[\substack{R_4-2R_1 \\ R_5-R_1}]{\substack{R_2-2R_1 \\ R_4-2R_1}} \left[\begin{array}{ccc} 1 & 0 & 1 \\ 0 & 1 & -1 \\ 0 & 1 & -1 \\ 0 & 0 & 0 \end{array}\right] \xrightarrow[\substack{R_4-R_2}]{\substack{R_3-R_2}} \left[\begin{array}{ccc} 1 & 0 & 1 \\ 0 & 1 & -1 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{array}\right]$.	$\left[\begin{array}{ccc} 1 & 0 & 1 \\ 2 & 1 & 1 \\ 0 & 1 & -1 \\ 2 & 1 & 1 \\ 1 & 0 & 1 \end{array}\right] \xrightarrow[\substack{R_4-2R_1 \\ R_5-R_1}]{\substack{R_2-2R_1 \\ R_4-2R_1}} \left[\begin{array}{ccc} 1 & 0 & 1 \\ 0 & 1 & -1 \\ 0 & 1 & -1 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{array}\right] \xrightarrow[\substack{R_4-R_2}]{\substack{R_3-R_2}} \left[\begin{array}{ccc} 1 & 0 & 1 \\ 0 & 1 & -1 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{array}\right]$.	$\left[\begin{array}{ccc} 1 & 0 & 1 \\ 2 & 1 & 1 \\ 0 & 1 & -1 \\ 2 & 1 & 1 \\ 1 & 0 & 1 \end{array}\right] \xrightarrow[\substack{R_4-2R_1 \\ R_5-R_1}]{\substack{R_2-2R_1 \\ R_4-2R_1}} \left[\begin{array}{ccc} 1 & 0 & 1 \\ 0 & 1 & -1 \\ 0 & 1 & -1 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{array}\right] \xrightarrow[\substack{R_4-R_2}]{\substack{R_3-R_2}} \left[\begin{array}{ccc} 1 & 0 & 1 \\ 0 & 1 & -1 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{array}\right]$.

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johnston-linear-matrix-algebra-EQ0667	254	1.000	$B=\{(1,2,0,2,1),(0,1,1,1,0)\}$.	$B = \{(1, 2, 0, 2, 1), (0, 1, 1, 1, 0)\}.$	$B = \{(1, 2, 0, 2, 1), (0, 1, 1, 1, 0)\}.$
johnston-linear-matrix-algebra-EQ0668	255	1.000	$(2,1,-3,1,2)=c_{\{1\}}(1,2,0,2,1)+c_{\{2\}}(0,1,1,1,0)$	$(2, 1, -3, 1, 2) = c_1(1, 2, 0, 2, 1) + c_2(0, 1, 1, 1, 0)$	$(2, 1, -3, 1, 2) = c_1(1, 2, 0, 2, 1) + c_2(0, 1, 1, 1, 0)$
johnston-linear-matrix-algebra-EQ0669	255	1.000	$\left[\begin{array}{cc c} 1 & 0 & 2 \\ 2 & 1 & 1 \\ 0 & 1 & -3 \\ 2 & 1 & 1 \\ 1 & 0 & 2 \end{array}\right] \xrightarrow[\substack{R_4-2R_1 \\ R_5-R_1}]{R_2-2R_1} \left[\begin{array}{cc c} 1 & 0 & 2 \\ 0 & 1 & -3 \\ 0 & 1 & -3 \\ 0 & 1 & -3 \\ 0 & 0 & 0 \end{array}\right] \xrightarrow[\substack{R_4-R_2}]{R_3-R_2} \left[\begin{array}{cc c} 1 & 0 & 2 \\ 0 & 1 & -3 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{array}\right] .$	$\left[\begin{array}{cc c} 1 & 0 & 2 \\ 2 & 1 & 1 \\ 0 & 1 & -3 \\ 2 & 1 & 1 \\ 1 & 0 & 2 \end{array}\right] \xrightarrow[\substack{R_4-2R_1 \\ R_5-R_1}]{R_2-2R_1} \left[\begin{array}{cc c} 1 & 0 & 2 \\ 0 & 1 & -3 \\ 0 & 1 & -3 \\ 0 & 1 & -3 \\ 0 & 0 & 0 \end{array}\right] \xrightarrow[\substack{R_4-R_2}]{R_3-R_2} \left[\begin{array}{cc c} 1 & 0 & 2 \\ 0 & 1 & -3 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{array}\right] .$	$\left[\begin{array}{cc c} 1 & 0 & 2 \\ 2 & 1 & 1 \\ 0 & 1 & -3 \\ 2 & 1 & 1 \\ 1 & 0 & 2 \end{array}\right] \xrightarrow[\substack{R_4-2R_1 \\ R_5-R_1}]{R_2-2R_1} \left[\begin{array}{cc c} 1 & 0 & 2 \\ 0 & 1 & -3 \\ 0 & 1 & -3 \\ 0 & 1 & -3 \\ 0 & 0 & 0 \end{array}\right] \xrightarrow[\substack{R_4-R_2}]{R_3-R_2} \left[\begin{array}{cc c} 1 & 0 & 2 \\ 0 & 1 & -3 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{array}\right] .$
johnston-linear-matrix-algebra-EQ0670	255	1.000	$\mathcal{S}=\operatorname{span}(B) \subset \mathbb{R}^5$	$\mathcal{S} = \operatorname{span}(B) \subset \mathbb{R}^5,$	$\mathcal{S} = \operatorname{span}(B) \subset \mathbb{R}^5,$
johnston-linear-matrix-algebra-EQ0671	255	1.000	$2\mathbf{v}-5\mathbf{w}=2(2,1,-3,1,2)-5(1,0,-2,0,1)=(-1,2,4,2,-1)$.	$2\mathbf{v}-5\mathbf{w} = 2(2, 1, -3, 1, 2) - 5(1, 0, -2, 0, 1) = (-1, 2, 4, 2, -1).$	$2\mathbf{v}-5\mathbf{w} = 2(2, 1, -3, 1, 2) - 5(1, 0, -2, 0, 1) = (-1, 2, 4, 2, -1).$
johnston-linear-matrix-algebra-EQ0672	256	1.000	$\left[\begin{array}{cc c} 1 & 0 & -1 \\ 2 & 1 & 2 \\ 0 & 1 & 4 \\ 2 & 1 & 2 \\ 1 & 0 & -1 \end{array}\right] \xrightarrow[\substack{R_4-2R_1 \\ R_5-R_1}]{R_2-2R_1} \left[\begin{array}{cc c} 1 & 0 & -1 \\ 0 & 1 & 4 \\ 0 & 1 & 4 \\ 0 & 1 & 4 \\ 0 & 0 & 0 \end{array}\right] \xrightarrow[\substack{R_4-R_2}]{R_3-R_2} \left[\begin{array}{cc c} 1 & 0 & -1 \\ 0 & 1 & 4 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{array}\right] .$	$\left[\begin{array}{cc c} 1 & 0 & -1 \\ 2 & 1 & 2 \\ 0 & 1 & 4 \\ 2 & 1 & 2 \\ 1 & 0 & -1 \end{array}\right] \xrightarrow[\substack{R_4-2R_1 \\ R_5-R_1}]{R_2-2R_1} \left[\begin{array}{cc c} 1 & 0 & -1 \\ 0 & 1 & 4 \\ 0 & 1 & 4 \\ 0 & 1 & 4 \\ 0 & 0 & 0 \end{array}\right] \xrightarrow[\substack{R_4-R_2}]{R_3-R_2} \left[\begin{array}{cc c} 1 & 0 & -1 \\ 0 & 1 & 4 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{array}\right] .$	$\left[\begin{array}{cc c} 1 & 0 & -1 \\ 2 & 1 & 2 \\ 0 & 1 & 4 \\ 2 & 1 & 2 \\ 1 & 0 & -1 \end{array}\right] \xrightarrow[\substack{R_4-2R_1 \\ R_5-R_1}]{R_2-2R_1} \left[\begin{array}{cc c} 1 & 0 & -1 \\ 0 & 1 & 4 \\ 0 & 1 & 4 \\ 0 & 1 & 4 \\ 0 & 0 & 0 \end{array}\right] \xrightarrow[\substack{R_4-R_2}]{R_3-R_2} \left[\begin{array}{cc c} 1 & 0 & -1 \\ 0 & 1 & 4 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{array}\right] .$
johnston-linear-matrix-algebra-EQ0673	256	1.000	$2\mathbf{v}-5\mathbf{w}=(-1,2,4,2,-1)=-(1,2,0,2,1)+4(0,1,1,1,0)$,	$2\mathbf{v}-5\mathbf{w} = (-1, 2, 4, 2, -1) = -(1, 2, 0, 2, 1) + 4(0, 1, 1, 1, 0),$	$2\mathbf{v}-5\mathbf{w} = (-1, 2, 4, 2, -1) = -(1, 2, 0, 2, 1) + 4(0, 1, 1, 1, 0),$
johnston-linear-matrix-algebra-EQ0674	256	1.000	$\left[\begin{array}{cc c} 1 & 0 & 1 \\ 2 & 1 & 0 \\ 0 & 1 & -2 \\ 2 & 1 & 0 \\ 1 & 0 & 1 \end{array}\right] \xrightarrow[\substack{R_4-2R_1 \\ R_5-R_1}]{R_2-2R_1} \left[\begin{array}{cc c} 1 & 0 & 1 \\ 0 & 1 & -2 \\ 0 & 1 & -2 \\ 0 & 1 & -2 \\ 0 & 0 & 0 \end{array}\right] \xrightarrow[\substack{R_4-R_2}]{R_3-R_2} \left[\begin{array}{cc c} 1 & 0 & 1 \\ 0 & 1 & -2 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{array}\right] .$	$\left[\begin{array}{cc c} 1 & 0 & 1 \\ 2 & 1 & 0 \\ 0 & 1 & -2 \\ 2 & 1 & 0 \\ 1 & 0 & 1 \end{array}\right] \xrightarrow[\substack{R_4-2R_1 \\ R_5-R_1}]{R_2-2R_1} \left[\begin{array}{cc c} 1 & 0 & 1 \\ 0 & 1 & -2 \\ 0 & 1 & -2 \\ 0 & 1 & -2 \\ 0 & 0 & 0 \end{array}\right] \xrightarrow[\substack{R_4-R_2}]{R_3-R_2} \left[\begin{array}{cc c} 1 & 0 & 1 \\ 0 & 1 & -2 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{array}\right] .$	$\left[\begin{array}{cc c} 1 & 0 & 1 \\ 2 & 1 & 0 \\ 0 & 1 & -2 \\ 2 & 1 & 0 \\ 1 & 0 & 1 \end{array}\right] \xrightarrow[\substack{R_4-2R_1 \\ R_5-R_1}]{R_2-2R_1} \left[\begin{array}{cc c} 1 & 0 & 1 \\ 0 & 1 & -2 \\ 0 & 1 & -2 \\ 0 & 1 & -2 \\ 0 & 0 & 0 \end{array}\right] \xrightarrow[\substack{R_4-R_2}]{R_3-R_2} \left[\begin{array}{cc c} 1 & 0 & 1 \\ 0 & 1 & -2 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{array}\right] .$
johnston-linear-matrix-algebra-EQ0675	256	1.000	$2[\mathbf{v}]_B-5[\mathbf{w}]_B=2(2,-3)-5(1,-2)=(-1,4)$,	$2[\mathbf{v}]_B-5[\mathbf{w}]_B = 2(2, -3) - 5(1, -2) = (-1, 4),$	$2[\mathbf{v}]_B-5[\mathbf{w}]_B = 2(2, -3) - 5(1, -2) = (-1, 4),$
johnston-linear-matrix-algebra-EQ0676	256	1.000	$(-1,1)=11\mathbf{v}_{\{1\}}-12\mathbf{v}_{\{2\}}, \quad \text{so } \quad [\text{so }]_{\{(-1,1)\}}_B=(11,-12)$.	$(-1, 1) = 11\mathbf{v}_1 - 12\mathbf{v}_2, \quad \text{so } \quad [(-1, 1)]_B = (11, -12).$	$(-1, 1) = 11\mathbf{v}_1 - 12\mathbf{v}_2, \quad \text{so } \quad [(-1, 1)]_B = (11, -12).$

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johnston-linear-matrix-algebra-EQ0677	257	1.000	$\begin{array}{l} \left[\begin{array}{ll l} 3 & 0 & 6 \\ 1 & 1 & 4 \end{array} \right] \xrightarrow{\frac{1}{3}R_1} \left[\begin{array}{ll l} 1 & 0 & 2 \\ 1 & 1 & 4 \end{array} \right] \xrightarrow{R_2-R_1} \left[\begin{array}{ll l} 1 & 0 & 2 \\ 0 & 1 & 2 \end{array} \right] . \\ \left[\begin{array}{ll l} 1 & 0 & 2 \\ 1 & 1 & 4 \end{array} \right] \xrightarrow{R_2-R_1} \left[\begin{array}{ll l} 1 & 0 & 2 \\ 0 & 1 & 2 \end{array} \right] . \end{array}$	$\left[\begin{array}{ll l} 3 & 0 & 6 \\ 1 & 1 & 4 \end{array} \right] \xrightarrow{\frac{1}{3}R_1} \left[\begin{array}{ll l} 1 & 0 & 2 \\ 1 & 1 & 4 \end{array} \right] \xrightarrow{R_2-R_1} \left[\begin{array}{ll l} 1 & 0 & 2 \\ 0 & 1 & 2 \end{array} \right] .$	
johnston-linear-matrix-algebra-EQ0678	258	1.000	$P_{C \leftarrow B} \stackrel{\text{def}}{=} \left[[\mathbf{v}_1]_C \mid [\mathbf{v}_2]_C \mid \cdots \mid [\mathbf{v}_k]_C \right] .$	$P_{C \leftarrow B} \stackrel{\text{def}}{=} \left[[\mathbf{v}_1]_C \mid [\mathbf{v}_2]_C \mid \cdots \mid [\mathbf{v}_k]_C \right] .$	
johnston-linear-matrix-algebra-EQ0679	258	1.000	$\begin{array}{l} \mathbf{v}_1 = (1, 1) = \frac{1}{3}(3, 1) + \frac{2}{3}(0, 1) \\ \quad \text{and} \quad \mathbf{v}_2 = (1, -1) = \frac{1}{3}(3, 1) - \frac{4}{3}(0, 1) . \end{array}$	$\mathbf{v}_1 = (1, 1) = \frac{1}{3}(3, 1) + \frac{2}{3}(0, 1) \quad \text{and} \quad \mathbf{v}_2 = (1, -1) = \frac{1}{3}(3, 1) - \frac{4}{3}(0, 1) .$	
johnston-linear-matrix-algebra-EQ0680	258	1.000	$\begin{array}{l} \left[\begin{array}{l} \mathbf{v}_1 \\ \mathbf{v}_2 \end{array} \right]_C = (1, 2) / 3 \quad \text{and} \quad \left[\begin{array}{l} \mathbf{v}_1 \\ \mathbf{v}_2 \end{array} \right]_C = (1, -4) / 3, \\ \text{so } P_{C \leftarrow B} = \frac{1}{3} \begin{bmatrix} 1 & 1 \\ 2 & -4 \end{bmatrix} . \end{array}$	$[\mathbf{v}_1]_C = (1, 2)/3 \quad \text{and} \quad [\mathbf{v}_2]_C = (1, -4)/3, \quad \text{so } P_{C \leftarrow B} = \frac{1}{3} \begin{bmatrix} 1 & 1 \\ 2 & -4 \end{bmatrix} .$	
johnston-linear-matrix-algebra-EQ0681 (3.1.1)	258	1.000	$\begin{array}{l} [\mathbf{v}]_C = P_{C \leftarrow B} [\mathbf{v}]_B = \frac{1}{3} \begin{bmatrix} 1 & 1 \\ 2 & -4 \end{bmatrix} \begin{bmatrix} 5 \\ 1 \end{bmatrix} = \begin{bmatrix} 2 \\ 2 \end{bmatrix} , \\ \left[\begin{array}{l} \mathbf{v}_1 \\ \mathbf{v}_2 \end{array} \right]_C = \left[\begin{array}{l} 2 \\ 2 \end{array} \right] , \end{array}$	$[\mathbf{v}]_C = P_{C \leftarrow B} [\mathbf{v}]_B = \frac{1}{3} \begin{bmatrix} 1 & 1 \\ 2 & -4 \end{bmatrix} \begin{bmatrix} 5 \\ 1 \end{bmatrix} = \begin{bmatrix} 2 \\ 2 \end{bmatrix} ,$	
johnston-linear-matrix-algebra-EQ0683	259	1.000	$\begin{array}{l} (P_{B \leftarrow C} P_{C \leftarrow B}) \mathbf{e}_j = P_{B \leftarrow C} (P_{C \leftarrow B} [\mathbf{v}_j]_B) = P_{B \leftarrow C} [\mathbf{v}_j]_C = [\mathbf{v}_j]_B = \mathbf{e}_j . \\ \left(P_{B \leftarrow C} P_{C \leftarrow B} \right) \mathbf{e}_j = P_{B \leftarrow C} (P_{C \leftarrow B} [\mathbf{v}_j]_B) = P_{B \leftarrow C} [\mathbf{v}_j]_C = [\mathbf{v}_j]_B = \mathbf{e}_j . \end{array}$	$(P_{B \leftarrow C} P_{C \leftarrow B}) \mathbf{e}_j = P_{B \leftarrow C} (P_{C \leftarrow B} [\mathbf{v}_j]_B) = P_{B \leftarrow C} [\mathbf{v}_j]_C = [\mathbf{v}_j]_B = \mathbf{e}_j .$	
johnston-linear-matrix-algebra-EQ0684	259	1.000	$\begin{array}{l} [\mathbf{w}]_C = P_{C \leftarrow B} [\mathbf{w}]_B = \frac{1}{3} \begin{bmatrix} 1 & 1 \\ 2 & -4 \end{bmatrix} \begin{bmatrix} -1 \\ 4 \end{bmatrix} = \begin{bmatrix} 1 \\ -6 \end{bmatrix} . \\ \left[\begin{array}{l} \mathbf{w}_1 \\ \mathbf{w}_2 \end{array} \right]_C = \left[\begin{array}{l} 1 \\ -6 \end{array} \right] . \end{array}$	$[\mathbf{w}]_C = P_{C \leftarrow B} [\mathbf{w}]_B = \frac{1}{3} \begin{bmatrix} 1 & 1 \\ 2 & -4 \end{bmatrix} \begin{bmatrix} -1 \\ 4 \end{bmatrix} = \begin{bmatrix} 1 \\ -6 \end{bmatrix} .$	
johnston-linear-matrix-algebra-EQ0685	260	1.000	$\begin{array}{l} \begin{aligned} & (2, 1, 0, 1, 2) \\ &= c_1(1, -1, 1, 0, 2) + c_2(1, 1, 2, 0, -1) + c_3(2, 3, 1, 1, -1) \end{aligned} \end{array}$	$\begin{array}{l} (2, 1, 0, 1, 2) \\ = c_1(1, -1, 1, 0, 2) + c_2(1, 1, 2, 0, -1) + c_3(2, 3, 1, 1, -1) \end{array}$	
johnston-linear-matrix-algebra-EQ0687	260	1.000	$P_{C \leftarrow B} = \left[[\mathbf{v}_1]_C \mid [\mathbf{v}_2]_C \mid [\mathbf{v}_3]_C \right] = \begin{bmatrix} 1 & 0 & 0 \\ -1 & -1 & -2 \\ 1 & 1 & 1 \end{bmatrix} .$	$P_{C \leftarrow B} = \left[[\mathbf{v}_1]_C \mid [\mathbf{v}_2]_C \mid [\mathbf{v}_3]_C \right] = \begin{bmatrix} 1 & 0 & 0 \\ -1 & -1 & -2 \\ 1 & 1 & 1 \end{bmatrix} .$	

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johnston-linear-matrix-algebra-EQ0689	260	1.000	$[\mathbf{w}]_C=P_{C\leftarrow B}[\mathbf{w}]_B=\left[\begin{array}{ccc}1&0&0\\-1&-1&-2\\1&1&1\end{array}\right]\left[\begin{array}{c}2\\0\\-1\end{array}\right]=\left[\begin{array}{c}2\\0\\1\end{array}\right].$	$[\mathbf{w}]_C=P_{C\leftarrow B}[\mathbf{w}]_B=\left[\begin{array}{ccc}1&0&0\\-1&-1&-2\\1&1&1\end{array}\right]\left[\begin{array}{c}2\\0\\-1\end{array}\right]=\left[\begin{array}{c}2\\0\\1\end{array}\right].$	
johnston-linear-matrix-algebra-EQ0691	261	1.000	$\mathbf{v}_1=(1,1,-1),\quad\mathbf{v}_2=(2,2,-1),\quad\mathbf{v}_3=(-1,-2,1).$	$\mathbf{v}_1=(1,1,-1),\quad\mathbf{v}_2=(2,2,-1),\quad\mathbf{v}_3=(-1,-2,1).$	$\mathbf{v}_1=(1,1,-1),\quad\mathbf{v}_2=(2,2,-1),\quad\mathbf{v}_3=(-1,-2,1).$
johnston-linear-matrix-algebra-EQ0693	262	1.000	$P_{E\leftarrow B}=\left[\left[\mathbf{v}_1\right]_E\left[\mathbf{v}_2\right]_E\left[\mathbf{v}_3\right]_E\right]=\left[\begin{array}{ccc}1&2&-1\\1&2&-2\\-1&-1&1\end{array}\right].$	$P_{E\leftarrow B}=[[\mathbf{v}_1]_E\mid[\mathbf{v}_2]_E\mid[\mathbf{v}_3]_E]=\left[\begin{array}{ccc}1&2&-1\\1&2&-2\\-1&-1&1\end{array}\right].$	$P_{E\leftarrow B}=\left[\left[\mathbf{v}_1\right]_E\mid\left[\mathbf{v}_2\right]_E\mid\left[\mathbf{v}_3\right]_E\right]=\left[\begin{array}{ccc}1&2&-1\\1&2&-2\\-1&-1&1\end{array}\right].$
johnston-linear-matrix-algebra-EQ0695	262	1.000	$[T(\mathbf{v})]_B=[T]_B[\mathbf{v}]_B\quad\text{for all}\quad\mathbf{v}\in\mathbb{R}^n.$	$[T(\mathbf{v})]_B=[T]_B[\mathbf{v}]_B\quad\text{for all}\quad\mathbf{v}\in\mathbb{R}^n.$	$[T(\mathbf{v})]_B=[T]_B[\mathbf{v}]_B\quad\text{for all}\quad\mathbf{v}\in\mathbb{R}^n.$
johnston-linear-matrix-algebra-EQ0698	263	1.000	$T(1,1)=(3,3)\quad\text{and}\quad T(1,-1)=(1,-1).$	$T(1,1)=(3,3)\quad\text{and}\quad T(1,-1)=(1,-1).$	$T(1,1)=(3,3)\quad\text{and}\quad T(1,-1)=(1,-1).$
johnston-linear-matrix-algebra-EQ0701	265	1.000	$[T]_C=P_{C\leftarrow B}[T]_BP_{B\leftarrow C},$	$[T]_C=P_{C\leftarrow B}[T]_BP_{B\leftarrow C},$	$[T]_C=P_{C\leftarrow B}[T]_BP_{B\leftarrow C},$
johnston-linear-matrix-algebra-EQ0705	266	1.000	$\begin{aligned} [T]_B&=P_{B\leftarrow E}[T]P_{E\leftarrow B} \\&=\frac{1}{7}\begin{bmatrix}2&4&-1\\1&-5&3\\-3&-6&5\end{bmatrix}\begin{bmatrix}2&1&-3\\1&0&1\\0&1&2\end{bmatrix}\begin{bmatrix}1&2&-1\\2&-1&1\\3&0&2\end{bmatrix} \\&=\frac{1}{7}\begin{bmatrix}2&4&-1\\1&-5&3\\-3&-6&5\end{bmatrix}\begin{bmatrix}-5&3&-7\\4&2&1\\8&-1&5\end{bmatrix} \\&=\frac{1}{7}\begin{bmatrix}-2&15&-15\\-1&-10&3\\31&-26&40\end{bmatrix}. \end{aligned}$	$\begin{aligned} [T]_B&=P_{B\leftarrow E}[T]P_{E\leftarrow B} \\&=\frac{1}{7}\begin{bmatrix}2&4&-1\\1&-5&3\\-3&-6&5\end{bmatrix}\begin{bmatrix}2&1&-3\\1&0&1\\0&1&2\end{bmatrix}\begin{bmatrix}1&2&-1\\2&-1&1\\3&0&2\end{bmatrix} \\&=\frac{1}{7}\begin{bmatrix}2&4&-1\\1&-5&3\\-3&-6&5\end{bmatrix}\begin{bmatrix}-5&3&-7\\4&2&1\\8&-1&5\end{bmatrix} \\&=\frac{1}{7}\begin{bmatrix}-2&15&-15\\-1&-10&3\\31&-26&40\end{bmatrix}. \end{aligned}$	$\begin{aligned} [T]_B&=P_{B\leftarrow E}[T]P_{E\leftarrow B} \\&=\frac{1}{7}\begin{bmatrix}2&4&-1\\1&-5&3\\-3&-6&5\end{bmatrix}\begin{bmatrix}2&1&-3\\1&0&1\\0&1&2\end{bmatrix}\begin{bmatrix}1&2&-1\\2&-1&1\\3&0&2\end{bmatrix} \\&=\frac{1}{7}\begin{bmatrix}2&4&-1\\1&-5&3\\-3&-6&5\end{bmatrix}\begin{bmatrix}-5&3&-7\\4&2&1\\8&-1&5\end{bmatrix} \\&=\frac{1}{7}\begin{bmatrix}-2&15&-15\\-1&-10&3\\31&-26&40\end{bmatrix}. \end{aligned}$
johnston-linear-matrix-algebra-EQ0706	266	1.000	$B=P_{D\leftarrow C}AP_{D\leftarrow C}^{-1}.$	$B=P_{D\leftarrow C}AP_{D\leftarrow C}^{-1}.$	$B=P_{D\leftarrow C}AP_{D\leftarrow C}^{-1}.$
johnston-linear-matrix-algebra-EQ0707 (3.1.2)	267	1.000	$A=\left[\begin{array}{cc}1&2\\3&4\end{array}\right]\quad\text{and}\quad B=\left[\begin{array}{cc}5&-1\\-2&0\end{array}\right]$	$A=\left[\begin{array}{cc}1&2\\3&4\end{array}\right]\quad\text{and}\quad B=\left[\begin{array}{cc}5&-1\\-2&0\end{array}\right]$	$A=\left[\begin{array}{cc}1&2\\3&4\end{array}\right]\quad\text{and}\quad B=\left[\begin{array}{cc}5&-1\\-2&0\end{array}\right]$
johnston-linear-matrix-algebra-EQ0709	267	1.000	$\operatorname{rank}(A)=\operatorname{rank}\left(PBP^{-1}\right)=\operatorname{rank}(PB)=\operatorname{rank}(B).$	$\operatorname{rank}(A)=\operatorname{rank}\left(PBP^{-1}\right)=\operatorname{rank}(PB)=\operatorname{rank}(B).$	$\operatorname{rank}(A)=\operatorname{rank}(PBP^{-1})=\operatorname{rank}(PB)=\operatorname{rank}(B).$

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johnston-linear-matrix-algebra-EQ0710	267	1.000	$\left[\begin{array}{ll} 1 & 2 \\ 2 & 4 \end{array}\right] \xrightarrow{R_2-2R_1} \left[\begin{array}{cc} 1 & 2 \\ 0 & 0 \end{array}\right] \quad \text{and} \quad \left[\begin{array}{ll} 1 & 2 \\ 3 & 4 \end{array}\right] \xrightarrow{R_2-3R_1} \left[\begin{array}{cc} 1 & 2 \\ 0 & -2 \end{array}\right].$	$\left[\begin{array}{ll} 1 & 2 \\ 2 & 4 \end{array}\right] \xrightarrow{R_2-2R_1} \left[\begin{array}{cc} 1 & 2 \\ 0 & 0 \end{array}\right] \quad \text{and} \quad \left[\begin{array}{ll} 1 & 2 \\ 3 & 4 \end{array}\right] \xrightarrow{R_2-3R_1} \left[\begin{array}{cc} 1 & 2 \\ 0 & -2 \end{array}\right].$	
johnston-linear-matrix-algebra-EQ0711	268	1.000	$\operatorname{tr}(A) \stackrel{\text{def}}{=} a_{1,1} + a_{2,2} + \cdots + a_{n,n}.$	$\operatorname{tr}(A) \stackrel{\text{def}}{=} a_{1,1} + a_{2,2} + \cdots + a_{n,n}.$	
johnston-linear-matrix-algebra-EQ0718	270	1.000	$\operatorname{tr}(A B C D) = \operatorname{tr}(B C D A) = \operatorname{tr}(C D A B) = \operatorname{tr}(D A B C).$	$\operatorname{tr}(A B C D) = \operatorname{tr}(B C D A) = \operatorname{tr}(C D A B) = \operatorname{tr}(D A B C).$	
johnston-linear-matrix-algebra-EQ0719	270	1.000	$B = \{(3, 1, 4, -2, 1), (2, 3, -1, 1, 2), (-1, 2, -2, 4, -2)\}$	$B = \{(3, 1, 4, -2, 1), (2, 3, -1, 1, 2), (-1, 2, -2, 4, -2)\}$	
johnston-linear-matrix-algebra-EQ0720	271	1.000	$\left(2 v_2 + 3 v_3, 3 v_1 + v_2 + 3 v_3, 3 v_1 + 2 v_2 + 2 v_3\right).$	$(2v_2 + 3v_3, 3v_1 + v_2 + 3v_3, 3v_1 + 2v_2 + 2v_3).$	
johnston-linear-matrix-algebra-EQ0721	271	1.000	$\left(4 v_1 + v_2 - v_3, 2 v_1 - v_2 - v_3, 2 v_1 - v_2 + v_3\right).$	$(4v_1 + v_2 - v_3, 2v_1 - v_2 - v_3, 2v_1 - v_2 + v_3).$	
johnston-linear-matrix-algebra-EQ0724	272	1.000	$A = \left[\mathbf{v}_1 \left \mathbf{v}_2 \right. \right] \cdots \left \mathbf{v}_n \right .$	$A = \left[\mathbf{v}_1 \left \mathbf{v}_2 \right. \right] \cdots \left \mathbf{v}_n \right .$	
johnston-linear-matrix-algebra-EQ0726	273	1.000	$\operatorname{det}(A B) = \operatorname{det}(A) \operatorname{det}(B) \quad \text{for all } A, B \in \mathcal{M}_n.$	$\operatorname{det}(A B) = \operatorname{det}(A) \operatorname{det}(B) \quad \text{for all } A, B \in \mathcal{M}_n.$	
johnston-linear-matrix-algebra-EQ0727	273	1.000	$\operatorname{det}\left(\left[\mathbf{a}_1 \left \cdots \left \mathbf{c} \mathbf{a}_j \right. \right] \cdots \left \mathbf{a}_n \right. \right] \right) = c \cdot \operatorname{det}\left(\left[\mathbf{a}_1 \left \cdots \left \mathbf{a}_j \right. \right] \cdots \left \mathbf{a}_n \right. \right] \right).$	$\operatorname{det}\left(\left[\mathbf{a}_1 \left \cdots \left \mathbf{c} \mathbf{a}_j \right. \right] \cdots \left \mathbf{a}_n \right. \right] \right) = c \cdot \operatorname{det}\left(\left[\mathbf{a}_1 \left \cdots \left \mathbf{a}_j \right. \right] \cdots \left \mathbf{a}_n \right. \right] \right).$	
johnston-linear-matrix-algebra-EQ0730	275	1.000	$1 = \operatorname{det}(I) = \operatorname{det}(A A^{-1}) = \operatorname{det}(A) \operatorname{det}(A^{-1}).$	$1 = \operatorname{det}(I) = \operatorname{det}(A A^{-1}) = \operatorname{det}(A) \operatorname{det}(A^{-1}).$	
johnston-linear-matrix-algebra-EQ0738	279	1.000	$0 = 0 + 1 + \operatorname{det}\left(\left[\mathbf{e}_1 \left \cdots \left \mathbf{e}_j \right. \right] \cdots \left \mathbf{e}_i \right. \right] \cdots \left \mathbf{e}_n \right. \right) + 0,$	$0 = 0 + 1 + \operatorname{det}\left(\left[\mathbf{e}_1 \left \cdots \left \mathbf{e}_j \right. \right] \cdots \left \mathbf{e}_i \right. \right] \cdots \left \mathbf{e}_n \right. \right) + 0,$	
johnston-linear-matrix-algebra-EQ0739	280	1.000	$\int_a^b f(x) \, dx$	$\int_a^b f(x) \, dx$	
johnston-linear-matrix-algebra-EQ0740	280	1.000	$\int_a^b f(x) \, dx + \int_b^c f(x) \, dx = \int_a^c f(x) \, dx,$	$\int_a^b f(x) \, dx + \int_b^c f(x) \, dx = \int_a^c f(x) \, dx,$	
johnston-linear-matrix-algebra-EQ0741	281	1.000	$\operatorname{det}(A) = \frac{(-1)^s}{c_1 c_2 \cdots c_k}.$	$\operatorname{det}(A) = \frac{(-1)^s}{c_1 c_2 \cdots c_k}.$	
johnston-linear-matrix-algebra-EQ0742	281	1.000	$\operatorname{det}(E_m \cdots E_2 E_1 A) = \operatorname{det}(I) = 1.$	$\operatorname{det}(E_m \cdots E_2 E_1 A) = \operatorname{det}(I) = 1.$	
johnston-linear-matrix-algebra-EQ0743	281	1.000	$\operatorname{det}(A) = \frac{1}{\operatorname{det}(E_1) \operatorname{det}(E_2) \cdots \operatorname{det}(E_m)}.$	$\operatorname{det}(A) = \frac{1}{\operatorname{det}(E_1) \operatorname{det}(E_2) \cdots \operatorname{det}(E_m)}.$	
johnston-linear-matrix-algebra-EQ0744	281	1.000	$\operatorname{det}(A) = \frac{1}{(-1)^s c_1 c_2 \cdots c_k},$	$\operatorname{det}(A) = \frac{1}{(-1)^s c_1 c_2 \cdots c_k},$	

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johnston-linear-matrix-algebra-EQ0745	282	1.000	$\left[\begin{array}{ll} 2 & 2 \\ 4 & 5 \end{array}\right] \xrightarrow{R_2-2R_1} \left[\begin{array}{ll} 2 & 2 \\ 0 & 1 \end{array}\right] \xrightarrow{R_1-2R_2} \left[\begin{array}{ll} 2 & 0 \\ 0 & 1 \end{array}\right] \xrightarrow{\frac{1}{2}R_1} \left[\begin{array}{ll} 1 & 0 \\ 0 & 1 \end{array}\right].$	$\left[\begin{array}{ll} 2 & 2 \\ 4 & 5 \end{array}\right] \xrightarrow{R_2-2R_1} \left[\begin{array}{ll} 2 & 2 \\ 0 & 1 \end{array}\right] \xrightarrow{R_1-2R_2} \left[\begin{array}{ll} 2 & 0 \\ 0 & 1 \end{array}\right] \xrightarrow{\frac{1}{2}R_1} \left[\begin{array}{ll} 1 & 0 \\ 0 & 1 \end{array}\right].$	
johnston-linear-matrix-algebra-EQ0749	283	1.000	$\operatorname{det}(A)=a_{1,1} \ a_{2,2} \ \cdots \ a_{n, \ n}.$	$\det(A) = a_{1,1}a_{2,2}\cdots a_{n,n}.$	$\det(A) = a_{1,1}a_{2,2}\cdots a_{n,n}.$
johnston-linear-matrix-algebra-EQ0751	283	1.000	$\frac{1}{a_{n, \ n}} R_n, \ \frac{1}{a_{n-1, \ n-1}} R_{n-1}, \dots, \frac{1}{a_{1,1}} R_1, \ \ldots, \ \frac{1}{a_{1,1}} R_1.$	$\frac{1}{a_{n,n}}R_n, \frac{1}{a_{n-1,n-1}}R_{n-1}, \dots, \frac{1}{a_{1,1}}R_1.$	$\frac{1}{a_{n,n}}R_n, \frac{1}{a_{n-1,n-1}}R_{n-1}, \dots, \frac{1}{a_{1,1}}R_1.$
johnston-linear-matrix-algebra-EQ0752	284	1.000	$\operatorname{det}(A)=\frac{1}{\frac{1}{a_{1,1}} \ \cdots \ \frac{1}{a_{n, \ n}}}=\frac{1}{\frac{1}{a_{1,1}} \cdot \frac{1}{a_{2,2}} \ \cdots \ \frac{1}{a_{n, \ n}}} = a_{1,1}a_{2,2}\cdots a_{n,n},$	$\det(A) = \frac{1}{\frac{1}{a_{1,1}} \cdot \frac{1}{a_{2,2}} \ \cdots \ \frac{1}{a_{n,n}}} = a_{1,1}a_{2,2}\cdots a_{n,n},$	$\det(A) = \frac{1}{\frac{1}{a_{1,1}} \cdot \frac{1}{a_{2,2}} \ \cdots \ \frac{1}{a_{n,n}}} = a_{1,1}a_{2,2}\cdots a_{n,n},$
johnston-linear-matrix-algebra-EQ0753	284	1.000	$\operatorname{det}(A)=\frac{(-1)^s}{c_1c_2\cdots c_k}r_{1,1}r_{2,2}\cdots r_{n,n},$	$\det(A) = \frac{(-1)^sr_{1,1}r_{2,2}\cdots r_{n,n}}{c_1c_2\cdots c_k},$	$\det(A) = \frac{(-1)^sr_{1,1}r_{2,2}\cdots r_{n,n}}{c_1c_2\cdots c_k},$
johnston-linear-matrix-algebra-EQ0754	284	1.000	$\left[\begin{array}{ll} 2 & 2 \\ 4 & 5 \end{array}\right] \xrightarrow{R_2-2R_1} \left[\begin{array}{ll} 2 & 2 \\ 0 & 1 \end{array}\right].$	$\left[\begin{array}{ll} 2 & 2 \\ 4 & 5 \end{array}\right] \xrightarrow{R_2-2R_1} \left[\begin{array}{ll} 2 & 2 \\ 0 & 1 \end{array}\right].$	$\left[\begin{array}{ll} 2 & 2 \\ 4 & 5 \end{array}\right] \xrightarrow{R_2-2R_1} \left[\begin{array}{ll} 2 & 2 \\ 0 & 1 \end{array}\right].$
johnston-linear-matrix-algebra-EQ0755	285	1.000	$\left[\begin{array}{cc} 1 & -1 \\ -1 & 1 \end{array}\right] \xrightarrow{R_2+R_1} \left[\begin{array}{cc} 1 & -1 \\ 0 & 0 \end{array}\right].$	$\left[\begin{array}{cc} 1 & -1 \\ -1 & 1 \end{array}\right] \xrightarrow{R_2+R_1} \left[\begin{array}{cc} 1 & -1 \\ 0 & 0 \end{array}\right].$	$\left[\begin{array}{cc} 1 & -1 \\ -1 & 1 \end{array}\right] \xrightarrow{R_2+R_1} \left[\begin{array}{cc} 1 & -1 \\ 0 & 0 \end{array}\right].$
johnston-linear-matrix-algebra-EQ0756 (1)	285	1.000	$\left[\begin{array}{ccc} 3 & 6 & -3 \\ 0 & -2 & 3 \\ 2 & 4 & -1 \end{array}\right] \xrightarrow{\frac{1}{3}R_1} \left[\begin{array}{ccc} 1 & 2 & -1 \\ 0 & -2 & 3 \\ 2 & 4 & -1 \end{array}\right] \xrightarrow{R_3-2R_1} \left[\begin{array}{ccc} 1 & 2 & -1 \\ 0 & -2 & 3 \\ 0 & 0 & 1 \end{array}\right].$	$\left[\begin{array}{ccc} 3 & 6 & -3 \\ 0 & -2 & 3 \\ 2 & 4 & -1 \end{array}\right] \xrightarrow{\frac{1}{3}R_1} \left[\begin{array}{ccc} 1 & 2 & -1 \\ 0 & -2 & 3 \\ 2 & 4 & -1 \end{array}\right] \xrightarrow{R_3-2R_1} \left[\begin{array}{ccc} 1 & 2 & -1 \\ 0 & -2 & 3 \\ 0 & 0 & 1 \end{array}\right].$	$\left[\begin{array}{ccc} 3 & 6 & -3 \\ 0 & -2 & 3 \\ 2 & 4 & -1 \end{array}\right] \xrightarrow{\frac{1}{3}R_1} \left[\begin{array}{ccc} 1 & 2 & -1 \\ 0 & -2 & 3 \\ 2 & 4 & -1 \end{array}\right] \xrightarrow{R_3-2R_1} \left[\begin{array}{ccc} 1 & 2 & -1 \\ 0 & -2 & 3 \\ 0 & 0 & 1 \end{array}\right].$
johnston-linear-matrix-algebra-EQ0770	288	1.000	$1 \cdot 1 - (-1) \cdot (-1) = 1 - 1 = 0.$	$1 \cdot 1 - (-1) \cdot (-1) = 1 - 1 = 0.$	$1 \cdot 1 - (-1) \cdot (-1) = 1 - 1 = 0.$
johnston-linear-matrix-algebra-EQ0773	289	1.000	$A=\left[\begin{array}{lll} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{array}\right],$	$A = \left[\begin{array}{lll} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{array}\right],$	$A = \left[\begin{array}{lll} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{array}\right],$
johnston-linear-matrix-algebra-EQ0777	290	1.000	$m_{1,2}=4, \quad m_{2,1}=2, \quad \text{and} \quad m_{2,2}=2.$	$m_{1,2} = 4, \quad m_{2,1} = 2, \quad \text{and} \quad m_{2,2} = 2.$	$m_{1,2} = 4, \quad m_{2,1} = 2, \quad \text{and} \quad m_{2,2} = 2.$
johnston-linear-matrix-algebra-EQ0778	290	1.000	$c_{1,1}=5, \quad c_{1,2}=-4, \quad c_{2,1}=-2, \quad \text{and} \quad c_{2,2}=2.$	$c_{1,1} = 5, \quad c_{1,2} = -4, \quad c_{2,1} = -2, \quad \text{and} \quad c_{2,2} = 2.$	$c_{1,1} = 5, \quad c_{1,2} = -4, \quad c_{2,1} = -2, \quad \text{and} \quad c_{2,2} = 2.$
johnston-linear-matrix-algebra-EQ0780	290	1.000	$\begin{array}{lll} m_{1,1}=-1 & m_{1,2}=-2 & m_{1,3}=2 \\ m_{2,1}=-1 & m_{2,2}=-2 & m_{2,3}=5 \\ m_{3,1}=3 & m_{3,2}=-1 & m_{3,3}=-2. \end{array}$	$\begin{array}{lll} m_{1,1} = -1 & m_{1,2} = -2 & m_{1,3} = 2 \\ m_{2,1} = -1 & m_{2,2} = -2 & m_{2,3} = 5 \\ m_{3,1} = 3 & m_{3,2} = -1 & m_{3,3} = -2. \end{array}$	$\begin{array}{lll} m_{1,1} = -1 & m_{1,2} = -2 & m_{1,3} = 2 \\ m_{2,1} = -1 & m_{2,2} = -2 & m_{2,3} = 5 \\ m_{3,1} = 3 & m_{3,2} = -1 & m_{3,3} = -2. \end{array}$

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johnston-linear-matrix-algebra-EQ0795	294	1.000	$\begin{aligned} & \operatorname{det}\left(\begin{array}{cccc} 0 & a_{1,2} & \cdots & a_{1,n} \\ a_{2,1} & a_{2,2} & \cdots & a_{2,n} \\ \vdots & \vdots & \ddots & \vdots \\ 0 & a_{n,2} & \cdots & a_{n,n} \end{array}\right) \\ & = -\operatorname{det}\left(\begin{array}{cccc} a_{2,1} & a_{2,2} & a_{2,3} & \cdots & a_{2,n} \\ 0 & a_{1,2} & a_{1,3} & \cdots & a_{1,n} \\ 0 & a_{3,2} & a_{3,3} & \cdots & a_{3,n} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & a_{n,2} & a_{n,3} & \cdots & a_{n,n} \end{array}\right) \\ & = -a_{2,1} \operatorname{det}\left(\begin{array}{cccc} a_{1,2} & a_{1,3} & \cdots & a_{1,n} \\ a_{3,2} & a_{3,3} & \cdots & a_{3,n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{n,2} & a_{n,3} & \cdots & a_{n,n} \end{array}\right) \\ & = a_{2,1} c_{2,1} \end{aligned}$	$\det\left(\begin{bmatrix} 0 & a_{1,2} & \cdots & a_{1,n} \\ a_{2,1} & a_{2,2} & \cdots & a_{2,n} \\ \vdots & \vdots & \ddots & \vdots \\ 0 & a_{n,2} & \cdots & a_{n,n} \end{bmatrix}\right) = -\det\left(\begin{bmatrix} a_{2,1} & a_{2,2} & a_{2,3} & \cdots & a_{2,n} \\ 0 & a_{1,2} & a_{1,3} & \cdots & a_{1,n} \\ 0 & a_{3,2} & a_{3,3} & \cdots & a_{3,n} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & a_{n,2} & a_{n,3} & \cdots & a_{n,n} \end{bmatrix}\right)$ $= -a_{2,1} \det\left(\begin{bmatrix} a_{1,2} & a_{1,3} & \cdots & a_{1,n} \\ a_{3,2} & a_{3,3} & \cdots & a_{3,n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{n,2} & a_{n,3} & \cdots & a_{n,n} \end{bmatrix}\right)$ $= a_{2,1} c_{2,1}.$	
johnston-linear-matrix-algebra-EQ0796	294	1.000	$\operatorname{det}(A)=a_{1,1} \ c_{1,1}+a_{2,1} \ c_{2,1}+\cdots+a_{n,1} c_{n,1}.$	$\det(A)=a_{1,1} c_{1,1}+a_{2,1} c_{2,1}+\cdots+a_{n,1} c_{n,1}.$	$\det(A)=a_{1,1} c_{1,1}+a_{2,1} c_{2,1}+\cdots+a_{n,1} c_{n,1}.$
johnston-linear-matrix-algebra-EQ0797	295	1.000	$\begin{aligned} & \begin{array}{ccccccc} 0 & -1 & 2 & 1 & 3 \\ 0 & 0 & 0 & 2 & 0 \\ -2 & 1 & 1 & -1 & 0 \\ 1 & 0 & -3 & 1 & 0 \\ 2 & 1 & -1 & 0 & 0 \end{array} \end{aligned}$	$\begin{bmatrix} 0 & -1 & 2 & 1 & 3 \\ 0 & 0 & 0 & 2 & 0 \\ -2 & 1 & 1 & -1 & 0 \\ 1 & 0 & -3 & 1 & 0 \\ 2 & 1 & -1 & 0 & 0 \end{bmatrix}$	$\begin{bmatrix} 0 & -1 & 2 & 1 & 3 \\ 0 & 0 & 0 & 2 & 0 \\ -2 & 1 & 1 & -1 & 0 \\ 1 & 0 & -3 & 1 & 0 \\ 2 & 1 & -1 & 0 & 0 \end{bmatrix}$
johnston-linear-matrix-algebra-EQ0801	296	1.000	$2=f(I)=f(I) \ f(I)=2 \ \cdot 2=4,$	$2=f(I)=f(I)f(I)=2 \cdot 2=4,$	$2=f(I)=f(I)f(I)=2 \cdot 2=4,$
johnston-linear-matrix-algebra-EQ0804	297	1.000	$\begin{aligned} & \cos(\theta) \ \sin(\theta) \\ & \sin(\theta) \ \cos(\theta) \end{aligned}$	$\begin{bmatrix} \cos(\theta) & -\sin(\theta) \\ \sin(\theta) & \cos(\theta) \end{bmatrix}$	$\begin{bmatrix} \cos(\theta) & -\sin(\theta) \\ \sin(\theta) & \cos(\theta) \end{bmatrix}$
johnston-linear-matrix-algebra-EQ0805	298	1.000	$\begin{aligned} & \begin{array}{cccc} a & b & c & d \end{array} \\ & \begin{array}{cccc} c & a & d & b \end{array} \\ & \begin{array}{cccc} a & b & c & d \end{array} \\ & \begin{array}{cccc} d & e & f & g \end{array} \\ & \begin{array}{cccc} g & d & a & h \end{array} \\ & \begin{array}{cccc} h & e & b & i \end{array} \\ & \begin{array}{cccc} i & f & c & \end{array} \end{aligned}$	$\begin{bmatrix} a & b \\ c & d \end{bmatrix}^R = \begin{bmatrix} c & a \\ d & b \end{bmatrix}, \quad \begin{bmatrix} a & b & c \\ d & e & f \\ g & h & i \end{bmatrix}^R = \begin{bmatrix} g & d & a \\ h & e & b \\ i & f & c \end{bmatrix}.$	$\begin{bmatrix} a & b \\ c & d \end{bmatrix}^R = \begin{bmatrix} c & a \\ d & b \end{bmatrix}, \quad \begin{bmatrix} a & b & c \\ d & e & f \\ g & h & i \end{bmatrix}^R = \begin{bmatrix} g & d & a \\ h & e & b \\ i & f & c \end{bmatrix}.$

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johnston-linear-matrix-algebra EQ0806	298	1.000	$A_n = \left[\begin{array}{ccccc} 1 & 1 & 1 & \cdots & 1 \\ 1 & 2 & 1 & \cdots & 1 \\ 1 & 1 & 3 & \cdots & 1 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 1 & 1 & 1 & \cdots & n \end{array} \right] .$	$A_n = \begin{bmatrix} 1 & 1 & 1 & \cdots & 1 \\ 1 & 2 & 1 & \cdots & 1 \\ 1 & 1 & 3 & \cdots & 1 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 1 & 1 & 1 & \cdots & n \end{bmatrix} .$	$A_n = \begin{bmatrix} 1 & 1 & 1 & \cdots & 1 \\ 1 & 2 & 1 & \cdots & 1 \\ 1 & 1 & 3 & \cdots & 1 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 1 & 1 & 1 & \cdots & n \end{bmatrix} .$
johnston-linear-matrix-algebra EQ0808	298	1.000	$\operatorname{det}\left(\left[\begin{array}{cccc} A_1 & * & \cdots & * \\ O & A_2 & \cdots & * \\ \vdots & \vdots & \ddots & \vdots \\ O & O & \cdots & A_n \end{array}\right]\right) = \prod_{j=1}^n \operatorname{det}(A_j),$	$\det \left(\begin{bmatrix} A_1 & * & \cdots & * \\ O & A_2 & \cdots & * \\ \vdots & \vdots & \ddots & \vdots \\ O & O & \cdots & A_n \end{bmatrix} \right) = \prod_{j=1}^n \det(A_j),$	$\det \left(\begin{bmatrix} A_1 & * & \cdots & * \\ O & A_2 & \cdots & * \\ \vdots & \vdots & \ddots & \vdots \\ O & O & \cdots & A_n \end{bmatrix} \right) = \prod_{j=1}^n \det(A_j),$
johnston-linear-matrix-algebra EQ0810	298	1.000	$\operatorname{det}\left(I_m + A B\right) = \operatorname{det}\left(I_n + B A\right) .$	$\det \left(I_m + A B \right) = \det \left(I_n + B A \right) .$	$\det(I_m + AB) = \det(I_n + BA).$
johnston-linear-matrix-algebra EQ0811	298	1.000	$\left[\begin{array}{cc} I_m & -A \\ B & I_n \end{array}\right] \left[\begin{array}{cc} I_m & A \\ O & I_n \end{array}\right]$	$\left[\begin{array}{cc} I_m & -A \\ B & I_n \end{array} \right] \left[\begin{array}{cc} I_m & A \\ O & I_n \end{array} \right]$	$\left[\begin{array}{cc} I_m & -A \\ B & I_n \end{array} \right] \left[\begin{array}{cc} I_m & A \\ O & I_n \end{array} \right]$
johnston-linear-matrix-algebra EQ0812	298	1.000	$V = \left[\begin{array}{cccccc} 1 & a_0 & a_0^2 & \cdots & a_0^n \\ 1 & a_1 & a_1^2 & \cdots & a_1^n \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 1 & a_n & a_n^2 & \cdots & a_n^n \end{array} \right] ,$	$V = \begin{bmatrix} 1 & a_0 & a_0^2 & \cdots & a_0^n \\ 1 & a_1 & a_1^2 & \cdots & a_1^n \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 1 & a_n & a_n^2 & \cdots & a_n^n \end{bmatrix} ,$	$V = \begin{bmatrix} 1 & a_0 & a_0^2 & \cdots & a_0^n \\ 1 & a_1 & a_1^2 & \cdots & a_1^n \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 1 & a_n & a_n^2 & \cdots & a_n^n \end{bmatrix} ,$
johnston-linear-matrix-algebra EQ0813	298	1.000	$\operatorname{det}(V) = \prod_{0 \leq i < j \leq n} (a_j - a_i) .$	$\det(V) = \prod_{0 \leq i < j \leq n} (a_j - a_i) .$	$\det(V) = \prod_{0 \leq i < j \leq n} (a_j - a_i) .$
johnston-linear-matrix-algebra EQ0816	299	1.000	$A \, \mathbf{v} = \lambda \mathbf{v} .$	$A \mathbf{v} = \lambda \mathbf{v} .$	$A \mathbf{v} = \lambda \mathbf{v} .$
johnston-linear-matrix-algebra EQ0817	300	1.000	$\begin{aligned} & v_1 + v_2 = 0 \\ & v_1 + v_2 = 0 \end{aligned}$	$\begin{aligned} v_1 + v_2 &= 0 \\ v_1 + v_2 &= 0 \end{aligned}$	$\begin{aligned} v_1 + v_2 &= 0 \\ v_1 + v_2 &= 0 \end{aligned}$
johnston-linear-matrix-algebra EQ0818	300	1.000	$A = \left[\begin{array}{ll} 2 & 1 \\ 1 & 2 \end{array} \right] \quad \text{and} \quad \mathbf{v} = \left[\begin{array}{l} 1 \\ 1 \end{array} \right] .$	$A = \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix} \quad \text{and} \quad \mathbf{v} = \begin{bmatrix} 1 \\ 1 \end{bmatrix} .$	$A = \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix} \quad \text{and} \quad \mathbf{v} = \begin{bmatrix} 1 \\ 1 \end{bmatrix} .$
johnston-linear-matrix-algebra EQ0819	300	1.000	$A \, \mathbf{v} = \left[\begin{array}{l} 2 & 1 \\ 1 & 2 \end{array} \right] \left[\begin{array}{l} 1 \\ 1 \end{array} \right] = \left[\begin{array}{l} 3 \\ 3 \end{array} \right] = 3 \mathbf{v} .$	$A \mathbf{v} = \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix} \begin{bmatrix} 1 \\ 1 \end{bmatrix} = \begin{bmatrix} 3 \\ 3 \end{bmatrix} = 3 \mathbf{v} .$	$A \mathbf{v} = \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix} \begin{bmatrix} 1 \\ 1 \end{bmatrix} = \begin{bmatrix} 3 \\ 3 \end{bmatrix} = 3 \mathbf{v} .$
johnston-linear-matrix-algebra EQ0820	300	1.000	$\begin{aligned} & 2 v_1 + v_2 = v_1 \\ & v_1 + 2 v_2 = v_2 \end{aligned}$	$\begin{aligned} 2 v_1 + v_2 &= v_1 \\ v_1 + 2 v_2 &= v_2 \end{aligned}$	$\begin{aligned} 2 v_1 + v_2 &= v_1 \\ v_1 + 2 v_2 &= v_2 \end{aligned}$
johnston-linear-matrix-algebra EQ0821	300	1.000	$\left[\begin{array}{lll} 1 & 1 & 0 \\ 1 & 1 & 0 \end{array} \right] \xrightarrow{R_2 - R_1} \left[\begin{array}{lll} 1 & 1 & 0 \\ 0 & 0 & 0 \end{array} \right]$	$\left[\begin{array}{lll} 1 & 1 & 0 \\ 1 & 1 & 0 \end{array} \right] \xrightarrow{R_2 - R_1} \left[\begin{array}{lll} 1 & 1 & 0 \\ 0 & 0 & 0 \end{array} \right]$	$\left[\begin{array}{lll} 1 & 1 & 0 \\ 1 & 1 & 0 \end{array} \right] \xrightarrow{R_2 - R_1} \left[\begin{array}{lll} 1 & 1 & 0 \\ 0 & 0 & 0 \end{array} \right]$

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johnston-linear-matrix-algebra-EQ0823	301	1.000	$A \mathbf{v} = \lambda \mathbf{v} \iff A \mathbf{v} - \lambda \mathbf{v} = \mathbf{0} \iff (A - \lambda I) \mathbf{v} = \mathbf{0}.$		$A \mathbf{v} = \lambda \mathbf{v} \iff A \mathbf{v} - \lambda \mathbf{v} = \mathbf{0} \iff (A - \lambda I) \mathbf{v} = \mathbf{0}.$
johnston-linear-matrix-algebra-EQ0826	302	1.000	$\lambda = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}.$		$\lambda = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}.$
johnston-linear-matrix-algebra-EQ0827	302	1.000	$\begin{aligned} 2v_1 + 2v_2 &= 0 \\ 5v_1 + 5v_2 &= 0 \end{aligned}$	$\begin{aligned} 2v_1 + 2v_2 &= 0 \\ 5v_1 + 5v_2 &= 0 \end{aligned}$	$\begin{aligned} 2v_1 + 2v_2 &= 0 \\ 5v_1 + 5v_2 &= 0 \end{aligned}$
johnston-linear-matrix-algebra-EQ0828	302	1.000	$\left[\begin{array}{ll l} 2 & 2 & 0 \\ 5 & 5 & 0 \end{array} \right] \xrightarrow{R_2 - \frac{5}{2}R_1} \left[\begin{array}{ll l} 2 & 2 & 0 \\ 0 & 0 & 0 \end{array} \right],$	$\left[\begin{array}{ll l} 2 & 2 & 0 \\ 5 & 5 & 0 \end{array} \right] \xrightarrow{R_2 - \frac{5}{2}R_1} \left[\begin{array}{ll l} 2 & 2 & 0 \\ 0 & 0 & 0 \end{array} \right],$	$\left[\begin{array}{ll l} 2 & 2 & 0 \\ 5 & 5 & 0 \end{array} \right] \xrightarrow{R_2 - \frac{5}{2}R_1} \left[\begin{array}{ll l} 2 & 2 & 0 \\ 0 & 0 & 0 \end{array} \right],$
johnston-linear-matrix-algebra-EQ0829	303	1.000	$\left[\begin{array}{cc c} -5 & 2 & 0 \\ 5 & -2 & 0 \end{array} \right] \xrightarrow{R_2 + R_1} \left[\begin{array}{cc c} -5 & 2 & 0 \\ 0 & 0 & 0 \end{array} \right]$	$\left[\begin{array}{cc c} -5 & 2 & 0 \\ 5 & -2 & 0 \end{array} \right] \xrightarrow{R_2 + R_1} \left[\begin{array}{cc c} -5 & 2 & 0 \\ 0 & 0 & 0 \end{array} \right]$	$\left[\begin{array}{cc c} -5 & 2 & 0 \\ 5 & -2 & 0 \end{array} \right] \xrightarrow{R_2 + R_1} \left[\begin{array}{cc c} -5 & 2 & 0 \\ 0 & 0 & 0 \end{array} \right],$
johnston-linear-matrix-algebra-EQ0830	303	1.000	$A = \begin{bmatrix} 1 & 2 \\ 5 & 4 \end{bmatrix}$	$A = \begin{bmatrix} 1 & 2 \\ 5 & 4 \end{bmatrix}$	$A = \begin{bmatrix} 1 & 2 \\ 5 & 4 \end{bmatrix}$
johnston-linear-matrix-algebra-EQ0831	303	1.000	$A = \begin{bmatrix} 1 & 3 & 3 \\ 3 & 1 & -1 \\ 0 & 0 & 2 \end{bmatrix}$	$A = \begin{bmatrix} 1 & 3 & 3 \\ 3 & 1 & -1 \\ 0 & 0 & 2 \end{bmatrix}$	$A = \begin{bmatrix} 1 & 3 & 3 \\ 3 & 1 & -1 \\ 0 & 0 & 2 \end{bmatrix}.$
johnston-linear-matrix-algebra-EQ0835	305	1.000	$\left[\begin{array}{lll} 1 & 0 & 0 \\ 0 & 1 & 1 \\ 0 & 0 & 0 \end{array} \right],$	$\left[\begin{array}{lll} 1 & 0 & 0 \\ 0 & 1 & 1 \\ 0 & 0 & 0 \end{array} \right],$	$\left[\begin{array}{lll} 1 & 0 & 0 \\ 0 & 1 & 1 \\ 0 & 0 & 0 \end{array} \right],$
johnston-linear-matrix-algebra-EQ0836	305	1.000	$\left[\begin{array}{lll} 1 & -1 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{array} \right],$	$\left[\begin{array}{lll} 1 & -1 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{array} \right],$	$\left[\begin{array}{lll} 1 & -1 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{array} \right],$
johnston-linear-matrix-algebra-EQ0839	306	1.000	$\lambda = \frac{-2 \pm \sqrt{4 + 48}}{-2} = 1 \pm \sqrt{13}.$	$\lambda = \frac{-2 \pm \sqrt{4 + 48}}{-2} = 1 \pm \sqrt{13}.$	$\lambda = \frac{-2 \pm \sqrt{4 + 48}}{-2} = 1 \pm \sqrt{13}.$
johnston-linear-matrix-algebra-EQ0840	307	1.000	$p_A(\lambda) = \det(A - \lambda I)$	$p_A(\lambda) = \det(A - \lambda I)$	$p_A(\lambda) = \det(A - \lambda I)$
johnston-linear-matrix-algebra-EQ0841	307	1.000	$A = \begin{bmatrix} 1 & 2 \\ 5 & 4 \end{bmatrix} \quad \text{and} \quad B = \begin{bmatrix} 1 & 2 & 3 \\ 1 & -2 & 1 \\ 3 & 2 & 1 \end{bmatrix}$	$A = \begin{bmatrix} 1 & 2 \\ 5 & 4 \end{bmatrix} \quad \text{and} \quad B = \begin{bmatrix} 1 & 2 & 3 \\ 1 & -2 & 1 \\ 3 & 2 & 1 \end{bmatrix}$	$A = \begin{bmatrix} 1 & 2 \\ 5 & 4 \end{bmatrix} \quad \text{and} \quad B = \begin{bmatrix} 1 & 2 & 3 \\ 1 & -2 & 1 \\ 3 & 2 & 1 \end{bmatrix}$
johnston-linear-matrix-algebra-EQ0843	307	1.000	$A = \begin{bmatrix} 2 & 0 & -3 \\ 1 & -1 & -1 \\ 0 & 0 & -1 \end{bmatrix}$	$A = \begin{bmatrix} 2 & 0 & -3 \\ 1 & -1 & -1 \\ 0 & 0 & -1 \end{bmatrix}$	$A = \begin{bmatrix} 2 & 0 & -3 \\ 1 & -1 & -1 \\ 0 & 0 & -1 \end{bmatrix}.$

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johnston-linear-matrix-algebra-EQ0844	308	1.000	$\begin{aligned} p_A(\lambda) &= \det(A - \lambda I) = \det \left(\begin{bmatrix} 2-\lambda & 0 & -3 \\ 1 & -1-\lambda & -1 \\ 0 & 0 & -1-\lambda \end{bmatrix} \right) \\ &= (2-\lambda)(-1-\lambda)(-1-\lambda) + 0 + 0 - 0 - 0 \\ &= (2-\lambda)(-1-\lambda)^2. \end{aligned}$	$p_A(\lambda) = \det(A - \lambda I) = \det \left(\begin{bmatrix} 2-\lambda & 0 & -3 \\ 1 & -1-\lambda & -1 \\ 0 & 0 & -1-\lambda \end{bmatrix} \right) \\ &= (2-\lambda)(-1-\lambda)(-1-\lambda) + 0 + 0 - 0 - 0 \\ &= (2-\lambda)(-1-\lambda)^2$	$p_A(\lambda) = \det(A - \lambda I) = \det \left(\begin{bmatrix} 2-\lambda & 0 & -3 \\ 1 & -1-\lambda & -1 \\ 0 & 0 & -1-\lambda \end{bmatrix} \right) \\ &= (2-\lambda)(-1-\lambda)(-1-\lambda) + 0 + 0 - 0 - 0 \\ &= (2-\lambda)(-1-\lambda)^2.$
johnston-linear-matrix-algebra-EQ0847	308	1.000	$\lambda = \frac{-2 \pm \sqrt{4-20}}{2} = -1 \pm \sqrt{-4} = -1 \pm 2i.$	$\lambda = \frac{-2 \pm \sqrt{4-20}}{2} = -1 \pm \sqrt{-4} = -1 \pm 2i.$	$\lambda = \frac{-2 \pm \sqrt{4-20}}{2} = -1 \pm \sqrt{-4} = -1 \pm 2i.$
johnston-linear-matrix-algebra-EQ0848	309	1.000	$\begin{bmatrix} 1 & 2 \\ -4 & -3 \end{bmatrix}$	$\begin{bmatrix} 1 & 2 \\ -4 & -3 \end{bmatrix}$	$\begin{bmatrix} 1 & 2 \\ -4 & -3 \end{bmatrix}$
johnston-linear-matrix-algebra-EQ0852	310	1.000	$p_A(\lambda) = (-1)^n \lambda^n + c_{n-1} \lambda^{n-1} + \cdots + c_1 \lambda + c_0.$	$p_A(\lambda) = (-1)^n \lambda^n + c_{n-1} \lambda^{n-1} + \cdots + c_1 \lambda + c_0.$	$p_A(\lambda) = (-1)^n \lambda^n + c_{n-1} \lambda^{n-1} + \cdots + c_1 \lambda + c_0.$
johnston-linear-matrix-algebra-EQ0855	311	1.000	$p_A(0) = \det(A) = c_0 = \lambda_1 \lambda_2 \cdots \lambda_n,$	$p_A(0) = \det(A) = c_0 = \lambda_1 \lambda_2 \cdots \lambda_n,$	$p_A(0) = \det(A) = c_0 = \lambda_1 \lambda_2 \cdots \lambda_n,$
johnston-linear-matrix-algebra-EQ0856	311	1.000	$p_A(\lambda) = (\lambda_1 - \lambda)(\lambda_2 - \lambda) \cdots (\lambda_n - \lambda)$	$p_A(\lambda) = (\lambda_1 - \lambda)(\lambda_2 - \lambda) \cdots (\lambda_n - \lambda)$	$p_A(\lambda) = (\lambda_1 - \lambda)(\lambda_2 - \lambda) \cdots (\lambda_n - \lambda)$
johnston-linear-matrix-algebra-EQ0857	311	1.000	$p_A(\lambda) = \det(A - \lambda I) = \det \left(\begin{bmatrix} a_{1,1} - \lambda & a_{1,2} & \cdots & a_{1,n} \\ a_{2,1} & a_{2,2} - \lambda & \cdots & a_{2,n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{n,1} & a_{n,2} & \cdots & a_{n,n} - \lambda \end{bmatrix} \right).$	$p_A(\lambda) = \det(A - \lambda I) = \det \left(\begin{bmatrix} a_{1,1} - \lambda & a_{1,2} & \cdots & a_{1,n} \\ a_{2,1} & a_{2,2} - \lambda & \cdots & a_{2,n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{n,1} & a_{n,2} & \cdots & a_{n,n} - \lambda \end{bmatrix} \right).$	$p_A(\lambda) = \det(A - \lambda I) = \det \left(\begin{bmatrix} a_{1,1} - \lambda & a_{1,2} & \cdots & a_{1,n} \\ a_{2,1} & a_{2,2} - \lambda & \cdots & a_{2,n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{n,1} & a_{n,2} & \cdots & a_{n,n} - \lambda \end{bmatrix} \right).$

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johnston-linear-matrix-algebra EQ0858	311	1.000	$\begin{gathered} \operatorname{det}(A-\lambda I)= \\ \left(a_{1,1}-\lambda\right) \operatorname{det}\left(\begin{array}{cccc} a_{2,2}-\lambda & a_{2,3} & \cdots & a_{2,n} \\ a_{3,2} & a_{3,3}-\lambda & \cdots & a_{3,n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{n,2} & a_{n,3} & \cdots & a_{n,n}-\lambda \end{array}\right) \\ -a_{2,1} \operatorname{det}\left(\begin{array}{cccc} a_{1,2} & a_{1,3} & \cdots & a_{1,n} \\ a_{3,2} & a_{3,3}-\lambda & \cdots & a_{3,n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{n,2} & a_{n,3} & \cdots & a_{n,n}-\lambda \end{array}\right) \\ +(-1)^{n+1} a_{n,1} \operatorname{det}\left(\begin{array}{cccc} a_{1,2} & a_{1,3} & \cdots & a_{1,n} \\ a_{2,2}-\lambda & a_{2,3} & \cdots & a_{2,n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{n-1,2} & a_{n-1,3} & \cdots & a_{n-1,n} \end{array}\right) . \end{gathered}$	$\det(A-\lambda I)=(a_{1,1}-\lambda) \det \left(\begin{bmatrix} a_{2,2}-\lambda & a_{2,3} & \cdots & a_{2,n} \\ a_{3,2} & a_{3,3}-\lambda & \cdots & a_{3,n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{n,2} & a_{n,3} & \cdots & a_{n,n}-\lambda \end{bmatrix}\right) \\ -a_{2,1} \det \left(\begin{bmatrix} a_{1,2} & a_{1,3} & \cdots & a_{1,n} \\ a_{3,2} & a_{3,3}-\lambda & \cdots & a_{3,n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{n,2} & a_{n,3} & \cdots & a_{n,n}-\lambda \end{bmatrix}\right) \\ +(-1)^{n+1} a_{n,1} \det \left(\begin{bmatrix} a_{1,2} & a_{1,3} & \cdots & a_{1,n} \\ a_{2,2}-\lambda & a_{2,3} & \cdots & a_{2,n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{n-1,2} & a_{n-1,3} & \cdots & a_{n-1,n} \end{bmatrix}\right) .$	
johnston-linear-matrix-algebra EQ0859	312	1.000	$\left(a_{1,1}-\lambda\right) \operatorname{det}\left(\begin{array}{cccc} a_{2,2}-\lambda & a_{2,3} & \cdots & a_{2,n} \\ a_{3,2} & a_{3,3}-\lambda & \cdots & a_{3,n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{n,2} & a_{n,3} & \cdots & a_{n,n}-\lambda \end{array}\right) .$	$(a_{1,1}-\lambda) \det \left(\begin{bmatrix} a_{2,2}-\lambda & a_{2,3} & \cdots & a_{2,n} \\ a_{3,2} & a_{3,3}-\lambda & \cdots & a_{3,n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{n,2} & a_{n,3} & \cdots & a_{n,n}-\lambda \end{bmatrix}\right) .$	$(a_{1,1}-\lambda) \det \left(\begin{bmatrix} a_{2,2}-\lambda & a_{2,3} & \cdots & a_{2,n} \\ a_{3,2} & a_{3,3}-\lambda & \cdots & a_{3,n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{n,2} & a_{n,3} & \cdots & a_{n,n}-\lambda \end{bmatrix}\right) .$
johnston-linear-matrix-algebra EQ0860	312	1.000	$\left(a_{1,1}-\lambda\right) \left(a_{2,2}-\lambda\right) \cdots\left(a_{2,2}-\lambda\right) .$	$(a_{1,1}-\lambda)\left(a_{2,2}-\lambda\right) \cdots\left(a_{2,2}-\lambda\right) .$	$(a_{1,1}-\lambda)\left(a_{2,2}-\lambda\right) \cdots\left(a_{2,2}-\lambda\right) .$
johnston-linear-matrix-algebra EQ0861	312	1.000	$c_{n-1}=(-1)^{n-1}\left(a_{1,1}+a_{2,2}+\cdots+a_{n,n}\right) \operatorname{tr}(A),$	$c_{n-1}=(-1)^{n-1}\left(a_{1,1}+a_{2,2}+\cdots+a_{n,n}\right)=(-1)^{n-1} \operatorname{tr}(A),$	$c_{n-1}=(-1)^{n-1}\left(a_{1,1}+a_{2,2}+\cdots+a_{n,n}\right)=(-1)^{n-1} \operatorname{tr}(A),$
johnston-linear-matrix-algebra EQ0862	312	1.000	$A=\left(\begin{array}{ccc} 3 & 1 & 1 \\ 0 & 1 & 0 \\ -2 & -1 & 0 \end{array}\right)$	$A=\left[\begin{array}{ccc} 3 & 1 & 1 \\ 0 & 1 & 0 \\ -2 & -1 & 0 \end{array}\right]$	$A=\left[\begin{array}{ccc} 3 & 1 & 1 \\ 0 & 1 & 0 \\ -2 & -1 & 0 \end{array}\right]$
johnston-linear-matrix-algebra EQ0863	312	1.000	$\operatorname{tr}(A)=3+1+0=4 \quad \text { and } \quad \operatorname{det}(A)=0+0+0-0-(-2)-0=2 .$	$\operatorname{tr}(A)=3+1+0=4 \quad \text { and } \quad \operatorname{det}(A)=0+0+0-0-(-2)-0=2 .$	$\operatorname{tr}(A)=3+1+0=4 \quad \text { and } \quad \operatorname{det}(A)=0+0+0-0-(-2)-0=2 .$
johnston-linear-matrix-algebra EQ0865	313	1.000	$p_A(\lambda)=\lambda^2-\operatorname{tr}(A) \lambda+\operatorname{det}(A) .$	$p_A(\lambda)=\lambda^2-\operatorname{tr}(A) \lambda+\operatorname{det}(A) .$	$p_A(\lambda)=\lambda^2-\operatorname{tr}(A) \lambda+\operatorname{det}(A) .$

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johnston-linear-matrix-algebra-EQ0866	313	1.000	$A^{*} \stackrel{\text{def}}{=} \bar{A}^T$.	$A^{*} \stackrel{\text{def}}{=} \bar{A}^T$.	$A^{*} \stackrel{\text{def}}{=} \bar{A}^T$.
johnston-linear-matrix-algebra-EQ0868	314	1.000	$\left[\begin{array}{cc} 0 & 2+3i \\ 2-3i & 4 \end{array}\right]^{*} = \left[\begin{array}{cc} 0 & 2+3i \\ 2-3i & 4 \end{array}\right]$.	$\left[\begin{array}{cc} 0 & 2+3i \\ 2-3i & 4 \end{array}\right]^{*} = \left[\begin{array}{cc} 0 & 2+3i \\ 2-3i & 4 \end{array}\right]$.	$\left[\begin{array}{cc} 0 & 2+3i \\ 2-3i & 4 \end{array}\right]^{*} = \left[\begin{array}{cc} 0 & 2+3i \\ 2-3i & 4 \end{array}\right]$.
johnston-linear-matrix-algebra-EQ0869	314	1.000	$\left[\begin{array}{lll} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{array}\right]^{*} = \left[\begin{array}{lll} 1 & 4 & 7 \\ 2 & 5 & 8 \\ 3 & 6 & 9 \end{array}\right]$.	$\left[\begin{array}{lll} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{array}\right]^{*} = \left[\begin{array}{lll} 1 & 4 & 7 \\ 2 & 5 & 8 \\ 3 & 6 & 9 \end{array}\right]$.	$\left[\begin{array}{lll} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{array}\right]^{*} = \left[\begin{array}{lll} 1 & 4 & 7 \\ 2 & 5 & 8 \\ 3 & 6 & 9 \end{array}\right]$.
johnston-linear-matrix-algebra-EQ0870	314	1.000	$\left[\begin{array}{cc} i & 1-i \\ -2 & -2-i \end{array}\right]^{*} = \left[\begin{array}{ccc} -i & -2 & 1-i \\ 1+i & -2+i & 2-3i \end{array}\right]$	$\left[\begin{array}{cc} i & 1-i \\ -2 & -2-i \\ 1+i & 2+3i \end{array}\right]^{*} = \left[\begin{array}{ccc} -i & -2 & 1-i \\ 1+i & -2+i & 2-3i \end{array}\right]$	$\left[\begin{array}{cc} i & 1-i \\ -2 & -2-i \\ 1+i & 2+3i \end{array}\right]^{*} = \left[\begin{array}{ccc} -i & -2 & 1-i \\ 1+i & -2+i & 2-3i \end{array}\right]$.
johnston-linear-matrix-algebra-EQ0873	314	1.000	$\lambda \mathbf{v} = \mathbf{v} A \mathbf{v} = (\mathbf{v}^* A \mathbf{v})^* = (\lambda \mathbf{v}^* \mathbf{v})^* = \bar{\lambda} \mathbf{v}^* \mathbf{v}$. $\mathbf{v} = \left(\mathbf{v}^* \mathbf{v}\right)^{*} A \mathbf{v} \left(\mathbf{v}^* \mathbf{v}\right)^{*} = \left(\lambda \mathbf{v}^* \mathbf{v}\right)^{*} \mathbf{v} = \bar{\lambda} \mathbf{v}$.	$\lambda \mathbf{v}^* \mathbf{v} = \mathbf{v}^* A \mathbf{v} = (\mathbf{v}^* A \mathbf{v})^* = (\lambda \mathbf{v}^* \mathbf{v})^* = \bar{\lambda} \mathbf{v}^* \mathbf{v}$.	$\lambda \mathbf{v}^* \mathbf{v} = \mathbf{v}^* A \mathbf{v} = (\mathbf{v}^* A \mathbf{v})^* = (\lambda \mathbf{v}^* \mathbf{v})^* = \bar{\lambda} \mathbf{v}^* \mathbf{v}$.
johnston-linear-matrix-algebra-EQ0878	316	1.000	$\lambda = \frac{4 \pm \sqrt{16+4}}{-2} = -2 \pm \sqrt{5}$.	$\lambda = \frac{4 \pm \sqrt{16+4}}{-2} = -2 \pm \sqrt{5}$.	$\lambda = \frac{4 \pm \sqrt{16+4}}{-2} = -2 \pm \sqrt{5}$.
johnston-linear-matrix-algebra-EQ0879	316	1.000	$A = \left[\begin{array}{ccc} 1 & 3 & 3 \\ 3 & 1 & -1 \\ 0 & 0 & 2 \end{array}\right]$	$A = \left[\begin{array}{ccc} 1 & 3 & 3 \\ 3 & 1 & -1 \\ 0 & 0 & 2 \end{array}\right]$	$A = \left[\begin{array}{ccc} 1 & 3 & 3 \\ 3 & 1 & -1 \\ 0 & 0 & 2 \end{array}\right]$
johnston-linear-matrix-algebra-EQ0880	317	1.000	$\left[\begin{array}{ccc c} 0 & 0 & -3 & 0 \\ 1 & -3 & -1 & 0 \\ 0 & 0 & -3 & 0 \end{array}\right] \xrightarrow{R_3-R_1} \left[\begin{array}{ccc c} 0 & 0 & -3 & 0 \\ 1 & -3 & -1 & 0 \\ 0 & 0 & 0 & 0 \end{array}\right]$.	$\left[\begin{array}{ccc c} 0 & 0 & -3 & 0 \\ 1 & -3 & -1 & 0 \\ 0 & 0 & -3 & 0 \end{array}\right] \xrightarrow{R_3-R_1} \left[\begin{array}{ccc c} 0 & 0 & -3 & 0 \\ 1 & -3 & -1 & 0 \\ 0 & 0 & 0 & 0 \end{array}\right]$.	$\left[\begin{array}{ccc c} 0 & 0 & -3 & 0 \\ 1 & -3 & -1 & 0 \\ 0 & 0 & -3 & 0 \end{array}\right] \xrightarrow{R_3-R_1} \left[\begin{array}{ccc c} 0 & 0 & -3 & 0 \\ 1 & -3 & -1 & 0 \\ 0 & 0 & 0 & 0 \end{array}\right]$.
johnston-linear-matrix-algebra-EQ0881	317	1.000	$\left[\begin{array}{ccc c} 3 & 0 & -3 & 0 \\ 1 & 0 & -1 & 0 \\ 0 & 0 & 0 & 0 \end{array}\right] \xrightarrow{R_2-\frac{1}{3}R_1} \left[\begin{array}{ccc c} 3 & 0 & -3 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{array}\right]$.	$\left[\begin{array}{ccc c} 3 & 0 & -3 & 0 \\ 1 & 0 & -1 & 0 \\ 0 & 0 & 0 & 0 \end{array}\right] \xrightarrow{R_2-\frac{1}{3}R_1} \left[\begin{array}{ccc c} 3 & 0 & -3 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{array}\right]$.	$\left[\begin{array}{ccc c} 3 & 0 & -3 & 0 \\ 1 & 0 & -1 & 0 \\ 0 & 0 & 0 & 0 \end{array}\right] \xrightarrow{R_2-\frac{1}{3}R_1} \left[\begin{array}{ccc c} 3 & 0 & -3 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{array}\right]$.
johnston-linear-matrix-algebra-EQ0882	317	1.000	$A = \left[\begin{array}{ccc} 2 & 0 & -3 \\ 1 & -1 & -1 \\ 0 & 0 & -1 \end{array}\right]$	$A = \left[\begin{array}{ccc} 2 & 0 & -3 \\ 1 & -1 & -1 \\ 0 & 0 & -1 \end{array}\right]$	$A = \left[\begin{array}{ccc} 2 & 0 & -3 \\ 1 & -1 & -1 \\ 0 & 0 & -1 \end{array}\right]$
johnston-linear-matrix-algebra-EQ0885	318	1.000	$\left[\begin{array}{lll l} 3 & 2 & 3 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \end{array}\right] \xrightarrow{R_1-2R_3} \left[\begin{array}{lll l} 3 & 0 & 3 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \end{array}\right]$.	$\left[\begin{array}{lll l} 3 & 2 & 3 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \end{array}\right] \xrightarrow{R_1-2R_3} \left[\begin{array}{lll l} 3 & 0 & 3 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \end{array}\right]$.	$\left[\begin{array}{lll l} 3 & 2 & 3 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \end{array}\right] \xrightarrow{R_1-2R_3} \left[\begin{array}{lll l} 3 & 0 & 3 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \end{array}\right]$.
johnston-linear-matrix-algebra-EQ0886	319	1.000	$P = [\mathbf{v}_1 \cdots \mathbf{v}_k V] \in \mathcal{M}_n$	$P = [\mathbf{v}_1 \cdots \mathbf{v}_k V] \in \mathcal{M}_n$	$P = [\mathbf{v}_1 \cdots \mathbf{v}_k V] \in \mathcal{M}_n$

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johnston-linear-matrix-algebra-EQ0888	319	1.000	$P^{-1}AP = \left[\begin{array}{cc} \lambda_1 I_k & B \\ 0 & C \end{array} \right],$	$P^{-1}AP = \begin{bmatrix} \lambda_1 I_k & B \\ O & C \end{bmatrix},$	$P^{-1}AP = \begin{bmatrix} \lambda_1 I_k & B \\ O & C \end{bmatrix},$
johnston-linear-matrix-algebra-EQ0890	320	1.000	$A = \left[\begin{array}{cccc} -6 & 0 & -2 & 1 \\ & 0 & -3 & -2 \\ -2 & 1 & 3 & 0 \\ & 1 & 1 & -3 \end{array} \right] \begin{array}{l} \\ \\ 0 & 2 & 2 \end{array} \end{array} \right]$	$A = \begin{bmatrix} -6 & 0 & -2 & 1 \\ 0 & -3 & -2 & 1 \\ 3 & 0 & 1 & 1 \\ -3 & 0 & 2 & 2 \end{bmatrix}$	$A = \begin{bmatrix} -6 & 0 & -2 & 1 \\ 0 & -3 & -2 & 1 \\ 3 & 0 & 1 & 1 \\ -3 & 0 & 2 & 2 \end{bmatrix}.$
johnston-linear-matrix-algebra-EQ0893	321	1.000	$A = \left[\begin{array}{lll} 1 & 2 & 3 \\ 0 & 4 & 5 \\ 0 & 6 & \end{array} \right] .$	$A = \begin{bmatrix} 1 & 2 & 3 \\ 0 & 4 & 5 \\ 0 & 0 & 6 \end{bmatrix} .$	$A = \begin{bmatrix} 1 & 2 & 3 \\ 0 & 4 & 5 \\ 0 & 0 & 6 \end{bmatrix} .$
johnston-linear-matrix-algebra-EQ0895	321	1.000	$\lambda = 1$ \text { : We want to find the null space of the matrix } A - \lambda I =	$\lambda = 1$: We want to find the null space of the matrix $A - \lambda I =$	$\lambda = 1$: We want to find the null space of the matrix $A - \lambda I =$
johnston-linear-matrix-algebra-EQ0896	322	1.000	$A - I :$	$A - I :$	$A - I :$
johnston-linear-matrix-algebra-EQ0898	322	1.000	$\left[\begin{array}{ccc c} -3 & 2 & 3 & 0 \\ 0 & 0 & 5 & 0 \\ 0 & 0 & 2 & 0 \end{array} \right] \xrightarrow{R_3 - \frac{2}{5}R_2} \left[\begin{array}{ccc c} -3 & 2 & 3 & 0 \\ 0 & 0 & 5 & 0 \\ 0 & 0 & 0 & 0 \end{array} \right] .$	$\left[\begin{array}{ccc c} -3 & 2 & 3 & 0 \\ 0 & 0 & 5 & 0 \\ 0 & 0 & 2 & 0 \end{array} \right] \xrightarrow{R_3 - \frac{2}{5}R_2} \left[\begin{array}{ccc c} -3 & 2 & 3 & 0 \\ 0 & 0 & 5 & 0 \\ 0 & 0 & 0 & 0 \end{array} \right] .$	$\left[\begin{array}{ccc c} -3 & 2 & 3 & 0 \\ 0 & 0 & 5 & 0 \\ 0 & 0 & 2 & 0 \end{array} \right] \xrightarrow{R_3 - \frac{2}{5}R_2} \left[\begin{array}{ccc c} -3 & 2 & 3 & 0 \\ 0 & 0 & 5 & 0 \\ 0 & 0 & 0 & 0 \end{array} \right] .$
johnston-linear-matrix-algebra-EQ0899	322	1.000	$\left[\begin{array}{ccc c} -5 & 2 & 3 & 0 \\ 0 & -2 & 5 & 0 \\ 0 & 0 & 0 & 0 \end{array} \right] \xrightarrow{R_1 + R_2} \left[\begin{array}{ccc c} -5 & 0 & 8 & 0 \\ 0 & -2 & 5 & 0 \\ 0 & 0 & 0 & 0 \end{array} \right] .$	$\left[\begin{array}{ccc c} -5 & 2 & 3 & 0 \\ 0 & -2 & 5 & 0 \\ 0 & 0 & 0 & 0 \end{array} \right] \xrightarrow{R_1 + R_2} \left[\begin{array}{ccc c} -5 & 0 & 8 & 0 \\ 0 & -2 & 5 & 0 \\ 0 & 0 & 0 & 0 \end{array} \right] .$	$\left[\begin{array}{ccc c} -5 & 2 & 3 & 0 \\ 0 & -2 & 5 & 0 \\ 0 & 0 & 0 & 0 \end{array} \right] \xrightarrow{R_1 + R_2} \left[\begin{array}{ccc c} -5 & 0 & 8 & 0 \\ 0 & -2 & 5 & 0 \\ 0 & 0 & 0 & 0 \end{array} \right] .$
johnston-linear-matrix-algebra-EQ0900	322	1.000	$\left[\begin{array}{lll} 2 & 1 & 0 \\ 0 & 2 & 1 \\ 0 & 0 & 2 \end{array} \right] - 2 \left[\begin{array}{lll} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{array} \right] = \left[\begin{array}{lll} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{array} \right] .$	$\left[\begin{array}{lll} 2 & 1 & 0 \\ 0 & 2 & 1 \\ 0 & 0 & 2 \end{array} \right] - 2 \left[\begin{array}{lll} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{array} \right] = \left[\begin{array}{lll} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{array} \right] .$	$\left[\begin{array}{lll} 2 & 1 & 0 \\ 0 & 2 & 1 \\ 0 & 0 & 2 \end{array} \right] - 2 \left[\begin{array}{lll} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{array} \right] = \left[\begin{array}{lll} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{array} \right] .$
johnston-linear-matrix-algebra-EQ0901	323	1.000	$\left[\begin{array}{cccc} -4 & -12 & 28 & 44 \\ 1 & 8 & -16 & -30 \\ -10 & -12 & 30 & 36 \\ 5 & 6 & -14 & -16 \end{array} \right] \begin{array}{l} \\ \\ -2 \\ 4 \end{array} \end{array} \right]$	$\left[\begin{array}{cccc} -4 & -12 & 28 & 44 \\ 1 & 8 & -16 & -30 \\ -10 & -12 & 30 & 36 \\ 5 & 6 & -14 & -16 \\ -2 & -2 & 0 & -4 \end{array} \right] \begin{array}{l} \\ \\ \\ 4 \end{array}$	$\left[\begin{array}{cccc} -4 & -12 & 28 & 44 \\ 1 & 8 & -16 & -30 \\ -10 & -12 & 30 & 36 \\ 5 & 6 & -14 & -16 \\ -2 & -2 & 0 & -4 \end{array} \right] \begin{array}{l} \\ \\ \\ 4 \end{array}$
johnston-linear-matrix-algebra-EQ0902 (1)	323	1.000	$A = \left[\begin{array}{cc} k & 1 \\ -1 & 1 \end{array} \right] .$	$A = \begin{bmatrix} k & 1 \\ -1 & 1 \end{bmatrix} .$	$A = \begin{bmatrix} k & 1 \\ -1 & 1 \end{bmatrix} .$
johnston-linear-matrix-algebra-EQ0903	324	1.000	$[R^\theta] = \left[\begin{array}{cc} \cos(\theta) & -\sin(\theta) \\ \sin(\theta) & \cos(\theta) \end{array} \right] .$	$[R^\theta] = \begin{bmatrix} \cos(\theta) & -\sin(\theta) \\ \sin(\theta) & \cos(\theta) \end{bmatrix} .$	$[R^\theta] = \begin{bmatrix} \cos(\theta) & -\sin(\theta) \\ \sin(\theta) & \cos(\theta) \end{bmatrix} .$
johnston-linear-matrix-algebra-EQ0904	324	1.000	$\mathbf{v} \cdot \mathbf{w} \stackrel{\text{def}}{=} \overline{v_1} w_1 + \overline{v_2} w_2 + \cdots + \overline{v_n} w_n,$	$\mathbf{v} \cdot \mathbf{w} \stackrel{\text{def}}{=} \overline{v_1} w_1 + \overline{v_2} w_2 + \cdots + \overline{v_n} w_n,$	$\mathbf{v} \cdot \mathbf{w} \stackrel{\text{def}}{=} \overline{v_1} w_1 + \overline{v_2} w_2 + \cdots + \overline{v_n} w_n,$

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johnston-linear-matrix-algebra-EQ9905	324	1.000	$C=\left[\begin{array}{ccc} 0 & 1 & 0 \\ 0 & 0 & 1 \\ -a_0 & -a_1 & -a_2 \end{array}\right]$	$C = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ -a_0 & -a_1 & -a_2 \end{bmatrix}$	$C = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ -a_0 & -a_1 & -a_2 \end{bmatrix}$
johnston-linear-matrix-algebra-EQ9906 (0)	324	1.000	$C=\left[\begin{array}{ccccc} 0 & 1 & 0 & \cdots & 0 \\ & 0 & 1 & \cdots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & 0 & \cdots & 1 \\ -a_0 & -a_1 & -a_2 & \cdots & -a_{n-1} \end{array}\right]$	$C = \begin{bmatrix} 0 & 1 & 0 & \cdots & 0 \\ 0 & 0 & 1 & \cdots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & 0 & \cdots & 1 \\ -a_0 & -a_1 & -a_2 & \cdots & -a_{n-1} \end{bmatrix}$	$C = \begin{bmatrix} 0 & 1 & 0 & \cdots & 0 \\ 0 & 0 & 1 & \cdots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & 0 & \cdots & 1 \\ -a_0 & -a_1 & -a_2 & \cdots & -a_{n-1} \end{bmatrix}$
johnston-linear-matrix-algebra-EQ9908	325	1.000	$A^k=\left[\begin{array}{cccc} a_{1,1} & 0 & \cdots & 0 \\ 0 & a_{2,2} & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & a_{n,n} \end{array}\right]^k=\left[\begin{array}{cccc} a_{1,1}^k & 0 & \cdots & 0 \\ 0 & a_{2,2}^k & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & a_{n,n}^k \end{array}\right].$	$A^k = \begin{bmatrix} a_{1,1} & 0 & \cdots & 0 \\ 0 & a_{2,2} & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & a_{n,n} \end{bmatrix}^k = \begin{bmatrix} a_{1,1}^k & 0 & \cdots & 0 \\ 0 & a_{2,2}^k & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & a_{n,n}^k \end{bmatrix}.$	$A^k = \begin{bmatrix} a_{1,1} & 0 & \cdots & 0 \\ 0 & a_{2,2} & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & a_{n,n} \end{bmatrix}^k = \begin{bmatrix} a_{1,1}^k & 0 & \cdots & 0 \\ 0 & a_{2,2}^k & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & a_{n,n}^k \end{bmatrix}.$
johnston-linear-matrix-algebra-EQ9910	326	1.000	$A\,P=A\left[\mathbf{v}_1\left \mathbf{v}_2\right.\right]\cdots\left \mathbf{v}_n\right.=\left[A\,\mathbf{v}_1\left \mathbf{v}_2\right.\right]\cdots\left \mathbf{v}_n\right.$	$AP = A[\mathbf{v}_1 \mathbf{v}_2 \cdots \mathbf{v}_n] = [A\mathbf{v}_1 A\mathbf{v}_2 \cdots A\mathbf{v}_n]$	$AP = A[\mathbf{v}_1 \mathbf{v}_2 \cdots \mathbf{v}_n] = [A\mathbf{v}_1 A\mathbf{v}_2 \cdots A\mathbf{v}_n]$
johnston-linear-matrix-algebra-EQ9911	326	1.000	$P\,D=\left[\mathbf{v}_1\left \mathbf{v}_2\right.\right]\cdots\left \mathbf{v}_n\right.D=\left[d_{1,1}\mathbf{v}_1\left d_{2,2}\mathbf{v}_2\right.\right]\cdots\left d_{n,n}\mathbf{v}_n\right.$	$PD = [\mathbf{v}_1 \mathbf{v}_2 \cdots \mathbf{v}_n] D = [d_{1,1}\mathbf{v}_1 d_{2,2}\mathbf{v}_2 \cdots d_{n,n}\mathbf{v}_n].$	$PD = [\mathbf{v}_1 \mathbf{v}_2 \cdots \mathbf{v}_n] D = [d_{1,1}\mathbf{v}_1 d_{2,2}\mathbf{v}_2 \cdots d_{n,n}\mathbf{v}_n].$
johnston-linear-matrix-algebra-EQ9912	326	1.000	$D=\left[\begin{array}{cc} \lambda_1 & 0 \\ 0 & \lambda_2 \end{array}\right]=\left[\begin{array}{cc} -1 & 0 \\ 0 & 6 \end{array}\right]\quad\text{and}\quad P=\left[\mathbf{v}_1\left \mathbf{v}_2\right.\right]=\left[\begin{array}{cc} -1 & 2 \\ 1 & 5 \end{array}\right].$	$D = \begin{bmatrix} \lambda_1 & 0 \\ 0 & \lambda_2 \end{bmatrix} = \begin{bmatrix} -1 & 0 \\ 0 & 6 \end{bmatrix} \quad \text{and} \quad P = [\mathbf{v}_1 \mathbf{v}_2] = \begin{bmatrix} -1 & 2 \\ 1 & 5 \end{bmatrix}.$	$D = \begin{bmatrix} \lambda_1 & 0 \\ 0 & \lambda_2 \end{bmatrix} = \begin{bmatrix} -1 & 0 \\ 0 & 6 \end{bmatrix} \quad \text{and} \quad P = [\mathbf{v}_1 \mathbf{v}_2] = \begin{bmatrix} -1 & 2 \\ 1 & 5 \end{bmatrix}.$
johnston-linear-matrix-algebra-EQ9913	327	1.000	$P^{-1}=\frac{1}{7}\left[\begin{array}{cc} -5 & 2 \\ 1 & 1 \end{array}\right],$	$P^{-1} = \frac{1}{7} \begin{bmatrix} -5 & 2 \\ 1 & 1 \end{bmatrix},$	$P^{-1} = \frac{1}{7} \begin{bmatrix} -5 & 2 \\ 1 & 1 \end{bmatrix},$
johnston-linear-matrix-algebra-EQ9916	328	1.000	$P^{-1}=\frac{1}{3}\left[\begin{array}{ccc} 1 & 0 & -1 \\ -1 & 3 & 1 \\ 0 & 0 & 3 \end{array}\right],$	$P^{-1} = \frac{1}{3} \begin{bmatrix} 1 & 0 & -1 \\ -1 & 3 & 1 \\ 0 & 0 & 3 \end{bmatrix},$	$P^{-1} = \frac{1}{3} \begin{bmatrix} 1 & 0 & -1 \\ -1 & 3 & 1 \\ 0 & 0 & 3 \end{bmatrix},$

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johnston-linear-matrix-algebra-EQ0917	328	1.000	$\begin{aligned} P D P^{-1} &= \frac{1}{3} \left[\begin{array}{lll} 3 & 0 & 1 \\ 1 & 1 & 0 \\ 0 & 0 & 1 \end{array} \right] \left[\begin{array}{lll} 2 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & -1 \end{array} \right] \left[\begin{array}{lll} 1 & 0 & -1 \\ -1 & 3 & 1 \\ 0 & 0 & 3 \end{array} \right] \\ &= \frac{1}{3} \left[\begin{array}{lll} 3 & 0 & 1 \\ 1 & 1 & 0 \\ 0 & 0 & 1 \end{array} \right] \left[\begin{array}{lll} 2 & 0 & -2 \\ 1 & -3 & -1 \\ 0 & 0 & -3 \end{array} \right] = \left[\begin{array}{lll} 2 & 0 & -3 \\ 1 & -1 & -1 \\ 0 & 0 & -1 \end{array} \right] = A. \end{aligned}$	$PDP^{-1} = \frac{1}{3} \begin{bmatrix} 3 & 0 & 1 \\ 1 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 2 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & -1 \end{bmatrix} \begin{bmatrix} 1 & 0 & -1 \\ -1 & 3 & 1 \\ 0 & 0 & 3 \end{bmatrix}$ $= \frac{1}{3} \begin{bmatrix} 3 & 0 & 1 \\ 1 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 2 & 0 & -2 \\ 1 & -3 & -1 \\ 0 & 0 & -3 \end{bmatrix} = \begin{bmatrix} 2 & 0 & -3 \\ 1 & -1 & -1 \\ 0 & 0 & -1 \end{bmatrix} = A.$	
johnston-linear-matrix-algebra-EQ0918	328	1.000	$B_{\{1\}} \cup B_{\{2\}} \cup \cdots \cup B_{\{m\}}$	$B_1 \cup B_2 \cup \cdots \cup B_m$	$B_1 \cup B_2 \cup \cdots \cup B_m$
johnston-linear-matrix-algebra-EQ0919 (3.4.1)	329	1.000	$\mathbf{v}_k = c_1 \mathbf{v}_1 + c_2 \mathbf{v}_2 + \cdots + c_{k-1} \mathbf{v}_{k-1} + \cdots + c_{k-1} \mathbf{v}_{k-1} .$	$\mathbf{v}_k = c_1 \mathbf{v}_1 + c_2 \mathbf{v}_2 + \cdots + c_{k-1} \mathbf{v}_{k-1} .$	$\mathbf{v}_k = c_1 \mathbf{v}_1 + c_2 \mathbf{v}_2 + \cdots + c_{k-1} \mathbf{v}_{k-1} .$
johnston-linear-matrix-algebra-EQ0923	329	1.000	$\mathbf{v}_1 + \mathbf{v}_2 + \cdots + \mathbf{v}_{m-1} + \mathbf{v}_m = \mathbf{v}_1 + \mathbf{v}_2 + \cdots + \mathbf{v}_{m-1} + \mathbf{0} = \mathbf{0},$	$\mathbf{v}_1 + \mathbf{v}_2 + \cdots + \mathbf{v}_{m-1} + \mathbf{v}_m = \mathbf{v}_1 + \mathbf{v}_2 + \cdots + \mathbf{v}_{m-1} + \mathbf{0} = \mathbf{0},$	$\mathbf{v}_1 + \mathbf{v}_2 + \cdots + \mathbf{v}_{m-1} + \mathbf{v}_m = \mathbf{v}_1 + \mathbf{v}_2 + \cdots + \mathbf{v}_{m-1} + \mathbf{0} = \mathbf{0},$
johnston-linear-matrix-algebra-EQ0924 (3.4.4)	329	1.000	$\underbrace{\sum_{\ell=1}^{\gamma_1} c_{1,\ell} \mathbf{v}_{1,\ell}}_{\text{call this } \mathbf{v}_1} + \underbrace{\sum_{\ell=1}^{\gamma_2} c_{2,\ell} \mathbf{v}_{2,\ell}}_{\text{call this } \mathbf{v}_2} + \cdots + \underbrace{\sum_{\ell=1}^{\gamma_m} c_{m,\ell} \mathbf{v}_{m,\ell}}_{\text{call this } \mathbf{v}_m} = \mathbf{0}.$	$\sum_{\ell=1}^{\gamma_1} c_{1,\ell} \mathbf{v}_{1,\ell} + \sum_{\ell=1}^{\gamma_2} c_{2,\ell} \mathbf{v}_{2,\ell} + \cdots + \sum_{\ell=1}^{\gamma_m} c_{m,\ell} \mathbf{v}_{m,\ell} = \mathbf{0}.$	$\underbrace{\sum_{\ell=1}^{\gamma_1} c_{1,\ell} \mathbf{v}_{1,\ell}}_{\text{call this } \mathbf{v}_1} + \underbrace{\sum_{\ell=1}^{\gamma_2} c_{2,\ell} \mathbf{v}_{2,\ell}}_{\text{call this } \mathbf{v}_2} + \cdots + \underbrace{\sum_{\ell=1}^{\gamma_m} c_{m,\ell} \mathbf{v}_{m,\ell}}_{\text{call this } \mathbf{v}_m} = \mathbf{0}.$
johnston-linear-matrix-algebra-EQ0925	329	1.000	$\mathbf{v}_j = \sum_{\ell=1}^{\gamma_j} c_{j,\ell} \mathbf{v}_{j,\ell} = \mathbf{0} \quad \text{for all } 1 \leq j \leq m$	$\mathbf{v}_j = \sum_{\ell=1}^{\gamma_j} c_{j,\ell} \mathbf{v}_{j,\ell} = \mathbf{0} \quad \text{for all } 1 \leq j \leq m$	$\mathbf{v}_j = \sum_{\ell=1}^{\gamma_j} c_{j,\ell} \mathbf{v}_{j,\ell} = \mathbf{0} \quad \text{for all } 1 \leq j \leq m$
johnston-linear-matrix-algebra-EQ0926	330	1.000	$\begin{aligned} p_A(\lambda) &= \operatorname{det} \left(\begin{array}{cc} 3-\lambda & -1 \\ 1 & 1-\lambda \end{array} \right) \\ &= (3-\lambda)(1-\lambda) + 1 \\ &= \lambda^2 - 4\lambda + 4 \\ &= (\lambda - 2)^2. \end{aligned}$	$p_A(\lambda) = \det \left(\begin{bmatrix} 3-\lambda & -1 \\ 1 & 1-\lambda \end{bmatrix} \right) = (3-\lambda)(1-\lambda) + 1$ $= \lambda^2 - 4\lambda + 4$ $= (\lambda - 2)^2.$	$p_A(\lambda) = \det \left(\begin{bmatrix} 3-\lambda & -1 \\ 1 & 1-\lambda \end{bmatrix} \right) = (3-\lambda)(1-\lambda) + 1$ $= \lambda^2 - 4\lambda + 4$ $= (\lambda - 2)^2.$

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johnston-linear-matrix-aligned- EQ0928	332	1.000	$\begin{aligned} \operatorname{det}(A-\lambda I) &= \operatorname{det}\left(\begin{bmatrix} 5-\lambda & 1 & -1 \\ 1 & 3-\lambda & -1 \\ 2 & 0 & 2-\lambda \end{bmatrix}\right) \\ &= (5-\lambda)(3-\lambda)(2-\lambda) - 2 + 0 - 0 + 2(3-\lambda) - (2-\lambda) \\ &= (5-\lambda)(3-\lambda)(2-\lambda) + (2-\lambda) \\ &= (2-\lambda)((5-\lambda)(3-\lambda) + 1) \\ &= (2-\lambda)(\lambda^2 - 8\lambda + 16) \\ &= (2-\lambda)(4-\lambda)^2. \end{aligned}$	$\det(A - \lambda I) = \det\left(\begin{bmatrix} 5-\lambda & 1 & -1 \\ 1 & 3-\lambda & -1 \\ 2 & 0 & 2-\lambda \end{bmatrix}\right)$ $= (5-\lambda)(3-\lambda)(2-\lambda) - 2 + 0 - 0 + 2(3-\lambda) - (2-\lambda)$ $= (5-\lambda)(3-\lambda)(2-\lambda) + (2-\lambda)$ $= (2-\lambda)((5-\lambda)(3-\lambda) + 1)$ $= (2-\lambda)(\lambda^2 - 8\lambda + 16)$ $= (2-\lambda)(4-\lambda)^2.$	
johnston-linear-matrix-aligned- EQ0930	333	1.000	$\lambda=\frac{-b\pm\sqrt{b^2-4\,a\,c}}{2\,a}.$	$\lambda=\frac{-b\pm\sqrt{b^2-4ac}}{2a}.$	$\lambda=\frac{-b\pm\sqrt{b^2-4ac}}{2a}.$
johnston-linear-matrix-aligned- EQ0932	333	1.000	$\lambda=\frac{1\pm\sqrt{1+4}}{2}=\frac{1\pm\sqrt{5}}{2}.$	$\lambda=\frac{1\pm\sqrt{1+4}}{2}=\frac{1\pm\sqrt{5}}{2}.$	$\lambda=\frac{1\pm\sqrt{1+4}}{2}=\frac{1\pm\sqrt{5}}{2}.$
johnston-linear-matrix-aligned- EQ0933	334	1.000	$A=\left[\begin{array}{cc} 3 & -1 \\ 1 & 1 \end{array}\right]$	$A=\begin{bmatrix} 3 & -1 \\ 1 & 1 \end{bmatrix}$	$A=\begin{bmatrix} 3 & -1 \\ 1 & 1 \end{bmatrix}$
johnston-linear-matrix-aligned- EQ0934	334	1.000	$\lambda=\frac{4+\varepsilon\pm\sqrt{\varepsilon^2-4\varepsilon}}{2},$	$\lambda=\frac{4+\varepsilon\pm\sqrt{\varepsilon^2-4\varepsilon}}{2},$	$\lambda=\frac{4+\varepsilon\pm\sqrt{\varepsilon^2-4\varepsilon}}{2},$
johnston-linear-matrix-aligned- EQ0935	334	1.000	$B=\left[\begin{array}{cc} 3 & -1 \\ 1 & 1.0001 \end{array}\right]$	$B=\begin{bmatrix} 3 & -1 \\ 1 & 1.0001 \end{bmatrix}$	$B=\begin{bmatrix} 3 & -1 \\ 1 & 1.0001 \end{bmatrix}$
johnston-linear-matrix-aligned- EQ0943	336	1.000	$A=\sum_{j=1}^n\lambda_j\,\mathbf{p}_j\,\mathbf{q}_j^*\,\mathbf{p}_j\,\mathbf{q}_j^*$	$A=\sum_{j=1}^n\lambda_j\mathbf{p}_j\mathbf{q}_j^*$	$A=\sum_{j=1}^n\lambda_j\mathbf{p}_j\mathbf{q}_j^*,$
johnston-linear-matrix-aligned- EQ0944 (3.4.6)	336	1.000	$A^k=\sum_{j=1}^n\lambda_j^k\,\mathbf{p}_j\,\mathbf{q}_j^*\,\mathbf{p}_j\,\mathbf{q}_j^*\quad\text{for all integers } k\geq 0.$	$A^k=\sum_{j=1}^n\lambda_j^k\mathbf{p}_j\mathbf{q}_j^*\quad\text{for all integers } k\geq 0.$	$A^k=\sum_{j=1}^n\lambda_j^k\mathbf{p}_j\mathbf{q}_j^*\quad\text{for all integers } k\geq 0.$
johnston-linear-matrix-aligned- EQ0948	336	1.000	$A^k=\frac{1}{3}\left(2^k\left[\begin{array}{ccc} 3 & 0 & -3 \\ 1 & 0 & -1 \\ 0 & 0 & 0 \end{array}\right]+(-1)^k\left[\begin{array}{ccc} 0 & 0 & 0 \\ -1 & 3 & 1 \\ 0 & 0 & 0 \end{array}\right]+(-1)^k\left[\begin{array}{ccc} 0 & 0 & 3 \\ 0 & 0 & 0 \\ 0 & 0 & 3 \end{array}\right]\right)$	$A^k=\frac{1}{3}\left(2^k\begin{bmatrix} 3 & 0 & -3 \\ 1 & 0 & -1 \\ 0 & 0 & 0 \end{bmatrix}+(-1)^k\begin{bmatrix} 0 & 0 & 0 \\ -1 & 3 & 1 \\ 0 & 0 & 0 \end{bmatrix}+(-1)^k\begin{bmatrix} 0 & 0 & 3 \\ 0 & 0 & 0 \\ 0 & 0 & 3 \end{bmatrix}\right)$	$A^k=\frac{1}{3}\left(2^k\begin{bmatrix} 3 & 0 & -3 \\ 1 & 0 & -1 \\ 0 & 0 & 0 \end{bmatrix}+(-1)^k\begin{bmatrix} 0 & 0 & 0 \\ -1 & 3 & 1 \\ 0 & 0 & 0 \end{bmatrix}+(-1)^k\begin{bmatrix} 0 & 0 & 3 \\ 0 & 0 & 0 \\ 0 & 0 & 3 \end{bmatrix}\right)$
johnston-linear-matrix-aligned- EQ0949	337	1.000	$F_{\,0}=0,\quad F_{\,1}=1,\quad F_{\,n+1}=F_{\,n}+F_{\,n-1}\quad\text{for all } n\geq 1.$	$F_0=0,\quad F_1=1,\quad F_{n+1}=F_n+F_{n-1}\quad\text{for all } n\geq 1.$	$F_0=0,\quad F_1=1,\quad F_{n+1}=F_n+F_{n-1}\quad\text{for all } n\geq 1.$
johnston-linear-matrix-aligned- EQ0952	337	1.000	$A=\left[\begin{array}{ll} 1 & 1 \\ 1 & 0 \end{array}\right],$	$A=\begin{bmatrix} 1 & 1 \\ 1 & 0 \end{bmatrix},$	$A=\begin{bmatrix} 1 & 1 \\ 1 & 0 \end{bmatrix},$

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johnston-linear-matrix-algebra-EQ0953	337	1.000	$\lambda=\frac{1}{2}(1\pm\sqrt{5})$.	$\lambda=\frac{1}{2}(1\pm\sqrt{5})$.	$\lambda=\frac{1}{2}(1\pm\sqrt{5})$.
johnston-linear-matrix-algebra-EQ0956	338	1.000	$D=\left[\begin{array}{cc}\phi&0\\0&1-\phi\end{array}\right],\quad P=\left[\begin{array}{cc}\phi&1-\phi\\1&1\end{array}\right],\quad\text{and}\quad P^{-1}=\frac{1}{\sqrt{5}}\left[\begin{array}{cc}1&\phi-1\\-1&\phi\end{array}\right]$.	$D=\left[\begin{array}{cc}\phi&0\\0&1-\phi\end{array}\right],\quad P=\left[\begin{array}{cc}\phi&1-\phi\\1&1\end{array}\right],\quad\text{and}\quad P^{-1}=\frac{1}{\sqrt{5}}\left[\begin{array}{cc}1&\phi-1\\-1&\phi\end{array}\right]$.	$D=\left[\begin{array}{cc}\phi&0\\0&1-\phi\end{array}\right],\quad P=\left[\begin{array}{cc}\phi&1-\phi\\1&1\end{array}\right],\quad\text{and}\quad P^{-1}=\frac{1}{\sqrt{5}}\left[\begin{array}{cc}1&\phi-1\\-1&\phi\end{array}\right]$.
johnston-linear-matrix-algebra-EQ0958	338	1.000	$F_n=\frac{1}{\sqrt{5}}\left(\phi^n-(1-\phi)^n\right)$.	$F_n=\frac{1}{\sqrt{5}}(\phi^n-(1-\phi)^n)$.	$F_n=\frac{1}{\sqrt{5}}(\phi^n-(1-\phi)^n)$.
johnston-linear-matrix-algebra-EQ0959	339	1.000	$A=\left[\begin{array}{lllll}0&1&1&1&1\\1&0&1&1&1\\1&1&0&0&0\\1&1&0&0&0\\1&1&0&0&0\end{array}\right]$.	$A=\left[\begin{array}{ccccc}0&1&1&1&1\\1&0&1&1&1\\1&1&0&0&0\\1&1&0&0&0\\1&1&0&0&0\end{array}\right]$.	$A=\left[\begin{array}{ccccc}0&1&1&1&1\\1&0&1&1&1\\1&1&0&0&0\\1&1&0&0&0\\1&1&0&0&0\end{array}\right]$.
johnston-linear-matrix-algebra-EQ0964	340	1.000	$A^{\{r\}}\stackrel{\text{def}}{=}P\,D^{\{r\}}\,P^{-1}\quad\text{for all }r\in\mathbb{R},$	$A^r\stackrel{\text{def}}{=}PD^rP^{-1}\quad\text{for all }r\in\mathbb{R},$	$A^r\stackrel{\text{def}}{=}PD^rP^{-1}\quad\text{for all }r\in\mathbb{R},$
johnston-linear-matrix-algebra-EQ0965	341	1.000	$A=\underbrace{\left[\begin{array}{ll}4&1\\1&1\end{array}\right]}_P\,\underbrace{\left[\begin{array}{ll}1&0\\0&4\end{array}\right]}_D\,\underbrace{\left(\frac{1}{3}\left[\begin{array}{ll}1&-1\\-1&4\end{array}\right]\right)}_{P^{-1}}$.	$A=\underbrace{\left[\begin{array}{cc}4&1\\1&1\end{array}\right]}_P\,\underbrace{\left[\begin{array}{cc}1&0\\0&4\end{array}\right]}_D\,\underbrace{\left(\frac{1}{3}\left[\begin{array}{cc}1&-1\\-1&4\end{array}\right]\right)}_{P^{-1}}$.	$A=\underbrace{\left[\begin{array}{cc}4&1\\1&1\end{array}\right]}_P\,\underbrace{\left[\begin{array}{cc}1&0\\0&4\end{array}\right]}_D\,\underbrace{\left(\frac{1}{3}\left[\begin{array}{cc}1&-1\\-1&4\end{array}\right]\right)}_{P^{-1}}$.
johnston-linear-matrix-algebra-EQ0969	341	1.000	$A^{-1}A=\frac{1}{4}\left[\begin{array}{cc}5&-4\\1&0\end{array}\right]\left[\begin{array}{cc}0&4\\-1&5\end{array}\right]=\left[\begin{array}{cc}1&0\\0&1\end{array}\right]=I$.	$A^{-1}A=\frac{1}{4}\left[\begin{array}{cc}5&-4\\1&0\end{array}\right]\left[\begin{array}{cc}0&4\\-1&5\end{array}\right]=\left[\begin{array}{cc}1&0\\0&1\end{array}\right]=I$.	$A^{-1}A=\frac{1}{4}\left[\begin{array}{cc}5&-4\\1&0\end{array}\right]\left[\begin{array}{cc}0&4\\-1&5\end{array}\right]=\left[\begin{array}{cc}1&0\\0&1\end{array}\right]=I$.
johnston-linear-matrix-algebra-EQ0971	342	1.000	$\begin{aligned}A^{-1}A&=\left(P\left[\begin{array}{ccc}\lambda_1^{-1}&\cdots&0\\ \vdots&\ddots&\vdots\\ 0&\cdots&\lambda_n^{-1}\end{array}\right]P^{-1}\right)\left(P\left[\begin{array}{ccc}\lambda_1&\cdots&0\\ \vdots&\ddots&\vdots\\ 0&\cdots&\lambda_n\end{array}\right]P^{-1}\right)\\&=P\left[\begin{array}{ccc}\lambda_1^{-1}\lambda_1&\cdots&0\\ \vdots&\ddots&\vdots\\ 0&\cdots&\lambda_n^{-1}\lambda_n\end{array}\right]P^{-1}=PIP^{-1}=I.\end{aligned}$	$\begin{aligned}A^{-1}A&=\left(P\left[\begin{array}{ccc}\lambda_1^{-1}&\cdots&0\\ \vdots&\ddots&\vdots\\ 0&\cdots&\lambda_n^{-1}\end{array}\right]P^{-1}\right)\left(P\left[\begin{array}{ccc}\lambda_1&\cdots&0\\ \vdots&\ddots&\vdots\\ 0&\cdots&\lambda_n\end{array}\right]P^{-1}\right)\\&=P\left[\begin{array}{ccc}\lambda_1^{-1}\lambda_1&\cdots&0\\ \vdots&\ddots&\vdots\\ 0&\cdots&\lambda_n^{-1}\lambda_n\end{array}\right]P^{-1}=PIP^{-1}=I.\end{aligned}$	$\begin{aligned}A^{-1}A&=\left(P\left[\begin{array}{ccc}\lambda_1^{-1}&\cdots&0\\ \vdots&\ddots&\vdots\\ 0&\cdots&\lambda_n^{-1}\end{array}\right]P^{-1}\right)\left(P\left[\begin{array}{ccc}\lambda_1&\cdots&0\\ \vdots&\ddots&\vdots\\ 0&\cdots&\lambda_n\end{array}\right]P^{-1}\right)\\&=P\left[\begin{array}{ccc}\lambda_1^{-1}\lambda_1&\cdots&0\\ \vdots&\ddots&\vdots\\ 0&\cdots&\lambda_n^{-1}\lambda_n\end{array}\right]P^{-1}=PIP^{-1}=I.\end{aligned}$

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johnston-linear-matrix-algebra-EQ0974	342	1.000	$A=\underbrace{\left[\begin{array}{cc} 2 & 1 \\ -1 & 0 \end{array}\right]}_P \underbrace{\left[\begin{array}{cc} 1 & 0 \\ 0 & -8 \end{array}\right]}_D \underbrace{\left(\frac{1}{3}\left[\begin{array}{cc} 1 & 1 \\ 1 & -2 \end{array}\right]\right)}_{P^{-1}}.$	$A=\underbrace{\left[\begin{array}{cc} 2 & 1 \\ 1 & -1 \end{array}\right]}_P \underbrace{\left[\begin{array}{cc} 1 & 0 \\ 0 & -8 \end{array}\right]}_D \underbrace{\left(\frac{1}{3}\left[\begin{array}{cc} 1 & 1 \\ 1 & -2 \end{array}\right]\right)}_{P^{-1}}.$	$A=\underbrace{\left[\begin{array}{cc} 2 & 1 \\ 1 & -1 \end{array}\right]}_P \underbrace{\left[\begin{array}{cc} 1 & 0 \\ 0 & -8 \end{array}\right]}_D \underbrace{\left(\frac{1}{3}\left[\begin{array}{cc} 1 & 1 \\ 1 & -2 \end{array}\right]\right)}_{P^{-1}}.$
johnston-linear-matrix-algebra-EQ0975	343	1.000	$B=\frac{1}{3}\left[\begin{array}{cc} 2 & 1 \\ 1 & -1 \end{array}\right]\left[\begin{array}{cc} 1 & 0 \\ 0 & -2 \end{array}\right]\left[\begin{array}{cc} 1 & 1 \\ 1 & -2 \end{array}\right]=\left[\begin{array}{cc} 0 & 2 \\ 1 & -1 \end{array}\right].$	$B=\frac{1}{3}\left[\begin{array}{cc} 2 & 1 \\ 1 & -1 \end{array}\right]\left[\begin{array}{cc} 1 & 0 \\ 0 & -2 \end{array}\right]\left[\begin{array}{cc} 1 & 1 \\ 1 & -2 \end{array}\right]=\left[\begin{array}{cc} 0 & 2 \\ 1 & -1 \end{array}\right].$	$B=\frac{1}{3}\left[\begin{array}{cc} 2 & 1 \\ 1 & -1 \end{array}\right]\left[\begin{array}{cc} 1 & 0 \\ 0 & -2 \end{array}\right]\left[\begin{array}{cc} 1 & 1 \\ 1 & -2 \end{array}\right]=\left[\begin{array}{cc} 0 & 2 \\ 1 & -1 \end{array}\right].$
johnston-linear-matrix-algebra-EQ0977	343	1.000	$A^{\{1 / 2\}}=\frac{1}{3}\left[\begin{array}{cc} 2 & 4 \\ -1 & 7 \end{array}\right]$	$A^{1 / 2}=\frac{1}{3}\left[\begin{array}{cc} 2 & 4 \\ -1 & 7 \end{array}\right]$	$A^{1 / 2}=\frac{1}{3}\left[\begin{array}{cc} 2 & 4 \\ -1 & 7 \end{array}\right]$
johnston-linear-matrix-algebra-EQ0978	343	1.000	$A=\frac{1}{3}\left[\begin{array}{cc} 4 & 1 \\ 1 & 1 \end{array}\right]\left[\begin{array}{cc} 1 & 0 \\ 0 & 4 \end{array}\right]\left[\begin{array}{cc} 1 & -1 \\ -1 & 4 \end{array}\right],$	$A=\frac{1}{3}\left[\begin{array}{cc} 4 & 1 \\ 1 & 1 \end{array}\right]\left[\begin{array}{cc} 1 & 0 \\ 0 & 4 \end{array}\right]\left[\begin{array}{cc} 1 & -1 \\ -1 & 4 \end{array}\right],$	$A=\frac{1}{3}\left[\begin{array}{cc} 4 & 1 \\ 1 & 1 \end{array}\right]\left[\begin{array}{cc} 1 & 0 \\ 0 & 4 \end{array}\right]\left[\begin{array}{cc} 1 & -1 \\ -1 & 4 \end{array}\right],$
johnston-linear-matrix-algebra-EQ0980	344	1.000	$\left((-1)^{\{1 / 2\}}\right)^2=i^2=-1, \quad \text { but } \quad\left((-1)^2\right)^{1 / 2}=1^{1 / 2}=1.$	$\left((-1)^{1 / 2}\right)^2=i^2=-1, \quad \text { but } \quad\left((-1)^2\right)^{1 / 2}=1^{1 / 2}=1.$	$\left((-1)^{1 / 2}\right)^2=i^2=-1, \quad \text { but } \quad\left((-1)^2\right)^{1 / 2}=1^{1 / 2}=1.$
johnston-linear-matrix-algebra-EQ0982	345	1.000	$p(A)=c_{\{k\}} A^{\{k\}}+\cdots+c_{\{2\}} A^{\{2\}}+c_{\{1\}} A+c_{\{0\}} I .$	$p(A)=c_k A^k+\cdots+c_2 A^2+c_1 A+c_0 I .$	$p(A)=c_k A^k+\cdots+c_2 A^2+c_1 A+c_0 I .$
johnston-linear-matrix-algebra-EQ0983	345	1.000	$p(A)=P p(D) P^{-1},$	$p(A)=P p(D) P^{-1},$	$p(A)=P p(D) P^{-1},$
johnston-linear-matrix-algebra-EQ0987	346	1.000	$A=\frac{1}{3}\left[\begin{array}{cc} 4 & 1 \\ 1 & 1 \end{array}\right] \underbrace{\left[\begin{array}{cc} 1 & 0 \\ 0 & 4 \end{array}\right]}_D\left[\begin{array}{cc} 1 & -1 \\ -1 & 4 \end{array}\right]$	$A=\frac{1}{3}\left[\begin{array}{cc} 4 & 1 \\ 1 & 1 \end{array}\right] \underbrace{\left[\begin{array}{cc} 1 & 0 \\ 0 & 4 \end{array}\right]}_D\left[\begin{array}{cc} 1 & -1 \\ -1 & 4 \end{array}\right]$	$A=\frac{1}{3}\left[\begin{array}{cc} 4 & 1 \\ 1 & 1 \end{array}\right] \underbrace{\left[\begin{array}{cc} 1 & 0 \\ 0 & 4 \end{array}\right]}_D\left[\begin{array}{cc} 1 & -1 \\ -1 & 4 \end{array}\right]$
johnston-linear-matrix-algebra-EQ0989	346	1.000	$f(A) \stackrel{\text { def }}{=} P f(D) P^{-1},$	$f(A) \stackrel{\text { def }}{=} P f(D) P^{-1},$	$f(A) \stackrel{\text { def }}{=} P f(D) P^{-1},$
johnston-linear-matrix-algebra-EQ0990	347	1.000	$A=\frac{1}{3}\left[\begin{array}{cc} 4 & 1 \\ 1 & 1 \end{array}\right] \underbrace{\left[\begin{array}{cc} 1 & 0 \\ 0 & 4 \end{array}\right]}_D\left[\begin{array}{cc} 1 & -1 \\ -1 & 4 \end{array}\right]$	$A=\frac{1}{3}\left[\begin{array}{cc} 4 & 1 \\ 1 & 1 \end{array}\right] \underbrace{\left[\begin{array}{cc} 1 & 0 \\ 0 & 4 \end{array}\right]}_D\left[\begin{array}{cc} 1 & -1 \\ -1 & 4 \end{array}\right]$	$A=\frac{1}{3}\left[\begin{array}{cc} 4 & 1 \\ 1 & 1 \end{array}\right] \underbrace{\left[\begin{array}{cc} 1 & 0 \\ 0 & 4 \end{array}\right]}_D\left[\begin{array}{cc} 1 & -1 \\ -1 & 4 \end{array}\right]$
johnston-linear-matrix-algebra-EQ0993	347	1.000	$f(x)=c_{\{0\}}+c_{\{1\}} x+c_{\{2\}} x^2+c_{\{3\}} x^3+\cdots=\sum_{\{k=0\}^{\{\infty\}}} c_{\{k\}} x^{\{k\}} \quad \text { for all } x \in(a, b) .$	$f(x)=c_0+c_1 x+c_2 x^2+c_3 x^3+\cdots=\sum_{k=0}^{\infty} c_k x^k \quad \text { for all } x \in(a, b) .$	$f(x)=c_0+c_1 x+c_2 x^2+c_3 x^3+\cdots=\sum_{k=0}^{\infty} c_k x^k \quad \text { for all } x \in(a, b) .$
johnston-linear-matrix-algebra-EQ0996	348	1.000	$f(A)=f\left(P D P^{-1}\right)=\sum_{\{k=0\}^{\{\infty\}}} c_{\{k\}} \left(P D P^{-1}\right)^{\{k\}}=P \left(\sum_{\{k=0\}^{\{\infty\}}} c_{\{k\}} D^{\{k\}}\right) P^{-1}=P f(D) P^{-1}$	$f(A)=f\left(P D P^{-1}\right)=\sum_{k=0}^{\infty} c_k\left(P D P^{-1}\right)^k=P\left(\sum_{k=0}^{\infty} c_k D^k\right) P^{-1}=P f(D) P^{-1}$	$f(A)=f\left(P D P^{-1}\right)=\sum_{k=0}^{\infty} c_k\left(P D P^{-1}\right)^k=P\left(\sum_{k=0}^{\infty} c_k D^k\right) P^{-1}=P f(D) P^{-1}$

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johnston-linear-matrix-algebra EQ0998	348	1.000	$\begin{aligned} e^A e^{-A} &= \left(P e^D P^{-1} \right) \left(P e^{-D} P^{-1} \right) \\ &= P \left[\begin{array}{cccc} e^{d_{1,1}} & 0 & \cdots & 0 \\ 0 & e^{d_{2,2}} & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & e^{d_{n,n}} \end{array} \right] \left[\begin{array}{cccc} e^{-d_{1,1}} & 0 & \cdots & 0 \\ 0 & e^{-d_{2,2}} & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & e^{-d_{n,n}} \end{array} \right] P^{-1} \end{aligned}$	$e^A e^{-A} = \left(P e^D P^{-1} \right) \left(P e^{-D} P^{-1} \right)$ $= P \left[\begin{array}{cccc} e^{d_{1,1}} & 0 & \cdots & 0 \\ 0 & e^{d_{2,2}} & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & e^{d_{n,n}} \end{array} \right] \left[\begin{array}{cccc} e^{-d_{1,1}} & 0 & \cdots & 0 \\ 0 & e^{-d_{2,2}} & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & e^{-d_{n,n}} \end{array} \right] P^{-1}$	
johnston-linear-matrix-algebra EQ0999	349	1.000	$\begin{aligned} &= P \left[\begin{array}{cccc} 1 & 0 & \cdots & 0 \\ 0 & 1 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & 1 \end{array} \right] P^{-1} \\ &= I \end{aligned}$	$= P \left[\begin{array}{cccc} 1 & 0 & \cdots & 0 \\ 0 & 1 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & 1 \end{array} \right] P^{-1}$ $= I.$	
johnston-linear-matrix-algebra EQ1002	349	1.000	$\operatorname{det}(A) = \lambda_1 \lambda_2 \cdots \lambda_n \quad \text{and} \quad \operatorname{tr}(A) = \lambda_1 + \lambda_2 + \cdots + \lambda_n.$	$\det(A) = \lambda_1 \lambda_2 \cdots \lambda_n \quad \text{and} \quad \operatorname{tr}(A) = \lambda_1 + \lambda_2 + \cdots + \lambda_n.$	
johnston-linear-matrix-algebra EQ1003	349	1.000	$\operatorname{det}(e^A) = e^{\lambda_1} e^{\lambda_2} \cdots e^{\lambda_n} = e^{\lambda_1 + \lambda_2 + \cdots + \lambda_n} = e^{\operatorname{tr}(A)}.$	$\det(e^A) = e^{\lambda_1} e^{\lambda_2} \cdots e^{\lambda_n} = e^{\lambda_1 + \lambda_2 + \cdots + \lambda_n} = e^{\operatorname{tr}(A)}.$	
johnston-linear-matrix-algebra EQ1004	350	1.000	$\left[\begin{array}{ccccc} 4 & -2 & 2 & -3 & 2 \\ 0 & 4 & -1 & 1 & 0 \\ 2 & 0 & 3 & -2 & 2 \\ 2 & 0 & 0 & 1 & 2 \\ -1 & 2 & -2 & 3 & 1 \end{array} \right]$	$\left[\begin{array}{ccccc} 4 & -2 & 2 & -3 & 2 \\ 0 & 4 & -1 & 1 & 0 \\ 2 & 0 & 3 & -2 & 2 \\ 2 & 0 & 0 & 1 & 2 \\ -1 & 2 & -2 & 3 & 1 \end{array} \right]$	
johnston-linear-matrix-algebra EQ1005 (5)	350	1.000	$\left[\begin{array}{ccccc} -2 & 7 & 9 & 3 & 10 \\ -1 & 4 & 3 & 2 & 5 \\ 2 & -4 & -5 & -3 & -7 \\ 1 & -1 & -1 & 0 & -2 \\ -2 & 3 & 5 & 2 & 6 \end{array} \right]$	$\left[\begin{array}{ccccc} -2 & 7 & 9 & 3 & 10 \\ -1 & 4 & 3 & 2 & 5 \\ 2 & -4 & -5 & -3 & -7 \\ 1 & -1 & -1 & 0 & -2 \\ -2 & 3 & 5 & 2 & 6 \end{array} \right]$	
johnston-linear-matrix-algebra EQ1006	350	1.000	$\left[\begin{array}{ccccc} 11 & 4 & 7 & -18 & 5 \\ 7 & 3 & 8 & -15 & 5 \\ 20 & 5 & 23 & -43 & 15 \\ 15 & 5 & 14 & -29 & 10 \\ -1 & 1 & -3 & 4 & 0 \end{array} \right]$	$\left[\begin{array}{ccccc} 11 & 4 & 7 & -18 & 5 \\ 7 & 3 & 8 & -15 & 5 \\ 20 & 5 & 23 & -43 & 15 \\ 15 & 5 & 14 & -29 & 10 \\ -1 & 1 & -3 & 4 & 0 \end{array} \right]$	
johnston-linear-matrix-algebra EQ1007	350	1.000	$\left[\begin{array}{ccccc} -4 & 4 & 2 & 2 & 1 \\ -5 & 5 & 2 & 2 & 1 \\ -5 & 4 & 3 & 2 & 1 \\ 5 & -4 & -2 & -1 & -1 \\ -5 & 4 & 2 & 2 & 2 \end{array} \right]$	$\left[\begin{array}{ccccc} -4 & 4 & 2 & 2 & 1 \\ -5 & 5 & 2 & 2 & 1 \\ -5 & 4 & 3 & 2 & 1 \\ 5 & -4 & -2 & -1 & -1 \\ -5 & 4 & 2 & 2 & 2 \end{array} \right]$	

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johnston-linear-matrix-algebra-EQ1008	350	1.000	$\left[\begin{array}{ccccc} 18 & 5 & 4 & 9 & -8 \\ -17 & -4 & -4 & -9 & 8 \\ 20 & 8 & 5 & 12 & -8 \\ 3 & 3 & 0 & 4 & 0 \\ 20 & 8 & 4 & 12 & -7 \end{array}\right]$	$\left[\begin{array}{ccccc} 18 & 5 & 4 & 9 & -8 \\ -17 & -4 & -4 & -9 & 8 \\ 20 & 8 & 5 & 12 & -8 \\ 3 & 3 & 0 & 4 & 0 \\ 20 & 8 & 4 & 12 & -7 \end{array}\right]$	$\left[\begin{array}{ccccc} 18 & 5 & 4 & 9 & -8 \\ -17 & -4 & -4 & -9 & 8 \\ 20 & 8 & 5 & 12 & -8 \\ 3 & 3 & 0 & 4 & 0 \\ 20 & 8 & 4 & 12 & -7 \end{array}\right]$
johnston-linear-matrix-algebra-EQ1009	350	1.000	$\left[\begin{array}{ccccc} -2 & 3 & 11 & -1 & 3 \\ -14 & -6 & -4 & -7 & 3 \\ 6 & -9 & 3 & -15 & -9 \\ 1 & 12 & -1 & 5 & 12 \\ 3 & 0 & 15 & -12 & -9 \end{array}\right]$	$\left[\begin{array}{ccccc} -2 & 3 & 11 & -1 & 3 \\ -14 & -6 & -4 & -7 & 3 \\ 6 & -9 & 3 & -15 & -9 \\ 1 & 12 & -1 & 5 & 12 \\ 3 & 0 & 15 & -12 & -9 \end{array}\right]$	$\left[\begin{array}{ccccc} -2 & 3 & 11 & -1 & 3 \\ -14 & -6 & -4 & -7 & 3 \\ 6 & -9 & 3 & -15 & -9 \\ 1 & 12 & -1 & 5 & 12 \\ 3 & 0 & 15 & -12 & -9 \end{array}\right]$
johnston-linear-matrix-algebra-EQ1010 (5)	350	1.000	$A=\left[\begin{array}{cc} 5 & 3 \\ k & -1 \end{array}\right]$.	$A=\left[\begin{array}{cc} 5 & 3 \\ k & -1 \end{array}\right]$.	$A=\left[\begin{array}{cc} 5 & 3 \\ k & -1 \end{array}\right]$.
johnston-linear-matrix-algebra-EQ1011	350	1.000	$A=\left[\begin{array}{cc} 5 & -6 \\ -3 & 2 \end{array}\right]$.	$A=\left[\begin{array}{cc} 5 & -6 \\ -3 & 2 \end{array}\right]$.	$A=\left[\begin{array}{cc} 5 & -6 \\ -3 & 2 \end{array}\right]$.
johnston-linear-matrix-algebra-EQ1015	351	1.000	$\operatorname{det}(e^{A+B})=\operatorname{det}(e^A e^B)$.	$\det(e^{A+B})=\det(e^A e^B)$.	$\det(e^{A+B})=\det(e^A e^B)$.
johnston-linear-matrix-algebra-EQ1016 (0)	351	1.000	$C=\left[\begin{array}{ccccc} 0 & 1 & 0 & \cdots & 0 \\ 0 & 0 & 1 & \cdots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & 0 & \cdots & 1 \\ -a_0 & -a_1 & -a_2 & \cdots & -a_{n-1} \end{array}\right]$.	$C=\left[\begin{array}{ccccc} 0 & 1 & 0 & \cdots & 0 \\ 0 & 0 & 1 & \cdots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & 0 & \cdots & 1 \\ -a_0 & -a_1 & -a_2 & \cdots & -a_{n-1} \end{array}\right]$.	$C=\left[\begin{array}{ccccc} 0 & 1 & 0 & \cdots & 0 \\ 0 & 0 & 1 & \cdots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & 0 & \cdots & 1 \\ -a_0 & -a_1 & -a_2 & \cdots & -a_{n-1} \end{array}\right]$.
johnston-linear-matrix-algebra-EQ1018	353	1.000	$\operatorname{det}(A)=\lambda_1 \lambda_2 \cdots \lambda_n$ and $\operatorname{tr}(A)=\lambda_1+\lambda_2+\cdots+\lambda_n$,	$\det(A)=\lambda_1 \lambda_2 \cdots \lambda_n$ and $\operatorname{tr}(A)=\lambda_1+\lambda_2+\cdots+\lambda_n$,	$\det(A)=\lambda_1 \lambda_2 \cdots \lambda_n$ and $\operatorname{tr}(A)=\lambda_1+\lambda_2+\cdots+\lambda_n$,
johnston-linear-matrix-algebra-EQ1019	353	1.000	$\begin{aligned} \operatorname{det}(A) &= \operatorname{det}(P D P^{-1})= \\ \operatorname{det}(D) &= \lambda_1 \lambda_2 \cdots \lambda_n \\ \operatorname{tr}(A) &= \operatorname{tr}(P D P^{-1})= \\ \operatorname{tr}(D) &= \lambda_1+\lambda_2+\cdots+\lambda_n, \end{aligned}$	$\det(A)=\det(PDP^{-1})=\det(D)=\lambda_1 \lambda_2 \cdots \lambda_n$ and $\operatorname{tr}(A)=\operatorname{tr}(PDP^{-1})=\operatorname{tr}(D)=\lambda_1+\lambda_2+\cdots+\lambda_n$,	$\det(A)=\det(PDP^{-1})=\det(D)=\lambda_1 \lambda_2 \cdots \lambda_n$ and $\operatorname{tr}(A)=\operatorname{tr}(PDP^{-1})=\operatorname{tr}(D)=\lambda_1+\lambda_2+\cdots+\lambda_n$,
johnston-linear-matrix-algebra-EQ1020	353	1.000	$N=\left[\begin{array}{ll} 0 & 1 \\ 0 & 0 \end{array}\right]$	$N=\left[\begin{array}{ll} 0 & 1 \\ 0 & 0 \end{array}\right]$	$N=\left[\begin{array}{ll} 0 & 1 \\ 0 & 0 \end{array}\right]$
johnston-linear-matrix-algebra-EQ1022	354	1.000	$A=P D P^{-1}=P\left(Q^{-1} B Q\right) P^{-1}=P\left(Q^{-1} B\left(P Q^{-1}\right)^{-1}\right) P^{-1}$,	$A=PDP^{-1}=P\left(Q^{-1} B Q\right) P^{-1}=\left(P Q^{-1}\right) B\left(P Q^{-1}\right)^{-1}$,	$A=PDP^{-1}=P\left(Q^{-1} B Q\right) P^{-1}=\left(P Q^{-1}\right) B\left(P Q^{-1}\right)^{-1}$,
johnston-linear-matrix-algebra-EQ1025	355	1.000	$\operatorname{cof}(A) \stackrel{\text { def }}{=} \left[\begin{array}{cccc} c_{1,1} & c_{1,2} & \cdots & c_{1,n} \\ c_{2,1} & c_{2,2} & \cdots & c_{2,n} \\ \vdots & \vdots & \ddots & \vdots \\ c_{n,1} & c_{n,2} & \cdots & c_{n,n} \end{array}\right] \in \mathcal{M}_n$.	$\operatorname{cof}(A) \stackrel{\text { def }}{=} \left[\begin{array}{cccc} c_{1,1} & c_{1,2} & \cdots & c_{1,n} \\ c_{2,1} & c_{2,2} & \cdots & c_{2,n} \\ \vdots & \vdots & \ddots & \vdots \\ c_{n,1} & c_{n,2} & \cdots & c_{n,n} \end{array}\right] \in \mathcal{M}_n$.	$\operatorname{cof}(A) \stackrel{\text { def }}{=} \left[\begin{array}{cccc} c_{1,1} & c_{1,2} & \cdots & c_{1,n} \\ c_{2,1} & c_{2,2} & \cdots & c_{2,n} \\ \vdots & \vdots & \ddots & \vdots \\ c_{n,1} & c_{n,2} & \cdots & c_{n,n} \end{array}\right] \in \mathcal{M}_n$.

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johnston-linear-matrix-algebra-EQ1027	356	1.000	$\operatorname{cof}\left(\left[\begin{array}{cc} 2 & 3 \\ -1 & 4 \end{array}\right]\right)=\left[\begin{array}{cc} c_{1,1} & c_{1,2} \\ c_{2,1} & c_{2,2} \end{array}\right]=\left[\begin{array}{cc} 4 & 1 \\ -3 & 2 \end{array}\right].$	$\operatorname{cof}\left(\left[\begin{array}{cc} 2 & 3 \\ -1 & 4 \end{array}\right]\right)=\left[\begin{array}{cc} c_{1,1} & c_{1,2} \\ c_{2,1} & c_{2,2} \end{array}\right]=\left[\begin{array}{cc} 4 & 1 \\ -3 & 2 \end{array}\right].$	$\operatorname{cof}\left(\left[\begin{array}{cc} 2 & 3 \\ -1 & 4 \end{array}\right]\right)=\left[\begin{array}{cc} c_{1,1} & c_{1,2} \\ c_{2,1} & c_{2,2} \end{array}\right]=\left[\begin{array}{cc} 4 & 1 \\ -3 & 2 \end{array}\right].$
johnston-linear-matrix-algebra-EQ1028	356	1.000	$c_{1,1}=\operatorname{det}\left(\left[\begin{array}{c} 5 \\ 8 \end{array}\right]\right)=5 \cdot 9-6 \cdot 8=45-48=-3.$	$c_{1,1}=\operatorname{det}\left(\left[\begin{array}{cc} 5 & 6 \\ 8 & 9 \end{array}\right]\right)=5 \cdot 9-6 \cdot 8=45-48=-3.$	$c_{1,1}=\operatorname{det}\left(\left[\begin{array}{cc} 5 & 6 \\ 8 & 9 \end{array}\right]\right)=5 \cdot 9-6 \cdot 8=45-48=-3.$
johnston-linear-matrix-algebra-EQ1029	356	1.000	$\begin{array}{l} c_{1,1}=-3 \text { \& } c_{1,2}=6 \text { \& } c_{1,3}=-3 \\ c_{2,1}=6 \text { \& } c_{2,2}=-12 \text { \& } c_{2,3}=6 \\ c_{3,1}=-3 \text { \& } c_{3,2}=6 \text { \& } c_{3,3}=-3 \end{array}$	$\begin{array}{ccc} c_{1,1}=-3 & c_{1,2}=6 & c_{1,3}=-3 \\ c_{2,1}=6 & c_{2,2}=-12 & c_{2,3}=6 \\ c_{3,1}=-3 & c_{3,2}=6 & c_{3,3}=-3 \end{array}$	$\begin{array}{ccc} c_{1,1}=-3 & c_{1,2}=6 & c_{1,3}=-3 \\ c_{2,1}=6 & c_{2,2}=-12 & c_{2,3}=6 \\ c_{3,1}=-3 & c_{3,2}=6 & c_{3,3}=-3, \end{array}$
johnston-linear-matrix-algebra-EQ1030	356	1.000	$\operatorname{cof}\left(\left[\begin{array}{ccc} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{array}\right]\right)=\left[\begin{array}{ccc} -3 & 6 & -3 \\ 6 & -12 & 6 \\ -3 & 6 & -3 \end{array}\right].$	$\operatorname{cof}\left(\left[\begin{array}{ccc} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{array}\right]\right)=\left[\begin{array}{ccc} -3 & 6 & -3 \\ 6 & -12 & 6 \\ -3 & 6 & -3 \end{array}\right].$	$\operatorname{cof}\left(\left[\begin{array}{ccc} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{array}\right]\right)=\left[\begin{array}{ccc} -3 & 6 & -3 \\ 6 & -12 & 6 \\ -3 & 6 & -3 \end{array}\right].$
johnston-linear-matrix-algebra-EQ1031	356	1.000	$A^{-1}=\frac{1}{\operatorname{det}(A)} \operatorname{cof}(A)^T.$	$A^{-1}=\frac{1}{\operatorname{det}(A)} \operatorname{cof}(A)^T.$	$A^{-1}=\frac{1}{\operatorname{det}(A)} \operatorname{cof}(A)^T.$
johnston-linear-matrix-algebra-EQ1033 (3.A.1)	356	1.000	$\left[A \operatorname{cof}(A)^T\right]_{i, j}=a_{i, 1} c_{j, 1}+a_{i, 2} c_{j, 2}+\cdots+a_{i, n} c_{j, n}.$	$\left[A \operatorname{cof}(A)^T\right]_{i, j}=a_{i, 1} c_{j, 1}+a_{i, 2} c_{j, 2}+\cdots+a_{i, n} c_{j, n}.$	$\left[A \operatorname{cof}(A)^T\right]_{i, j}=a_{i, 1} c_{j, 1}+a_{i, 2} c_{j, 2}+\cdots+a_{i, n} c_{j, n}.$
johnston-linear-matrix-algebra-EQ1034	357	1.000	$A=\left[\begin{array}{cccc} a_{1,1} & a_{1,2} & \cdots & a_{1,n} \\ a_{2,1} & a_{2,2} & \cdots & a_{2,n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{i,1} & a_{i,2} & \cdots & a_{i,n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{j,1} & a_{j,2} & \cdots & a_{j,n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{n,1} & a_{n,2} & \cdots & a_{n,n} \end{array}\right], B=\left[\begin{array}{cccc} a_{1,1} & a_{1,2} & \cdots & a_{1,n} \\ a_{2,1} & a_{2,2} & \cdots & a_{2,n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{i,1} & a_{i,2} & \cdots & a_{i,n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{j,1} & a_{j,2} & \cdots & a_{j,n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{n,1} & a_{n,2} & \cdots & a_{n,n} \end{array}\right].$	$A=\left[\begin{array}{cccc} a_{1,1} & a_{1,2} & \cdots & a_{1,n} \\ a_{2,1} & a_{2,2} & \cdots & a_{2,n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{i,1} & a_{i,2} & \cdots & a_{i,n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{j,1} & a_{j,2} & \cdots & a_{j,n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{n,1} & a_{n,2} & \cdots & a_{n,n} \end{array}\right], B=\left[\begin{array}{cccc} a_{1,1} & a_{1,2} & \cdots & a_{1,n} \\ a_{2,1} & a_{2,2} & \cdots & a_{2,n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{i,1} & a_{i,2} & \cdots & a_{i,n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{j,1} & a_{j,2} & \cdots & a_{j,n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{n,1} & a_{n,2} & \cdots & a_{n,n} \end{array}\right].$	$A=\left[\begin{array}{cccc} a_{1,1} & a_{1,2} & \cdots & a_{1,n} \\ a_{2,1} & a_{2,2} & \cdots & a_{2,n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{i,1} & a_{i,2} & \cdots & a_{i,n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{j,1} & a_{j,2} & \cdots & a_{j,n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{n,1} & a_{n,2} & \cdots & a_{n,n} \end{array}\right], B=\left[\begin{array}{cccc} a_{1,1} & a_{1,2} & \cdots & a_{1,n} \\ a_{2,1} & a_{2,2} & \cdots & a_{2,n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{i,1} & a_{i,2} & \cdots & a_{i,n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{j,1} & a_{j,2} & \cdots & a_{j,n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{n,1} & a_{n,2} & \cdots & a_{n,n} \end{array}\right].$ cofactor expansion
johnston-linear-matrix-algebra-EQ1035	357	1.000	$\operatorname{det}(B)=a_{i, 1} c_{j, 1}+a_{i, 2} c_{j, 2}+\cdots+a_{i, n} c_{j, n},$	$\operatorname{det}(B)=a_{i, 1} c_{j, 1}+a_{i, 2} c_{j, 2}+\cdots+a_{i, n} c_{j, n},$	$\operatorname{det}(B)=a_{i, 1} c_{j, 1}+a_{i, 2} c_{j, 2}+\cdots+a_{i, n} c_{j, n},$

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johnston-linear-matrix-algebra-EQ1037	357	1.000	$A=\left[\begin{array}{ll} a & b \\ c & d \end{array}\right] \quad \text{is} \quad \operatorname{cof}(A)=\left[\begin{array}{cc} d & -b \\ -c & a \end{array}\right] .$	$A=\left[\begin{array}{cc} a & b \\ c & d \end{array}\right] \quad \text{is} \quad \operatorname{cof}(A)=\left[\begin{array}{cc} d & -b \\ -c & a \end{array}\right] .$	
johnston-linear-matrix-algebra-EQ1038	357	1.000	$A^{-1}=\frac{1}{\operatorname{det}(A)}\operatorname{cof}(A)^T=\frac{1}{ad-bc}\left[\begin{array}{cc} d & -b \\ -c & a \end{array}\right]$	$A^{-1}=\frac{1}{\operatorname{det}(A)}\operatorname{cof}(A)^T=\frac{1}{ad-bc}\left[\begin{array}{cc} d & -b \\ -c & a \end{array}\right],$	
johnston-linear-matrix-algebra-EQ1039	357	1.000	$A=\left[\begin{array}{lll} 1 & 1 & 1 \\ 1 & 2 & 4 \\ 1 & 3 & 9 \end{array}\right] .$	$A=\left[\begin{array}{ccc} 1 & 1 & 1 \\ 1 & 2 & 4 \\ 1 & 3 & 9 \end{array}\right] .$	$A=\left[\begin{array}{ccc} 1 & 1 & 1 \\ 1 & 2 & 4 \\ 1 & 3 & 9 \end{array}\right] .$
johnston-linear-matrix-algebra-EQ1041	358	1.000	$\operatorname{cof}(A)=\left[\begin{array}{ccc} 6 & -5 & 1 \\ -6 & 8 & -2 \\ 2 & -3 & 1 \end{array}\right], \quad \text{so} \quad A^{-1}=\frac{1}{2}\left[\begin{array}{ccc} 6 & -6 & 2 \\ -5 & 8 & -3 \\ 1 & -2 & 1 \end{array}\right] .$	$\operatorname{cof}(A)=\left[\begin{array}{ccc} 6 & -5 & 1 \\ -6 & 8 & -2 \\ 2 & -3 & 1 \end{array}\right], \quad \text{so} \quad A^{-1}=\frac{1}{2}\left[\begin{array}{ccc} 6 & -6 & 2 \\ -5 & 8 & -3 \\ 1 & -2 & 1 \end{array}\right] .$	$\operatorname{cof}(A)=\left[\begin{array}{ccc} 6 & -5 & 1 \\ -6 & 8 & -2 \\ 2 & -3 & 1 \end{array}\right], \quad \text{so} \quad A^{-1}=\frac{1}{2}\left[\begin{array}{ccc} 6 & -6 & 2 \\ -5 & 8 & -3 \\ 1 & -2 & 1 \end{array}\right] .$
johnston-linear-matrix-algebra-EQ1044	358	1.000	$x_j=\frac{\operatorname{det}\left(A_j\right)}{\operatorname{det}(A)} \quad \text{for all} \quad 1 \leq j \leq n .$	$x_j=\frac{\operatorname{det}\left(A_j\right)}{\operatorname{det}(A)} \quad \text{for all} \quad 1 \leq j \leq n .$	$x_j=\frac{\operatorname{det}(A_j)}{\operatorname{det}(A)} \quad \text{for all} \quad 1 \leq j \leq n .$
johnston-linear-matrix-algebra-EQ1045	359	1.000	$x_j=\left[A^{-1} \mathbf{b}\right]_j=\left[\frac{1}{\operatorname{det}(A)} \operatorname{cof}(A)^T \mathbf{b}\right]_j=\frac{1}{\operatorname{det}(A)}\left(c_{1, j} b_1+c_{2, j} b_2+\cdots+c_{n, j} b_n\right),$	$x_j=\left[A^{-1} \mathbf{b}\right]_j=\left[\frac{1}{\operatorname{det}(A)} \operatorname{cof}(A)^T \mathbf{b}\right]_j=\frac{1}{\operatorname{det}(A)}\left(c_{1, j} b_1+c_{2, j} b_2+\cdots+c_{n, j} b_n\right),$	$x_j=\left[A^{-1} \mathbf{b}\right]_j=\left[\frac{1}{\operatorname{det}(A)} \operatorname{cof}(A)^T \mathbf{b}\right]_j=\frac{1}{\operatorname{det}(A)}\left(c_{1, j} b_1+c_{2, j} b_2+\cdots+c_{n, j} b_n\right),$
johnston-linear-matrix-algebra-EQ1048	359	1.000	$\begin{array}{r} x+2 y=4 \\ 3 x+4 y=6 \end{array}$	$\begin{array}{l} x+2 y=4 \\ 3 x+4 y=6 \end{array}$	$\begin{array}{l} x+2 y=4 \\ 3 x+4 y=6 \end{array}$
johnston-linear-matrix-algebra-EQ1049	359	1.000	$\begin{array}{l} 3 x-2 y+z=-3 \\ 2 x+3 y-2 z=5 \\ y+z=4 \end{array}$	$\begin{array}{l} 3 x-2 y+z=-3 \\ 2 x+3 y-2 z=5 \\ y+z=4 \end{array}$	$\begin{array}{l} 3 x-2 y+z=-3 \\ 2 x+3 y-2 z=5 \\ y+z=4 \end{array}$
johnston-linear-matrix-algebra-EQ1050	359	1.000	$x=\frac{4 \cdot 4-2 \cdot 6}{1 \cdot 4-2 \cdot 3}=\frac{4}{-2}=-2 \quad y=\frac{1 \cdot 6-4 \cdot 3}{1 \cdot 4-2 \cdot 3}=\frac{-6}{-2}=3 .$	$x=\frac{4 \cdot 4-2 \cdot 6}{1 \cdot 4-2 \cdot 3}=\frac{4}{-2}=-2 \quad y=\frac{1 \cdot 6-4 \cdot 3}{1 \cdot 4-2 \cdot 3}=\frac{-6}{-2}=3 .$	$x=\frac{4 \cdot 4-2 \cdot 6}{1 \cdot 4-2 \cdot 3}=\frac{4}{-2}=-2 \quad y=\frac{1 \cdot 6-4 \cdot 3}{1 \cdot 4-2 \cdot 3}=\frac{-6}{-2}=3 .$
johnston-linear-matrix-algebra-EQ1052	360	1.000	$\operatorname{det}(A)=21, \quad \operatorname{det}\left(A_1\right)=4, \quad \operatorname{det}\left(A_2\right)=53, \quad \text{and} \quad \operatorname{det}\left(A_3\right)=31 .$	$\operatorname{det}(A)=21, \quad \operatorname{det}\left(A_1\right)=4, \quad \operatorname{det}\left(A_2\right)=53, \quad \text{and} \quad \operatorname{det}\left(A_3\right)=31 .$	$\operatorname{det}(A)=21, \quad \operatorname{det}\left(A_1\right)=4, \quad \operatorname{det}\left(A_2\right)=53, \quad \text{and} \quad \operatorname{det}\left(A_3\right)=31 .$
johnston-linear-matrix-algebra-EQ1053	360	1.000	$x=\frac{\operatorname{det}\left(A_1\right)}{\operatorname{det}(A)}=\frac{4}{21}, \quad y=\frac{\operatorname{det}\left(A_2\right)}{\operatorname{det}(A)}=\frac{53}{21}, \quad z=\frac{\operatorname{det}\left(A_3\right)}{\operatorname{det}(A)}=\frac{31}{21} .$	$x=\frac{\operatorname{det}\left(A_1\right)}{\operatorname{det}(A)}=\frac{4}{21}, \quad y=\frac{\operatorname{det}\left(A_2\right)}{\operatorname{det}(A)}=\frac{53}{21}, \quad z=\frac{\operatorname{det}\left(A_3\right)}{\operatorname{det}(A)}=\frac{31}{21} .$	$x=\frac{\operatorname{det}\left(A_1\right)}{\operatorname{det}(A)}=\frac{4}{21}, \quad y=\frac{\operatorname{det}\left(A_2\right)}{\operatorname{det}(A)}=\frac{53}{21}, \quad z=\frac{\operatorname{det}\left(A_3\right)}{\operatorname{det}(A)}=\frac{31}{21} .$

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johnston-linear-matrix-algebra-EQ1060	362	1.000	$P_{\{\sigma\}}=\left[\mathbf{e}_{\{\sigma(1)\}}\left \mathbf{e}_{\{\sigma(2)\}}\right \cdots\mid\mathbf{e}_{\{\sigma(n)\}}\right]\right]$.	$P_{\sigma} = \left[\mathbf{e}_{\sigma(1)} \left \mathbf{e}_{\sigma(2)} \right \cdots \left \mathbf{e}_{\sigma(n)} \right] .$	$P_{\sigma} = \left[\mathbf{e}_{\sigma(1)} \left \mathbf{e}_{\sigma(2)} \right \cdots \left \mathbf{e}_{\sigma(n)} \right] .$
johnston-linear-matrix-algebra-EQ1066	364	1.000	$\operatorname{sgn}(\sigma)\stackrel{\text{def}}{=}\operatorname{det}\left(P_{\{\sigma\}}\right)$.	$\operatorname{sgn}(\sigma)\stackrel{\text{def}}{=}\operatorname{det}\left(P_{\sigma}\right).$	$\operatorname{sgn}(\sigma)\stackrel{\text{def}}{=}\operatorname{det}(P_{\sigma}).$
johnston-linear-matrix-algebra-EQ1067	364	1.000	$\operatorname{sgn}(\sigma)=\operatorname{det}\left(P_{\{\sigma\}}\right)=\operatorname{det}\left(\left(\begin{array}{llll}0&0&0&1\\0&0&1&0\\1&0&0&0\\0&1&0&0\end{array}\right)\right)=-1$.	$\operatorname{sgn}(\sigma)=\operatorname{det}\left(P_{\sigma}\right)=\operatorname{det}\left(\left(\begin{array}{llll}0&0&0&1\\0&0&1&0\\1&0&0&0\\0&1&0&0\end{array}\right)\right)=-1.$	$\operatorname{sgn}(\sigma)=\operatorname{det}(P_{\sigma})=\operatorname{det}\left(\left(\begin{array}{llll}0&0&0&1\\0&0&1&0\\1&0&0&0\\0&1&0&0\end{array}\right)\right)=-1.$
johnston-linear-matrix-algebra-EQ1071	366	1.000	$\mathbf{a}_{\{j\}}=\sum_{i=1}^n a_{\{i,j\}}\mathbf{e}_{\{i\}}$.	$\mathbf{a}_j = \sum_{i=1}^n a_{i,j} \mathbf{e}_i.$	$\mathbf{a}_j = \sum_{i=1}^n a_{i,j} \mathbf{e}_i.$
johnston-linear-matrix-algebra-EQ1073	366	1.000	$\operatorname{det}(A)=\sum_{\sigma\in S_n}\left(a_{\sigma(1),1}a_{\sigma(2),2}\cdots a_{\sigma(n),n}\right)\operatorname{det}\left(\left[\mathbf{e}_{\sigma(1)}\left \mathbf{e}_{\sigma(2)}\right \cdots\left \mathbf{e}_{\sigma(n)}\right]\right)$ $=\sum_{\sigma\in S_n}\left(a_{\sigma(1),1}a_{\sigma(2),2}\cdots a_{\sigma(n),n}\right)\operatorname{det}(P_{\sigma})$ $=\sum_{\sigma\in S_n}\operatorname{sgn}(\sigma)a_{\sigma(1),1}a_{\sigma(2),2}\cdots a_{\sigma(n),n}$	$\operatorname{det}(A)=\sum_{\sigma\in S_n}\left(a_{\sigma(1),1}a_{\sigma(2),2}\cdots a_{\sigma(n),n}\right)\operatorname{det}\left(\left[\mathbf{e}_{\sigma(1)}\left \mathbf{e}_{\sigma(2)}\right \cdots\left \mathbf{e}_{\sigma(n)}\right]\right)$ $=\sum_{\sigma\in S_n}\left(a_{\sigma(1),1}a_{\sigma(2),2}\cdots a_{\sigma(n),n}\right)\operatorname{det}(P_{\sigma})$ $=\sum_{\sigma\in S_n}\operatorname{sgn}(\sigma)a_{\sigma(1),1}a_{\sigma(2),2}\cdots a_{\sigma(n),n}$	$\operatorname{det}(A)=\sum_{\sigma\in S_n}\left(a_{\sigma(1),1}a_{\sigma(2),2}\cdots a_{\sigma(n),n}\right)\operatorname{det}\left(\left[\mathbf{e}_{\sigma(1)}\left \mathbf{e}_{\sigma(2)}\right \cdots\left \mathbf{e}_{\sigma(n)}\right]\right)$ $=\sum_{\sigma\in S_n}\left(a_{\sigma(1),1}a_{\sigma(2),2}\cdots a_{\sigma(n),n}\right)\operatorname{det}(P_{\sigma})$ $=\sum_{\sigma\in S_n}\operatorname{sgn}(\sigma)a_{\sigma(1),1}a_{\sigma(2),2}\cdots a_{\sigma(n),n},$
johnston-linear-matrix-algebra-EQ1074	367	1.000	$\operatorname{det}(A\,B)=\sum_{1\leq j_1\leq\cdots\leq j_m\leq n}\left(a_{j_1,1}\cdots a_{j_m,m}\right)\operatorname{det}(B_{j_1,\ldots,j_m})$.	$\operatorname{det}(AB)=\sum_{1\leq j_1<\cdots<j_m\leq n}\operatorname{det}\left(A_{j_1,\ldots,j_m}\right)\operatorname{det}\left(B_{j_1,\ldots,j_m}\right).$	$\operatorname{det}(AB)=\sum_{1\leq j_1<\cdots<j_m\leq n}\operatorname{det}\left(A_{j_1,\ldots,j_m}\right)\operatorname{det}\left(B_{j_1,\ldots,j_m}\right).$
johnston-linear-matrix-algebra-EQ1077	368	1.000	$\operatorname{det}(A\,B)=a_{1,1}b_{1,1}+a_{1,2}b_{2,1}+\cdots+a_{1,n}b_{n,1},$	$\operatorname{det}(AB)=a_{1,1}b_{1,1}+a_{1,2}b_{2,1}+\cdots+a_{1,n}b_{n,1},$	$\operatorname{det}(AB)=a_{1,1}b_{1,1}+a_{1,2}b_{2,1}+\cdots+a_{1,n}b_{n,1},$
johnston-linear-matrix-algebra-EQ1078	368	1.000	$A=\left(\begin{array}{ccc}1&0&2\\2&1&-1\end{array}\right)\quad\text{and}\quad B=\left(\begin{array}{cc}3&1\\-2&3\end{array}\right)$.	$A=\left[\begin{array}{lll}1&0&2\\2&1&-1\end{array}\right]\quad\text{and}\quad B=\left[\begin{array}{ll}3&1\\-2&3\\-1&2\end{array}\right].$	$A=\left[\begin{array}{lll}1&0&2\\2&1&-1\end{array}\right]\quad\text{and}\quad B=\left[\begin{array}{ll}3&1\\-2&3\\-1&2\end{array}\right].$
johnston-linear-matrix-algebra-EQ1081	368	1.000	$\operatorname{det}(A\,B)=1\cdot 11+(-5)\cdot 7+(-2)\cdot (-1)=11-35+2=-22$.	$\operatorname{det}(AB)=1\cdot 11+(-5)\cdot 7+(-2)\cdot (-1)=11-35+2=-22.$	$\operatorname{det}(AB)=1\cdot 11+(-5)\cdot 7+(-2)\cdot (-1)=11-35+2=-22.$
johnston-linear-matrix-algebra-EQ1082	368	1.000	$\operatorname{det}(A\,B)=\operatorname{det}\left(\left(\begin{array}{ll}1&5\\5&3\end{array}\right)\right)=-22$.	$\operatorname{det}(AB)=\operatorname{det}\left(\left(\begin{array}{ll}1&5\\5&3\end{array}\right)\right)=-22.$	$\operatorname{det}(AB)=\operatorname{det}\left(\left(\begin{array}{ll}1&5\\5&3\end{array}\right)\right)=-22.$
johnston-linear-matrix-algebra-EQ1085	369	1.000	$x+y+z=4$ $x-y+z=0$ $x+y-z=1$	$x+y+z=4$ $x-y+z=0$ $x+y-z=1$	$x+y+z=4$ $x-y+z=0$ $x+y-z=1$

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johnston-linear-matrix-algebra-EQ1086	369	1.000	$\begin{aligned} 2x+y-z &= 1 \\ x-3y+z &= -2 \\ y-z &= 6 \end{aligned}$	$\begin{aligned} 2x+y-z &= 1 \\ x-3y+z &= -2 \\ 2x-2y-z &= 6 \end{aligned}$	$\begin{aligned} 2x+y-z &= 1 \\ x-3y+z &= -2 \\ 2x-2y-z &= 6 \end{aligned}$
johnston-linear-matrix-algebra-EQ1089	369	1.000	$\operatorname{det}\left(A\ A^T\right)=\sum_{1\leq j_1<\cdots<j_m\leq n}\operatorname{det}\left(A_{j_1},\ldots,j_m\right)^2.$	$\det\left(AA^T\right)=\sum_{1\leq j_1<\cdots<j_m\leq n}\det\left(A_{j_1},\ldots,j_m\right)^2.$	$\det(AA^T)=\sum_{1\leq j_1<\cdots<j_m\leq n}\det\left(A_{j_1},\ldots,j_m\right)^2.$
johnston-linear-matrix-algebra-EQ1093	373	1.000	$A=\left[\begin{array}{ccccccc} 0 & -1 & 1 & 0 & 1 & 0 & 1 \\ -1 & 0 & 1 & 1 & 1 & -1 & 0 \\ 1 & 1 & 0 & -1 & -2 & 1 & 0 \\ 0 & 1 & -1 & 0 & -1 & 1 & 1 \\ 1 & 1 & -2 & -1 & 0 & 1 & 0 \\ 0 & -1 & 1 & 1 & 1 & 0 & -1 \\ 1 & 0 & 0 & 1 & 0 & -1 & 0 \end{array}\right].$	$A=\left[\begin{array}{ccccccc} 0 & -1 & 1 & 0 & 1 & 0 & 1 \\ -1 & 0 & 1 & 1 & 1 & -1 & 0 \\ 1 & 1 & 0 & -1 & -2 & 1 & 0 \\ 0 & 1 & -1 & 0 & -1 & 1 & 1 \\ 1 & 1 & -2 & -1 & 0 & 1 & 0 \\ 0 & -1 & 1 & 1 & 1 & 0 & -1 \\ 1 & 0 & 0 & 1 & 0 & -1 & 0 \end{array}\right].$	$A=\left[\begin{array}{ccccccc} 0 & -1 & 1 & 0 & 1 & 0 & 1 \\ -1 & 0 & 1 & 1 & 1 & -1 & 0 \\ 1 & 1 & 0 & -1 & -2 & 1 & 0 \\ 0 & 1 & -1 & 0 & -1 & 1 & 1 \\ 1 & 1 & -2 & -1 & 0 & 1 & 0 \\ 0 & -1 & 1 & 1 & 1 & 0 & -1 \\ 1 & 0 & 0 & 1 & 0 & -1 & 0 \end{array}\right].$
johnston-linear-matrix-algebra-EQ1094	373	1.000	$p_A(\lambda)=-\lambda^7+19\lambda^5-36\lambda^4-24\lambda^3+116\lambda^2-102\lambda+28.$	$p_A(\lambda)=-\lambda^7+19\lambda^5-36\lambda^4-24\lambda^3+116\lambda^2-102\lambda+28.$	$p_A(\lambda)=-\lambda^7+19\lambda^5-36\lambda^4-24\lambda^3+116\lambda^2-102\lambda+28.$
johnston-linear-matrix-algebra-EQ1095	374	1.000	$C=\left[\begin{array}{ccccc} 0 & 1 & 0 & \cdots & 0 \\ 0 & 0 & 1 & \cdots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & 0 & \cdots & 1 \\ -a_0 & -a_1 & -a_2 & \cdots & -a_{n-1} \end{array}\right],$	$C=\left[\begin{array}{ccccc} 0 & 1 & 0 & \cdots & 0 \\ 0 & 0 & 1 & \cdots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & 0 & \cdots & 1 \\ -a_0 & -a_1 & -a_2 & \cdots & -a_{n-1} \end{array}\right],$	$C=\left[\begin{array}{ccccc} 0 & 1 & 0 & \cdots & 0 \\ 0 & 0 & 1 & \cdots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & 0 & \cdots & 1 \\ -a_0 & -a_1 & -a_2 & \cdots & -a_{n-1} \end{array}\right],$
johnston-linear-matrix-algebra-EQ1096	374	1.000	$C=\left[\begin{array}{cccccc} 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 \\ 2 & -1 & -1 & 3 & -2 & 2 \end{array}\right],$	$C=\left[\begin{array}{cccccc} 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 \\ 2 & -1 & -1 & 3 & -2 & 2 \end{array}\right],$	$C=\left[\begin{array}{cccccc} 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 \\ 2 & -1 & -1 & 3 & -2 & 2 \end{array}\right],$
johnston-linear-matrix-algebra-EQ1097	376	1.000	$\left[\begin{array}{cc} 1 & 1 \\ -1 & 1 \end{array}\right]$	$\left[\begin{array}{cc} 1 & 1 \\ -1 & 1 \end{array}\right]$	$\left[\begin{array}{cc} 1 & 1 \\ -1 & 1 \end{array}\right]$
johnston-linear-matrix-algebra-EQ1098	377	1.000	$\mathbf{v}_0=c_1\mathbf{w}_1+c_2\mathbf{w}_2+\cdots+c_n\mathbf{w}_n.$	$\mathbf{v}_0=c_1\mathbf{w}_1+c_2\mathbf{w}_2+\cdots+c_n\mathbf{w}_n.$	$\mathbf{v}_0=c_1\mathbf{w}_1+c_2\mathbf{w}_2+\cdots+c_n\mathbf{w}_n.$
johnston-linear-matrix-algebra-EQ1100	377	1.000	$\mathbf{r}_k=c_2\left(\frac{\lambda_2}{\lambda_1}\right)^k\mathbf{w}_2+\cdots+c_n\left(\frac{\lambda_n}{\lambda_1}\right)^k\mathbf{w}_n.$	$\mathbf{r}_k=c_2\left(\frac{\lambda_2}{\lambda_1}\right)^k\mathbf{w}_2+\cdots+c_n\left(\frac{\lambda_n}{\lambda_1}\right)^k\mathbf{w}_n.$	$\mathbf{r}_k=c_2\left(\frac{\lambda_2}{\lambda_1}\right)^k\mathbf{w}_2+\cdots+c_n\left(\frac{\lambda_n}{\lambda_1}\right)^k\mathbf{w}_n.$
johnston-linear-matrix-algebra-EQ1101	377	1.000	$\mathbf{v}_k=\frac{A^k\mathbf{v}_0}{\ A^k\mathbf{v}_0\ }=\frac{\lambda_1^k(c_1\mathbf{w}_1+\mathbf{r}_k)}{\ \lambda_1^k(c_1\mathbf{w}_1+\mathbf{r}_k)\ }\quad\text{for all }k\geq 0$	$\mathbf{v}_k=\frac{A^k\mathbf{v}_0}{\ A^k\mathbf{v}_0\ }=\frac{\lambda_1^k(c_1\mathbf{w}_1+\mathbf{r}_k)}{\ \lambda_1^k(c_1\mathbf{w}_1+\mathbf{r}_k)\ }\quad\text{for all }k\geq 0$	$\mathbf{v}_k=\frac{A^k\mathbf{v}_0}{\ A^k\mathbf{v}_0\ }=\frac{\lambda_1^k(c_1\mathbf{w}_1+\mathbf{r}_k)}{\ \lambda_1^k(c_1\mathbf{w}_1+\mathbf{r}_k)\ }\quad\text{for all }k\geq 0.$

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johnston-linear-matrix-algebra-EQ1103	378	1.000	$\mathbf{v}_k = \frac{\lambda_1^k (c_1 \mathbf{w}_1 + \mathbf{r}_k)}{\ \lambda_1^k (c_1 \mathbf{w}_1 + \mathbf{r}_k)\ } = \left(\frac{\lambda_1}{ \lambda_1 } \right)^k \frac{c_1 \mathbf{w}_1 + \mathbf{r}_k}{\ c_1 \mathbf{w}_1 + \mathbf{r}_k\ } \approx \frac{c_1}{ c_1 } \left(\frac{\lambda_1}{ \lambda_1 } \right)^k \mathbf{w}_1$		
johnston-linear-matrix-algebra-EQ1104	379	1.000	$A - \lambda_1 \mathbf{w}_1 \mathbf{w}_1^T$		
johnston-linear-matrix-algebra-EQ1107	381	1.000	$p_A(\lambda) = -\lambda^3 + 7\lambda^2 + 14\lambda - 8,$		
johnston-linear-matrix-algebra-EQ1108	381	1.000	$A = \begin{bmatrix} 1 & 2 & 2 & 2 \\ 3 & -1 & 2 & -1 \\ 2 & 3 & 2 & 2 \\ 3 & 3 & 0 & -1 \end{bmatrix}$		
johnston-linear-matrix-algebra-EQ1109 (3.B.1)	382	1.000	$L(\mathbf{v}) = \max \{c \in \mathbb{R} : A \mathbf{v} \geq c \mathbf{v}\}.$		
johnston-linear-matrix-algebra-EQ1110	382	1.000	$L(\mathbf{v}) = \min_{1 \leq j \leq n} \left\{ \frac{[A\mathbf{v}]_j}{v_j} : v_j \neq 0 \right\}$		
johnston-linear-matrix-algebra-EQ1111	383	1.000	$\frac{A(A \mathbf{v})}{\ A\mathbf{v}\ } > \frac{\mu A \mathbf{v}}{\ A\mathbf{v}\ }, \quad \text{so} \quad L\left(\frac{A \mathbf{v}}{\ A\mathbf{v}\ }\right) > \mu.$		
johnston-linear-matrix-algebra-EQ1112 (3.B.2)	383	1.000	$ \lambda w_i = [B\mathbf{w}]_i = \left \sum_{j=1}^n b_{i,j} w_j \right \leq \sum_{j=1}^n b_{i,j} w_j = [B \mathbf{w}]_i,$		
johnston-linear-matrix-algebra-EQ1113	384	1.000	$A(\mathbf{v}_1 - c\mathbf{v}) = A\mathbf{v}_1 - cA\mathbf{v} = \mu\mathbf{v}_1 - c\lambda\mathbf{v} \geq \mu\mathbf{v}_1 - c\mu\mathbf{v} = \mu(\mathbf{v}_1 - c\mathbf{v}).$		
johnston-linear-matrix-algebra-EQ1114 (3.B.3)	384	1.000	$\frac{A(\mathbf{v}_1 - c\mathbf{v})}{\ \mathbf{v}_1 - c\mathbf{v}\ } \geq \frac{\mu(\mathbf{v}_1 - c\mathbf{v})}{\ \mathbf{v}_1 - c\mathbf{v}\ }, \quad \text{so} \quad L\left(\frac{\mathbf{v}_1 - c\mathbf{v}}{\ \mathbf{v}_1 - c\mathbf{v}\ }\right) \geq \mu.$		
johnston-linear-matrix-algebra-EQ1115	385	1.000	$A = \begin{bmatrix} 2 & 1 & 4 \\ 4 & 1 & 2 \\ 1 & 2 & 4 \end{bmatrix}$		
johnston-linear-matrix-algebra-EQ1118 (3.B.4)	385	1.000	$r_i = a_{i,1}r_1 + a_{i,2}r_2 + \cdots + a_{i,n}r_n \quad \text{for all } 1 \leq i \leq n.$		
johnston-linear-matrix-algebra-EQ1119	386	1.000	$r_A = \frac{1}{3}r_B + r_C + \frac{1}{2}r_D + \frac{1}{4}r_E.$		

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johnston-linear-matrix-algebra-EQ1121	387	1.000	$A=\left[\begin{array}{ccccc} 0 & 1/3 & 1 & 1/2 & 1/4 \\ 1/2 & 0 & 0 & 0 & 1/4 \\ 1/2 & 0 & 0 & 0 & 1/4 \\ 0 & 1/3 & 0 & 0 & 1/4 \\ 0 & 1/3 & 0 & 1/2 & 0 \end{array}\right].$	$A=\left[\begin{array}{ccccc} 0 & 1/3 & 1 & 1/2 & 1/4 \\ 1/2 & 0 & 0 & 0 & 1/4 \\ 1/2 & 0 & 0 & 0 & 1/4 \\ 0 & 1/3 & 0 & 0 & 1/4 \\ 0 & 1/3 & 0 & 1/2 & 0 \end{array}\right].$	$A=\left[\begin{array}{ccccc} 0 & 1/3 & 1 & 1/2 & 1/4 \\ 1/2 & 0 & 0 & 0 & 1/4 \\ 1/2 & 0 & 0 & 0 & 1/4 \\ 0 & 1/3 & 0 & 0 & 1/4 \\ 0 & 1/3 & 0 & 1/2 & 0 \end{array}\right].$
johnston-linear-matrix-algebra-EQ1122	388	1.000	$\mathbf{r}=\left(r_{\{A\}},\,r_{\{B\}},\,r_{\{C\}},\,r_{\{D\}},\,r_{\{E\}}\right)\approx\left(0.74,0.42,0.42,0.21,0.24\right)$	$\mathbf{r}=\left(r_A,r_B,r_C,r_D,r_E\right)\approx\left(0.74,0.42,0.42,0.21,0.24\right)$	$\mathbf{r}=\left(r_A,r_B,r_C,r_D,r_E\right)\approx\left(0.74,0.42,0.42,0.21,0.24\right)$
johnston-linear-matrix-algebra-EQ1124	389	1.000	$\left[\begin{array}{cccccc} -3 & -4 & 3 & 7 & 1 & -2 \\ 4 & 6 & 4 & -1 & -4 & 3 \\ -2 & 1 & -3 & -1 & -3 & 1 \\ 0 & 4 & 1 & 5 & -3 & 5 \\ 0 & 3 & 4 & 0 & 2 & -1 \\ 3 & -1 & 3 & 5 & 6 & 0 \\ 4 & 8 & 3 & 7 & 7 & 6 \end{array}\right]$	$\left[\begin{array}{cccccc} -3 & -4 & 3 & 7 & 1 & -2 \\ 4 & 6 & 4 & -1 & -4 & 3 \\ -2 & 1 & -3 & -1 & -3 & 1 \\ 0 & 4 & 1 & 5 & -3 & 5 \\ 0 & 3 & 4 & 0 & 2 & -1 \\ 3 & -1 & 3 & 5 & 6 & 0 \\ 4 & 8 & 3 & 7 & 7 & 6 \end{array}\right]$	$\left[\begin{array}{cccccc} -3 & -4 & 3 & 7 & 1 & -2 \\ 4 & 6 & 4 & -1 & -4 & 3 \\ -2 & 1 & -3 & -1 & -3 & 1 \\ 0 & 4 & 1 & 5 & -3 & 5 \\ 0 & 3 & 4 & 0 & 2 & -1 \\ 3 & -1 & 3 & 5 & 6 & 0 \\ 4 & 8 & 3 & 7 & 7 & 6 \end{array}\right]$
johnston-linear-matrix-algebra-EQ1125	389	1.000	$\left[\begin{array}{llllllll} 0 & 1 & 2 & 3 & 4 & 5 & 6 & 7 \\ 7 & 1 & 2 & 3 & 4 & 5 & 6 & 7 \\ 7 & 6 & 5 & 3 & 4 & 5 & 6 & 7 \\ 7 & 6 & 5 & 4 & 3 & 5 & 6 & 7 \\ 7 & 6 & 5 & 4 & 3 & 2 & 6 & 7 \\ 7 & 6 & 5 & 4 & 3 & 2 & 1 & 7 \\ 7 & 6 & 5 & 4 & 3 & 2 & 1 & 0 \end{array}\right]$	$\left[\begin{array}{cccccc} 0 & 1 & 2 & 3 & 4 & 5 \\ 1 & 2 & 3 & 4 & 5 & 6 \\ 2 & 3 & 4 & 5 & 6 & 7 \\ 3 & 4 & 5 & 6 & 7 & 6 \\ 4 & 5 & 6 & 7 & 6 & 5 \\ 5 & 6 & 7 & 6 & 5 & 4 \\ 6 & 7 & 6 & 5 & 4 & 3 \\ 7 & 6 & 5 & 4 & 3 & 2 \end{array}\right]$	$\left[\begin{array}{cccccc} 0 & 1 & 2 & 3 & 4 & 5 \\ 1 & 2 & 3 & 4 & 5 & 6 \\ 2 & 3 & 4 & 5 & 6 & 7 \\ 3 & 4 & 5 & 6 & 7 & 6 \\ 4 & 5 & 6 & 7 & 6 & 5 \\ 5 & 6 & 7 & 6 & 5 & 4 \\ 6 & 7 & 6 & 5 & 4 & 3 \\ 7 & 6 & 5 & 4 & 3 & 2 \end{array}\right]$
johnston-linear-matrix-algebra-EQ1126	389	1.000	$\mathbf{v}_{\{0\}}^{\{T\}}\,A\,\mathbf{v}_{\{0\}},\,\mathbf{v}_{\{1\}}^{\{T\}}\,A\,\mathbf{v}_{\{1\}},\,\mathbf{v}_{\{2\}}^{\{T\}}\,A\,\mathbf{v}_{\{2\}},\,\ldots$	$\mathbf{v}_0^T A \mathbf{v}_0, \mathbf{v}_1^T A \mathbf{v}_1, \mathbf{v}_2^T A \mathbf{v}_2, \dots$	$\mathbf{v}_0^T A \mathbf{v}_0, \mathbf{v}_1^T A \mathbf{v}_1, \mathbf{v}_2^T A \mathbf{v}_2, \dots$
johnston-linear-matrix-algebra-EQ1128	390	1.000	$\left[\begin{array}{ll} 0 & 1 \\ 1 & 0 \end{array}\right].$	$\left[\begin{array}{cc} 0 & 1 \\ 1 & 0 \end{array}\right].$	$\left[\begin{array}{cc} 0 & 1 \\ 1 & 0 \end{array}\right].$
johnston-linear-matrix-algebra-EQ1129	391	1.000	$\left[\begin{array}{cc c} -2-2i & -2 & 0 \\ 4 & 2-2i & 0 \end{array}\right] \xrightarrow{R_2+(1-i)R_1} \left[\begin{array}{cc c} -2-2i & -2 & 0 \\ 0 & 0 & 0 \end{array}\right].$	$\left[\begin{array}{cc c} -2-2i & -2 & 0 \\ 4 & 2-2i & 0 \end{array}\right] \xrightarrow{R_2+(1-i)R_1} \left[\begin{array}{cc c} -2-2i & -2 & 0 \\ 0 & 0 & 0 \end{array}\right].$	$\left[\begin{array}{cc c} -2-2i & -2 & 0 \\ 4 & 2-2i & 0 \end{array}\right] \xrightarrow{R_2+(1-i)R_1} \left[\begin{array}{cc c} -2-2i & -2 & 0 \\ 0 & 0 & 0 \end{array}\right].$
johnston-linear-matrix-algebra-EQ1131	391	1.000	$\operatorname{Re}(\mathbf{v})=\frac{1}{2}(\mathbf{v}+\overline{\mathbf{v}}) \quad \text{and} \quad \operatorname{Im}(\mathbf{v})=\frac{1}{2i}(\mathbf{v}-\overline{\mathbf{v}}).$	$\operatorname{Re}(\mathbf{v})=\frac{1}{2}(\mathbf{v}+\overline{\mathbf{v}}) \quad \text{and} \quad \operatorname{Im}(\mathbf{v})=\frac{1}{2i}(\mathbf{v}-\overline{\mathbf{v}}).$	$\operatorname{Re}(\mathbf{v})=\frac{1}{2}(\mathbf{v}+\overline{\mathbf{v}}) \quad \text{and} \quad \operatorname{Im}(\mathbf{v})=\frac{1}{2i}(\mathbf{v}-\overline{\mathbf{v}}).$
johnston-linear-matrix-algebra-EQ1132	392	1.000	$[R^\theta]=\left[\begin{array}{cc} \cos(\theta) & -\sin(\theta) \\ \sin(\theta) & \cos(\theta) \end{array}\right].$	$[R^\theta]=\left[\begin{array}{cc} \cos(\theta) & -\sin(\theta) \\ \sin(\theta) & \cos(\theta) \end{array}\right].$	$[R^\theta]=\left[\begin{array}{cc} \cos(\theta) & -\sin(\theta) \\ \sin(\theta) & \cos(\theta) \end{array}\right].$
johnston-linear-matrix-algebra-EQ1133	392	1.000	$A=Q\left(r\left[R^\theta\right]\right)Q^{-1}.$	$A=Q\left(r\left[R^\theta\right]\right)Q^{-1}.$	$A=Q\left(r\left[R^\theta\right]\right)Q^{-1}.$

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johnston-linear-matrix-algebra-EQ1135	393	1.000	$D=\left[\begin{array}{cc} re^{i\theta} & 0 \\ 0 & re^{-i\theta} \end{array}\right] \quad \text{and} \quad P=[\mathbf{v} \mid \overline{\mathbf{v}}]$	$D = \begin{bmatrix} re^{i\theta} & 0 \\ 0 & re^{-i\theta} \end{bmatrix} \quad \text{and} \quad P = [\mathbf{v} \mid \overline{\mathbf{v}}]$	$D = \begin{bmatrix} re^{i\theta} & 0 \\ 0 & re^{-i\theta} \end{bmatrix} \quad \text{and} \quad P = [\mathbf{v} \mid \overline{\mathbf{v}}]$
johnston-linear-matrix-algebra-EQ1136 (3.C.1)	393	1.000	$H=\left[\begin{array}{cc} 1 & 1 \\ -i & i \end{array}\right] \quad \text{with inverse} \quad H^{-1}=\frac{1}{2}\left[\begin{array}{cc} 1 & i \\ 1 & -i \end{array}\right].$	$H = \begin{bmatrix} 1 & 1 \\ -i & i \end{bmatrix} \quad \text{with inverse} \quad H^{-1} = \frac{1}{2} \begin{bmatrix} 1 & i \\ 1 & -i \end{bmatrix}.$	$H = \begin{bmatrix} 1 & 1 \\ -i & i \end{bmatrix} \quad \text{with inverse} \quad H^{-1} = \frac{1}{2} \begin{bmatrix} 1 & i \\ 1 & -i \end{bmatrix}.$
johnston-linear-matrix-algebra-EQ1140	394	1.000	$Q^{-1}AQ = HDH^{-1} = r[R^\theta]$	$Q^{-1}AQ = HDH^{-1} = r[R^\theta]$	$Q^{-1}AQ = HDH^{-1} = r[R^\theta],$
johnston-linear-matrix-algebra-EQ1141	394	1.000	$A=\left[\begin{array}{cc} -3 & -2 \\ 4 & 1 \end{array}\right]$	$A = \begin{bmatrix} -3 & -2 \\ 4 & 1 \end{bmatrix}$	$A = \begin{bmatrix} -3 & -2 \\ 4 & 1 \end{bmatrix}$
johnston-linear-matrix-algebra-EQ1143	394	1.000	$Q=PH^{-1}=\frac{1}{2}\left[\begin{array}{cc} 1 & 1 \\ -1-i & -1+i \end{array}\right]\left[\begin{array}{cc} 1 & i \\ 1 & -i \end{array}\right]=\left[\begin{array}{cc} 1 & 0 \\ -1 & 1 \end{array}\right],$	$Q = PH^{-1} = \frac{1}{2} \begin{bmatrix} 1 & 1 \\ -1-i & -1+i \end{bmatrix} \begin{bmatrix} 1 & i \\ 1 & -i \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ -1 & 1 \end{bmatrix},$	$Q = PH^{-1} = \frac{1}{2} \begin{bmatrix} 1 & 1 \\ -1-i & -1+i \end{bmatrix} \begin{bmatrix} 1 & i \\ 1 & -i \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ -1 & 1 \end{bmatrix},$
johnston-linear-matrix-algebra-EQ1144	395	1.000	$H D H^{-1}=\frac{1}{2}\left[\begin{array}{cc} 1 & 1 \\ -i & i \end{array}\right]\left[\begin{array}{cc} -1+2i & 0 \\ 0 & -1-2i \end{array}\right]\left[\begin{array}{cc} 1 & i \\ 1 & -i \end{array}\right]=\left[\begin{array}{cc} -1 & -2 \\ 2 & -1 \end{array}\right],$	$HDH^{-1} = \frac{1}{2} \begin{bmatrix} 1 & 1 \\ -i & i \end{bmatrix} \begin{bmatrix} -1+2i & 0 \\ 0 & -1-2i \end{bmatrix} \begin{bmatrix} 1 & i \\ 1 & -i \end{bmatrix} = \begin{bmatrix} -1 & -2 \\ 2 & -1 \end{bmatrix},$	$HDH^{-1} = \frac{1}{2} \begin{bmatrix} 1 & 1 \\ -i & i \end{bmatrix} \begin{bmatrix} -1+2i & 0 \\ 0 & -1-2i \end{bmatrix} \begin{bmatrix} 1 & i \\ 1 & -i \end{bmatrix} = \begin{bmatrix} -1 & -2 \\ 2 & -1 \end{bmatrix},$
johnston-linear-matrix-algebra-EQ1148	395	1.000	$A=\frac{1}{4}\left[\begin{array}{ccc} 3 & 2 & -1 \\ -5 & 2 & 3 \\ -5 & 2 & 7 \end{array}\right]$	$A = \frac{1}{4} \begin{bmatrix} 3 & 2 & -1 \\ -5 & 2 & 3 \\ -5 & 2 & 7 \end{bmatrix}$	$A = \frac{1}{4} \begin{bmatrix} 3 & 2 & -1 \\ -5 & 2 & 3 \\ -5 & 2 & 7 \end{bmatrix}$
johnston-linear-matrix-algebra-EQ1150	396	1.000	$\widetilde{H}=\operatorname{diag}(\underbrace{H, H, \ldots, H}_{\ell \text{ copies}}, \underbrace{1, 1, \ldots, 1}_m)$	$\widetilde{H} = \operatorname{diag}(\underbrace{H, H, \ldots, H}_{\ell \text{ copies}}, \underbrace{1, 1, \ldots, 1}_m).$	$\widetilde{H} = \operatorname{diag}(\underbrace{H, H, \ldots, H}_{\ell \text{ copies}}, \underbrace{1, 1, \ldots, 1}_m).$
johnston-linear-matrix-algebra-EQ1153	396	1.000	$QBQ^{-1}=Q\left(\widetilde{H}D\widetilde{H}^{-1}\right)Q^{-1}=(Q\widetilde{H})D(Q\widetilde{H})^{-1}=PDP^{-1}=A,$	$QBQ^{-1} = Q\left(\widetilde{H}D\widetilde{H}^{-1}\right)Q^{-1} = (Q\widetilde{H})D(Q\widetilde{H})^{-1} = PDP^{-1} = A,$	$QBQ^{-1} = Q(\widetilde{H}D\widetilde{H}^{-1})Q^{-1} = (Q\widetilde{H})D(Q\widetilde{H})^{-1} = PDP^{-1} = A,$
johnston-linear-matrix-algebra-EQ1157	397	1.000	$\mathcal{S}=\operatorname{span}(\operatorname{Re}(\mathbf{v}),-\operatorname{Im}(\mathbf{v}))=\operatorname{span}((1,0,1),(0,-1,0))$	$\mathcal{S} = \operatorname{span}(\operatorname{Re}(\mathbf{v}), -\operatorname{Im}(\mathbf{v})) = \operatorname{span}((1,0,1), (0, -1, 0))$	$\mathcal{S} = \operatorname{span}(\operatorname{Re}(\mathbf{v}), -\operatorname{Im}(\mathbf{v})) = \operatorname{span}((1,0,1), (0, -1, 0))$
johnston-linear-matrix-algebra-EQ1158	398	1.000	$A^k=\left(QBQ^{-1}\right)^k=QB^kQ^{-1}=Q\operatorname{diag}\left(r_1^k\left[R^{k\theta_1}\right],\ldots,r_\ell^k\left[R^{k\theta_\ell}\right],\lambda_1^k,\ldots,\lambda_m^k\right)Q^{-1}.$	$A^k = (QBQ^{-1})^k = QB^kQ^{-1} = Q\operatorname{diag}\left(r_1^k\left[R^{k\theta_1}\right], \ldots, r_\ell^k\left[R^{k\theta_\ell}\right], \lambda_1^k, \ldots, \lambda_m^k\right)Q^{-1}.$	$A^k = (QBQ^{-1})^k = QB^kQ^{-1} = Q\operatorname{diag}\left(r_1^k\left[R^{k\theta_1}\right], \ldots, r_\ell^k\left[R^{k\theta_\ell}\right], \lambda_1^k, \ldots, \lambda_m^k\right)Q^{-1}.$
johnston-linear-matrix-algebra-EQ1159	398	1.000	$B=\left[\begin{array}{c c} \frac{1}{\sqrt{2}}\left[R^{\pi/4}\right] & \mathbf{0} \\ \hline \mathbf{0}^T & 2 \end{array}\right] \quad \text{and} \quad Q=\left[\begin{array}{ccc} 1 & 0 & 0 \\ 0 & -1 & 1 \\ 1 & 0 & 2 \end{array}\right].$	$B = \left[\begin{array}{c c} \frac{1}{\sqrt{2}}\left[R^{\pi/4}\right] & \mathbf{0} \\ \hline \mathbf{0}^T & 2 \end{array}\right] \quad \text{and} \quad Q = \begin{bmatrix} 1 & 0 & 0 \\ 0 & -1 & 1 \\ 1 & 0 & 2 \end{bmatrix}.$	$B = \left[\begin{array}{c c} \frac{1}{\sqrt{2}}\left[R^{\pi/4}\right] & \mathbf{0} \\ \hline \mathbf{0}^T & 2 \end{array}\right] \quad \text{and} \quad Q = \begin{bmatrix} 1 & 0 & 0 \\ 0 & -1 & 1 \\ 1 & 0 & 2 \end{bmatrix}.$

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johnston-linear-matrix-algebra-EQ1161 (2)	399	1.000	$\left[\begin{array}{ccccc} 2 & 0 & 0 & 0 & -2 \\ 0 & 2 & 1 & 2 & -1 \\ 2 & 0 & 2 & 0 & 0 \\ 0 & -2 & -1 & 2 & 1 \\ 2 & 0 & -2 & 0 & 4 \end{array}\right]$	$\begin{bmatrix} 2 & 0 & 0 & 0 & -2 \\ 0 & 2 & 1 & 2 & -1 \\ 2 & 0 & 2 & 0 & 0 \\ 0 & -2 & -1 & 2 & 1 \\ 2 & 0 & -2 & 0 & 4 \end{bmatrix}$	$\begin{bmatrix} 2 & 0 & 0 & 0 & -2 \\ 0 & 2 & 1 & 2 & -1 \\ 2 & 0 & 2 & 0 & 0 \\ 0 & -2 & -1 & 2 & 1 \\ 2 & 0 & -2 & 0 & 4 \end{bmatrix}$
johnston-linear-matrix-algebra-EQ1162	399	1.000	$\left[\begin{array}{cccccc} -2 & -3 & 7 & -1 & -7 & 3 \\ 2 & -9 & -9 & 1 & 13 & -3 \\ -9 & 16 & 31 & -5 & -39 & 9 \\ -13 & 38 & 50 & -8 & -67 & 14 \\ -2 & -1 & 5 & -1 & -5 & 1 \\ 8 & -20 & -29 & 5 & 38 & -9 \end{array}\right]$	$\begin{bmatrix} -2 & -3 & 7 & -1 & -7 & 3 \\ 2 & -9 & -9 & 1 & 13 & -3 \\ -9 & 16 & 31 & -5 & -39 & 9 \\ -13 & 38 & 50 & -8 & -67 & 14 \\ -2 & -1 & 5 & -1 & -5 & 1 \\ 8 & -20 & -29 & 5 & 38 & -9 \end{bmatrix}$	$\begin{bmatrix} -2 & -3 & 7 & -1 & -7 & 3 \\ 2 & -9 & -9 & 1 & 13 & -3 \\ -9 & 16 & 31 & -5 & -39 & 9 \\ -13 & 38 & 50 & -8 & -67 & 14 \\ -2 & -1 & 5 & -1 & -5 & 1 \\ 8 & -20 & -29 & 5 & 38 & -9 \end{bmatrix}$
johnston-linear-matrix-algebra-EQ1163	399	1.000	$A=\left[\begin{array}{cc} \sqrt{3}-1 & 1 \\ -2 & \sqrt{3}+1 \end{array}\right]$	$A = \begin{bmatrix} \sqrt{3}-1 & 1 \\ -2 & \sqrt{3}+1 \end{bmatrix}$	$A = \begin{bmatrix} \sqrt{3}-1 & 1 \\ -2 & \sqrt{3}+1 \end{bmatrix}$
johnston-linear-matrix-algebra-EQ1168	401	1.000	$x_{\{n\}}=a_{\{k-1\}}\ x_{\{n-1\}}+a_{\{k-2\}}\ x_{\{n-2\}}+\cdots+a_{\{0\}}\ x_{\{n-k\}}$	$x_n = a_{k-1}x_{n-1} + a_{k-2}x_{n-2} + \cdots + a_0x_{n-k}$	$x_n = a_{k-1}x_{n-1} + a_{k-2}x_{n-2} + \cdots + a_0x_{n-k}$
johnston-linear-matrix-algebra-EQ1169	401	1.000	$C=\left[\begin{array}{cccccc} 0 & 1 & 0 & \cdots & 0 \\ 0 & 0 & 1 & \cdots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & 0 & \cdots & 1 \\ a_0 & a_1 & a_2 & \cdots & a_{k-1} \end{array}\right]$	$C = \begin{bmatrix} 0 & 1 & 0 & \cdots & 0 \\ 0 & 0 & 1 & \cdots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & 0 & \cdots & 1 \\ a_0 & a_1 & a_2 & \cdots & a_{k-1} \end{bmatrix}.$	$C = \begin{bmatrix} 0 & 1 & 0 & \cdots & 0 \\ 0 & 0 & 1 & \cdots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & 0 & \cdots & 1 \\ a_0 & a_1 & a_2 & \cdots & a_{k-1} \end{bmatrix}.$
johnston-linear-matrix-algebra-EQ1170	401	1.000	$C=\left[\begin{array}{ll} 0 & 1 \\ 1 & 1 \end{array}\right]$	$C = \begin{bmatrix} 0 & 1 \\ 1 & 1 \end{bmatrix}.$	$C = \begin{bmatrix} 0 & 1 \\ 1 & 1 \end{bmatrix}.$
johnston-linear-matrix-algebra-EQ1171	401	1.000	$C=\left[\begin{array}{ccc} 0 & 1 & 0 \\ 0 & 0 & 1 \\ -6 & -1 & 4 \end{array}\right]$	$C = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ -6 & -1 & 4 \end{bmatrix}.$	$C = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ -6 & -1 & 4 \end{bmatrix}.$
johnston-linear-matrix-algebra-EQ1172	401	1.000	$C=\left[\begin{array}{llll} 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 2 & 4 & 0 & 1 \end{array}\right]$	$C = \begin{bmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 2 & 4 & 0 & 1 \end{bmatrix}.$	$C = \begin{bmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 2 & 4 & 0 & 1 \end{bmatrix}.$
johnston-linear-matrix-algebra-EQ1173	401	1.000	$\mathbf{x}_{\{n\}}=C^{\{n\}}\ \mathbf{x}_{\{0\}}\ \text{for all } n \geq 0.$	$\mathbf{x}_n = C^n\mathbf{x}_0 \quad \text{for all } n \geq 0.$	$\mathbf{x}_n = C^n\mathbf{x}_0 \quad \text{for all } n \geq 0.$

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johnston-linear-matrix-align-402 EQ1174	402	1.000	$\begin{aligned} C \mathbf{x}_0 &= \left[\begin{array}{ccccc} 0 & 1 & 0 & \cdots & 0 \\ 0 & 0 & 1 & \cdots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & 0 & \cdots & 1 \\ a_0 & a_1 & a_2 & \cdots & a_{k-1} \end{array} \right] \begin{bmatrix} x_0 \\ x_1 \\ \vdots \\ x_{k-2} \\ x_{k-1} \end{bmatrix} \\ &= \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_{k-1} \\ a_0x_0 + a_1x_1 + \cdots + a_{k-1}x_{k-1} \end{bmatrix} = \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_{k-1} \\ x_k \end{bmatrix} = \mathbf{x}_1, \end{aligned}$	$C\mathbf{x}_0 = \begin{bmatrix} 0 & 1 & 0 & \cdots & 0 \\ 0 & 0 & 1 & \cdots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & 0 & \cdots & 1 \\ a_0 & a_1 & a_2 & \cdots & a_{k-1} \end{bmatrix} \begin{bmatrix} x_0 \\ x_1 \\ \vdots \\ x_{k-2} \\ x_{k-1} \end{bmatrix} \\ = \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_{k-1} \\ a_0x_0 + a_1x_1 + \cdots + a_{k-1}x_{k-1} \end{bmatrix} = \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_{k-1} \\ x_k \end{bmatrix} = \mathbf{x}_1,$	$C\mathbf{x}_0 = \begin{bmatrix} 0 & 1 & 0 & \cdots & 0 \\ 0 & 0 & 1 & \cdots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & 0 & \cdots & 1 \\ a_0 & a_1 & a_2 & \cdots & a_{k-1} \end{bmatrix} \begin{bmatrix} x_0 \\ x_1 \\ \vdots \\ x_{k-2} \\ x_{k-1} \end{bmatrix} \\ = \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_{k-1} \\ a_0x_0 + a_1x_1 + \cdots + a_{k-1}x_{k-1} \end{bmatrix} = \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_{k-1} \\ x_k \end{bmatrix} = \mathbf{x}_1,$
johnston-linear-matrix-align-402 EQ1175	402	1.000	$C^n \mathbf{x}_0 = C^{n-1} (C\mathbf{x}_0) = C^{n-1} \mathbf{x}_1 = C^{n-2} (C\mathbf{x}_1) = C^{n-2} \mathbf{x}_2 = \cdots = C\mathbf{x}_{n-1} = \mathbf{x}_n,$	$C^n \mathbf{x}_0 = C^{n-1} (C\mathbf{x}_0) = C^{n-1} \mathbf{x}_1 = C^{n-2} (C\mathbf{x}_1) = C^{n-2} \mathbf{x}_2 = \cdots = C\mathbf{x}_{n-1} = \mathbf{x}_n,$	$C^n \mathbf{x}_0 = C^{n-1} (C\mathbf{x}_0) = C^{n-1} \mathbf{x}_1 = C^{n-2} (C\mathbf{x}_1) = C^{n-2} \mathbf{x}_2 = \cdots = C\mathbf{x}_{n-1} = \mathbf{x}_n,$
johnston-linear-matrix-align-402 EQ1177	402	1.000	$\left[\begin{array}{ccc} 0 & 1 & 0 \\ 0 & 0 & 1 \\ -6 & -1 & 4 \end{array} \right]^n \left[\begin{array}{c} 3 \\ -2 \\ 8 \end{array} \right].$	$\left[\begin{array}{ccc} 0 & 1 & 0 \\ 0 & 0 & 1 \\ -6 & -1 & 4 \end{array} \right]^n \left[\begin{array}{c} 3 \\ -2 \\ 8 \end{array} \right].$	$\left[\begin{array}{ccc} 0 & 1 & 0 \\ 0 & 0 & 1 \\ -6 & -1 & 4 \end{array} \right]^n \left[\begin{array}{c} 3 \\ -2 \\ 8 \end{array} \right].$
johnston-linear-matrix-align-403 EQ1178	403	1.000	$D = \left[\begin{array}{ccc} 3 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & -1 \end{array} \right], \quad P = \left[\begin{array}{ccc} 1 & 1 & 1 \\ 3 & 2 & -1 \\ 9 & 4 & 1 \end{array} \right], \quad P^{-1} = \frac{1}{12} \left[\begin{array}{ccc} -6 & -3 & 3 \\ 12 & 8 & -4 \\ 6 & -5 & 1 \end{array} \right].$	$D = \left[\begin{array}{ccc} 3 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & -1 \end{array} \right], \quad P = \left[\begin{array}{ccc} 1 & 1 & 1 \\ 3 & 2 & -1 \\ 9 & 4 & 1 \end{array} \right], \quad P^{-1} = \frac{1}{12} \left[\begin{array}{ccc} -6 & -3 & 3 \\ 12 & 8 & -4 \\ 6 & -5 & 1 \end{array} \right].$	$D = \left[\begin{array}{ccc} 3 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & -1 \end{array} \right], \quad P = \left[\begin{array}{ccc} 1 & 1 & 1 \\ 3 & 2 & -1 \\ 9 & 4 & 1 \end{array} \right], \quad P^{-1} = \frac{1}{12} \left[\begin{array}{ccc} -6 & -3 & 3 \\ 12 & 8 & -4 \\ 6 & -5 & 1 \end{array} \right].$
johnston-linear-matrix-align-403 EQ1180	403	1.000	$\left[\begin{array}{ccccc} 0 & 1 & 0 & \cdots & 0 \\ 0 & 0 & 1 & \cdots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & 0 & \cdots & 1 \\ a_0 & a_1 & a_2 & \cdots & a_{k-1} \end{array} \right]$	$\left[\begin{array}{ccccc} 0 & 1 & 0 & \cdots & 0 \\ 0 & 0 & 1 & \cdots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & 0 & \cdots & 1 \\ a_0 & a_1 & a_2 & \cdots & a_{k-1} \end{array} \right]$	$\left[\begin{array}{ccccc} 0 & 1 & 0 & \cdots & 0 \\ 0 & 0 & 1 & \cdots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & 0 & \cdots & 1 \\ a_0 & a_1 & a_2 & \cdots & a_{k-1} \end{array} \right]$
johnston-linear-matrix-align-404 EQ1181	404	1.000	$x_n = a_{k-1}x_{n-1} + a_{k-2}x_{n-2} + \cdots + a_0x_{n-k}$	$x_n = a_{k-1}x_{n-1} + a_{k-2}x_{n-2} + \cdots + a_0x_{n-k}$	$x_n = a_{k-1}x_{n-1} + a_{k-2}x_{n-2} + \cdots + a_0x_{n-k}$
johnston-linear-matrix-align-404 EQ1182	404	1.000	$p(\lambda) = \lambda^k - a_{k-1}\lambda^{k-1} - a_{k-2}\lambda^{k-2} - \cdots - a_1\lambda - a_0.$	$p(\lambda) = \lambda^k - a_{k-1}\lambda^{k-1} - a_{k-2}\lambda^{k-2} - \cdots - a_1\lambda - a_0.$	$p(\lambda) = \lambda^k - a_{k-1}\lambda^{k-1} - a_{k-2}\lambda^{k-2} - \cdots - a_1\lambda - a_0.$
johnston-linear-matrix-align-404 EQ1183 (3.D.2)	404	1.000	$x_n = c_0\lambda_0^n + c_1\lambda_1^n + \cdots + c_{k-1}\lambda_{k-1}^n \quad \text{when } 0 \leq n < k.$	$x_n = c_0\lambda_0^n + c_1\lambda_1^n + \cdots + c_{k-1}\lambda_{k-1}^n \quad \text{when } 0 \leq n < k.$	$x_n = c_0\lambda_0^n + c_1\lambda_1^n + \cdots + c_{k-1}\lambda_{k-1}^n \quad \text{when } 0 \leq n < k.$

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johnston-linear-matrix-atqobvEQ1184	404	1.000	$x_{\{n\}}=a_{\{k-1\}}\ x_{\{n-1\}}+a_{\{k-2\}}\ x_{\{n-2\}}+\cdots+a_{\{0\}}\ x_{\{n-k\}}$	$x_n = a_{k-1}x_{n-1} + a_{k-2}x_{n-2} + \cdots + a_0x_{n-k}$	$x_n = a_{k-1}x_{n-1} + a_{k-2}x_{n-2} + \cdots + a_0x_{n-k}$
johnston-linear-matrix-atqobvEQ1186	405	1.000	$x_{\{n\}}=c_{\{0\}}\ \lambda_{\{0\}}^{\{n\}}+c_{\{1\}}\ \lambda_{\{1\}}^{\{n\}}+\cdots+c_{\{k-1\}}\ \lambda_{\{k-1\}}^{\{n\}}\quad\text{for}\quad 0\leq n<k,$	$x_n = c_0\lambda_0^n + c_1\lambda_1^n + \cdots + c_{k-1}\lambda_{k-1}^n\quad\text{for}\quad 0\leq n < k,$	$x_n = c_0\lambda_0^n + c_1\lambda_1^n + \cdots + c_{k-1}\lambda_{k-1}^n\quad\text{for}\quad 0\leq n < k,$
johnston-linear-matrix-atqobvEQ1189	405	1.000	$p(\lambda)=\lambda^3-8\lambda^2+17\lambda-10=(\lambda-5)(\lambda-2)(\lambda-1),$	$p(\lambda) = \lambda^3 - 8\lambda^2 + 17\lambda - 10 = (\lambda - 5)(\lambda - 2)(\lambda - 1),$	$p(\lambda) = \lambda^3 - 8\lambda^2 + 17\lambda - 10 = (\lambda - 5)(\lambda - 2)(\lambda - 1),$
johnston-linear-matrix-atqobvEQ1196	406	1.000	$p(\lambda)=\lambda^3-2\lambda^2+\lambda-2=(\lambda-2)(\lambda^2+1),$	$p(\lambda) = \lambda^3 - 2\lambda^2 + \lambda - 2 = (\lambda - 2)(\lambda^2 + 1),$	$p(\lambda) = \lambda^3 - 2\lambda^2 + \lambda - 2 = (\lambda - 2)(\lambda^2 + 1),$
johnston-linear-matrix-atqobvEQ1198	407	1.000	$x_{\{3\}}=2^{\{3\}}+(1-3\ i)\ i^{\{3\}}+(1+3\ i)(-i)^{\{3\}}=8+(-3-i)+(-3+i)=2\ .$	$x_3 = 2^3 + (1 - 3i)i^3 + (1 + 3i)(-i)^3 = 8 + (-3 - i) + (-3 + i) = 2.$	$x_3 = 2^3 + (1 - 3i)i^3 + (1 + 3i)(-i)^3 = 8 + (-3 - i) + (-3 + i) = 2.$
johnston-linear-matrix-atqobvEQ1200 (3.D.3)	407	1.000	$x_{\{n\}}=\sum_{j=0}^{\{m-1\}}\ q_{\{j\}}(n)\ \lambda_{\{j\}}^{\{n\}}\quad\text{when}\quad 0\leq n<k\ .$	$x_n = \sum_{j=0}^{m-1} q_j(n)\lambda_j^n\quad\text{when}\quad 0\leq n < k.$	$x_n = \sum_{j=0}^{m-1} q_j(n)\lambda_j^n\quad\text{when}\quad 0\leq n < k.$
johnston-linear-matrix-atqobvEQ1202	408	1.000	$p(\lambda)=\lambda^3-3\lambda-2=(\lambda-2)(\lambda+1)^2,$	$p(\lambda) = \lambda^3 - 3\lambda - 2 = (\lambda - 2)(\lambda + 1)^2,$	$p(\lambda) = \lambda^3 - 3\lambda - 2 = (\lambda - 2)(\lambda + 1)^2,$
johnston-linear-matrix-atqobvEQ1203	408	1.000	$c_{\{0\}}\ 2^{\{n\}}+\left(c_{\{1\}}+c_{\{2\}}\ n\right)(-1)^{\{n\}}=x_{\{n\}}\quad\text{when}\quad 0\leq n<3\ .$	$c_02^n + (c_1 + c_2n)(-1)^n = x_n\quad\text{when}\quad 0\leq n < 3.$	$c_02^n + (c_1 + c_2n)(-1)^n = x_n\quad\text{when}\quad 0\leq n < 3.$
johnston-linear-matrix-atqobvEQ1204	408	1.000	$\left[\begin{array}{ccc} 1 & 1 & 0 \\ 2 & -1 & -1 \\ 4 & 1 & 2 \end{array}\right]\left[\begin{array}{c} c_0 \\ c_1 \\ c_2 \end{array}\right]=\left[\begin{array}{c} 6 \\ 0 \\ 3 \end{array}\right],$	$\left[\begin{array}{ccc} 1 & 1 & 0 \\ 2 & -1 & -1 \\ 4 & 1 & 2 \end{array}\right]\left[\begin{array}{c} c_0 \\ c_1 \\ c_2 \end{array}\right]=\left[\begin{array}{c} 6 \\ 0 \\ 3 \end{array}\right],$	$\left[\begin{array}{ccc} 1 & 1 & 0 \\ 2 & -1 & -1 \\ 4 & 1 & 2 \end{array}\right]\left[\begin{array}{c} c_0 \\ c_1 \\ c_2 \end{array}\right]=\left[\begin{array}{c} 6 \\ 0 \\ 3 \end{array}\right],$
johnston-linear-matrix-atqobvEQ1205	408	1.000	$x_{\{n\}}=c_{\{0\}}\ 2^{\{n\}}+\left(c_{\{1\}}+c_{\{2\}}\ n\right)(-1)^{\{n\}}=2^{\{n\}}+(5-3\ n)(-1)^{\{n\}}\quad\text{for all}\quad n\geq 0\ .$	$x_n = c_02^n + (c_1 + c_2n)(-1)^n = 2^n + (5 - 3n)(-1)^n\quad\text{for all}\quad n\geq 0.$	$x_n = c_02^n + (c_1 + c_2n)(-1)^n = 2^n + (5 - 3n)(-1)^n\quad\text{for all}\quad n\geq 0.$
johnston-linear-matrix-atqobvEQ1206	408	1.000	$V=\left[V_{\{0\}}\left V_{\{1\}}\right \cdots\mid V_{\{m-1\}}\right],$	$V = [V_0 V_1 \cdots V_{m-1}],$	$V = [V_0 V_1 \cdots V_{m-1}],$
johnston-linear-matrix-atqobvEQ1207	408	1.000	$V_{\{j\}}=\left[\begin{array}{cccccc} 1 & 0 & 0 & \cdots & 0 \\ \lambda_{\{j\}} & \lambda_{\{j\}} & \lambda_{\{j\}} & \cdots & \lambda_{\{j\}} \\ \lambda_{\{j\}}^2 & 2\lambda_{\{j\}}^2 & 4\lambda_{\{j\}}^2 & \cdots & 2^{r_j-1}\lambda_{\{j\}}^2 \\ \lambda_{\{j\}}^3 & 3\lambda_{\{j\}}^3 & 9\lambda_{\{j\}}^3 & \cdots & 3^{r_j-1}\lambda_{\{j\}}^3 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ \lambda_{\{j\}}^{k-1} & (k-1)\lambda_{\{j\}}^{k-1} & (k-1)^2\lambda_{\{j\}}^{k-1} & \cdots & (k-1)^{r_j-1}\lambda_{\{j\}}^{k-1} \end{array}\right].$	$V_j = \left[\begin{array}{ccccc} 1 & 0 & 0 & \cdots & 0 \\ \lambda_j & \lambda_j & \lambda_j & \cdots & \lambda_j \\ \lambda_j^2 & 2\lambda_j^2 & 4\lambda_j^2 & \cdots & 2^{r_j-1}\lambda_j^2 \\ \lambda_j^3 & 3\lambda_j^3 & 9\lambda_j^3 & \cdots & 3^{r_j-1}\lambda_j^3 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ \lambda_j^{k-1} & (k-1)\lambda_j^{k-1} & (k-1)^2\lambda_j^{k-1} & \cdots & (k-1)^{r_j-1}\lambda_j^{k-1} \end{array}\right].$	$V_j = \left[\begin{array}{ccccc} 1 & 0 & 0 & \cdots & 0 \\ \lambda_j & \lambda_j & \lambda_j & \cdots & \lambda_j \\ \lambda_j^2 & 2\lambda_j^2 & 4\lambda_j^2 & \cdots & 2^{r_j-1}\lambda_j^2 \\ \lambda_j^3 & 3\lambda_j^3 & 9\lambda_j^3 & \cdots & 3^{r_j-1}\lambda_j^3 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ \lambda_j^{k-1} & (k-1)\lambda_j^{k-1} & (k-1)^2\lambda_j^{k-1} & \cdots & (k-1)^{r_j-1}\lambda_j^{k-1} \end{array}\right].$
johnston-linear-matrix-atqobvEQ1209	409	1.000	$p(\lambda)=d_{\{0\}}+d_{\{1\}}\ \lambda+\cdots+d_{\{k-1\}}\ \lambda^{\{k-1\}}$	$p(\lambda) = d_0 + d_1\lambda + \cdots + d_{k-1}\lambda^{k-1}$	$p(\lambda) = d_0 + d_1\lambda + \cdots + d_{k-1}\lambda^{k-1}$
johnston-linear-matrix-atqobvEQ1210	409	1.000	$x_{\{n\}}=q_{\{0\}}(n)\ \lambda_{\{0\}}^{\{n\}}+q_{\{1\}}(n)\ \lambda_{\{1\}}^{\{n\}}+\cdots+q_{\{m-1\}}(n)\ \lambda_{\{m-1\}}^{\{n\}}\quad\text{when}\quad 0\leq n<k\ .$	$x_n = q_0(n)\lambda_0^n + q_1(n)\lambda_1^n + \cdots + q_{m-1}(n)\lambda_{m-1}^n\quad\text{when}\quad 0\leq n < k.$	$x_n = q_0(n)\lambda_0^n + q_1(n)\lambda_1^n + \cdots + q_{m-1}(n)\lambda_{m-1}^n\quad\text{when}\quad 0\leq n < k.$
johnston-linear-matrix-atqobvEQ1213	410	1.000	$p(\lambda)=\lambda^4-6\lambda^3+12\lambda^2-10\lambda+3=(\lambda-1)^3(\lambda-3),$	$p(\lambda) = \lambda^4 - 6\lambda^3 + 12\lambda^2 - 10\lambda + 3 = (\lambda - 1)^3(\lambda - 3),$	$p(\lambda) = \lambda^4 - 6\lambda^3 + 12\lambda^2 - 10\lambda + 3 = (\lambda - 1)^3(\lambda - 3),$

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johnston-linear-matrix-atqelrE01214	410	■ 1.000	$\text{c}_{\{0\}}+\text{c}_{\{1\}}\,n+\text{c}_{\{2\}}\,n^2+\text{c}_{\{3\}}\,3^{\wedge}\{n\}=\text{x}_{\{n\}}\,\,\,\text{when }\,0\leq n<4\,.$	$c_0+c_1n+c_2n^2+c_33^n=x_n\quad\text{when}\quad 0\leq n<4.$	$c_0+c_1n+c_2n^2+c_33^n=x_n\quad\text{when}\quad 0\leq n<4.$
johnston-linear-matrix-atqelrE01217	410	■ 1.000	$F_{\{n\}}=\frac{1}{\sqrt{5}}\left(\phi^{\wedge}\{n\}-(1-\phi)^{\wedge}\{n\}\right),\,\,\,\text{where}\,\,\,\phi=\frac{1+\sqrt{5}}{2}\approx 1.6180\,.$	$F_n=\frac{1}{\sqrt{5}}(\phi^n-(1-\phi)^n),\quad\text{where}\quad\phi=\frac{1+\sqrt{5}}{2}\approx 1.6180.$	$F_n=\frac{1}{\sqrt{5}}(\phi^n-(1-\phi)^n),\quad\text{where}\quad\phi=\frac{1+\sqrt{5}}{2}\approx 1.6180.$
johnston-linear-matrix-atqelrE01220	411	■ 1.000	$\text{x}_{\{n\}}=\text{q}_{\{0\}}(n)\,\,\lambda_{\{0\}}^{\wedge}\{n\}+\text{q}_{\{1\}}(n)\,\,\lambda_{\{1\}}^{\wedge}\{n\}+\cdots+\text{q}_{\{m-1\}}(n)\,\,\lambda_{\{m-1\}}^{\wedge}\{n\},$	$x_n=q_0(n)\lambda_0^n+q_1(n)\lambda_1^n+\cdots+q_{m-1}(n)\lambda_{m-1}^n,$	$x_n=q_0(n)\lambda_0^n+q_1(n)\lambda_1^n+\cdots+q_{m-1}(n)\lambda_{m-1}^n,$
johnston-linear-matrix-atqelrE01223	412	■ 1.000	$\text{p}(\lambda)=\lambda_{\{3\}}-\lambda_{\{2\}}^2+4\,\,\lambda_{\{4\}}-(\lambda_{\{1\}}-1)\left(\lambda_{\{2\}}^2+4\right),$	$p(\lambda)=\lambda^3-\lambda^2+4\lambda-4=(\lambda-1)(\lambda^2+4),$	$p(\lambda)=\lambda^3-\lambda^2+4\lambda-4=(\lambda-1)(\lambda^2+4),$
johnston-linear-matrix-atqelrE01224	412	■ 1.000	$0,0,1,1,-3,-3,13,13,-51,-51,205,205,\,\,\ldots$	$0,0,1,1,-3,-3,13,13,-51,-51,205,205,\ldots$	$0,0,1,1,-3,-3,13,13,-51,-51,205,205,\ldots$
johnston-linear-matrix-atqelrE01225	412	■ 1.000	$\text{p}(\lambda)=\lambda_{\{3\}}^2-\lambda_{\{2\}}^2-9\,\,\lambda_{\{18\}}+(\lambda_{\{2\}}-2)(\lambda_{\{3\}}+3),$	$p(\lambda)=\lambda^3-2\lambda^2-9\lambda+18=(\lambda-2)(\lambda-3)(\lambda+3),$	$p(\lambda)=\lambda^3-2\lambda^2-9\lambda+18=(\lambda-2)(\lambda-3)(\lambda+3),$
johnston-linear-matrix-atqelrE01226	412	■ 1.000	$0,1,0,9,0,81,0,729,0,6561,0,59049,\,\,\ldots$	$0,1,0,9,0,81,0,729,0,6561,0,59049,\ldots$	$0,1,0,9,0,81,0,729,0,6561,0,59049,\ldots$
johnston-linear-matrix-atqelrE01231	414	■ 1.000	$\text{x}_{\{n\}}=\text{x}_{\{n-1\}}+6\,\,\text{x}_{\{n-2\}}\,\,\,\text{when }\,n\geq 2$	$x_n=x_{n-1}+6x_{n-2}\,\,\text{when}\,n\geq 2$	$x_n=x_{n-1}+6x_{n-2}\,\,\text{when}\,n\geq 2$
johnston-linear-matrix-atqelrE01232	414	■ 1.000	$\text{x}_{\{n\}}=6\,\,\text{x}_{\{n-1\}}-9\,\,\text{x}_{\{n-2\}}\,\,\,\text{when }\,n\geq 2$	$x_n=6x_{n-1}-9x_{n-2}\,\,\text{when}\,n\geq 2$	$x_n=6x_{n-1}-9x_{n-2}\,\,\text{when}\,n\geq 2$
johnston-linear-matrix-atqelrE01233	414	■ 1.000	$\text{x}_{\{n\}}=-\text{x}_{\{n-2\}}\,\,\,\text{when }\,n\geq 2$	$x_n=-x_{n-2}\,\,\text{when}\,n\geq 2$	$x_n=-x_{n-2}\,\,\text{when}\,n\geq 2$
johnston-linear-matrix-atqelrE01234	414	■ 1.000	$\text{x}_{\{n\}}=2\,\,\text{x}_{\{n-1\}}-\text{x}_{\{n-2\}}\,\,\,\text{when }\,n\geq 2$	$x_n=2x_{n-1}-x_{n-2}\,\,\text{when}\,n\geq 2$	$x_n=2x_{n-1}-x_{n-2}\,\,\text{when}\,n\geq 2$
johnston-linear-matrix-atqelrE01235	414	■ 1.000	$\text{x}_{\{n\}}=6\,\,\text{x}_{\{n-1\}}-11\,\,\text{x}_{\{n-2\}}+6\,\,\text{x}_{\{n-3\}}\,\,\,\text{when }\,n\geq 3$	$x_n=6x_{n-1}-11x_{n-2}+6x_{n-3}\,\,\text{when}\,n\geq 3$	$x_n=6x_{n-1}-11x_{n-2}+6x_{n-3}\,\,\text{when}\,n\geq 3$
johnston-linear-matrix-atqelrE01236	414	■ 1.000	$\text{x}_{\{n\}}=6\,\,\text{x}_{\{n-1\}}-12\,\,\text{x}_{\{n-2\}}+8\,\,\text{x}_{\{n-3\}}\,\,\,\text{when }\,n\geq 3$	$x_n=6x_{n-1}-12x_{n-2}+8x_{n-3}\,\,\text{when}\,n\geq 3$	$x_n=6x_{n-1}-12x_{n-2}+8x_{n-3}\,\,\text{when}\,n\geq 3$
johnston-linear-matrix-atqelrE01237	414	■ 1.000	$\text{x}_{\{n\}}=4\,\,\text{x}_{\{n-1\}}-5\,\,\text{x}_{\{n-2\}}+2\,\,\text{x}_{\{n-3\}}\,\,\,\text{when }\,n\geq 3$	$x_n=4x_{n-1}-5x_{n-2}+2x_{n-3}\,\,\text{when}\,n\geq 3$	$x_n=4x_{n-1}-5x_{n-2}+2x_{n-3}\,\,\text{when}\,n\geq 3$
johnston-linear-matrix-atqelrE01239	417	■ 1.000	$(3+7\,\,\text{i})+(2-4\,\,\text{i})=(3+2)+(7-4)\,\,\text{i}=5+3\,\,\text{i}\,.$	$(3+7i)+(2-4i)=(3+2)+(7-4)i=5+3i.$	$(3+7i)+(2-4i)=(3+2)+(7-4)i=5+3i.$
johnston-linear-matrix-atqelrE01240	417	■ 1.000	$(a+b\,\,\text{i})(c+d\,\,\text{i})=a\,\,\text{c}+b\,\,\text{c}\,\,\text{i}+a\,\,\text{d}\,\,\text{i}+b\,\,\text{d}\,\,\text{i}^2=(a\,\,\text{c}-b\,\,\text{d})+(a\,\,\text{d}+b\,\,\text{c})\,\,\text{i}\,.$	$(a+bi)(c+di)=ac+bci+adi+bdi^2=(ac-bd)+(ad+bc)i.$	$(a+bi)(c+di)=ac+bci+adi+bdi^2=(ac-bd)+(ad+bc)i.$
johnston-linear-matrix-atqelrE01241	417	■ 1.000	$(3+7\,\,\text{i})(4+2\,\,\text{i})=(12-14)+(6+28)\,\,\text{i}=-2+34\,\,\text{i}\,.$	$(3+7i)(4+2i)=(12-14)+(6+28)i=-2+34i.$	$(3+7i)(4+2i)=(12-14)+(6+28)i=-2+34i.$
johnston-linear-matrix-atqelrE01242	417	■ 1.000	$\begin{aligned} (1+2\,\,\text{i})^2-2(1+2\,\,\text{i})+5 &= (1+2\,\,\text{i})(1+2\,\,\text{i})-(2+4\,\,\text{i})+5 \\ &= (-3+4\,\,\text{i})-(2+4\,\,\text{i})+5 \\ &= (-5+5)+(4-4)\,\,\text{i} \\ &= 0\,.\end{aligned}$	$\begin{aligned} (1+2i)^2-2(1+2i)+5 &= (1+2i)(1+2i)-(2+4i)+5 \\ &= (-3+4i)-(2+4i)+5 \\ &= (-5+5)+(4-4)i \\ &= 0.\end{aligned}$	$\begin{aligned} (1+2i)^2-2(1+2i)+5 &= (1+2i)(1+2i)-(2+4i)+5 \\ &= (-3+4i)-(2+4i)+5 \\ &= (-5+5)+(4-4)i \\ &= 0.\end{aligned}$
johnston-linear-matrix-atqelrE01243	418	■ 1.000	$ a =\sqrt{a^{\wedge}\{2\}}\,.$	$ a =\sqrt{a^2}.$	$ a =\sqrt{a^2}.$
johnston-linear-matrix-atqelrE01245	418	■ 1.000	$ a+b\,\,\text{i} \,\,\stackrel{\text{def}}{=}\,\,\sqrt{a^{\wedge}\{2\}+b^{\wedge}\{2\}},$	$ a+bi \stackrel{\text{def}}{=}\sqrt{a^2+b^2},$	$ a+bi \stackrel{\text{def}}{=}\sqrt{a^2+b^2},$
johnston-linear-matrix-atqelrE01246	418	■ 1.000	$\overline{\{3+4\,\,\text{i}\}}=3-4\,\,\text{i},\,\,\text{quad}\,\,\overline{\{5-2\,\,\text{i}\}}=5+2\,\,\text{i},\,\,\text{quad}\,\,\overline{\{3\,\,\text{i}\}}=-3\,\,\text{i},\,\,\text{quad}\,\,\text{and}\,\,\text{quad}\,\,\overline{\{7\}}=7\,.$	$\overline{3+4i}=3-4i,\quad\overline{5-2i}=5+2i,\quad\overline{3i}=-3i,\quad\text{and}\quad\overline{7}=7.$	$\overline{3+4i}=3-4i,\quad\overline{5-2i}=5+2i,\quad\overline{3i}=-3i,\quad\text{and}\quad\overline{7}=7.$
johnston-linear-matrix-atqelrE01247	419	■ 1.000	$\begin{aligned} z\,\overline{\{z\}} &= (a+b\,\,\text{i})\,\,\overline{\{a+b\,\,\text{i}\}} \\ &= (a+b\,\,\text{i})(a-b\,\,\text{i}) \\ &= a^{\wedge}\{2\}+b^{\wedge}\{2\}= a+b\,\,\text{i} ^{\wedge}\{2\}= z ^{\wedge}\{2\}\,.\end{aligned}$	$\begin{aligned} z\bar{z} &= (a+bi)\overline{(a+bi)}=(a+bi)(a-bi) \\ &= a^2+b^2= a+bi ^2= z ^2.\end{aligned}$	$\begin{aligned} z\bar{z} &= (a+bi)\overline{(a+bi)}=(a+bi)(a-bi) \\ &= a^2+b^2= a+bi ^2= z ^2.\end{aligned}$

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johnston-linear-matrix-algebra-EQ1248	419	1.000	$\begin{aligned} \frac{a+b\,i}{c+d\,i} &= \left(\frac{a+b\,i}{c+d\,i}\right) \left(\frac{c-di}{c-di}\right) \\ &= \frac{(ac+bd) + (bc-ad)\,i}{c^2+d^2} = \left(\frac{ac+bd}{c^2+d^2}\right) + \left(\frac{bc-ad}{c^2+d^2}\right) i. \end{aligned}$	$\frac{a+bi}{c+di} = \left(\frac{a+bi}{c+di}\right) \left(\frac{c-di}{c-di}\right) = \frac{(ac+bd) + (bc-ad)i}{c^2+d^2} = \left(\frac{ac+bd}{c^2+d^2}\right) + \left(\frac{bc-ad}{c^2+d^2}\right) i.$	$\frac{a+bi}{c+di} = \left(\frac{a+bi}{c+di}\right) \left(\frac{c-di}{c-di}\right) = \frac{(ac+bd) + (bc-ad)i}{c^2+d^2} = \left(\frac{ac+bd}{c^2+d^2}\right) + \left(\frac{bc-ad}{c^2+d^2}\right) i.$
johnston-linear-matrix-algebra-EQ1249	419	1.000	$\operatorname{Re}(z) = \frac{z+\bar{z}}{2} \quad \text{and} \quad \operatorname{Im}(z) = \frac{z-\bar{z}}{2i},$	$\operatorname{Re}(z) = \frac{z+\bar{z}}{2} \quad \text{and} \quad \operatorname{Im}(z) = \frac{z-\bar{z}}{2i},$	$\operatorname{Re}(z) = \frac{z+\bar{z}}{2} \quad \text{and} \quad \operatorname{Im}(z) = \frac{z-\bar{z}}{2i},$
johnston-linear-matrix-algebra-EQ1255	421	1.000	$r^{1/n} e^{i\theta/n}, \quad r^{1/n} e^{i(\theta+2\pi)/n}, \quad r^{1/n} e^{i(\theta+4\pi)/n}, \quad \dots, \quad r^{1/n} e^{i(\theta+2(n-1)\pi)/n}.$	$r^{1/n} e^{i\theta/n}, \quad r^{1/n} e^{i(\theta+2\pi)/n}, \quad r^{1/n} e^{i(\theta+4\pi)/n}, \quad \dots, \quad r^{1/n} e^{i(\theta+2(n-1)\pi)/n}.$	$r^{1/n} e^{i\theta/n}, \quad r^{1/n} e^{i(\theta+2\pi)/n}, \quad r^{1/n} e^{i(\theta+4\pi)/n}, \quad \dots, \quad r^{1/n} e^{i(\theta+2(n-1)\pi)/n}.$
johnston-linear-matrix-algebra-EQ1257	421	1.000	$e^{\pi i/6} = (\sqrt{3}+i)/2, \quad e^{5\pi i/6} = (-\sqrt{3}+i)/2, \quad \text{and} \quad e^{9\pi i/6} = -i.$	$e^{\pi i/6} = (\sqrt{3}+i)/2, \quad e^{5\pi i/6} = (-\sqrt{3}+i)/2, \quad \text{and} \quad e^{9\pi i/6} = -i.$	$e^{\pi i/6} = (\sqrt{3}+i)/2, \quad e^{5\pi i/6} = (-\sqrt{3}+i)/2, \quad \text{and} \quad e^{9\pi i/6} = -i.$
johnston-linear-matrix-algebra-EQ1258	422	1.000	$p(x) = a_n x^n + a_{n-1} x^{n-1} + \cdots + a_2 x^2 + a_1 x + a_0,$	$p(x) = a_n x^n + a_{n-1} x^{n-1} + \cdots + a_2 x^2 + a_1 x + a_0,$	$p(x) = a_n x^n + a_{n-1} x^{n-1} + \cdots + a_2 x^2 + a_1 x + a_0,$
johnston-linear-matrix-algebra-EQ1259	422	1.000	$p(x) = ax^2 + bx + c,$	$p(x) = ax^2 + bx + c,$	$p(x) = ax^2 + bx + c,$
johnston-linear-matrix-algebra-EQ1260	423	1.000	$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}.$	$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}.$	$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}.$
johnston-linear-matrix-algebra-EQ1261	423	1.000	$x = \frac{6 \pm \sqrt{36 - 32}}{2} = \frac{6 \pm 2}{2} = 3 \pm 1.$	$x = \frac{6 \pm \sqrt{36 - 32}}{2} = \frac{6 \pm 2}{2} = 3 \pm 1.$	$x = \frac{6 \pm \sqrt{36 - 32}}{2} = \frac{6 \pm 2}{2} = 3 \pm 1.$
johnston-linear-matrix-algebra-EQ1262	423	1.000	$x = \frac{-8 \pm \sqrt{64 - 64}}{4} = \frac{-8 \pm 0}{4} = -2.$	$x = \frac{-8 \pm \sqrt{64 - 64}}{4} = \frac{-8 \pm 0}{4} = -2.$	$x = \frac{-8 \pm \sqrt{64 - 64}}{4} = \frac{-8 \pm 0}{4} = -2.$
johnston-linear-matrix-algebra-EQ1263	423	1.000	$x = \frac{-2 \pm \sqrt{4 - 12}}{2} = -1 \pm \sqrt{-2}.$	$x = \frac{-2 \pm \sqrt{4 - 12}}{2} = -1 \pm \sqrt{-2}.$	$x = \frac{-2 \pm \sqrt{4 - 12}}{2} = -1 \pm \sqrt{-2}.$
johnston-linear-matrix-algebra-EQ1264	424	1.000	$p(x) = a_n x^n + a_{n-1} x^{n-1} + \cdots + a_2 x^2 + a_1 x + a_0,$	$p(x) = a_n x^n + a_{n-1} x^{n-1} + \cdots + a_2 x^2 + a_1 x + a_0,$	$p(x) = a_n x^n + a_{n-1} x^{n-1} + \cdots + a_2 x^2 + a_1 x + a_0,$
johnston-linear-matrix-algebra-EQ1267 (A.2.1)	425	1.000	$p(x) = (x-2)q(x),$	$p(x) = (x-2)q(x),$	$p(x) = (x-2)q(x),$
johnston-linear-matrix-algebra-EQ1271	426	1.000	$\begin{array}{r} 3x^2 + 6x - 2 \\ x - 2 \overline{) 3x^3 + 0x^2 - 14x + 4} \\ \underline{3x^3 - 6x^2} \\ 6x^2 - 14x + 4 \\ \underline{6x^2 - 12x} \\ -2x + 4 \\ \underline{-2x + 4} \\ 0 \end{array}$	$\begin{array}{r} 3x^2 + 6x - 2 \\ x - 2 \overline{) 3x^3 + 0x^2 - 14x + 4} \\ \underline{3x^3 - 6x^2} \\ 6x^2 - 14x + 4 \\ \underline{6x^2 - 12x} \\ -2x + 4 \\ \underline{-2x + 4} \\ 0 \end{array}$	$\begin{array}{r} 3x^2 + 6x - 2 \\ x - 2 \overline{) 3x^3 + 0x^2 - 14x + 4} \\ \underline{3x^3 - 6x^2} \\ 6x^2 - 14x + 4 \\ \underline{6x^2 - 12x} \\ -2x + 4 \\ \underline{-2x + 4} \\ 0 \end{array}$
johnston-linear-matrix-algebra-EQ1273 (3)	426	1.000	$x = \frac{-6 \pm \sqrt{36 + 24}}{6} = -1 \pm \sqrt{\frac{5}{3}}.$	$x = \frac{-6 \pm \sqrt{36 + 24}}{6} = -1 \pm \sqrt{\frac{5}{3}}.$	$x = \frac{-6 \pm \sqrt{36 + 24}}{6} = -1 \pm \sqrt{\frac{5}{3}}.$
johnston-linear-matrix-algebra-EQ1277	427	1.000	$p(x) = 3x^4 + 4x^3 - 14x^2 - 11x - 2 = 3(x-2)(x+1/3)(x^2+3x+1),$	$p(x) = 3x^4 + 4x^3 - 14x^2 - 11x - 2 = 3(x-2)(x+1/3)(x^2+3x+1),$	$p(x) = 3x^4 + 4x^3 - 14x^2 - 11x - 2 = 3(x-2)(x+1/3)(x^2+3x+1),$

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johnston-linear-matrix-atqdw7 EQ1278 (1)	427	■ 1.000	$x=\frac{-3\pm\sqrt{9-4}}{2}=\frac{1}{2}(-3\pm\sqrt{5})$.	$x = \frac{-3 \pm \sqrt{9 - 4}}{2} = \frac{1}{2}(-3 \pm \sqrt{5}).$	$x = \frac{-3 \pm \sqrt{9 - 4}}{2} = \frac{1}{2}(-3 \pm \sqrt{5}).$
johnston-linear-matrix-atqdw7 EQ1280	427	■ 1.000	$q=\frac{-b}{3a}$, $\quad r=q^3+\frac{bc-3ad}{6a^2}$, $\quad \text{and} \quad s=\frac{c}{3a}$	$q = \frac{-b}{3a}, \quad r = q^3 + \frac{bc - 3ad}{6a^2}, \quad \text{and} \quad s = \frac{c}{3a}$	$q = \frac{-b}{3a}, \quad r = q^3 + \frac{bc - 3ad}{6a^2}, \quad \text{and} \quad s = \frac{c}{3a}$
johnston-linear-matrix-atqdw7 EQ1281	427	■ 1.000	$x=q+\sqrt[3]{r+\sqrt{r^2+\left(s-q^2\right)^3}}+\sqrt[3]{r-\sqrt{r^2+\left(s-q^2\right)^3}}$.	$x = q + \sqrt[3]{r + \sqrt{r^2 + (s - q^2)^3}} + \sqrt[3]{r - \sqrt{r^2 + (s - q^2)^3}}.$	$x = q + \sqrt[3]{r + \sqrt{r^2 + (s - q^2)^3}} + \sqrt[3]{r - \sqrt{r^2 + (s - q^2)^3}}.$
johnston-linear-matrix-atqdw7 EQ1282	428	■ 1.000	$p(x)=a\left(x-r_1\right)\left(x-r_2\right)\cdots\left(x-r_n\right)$,	$p(x) = a(x - r_1)(x - r_2)\cdots(x - r_n),$	$p(x) = a(x - r_1)(x - r_2)\cdots(x - r_n),$
johnston-linear-matrix-atqdw7 EQ1283	429	■ 1.000	$m^2=(2\,k)^2=4\,k^2=2\left(2\,k^2\right)$,	$m^2 = (2k)^2 = 4k^2 = 2\left(2k^2\right),$	$m^2 = (2k)^2 = 4k^2 = 2(2k^2),$
johnston-linear-matrix-atqdw7 EQ1284	429	■ 1.000	$0\leq (x-y)^2=x^2-2\,x\,y+y^2$.	$0 \leq (x - y)^2 = x^2 - 2xy + y^2.$	$0 \leq (x - y)^2 = x^2 - 2xy + y^2.$
johnston-linear-matrix-atqdw7 EQ1285	430	■ 1.000	$x=\frac{-b\pm\sqrt{b^2-4\,a\,c}}{2\,a}$.	$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}.$	$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}.$
johnston-linear-matrix-atqdw7 EQ1288	431	■ 1.000	$m^2=(2\,k+1)^2=4\,k^2+4\,k+1=2\left(2\,k^2+2\,k\right)+1$,	$m^2 = (2k + 1)^2 = 4k^2 + 4k + 1 = 2\left(2k^2 + 2k\right) + 1,$	$m^2 = (2k + 1)^2 = 4k^2 + 4k + 1 = 2(2k^2 + 2k) + 1,$
johnston-linear-matrix-atqdw7 EQ1289 (A.3.1)	433	■ 1.000	$x=\frac{-b\pm\sqrt{b^2-4\,a\,c}}{2\,a}$.	$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}.$	$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}.$
johnston-linear-matrix-atqdw7 EQ1294	437	■ 1.000	$A_{\{i,\,1\}}B_{\{1,\,j\}}+A_{\{i,\,2\}}B_{\{2,\,j\}}+\cdots+A_{\{i,\,n\}}B_{\{n,\,j\}},$	$A_{i,1}B_{1,j} + A_{i,2}B_{2,j} + \cdots + A_{i,n}B_{n,j},$	$A_{i,1}B_{1,j} + A_{i,2}B_{2,j} + \cdots + A_{i,n}B_{n,j},$
johnston-linear-matrix-atqdw7 EQ1296 (B.1.1)	438	■ 1.000	$\sum_{a=1}^n\sum_b\left[A_{\{i,\,a\}}\right]_{\{k,\,b\}}\left[B_{\{a,\,j\}}\right]_{\{b,\,\ell\}}$	$\sum_{a=1}^n\sum_b[A_{i,a}]_{k,b}[B_{a,j}]_{b,\ell}$	$\sum_{a=1}^n\sum_b[A_{i,a}]_{k,b}[B_{a,j}]_{b,\ell}$
johnston-linear-matrix-atqdw7 EQ1297	438	■ 1.000	$\left[\sum_{a=1}^nA_{\{i,\,a\}}B_{\{a,\,j\}}\right]_{\{k,\,\ell\}}=\sum_{a=1}^n\left[A_{i,a}B_{a,j}\right]_{k,\ell}=\sum_{a=1}^n\left(\sum_b\left[A_{i,a}\right]_{k,b}\left[B_{a,j}\right]_{b,\ell}\right),$	$\left[\sum_{a=1}^nA_{i,a}B_{a,j}\right]_{k,\ell}=\sum_{a=1}^n[A_{i,a}B_{a,j}]_{k,\ell}=\sum_{a=1}^n\left(\sum_b[A_{i,a}]_{k,b}[B_{a,j}]_{b,\ell}\right),$	$\left[\sum_{a=1}^nA_{i,a}B_{a,j}\right]_{k,\ell}=\sum_{a=1}^n[A_{i,a}B_{a,j}]_{k,\ell}=\sum_{a=1}^n\left(\sum_b[A_{i,a}]_{k,b}[B_{a,j}]_{b,\ell}\right),$
johnston-linear-matrix-atqdw7 EQ1301	440	■ 1.000	$E=\left[\mathbf{e}_1\right]\cdots\left[\mathbf{e}_j+c\mathbf{e}_i\right]\cdots\left[\mathbf{e}_m\right],$	$E = [\mathbf{e}_1 \cdots \mathbf{e}_j + c\mathbf{e}_i \cdots \mathbf{e}_m],$	$E = [\mathbf{e}_1 \cdots \mathbf{e}_j + c\mathbf{e}_i \cdots \mathbf{e}_m],$
johnston-linear-matrix-atqdw7 EQ1303	440	■ 1.000	$\left[\mathbf{e}_1\right]\cdots\left[\mathbf{e}_j\right]\cdots\left[\mathbf{e}_i\right]\cdots\left[\mathbf{e}_m\right],$	$[\mathbf{e}_1 \cdots \mathbf{e}_j \cdots \mathbf{e}_i \cdots \mathbf{e}_m],$	$[\mathbf{e}_1 \cdots \mathbf{e}_j \cdots \mathbf{e}_i \cdots \mathbf{e}_m],$
johnston-linear-matrix-atqdw7 EQ1306	441	■ 1.000	$\operatorname{sgn}(\sigma)\,a_{\sigma(1),1}a_{\sigma(2),2}\cdots a_{\sigma(n),n}=a_{1,1}a_{2,2}\cdots a_{n,n}=1,$	$\operatorname{sgn}(\sigma)a_{\sigma(1),1}a_{\sigma(2),2}\cdots a_{\sigma(n),n} = a_{1,1}a_{2,2}\cdots a_{n,n} = 1,$	$\operatorname{sgn}(\sigma)a_{\sigma(1),1}a_{\sigma(2),2}\cdots a_{\sigma(n),n} = a_{1,1}a_{2,2}\cdots a_{n,n} = 1,$
johnston-linear-matrix-atqdw7 EQ1308	442	■ 1.000	$(A-B) \mathbf{w} \leq\mu \mathbf{w} - \mu \mathbf{w} =\mathbf{0},$	$(A - B) \mathbf{w} \leq \mu \mathbf{w} - \mu \mathbf{w} = \mathbf{0},$	$(A - B) \mathbf{w} \leq \mu \mathbf{w} - \mu \mathbf{w} = \mathbf{0},$
johnston-linear-matrix-atqdw7 EQ1311 (B.5.1)	442	■ 1.000	$\frac{\partial}{\partial\lambda_i}\det(A-\Lambda)=-c_{\{i,\,i\}}=-\operatorname{det}\left(A_{\{i\}}-\Lambda_{\{i\}}\right)$.	$\frac{\partial}{\partial\lambda_i}\det(A - \Lambda) = -c_{i,i} = -\det(A_i - \Lambda_i).$	$\frac{\partial}{\partial\lambda_i}\det(A - \Lambda) = -c_{i,i} = -\det(A_i - \Lambda_i).$
johnston-linear-matrix-atqdw7 EQ1314	443	■ 1.000	$p(r)=p^{\prime}(r)=p^{\prime\prime}(r)=\cdots=p^{(m-1)}(r)=\cdots=p^{\{m-1\}}(r)=\theta$.	$p(r) = p'(r) = p''(r) = \cdots = p^{(m-1)}(r) = 0.$	$p(r) = p'(r) = p''(r) = \cdots = p^{(m-1)}(r) = 0.$
johnston-linear-matrix-atqdw7 EQ1316	443	■ 1.000	$p^{\prime}(x)=m(x-r)^{m-1}q(x)+(x-r)^mq^{\prime}(x)$	$p'(x) = m(x - r)^{m-1}q(x) + (x - r)^mq'(x) = (x - r)^{m-1}(mq(x) + (x - r)q'(x)),$	$p'(x) = m(x - r)^{m-1}q(x) + (x - r)^mq'(x) = (x - r)^{m-1}(mq(x) + (x - r)q'(x)),$

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johnston-linear-matrix-attrib EQ1354	449	1.000	$\begin{aligned} [c(A+B)]_{i,j} &= c\left(a_{i,j} + b_{i,j}\right) \\ &= ca_{i,j} + cb_{i,j} \\ &= [cA + cB]_{i,j}. \end{aligned}$	$\begin{aligned} [c(A+B)]_{i,j} &= c(a_{i,j} + b_{i,j}) \\ &= ca_{i,j} + cb_{i,j} \\ &= [cA + cB]_{i,j}. \end{aligned}$	
johnston-linear-matrix-attrib EQ1355	450	1.000	$\begin{aligned} [(c+d)A]_{i,j} &= (c+d)a_{i,j} \\ &= ca_{i,j} + da_{i,j} \\ &= [cA + dA]_{i,j}. \end{aligned}$	$\begin{aligned} [(c+d)A]_{i,j} &= (c+d)a_{i,j} \\ &= ca_{i,j} + da_{i,j} \\ &= [cA + dA]_{i,j}. \end{aligned}$	
johnston-linear-matrix-attrib EQ1356	450	1.000	$[c(dA)]_{i,j} = c\left(da_{i,j}\right) = (cd)a_{i,j} = [(cd)A]_{i,j}.$	$[c(dA)]_{i,j} = c\left(da_{i,j}\right) = (cd)a_{i,j} = [(cd)A]_{i,j}.$	
johnston-linear-matrix-attrib EQ1358	450	1.000	$\begin{aligned} [A(B+C)]_{i,j} &= \sum_{k=1}^n a_{i,k} (b_{k,j} + c_{k,j}) \\ &= \sum_{k=1}^n a_{i,k} b_{k,j} + \sum_{k=1}^n a_{i,k} c_{k,j} \\ &= [AB + AC]_{i,j}. \end{aligned}$	$\begin{aligned} [A(B+C)]_{i,j} &= \sum_{k=1}^n a_{i,k} (b_{k,j} + c_{k,j}) \\ &= \sum_{k=1}^n a_{i,k} b_{k,j} + \sum_{k=1}^n a_{i,k} c_{k,j} \\ &= [AB + AC]_{i,j}. \end{aligned}$	
johnston-linear-matrix-attrib EQ1360	450	1.000	$\left(A^k\right)^\ell = (A A \cdots A)(A A \cdots A) \cdots (A A \cdots A),$	$\left(A^k\right)^\ell = (A A \cdots A)(A A \cdots A) \cdots (A A \cdots A),$	
johnston-linear-matrix-attrib EQ1362	450	1.000	$A B = \left[A \mid \mathbf{b}_1 \mid \mathbf{b}_2 \mid \cdots \mid \mathbf{b}_p\right].$	$AB = [A\mathbf{b}_1 \mid A\mathbf{b}_2 \mid \cdots \mid A\mathbf{b}_p].$	
johnston-linear-matrix-attrib EQ1363	450	1.000	$A \, \mathbf{b}_j = b_{1,j} \, \mathbf{a}_1 + b_{2,j} \, \mathbf{a}_2 + \cdots + b_{n,j} \, \mathbf{a}_n,$	$A\mathbf{b}_j = b_{1,j}\mathbf{a}_1 + b_{2,j}\mathbf{a}_2 + \cdots + b_{n,j}\mathbf{a}_n,$	
johnston-linear-matrix-attrib EQ1364	450	1.000	$[T] = \left[\begin{array}{cc} 3 & 1 \\ -1 & 2 \end{array}\right].$	$[T] = \left[\begin{array}{cc} 3 & 1 \\ -1 & 2 \end{array}\right].$	
johnston-linear-matrix-attrib EQ1365	451	1.000	$[T] = \left[\begin{array}{cc} -1 & 2 \\ 0 & 3 \\ 1 & 0 \end{array}\right]$	$[T] = \left[\begin{array}{cc} -1 & 2 \\ 0 & 3 \\ 1 & 0 \end{array}\right]$	
johnston-linear-matrix-attrib EQ1366	451	1.000	$[T] = \left[\begin{array}{ccc} 1 & -1 & 0 \\ 2 & -1 & -1 \\ 3 & -1 & -1 \end{array}\right].$	$[T] = \left[\begin{array}{ccc} 1 & -1 & 0 \\ 2 & -1 & -1 \\ 3 & -1 & -1 \end{array}\right].$	
johnston-linear-matrix-attrib EQ1367	451	1.000	$[S] = \left[\begin{array}{ll} 0 & 2 \\ 1 & 1 \end{array}\right], \quad [T] = \left[\begin{array}{cc} 1 & 2 \\ 3 & -1 \end{array}\right],$	$[S] = \left[\begin{array}{cc} 0 & 2 \\ 1 & 1 \end{array}\right], \quad [T] = \left[\begin{array}{cc} 1 & 2 \\ 3 & -1 \end{array}\right],$	
johnston-linear-matrix-attrib EQ1368	451	1.000	$[S \circ T] = [S][T] = \left[\begin{array}{cc} 6 & -2 \\ 4 & 1 \end{array}\right].$	$[S \circ T] = [S][T] = \left[\begin{array}{cc} 6 & -2 \\ 4 & 1 \end{array}\right].$	
johnston-linear-matrix-attrib EQ1371	451	1.000	$\left[P_{\mathbf{u}}\right] = \frac{1}{2} \left[\begin{array}{cc} 1 & 1 \\ 1 & 1 \end{array}\right], \quad \left[R^{-\pi/4}\right] = \frac{1}{\sqrt{2}} \left[\begin{array}{cc} 1 & 1 \\ -1 & 1 \end{array}\right].$	$[P_{\mathbf{u}}] = \frac{1}{2} \left[\begin{array}{cc} 1 & 1 \\ 1 & 1 \end{array}\right], \quad \left[R^{-\pi/4}\right] = \frac{1}{\sqrt{2}} \left[\begin{array}{cc} 1 & 1 \\ -1 & 1 \end{array}\right].$	
johnston-linear-matrix-attrib EQ1375	452	1.000	$A^T A = \left[\begin{array}{cc} \cos(\theta) & \sin(\theta) \\ -\sin(\theta) & \cos(\theta) \end{array}\right] \left[\begin{array}{cc} \cos(\theta) & -\sin(\theta) \\ \sin(\theta) & \cos(\theta) \end{array}\right].$	$A^T A = \left[\begin{array}{cc} \cos(\theta) & \sin(\theta) \\ -\sin(\theta) & \cos(\theta) \end{array}\right] \left[\begin{array}{cc} \cos(\theta) & -\sin(\theta) \\ \sin(\theta) & \cos(\theta) \end{array}\right].$	

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johnston-linear-matrix-atlas- EQ1379	452	■ 1.000	$\left[R_{\mathbf{u}}^{\theta}\right]=\left[\begin{array}{ccc} 1 & 0 & 0 \\ 0 & \cos (\theta) & -\sin (\theta) \\ 0 & \sin (\theta) & \cos (\theta) \end{array}\right] \cdot$ $\left[R_{\mathbf{y} \ z}^{\theta}\right]=\left[R_{\mathbf{y} \ z}^{\theta}\right] \cdot$	$\left[R_{\mathbf{u}}^{\theta}\right]=\left[\begin{array}{ccc} 1 & 0 & 0 \\ 0 & \cos (\theta) & -\sin (\theta) \\ 0 & \sin (\theta) & \cos (\theta) \end{array}\right] =\left[R_{\mathbf{y} \ z}^{\theta}\right] \cdot$	$\left[R_{\mathbf{u}}^{\theta}\right]=\left[\begin{array}{ccc} 1 & 0 & 0 \\ 0 & \cos (\theta) & -\sin (\theta) \\ 0 & \sin (\theta) & \cos (\theta) \end{array}\right] =\left[R_{\mathbf{y} \ z}^{\theta}\right] \cdot$
johnston-linear-matrix-atlas- EQ1381	452	■ 1.000	$\begin{aligned} & R_{\mathbf{u}}^{\pi / 4}(3,2,1)= \\ & (3+2 \sqrt{2}, 6+5 / \sqrt{2}, 6-7 / \sqrt{2}) / 3 . \end{aligned}$	$R_{\mathbf{u}}^{\pi / 4}(3,2,1)=$ $(3+2 \sqrt{2}, 6+5 / \sqrt{2}, 6-7 / \sqrt{2}) / 3 .$	$R_{\mathbf{u}}^{\pi / 4}(3,2,1)=$ $(3+2 \sqrt{2}, 6+5 / \sqrt{2}, 6-7 / \sqrt{2}) / 3 .$
johnston-linear-matrix-atlas- EQ1382	452	■ 1.000	$\left[R_{\mathbf{y} \ z}^{\pi / 3}\right] \circ R_{\mathbf{x} \ y}^{\pi / 3}=\left[\begin{array}{ccc} 1 / 2 & -\sqrt{3} / 2 & 0 \\ \sqrt{3} / 4 & 1 / 4 & -\sqrt{3} / 2 \\ 3 / 4 & \sqrt{3} / 4 & 1 / 2 \end{array}\right] \cdot$	$\left[R_{\mathbf{y} \ z}^{\pi / 3} \circ R_{\mathbf{x} \ y}^{\pi / 3}\right]=\left[\begin{array}{ccc} 1 / 2 & -\sqrt{3} / 2 & 0 \\ \sqrt{3} / 4 & 1 / 4 & -\sqrt{3} / 2 \\ 3 / 4 & \sqrt{3} / 4 & 1 / 2 \end{array}\right] \cdot$	$\left[R_{\mathbf{y} \ z}^{\pi / 3} \circ R_{\mathbf{x} \ y}^{\pi / 3}\right]=\left[\begin{array}{ccc} 1 / 2 & -\sqrt{3} / 2 & 0 \\ \sqrt{3} / 4 & 1 / 4 & -\sqrt{3} / 2 \\ 3 / 4 & \sqrt{3} / 4 & 1 / 2 \end{array}\right] \cdot$
johnston-linear-matrix-atlas- EQ1383	453	■ 1.000	$A=\left[\begin{array}{ll} 0 & 1 \\ 0 & 0 \end{array}\right] \cdot$	$A=\left[\begin{array}{cc} 0 & 1 \\ 0 & 0 \end{array}\right] \cdot$	$A=\left[\begin{array}{cc} 0 & 1 \\ 0 & 0 \end{array}\right] \cdot$
johnston-linear-matrix-atlas- EQ1384	453	■ 1.000	$\left[\begin{array}{cc} \cos (\theta) & -\sin (\theta) \\ \sin (\theta) & \cos (\theta) \end{array}\right]$	$\left[\begin{array}{cc} \cos (\theta) & -\sin (\theta) \\ \sin (\theta) & \cos (\theta) \end{array}\right]$	$\left[\begin{array}{cc} \cos (\theta) & -\sin (\theta) \\ \sin (\theta) & \cos (\theta) \end{array}\right]$
johnston-linear-matrix-atlas- EQ1385	453	■ 1.000	$\left[R^{\theta}\right]^n=\left[R^{\theta} \circ R^{\theta} \circ \cdots \circ R^{\theta}\right]=\left[R^{n \theta}\right],$	$\left[R^{\theta}\right]^n=\left[R^{\theta} \circ R^{\theta} \circ \cdots \circ R^{\theta}\right]=\left[R^{n \theta}\right],$	$\left[R^{\theta}\right]^n=\left[R^{\theta} \circ R^{\theta} \circ \cdots \circ R^{\theta}\right]=\left[R^{n \theta}\right],$
johnston-linear-matrix-atlas- EQ1387	453	■ 1.000	$\mathbf{v} \times(\mathbf{w}+\mathbf{x})=\left[\begin{array}{l} v_2\left(w_3+x_3\right)-v_3\left(w_2+x_2\right) \\ v_3\left(w_1+x_1\right)-v_1\left(w_3+x_3\right) \\ v_1\left(w_2+x_2\right)-v_2\left(w_1+x_1\right) \end{array}\right] \cdot$	$\mathbf{v} \times(\mathbf{w}+\mathbf{x})=\left[\begin{array}{l} v_2\left(w_3+x_3\right)-v_3\left(w_2+x_2\right) \\ v_3\left(w_1+x_1\right)-v_1\left(w_3+x_3\right) \\ v_1\left(w_2+x_2\right)-v_2\left(w_1+x_1\right) \end{array}\right] \cdot$	$\mathbf{v} \times(\mathbf{w}+\mathbf{x})=\left[\begin{array}{l} v_2\left(w_3+x_3\right)-v_3\left(w_2+x_2\right) \\ v_3\left(w_1+x_1\right)-v_1\left(w_3+x_3\right) \\ v_1\left(w_2+x_2\right)-v_2\left(w_1+x_1\right) \end{array}\right] \cdot$
johnston-linear-matrix-atlas- EQ1389	453	■ 1.000	$\begin{aligned} & (c \mathbf{v}) \times \mathbf{w}=\left[\begin{array}{l} \left(c v_2\right) w_3-\left(c v_3\right) w_2 \\ \left(c v_3\right) w_1-\left(c v_1\right) w_3 \\ \left(c v_1\right) w_2-\left(c v_2\right) w_1 \end{array}\right] \\ & =c\left[\begin{array}{l} v_2 w_3-v_3 w_2 \\ v_3 w_1-v_1 w_3 \\ v_1 w_2-v_2 w_1 \end{array}\right]=c(\mathbf{v} \times \mathbf{w}) . \end{aligned}$	$(c \mathbf{v}) \times \mathbf{w}=\left[\begin{array}{l} \left(c v_2\right) w_3-\left(c v_3\right) w_2 \\ \left(c v_3\right) w_1-\left(c v_1\right) w_3 \\ \left(c v_1\right) w_2-\left(c v_2\right) w_1 \end{array}\right]$ $=c\left[\begin{array}{l} v_2 w_3-v_3 w_2 \\ v_3 w_1-v_1 w_3 \\ v_1 w_2-v_2 w_1 \end{array}\right]=c(\mathbf{v} \times \mathbf{w}) .$	$(c \mathbf{v}) \times \mathbf{w}=\left[\begin{array}{l} \left(c v_2\right) w_3-\left(c v_3\right) w_2 \\ \left(c v_3\right) w_1-\left(c v_1\right) w_3 \\ \left(c v_1\right) w_2-\left(c v_2\right) w_1 \end{array}\right]$ $=c\left[\begin{array}{l} v_2 w_3-v_3 w_2 \\ v_3 w_1-v_1 w_3 \\ v_1 w_2-v_2 w_1 \end{array}\right]=c(\mathbf{v} \times \mathbf{w}) .$
johnston-linear-matrix-atlas- EQ1392	454	■ 1.000	$\begin{aligned} & \mathbf{v} \cdot(\mathbf{w} \times \mathbf{x})=-\mathbf{v} \cdot(\mathbf{x} \times \mathbf{w}), \\ & \mathbf{v} \cdot(\mathbf{w} \times \mathbf{x})=\mathbf{w} \cdot(\mathbf{x} \times \mathbf{v})=-\mathbf{w} \cdot(\mathbf{v} \times \mathbf{x}), \quad \text { and } \\ & \mathbf{v} \cdot(\mathbf{w} \times \mathbf{x})=\mathbf{x} \cdot(\mathbf{v} \times \mathbf{w})=-\mathbf{x} \cdot(\mathbf{w} \times \mathbf{v}) . \end{aligned}$	$\mathbf{v} \cdot(\mathbf{w} \times \mathbf{x})=-\mathbf{v} \cdot(\mathbf{x} \times \mathbf{w}),$ $\mathbf{v} \cdot(\mathbf{w} \times \mathbf{x})=\mathbf{w} \cdot(\mathbf{x} \times \mathbf{v})=-\mathbf{w} \cdot(\mathbf{v} \times \mathbf{x}), \quad \text { and }$ $\mathbf{v} \cdot(\mathbf{w} \times \mathbf{x})=\mathbf{x} \cdot(\mathbf{v} \times \mathbf{w})=-\mathbf{x} \cdot(\mathbf{w} \times \mathbf{v}) .$	$\mathbf{v} \cdot(\mathbf{w} \times \mathbf{x})=-\mathbf{v} \cdot(\mathbf{x} \times \mathbf{w}),$ $\mathbf{v} \cdot(\mathbf{w} \times \mathbf{x})=\mathbf{w} \cdot(\mathbf{x} \times \mathbf{v})=-\mathbf{w} \cdot(\mathbf{v} \times \mathbf{x}), \quad \text { and }$ $\mathbf{v} \cdot(\mathbf{w} \times \mathbf{x})=\mathbf{x} \cdot(\mathbf{v} \times \mathbf{w})=-\mathbf{x} \cdot(\mathbf{w} \times \mathbf{v}) .$

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johnston-linear-matrix-aligned EQ1399	454	1.000	$\left[\begin{array}{llllll} 0 & 1 & 0 & 1 & 0 & 0 \\ 0 & 2 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 \\ 1 & 0 & 0 & 0 & 2 & 1 \\ 0 & 0 & 0 & 0 & 2 & 1 \\ 0 & 0 & 1 & 0 & 0 & 0 \end{array}\right]$.	$\begin{bmatrix} 0 & 1 & 0 & 1 & 0 & 0 \\ 0 & 0 & 2 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 \\ 1 & 0 & 0 & 0 & 2 & 1 \\ 0 & 0 & 0 & 0 & 2 & 1 \\ 0 & 0 & 1 & 0 & 0 & 0 \end{bmatrix}$.	
johnston-linear-matrix-aligned EQ1400	455	1.000	$A=\left[\begin{array}{llll} 0 & 1 & 0 & 1 \\ 1 & 0 & 1 & 1 \\ 0 & 1 & 0 & 1 \\ 1 & 1 & 1 & 0 \end{array}\right]$.	$A = \begin{bmatrix} 0 & 1 & 0 & 1 \\ 1 & 0 & 1 & 1 \\ 0 & 1 & 0 & 1 \\ 1 & 1 & 1 & 0 \end{bmatrix}$.	$A = \begin{bmatrix} 0 & 1 & 0 & 1 \\ 1 & 0 & 1 & 1 \\ 0 & 1 & 0 & 1 \\ 1 & 1 & 1 & 0 \end{bmatrix}$.
johnston-linear-matrix-aligned EQ1401	455	1.000	$A^2=\left[\begin{array}{llll} 2 & 1 & 2 & 1 \\ 1 & 2 & 1 & 2 \\ 2 & 1 & 2 & 1 \\ 1 & 2 & 1 & 3 \end{array}\right]$.	$A^2 = \begin{bmatrix} 2 & 1 & 2 & 1 \\ 1 & 3 & 1 & 2 \\ 2 & 1 & 2 & 1 \\ 1 & 2 & 1 & 3 \end{bmatrix}$.	$A^2 = \begin{bmatrix} 2 & 1 & 2 & 1 \\ 1 & 3 & 1 & 2 \\ 2 & 1 & 2 & 1 \\ 1 & 2 & 1 & 3 \end{bmatrix}$.
johnston-linear-matrix-aligned EQ1402	455	1.000	$A^4=\left[\begin{array}{cccc} 10 & 9 & 10 & 9 \\ 9 & 15 & 9 & 14 \\ 10 & 9 & 10 & 9 \\ 9 & 14 & 9 & 15 \end{array}\right]$.	$A^4 = \begin{bmatrix} 10 & 9 & 10 & 9 \\ 9 & 15 & 9 & 14 \\ 10 & 9 & 10 & 9 \\ 9 & 14 & 9 & 15 \end{bmatrix}$.	$A^4 = \begin{bmatrix} 10 & 9 & 10 & 9 \\ 9 & 15 & 9 & 14 \\ 10 & 9 & 10 & 9 \\ 9 & 14 & 9 & 15 \end{bmatrix}$.
johnston-linear-matrix-aligned EQ1405	455	1.000	$\left[\begin{array}{cccc} 1 & 0 & 0 & -1 \\ 0 & 1 & 0 & -1/2 \\ 0 & 0 & 1 & -1/2 \\ 0 & 0 & 0 & 0 \end{array}\right]$	$\begin{bmatrix} 1 & 0 & 0 & -1 \\ 0 & 1 & 0 & -1/2 \\ 0 & 0 & 1 & -1/2 \\ 0 & 0 & 0 & 0 \end{bmatrix}$	$\begin{bmatrix} 1 & 0 & 0 & -1 \\ 0 & 1 & 0 & -1/2 \\ 0 & 0 & 1 & -1/2 \\ 0 & 0 & 0 & 0 \end{bmatrix}$
johnston-linear-matrix-aligned EQ1407	456	1.000	$\begin{array}{r} x+y=1 \\ 2x+2y=2 \\ 3x+3y=3 \end{array}$	$\begin{array}{l} x+y=1 \\ 2x+2y=2 \\ 3x+3y=3 \end{array}$	$\begin{array}{l} x+y=1 \\ 2x+2y=2 \\ 3x+3y=3 \end{array}$
johnston-linear-matrix-aligned EQ1408	456	1.000	$\begin{array}{l} x+y+z=1 \\ x+y+z=2 \end{array}$	$\begin{array}{l} x+y+z=1 \\ x+y+z=2 \end{array}$	$\begin{array}{l} x+y+z=1 \\ x+y+z=2 \end{array}$
johnston-linear-matrix-aligned EQ1409	456	1.000	$\left[\begin{array}{cc c} 1 & -1 & 0 \\ 0 & 1 & 1/2 \\ 0 & 0 & t-1/2 \end{array}\right]$.	$\left[\begin{array}{cc c} 1 & -1 & 0 \\ 0 & 1 & 1/2 \\ 0 & 0 & t-1/2 \end{array}\right]$.	$\left[\begin{array}{cc c} 1 & -1 & 0 \\ 0 & 1 & 1/2 \\ 0 & 0 & t-1/2 \end{array}\right]$.
johnston-linear-matrix-aligned EQ1412	456	1.000	$\left[\begin{array}{ccc c} 1 & 0 & -1 & 0 \\ 0 & 1 & 2 & 0 \end{array}\right],$	$\left[\begin{array}{ccc c} 1 & 0 & -1 & 0 \\ 0 & 1 & 2 & 0 \end{array}\right],$	$\left[\begin{array}{ccc c} 1 & 0 & -1 & 0 \\ 0 & 1 & 2 & 0 \end{array}\right],$
johnston-linear-matrix-aligned EQ1413	456	1.000	$\begin{array}{l} v_1+2v_2+v_4=0 \\ -2v_1+v_2+v_3+3v_4=0 \\ -v_1-v_2+2v_3+v_4=0 \end{array}$	$\begin{array}{l} v_1+2v_2+v_4=0 \\ -2v_1+v_2+v_3+3v_4=0 \\ -v_1-v_2+2v_3+v_4=0 \end{array}$	$\begin{array}{l} v_1+2v_2+v_4=0 \\ -2v_1+v_2+v_3+3v_4=0 \\ -v_1-v_2+2v_3+v_4=0 \end{array}$
johnston-linear-matrix-aligned EQ1414	456	1.000	$\left[\begin{array}{cccc c} 1 & 0 & 0 & -7/9 & 0 \\ 0 & 1 & 0 & 8/9 & 0 \\ 0 & 0 & 1 & 5/9 & 0 \end{array}\right],$	$\left[\begin{array}{cccc c} 1 & 0 & 0 & -7/9 & 0 \\ 0 & 1 & 0 & 8/9 & 0 \\ 0 & 0 & 1 & 5/9 & 0 \end{array}\right],$	$\left[\begin{array}{cccc c} 1 & 0 & 0 & -7/9 & 0 \\ 0 & 1 & 0 & 8/9 & 0 \\ 0 & 0 & 1 & 5/9 & 0 \end{array}\right],$
johnston-linear-matrix-aligned EQ1415	456	1.000	$\left[\begin{array}{ll l} 1 & 0 & 1 \\ 0 & 1 & 2 \end{array}\right],$	$\left[\begin{array}{ll l} 1 & 0 & 1 \\ 0 & 1 & 2 \end{array}\right],$	$\left[\begin{array}{ll l} 1 & 0 & 1 \\ 0 & 1 & 2 \end{array}\right],$

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johnston-linear-matrix-atqsdv-EQ1416	456	1.000	$\left[\begin{array}{cc c} 0 & -1 & -2 \\ 0 & 1 & 2 \end{array}\right]$.	$\left[\begin{array}{cc c} 0 & -1 & -2 \\ 0 & 1 & 2 \end{array}\right]$.	$\left[\begin{array}{cc c} 0 & -1 & -2 \\ 0 & 1 & 2 \end{array}\right]$.
johnston-linear-matrix-atqsdv-EQ1417	457	1.000	$\left[\begin{array}{ccc c} v_1 & v_2 & v_3 & 0 \\ w_1 & w_2 & w_3 & 0 \end{array}\right]$	$\left[\begin{array}{ccc c} v_1 & v_2 & v_3 & 0 \\ w_1 & w_2 & w_3 & 0 \end{array}\right]$	$\left[\begin{array}{ccc c} v_1 & v_2 & v_3 & 0 \\ w_1 & w_2 & w_3 & 0 \end{array}\right]$
johnston-linear-matrix-atqsdv-EQ1418	457	1.000	$\left[\begin{array}{ccc c} w_1 & v_1 & w_1 & v_2 \\ w_1 & v_3 & 0 & 0 \\ 0 & w_1 & v_2-v_1 & w_2 \\ v_3-v_1 & w_3 & 0 & 0 \end{array}\right]$	$\left[\begin{array}{ccc c} w_1 v_1 & w_1 v_2 & w_1 v_3 & 0 \\ 0 & w_1 v_2 - v_1 w_2 & w_1 v_3 - v_1 w_3 & 0 \end{array}\right]$	$\left[\begin{array}{ccc c} w_1 v_1 & w_1 v_2 & w_1 v_3 & 0 \\ 0 & w_1 v_2 - v_1 w_2 & w_1 v_3 - v_1 w_3 & 0 \end{array}\right]$
johnston-linear-matrix-atqsdv-EQ1420	457	1.000	$\begin{aligned} & \left(w_1 v_2 - v_1 w_2\right) x_1 = \left(v_3 w_2 - v_2 w_3\right) x_3 \\ x_1 & = \left(v_3 w_2 - v_2 w_3\right) x_3 \\ & \left(w_1 v_2 - v_1 w_2\right) x_2 = \left(v_1 w_3 - w_1 v_3\right) x_3 \\ & \left(w_1 v_2 - v_1 w_2\right) x_2 = \left(v_1 w_3 - w_1 v_3\right) x_3 \end{aligned}$	$\begin{aligned} & \left(w_1 v_2 - v_1 w_2\right) x_1 = \left(v_3 w_2 - v_2 w_3\right) x_3 \\ & \left(w_1 v_2 - v_1 w_2\right) x_2 = \left(v_1 w_3 - w_1 v_3\right) x_3 \end{aligned}$	$\begin{aligned} & \left(w_1 v_2 - v_1 w_2\right) x_1 = \left(v_3 w_2 - v_2 w_3\right) x_3 \\ & \left(w_1 v_2 - v_1 w_2\right) x_2 = \left(v_1 w_3 - w_1 v_3\right) x_3 \end{aligned}$
johnston-linear-matrix-atqsdv-EQ1421	457	1.000	$\left(x_1, x_2, x_3\right) = c \left(v_2 w_3 - v_3 w_2, v_3 w_1 - v_1 w_3, v_1 w_2 - v_2 w_1\right)$,	$\left(x_1, x_2, x_3\right) = c \left(v_2 w_3 - v_3 w_2, v_3 w_1 - v_1 w_3, v_1 w_2 - v_2 w_1\right)$,	$\left(x_1, x_2, x_3\right) = c \left(v_2 w_3 - v_3 w_2, v_3 w_1 - v_1 w_3, v_1 w_2 - v_2 w_1\right)$,
johnston-linear-matrix-atqsdv-EQ1422	457	1.000	$\begin{aligned} & x + y = 120 \\ & 0.13 x + 0.05 y = 9.6 \end{aligned}$	$\begin{aligned} & x + y = 120 \\ & 0.13 x + 0.05 y = 9.6 \end{aligned}$	$\begin{aligned} & x + y = 120 \\ & 0.13 x + 0.05 y = 9.6 \end{aligned}$
johnston-linear-matrix-atqsdv-EQ1424	457	1.000	$2 \text{ZnS} + 3 \text{O}_2 \rightarrow 2 \text{ZnO} + 2 \text{SO}_2$.	$2 \text{ZnS} + 3 \text{O}_2 \rightarrow 2 \text{ZnO} + 2 \text{SO}_2$.	$2 \text{ZnS} + 3 \text{O}_2 \rightarrow 2 \text{ZnO} + 2 \text{SO}_2$.
johnston-linear-matrix-atqsdv-EQ1427	457	1.000	$\left[\begin{array}{lll l} 1 & / & 9 & 2 / 9 & 2 / 9 & 2 \\ 2 & / & 9 & 4 / 9 & 4 / 9 & 4 \\ 2 & / & 9 & 4 / 9 & 4 / 9 & 4 \end{array}\right]$,	$\left[\begin{array}{lll l} 1/9 & 2/9 & 2/9 & 2 \\ 2/9 & 4/9 & 4/9 & 4 \\ 2/9 & 4/9 & 4/9 & 4 \end{array}\right]$,	$\left[\begin{array}{lll l} 1/9 & 2/9 & 2/9 & 2 \\ 2/9 & 4/9 & 4/9 & 4 \\ 2/9 & 4/9 & 4/9 & 4 \end{array}\right]$,
johnston-linear-matrix-atqsdv-EQ1428	457	1.000	$\left[\begin{array}{ccc c} 1 & 2 & 2 & 18 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{array}\right]$.	$\left[\begin{array}{ccc c} 1 & 2 & 2 & 18 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{array}\right]$.	$\left[\begin{array}{ccc c} 1 & 2 & 2 & 18 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{array}\right]$.
johnston-linear-matrix-atqsdv-EQ1429	457	1.000	$\begin{aligned} & c_1 + 2 c_2 = 3 \\ & 4 c_1 - c_2 = -2 \\ & 2 c_1 + c_2 = 1 \end{aligned}$	$\begin{aligned} & c_1 + 2 c_2 = 3 \\ & 4 c_1 - c_2 = -2 \\ & 2 c_1 + c_2 = 1 \end{aligned}$	$\begin{aligned} & c_1 + 2 c_2 = 3 \\ & 4 c_1 - c_2 = -2 \\ & 2 c_1 + c_2 = 1 \end{aligned}$
johnston-linear-matrix-atqsdv-EQ1430	457	1.000	$\left[\begin{array}{cc c} 1 & 2 & 3 \\ 0 & -9 & -14 \\ 0 & 0 & -1/3 \end{array}\right]$.	$\left[\begin{array}{cc c} 1 & 2 & 3 \\ 0 & -9 & -14 \\ 0 & 0 & -1/3 \end{array}\right]$.	$\left[\begin{array}{cc c} 1 & 2 & 3 \\ 0 & -9 & -14 \\ 0 & 0 & -1/3 \end{array}\right]$.
johnston-linear-matrix-atqsdv-EQ1432	457	1.000	$\begin{aligned} & T(1,0) = T\left(\frac{1}{2}(1,1) + \frac{1}{2}(1,-1)\right) \\ & = \frac{1}{2}T(1,1) + \frac{1}{2}T(1,-1) \\ & = \frac{1}{2}(3,7) + \frac{1}{2}(-1,-1) \\ & = (1,3) . \end{aligned}$	$\begin{aligned} & T(1,0) = T\left(\frac{1}{2}(1,1) + \frac{1}{2}(1,-1)\right) \\ & = \frac{1}{2}T(1,1) + \frac{1}{2}T(1,-1) \\ & = \frac{1}{2}(3,7) + \frac{1}{2}(-1,-1) \\ & = (1,3) . \end{aligned}$	$\begin{aligned} & T(1,0) = T\left(\frac{1}{2}(1,1) + \frac{1}{2}(1,-1)\right) \\ & = \frac{1}{2}T(1,1) + \frac{1}{2}T(1,-1) \\ & = \frac{1}{2}(3,7) + \frac{1}{2}(-1,-1) \\ & = (1,3) . \end{aligned}$
johnston-linear-matrix-atqsdv-EQ1433	457	1.000	$[T] = \left[\begin{array}{ll} 1 & 2 \\ 3 & 4 \end{array}\right]$.	$[T] = \left[\begin{array}{ll} 1 & 2 \\ 3 & 4 \end{array}\right]$.	$[T] = \left[\begin{array}{ll} 1 & 2 \\ 3 & 4 \end{array}\right]$.
johnston-linear-matrix-atqsdv-EQ1434	458	1.000	$[T] = \left[\begin{array}{ccc} 1 & 2 & 1 \\ 1 & 3 & 2 \\ -1 & 2 & 0 \end{array}\right]$.	$[T] = \left[\begin{array}{ccc} 1 & 2 & 1 \\ 1 & 3 & 2 \\ -1 & 2 & 0 \end{array}\right]$.	$[T] = \left[\begin{array}{ccc} 1 & 2 & 1 \\ 1 & 3 & 2 \\ -1 & 2 & 0 \end{array}\right]$.
johnston-linear-matrix-atqsdv-EQ1436	458	1.000	$\begin{aligned} & u - 2 w = -4 \\ & u + v + w = 4 \\ & 2 u - 6 v - 2 w = 4 \end{aligned}$	$\begin{aligned} & u - 2 w = -4 \\ & u + v + w = 4 \\ & 2 u - 6 v - 2 w = 4 \end{aligned}$	$\begin{aligned} & u - 2 w = -4 \\ & u + v + w = 4 \\ & 2 u - 6 v - 2 w = 4 \end{aligned}$

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johnston-linear-matrix-algebraEQ1437	458	1.000	$\left[\begin{array}{ll} 0 & 1 \\ 1 & 0 \end{array}\right] \left[\begin{array}{ll} a & b \\ c & d \end{array}\right] = \left[\begin{array}{ll} a & b \\ c & d \end{array}\right] \left[\begin{array}{ll} 0 & 1 \\ 1 & 0 \end{array}\right].$		
johnston-linear-matrix-algebraEQ1442	458	1.000	$\begin{aligned} a+b+c &= 3 \\ 4a+2b+c &= 6 \\ 9a+3b+c &= 13 \end{aligned}$		
johnston-linear-matrix-algebraEQ1443	458	1.000	$\begin{aligned} b+m &= 3 \\ b+2m &= 6 \\ b+3m &= 13 \end{aligned}$		
johnston-linear-matrix-algebraEQ1444	458	1.000	$\begin{aligned} a+b+c+d &= 3 \\ 8a+4b+2c+d &= 6 \\ 27a+9b+3c+d &= 13 \end{aligned}$		
johnston-linear-matrix-algebraEQ1445	458	1.000	$\left[\begin{array}{cccc c} 1 & 0 & 0 & 1/6 & 2/3 \\ 0 & 1 & 0 & -1 & -2 \\ 0 & 0 & 1 & 11/6 & 13/3 \end{array}\right].$		
johnston-linear-matrix-algebraEQ1449	459	1.000	$\left[\begin{array}{cccccc} 1 & 1 & 1 & \cdots & 1 \\ 0 & 0 & 0 & \cdots & 0 \\ 0 & 0 & 0 & \cdots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & 0 & \cdots & 0 \end{array}\right],$		
johnston-linear-matrix-algebraEQ1451	460	1.000	$A^{-1} = \frac{1}{a-b} (I_n - b(nd + 1/(a-b))J_n).$		
johnston-linear-matrix-algebraEQ1453	460	1.000	$E_{\{k\}} \cdots E_{\{2\}} E_{\{1\}}[P \mid Q] = [E_{\{k\}} \cdots E_{\{2\}} E_{\{1\}} P \mid E_{\{k\}} \cdots E_{\{2\}} E_{\{1\}} Q].$		
johnston-linear-matrix-algebraEQ1454	460	1.000	$A^{-1} = [\mathbf{b}_1 \mid \mathbf{b}_2 \mid \cdots \mid \mathbf{b}_n].$		
johnston-linear-matrix-algebraEQ1457	460	1.000	$A^k (A^{-1})^k = (AA \cdots A) (A^{-1} A^{-1} \cdots A^{-1}).$		

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johnston-linear-matrix-algebra-EQ1459	461	1.000	$\begin{aligned} & \left[\begin{array}{cccc} A_1 & 0 & \cdots & 0 \\ 0 & A_2 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & A_n \end{array}\right] \left[\begin{array}{cccc} B_{1,1} & B_{1,2} & \cdots & B_{1,n} \\ B_{2,1} & B_{2,2} & \cdots & B_{2,n} \\ \vdots & \vdots & \ddots & \vdots \\ B_{n,1} & B_{n,2} & \cdots & B_{n,n} \end{array}\right] \\ & = \left[\begin{array}{cccc} A_1 B_{1,1} & A_1 B_{1,2} & \cdots & A_1 B_{1,n} \\ A_2 B_{2,1} & A_2 B_{2,2} & \cdots & A_2 B_{2,n} \\ \vdots & \vdots & \ddots & \vdots \\ A_n B_{n,1} & A_n B_{n,2} & \cdots & A_n B_{n,n} \end{array}\right] . \end{aligned}$	$\begin{bmatrix} A_1 & O & \cdots & O \\ O & A_2 & \cdots & O \\ \vdots & \vdots & \ddots & \vdots \\ O & O & \cdots & A_n \end{bmatrix} \begin{bmatrix} B_{1,1} & B_{1,2} & \cdots & B_{1,n} \\ B_{2,1} & B_{2,2} & \cdots & B_{2,n} \\ \vdots & \vdots & \ddots & \vdots \\ B_{n,1} & B_{n,2} & \cdots & B_{n,n} \end{bmatrix} = \begin{bmatrix} A_1 B_{1,1} & A_1 B_{1,2} & \cdots & A_1 B_{1,n} \\ A_2 B_{2,1} & A_2 B_{2,2} & \cdots & A_2 B_{2,n} \\ \vdots & \vdots & \ddots & \vdots \\ A_n B_{n,1} & A_n B_{n,2} & \cdots & A_n B_{n,n} \end{bmatrix} .$	
johnston-linear-matrix-algebra-EQ1464	461	1.000	$\left[\begin{array}{cccc} 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 1 & 0 & -1 & -2 \\ 0 & 1 & -2 & -3 \end{array}\right] .$	$\left[\begin{array}{cccc} 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 1 & 0 & -1 & -2 \\ 0 & 1 & -2 & -3 \end{array}\right] .$	$\begin{bmatrix} 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 1 & 0 & -1 & -2 \\ 0 & 1 & -2 & -3 \end{bmatrix} .$
johnston-linear-matrix-algebra-EQ1466	461	1.000	$\frac{1}{6} \left[\begin{array}{ccccc} 12 & -6 & -12 & 6 & 2 \\ -6 & 6 & 0 & -3 & -2 \\ 0 & 0 & 6 & 0 & 0 \\ 0 & 0 & 0 & 3 & 0 \\ 0 & 0 & 0 & 0 & 2 \end{array}\right] .$	$\frac{1}{6} \left[\begin{array}{ccccc} 12 & -6 & -12 & 6 & 2 \\ -6 & 6 & 0 & -3 & -2 \\ 0 & 0 & 6 & 0 & 0 \\ 0 & 0 & 0 & 3 & 0 \\ 0 & 0 & 0 & 0 & 2 \end{array}\right] .$	$\frac{1}{6} \begin{bmatrix} 12 & -6 & -12 & 6 & 2 \\ -6 & 6 & 0 & -3 & -2 \\ 0 & 0 & 6 & 0 & 0 \\ 0 & 0 & 0 & 3 & 0 \\ 0 & 0 & 0 & 0 & 2 \end{bmatrix} .$
johnston-linear-matrix-algebra-EQ1469 (1)	462	1.000	$\left[\begin{array}{lll l} 1 & 1 & 2 & 0 \\ 1 & 1 & 2 & 0 \\ 2 & 2 & 4 & 0 \end{array}\right]$	$\left[\begin{array}{lll l} 1 & 1 & 2 & 0 \\ 1 & 1 & 2 & 0 \\ 2 & 2 & 4 & 0 \end{array}\right]$	$\left[\begin{array}{lll l} 1 & 1 & 2 & 0 \\ 1 & 1 & 2 & 0 \\ 2 & 2 & 4 & 0 \end{array}\right]$
johnston-linear-matrix-algebra-EQ1470	462	1.000	$\left[\begin{array}{lll l} 1 & 1 & 2 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{array}\right] .$	$\left[\begin{array}{lll l} 1 & 1 & 2 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{array}\right] .$	$\left[\begin{array}{lll l} 1 & 1 & 2 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{array}\right] .$
johnston-linear-matrix-algebra-EQ1472	462	1.000	$(4,3,2,1)=(3,1,4,2)+(2,4,1,3)-(1,2,3,4) \text{ .}$	$(4,3,2,1) = (3,1,4,2) + (2,4,1,3) - (1,2,3,4).$	$(4,3,2,1) = (3,1,4,2) + (2,4,1,3) - (1,2,3,4).$
johnston-linear-matrix-algebra-EQ1474	463	1.000	$\left[\begin{array}{l} 1 & 2 \\ 1 & 5 \end{array}\right] \left[\begin{array}{l} c_0 \\ c_1 \end{array}\right] = \left[\begin{array}{l} 1 \\ 7 \end{array}\right] .$	$\left[\begin{array}{cc} 1 & 2 \\ 1 & 5 \end{array}\right] \left[\begin{array}{c} c_0 \\ c_1 \end{array}\right] = \left[\begin{array}{c} 1 \\ 7 \end{array}\right] .$	$\left[\begin{array}{cc} 1 & 2 \\ 1 & 5 \end{array}\right] \begin{bmatrix} c_0 \\ c_1 \end{bmatrix} = \begin{bmatrix} 1 \\ 7 \end{bmatrix} .$
johnston-linear-matrix-algebra-EQ1476	463	1.000	$A_2=\left[\begin{array}{cc} 1 & 2 \\ 3 & 4 \end{array}\right] \quad \text{so} \quad A_2^{-1}=\frac{1}{2} \left[\begin{array}{cc} -4 & 2 \\ 3 & -1 \end{array}\right] .$	$A_2 = \left[\begin{array}{cc} 1 & 2 \\ 3 & 4 \end{array}\right] \quad \text{so} \quad A_2^{-1} = \frac{1}{2} \left[\begin{array}{cc} -4 & 2 \\ 3 & -1 \end{array}\right] .$	$A_2 = \left[\begin{array}{cc} 1 & 2 \\ 3 & 4 \end{array}\right] \quad \text{so} \quad A_2^{-1} = \frac{1}{2} \left[\begin{array}{cc} -4 & 2 \\ 3 & -1 \end{array}\right] .$
johnston-linear-matrix-algebra-EQ1477	463	1.000	$\begin{aligned} & 2(n+1, n+2, \ldots, 2n) - (1, 2, \ldots, n) \\ & = (2n+1, 2n+2, \ldots, 3n) \end{aligned}$	$\begin{aligned} & 2(n+1, n+2, \ldots, 2n) - (1, 2, \ldots, n) \\ & = (2n+1, 2n+2, \ldots, 3n) \end{aligned}$	$\begin{aligned} & 2(n+1, n+2, \ldots, 2n) - (1, 2, \ldots, n) \\ & = (2n+1, 2n+2, \ldots, 3n) \end{aligned}$
johnston-linear-matrix-algebra-EQ1478	463	1.000	$c_1 \mathbf{v}_1 + \cdots + c_k \mathbf{v}_k \in \mathcal{S}$	$c_1 \mathbf{v}_1 + \cdots + c_k \mathbf{v}_k \in \mathcal{S}$	$c_1 \mathbf{v}_1 + \cdots + c_k \mathbf{v}_k \in \mathcal{S}$

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johnston-linear-matrix-atlas- EQ1479	463	1.000	$c_{\{1\}} \mathbf{v}_{\{1\}} + c_{\{2\}} \mathbf{v}_{\{2\}} + \cdots + c_{\{k\}} \mathbf{v}_{\{k\}} = \mathbf{0}$.	$c_1 \mathbf{v}_1 + c_2 \mathbf{v}_2 + \cdots + c_k \mathbf{v}_k = \mathbf{0}$.	$c_1 \mathbf{v}_1 + c_2 \mathbf{v}_2 + \cdots + c_k \mathbf{v}_k = \mathbf{0}$.
johnston-linear-matrix-atlas- EQ1480	463	1.000	$\mathbf{v}_{\{i\}} = c_{\{1\}} \mathbf{v}_{\{1\}} + \ldots + c_{i-1} \mathbf{v}_{i-1} + c_{i+1} \mathbf{v}_{i+1} + \ldots + c_k \mathbf{v}_k$. $\mathbf{v}_{\{i-1\}} + c_{\{i+1\}} \mathbf{v}_{\{i+1\}} + \ldots + c_{\{k\}} \mathbf{v}_{\{k\}}$.	$\mathbf{v}_i = c_1 \mathbf{v}_1 + \ldots + c_{i-1} \mathbf{v}_{i-1} + c_{i+1} \mathbf{v}_{i+1} + \ldots + c_k \mathbf{v}_k$.	$\mathbf{v}_i = c_1 \mathbf{v}_1 + \ldots + c_{i-1} \mathbf{v}_{i-1} + c_{i+1} \mathbf{v}_{i+1} + \ldots + c_k \mathbf{v}_k$.
johnston-linear-matrix-atlas- EQ1481	463	1.000	$c_{\{1\}} \mathbf{v}_{\{1\}} + \ldots + c_{i-1} \mathbf{v}_{i-1} - \mathbf{v}_{\{i-1\}} + c_{i+1} \mathbf{v}_{i+1} + \ldots + c_k \mathbf{v}_k = \mathbf{0}$, $\mathbf{v}_{\{i\}} + c_{\{i+1\}} \mathbf{v}_{\{i+1\}} + \ldots + c_{\{k\}} \mathbf{v}_{\{k\}} = \mathbf{0}$,	$c_1 \mathbf{v}_1 + \ldots + c_{i-1} \mathbf{v}_{i-1} - \mathbf{v}_i + c_{i+1} \mathbf{v}_{i+1} + \ldots + c_k \mathbf{v}_k = \mathbf{0}$,	$c_1 \mathbf{v}_1 + \ldots + c_{i-1} \mathbf{v}_{i-1} - \mathbf{v}_i + c_{i+1} \mathbf{v}_{i+1} + \ldots + c_k \mathbf{v}_k = \mathbf{0}$,
johnston-linear-matrix-atlas- EQ1482	463	1.000	$\mathbf{0} + \mathbf{0} \mathbf{v}_1 + \mathbf{0} \mathbf{v}_2 + \cdots + \mathbf{0} \mathbf{v}_k = \mathbf{0}$.	$\mathbf{0} + 0 \mathbf{v}_1 + 0 \mathbf{v}_2 + \cdots + 0 \mathbf{v}_k = \mathbf{0}$.	$\mathbf{0} + 0 \mathbf{v}_1 + 0 \mathbf{v}_2 + \cdots + 0 \mathbf{v}_k = \mathbf{0}$.
johnston-linear-matrix-atlas- EQ1483	463	1.000	$c_{\{1\}} \mathbf{v}_{\{1\}} + c_{\{2\}} \mathbf{v}_{\{2\}} + \cdots + c_{\{k\}} \mathbf{v}_{\{k\}} = \mathbf{0}$.	$c_1 \mathbf{v}_1 + c_2 \mathbf{v}_2 + \cdots + c_k \mathbf{v}_k = \mathbf{0}$.	$c_1 \mathbf{v}_1 + c_2 \mathbf{v}_2 + \cdots + c_k \mathbf{v}_k = \mathbf{0}$.
johnston-linear-matrix-atlas- EQ1486	464	1.000	$A = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}$,	$AB = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}$,	$AB = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}$,
johnston-linear-matrix-atlas- EQ1490 (0)	464	1.000	$\begin{bmatrix} 1 & 0 & -1 & -3 & -3/2 \\ 0 & 1 & 0 & -1/2 & -1/4 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}$.	$\begin{bmatrix} 1 & 0 & -1 & -3 & -3/2 \\ 0 & 1 & 0 & -1/2 & -1/4 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}$.	$\begin{bmatrix} 1 & 0 & -1 & -3 & -3/2 \\ 0 & 1 & 0 & -1/2 & -1/4 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}$.
johnston-linear-matrix-atlas- EQ1491	465	1.000	$A = \begin{bmatrix} 1 & 1 \\ 2 & 2 \\ 1 & 2 \end{bmatrix}$ is $\begin{bmatrix} 1 & 0 \\ 0 & 1 \\ 0 & 0 \end{bmatrix}$.	$A = \begin{bmatrix} 1 & 1 \\ 2 & 2 \\ 1 & 2 \end{bmatrix}$ is $\begin{bmatrix} 1 & 0 \\ 0 & 1 \\ 0 & 0 \end{bmatrix}$.	$A = \begin{bmatrix} 1 & 1 \\ 2 & 2 \\ 1 & 2 \end{bmatrix}$ is $\begin{bmatrix} 1 & 0 \\ 0 & 1 \\ 0 & 0 \end{bmatrix}$.
johnston-linear-matrix-atlas- EQ1494	465	1.000	$\begin{bmatrix} 1 & x \\ 0 & 4-2x \end{bmatrix}$.	$\begin{bmatrix} 1 & x \\ 0 & 4-2x \end{bmatrix}$.	$\begin{bmatrix} 1 & x \\ 0 & 4-2x \end{bmatrix}$.
johnston-linear-matrix-atlas- EQ1495	465	1.000	$\begin{bmatrix} 1 & 2 & x \\ 0 & -1-x & x+1 \\ 0 & 0 & x^2-1 \end{bmatrix}$.	$\begin{bmatrix} 1 & 2 & x \\ 0 & -1-x & x+1 \\ 0 & 0 & x^2-1 \end{bmatrix}$.	$\begin{bmatrix} 1 & 2 & x \\ 0 & -1-x & x+1 \\ 0 & 0 & x^2-1 \end{bmatrix}$.
johnston-linear-matrix-atlas- EQ1496	465	1.000	$A = \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}$,	$A = \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}$,	$A = \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}$,
johnston-linear-matrix-atlas- EQ1497	466	1.000	$2(1, -1, 2) + 0(-1, 2, 3) - (2, -2, 4) = (0, 0, 0)$.	$2(1, -1, 2) + 0(-1, 2, 3) - (2, -2, 4) = (0, 0, 0)$.	$2(1, -1, 2) + 0(-1, 2, 3) - (2, -2, 4) = (0, 0, 0)$.
johnston-linear-matrix-atlas- EQ1498	466	1.000	$\mathbf{v} \cdot \mathbf{w} = (A\mathbf{x}) \cdot \mathbf{w} = \mathbf{x} \cdot (A^T \mathbf{w}) = \mathbf{x} \cdot \mathbf{0} = 0$.	$\mathbf{v} \cdot \mathbf{w} = (A\mathbf{x}) \cdot \mathbf{w} = \mathbf{x} \cdot (A^T \mathbf{w}) = \mathbf{x} \cdot \mathbf{0} = 0$.	$\mathbf{v} \cdot \mathbf{w} = (A\mathbf{x}) \cdot \mathbf{w} = \mathbf{x} \cdot (A^T \mathbf{w}) = \mathbf{x} \cdot \mathbf{0} = 0$.
johnston-linear-matrix-atlas- EQ1499	466	1.000	$\mathbf{v} \cdot \mathbf{w} = \mathbf{v} \cdot (A^T \mathbf{x}) = (A\mathbf{v}) \cdot \mathbf{x} = \mathbf{0} \cdot \mathbf{x} = 0$.	$\mathbf{v} \cdot \mathbf{w} = \mathbf{v} \cdot (A^T \mathbf{x}) = (A\mathbf{v}) \cdot \mathbf{x} = \mathbf{0} \cdot \mathbf{x} = 0$.	$\mathbf{v} \cdot \mathbf{w} = \mathbf{v} \cdot (A^T \mathbf{x}) = (A\mathbf{v}) \cdot \mathbf{x} = \mathbf{0} \cdot \mathbf{x} = 0$.
johnston-linear-matrix-atlas- EQ1500	466	1.000	$A = [\mathbf{v} \mid w_2 \mathbf{v} \mid \cdots \mid w_n \mathbf{v}]$.	$A = [\mathbf{v} \mid w_2 \mathbf{v} \mid \cdots \mid w_n \mathbf{v}]$.	$A = [\mathbf{v} \mid w_2 \mathbf{v} \mid \cdots \mid w_n \mathbf{v}]$.
johnston-linear-matrix-atlas- EQ1502	467	1.000	$\begin{bmatrix} 1 & 0 & 1/3 \\ 0 & 1 & 1/3 \\ 0 & 0 & 0 \end{bmatrix}$.	$\begin{bmatrix} 1 & 0 & 1/3 \\ 0 & 1 & 1/3 \\ 0 & 0 & 0 \end{bmatrix}$.	$\begin{bmatrix} 1 & 0 & 1/3 \\ 0 & 1 & 1/3 \\ 0 & 0 & 0 \end{bmatrix}$.
johnston-linear-matrix-atlas- EQ1503	467	1.000	$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{bmatrix}$.	$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{bmatrix}$.	$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{bmatrix}$.
johnston-linear-matrix-atlas- EQ1504	467	1.000	$\mathbf{v} = c_1 \mathbf{v}_1 + c_2 \mathbf{v}_2 + \cdots + c_n \mathbf{v}_n$.	$\mathbf{v} = c_1 \mathbf{v}_1 + c_2 \mathbf{v}_2 + \cdots + c_n \mathbf{v}_n$.	$\mathbf{v} = c_1 \mathbf{v}_1 + c_2 \mathbf{v}_2 + \cdots + c_n \mathbf{v}_n$.

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johnston-linear-matrix-align- EQ1507	467	1.000	$\begin{aligned} \{[R \mid E] \} &= E_{\{k\}} \cdots E_{\{2\}} E_{\{1\}} [A \mid I] \\ &= [E_k \cdots E_2 E_1 A \mid E_k \cdots E_2 E_1] . \end{aligned}$	$\begin{aligned} [R \mid E] &= E_k \cdots E_2 E_1 [A \mid I] \\ &= [E_k \cdots E_2 E_1 A \mid E_k \cdots E_2 E_1] . \end{aligned}$	$\begin{aligned} [R \mid E] &= E_k \cdots E_2 E_1 [A \mid I] \\ &= [E_k \cdots E_2 E_1 A \mid E_k \cdots E_2 E_1] . \end{aligned}$
johnston-linear-matrix-align- EQ1508	467	1.000	$P_{\{1\}} P_{\{2\}}^{-1} B = \left(P_{\{1\}} P_{\{2\}}^{-1} \right) P_2 R = P_1 R = A,$	$P_1 P_2^{-1} B = (P_1 P_2^{-1}) P_2 R = P_1 R = A,$	$P_1 P_2^{-1} B = (P_1 P_2^{-1}) P_2 R = P_1 R = A,$
johnston-linear-matrix-align- EQ1510	468	1.000	$m_{\{1\}} + m_{\{2\}} + \cdots + m_{\{k\}}$	$m_1 + m_2 + \cdots + m_k$	$m_1 + m_2 + \cdots + m_k$
johnston-linear-matrix-align- EQ1512	468	1.000	$\begin{aligned} &3 y_{\{1\}} + y_{\{2\}} + 3 y_{\{3\}} && \& \& 2 y_{\{2\}} + y_{\{3\}} \\ &\& \geq 1 && \& \geq 2 \\ &2 y_{\{2\}} + y_{\{3\}} && \& \geq 2 \\ &y_{\{1\}} + 2 y_{\{2\}} + 2 y_{\{3\}} && \& \geq 2 \\ &- y_{\{1\}} - y_{\{2\}} + 2 y_{\{3\}} && \& \geq -1 \\ &- y_{\{1\}}, && y_{\{3\}} \geq 0 \end{aligned}$	$\begin{aligned} &3 y_1 + y_2 + 3 y_3 \\ &2 y_2 + y_3 \geq 1 \\ &y_1 + 2 y_2 + 2 y_3 \geq 2 \\ &- y_1 - y_2 + 2 y_3 \geq -1 \\ &- y_1, \quad y_3 \geq 0 \end{aligned}$	$\begin{aligned} &3 y_1 + y_2 + 3 y_3 \\ &2 y_2 + y_3 \geq 1 \\ &y_1 + 2 y_2 + 2 y_3 \geq 2 \\ &- y_1 - y_2 + 2 y_3 \geq -1 \\ &- y_1, \quad y_3 \geq 0 \end{aligned}$
johnston-linear-matrix-align- EQ1514	468	1.000	$A = \left[\begin{array}{ll} 1 & 2 \\ 2 & 3 \end{array} \right],$	$A = \left[\begin{array}{ll} 1 & 2 \\ 2 & 3 \end{array} \right],$	$A = \left[\begin{array}{ll} 1 & 2 \\ 2 & 3 \end{array} \right],$
johnston-linear-matrix-align- EQ1516	469	1.000	$\mathbf{c} \cdot \mathbf{x} = \sum_{j=1}^n c_j x_j \geq \sum_{j=1}^n c_j y_j = \mathbf{c} \cdot \mathbf{y}.$	$\mathbf{c} \cdot \mathbf{x} = \sum_{j=1}^n c_j x_j \geq \sum_{j=1}^n c_j y_j = \mathbf{c} \cdot \mathbf{y}.$	$\mathbf{c} \cdot \mathbf{x} = \sum_{j=1}^n c_j x_j \geq \sum_{j=1}^n c_j y_j = \mathbf{c} \cdot \mathbf{y}.$
johnston-linear-matrix-align- EQ1520	469	1.000	$\begin{aligned} &\text{maximize: } \mathbf{c} \cdot \mathbf{x} \\ &\text{subject to: } \mathbf{A} \mathbf{x} \leq \mathbf{b} \\ &\mathbf{x} \geq \mathbf{0} \end{aligned}$	$\begin{aligned} &\text{maximize: } \mathbf{c} \cdot \mathbf{x} \\ &\text{subject to: } \mathbf{A} \mathbf{x} \leq \mathbf{b} \\ &\mathbf{x} \geq \mathbf{0} \end{aligned}$	$\begin{aligned} &\text{maximize: } \mathbf{c} \cdot \mathbf{x} \\ &\text{subject to: } \mathbf{A} \mathbf{x} \leq \mathbf{b} \\ &\mathbf{x} \geq \mathbf{0} \end{aligned}$
johnston-linear-matrix-align- EQ1524	470	1.000	$\left[\begin{array}{ll} 1 & 2 \\ 1 & 3 \end{array} \right] .$	$\left[\begin{array}{ll} 1 & 2 \\ 1 & 3 \end{array} \right] .$	$\left[\begin{array}{ll} 1 & 2 \\ 1 & 3 \end{array} \right] .$
johnston-linear-matrix-align- EQ1525	470	1.000	$\left[\begin{array}{ll} 1 & 2 \\ 4 & 5 \end{array} \right] .$	$\left[\begin{array}{ll} 1 & 2 \\ 4 & 5 \end{array} \right] .$	$\left[\begin{array}{ll} 1 & 2 \\ 4 & 5 \end{array} \right] .$
johnston-linear-matrix-align- EQ1526	470	1.000	$\left[\begin{array}{ccc} 4 & 5 & 0 \\ 2 & 2 & 1 \\ -4 & 0 & 3 \end{array} \right]$	$\left[\begin{array}{ccc} 4 & 5 & 0 \\ 2 & 2 & 1 \\ -4 & 0 & 3 \end{array} \right]$	$\left[\begin{array}{ccc} 4 & 5 & 0 \\ 2 & 2 & 1 \\ -4 & 0 & 3 \end{array} \right]$
johnston-linear-matrix-align- EQ1527	470	1.000	$\left[\begin{array}{lll} 1 & 1 & 1 \\ 1 & 0 & 0 \\ 1 & 0 & 0 \end{array} \right] .$	$\left[\begin{array}{lll} 1 & 1 & 1 \\ 1 & 0 & 0 \\ 1 & 0 & 0 \end{array} \right] .$	$\left[\begin{array}{lll} 1 & 1 & 1 \\ 1 & 0 & 0 \\ 1 & 0 & 0 \end{array} \right] .$
johnston-linear-matrix-align- EQ1528	470	1.000	$R = \left[\begin{array}{cc} I_{\text{rank}(A)} & O \\ O & O \end{array} \right] Q,$	$R = \left[\begin{array}{cc} I_{\text{rank}(A)} & O \\ O & O \end{array} \right] Q,$	$R = \left[\begin{array}{cc} I_{\text{rank}(A)} & O \\ O & O \end{array} \right] Q,$
johnston-linear-matrix-align- EQ1530	470	1.000	$A = P_{\{1\}} P_{\{2\}}^{-1} B Q_{\{2\}}^{-1} Q_{\{1\}},$	$A = P_1 P_2^{-1} B Q_2^{-1} Q_1,$	$A = P_1 P_2^{-1} B Q_2^{-1} Q_1,$
johnston-linear-matrix-align- EQ1534	471	1.000	$\begin{aligned} &[3], \quad \& \& \left[\begin{array}{cc} 3 & 1 \\ -1 & 1 \end{array} \right], \quad \& \& \text{and} \quad \& \& \left[\begin{array}{ccc} 3 & 1 & -1 \\ -1 & 1 & 3 \\ 2 & -2 & 1 \end{array} \right], \\ &\& \& \left[\begin{array}{ccc} 3 & 1 & -1 \\ -1 & 1 & 3 \\ 2 & -2 & 1 \end{array} \right], \end{aligned}$	$\begin{aligned} &[3], \quad \& \& \left[\begin{array}{cc} 3 & 1 \\ -1 & 1 \end{array} \right], \quad \& \& \text{and} \quad \& \& \left[\begin{array}{ccc} 3 & 1 & -1 \\ -1 & 1 & 3 \\ 2 & -2 & 1 \end{array} \right], \\ &\& \& \left[\begin{array}{ccc} 3 & 1 & -1 \\ -1 & 1 & 3 \\ 2 & -2 & 1 \end{array} \right], \end{aligned}$	$\begin{aligned} &[3], \quad \& \& \left[\begin{array}{cc} 3 & 1 \\ -1 & 1 \end{array} \right], \quad \& \& \text{and} \quad \& \& \left[\begin{array}{ccc} 3 & 1 & -1 \\ -1 & 1 & 3 \\ 2 & -2 & 1 \end{array} \right], \\ &\& \& \left[\begin{array}{ccc} 3 & 1 & -1 \\ -1 & 1 & 3 \\ 2 & -2 & 1 \end{array} \right], \end{aligned}$
johnston-linear-matrix-align- EQ1537	471	1.000	$A = \left[\begin{array}{ll} 0 & 0 \\ 1 & 1 \end{array} \right] \left[\begin{array}{ll} 1 & 0 \\ 0 & 1 \end{array} \right] .$	$A = \left[\begin{array}{ll} 0 & 0 \\ 1 & 1 \end{array} \right] \left[\begin{array}{ll} 1 & 0 \\ 0 & 1 \end{array} \right] .$	$A = \left[\begin{array}{ll} 0 & 0 \\ 1 & 1 \end{array} \right] \left[\begin{array}{ll} 1 & 0 \\ 0 & 1 \end{array} \right] .$

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johnston-linear-matrix-atgevr2 EQ1539	472	1.000	$P=\left[\begin{array}{ll} 0 & 1 \\ 1 & 0 \end{array}\right], \quad L=\left[\begin{array}{ll} 1 & 0 \\ 0 & 1 \end{array}\right], \quad U=\left[\begin{array}{ll} 1 & 1 \\ 0 & 0 \end{array}\right].$	$P=\begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}, \quad L=\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}, \quad U=\begin{bmatrix} 1 & 1 \\ 0 & 0 \end{bmatrix}.$	$P=\begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}, \quad L=\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}, \quad U=\begin{bmatrix} 1 & 1 \\ 0 & 0 \end{bmatrix}.$
johnston-linear-matrix-atgevr2 EQ1540	472	1.000	$A=\left[\begin{array}{ll} 1 & 0 \\ 2 & 1 \end{array}\right]\left[\begin{array}{cc} 2 & -1 \\ 0 & 1 \end{array}\right],$	$A=\begin{bmatrix} 1 & 0 \\ 2 & 1 \end{bmatrix}\begin{bmatrix} 2 & -1 \\ 0 & 1 \end{bmatrix},$	$A=\begin{bmatrix} 1 & 0 \\ 2 & 1 \end{bmatrix}\begin{bmatrix} 2 & -1 \\ 0 & 1 \end{bmatrix},$
johnston-linear-matrix-atgevr2 EQ1541	472	1.000	$A=\left[\begin{array}{ll} 0 & 1 \\ 1 & 0 \end{array}\right]\left[\begin{array}{cc} 1 & 0 \\ 1/2 & 1 \end{array}\right]\left[\begin{array}{cc} 4 & -1 \\ 0 & -1/2 \end{array}\right].$	$A=\begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}\begin{bmatrix} 1 & 0 \\ 1/2 & 1 \end{bmatrix}\begin{bmatrix} 4 & -1 \\ 0 & -1/2 \end{bmatrix}.$	$A=\begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}\begin{bmatrix} 1 & 0 \\ 1/2 & 1 \end{bmatrix}\begin{bmatrix} 4 & -1 \\ 0 & -1/2 \end{bmatrix}.$
johnston-linear-matrix-atgevr2 EQ1542	472	1.000	$\left[P^{\mathrm{T}}P\right]_{i,j}=\mathbf{p}_{\mathrm{i}}\cdot\mathbf{p}_{\mathrm{j}}.$	$\left[P^TP\right]_{i,j}=\mathbf{p}_i\cdot\mathbf{p}_j.$	$\left[P^TP\right]_{i,j}=\mathbf{p}_i\cdot\mathbf{p}_j.$
johnston-linear-matrix-atgevr2 EQ1544	472	1.000	$P=\left[\begin{array}{cc} 1 & -2 \\ -1 & 1 \end{array}\right].$	$P=\begin{bmatrix} 1 & -2 \\ -1 & 1 \end{bmatrix}.$	$P=\begin{bmatrix} 1 & -2 \\ -1 & 1 \end{bmatrix}.$
johnston-linear-matrix-atgevr2 EQ1545	472	1.000	$A=\left[\begin{array}{ll} 1 & 0 \\ 0 & 1 \end{array}\right], \quad B=\left[\begin{array}{ll} 1 & 1 \\ 0 & 1 \end{array}\right],$	$A=\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}, \quad B=\begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix},$	$A=\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}, \quad B=\begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix},$
johnston-linear-matrix-atgevr2 EQ1547	473	1.000	$\mathbf{v}=c_1\mathbf{v}_1+c_2\mathbf{v}_2+\cdots+c_k\mathbf{v}_k.$	$\mathbf{v}=c_1\mathbf{v}_1+c_2\mathbf{v}_2+\cdots+c_k\mathbf{v}_k.$	$\mathbf{v}=c_1\mathbf{v}_1+c_2\mathbf{v}_2+\cdots+c_k\mathbf{v}_k.$
johnston-linear-matrix-atgevr2 EQ1548	473	1.000	$\mathbf{0}=d_1\mathbf{v}_1+d_2\mathbf{v}_2+\cdots+d_k\mathbf{v}_k.$	$\mathbf{0}=d_1\mathbf{v}_1+d_2\mathbf{v}_2+\cdots+d_k\mathbf{v}_k.$	$\mathbf{0}=d_1\mathbf{v}_1+d_2\mathbf{v}_2+\cdots+d_k\mathbf{v}_k.$
johnston-linear-matrix-atgevr2 EQ1551	473	1.000	$\mathbf{v}+\mathbf{w}=\left(c_1+d_1\right)\mathbf{v}_1+\cdots+\left(c_k+d_k\right)\mathbf{v}_k,$	$\mathbf{v}+\mathbf{w}=\left(c_1+d_1\right)\mathbf{v}_1+\cdots+\left(c_k+d_k\right)\mathbf{v}_k,$	$\mathbf{v}+\mathbf{w}=\left(c_1+d_1\right)\mathbf{v}_1+\cdots+\left(c_k+d_k\right)\mathbf{v}_k,$
johnston-linear-matrix-atgevr2 EQ1552	473	1.000	$\mathbf{v}=d_1\mathbf{v}_1+d_2\mathbf{v}_2+\cdots+d_k\mathbf{v}_k.$	$\mathbf{v}=d_1\mathbf{v}_1+d_2\mathbf{v}_2+\cdots+d_k\mathbf{v}_k.$	$\mathbf{v}=d_1\mathbf{v}_1+d_2\mathbf{v}_2+\cdots+d_k\mathbf{v}_k.$
johnston-linear-matrix-atgevr2 EQ1553	473	1.000	$c\,\mathbf{v}=\left(c\,d_1\right)\mathbf{v}_1+\left(c\,d_2\right)\mathbf{v}_2+\cdots+\left(c\,d_k\right)\mathbf{v}_k,$	$c\mathbf{v}=\left(cd_1\right)\mathbf{v}_1+\left(cd_2\right)\mathbf{v}_2+\cdots+\left(cd_k\right)\mathbf{v}_k,$	$c\mathbf{v}=\left(cd_1\right)\mathbf{v}_1+\left(cd_2\right)\mathbf{v}_2+\cdots+\left(cd_k\right)\mathbf{v}_k,$
johnston-linear-matrix-atgevr2 EQ1554	473	1.000	$c_1\mathbf{w}_1+\cdots+c_m\mathbf{w}_m=\mathbf{0}$	$c_1\mathbf{w}_1+\cdots+c_m\mathbf{w}_m=\mathbf{0}$	$c_1\mathbf{w}_1+\cdots+c_m\mathbf{w}_m=\mathbf{0}$
johnston-linear-matrix-atgevr2 EQ1556	473	1.000	$c_1\mathbf{w}_1+\cdots+c_m\mathbf{w}_m=\mathbf{v}$	$c_1\mathbf{w}_1+\cdots+c_m\mathbf{w}_m=\mathbf{v}$	$c_1\mathbf{w}_1+\cdots+c_m\mathbf{w}_m=\mathbf{v}$
johnston-linear-matrix-atgevr2 EQ1558	473	1.000	$P\stackrel{\mathrm{def}}{=}[[\mathbf{v}_1]_B\, [\mathbf{v}_2]_B\, \cdots\, [\mathbf{v}_k]_B]$	$P\stackrel{\mathrm{def}}{=}[[\mathbf{v}_1]_B\, [\mathbf{v}_2]_B\, \cdots\, [\mathbf{v}_k]_B]$	$P\stackrel{\mathrm{def}}{=}[\, [\mathbf{v}_1]_B \, \, [\mathbf{v}_2]_B \, \, \cdots \, \, [\mathbf{v}_k]_B \,]$
johnston-linear-matrix-atgevr2 EQ1559	473	1.000	$\{\left[\mathbf{v}_1\right]_B,\left[\mathbf{v}_2\right]_B,\ldots,\left[\mathbf{v}_k\right]_B\}$	$\{[\mathbf{v}_1]_B,[\mathbf{v}_2]_B,\ldots,[\mathbf{v}_k]_B\}$	$\{[\mathbf{v}_1]_B,[\mathbf{v}_2]_B,\ldots,[\mathbf{v}_k]_B\}$
johnston-linear-matrix-atgevr2 EQ1561	473	1.000	$\left[I\mid P_{E\leftarrow C}^{-1}P_{E\leftarrow B}\right]=\left[I\mid P_{C\leftarrow E}P_{E\leftarrow B}\right],$	$\left[I\mid P_{E\leftarrow C}^{-1}P_{E\leftarrow B}\right]=\left[I\mid P_{C\leftarrow E}P_{E\leftarrow B}\right],$	$\left[I\mid P_{E\leftarrow C}^{-1}P_{E\leftarrow B}\right]=\left[I\mid P_{C\leftarrow E}P_{E\leftarrow B}\right],$
johnston-linear-matrix-atgevr2 EQ1562	473	1.000	$\left[\begin{array}{cccc} 1 & 1 & 5/6 & 3/2 \\ -2 & -1 & -2/3 & 1 \\ 1 & 2 & 4/3 & 3 \\ 1 & 1 & 1/12 & -1/4 \end{array}\right]$	$\begin{bmatrix} 1 & 1 & 5/6 & 3/2 \\ -2 & -1 & -2/3 & 1 \\ 1 & 2 & 4/3 & 3 \\ 1 & 1 & 1/12 & -1/4 \end{bmatrix}$	$\begin{bmatrix} 1 & 1 & 5/6 & 3/2 \\ -2 & -1 & -2/3 & 1 \\ 1 & 2 & 4/3 & 3 \\ 1 & 1 & 1/12 & -1/4 \end{bmatrix}$
johnston-linear-matrix-atgevr2 EQ1564	473	1.000	$n\text{-}\operatorname{nullity}(A)=n\text{-}\operatorname{nullity}(B),$	$n-\operatorname{nullity}(A)=n-\operatorname{nullity}(B),$	$n-\operatorname{nullity}(A)=n-\operatorname{nullity}(B),$

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johnston-linear-matrix-atdewire EQ1565	474	■ 1.000	$\begin{aligned} \operatorname{tr}(A+B) &= \sum_{j=1}^n [A+B]_{j,j} = \sum_{j=1}^n a_{j,j} + b_{j,j} \\ &= \sum_{j=1}^n a_{j,j} + \sum_{j=1}^n b_{j,j} = \operatorname{tr}(A) + \operatorname{tr}(B). \end{aligned}$	$\operatorname{tr}(A+B) = \sum_{j=1}^n [A+B]_{j,j} = \sum_{j=1}^n a_{j,j} + b_{j,j}$ $= \sum_{j=1}^n a_{j,j} + \sum_{j=1}^n b_{j,j} = \operatorname{tr}(A) + \operatorname{tr}(B).$	$\operatorname{tr}(A+B) = \sum_{j=1}^n [A+B]_{j,j} = \sum_{j=1}^n a_{j,j} + b_{j,j}$ $= \sum_{j=1}^n a_{j,j} + \sum_{j=1}^n b_{j,j} = \operatorname{tr}(A) + \operatorname{tr}(B).$
johnston-linear-matrix-atdewire EQ1566	474	■ 1.000	$\begin{aligned} \operatorname{tr}(cA) &= \sum_{j=1}^n [cA]_{j,j} = \sum_{j=1}^n ca_{j,j} \\ &= c \sum_{j=1}^n a_{j,j} = c \operatorname{tr}(A). \end{aligned}$	$\operatorname{tr}(cA) = \sum_{j=1}^n [cA]_{j,j} = \sum_{j=1}^n ca_{j,j}$ $= c \sum_{j=1}^n a_{j,j} = c \operatorname{tr}(A).$	$\operatorname{tr}(cA) = \sum_{j=1}^n [cA]_{j,j} = \sum_{j=1}^n ca_{j,j}$ $= c \sum_{j=1}^n a_{j,j} = \operatorname{ctr}(A).$
johnston-linear-matrix-atdewire EQ1569 (2)	474	■ 1.000	$f(A) = f\left(\sum_{j,k=1}^n a_{j,k} \mathbf{e}_j \mathbf{e}_k^T\right) = \sum_{j,k=1}^n a_{j,k} f(\mathbf{e}_j \mathbf{e}_k^T) = 0$	$f(A) = f\left(\sum_{j,k=1}^n a_{j,k} \mathbf{e}_j \mathbf{e}_k^T\right) = \sum_{j,k=1}^n a_{j,k} f(\mathbf{e}_j \mathbf{e}_k^T) = 0$	$f(A) = f\left(\sum_{j,k=1}^n a_{j,k} \mathbf{e}_j \mathbf{e}_k^T\right) = \sum_{j,k=1}^n a_{j,k} f(\mathbf{e}_j \mathbf{e}_k^T) = 0$
johnston-linear-matrix-atdewire EQ1571	474	■ 1.000	$[\mathbf{v}]_B \cdot [\mathbf{w}]_B = c_1 d_1 + c_2 d_2 + \cdots + c_n d_n$	$[\mathbf{v}]_B \cdot [\mathbf{w}]_B = c_1 d_1 + c_2 d_2 + \cdots + c_n d_n$	$[\mathbf{v}]_B \cdot [\mathbf{w}]_B = c_1 d_1 + c_2 d_2 + \cdots + c_n d_n$
johnston-linear-matrix-atdewire EQ1574	474	■ 1.000	$c_1 \mathbf{v}_1 + c_2 \mathbf{v}_2 + \cdots + c_k \mathbf{v}_k = \mathbf{0}.$	$c_1 \mathbf{v}_1 + c_2 \mathbf{v}_2 + \cdots + c_k \mathbf{v}_k = \mathbf{0}.$	$c_1 \mathbf{v}_1 + c_2 \mathbf{v}_2 + \cdots + c_k \mathbf{v}_k = \mathbf{0}.$
johnston-linear-matrix-atdewire EQ1575	474	■ 1.000	$\begin{aligned} 0 &= \mathbf{v}_1 \cdot \mathbf{0} \\ &= \mathbf{v}_1 \cdot (c_1 \mathbf{v}_1 + c_2 \mathbf{v}_2 + \cdots + c_k \mathbf{v}_k) \\ &= c_1 (\mathbf{v}_1 \cdot \mathbf{v}_1) + c_2 (\mathbf{v}_1 \cdot \mathbf{v}_2) + \cdots + c_k (\mathbf{v}_1 \cdot \mathbf{v}_k) \\ &= c_1 \ \mathbf{v}_1\ ^2 + 0 + \cdots + 0 \end{aligned}$	$0 = \mathbf{v}_1 \cdot \mathbf{0}$ $= \mathbf{v}_1 \cdot (c_1 \mathbf{v}_1 + c_2 \mathbf{v}_2 + \cdots + c_k \mathbf{v}_k)$ $= c_1 (\mathbf{v}_1 \cdot \mathbf{v}_1) + c_2 (\mathbf{v}_1 \cdot \mathbf{v}_2) + \cdots + c_k (\mathbf{v}_1 \cdot \mathbf{v}_k)$ $= c_1 \ \mathbf{v}_1\ ^2 + 0 + \cdots + 0$	$0 = \mathbf{v}_1 \cdot \mathbf{0}$ $= \mathbf{v}_1 \cdot (c_1 \mathbf{v}_1 + c_2 \mathbf{v}_2 + \cdots + c_k \mathbf{v}_k)$ $= c_1 (\mathbf{v}_1 \cdot \mathbf{v}_1) + c_2 (\mathbf{v}_1 \cdot \mathbf{v}_2) + \cdots + c_k (\mathbf{v}_1 \cdot \mathbf{v}_k)$ $= c_1 \ \mathbf{v}_1\ ^2 + 0 + \cdots + 0.$
johnston-linear-matrix-atdewire EQ1580	475	■ 1.000	$\begin{aligned} \operatorname{det}(cA) &= \operatorname{det}([\mathbf{c}\mathbf{a}_1 \mid \mathbf{c}\mathbf{a}_2 \mid \cdots \mid \mathbf{c}\mathbf{a}_n]) \\ &= c \operatorname{det}([\mathbf{a}_1 \mid \mathbf{a}_2 \mid \cdots \mid \mathbf{a}_n]) \\ &= c^2 \operatorname{det}([\mathbf{a}_1 \mid \mathbf{a}_2 \mid \cdots \mid \mathbf{a}_n]) \end{aligned}$	$\operatorname{det}(cA) = \operatorname{det}([\mathbf{c}\mathbf{a}_1 \mid \mathbf{c}\mathbf{a}_2 \mid \cdots \mid \mathbf{c}\mathbf{a}_n])$ $= c \operatorname{det}([\mathbf{a}_1 \mid \mathbf{a}_2 \mid \cdots \mid \mathbf{a}_n])$ $= c^2 \operatorname{det}([\mathbf{a}_1 \mid \mathbf{a}_2 \mid \cdots \mid \mathbf{a}_n])$	$\operatorname{det}(cA) = \operatorname{det}([\mathbf{c}\mathbf{a}_1 \mid \mathbf{c}\mathbf{a}_2 \mid \cdots \mid \mathbf{c}\mathbf{a}_n])$ $= c \operatorname{det}([\mathbf{a}_1 \mid \mathbf{a}_2 \mid \cdots \mid \mathbf{a}_n])$ $= c^2 \operatorname{det}([\mathbf{a}_1 \mid \mathbf{a}_2 \mid \cdots \mid \mathbf{a}_n])$
johnston-linear-matrix-atdewire EQ1584	476	■ 1.000	$\operatorname{det}\left(\left[\begin{array}{cc} I_k & B \\ 0 & C \end{array}\right]\right) = \operatorname{det}\left(\left[\begin{array}{cc} I_{k-1} & \tilde{B} \\ 0 & C \end{array}\right]\right),$	$\operatorname{det}\left(\left[\begin{array}{cc} I_k & B \\ 0 & C \end{array}\right]\right) = \operatorname{det}\left(\left[\begin{array}{cc} I_{k-1} & \tilde{B} \\ 0 & C \end{array}\right]\right),$	$\operatorname{det}\left(\left[\begin{array}{cc} I_k & B \\ 0 & C \end{array}\right]\right) = \operatorname{det}\left(\left[\begin{array}{cc} I_{k-1} & \tilde{B} \\ 0 & C \end{array}\right]\right),$

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johnston-linear-matrix-algebra-EQ1587	476	1.000	$\begin{aligned} & \left[\begin{array}{ccccc} A_1 & * & \cdots & * & * \\ O & A_2 & \cdots & * & * \\ \vdots & \vdots & \ddots & \vdots & \vdots \\ O & O & \cdots & A_n & * \\ O & O & \cdots & O & A_{n+1} \end{array} \right] \\ & = \left[\begin{array}{cccc c} A_1 & * & \cdots & * & * \\ O & A_2 & \cdots & * & * \\ \vdots & \vdots & \ddots & \vdots & \vdots \\ O & O & \cdots & A_n & * \\ \hline O & O & \cdots & O & A_{n+1} \end{array} \right] . \end{aligned}$	$\begin{bmatrix} A_1 & * & \cdots & * & * \\ O & A_2 & \cdots & * & * \\ \vdots & \vdots & \ddots & \vdots & \vdots \\ O & O & \cdots & A_n & * \\ O & O & \cdots & O & A_{n+1} \end{bmatrix} = \left[\begin{array}{cccc c} A_1 & * & \cdots & * & * \\ O & A_2 & \cdots & * & * \\ \vdots & \vdots & \ddots & \vdots & \vdots \\ O & O & \cdots & A_n & * \\ \hline O & O & \cdots & O & A_{n+1} \end{array} \right] .$	
johnston-linear-matrix-algebra-EQ1588	476	1.000	$\begin{aligned} & \operatorname{det} \left(\left[\begin{array}{cccc c} A_1 & * & \cdots & * & * \\ O & A_2 & \cdots & * & * \\ \vdots & \vdots & \ddots & \vdots & \vdots \\ O & O & \cdots & A_n & * \\ \hline O & O & \cdots & O & A_{n+1} \end{array} \right] \right) \\ & = \operatorname{det} \left(\left[\begin{array}{cccc c} A_1 & * & \cdots & * & * \\ O & A_2 & \cdots & * & * \\ \vdots & \vdots & \ddots & \vdots & \vdots \\ O & O & \cdots & A_n & * \\ \hline O & O & \cdots & O & A_{n+1} \end{array} \right] \right) \operatorname{det} (A_{n+1}) \\ & = \left(\prod_{j=1}^n \operatorname{det} (A_j) \right) \operatorname{det} (A_{n+1}) = \prod_{j=1}^{n+1} \operatorname{det} (A_j) , \end{aligned}$	$\det \left(\left[\begin{array}{cccc c} A_1 & * & \cdots & * & * \\ O & A_2 & \cdots & * & * \\ \vdots & \vdots & \ddots & \vdots & \vdots \\ O & O & \cdots & A_n & * \\ \hline O & O & \cdots & O & A_{n+1} \end{array} \right] \right) = \det \left(\left[\begin{array}{cccc c} A_1 & * & \cdots & * & * \\ O & A_2 & \cdots & * & * \\ \vdots & \vdots & \ddots & \vdots & \vdots \\ O & O & \cdots & A_n & * \\ \hline O & O & \cdots & O & A_{n+1} \end{array} \right] \right) \det(A_{n+1}) = \left(\prod_{j=1}^n \det(A_j) \right) \det(A_{n+1}) = \prod_{j=1}^{n+1} \det(A_j) ,$	
johnston-linear-matrix-algebra-EQ1589	476	1.000	$\left[\begin{array}{cc} I_m & -A \\ B & I_n \end{array} \right] \left[\begin{array}{cc} I_m & A \\ O & I_n \end{array} \right] = \left[\begin{array}{cc} I_m & O \\ B & I_n + BA \end{array} \right] .$	$\left[\begin{array}{cc} I_m & -A \\ B & I_n \end{array} \right] \left[\begin{array}{cc} I_m & A \\ O & I_n \end{array} \right] = \left[\begin{array}{cc} I_m & O \\ B & I_n + BA \end{array} \right] .$	$\left[\begin{array}{cc} I_m & -A \\ B & I_n \end{array} \right] \left[\begin{array}{cc} I_m & A \\ O & I_n \end{array} \right] = \left[\begin{array}{cc} I_m & O \\ B & I_n + BA \end{array} \right] .$
johnston-linear-matrix-algebra-EQ1594	476	1.000	$\operatorname{det} \left(\left[\begin{array}{cc} I_n & \mathbf{v} \\ \mathbf{0}^T & \mathbf{w}^T \mathbf{v} + 1 \end{array} \right] \right) = \mathbf{w}^T \mathbf{v} + 1 ,$	$\det \left(\left[\begin{array}{cc} I_n & \mathbf{v} \\ \mathbf{0}^T & \mathbf{w}^T \mathbf{v} + 1 \end{array} \right] \right) = \mathbf{w}^T \mathbf{v} + 1 ,$	$\det \left(\left[\begin{array}{cc} I_n & \mathbf{v} \\ \mathbf{0}^T & \mathbf{w}^T \mathbf{v} + 1 \end{array} \right] \right) = \mathbf{w}^T \mathbf{v} + 1 ,$
johnston-linear-matrix-algebra-EQ1597	477	1.000	$\left[\begin{array}{cc c} 1 & 2 & 0 \\ -1 & -2 & 0 \end{array} \right] \xrightarrow{R_2+R_1} \left[\begin{array}{cc c} 1 & 2 & 0 \\ 0 & 0 & 0 \end{array} \right] .$	$\left[\begin{array}{cc c} 1 & 2 & 0 \\ -1 & -2 & 0 \end{array} \right] \xrightarrow{R_2+R_1} \left[\begin{array}{cc c} 1 & 2 & 0 \\ 0 & 0 & 0 \end{array} \right] .$	$\left[\begin{array}{cc c} 1 & 2 & 0 \\ -1 & -2 & 0 \end{array} \right] \xrightarrow{R_2+R_1} \left[\begin{array}{cc c} 1 & 2 & 0 \\ 0 & 0 & 0 \end{array} \right] .$

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johnston-linear-matrix-algebra-EQ1599	477	1.000	$\left[\begin{array}{ll} 1 & 0 \\ 0 & 2 \end{array}\right], \quad \left[\begin{array}{ll} 3 & 0 \\ 0 & 4 \end{array}\right],$	$\left[\begin{array}{cc} 1 & 0 \\ 0 & 2 \end{array}\right], \quad \left[\begin{array}{cc} 3 & 0 \\ 0 & 4 \end{array}\right],$	$\left[\begin{array}{cc} 1 & 0 \\ 0 & 2 \end{array}\right], \quad \left[\begin{array}{cc} 3 & 0 \\ 0 & 4 \end{array}\right],$
johnston-linear-matrix-algebra-EQ1602 (2)	477	1.000	$\lambda=\frac{\operatorname{tr}(A) \pm \sqrt{\operatorname{tr}(A)^2-4 \operatorname{det}(A)}}{2}$.	$\lambda=\frac{\operatorname{tr}(A) \pm \sqrt{\operatorname{tr}(A)^2-4 \operatorname{det}(A)}}{2}.$	$\lambda=\frac{\operatorname{tr}(A) \pm \sqrt{\operatorname{tr}(A)^2-4 \operatorname{det}(A)}}{2}.$
johnston-linear-matrix-algebra-EQ1603	478	1.000	$\begin{aligned} & \operatorname{det}\left(\left[\begin{array}{cc} \cos (\theta)-\lambda & -\sin (\theta) \\ \sin (\theta) & \cos (\theta)-\lambda \end{array}\right]\right) \\ & =(\cos (\theta)-\lambda)^2+\sin ^2(\theta) \\ & =\lambda^2-2 \cos (\theta) \lambda+1 \\ & =0 \end{aligned}$	$\begin{aligned} & \operatorname{det}\left(\left[\begin{array}{cc} \cos (\theta)-\lambda & -\sin (\theta) \\ \sin (\theta) & \cos (\theta)-\lambda \end{array}\right]\right) \\ & =(\cos (\theta)-\lambda)^2+\sin ^2(\theta) \\ & =\lambda^2-2 \cos (\theta) \lambda+1 \\ & =0 \end{aligned}$	$\begin{aligned} & \operatorname{det}\left(\left[\begin{array}{cc} \cos (\theta)-\lambda & -\sin (\theta) \\ \sin (\theta) & \cos (\theta)-\lambda \end{array}\right]\right) \\ & =(\cos (\theta)-\lambda)^2+\sin ^2(\theta) \\ & =\lambda^2-2 \cos (\theta) \lambda+1 \\ & =0. \end{aligned}$
johnston-linear-matrix-algebra-EQ1609	478	1.000	$A=\left[\begin{array}{ll} 1 & 1 \\ 0 & 2 \end{array}\right],$	$A=\left[\begin{array}{cc} 1 & 1 \\ 0 & 2 \end{array}\right],$	$A=\left[\begin{array}{cc} 1 & 1 \\ 0 & 2 \end{array}\right],$
johnston-linear-matrix-algebra-EQ1610	478	1.000	$B=A-\lambda_1 \mathbf{v}_1 \mathbf{v}_1^*=\left[\begin{array}{ll} 0 & 1 \\ 0 & 2 \end{array}\right]$	$B=A-\lambda_1 \mathbf{v}_1 \mathbf{v}_1^*=\left[\begin{array}{cc} 0 & 1 \\ 0 & 2 \end{array}\right]$	$B=A-\lambda_1 \mathbf{v}_1 \mathbf{v}_1^*=\left[\begin{array}{cc} 0 & 1 \\ 0 & 2 \end{array}\right]$
johnston-linear-matrix-algebra-EQ1613	479	1.000	$\begin{aligned} & D=\left[\begin{array}{ll} 0 & 0 \\ 0 & 2 \end{array}\right], \quad P=\left[\begin{array}{cc} -1 & 1 \\ 1 & 1 \end{array}\right], \quad P^{-1}=\frac{1}{2}\left[\begin{array}{cc} -1 & 1 \\ 1 & 1 \end{array}\right]. \end{aligned}$	$D=\left[\begin{array}{cc} 0 & 0 \\ 0 & 2 \end{array}\right], \quad P=\left[\begin{array}{cc} -1 & 1 \\ 1 & 1 \end{array}\right], P^{-1}=\frac{1}{2}\left[\begin{array}{cc} -1 & 1 \\ 1 & 1 \end{array}\right].$	$D=\left[\begin{array}{cc} 0 & 0 \\ 0 & 2 \end{array}\right], \quad P=\left[\begin{array}{cc} -1 & 1 \\ 1 & 1 \end{array}\right], P^{-1}=\frac{1}{2}\left[\begin{array}{cc} -1 & 1 \\ 1 & 1 \end{array}\right].$
johnston-linear-matrix-algebra-EQ1617	479	1.000	$ \lambda =\left \sum_{j=1}^n a_{i, j} v_j\right \leq \sum_{j=1}^n\left a_{i, j} v_j\right \leq \sum_{j=1}^n\left a_{i, j}\right =1,$	$ \lambda =\left \sum_{j=1}^n a_{i, j} v_j\right \leq \sum_{j=1}^n\left a_{i, j} v_j\right \leq \sum_{j=1}^n\left a_{i, j}\right =1,$	$ \lambda =\left \sum_{j=1}^n a_{i, j} v_j\right \leq \sum_{j=1}^n\left a_{i, j} v_j\right \leq \sum_{j=1}^n\left a_{i, j}\right =1,$
johnston-linear-matrix-algebra-EQ1618	479	1.000	$\begin{aligned} & D=\left[\begin{array}{cc} 2+i & 0 \\ 0 & 2-i \end{array}\right], P=\left[\begin{array}{cc} 1 & 1 \\ i & -i \end{array}\right], \\ & P^{-1}=\frac{1}{2}\left[\begin{array}{cc} 1 & -i \\ 1 & i \end{array}\right]. \end{aligned}$	$D=\left[\begin{array}{cc} 2+i & 0 \\ 0 & 2-i \end{array}\right], P=\left[\begin{array}{cc} 1 & 1 \\ i & -i \end{array}\right], \\ P^{-1}=\frac{1}{2}\left[\begin{array}{cc} 1 & -i \\ 1 & i \end{array}\right].$	$D=\left[\begin{array}{cc} 2+i & 0 \\ 0 & 2-i \end{array}\right], P=\left[\begin{array}{cc} 1 & 1 \\ i & -i \end{array}\right], \\ P^{-1}=\frac{1}{2}\left[\begin{array}{cc} 1 & -i \\ 1 & i \end{array}\right].$
johnston-linear-matrix-algebra-EQ1620	480	1.000	$P=\left[\begin{array}{ccccc} -1 & -1 & 1 & -1 & -1 \\ 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 1 \\ -1 & 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & -1 & 0 \end{array}\right].$	$P=\left[\begin{array}{ccccc} -1 & -1 & 1 & -1 & -1 \\ 0 & 0 & 0 & 1 & 1 \\ -1 & 0 & 1 & 0 & 0 \\ -1 & 0 & 1 & -1 & 0 \\ 1 & 1 & 0 & 0 & 1 \end{array}\right].$	$P=\left[\begin{array}{ccccc} -1 & -1 & 1 & -1 & -1 \\ 0 & 0 & 0 & 1 & 1 \\ -1 & 0 & 1 & 0 & 0 \\ -1 & 0 & 1 & -1 & 0 \\ 1 & 1 & 0 & 0 & 1 \end{array}\right].$

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johnston-linear-matrix-align-0 EQ1621	480	1.000	P=\left[\begin{array}{ccccc} 2 & 2+i & 2 & 2-i & 2 \\ 1-i & -1 & 1+i & -1 & 0 \\ 2-i & -1-i & 2+i & -1+i & 2 \\ 2-i & 1 & 2+i & 1 & 2 \\ -i & 2 & i & 2 & 0 \end{array}\right] .	$P = \begin{bmatrix} 2 & 2+i & 2 & 2-i & 2 \\ 1-i & -1 & 1+i & -1 & 0 \\ 2-i & -1-i & 2+i & -1+i & 2 \\ 2-i & 1 & 2+i & 1 & 2 \\ -i & 2 & i & 2 & 0 \end{bmatrix} .$	$P = \begin{bmatrix} 2 & 2+i & 2 & 2-i & 2 \\ 1-i & -1 & 1+i & -1 & 0 \\ 2-i & -1-i & 2+i & -1+i & 2 \\ 2-i & 1 & 2+i & 1 & 2 \\ -i & 2 & i & 2 & 0 \end{bmatrix} .$
johnston-linear-matrix-align-0 EQ1622 (0)	480	1.000	\left[\begin{array}{ll} \left(e^2+1\right) / 2 & \left(e^2-1\right) / 2 \\ \left(e^2-1\right) / 2 & \left(e^2+1\right) / 2 \end{array}\right]	$\begin{bmatrix} \left(e^2+1\right) / 2 & \left(e^2-1\right) / 2 \\ \left(e^2-1\right) / 2 & \left(e^2+1\right) / 2 \end{bmatrix}$	$\begin{bmatrix} \left(e^2+1\right) / 2 & \left(e^2-1\right) / 2 \\ \left(e^2-1\right) / 2 & \left(e^2+1\right) / 2 \end{bmatrix}$
johnston-linear-matrix-align-0 EQ1623 (1)	480	1.000	\left[\begin{array}{ccccc} 5 & 1 & 1 & 2 & -2 \\ -4 & 0 & -1 & -2 & 2 \\ 5 & 2 & 2 & 3 & -2 \\ 1 & 1 & 0 & 2 & 0 \\ 5 & 2 & 1 & 3 & -1 \end{array}\right]	$\begin{bmatrix} 5 & 1 & 1 & 2 & -2 \\ -4 & 0 & -1 & -2 & 2 \\ 5 & 2 & 2 & 3 & -2 \\ 1 & 1 & 0 & 2 & 0 \\ 5 & 2 & 1 & 3 & -1 \end{bmatrix}$	$\begin{bmatrix} 5 & 1 & 1 & 2 & -2 \\ -4 & 0 & -1 & -2 & 2 \\ 5 & 2 & 2 & 3 & -2 \\ 1 & 1 & 0 & 2 & 0 \\ 5 & 2 & 1 & 3 & -1 \end{bmatrix}$
johnston-linear-matrix-align-0 EQ1624	480	1.000	\left[\begin{array}{ll} 1 & 0 \\ 0 & 0 \end{array}\right]	$\begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}$	$\begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}$
johnston-linear-matrix-align-0 EQ1625	480	1.000	D=\left[\begin{array}{cc} -1 & 0 \\ 0 & 8 \end{array}\right], P=\left[\begin{array}{cc} 1 & 2 \\ 1 & -1 \end{array}\right], P^{-1}=\frac{1}{3}\left[\begin{array}{cc} 1 & 2 \\ 1 & -1 \end{array}\right] .	$D = \begin{bmatrix} -1 & 0 \\ 0 & 8 \end{bmatrix}, P = \begin{bmatrix} 1 & 2 \\ 1 & -1 \end{bmatrix}, P^{-1} = \frac{1}{3} \begin{bmatrix} 1 & 2 \\ 1 & -1 \end{bmatrix} .$	$D = \begin{bmatrix} -1 & 0 \\ 0 & 8 \end{bmatrix}, P = \begin{bmatrix} 1 & 2 \\ 1 & -1 \end{bmatrix}, P^{-1} = \frac{1}{3} \begin{bmatrix} 1 & 2 \\ 1 & -1 \end{bmatrix} .$
johnston-linear-matrix-align-0 EQ1627 (0)	480	1.000	\begin{aligned} & \left[\begin{array}{c} L_{n+1} \\ L_n \end{array}\right] = \left[\begin{array}{cc} 1 & 1 \\ 1 & 0 \end{array}\right]^n \left[\begin{array}{c} 1 \\ 2 \end{array}\right] \\ & = PD^n P^{-1} \left[\begin{array}{c} 1 \\ 2 \end{array}\right] \\ & = \frac{1}{\sqrt{5}} PD^n \begin{bmatrix} 1 & \phi-1 \\ -1 & \phi \end{bmatrix} \begin{bmatrix} 1 \\ 2 \end{bmatrix} \\ & = P \begin{bmatrix} \phi^n & 0 \\ 0 & (1-\phi)^n \end{bmatrix} \begin{bmatrix} 1 \\ 1 \end{bmatrix} \\ & = \begin{bmatrix} \phi & 1-\phi \\ 1 & 1 \end{bmatrix} \begin{bmatrix} \phi^n \\ (1-\phi)^n \end{bmatrix} \\ & = \begin{bmatrix} \phi^{n+1} + (1-\phi)^{n+1} \\ \phi^n + (1-\phi)^n \end{bmatrix}, \end{aligned}	$\begin{aligned} \begin{bmatrix} L_{n+1} \\ L_n \end{bmatrix} &= \begin{bmatrix} 1 & 1 \\ 1 & 0 \end{bmatrix}^n \begin{bmatrix} 1 \\ 2 \end{bmatrix} \\ &= PD^n P^{-1} \begin{bmatrix} 1 \\ 2 \end{bmatrix} \\ &= \frac{1}{\sqrt{5}} PD^n \begin{bmatrix} 1 & \phi-1 \\ -1 & \phi \end{bmatrix} \begin{bmatrix} 1 \\ 2 \end{bmatrix} \\ &= P \begin{bmatrix} \phi^n & 0 \\ 0 & (1-\phi)^n \end{bmatrix} \begin{bmatrix} 1 \\ 1 \end{bmatrix} \\ &= \begin{bmatrix} \phi & 1-\phi \\ 1 & 1 \end{bmatrix} \begin{bmatrix} \phi^n \\ (1-\phi)^n \end{bmatrix} \\ &= \begin{bmatrix} \phi^{n+1} + (1-\phi)^{n+1} \\ \phi^n + (1-\phi)^n \end{bmatrix}, \end{aligned}$	$\begin{aligned} \begin{bmatrix} L_{n+1} \\ L_n \end{bmatrix} &= \begin{bmatrix} 1 & 1 \\ 1 & 0 \end{bmatrix}^n \begin{bmatrix} 1 \\ 2 \end{bmatrix} \\ &= PD^n P^{-1} \begin{bmatrix} 1 \\ 2 \end{bmatrix} \\ &= \frac{1}{\sqrt{5}} PD^n \begin{bmatrix} 1 & \phi-1 \\ -1 & \phi \end{bmatrix} \begin{bmatrix} 1 \\ 2 \end{bmatrix} \\ &= P \begin{bmatrix} \phi^n & 0 \\ 0 & (1-\phi)^n \end{bmatrix} \begin{bmatrix} 1 \\ 1 \end{bmatrix} \\ &= \begin{bmatrix} \phi & 1-\phi \\ 1 & 1 \end{bmatrix} \begin{bmatrix} \phi^n \\ (1-\phi)^n \end{bmatrix} \\ &= \begin{bmatrix} \phi^{n+1} + (1-\phi)^{n+1} \\ \phi^n + (1-\phi)^n \end{bmatrix}, \end{aligned}$
johnston-linear-matrix-align-0 EQ1628	480	1.000	L_n=\phi^n+(1-\phi)^n .	$L_n = \phi^n + (1-\phi)^n .$	$L_n = \phi^n + (1-\phi)^n .$

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johnston-linear-matrix-align-01 EQ1629	480	■ 1.000	$A=\left[\begin{array}{ll} 0 & 1 \\ 0 & 0 \end{array}\right]$	$A=\begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}$	$A=\begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}$
johnston-linear-matrix-align-01 EQ1631	481	■ 1.000	$Q=\left[c_{\{1\}}\ \mathbf{v}_{\{1\}}+c_{\{2\}}\ \mathbf{v}_{\{2\}}\right.\left.\left d_{\{1\}}\ \mathbf{v}_{\{1\}}+d_{\{2\}}\ \mathbf{v}_{\{2\}}\right P_{\{2\}}\right]$	$Q=[c_1\mathbf{v}_1+c_2\mathbf{v}_2 d_1\mathbf{v}_1+d_2\mathbf{v}_2 P_2]$	$Q=[c_1\mathbf{v}_1+c_2\mathbf{v}_2 d_1\mathbf{v}_1+d_2\mathbf{v}_2 P_2]$
johnston-linear-matrix-align-01 EQ1636	481	■ 1.000	$\begin{aligned} C\ \mathbf{v}_{\{j\}} &= \left[\begin{array}{cccc} 0 & 1 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & 1 \\ -a_0 & -a_1 & \cdots & -a_{n-1} \end{array}\right] \left[\begin{array}{c} 1 \\ \lambda_j \\ \vdots \\ \lambda_j^{n-2} \\ \lambda_j^{n-1} \end{array}\right] \\ &= \left[\begin{array}{c} \lambda_j \\ \lambda_j^2 \\ \vdots \\ \lambda_j^{n-1} \\ -a_0-a_1\lambda_j-\cdots-a_{n-1}\lambda_j^{n-1} \end{array}\right] = \lambda_j\mathbf{v}_j \end{aligned}$	$C\mathbf{v}_j=\begin{bmatrix} 0 & 1 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & 1 \\ -a_0 & -a_1 & \cdots & -a_{n-1} \end{bmatrix}\begin{bmatrix} 1 \\ \lambda_j \\ \vdots \\ \lambda_j^{n-2} \\ \lambda_j^{n-1} \end{bmatrix}$ $=\begin{bmatrix} \lambda_j \\ \lambda_j^2 \\ \vdots \\ \lambda_j^{n-1} \\ -a_0-a_1\lambda_j-\cdots-a_{n-1}\lambda_j^{n-1} \end{bmatrix}=\lambda_j\mathbf{v}_j$	$C\mathbf{v}_j=\begin{bmatrix} 0 & 1 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & 1 \\ -a_0 & -a_1 & \cdots & -a_{n-1} \end{bmatrix}\begin{bmatrix} 1 \\ \lambda_j \\ \vdots \\ \lambda_j^{n-2} \\ \lambda_j^{n-1} \end{bmatrix}$ $=\begin{bmatrix} \lambda_j \\ \lambda_j^2 \\ \vdots \\ \lambda_j^{n-1} \\ -a_0-a_1\lambda_j-\cdots-a_{n-1}\lambda_j^{n-1} \end{bmatrix}=\lambda_j\mathbf{v}_j$
johnston-linear-matrix-align-01 EQ1638	482	■ 1.000	$P=\frac{1}{5}\left[\begin{array}{cc} 2 & -1 \\ 1 & 2 \end{array}\right]$	$P=\frac{1}{5}\begin{bmatrix} 2 & -1 \\ 1 & 2 \end{bmatrix}.$	$P=\frac{1}{5}\begin{bmatrix} 2 & -1 \\ 1 & 2 \end{bmatrix}.$
johnston-linear-matrix-align-01 EQ1640	482	■ 1.000	$\begin{aligned} \lambda v_n &= -a_0v_1-a_1v_2-\cdots-a_{n-1}v_n \\ &= v_1(-a_0-a_1\lambda-\cdots-a_{n-1}\lambda^{n-1})=\lambda^nv_1, \end{aligned}$	$\lambda v_n=-a_0v_1-a_1v_2-\cdots-a_{n-1}v_n$ $=v_1(-a_0-a_1\lambda-\cdots-a_{n-1}\lambda^{n-1})=\lambda^nv_1,$	$\lambda v_n=-a_0v_1-a_1v_2-\cdots-a_{n-1}v_n$ $=v_1(-a_0-a_1\lambda-\cdots-a_{n-1}\lambda^{n-1})=\lambda^nv_1,$
johnston-linear-matrix-align-01 EQ1642 (0)	482	■ 1.000	$P=\left[\begin{array}{cccc} p_{\{1,1\}} & 0 & p_{\{1,3\}} & p_{\{1,4\}} \\ p_{\{2,1\}} & 0 & p_{\{2,3\}} & p_{\{2,4\}} \\ 0 & 0 & 0 & p_{1,3} \\ 0 & 0 & 0 & p_{2,3} \end{array}\right]$	$P=\begin{bmatrix} p_{1,1} & 0 & p_{1,3} & p_{1,4} \\ p_{2,1} & 0 & p_{2,3} & p_{2,4} \\ 0 & 0 & 0 & p_{1,3} \\ 0 & 0 & 0 & p_{2,3} \end{bmatrix}.$	$P=\begin{bmatrix} p_{1,1} & 0 & p_{1,3} & p_{1,4} \\ p_{2,1} & 0 & p_{2,3} & p_{2,4} \\ 0 & 0 & 0 & p_{1,3} \\ 0 & 0 & 0 & p_{2,3} \end{bmatrix}.$
johnston-linear-matrix-align-01 EQ1643	482	■ 1.000	$A=\left[\begin{array}{ll} a & b \\ c & d \end{array}\right]$	$A=\begin{bmatrix} a & b \\ c & d \end{bmatrix}$	$A=\begin{bmatrix} a & b \\ c & d \end{bmatrix}$
johnston-linear-matrix-align-01 EQ1644	482	■ 1.000	$\left[\begin{array}{cc} a^2+b\ c & b(a+d) \\ c(a+d) & bc+d^2 \end{array}\right]=\left[\begin{array}{ll} 0 & 1 \\ 0 & 0 \end{array}\right]$	$\begin{bmatrix} a^2+bc & b(a+d) \\ c(a+d) & bc+d^2 \end{bmatrix}=\begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}.$	$\begin{bmatrix} a^2+bc & b(a+d) \\ c(a+d) & bc+d^2 \end{bmatrix}=\begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}.$
johnston-linear-matrix-align-01 EQ1645	482	■ 1.000	$\left[\begin{array}{cc} a^2 & b(a+d) \\ 0 & d^2 \end{array}\right]=\left[\begin{array}{ll} 0 & 1 \\ 0 & 0 \end{array}\right]$	$\begin{bmatrix} a^2 & b(a+d) \\ 0 & d^2 \end{bmatrix}=\begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}.$	$\begin{bmatrix} a^2 & b(a+d) \\ 0 & d^2 \end{bmatrix}=\begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}.$
johnston-linear-matrix-align-01 EQ1646	483	■ 1.000	$\left[\begin{array}{ll} 1 & 0 \\ 0 & 1 \end{array}\right]$	$\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$	$\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$

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johnston-linear-matrix-atgbo4 EQ1647	483	1.000	$\left[\begin{array}{ccc} 2 & 1 & 4 \\ 4 & -2 & 8 \\ 0 & 0 & -8 \end{array}\right]$	$\begin{bmatrix} 2 & 1 & 4 \\ 4 & -2 & 8 \\ 0 & 0 & -8 \end{bmatrix}$	$\begin{bmatrix} 2 & 1 & 4 \\ 4 & -2 & 8 \\ 0 & 0 & -8 \end{bmatrix}$
johnston-linear-matrix-atgbo4 EQ1648	483	1.000	$\left[\begin{array}{cc} 2 & -3 \\ -3 & 2 \end{array}\right]$	$\begin{bmatrix} 2 & -3 \\ -3 & 2 \end{bmatrix}$	$\begin{bmatrix} 2 & -3 \\ -3 & 2 \end{bmatrix}$
johnston-linear-matrix-atgbo4 EQ1649	483	1.000	$\left[\begin{array}{ccc} -3 & 12 & -8 \\ 5 & -4 & 0 \\ 0 & 0 & 0 \end{array}\right]$	$\begin{bmatrix} -3 & 12 & -8 \\ 5 & -4 & 0 \\ -2 & 0 & 0 \end{bmatrix}$	$\begin{bmatrix} -3 & 12 & -8 \\ 5 & -4 & 0 \\ -2 & 0 & 0 \end{bmatrix}$
johnston-linear-matrix-atgbo4 EQ1650	483	1.000	$\left[\begin{array}{cccc} 3 & 27 & -6 & -6 \\ -5 & -31 & 17 & 3 \\ 5 & -11 & 4 & -3 \\ -1 & -2 & -5 & 9 \end{array}\right]$	$\begin{bmatrix} 3 & 27 & -6 & -6 \\ -5 & -31 & 17 & 3 \\ 5 & -11 & 4 & -3 \\ -1 & -2 & -5 & 9 \end{bmatrix}$	$\begin{bmatrix} 3 & 27 & -6 & -6 \\ -5 & -31 & 17 & 3 \\ 5 & -11 & 4 & -3 \\ -1 & -2 & -5 & 9 \end{bmatrix}$
johnston-linear-matrix-atgbo4 EQ1651	483	1.000	$(\operatorname{cof}(A))^{-1}=\frac{1}{\operatorname{det}(A)}A^T$.	$(\operatorname{cof}(A))^{-1}=\frac{1}{\operatorname{det}(A)}A^T$.	$(\operatorname{cof}(A))^{-1}=\frac{1}{\operatorname{det}(A)}A^T$.
johnston-linear-matrix-atgbo4 EQ1652	483	1.000	$(\operatorname{cof}(I))=\operatorname{det}(I)\left(I^T\right)^{-1}=I$.	$(\operatorname{cof}(I))=\operatorname{det}(I)\left(I^T\right)^{-1}=I$.	$(\operatorname{cof}(I))=\operatorname{det}(I)(I^T)^{-1}=I$.
johnston-linear-matrix-atgbo4 EQ1653	483	1.000	$\begin{aligned} & \operatorname{det}(\operatorname{cof}(A)) \\ & =\operatorname{det}\left(\operatorname{det}(A)\right)^{-1} \\ & \left(A^T\right)^{-1} \end{aligned} \quad \& \quad =\left(\operatorname{det}(A)\right)^n \operatorname{det}\left(\left(A^T\right)^{-1}\right) \\ & =\left(\operatorname{det}(A)\right)^n / \operatorname{det}(A) \\ & =\left(\operatorname{det}(A)\right)^{n-1}$	$\begin{aligned} \operatorname{det}(\operatorname{cof}(A)) & =\operatorname{det}\left(\operatorname{det}(A)\left(A^T\right)^{-1}\right) \\ & =\left(\operatorname{det}(A)\right)^n \operatorname{det}\left(\left(A^T\right)^{-1}\right) \\ & =\left(\operatorname{det}(A)\right)^n / \operatorname{det}(A) \\ & =\left(\operatorname{det}(A)\right)^{n-1} \end{aligned}$	$\begin{aligned} \operatorname{det}(\operatorname{cof}(A)) & =\operatorname{det}\left(\operatorname{det}(A)\left(A^T\right)^{-1}\right) \\ & =\left(\operatorname{det}(A)\right)^n \operatorname{det}\left(\left(A^T\right)^{-1}\right) \\ & =\left(\operatorname{det}(A)\right)^n / \operatorname{det}(A) \\ & =\left(\operatorname{det}(A)\right)^{n-1} \end{aligned}$
johnston-linear-matrix-atgbo4 EQ1657	484	1.000	$\lim _{k \rightarrow \infty}\left(A \mathbf{v}_k-\lambda_1 \mathbf{v}_k\right)=\mathbf{0}$	$\lim _{k \rightarrow \infty}\left(A \mathbf{v}_k-\lambda_1 \mathbf{v}_k\right)=\mathbf{0}$	$\lim _{k \rightarrow \infty}\left(A \mathbf{v}_k-\lambda_1 \mathbf{v}_k\right)=\mathbf{0},$
johnston-linear-matrix-atgbo4 EQ1658	484	1.000	$\lim _{k \rightarrow \infty}\left\ A \mathbf{v}_k-\lambda_1 \mathbf{v}_k\right\ =0$	$\lim _{k \rightarrow \infty}\left\ A \mathbf{v}_k-\lambda_1 \mathbf{v}_k\right\ =0$	$\lim _{k \rightarrow \infty}\left\ A \mathbf{v}_k-\lambda_1 \mathbf{v}_k\right\ =0.$
johnston-linear-matrix-atgbo4 EQ1661	484	1.000	$Q=\left[\begin{array}{cc} 1 & 0 \\ -1 & -1 \end{array}\right]$.	$Q=\left[\begin{array}{cc} 1 & 0 \\ -1 & -1 \end{array}\right]$.	$Q=\left[\begin{array}{cc} 1 & 0 \\ -1 & -1 \end{array}\right]$.
johnston-linear-matrix-atgbo4 EQ1662	484	1.000	$Q=\left[\begin{array}{cccc} 0 & 1 & 1 & 0 \\ 1 & 0 & 0 & 0 \\ 1 & 0 & 0 & -1 \\ 0 & 1 & 0 & 1 \end{array}\right]$.	$Q=\left[\begin{array}{cccc} 0 & 1 & 1 & 0 \\ 1 & 0 & 0 & 0 \\ 1 & 0 & 0 & -1 \\ 0 & 1 & 0 & 1 \end{array}\right]$.	$Q=\left[\begin{array}{cccc} 0 & 1 & 1 & 0 \\ 1 & 0 & 0 & 0 \\ 1 & 0 & 0 & -1 \\ 0 & 1 & 0 & 1 \end{array}\right]$.
johnston-linear-matrix-atgbo4 EQ1663	484	1.000	$Q=\left[\begin{array}{cccc} 1 & 0 & 0 & 1 \\ -1 & 1 & 0 & 1 \\ 0 & 1 & -1 & 0 \\ 1 & 1 & 1 & 0 \end{array}\right]$.	$Q=\left[\begin{array}{cccc} 1 & 0 & 0 & 1 \\ -1 & 1 & 0 & 1 \\ 0 & 1 & -1 & 0 \\ 1 & 1 & 1 & 0 \end{array}\right]$.	$Q=\left[\begin{array}{cccc} 1 & 0 & 0 & 1 \\ -1 & 1 & 0 & 1 \\ 0 & 1 & -1 & 0 \\ 1 & 1 & 1 & 0 \end{array}\right]$.
johnston-linear-matrix-atgbo4 EQ1664	485	1.000	$\left[\begin{array}{lll} 1 & 1 & 1 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{array}\right]$	$\left[\begin{array}{lll} 1 & 1 & 1 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{array}\right]$	$\left[\begin{array}{lll} 1 & 1 & 1 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{array}\right]$

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johnston-linear-matrix-algebra EQ1665	485	■ 1.000	$P=\left[\begin{array}{cc} 1 & 1 \\ 1+i & 1-i \end{array}\right], \quad D=\left[\begin{array}{cc} \sqrt{3}+i & 0 \\ 0 & \sqrt{3}-i \end{array}\right].$	$P=\left[\begin{array}{cc} 1 & 1 \\ 1+i & 1-i \end{array}\right], \quad D=\left[\begin{array}{cc} \sqrt{3}+i & 0 \\ 0 & \sqrt{3}-i \end{array}\right].$	$P=\left[\begin{array}{cc} 1 & 1 \\ 1+i & 1-i \end{array}\right], \quad D=\left[\begin{array}{cc} \sqrt{3}+i & 0 \\ 0 & \sqrt{3}-i \end{array}\right].$
johnston-linear-matrix-algebra EQ1666	485	■ 1.000	$\begin{aligned} &A^k=P D^k P^{-1}= \\ &\frac{1}{2}\left[\begin{array}{cc} 1 & 1 \\ 1+i & 1-i \end{array}\right]\left[\begin{array}{cc} (\sqrt{3}+i)^k & 0 \\ 0 & (\sqrt{3}-i)^k \end{array}\right]\left[\begin{array}{cc} 1+i & -i \\ 1-i & i \end{array}\right], \end{aligned}$	$A^k=P D^k P^{-1}= \\ \frac{1}{2}\left[\begin{array}{cc} 1 & 1 \\ 1+i & 1-i \end{array}\right]\left[\begin{array}{cc} (\sqrt{3}+i)^k & 0 \\ 0 & (\sqrt{3}-i)^k \end{array}\right]\left[\begin{array}{cc} 1+i & -i \\ 1-i & i \end{array}\right],$	$A^k=P D^k P^{-1}= \\ \frac{1}{2}\left[\begin{array}{cc} 1 & 1 \\ 1+i & 1-i \end{array}\right]\left[\begin{array}{cc} (\sqrt{3}+i)^k & 0 \\ 0 & (\sqrt{3}-i)^k \end{array}\right]\left[\begin{array}{cc} 1+i & -i \\ 1-i & i \end{array}\right],$
johnston-linear-matrix-algebra EQ1667	485	■ 1.000	$Q=\left[\begin{array}{cc} 1 & 0 \\ 1 & -1 \end{array}\right] \quad \text{and} \quad B=2\left[R^{\pi / 6}\right].$	$Q=\left[\begin{array}{cc} 1 & 0 \\ 1 & -1 \end{array}\right] \quad \text{and} \quad B=2\left[R^{\pi / 6}\right].$	$Q=\left[\begin{array}{cc} 1 & 0 \\ 1 & -1 \end{array}\right] \quad \text{and} \quad B=2\left[R^{\pi / 6}\right].$
johnston-linear-matrix-algebra EQ1668	485	■ 1.000	$\begin{aligned} &A^k \&=Q B^k Q^{-1} \\ &=2^k\left[\begin{array}{cc} 1 & 0 \\ 1 & -1 \end{array}\right]\left[R^{k \pi / 6}\right]\left[\begin{array}{cc} 1 & 0 \\ 1 & -1 \end{array}\right], \end{aligned}$	$A^k=Q B^k Q^{-1} \\ =2^k\left[\begin{array}{cc} 1 & 0 \\ 1 & -1 \end{array}\right]\left[R^{k \pi / 6}\right]\left[\begin{array}{cc} 1 & 0 \\ 1 & -1 \end{array}\right],$	$A^k=Q B^k Q^{-1} \\ =2^k\left[\begin{array}{cc} 1 & 0 \\ 1 & -1 \end{array}\right]\left[R^{k \pi / 6}\right]\left[\begin{array}{cc} 1 & 0 \\ 1 & -1 \end{array}\right],$
johnston-linear-matrix-algebra EQ1669	485	■ 1.000	$x_n=5 x_{n-1}-6 x_{n-2} .$	$x_n=5 x_{n-1}-6 x_{n-2} .$	$x_n=5 x_{n-1}-6 x_{n-2} .$
johnston-linear-matrix-algebra EQ0298 (0)	120	—	—	—	$\left[\begin{array}{cccc cccc} 0 & 2 & 4 & 0 & 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & -1 & 0 & 1 & 0 & 0 \\ 3 & 4 & 2 & 1 & 0 & 0 & 1 & 0 \end{array}\right] \xrightarrow{R_1 \leftrightarrow R_2} \left[\begin{array}{cccc cccc} 1 & 1 & 0 & -1 & 0 & 1 & 0 & 0 \\ 0 & 2 & 4 & 0 & 1 & 0 & 0 & 0 \\ 3 & 4 & 2 & 1 & 0 & 0 & 1 & 0 \end{array}\right]$ $\xrightarrow{R_3-3 R_1} \left[\begin{array}{cccc cccc} 1 & 1 & 0 & -1 & 0 & 1 & 0 & 0 \\ 0 & 2 & 4 & 0 & 1 & 0 & 0 & 0 \\ 0 & 1 & 2 & 4 & 0 & -3 & 1 & 0 \end{array}\right]$ $\xrightarrow{\frac{1}{2} R_2} \left[\begin{array}{cccc cccc} 1 & 1 & 0 & -1 & 0 & 1 & 0 & 0 \\ 0 & 1 & 2 & 0 & 1 / 2 & 0 & 0 & 0 \\ 0 & 1 & 2 & 4 & 0 & -3 & 1 & 0 \end{array}\right]$ $\xrightarrow{\begin{array}{l} R_1-R_2 \\ R_3-R_2 \end{array}} \left[\begin{array}{cccc cccc} 1 & 0 & -2 & -1 & -1 / 2 & 1 & 0 & 0 \\ 0 & 1 & 2 & 0 & 1 / 2 & 0 & 0 & 0 \\ 0 & 0 & 0 & 4 & -1 / 2 & -3 & 1 & 0 \end{array}\right].$
johnston-linear-matrix-algebra EQ0484	196	—	—	—	$\begin{array}{l} 3 w+x+y+z \text { is even } \\ -2 v-w+2 x+y \text { is odd } \\ v+w+y+z \text { is odd } \\ w-x+2 y-z \text { is even, and } \\ v+w-3 x+y+2 z \text { is even. } \end{array}$
johnston-linear-matrix-algebra EQ0520	207	—	—	—	$\begin{array}{cccccc c} z & s_1 & s_2 & s_3 & x_1 & x_2 & x_3 & \\ \left[\begin{array}{cccc ccc} 1 & 0 & 2 & 0 & 0 & -3 & 3 & 4 \\ 0 & 1 & 3 & 0 & 0 & -5 & 5 & 7 \\ 0 & 0 & 1 & 0 & 1 & -1 & 1 & 2 \\ 0 & 0 & 1 & 1 & 0 & 2 & -2 & 5 \end{array}\right] & \begin{array}{l} \text { new “} z \text {” value} \\ \downarrow \end{array} \\ \xrightarrow{\begin{array}{l} (1 / 2) R_4 \\ R_1+3 R_4 \\ R_2+5 R_4 \\ R_3+R_4 \end{array}} & \left[\begin{array}{cccc ccc} 1 & 0 & 7 / 2 & 3 / 2 & 0 & 0 & 0 & 23 / 2 \\ 0 & 1 & 11 / 2 & 5 / 2 & 0 & 0 & 0 & 39 / 2 \\ 0 & 0 & 3 / 2 & 1 / 2 & 1 & 0 & 0 & 9 / 2 \\ 0 & 0 & 1 / 2 & 1 / 2 & 0 & 1 & -1 & 5 / 2 \end{array}\right] & \begin{array}{l} \uparrow \text { new “leading” entry} \end{array} \end{array}$

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johnston-linear-matrix-atlas EQ1295	438	—	—	—	Diagram illustrating the block structure of matrix multiplication. Matrix A is partitioned into blocks $A_{i,1}, \dots, A_{i,n}$ in its i-th row and $A_{1,1}, \dots, A_{1,n}$ in its 1st column. Matrix B is partitioned into blocks $B_{1,1}, \dots, B_{1,s}$ in its 1st row and $B_{n,1}, \dots, B_{n,s}$ in its n-th column. The element $[A_{i,1}]_{r,1}$ is highlighted in blue, and the element $[B_{1,j}]_{c,1}$ is highlighted in blue. Arrows indicate that the k-th row of the i-th block row of A is multiplied by the ℓ-th column of the j-th block column of B.